Canon

imagePRESS V1000

Operation guide

PRISMA

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Data security

Introduction

Security is about safeguarding communication and data and keeping our online society and economy secure. It is critical for our customer’s business and we take it seriously at Canon Production Printing. Security is incorporated in the early stages of our product development process. Our hardware and software products are developed according to industry security standard working methods, and equipped with the required features to protect our printing products and PRISMA workflow and application software against cyber security threats.

Policy / Guidelines / Rules

Canon's Secure Development Life Cycle (SDLC) process is an agile based development process that also models itself comparing with the National Institute of Standards and Technology (NIST) and Open Web Application Security Project (aka OWASP).

It uses the practice of sprints to take advantage of the continuous delivery and flexibility that an agile process enables.

Our back office support program (development of service tools and online services) is ISO/IEC 27001:2013 certified.

Regulation notices

For more detailed information on the security of our products, services and solutions please refer to your local Canon representative or to *https://cpp.canon/security* .

Source code

Introduction

The listed open source components are governed by different licenses. Some of the licenses contain regulations regarding the provision of the pertaining source code.

Policy / Guidelines / Rules

If you would like to obtain a machine-readable copy of the source code of an open source component, which is made available under applicable license terms, you may request such copy of the source code by sending an e-mail to **opensource@cpp.canon** indicating the following:

1. Name and version of the Canon Production Printing software you licensed.
2. Name and version of open source software modules contained in the above identified Canon Production Printing software of which you legitimately would like to obtain the source code.

We will provide you with a find place where you can obtain a machine-readable copy of the source code or we will deliver a complete machine-readable copy of the corresponding source code of the respective open source software component, as available and as is, on a medium of our choice which can be customarily used for software interchange. You may be required to pay the cost incurred for the source code distribution, however, the cost will in no event excess the cost incurred for physically performing the source code distribution.

Regulation notices

This offer is valid to anyone in receipt of this information.

Contents

Chapter 1
Introduction

About the imagePRESS V1000

The imagePRESS V1000 is a sheet-fed digital press printing at 100 ipm. The printer platform supports a wide range of media and has a versatile in-line finishing portfolio.

The imagePRESS V1000 steered by PRISMAsync Print Server offers digital workflows, both black & white and full color, into one effective and productive platform.

Working with the imagePRESS V1000 is more than working with the printer alone. The most important **standard** software applications are the following.

- **PRISMAsync Print Server**
  PRISMAsync Print Server ensures you to get the most out of your printer. PRISMAsync Print Server is used for a wide range of Canon color and black & white sheet-fed printers.
- **Settings Editor**
  The web-based Settings Editor is the tool where you configure the printer according to new or changed requirements.
- **PRISMAsync Remote Manager Classic / PRISMAremote Manager**
  The web-based PRISMAsync Remote Manager / PRISMAremote Manager shows the job processing and the print queues of the printer remotely on your workstation. All upcoming events can be monitored from a distance. You can edit job properties or re-route jobs from a PRISMAsync printer to an other PRISMAsync printer.
- **PRISMAsync Remote Printer Driver**
  PRISMAsync Remote Printer Driver lets you prepare and submit documents to the PRISMAsync printer with a complete set of print job settings. This ensures that the job will suit both your output requirements and the features of the print system.

In addition, the following **cloud service applications** are available for imagePRESS V1000.

- **PRISMAlytics Dashboard**
  The PRISMAlytics Dashboard is a cloud service that delivers fact-based printer information. Printers that use the PRISMAsync Print Server send real-time production information, such as printing times, idle times, and the usage of media and consumables to the Dashboard. Key performance indicators and improvement areas help you to optimize and streamline the production across your printer fleet.
- **PRISMAprepare Go** and **PRISMAsubmit Go**
  PRISMAprepare Go and PRISMAsubmit Go offer a modern cloud-based application for job on-boarding, customer communication, impositioning, and production on Canon monochrome and color production printers and JDF compliant 3rd party printers.
- **PRISMAlytics Accounting**
  PRISMAlytics Accounting is a simple and secure cloud-based management information tool that provides state-of-the-art cost-charging for PRISMAsync-driven production devices. It is capable of collecting accounting information from PRISMAsync-driven devices and presenting this information in clear, standard reports.
- **PRISMAcolor Manager**

PRISMAcolor Manager offers a cloud-based color validation application. This application enables printer operators to measure and validate color on printed media, and compare quality and consistency across Canon and non-Canon CMYK printers over multiple locations.

Further, the following **optional** software applications can be part of the platform of the imagePRESS V1000.

- **PRISMAremote Monitoring**
  The PRISMAremote Monitoring app on your smartphone helps to stay informed about the print production on the imagePRESS V1000. Even at a distance.
- **PRISMAprepare**
  PRISMAprepare is a powerful all-in-one make-ready solution that accelerates document preparation, from composition through production.
- **PRISMAproduction**
  PRISMAproduction create a uniform workflow with the imagePRESS V1000. So you can realize end-to-end job automation, media catalog synchronization, and all-in-one job preparation.
- **PRISMAdirect**
  PRISMAdirect is a web-based end-to-end workflow management solution that streamlines the entire order and production process.

About this guide

This guide is written for worldwide markets. It may contain information related to accessories or licensed functionalities not supported by your local Canon organization or dealer.

Language

Original instructions are in English.

Read this operation guide to learn what this product can do for you, how to operate and maintain the print system and how to use it in a safe way.

Topic	Chapter
Introduction, information in this guide, users of the printer	1
Installation and handling precautions	2
Printer configuration and parts	3
Operation of the printer	4
Defaults for the control panel, printer, jobs and print quality	5
Handling of media, envelopes, tab paper and long sheets	6
Copy and scan job preparation	7
Transaction printing	8
Document printing: plan, change, check, print and monitor jobs	9
Printer calibration, media family calibration, G7© media families, color validation	10
Media catalog, media, media families	11
Color defaults, color presets, profiles, trapping presets, spot colors, CMYK curves, color mappings, color and information bars	12
Refill toner and staples, remove staple and punch waste, clean printer parts	13
Solutions for common problems, log files	14
Specifications, feed instructions and status indicators	15

Optionals described in this manual

This operation guide includes the functions and descriptions of the following paper input optionals:
- Stack Bypass-D
- Multi-drawer Paper Deck-E
- POD Deck Lite-C
- POD Deck Lite XL-A
- Document Insertion Unit-R

The functions and descriptions of the following finishing optionals are also included:
- Paper Folding Unit-K
- Staple Finisher-AF/Booklet Finisher-AF
- Finisher Bridge-A
- DFD Adapter-B

Other product information

On the downloads site **downloads.cpp.canon** you can find the following information for the imagePRESS V1000:
- This manual
- High Capacity Stacker-J operation guide
- Technical reference manual
- Safety guides of the PRISMAsync Print Server and options
- imagePRESS V1000 Third party software

The following optionals have a manual that is delivered together with the product:
- Multi Function Professional Puncher-C
- Perfect Binder-F
- Booklet Trimmer-G
- Two-Knife Booklet Trimmer-B

Users of the printer

Target group of the imagePRESS V1000 printer

Print environments where the imagePRESS V1000 printer is installed differ in many aspects. Therefore, it is difficult to describe the user roles that are applicable to all print environments. The imagePRESS V1000 has several default user accounts: system administrator, key operator, central operator, maintenance operator, operator, maintenance specialist, and advanced maintenance specialist.

Target group of this operation guide

This operation guide is developed to support users that are responsible for the job management, workflow management, color and media management, print quality and color validation, and printer maintenance.

The system administrator is responsible for the security, the connectivity, the user authorization and authentication, and several workflow configurations. These tasks are not part of this operation guide.

Notes for the reader

Introduction

This manual helps you to use the imagePRESS V1000. The manual contains a description of the product and guidelines to use and operate the imagePRESS V1000.

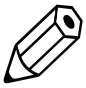
Definition

Attention Getters

Parts of this manual require your special attention. These parts can provide the following:
- Additional general information, for example information that is useful when you perform a task.
- Information to prevent personal injuries or property damage.

Symbols used in this manual

The following symbols are used in this manual to explain procedures, restrictions, handling precautions, and instructions that should be observed for safety.

Word	Icon	Indicates
WARNING		Indicates a warning concerning operations that may lead to death or injury to persons if not performed correctly. To use the machine safely, always pay attention to these warnings.
CAUTION		Indicates a caution concerning operations that may lead to injury to persons if not performed correctly. To use the machine safely, always pay attention to these cautions.
IMPORTANT		Indicates operational requirements and restrictions. Be sure to read these items carefully to operate the machine correctly and to avoid damaging the machine or property.
NOTE		Indicates a clarification of an operation or contains additional explanations for a procedure. Reading these notes is highly recommended.
-		Indicates an operation that must not be performed. Read these items carefully and make sure not to perform the described operations.

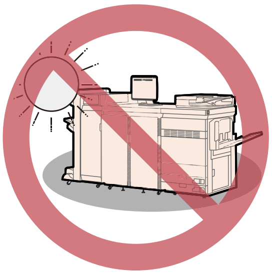
Overview of the attention-getters

Chapter 2
Precautions

Installation precautions

Avoid installing the machine in the following locations

- **Avoid locations subject to temperature and humidity extremes, whether low or high.**
  For example, avoid installing the machine near water faucets, hot water heaters, humidifiers, air conditioners, heaters, or stoves.
- **Avoid installing the machine in direct sunlight.**
  If this is unavoidable, use curtains to shade the machine. Be sure that the curtains do not block the machine's ventilation slots or louvers, or interfere with the electrical cord or power supply.

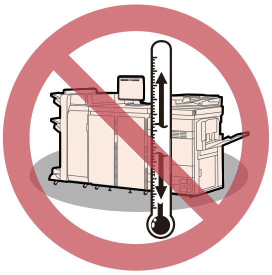


- **Avoid poorly ventilated locations.**
  This machine generates a slight amount of ozone and other gases during normal use. Although sensitivity to the gases may vary, this amount is not harmful. The presence of ozone and other gases may be more noticeable during extended use or long production runs, especially in poorly ventilated rooms. It is recommended that the room be appropriately ventilated, sufficient to maintain a comfortable working environment, in areas of machine operation. In addition, do not install this machine where it exhausts directly onto a person.

- **Avoid locations where a considerable amount of dust accumulates.**
- **Avoid locations where ammonia gas is emitted.**

- **Avoid locations near volatile or flammable materials, such as alcohol or paint thinner.**
- **Avoid locations that are subject to vibration.**
  For example, avoid installing the machine on unstable floors or stands.

- **Avoid exposing the machine to rapid changes in temperature.**
  If the room in which the machine is installed is cold but rapidly heated, water droplets (condensation) may form inside the machine. This may result in a noticeable degradation in the quality of the copied image, the inability to properly scan an original, or the copies having no printed image at all.



- **Avoid installing the machine near computers or other precision electronic equipment.**
  Electrical interference and vibrations generated by the machine during printing can adversely affect the operation of such equipment.
- **Avoid installing the machine near televisions, radios, or similar electronic equipment.**
  The machine might interfere with sound and picture signal reception. Insert the power plug into a dedicated power outlet, and maintain as much space as possible between the machine and other electronic equipment.
- **Do not remove the machine's levelling feet.**
  Do not remove the machine's leveling feet after the machine has been installed.
  If you put weight on the front of the machine while the drawers are pulled out, the machine may fall forward. To prevent this from happening, make sure that the machine's leveling feet are in place.
- **Avoid installing the machine at high altitudes of about 9,800 ft. (3,000 meters) above sea level or higher.**
  Machines with a hard disk may not operate properly when used at high altitudes of about 9,800 feet above sea level or higher.

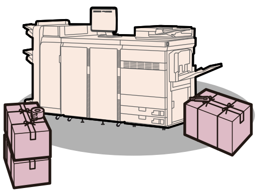
Select a safe power supply

- **This machine has three power cords. Connect each plug of the machine to the independent outlet that is 220 to 240 V AC 13 A or greater.**
- **Make sure that the power supply for the machine is safe, and has a steady voltage.**
- **Do not connect other electrical equipment to the same power outlet to which the machine is connected.**
- **Do not connect the power cord to a multiplug power strip, as this may cause a fire or electrical shock.**
- **The power cord may become damaged if it is stepped on, affixed with staples, or if heavy objects are placed on it. Continued use of a damaged power cord can lead to an accident, such as a fire or electrical shock.**

- **The power cord should not be taut, as this may lead to a loose connection and cause overheating, which could result in a fire.**
- **If excessive stress is applied to the connection part of the power cord, it may damage the power cord or the wires inside the machine may disconnect. This could result in a fire. Avoid the following situations:**
  - Connecting and disconnecting the power cord frequently.
  - Tripping over the power cord.
  - The power cord is bent near the connection part, and continuous stress is being applied to the power outlet or the connection part.
  - Applying excessive force on the power plug.

Moving the machine

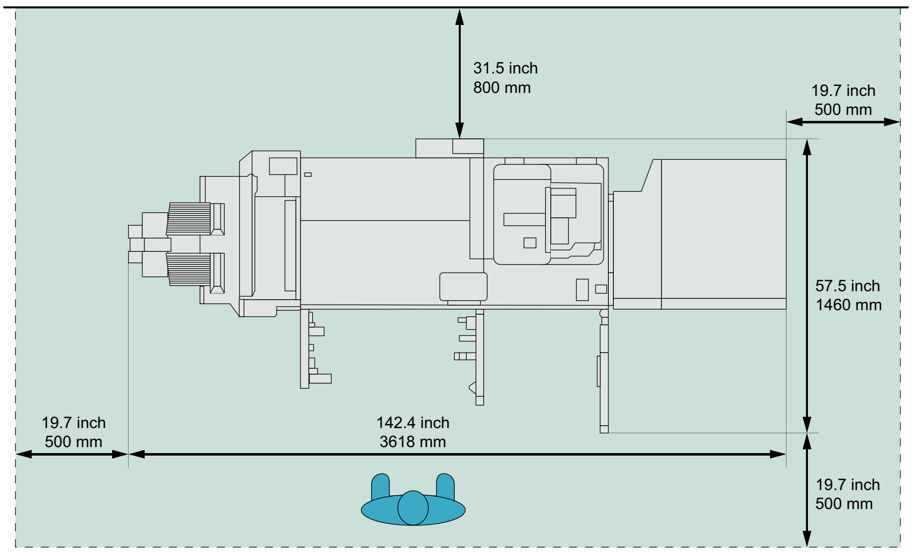
If you intend to move the machine, even to a location on the same floor of your building, contact your local authorized Canon dealer beforehand. Do not attempt to move the machine yourself.



Provide adequate installation space

Provide enough space on each side of the machine for unrestricted operation.

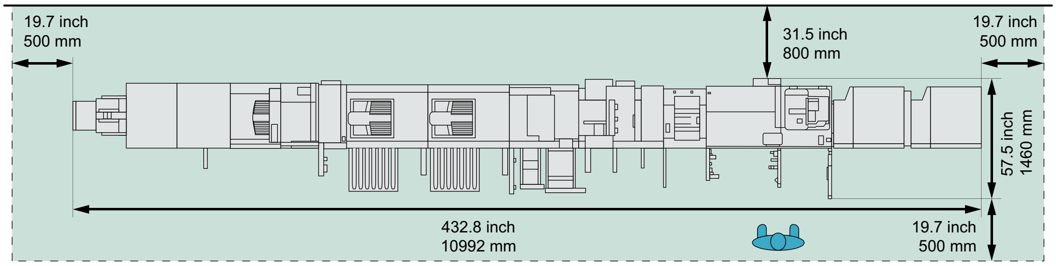
When the Staple Finisher-AF and Multi-drawer Paper Deck-E are attached:



Standard (minimum) configuration

When the Two-Knife Booklet Trimmer-B, Booklet Trimmer-G, Booklet Finisher-AF, Paper Folding Unit-K, High Capacity Stacker-J (2), Perfect Binder-F, Document Insertion Unit-R, Inspection

unit-B, Sensing unit-B, Multi-drawer Paper Deck-E (2) and Multi Function Professional Puncher-C are attached:



Full (maximum) configuration

Handling precautions

Do not attempt to disassemble or modify the machine.

Some parts inside the machine are subject to high voltages and temperatures. Take adequate precautions when inspecting the inside of the machine. Do not carry out any inspections that are not described in the manuals for this machine.

Be careful not to spill liquid or drop any foreign objects, such as paper clips or staples inside the machine. If a foreign object comes into contact with electrical parts inside the machine, it might cause a short circuit and result in a fire or electrical shock.

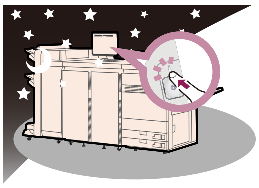
If there is smoke or unusual noise, immediately turn the main power switch OFF, disconnect the power cord from the power outlet, and then call your local authorized Canon dealer. Using the machine in this state may cause a fire or electrical shock. Also, avoid placing objects around the power plug so that the machine can be disconnected whenever necessary.

Do not turn the main power switch OFF or open the front covers while the machine is in operation. This may result in paper jams.

Do not use flammable sprays, such as spray glue, near the machine. There is a danger of ignition.

For safety reasons, press the sleep button ☾ when it will not be used for a long period of time, such as overnight. As an added safety measure, turn OFF the main power switch, and disconnect

the power cord when the machine will not be used for an extended period of time, such as during consecutive holidays.



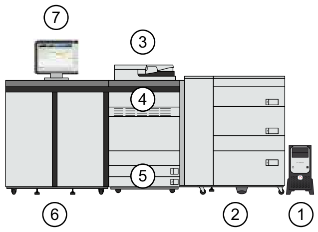
Paper that has just been output from the machine may be hot. Be careful when removing or aligning paper from the output tray. Touching paper right after its output may result in low-temperature burns.

Chapter 3
Explore the printer

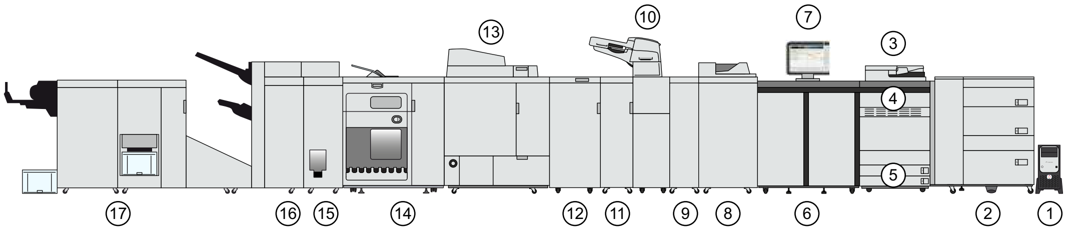
Main hardware parts

The print system can be given different configurations to match various print environments. The print system has basic hardware parts extended with optionals. In the table below you can find descriptions of the main parts.

NOTE
- This manual uses functional names for the print system parts. However, sometimes you may need the commercial name, for example, if you want to order an optional. In this case, use the table with the commercial names.
- The availability of optionals differs per country.



Base print system with bulk paper module



Example of a print system extended with several input and output optionals

	Description of main hardware parts
1	PRISMAsync Print Server steers the performance, workflow, image quality, and color quality of the print system.
2	Paper module to hold media
3	Automatic document feeder to copy and scan originals
4	Toner compartment cover to access the toner cartridges
5	Internal paper trays to hold media
6	Print module where printing takes place
7	Control panel, which is the central information point for the operator
8	Sensing unit to automatically perform media registration, color density adjustments and more
9	Inspection unit to automatically inspect the print quality of output based on a reference image
10	Inserter to feed (preprinted) covers and insert sheets
11	Static elimination unit to remove static electricity from media during printing

	Description of main hardware parts
12	Professional puncher with replaceable die sets to punch holes in the printed output or to crease the printed output
13	Perfect binder to create perfect-bound and trimmed books
14	High-capacity stacker to stack and order large piles of output
15	Folder to fold the output in various ways
16	Stacker/stapler to staple and stack the printed output. A booklet maker creates saddle-stitched booklets. A punch unit punches printed output.
17	Booklet trimmer to trim the leading edge of booklets and two-knife trimmer to trim the top- and bottom-edges of booklets

Commercial names of print module optionals

Optional	Commercial name
Automatic document feeder	Duplex Color Image Reader Unit-P
Printer cover	Printer Cover-P
Special feeder	Stack Bypass-D
Paper module	Multi-drawer Paper Deck-E
Bulk paper module	POD Deck Lite-C POD Deck Lite XL-A
Inserter	Document Insertion Unit-R
Sensing unit	Sensing unit-B
Inspection unit	Inspection unit-B

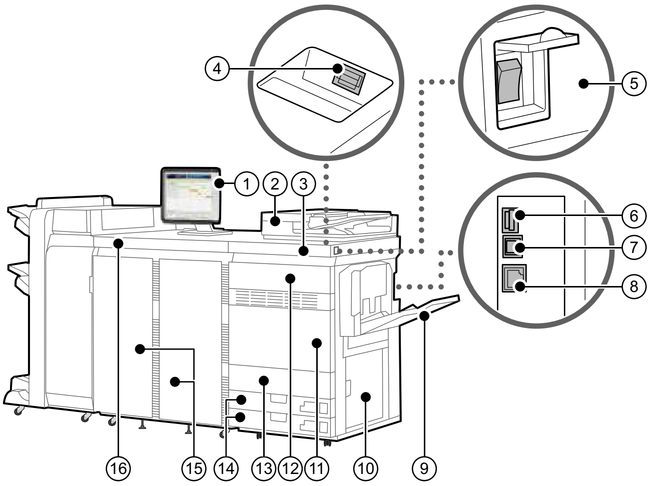
For more information, see *Paper input specifications* on page 618.

Commercial names of finishing optionals

Optional	Commercial name
Static elimination unit	Canon Static Eliminator-A1
Professional puncher	Multi Function Professional Puncher-C
Perfect binder	Perfect Binder-F
High capacity stacker	High Capacity Stacker-J
Folder	Paper Folding Unit-K
Stacker/stapler	Staple Finisher-AF or Booklet Finisher-AF
Booklet trimmer	Booklet Trimmer-G
Two-knife booklet trimmer	Two-Knife Booklet Trimmer-B
Finisher bridge	Finisher Bridge-A
DFD (Document Finishing Device) adapter	DFD Adapter-B
eWire	GBC eWire Pro

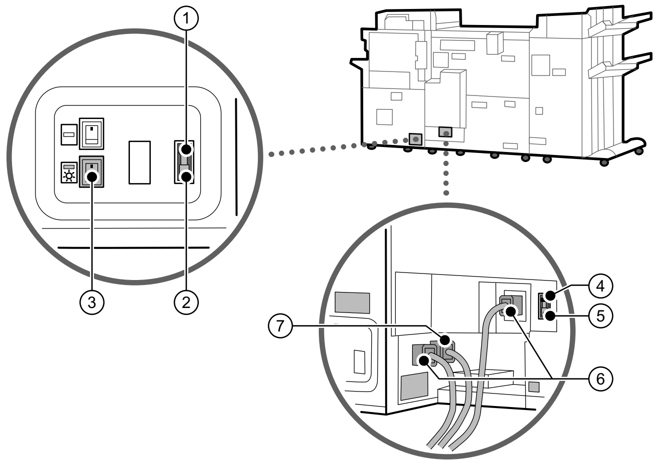
For more information, see *Finishers specifications* on page 624.

Print module parts



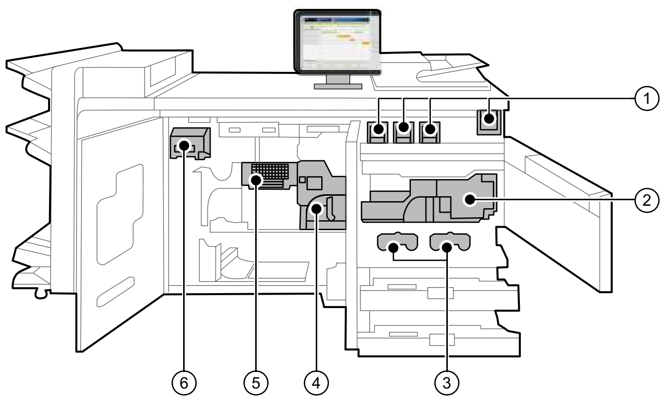
External view of the print module (front)

	Description print module
1	Control panel to operate the print system and perform daily tasks and maintenance
2	Automatic document feeder to copy and scan originals (optional)
3	Print module
4	USB port (front side) to connect devices such as a USB memory device to the printer NOTE The USB port supports USB 2.0.
5	Main power switch
6	USB port (rear side) to connect devices such as an external hard disk to the printer NOTE The USB port supports USB 3.0.
7	USB connector to connect a USB cable when connecting the printer and a computer NOTE The USB connector supports USB 2.0.
8	LAN port to connect the print module to the print server
9	Special feeder to manually feed special media, envelopes and tab paper
10	Lower right cover of the print module
11	Front cover of the print module
12	Toner compartment cover to access the toner cartridges
13	Waste toner container cover to access the waste toner containers
14	Internal paper trays to hold the media
15	Open the covers of the fixing station
16	Fixing station



External view of the printer module (back)

	Description print module
1	Breaker of the print module to detect excess current or leakage current
2	Test button of the print module to periodically test the circuit breaker
3	Dehumidification switch to dry media
4	Breaker of the fixing station to detect excess current or leakage current
5	Test button of the fixing station to periodically test the circuit breaker
6	Sub-power cords
7	Main power cord



Internal view of the printer module

	Description print module
1	Toner cartridge
2	Fixing transport unit
3	Waste toner containers
4	Fixing unit to fix the toner transferred to media
5	Cooling unit to cool down media

	Description print module
6	Decurler unit to correct media curling caused by heat

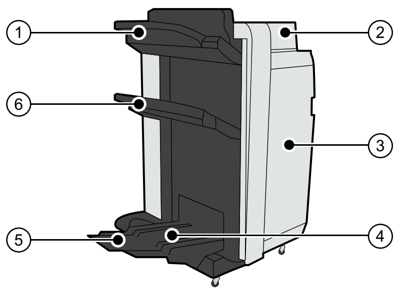
Finishers

This topic describes the parts of the following finishers:
- Stacker/stapler
- Folder
- Long sheet catch tray
- Bridge and finisher adapter for DFD (Document Finishing Device)

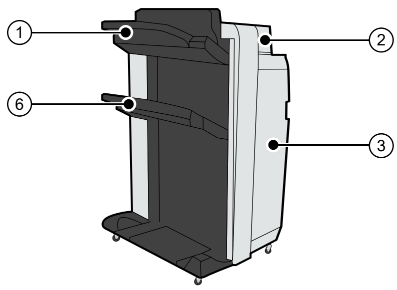
Stacker/stapler

There are two standard stacker/stapler configurations: the stacker/stapler with booklet maker (Booklet Finisher-AF) and without a booklet maker (Staple Finisher-AF). The booklet maker folds and staples booklets. Both finisher configurations can enable the optional punch unit to punch two, three or four holes in the printed output.

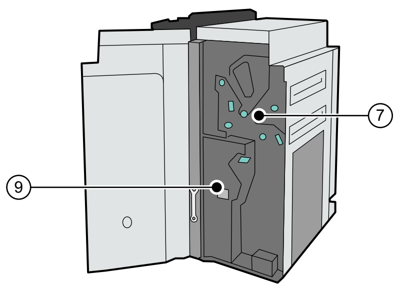
The dashboard of the control panel displays the status of the staple cartridges. When the color of the staple icon is orange or red, the staple cartridge needs replacement.



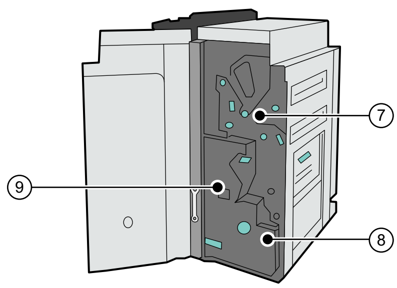
Stacker/stapler with booklet maker (external view)



Stacker/stapler without booklet maker (external view)



Stacker/stapler with booklet maker (internal view)



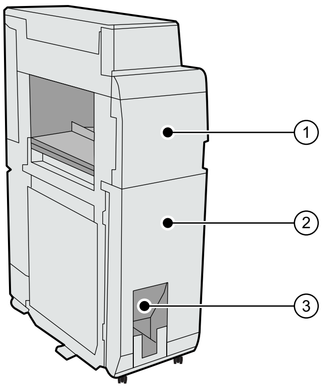
Stacker/stapler without booklet maker (internal view)

	Description stacker/stapler
1	Upper tray to collect finished output; the guides enable a correct placement of the printed output.
2	Top cover to access the area to clear a paper jam
3	Front cover to access the area where you replace staple cartridges, clear a paper jam or a staple jam
4	Booklet tray to collect booklets
5	Auxiliary booklet tray to collect large booklets
6	Lower tray to collect finished output; the guides enable a correct placement of the printed output.
7	Punch waste tray to collect the punch waste

	Description stacker/stapler
8	Staple waste tray to collect the staple waste
9	Saddle-stitch unit to access the area where you replace staple cartridges, clear a paper jam or a staple jam in a saddle-stitch unit

Folder

The folder (Paper Folding Unit-K) can fold the printed output in various ways: Z-fold, tri-fold in, half fold, tri-fold out, parallel fold.

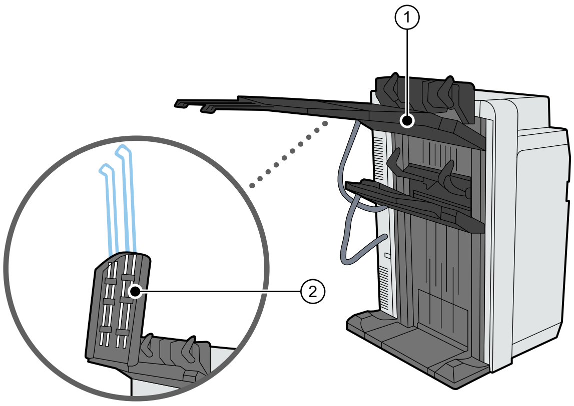


Folder

	Description folder
1	Front cover to access the area where a paper jam occurred
2	Folding unit that you can pull out to clear a paper jam
3	Exit slot where the printed output folded in tri-fold in, tri-fold out or parallel fold is output NOTE The printed output folded in Z-fold or in half fold is output to the output tray of the finisher.

Long sheet catch tray

When printing on media which is longer than 762 mm (30 inch), attach the long sheet catch tray to the stacker/stapler.

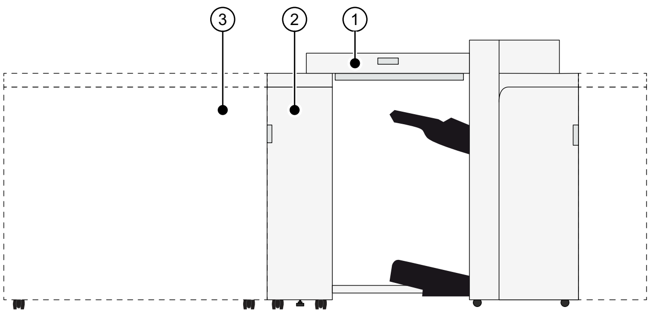


Long sheet catch tray

	Description long sheet catch tray
1	Long sheet output tray
2	Extension long sheet output tray to support long sheets

Bridge and finisher adapter for DFD (Document Finishing Device)

The DFD (Document Finishing Device) is a third-party external finisher such as a booklet maker or a perforator.



Bridge and finisher adapter

	Description
1	Bridge (Finisher Bridge-A) to connect the finisher adapter to the stacker/stapler
2	Finisher adapter (DFD Adapter-B) to connect the DFD to the bridge
3	DFD to finish printed output

Chapter 4
Getting started

Interfaces to access the printer

There are several starting points to use the imagePRESS V1000. Whether you access the printer via the control panel or remotely via PRISMAremote Manager / PRISMAsync Remote Manager Classic, you quickly find your way while operating the imagePRESS V1000. All interfaces have the same design approach and look and feel.

You can submit jobs with PRISMAsync Remote Printer Driver, PRISMAremote Manager / PRISMAsync Remote Manager Classic, PRISMAprepare, PRISMAproduction, UPD printer driver or Mac driver.

You edit or plan jobs on the control panel or with PRISMAsync Remote Manager. The control panel, PRISMAremote Manager / PRISMAsync Remote Manager Classic and PRISMAremote Monitoring show upcoming actions and messages. That way, you get informed about all upcoming and immediate tasks.

You configure the imagePRESS V1000 from the web-based configuration tool, the Settings Editor, and the control panel.

This topic describes the following interfaces:
- Control panel
- PRISMAsync Remote Printer Driver
- Settings Editor
- PRISMAremote Manager / PRISMAsync Remote Manager Classic
- PRISMAremote Monitoring

Below you find an introduction how to use the imagePRESS V1000 via remote printer driver, the control panel, PRISMAremote Manager / PRISMAsync Remote Manager Classic, PRISMAremote Monitoring, and the web-based configuration tool, the Settings Editor.

Control panel

The control panel is local point to operate the printer and the domain of the operator and the maintenance operator.

Many instructions in this operation guide explain job editing and job management from the control panel.

Operator panel

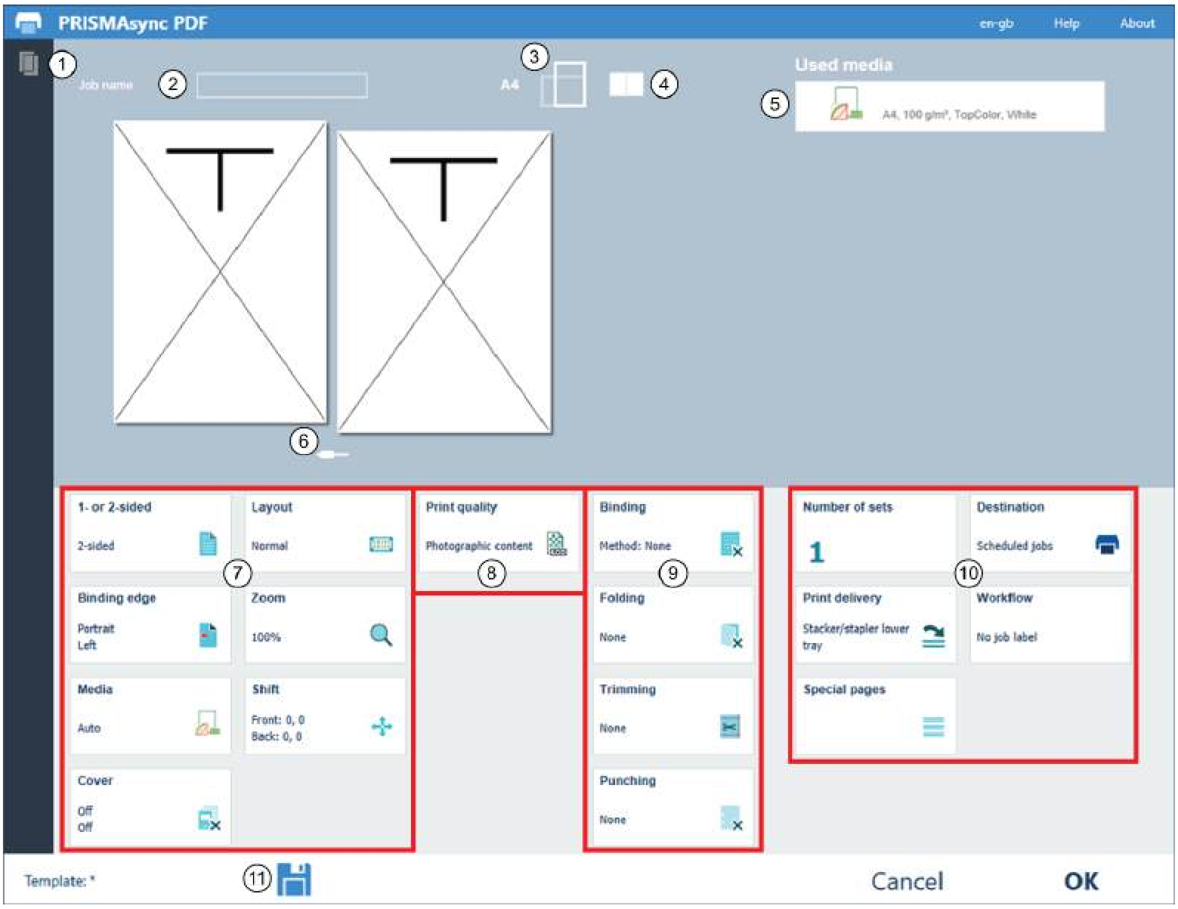
The table below describes the main parts of the control panel and their functions.

	Description control panel
1	Sleep button to bring the print system into the sleep mode or to awake the print system from the sleep mode
2	Stop button to stop the print process immediately, after a set, after a job, after a record, or as soon as possible
3	Paper tray button to access paper tray functions and overviews while the active window remains available
4	USB port, to enable printing from a USB drive or connecting the i1Pro3 spectrophotometer
5	[Schedule] button to access the schedule
6	[Jobs] button to access job locations: list of scheduled jobs, list of waiting jobs, list of printed jobs, list of scan jobs and the DocBox folders
7	[Trays] button to access paper tray functions and overviews (see the paper tray button (3))
8	[System] button, to access system and support features, maintenance tasks, consumable supplies, and workflow preferences
9	Access key button to display the user name that is currently logged in, to log in as another user, or to change your password
10	Dashboard to display all kinds of status information
11	Status LED to indicate the system status with a color

PRISMAsync Remote Printer Driver

The printer driver is the remote point to prepare a print job from a desktop application, such as Microsoft Office and Adobe Acrobat. The job properties in the printer driver match the job properties on the control panel and PRISMAsync Remote Manager. A job submitted with the printer driver can easily be changed when the job is visible in the print queue on the control panel of in PRISMAremote Manager / PRISMAsync Remote Manager Classic.

In the illustration and table below, you see the main parts of the printer driver. Use the online help of the printer driver if you need more information.



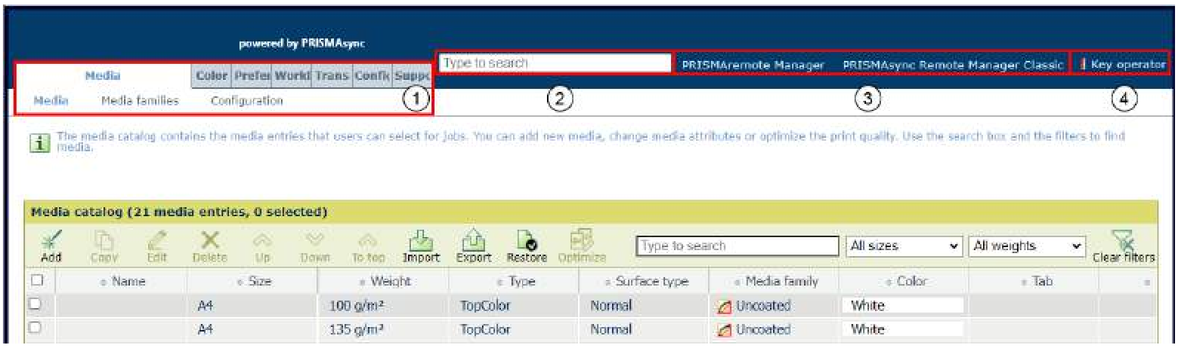
PRISMAsync Remote Printer Driver

	Description control panel
1	Printer status and available media
2	General job settings
3	
4	
5	
6	
7	Layout settings
8	Colour quality and print quality settings
9	Finishing settings

	Description control panel
10	Job production settings
11	Job templates

Settings Editor

The Settings Editor is the location to configure the printer. The Settings Editor uses the same language as the web browser.



Settings Editor

The table below describes the main parts of the Settings Editor.

	Description
1	The settings are grouped in main and sub tabs.
2	Use the search box to quickly find a setting.
3	Use the link to start PRISMAremote Manager / PRISMAsync Remote Manager Classic.
4	Click the user name to log in to the Settings Editor, log out or change the password. A red exclamation mark indicates that a default password is used. You are advised to change default passwords.

A setting can show an icon to provide more information on the use. The table below explains the icons.

	Setting is editable
	Setting not editable
	Setting is also available on the control panel
	Setting can be changed with key operator or system administrator credentials

PRISMAremote Manager / PRISMAsync Remote Manager Classic

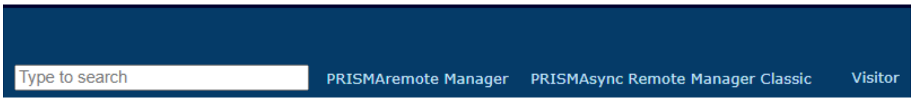
PRISMAremote Manager / PRISMAsync Remote Manager Classic is a multi-printer schedule and remote management console to control PRISMAsync Print Server-driven printers.

This web-based tool helps central operators to manage their print production and to provide insight into the production schedules.

There are two versions of the tool. The main difference between the versions is that PRISMAremote Manager has a fully customisable user interface.

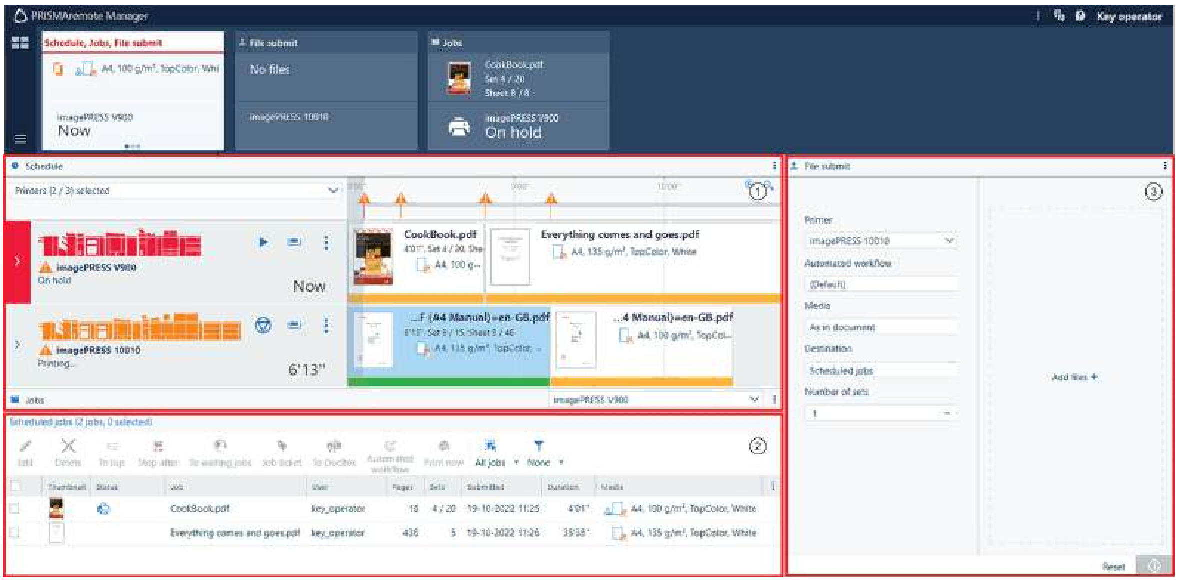
PRISMAremote Manager / PRISMAsync Remote Manager Classic enables to upload jobs, to edit job properties and to change the print queues from a remote location or workstation. Jobs can easily be rescheduled , relocated or changed before being printed.

You can start the tool from the Settings Editor.



Start Remote Manager

Use the help function or user guide of PRISMAremote Manager / PRISMAsync Remote Manager Classic for more information.



User interface of PRISMAremote Manager

	Description
1	A schedule that you can use to: • view the printers of the cluster, their status and activity; • monitor upcoming and immediate actions; • read the job status; • read or change the job settings in the schedule.
2	A list of scheduled jobs on a printer The toolbar has a number of buttons to edit, delete, move, apply automated workflow and print jobs.

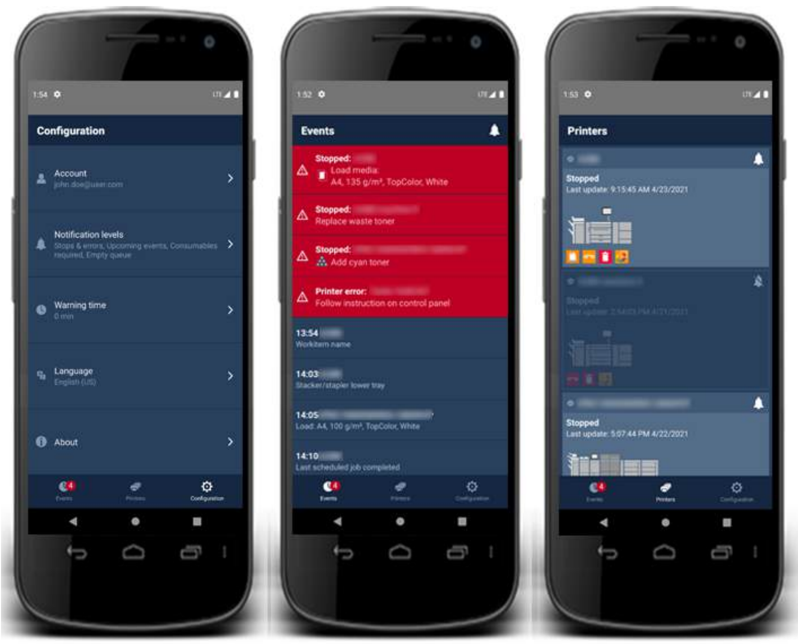
	Description
3	Submission of new print jobs

PRISMAremote Monitoring

The PRISMAremote Monitoring app helps the central operator to stay informed about the print production. Even at a distance.

You receive alerts when actions, such as loading media or adding consumables, are foreseen. Or, in case a problem at the printer asks for an immediate action.

The table below describes three screens of PRISMAremote Monitoring.



Screens with status and events on the smartphone

Refer to the quick reference guide of PRISMAremote Monitoring for more information.

More information

- *Learn about system configuration and maintenance* on page 500
- *Status indicators* on page 588
- *Work with the schedule in document printing mode* on page 243
- *Settings Editor specifications* on page 633

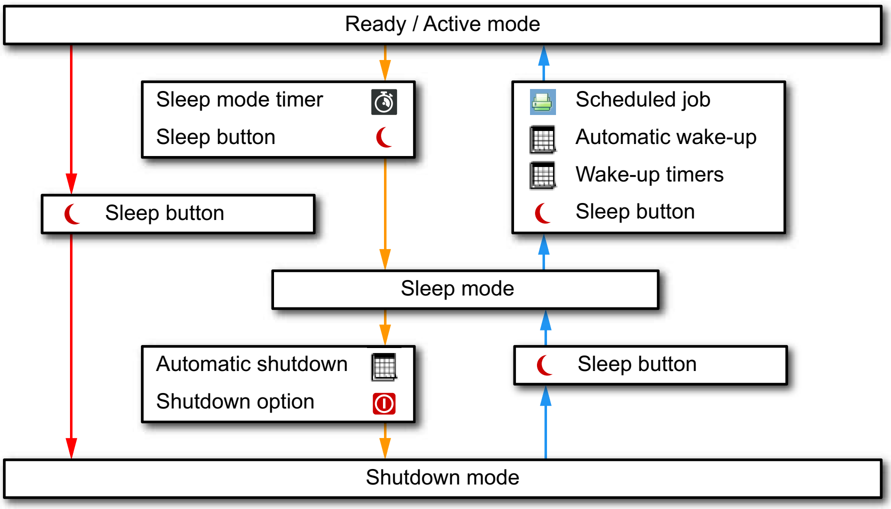
Learn about power states of the printer

Power states of the imagePRESS V1000

The imagePRESS V1000 has four power states.

- **Active**: the printer is printing jobs, processing jobs, or performing maintenance tasks.
- **Ready**: the printer is ready to accept user commands.
- **Sleep mode**: the printer uses an energy save level and cannot accept user commands. The sleep mode can be activated by the sleep button or the sleep mode timer.
- **Shutdown mode**: The printer uses an energy save level and cannot accept user commands. The shutdown mode can be activated via the control panel or the automatic shutdown function.

Overview power states



What is sleep mode

When you do not operate the printer, you can reduce the energy consumption with the **sleep mode**.

During the sleep mode the printer behaves as follows:

- The control panel is in energy save mode.
- PRISMAsync Print Server remains switched on.
- The print engine and the attached modules are switched off.

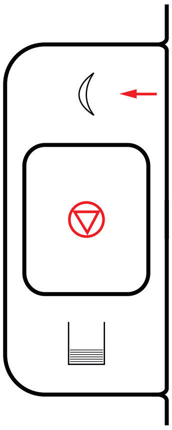
NOTE
It depends on the sleep mode configuration how the printer reacts to new scheduled jobs.

What is the sleep button

You use the sleep button in the following situations:

- Put the printer in sleep mode. When the printer is ready, the sleep mode is entered immediately. When the printer is active, the sleep mode is entered as soon as all scheduled jobs are printed.
- Put the printer in shutdown mode.
- Wake up the printer from sleep mode.

- Wake up the printer from shutdown mode.



Sleep button

What is the sleep mode timer

You can define the period of inactivity before the printer automatically enters the sleep mode. The default value of the sleep mode timer is 40 minutes. When the printer is ready, the sleep mode is entered immediately. When the printer is active, the sleep mode is entered as soon as all scheduled jobs are printed.

What is automatic wake-up from sleep mode

Use automatic wake-up to awake the printer from sleep mode at a configured time.

In the Settings Editor you define the days and times that the printer automatically wakes up.

What is shutdown mode

You can **shut down** the printer with the option or the sleep button to further reduce the energy consumption.

During the shutdown the printer behaves as follows:
- The control panel is in energy save mode.
- PRISMAsync Print Server is switched off.
- The print engine and the attached modules are switched off.

IMPORTANT
Do not use the main power switch.

What is automatic shutdown

To save energy during recurring periods that the printer is not productive, for example overnight or over the weekend, use **automatic shutdown**.

In the Settings Editor you define the days and times that the printer automatically shuts down.

The printer can only automatically shut down when it is in the sleep mode. Ensure that the printer is in sleep mode before the configured shutdown time.
- The sleep mode timer must have been expired, or
- You must press the sleep button and touch [To sleep mode].

The printer leaves the shutdown mode when you press the **sleep button** .

More information

Configure sleep mode, automatic wake-up, and shutdown mode on page 74

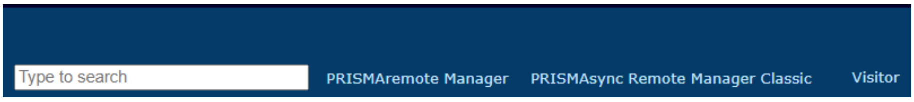
Log in, log out, and change passwords

Log in to the printer

Access without login

It depends on the Settings Editor setting [Configuration]→[Users]→[Configure user login]→[Access to control panel] if users are allowed to view and change the control panel settings without user authentication.

It depends on the Settings Editor setting [Configuration]→[Security]→[Passwords]→[Permission to view Settings Editor] if users are allowed to view the Settings Editor without user authentication. If you can view the Settings Editor without logging in, the name *Visitor* is shown.

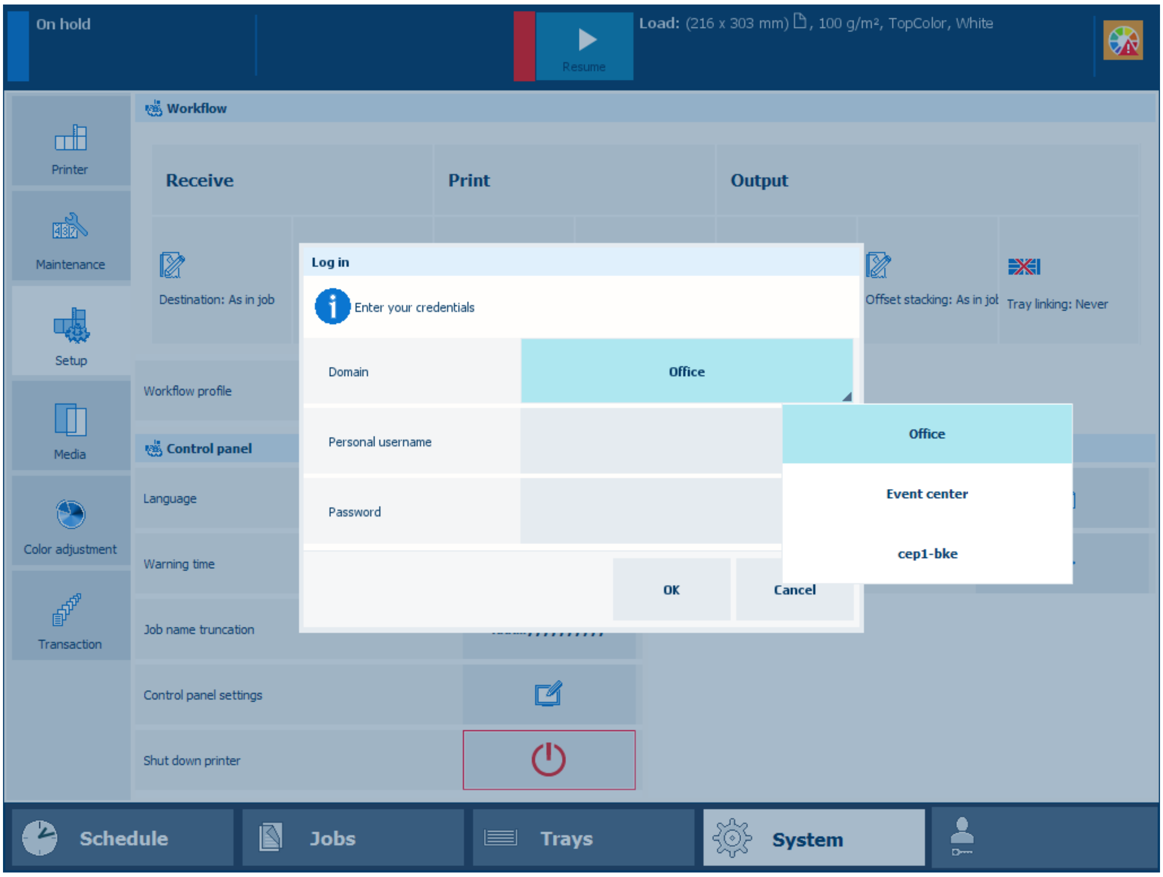


Visitor is logged in

Log in with a domain user account

When there are no configured domains, you do not see the domain selection on the login panel. When at least one domain has been added, a drop-down list is shown.

Your corporate username combined with the selected domain and the suffix defined for the domain make the Universal Personal Name (UPN). This UPN is needed to login to the LDAP directory server.

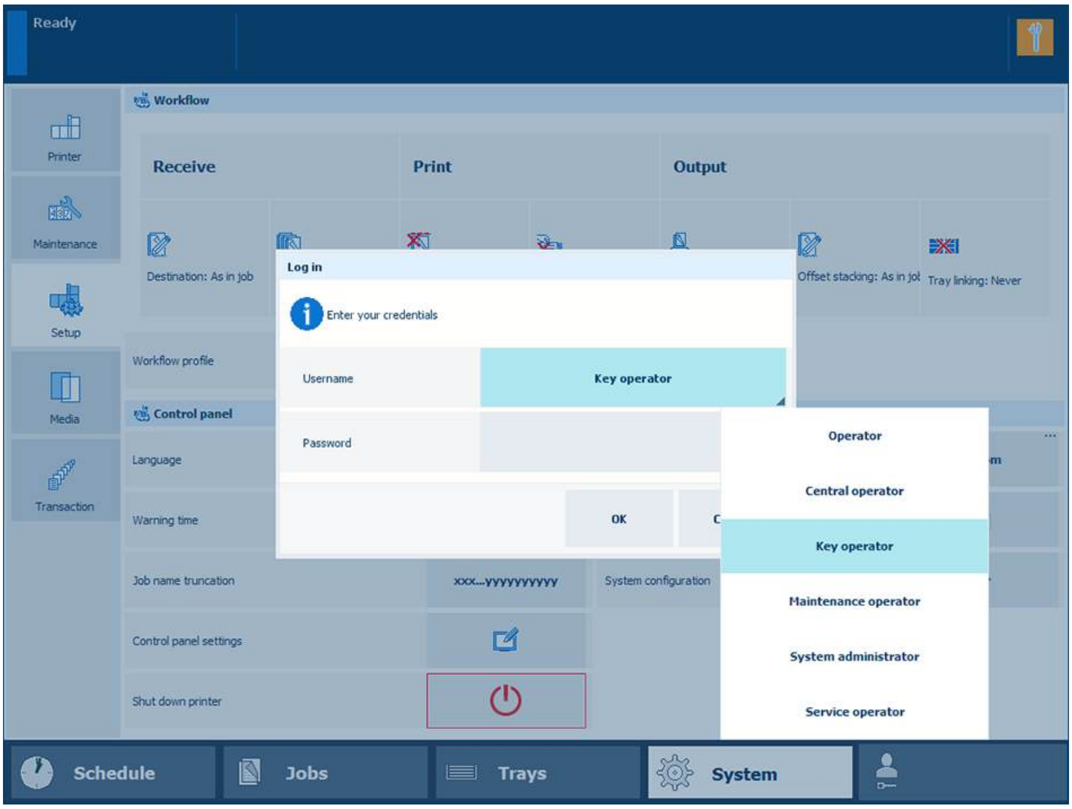


Log in with domain user account

1. Select the domain from the [Domain] drop-down list.
2. Enter your username and password.
3. Touch or click [OK].

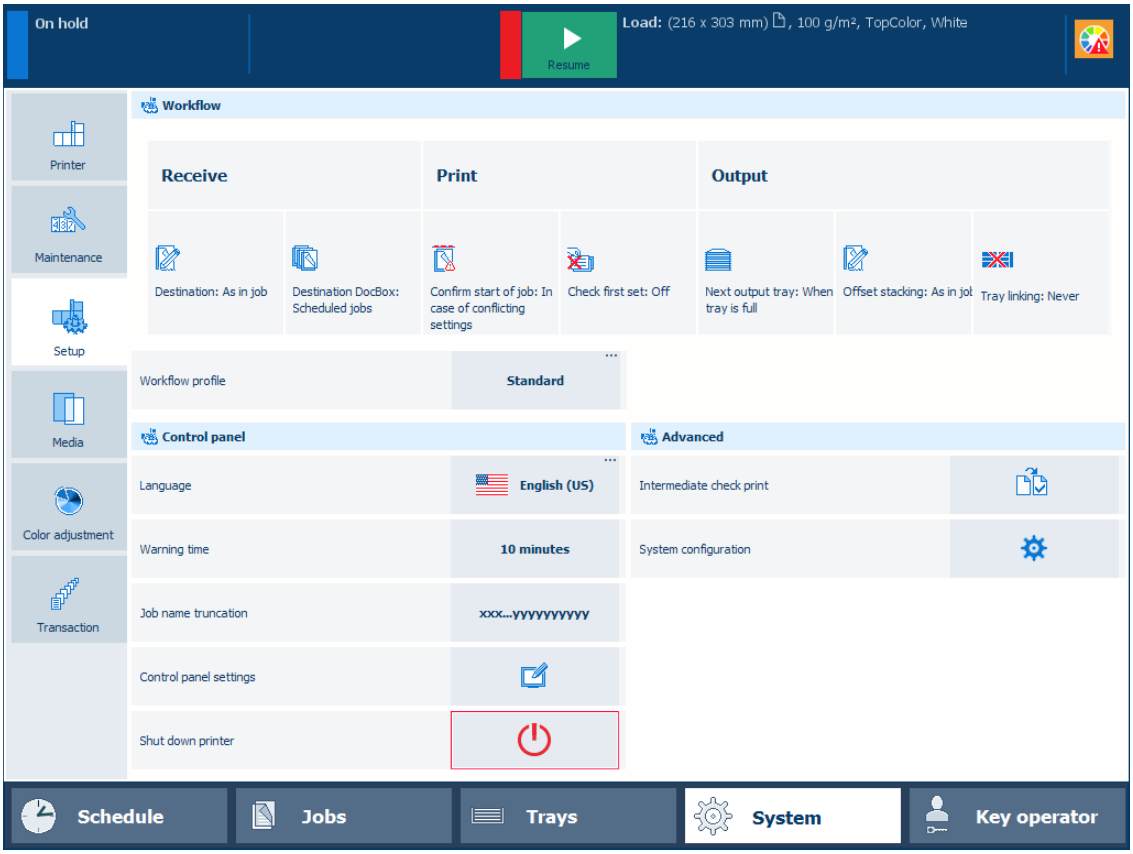
Log in with a factory defined user account

When you use a factory defined user account, you can select the user account from a list.



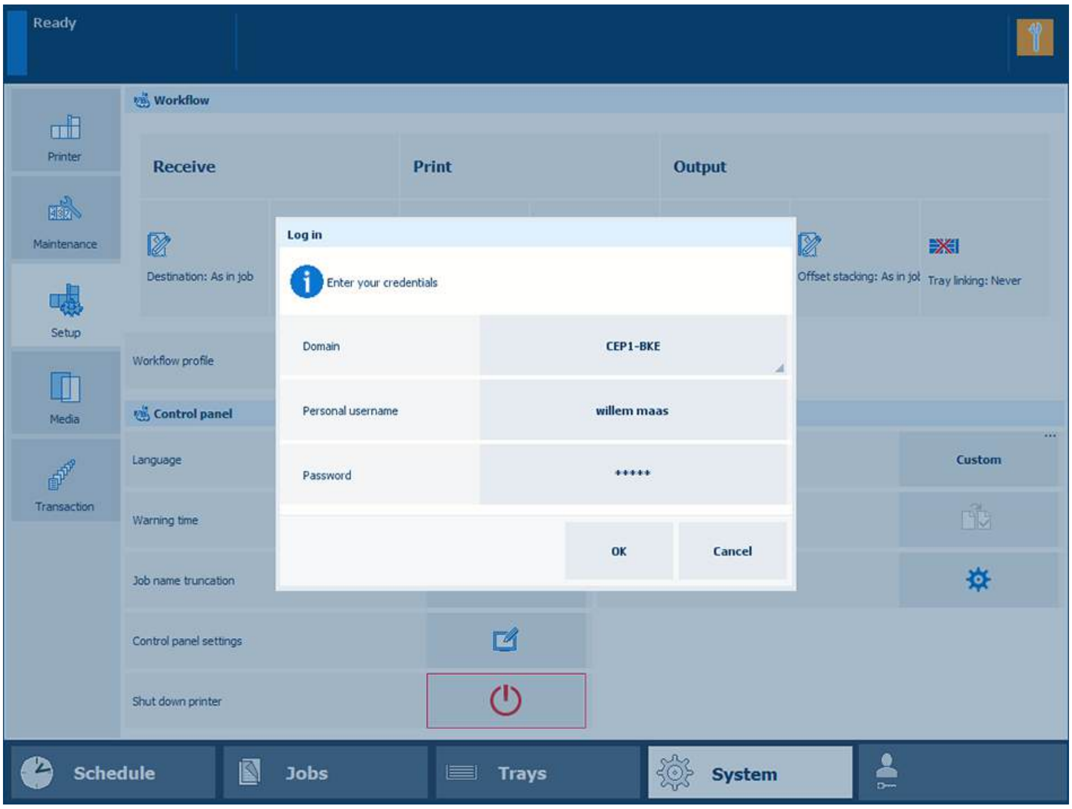
Log in with factory defined user account

1. When the [Domain] drop-down list is displayed, select the hostname or IP address of the printer.
2. Select the factory defined user account from the [Username] drop-down list.
3. Enter the password.
4. Touch or click [OK].
   The name of the user account is displayed.



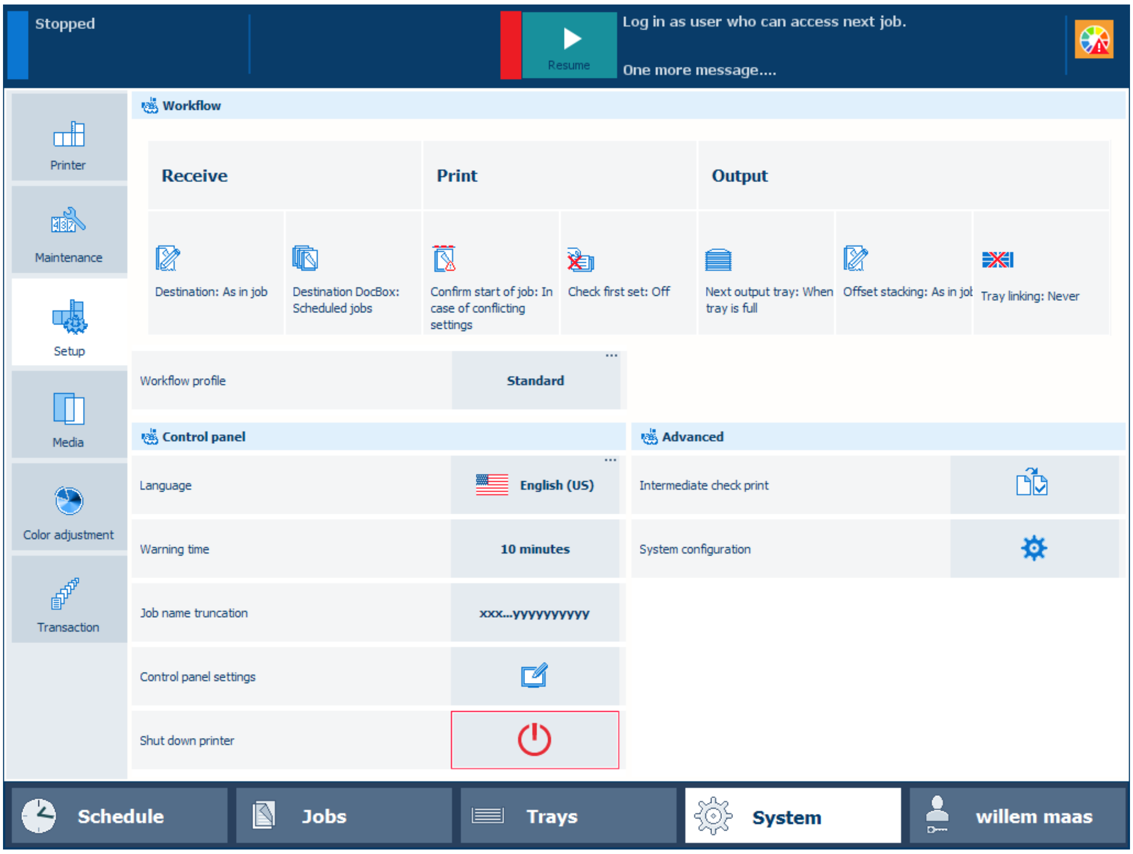
Log in with a custom user account

When you use a custom user account, you need to log in with a username.



Log in with custom user account

1. When the [Domain] drop-down list is displayed, select the hostname or IP address of the printer.
2. Select [Personal username] from the [Username] drop-down list.
3. Enter your username in the [Personal username] field.
4. Enter your password in the [Password] field.
5. Touch or click [OK].
   After a successful login, the name of your user account is displayed.

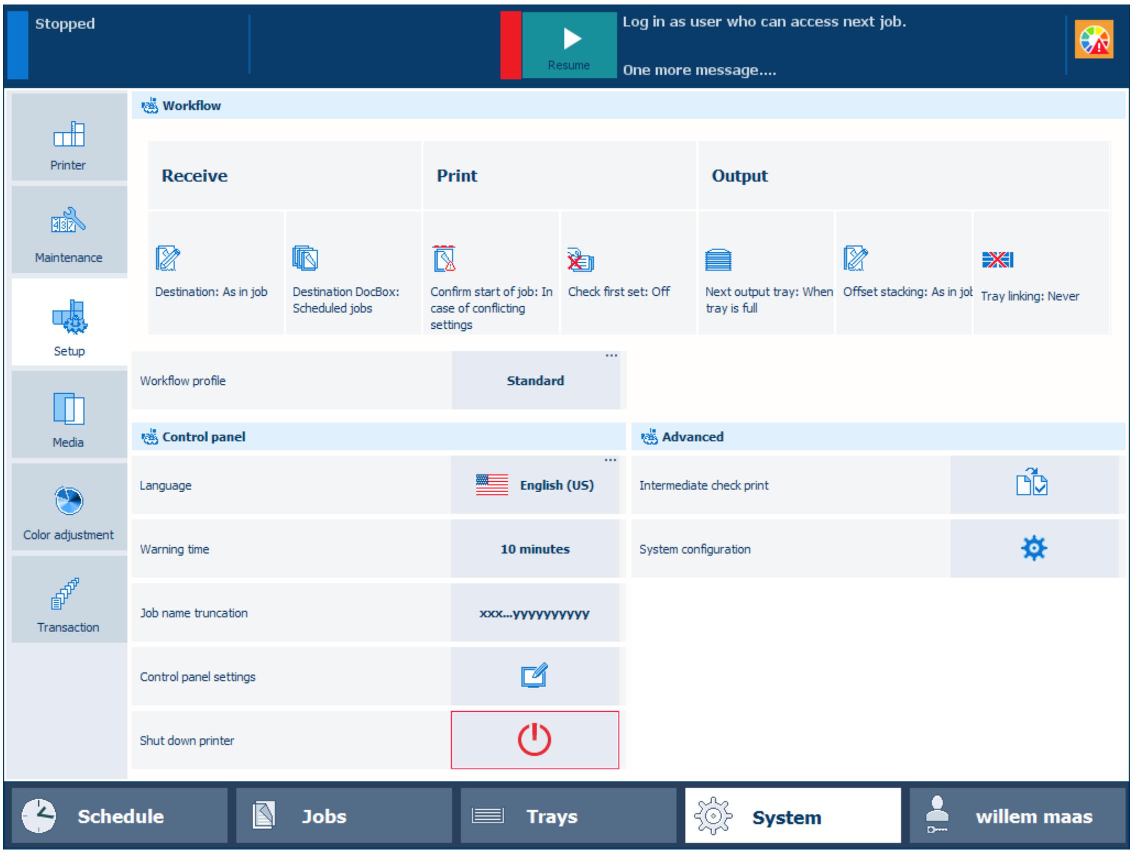


Custom user account is logged in

Log in with a smart card

Your organisation can use PKI or NFC smart cards to identify users of the printer. A single smart card can be used for the authentication of one or multiple user accounts. The user account credentials are configured on an LDAP directory server.

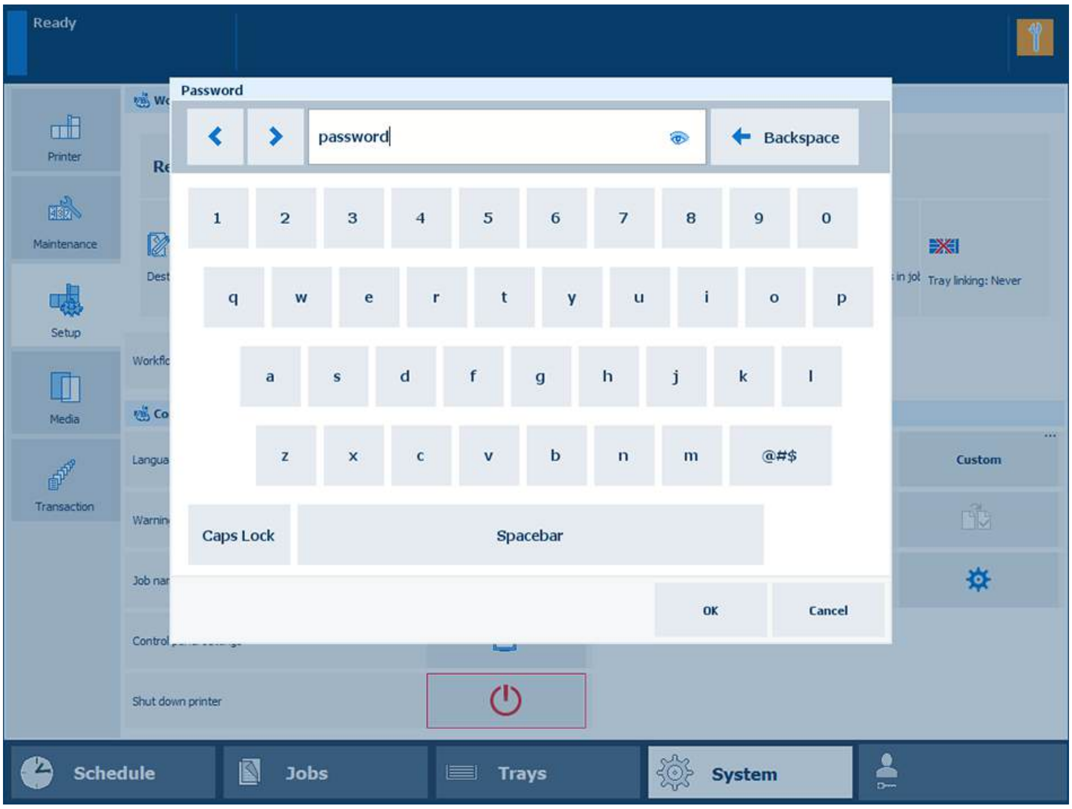
1. - Insert your PKI smart card into the reader attached to the printer.
   - Hold your NFC smart card next to the reader attached to the printer.
2. Select your username, if the smart card is configured for multiple user accounts.
3. Enter your password or PIN in the [Password] field, if required.
4. Touch [OK].
   After a successful login, the name of your user account is displayed.



Custom user account is logged in

See your password during login

The password field where you type your password hides the characters you enter. Touch the *eye* symbol in the text field to see and check the characters you entered.


Ready
Password
password
Backspace
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m @#$
Caps Lock
Spacebar
OK
Cancel
Printer
Maintenance
Setup
Media
Transaction
Tray linking: Never
Custom
Shut down printer
Schedule
Jobs
Trays
System
Password

Log out or switch roles

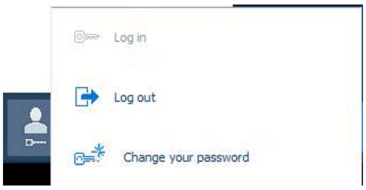
End the session

When you want to leave PRISMAsync Print Server, it is important to log out to end the session. It is also possible to authenticate again with other credentials without leaving PRISMAsync Print Server.

The **local session timeout** period determines how long you remain logged in without using the printer.

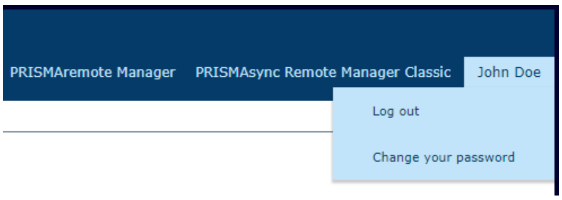
Log out on the control panel and the Settings Editor

1. • On the control panel touch the name of your user account.



Log out on the control panel

• In the Settings Editor click the name of your user account.



Log out in the Settings Editor

2. Touch or click [Log out].

Log out on control panel when you use a PKI smart card

Remove your PKI smart card from the reader.

Log out on control panel when you use an NFC smart card

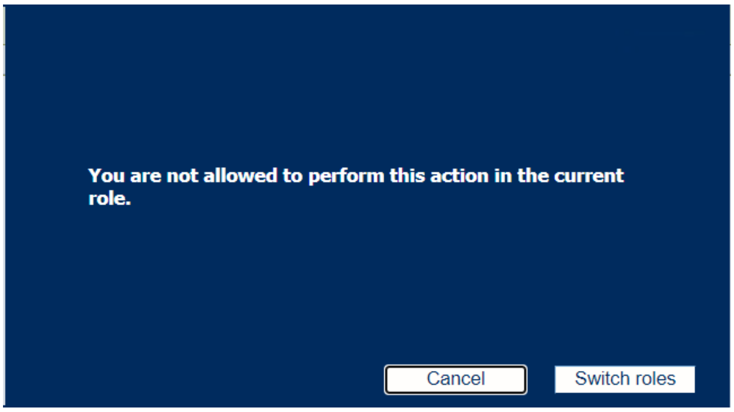
Hold your NFC smart card next to the reader. You can also use the [Log out] option.

Log out in PRISMAsync Remote Manager

Click [Log out].

Switch roles to change settings

When you are logged in to the Settings Editor or to the control panel and you want to perform a task for which you do not have sufficient rights, you can log in again with the appropriate credentials. Therefore, you need to have another user account that is authorized for this task.

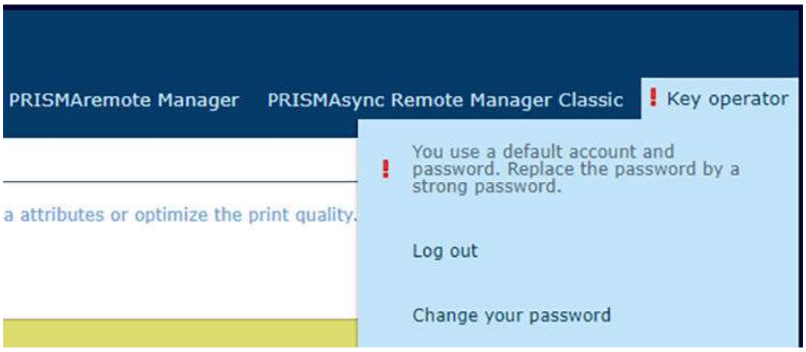


Switch roles

Change password

When do you need to change your password

You are strongly recommended to follow the security guidelines of your organization. The definition of your password is part of these security guidelines. You are warned when your password no longer meets the guidelines.

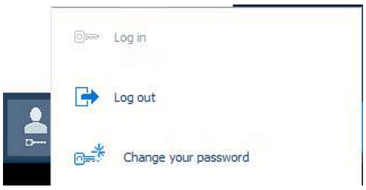


Expired password

NOTE
This instruction does not apply to the password of a domain user account because this password is defined on the LDAP directory server.

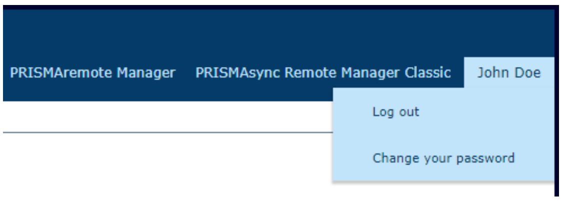
Change a password

1. • Log in from the control panel and touch the name of your user account.



Change password

• Log in from the Settings Editor and click the name of your user account.



Change password

2. Touch or click [Change your password].
3. Enter your current password.
4. Enter your new password and confirm the new password.

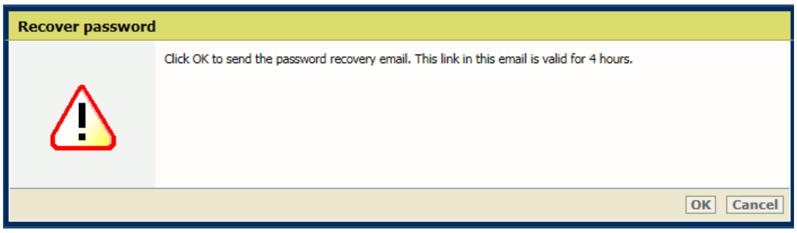
NOTE
It depends on the security guidelines of your organization if you can reuse a previously used password.

5. Touch or click [OK].

Recover password

What is a password recovery mail

When you have lost your password, you can define a new password via a password recovery email. The email that holds the link to define a new password, is sent to the email address that belongs to your user account. Activate the link within four hours after your received the email.



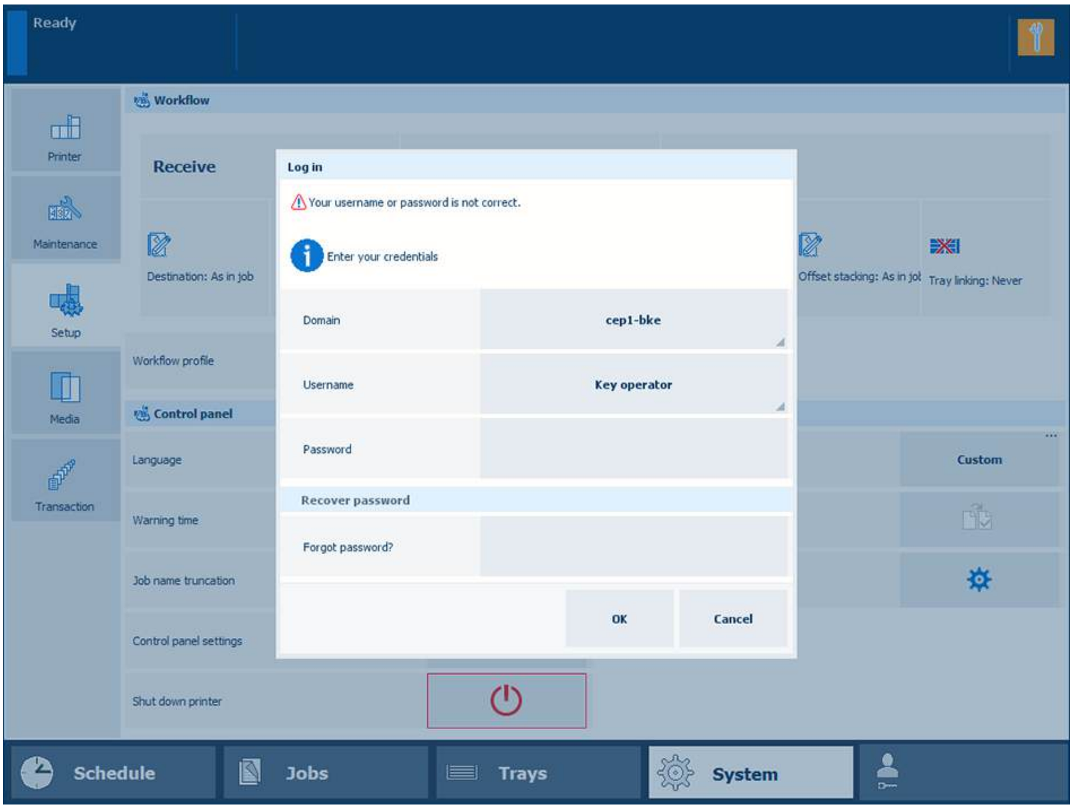
Recover password

IMPORTANT
Ensure that your email address is part of your user account properties. Otherwise, the password must be changed in the PRISMAsync Print Server user account options. In case you do not have the rights to change the PRISMAsync Print Server user account options, you need to provide your password to the system administrator. See the PRISMAsync Print Server administration guide for more information about user authorisation (https://downloads.cpp.canon).

NOTE
This recovery instruction does not apply to the password of a domain user account and the factory defined system administrator account.

Recover password from control panel

1. Go the control panel.
2. Enter your username without entering a password.
3. Touch [OK].

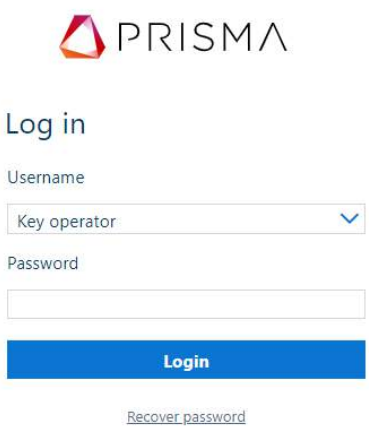


Login window on the control panel

4. Touch [Forgot password?].

Recover password from Settings Editor or PRISMAremote Manager / PRISMAsync Remote Manager Classic

1. Click [Recover password] at the login window.



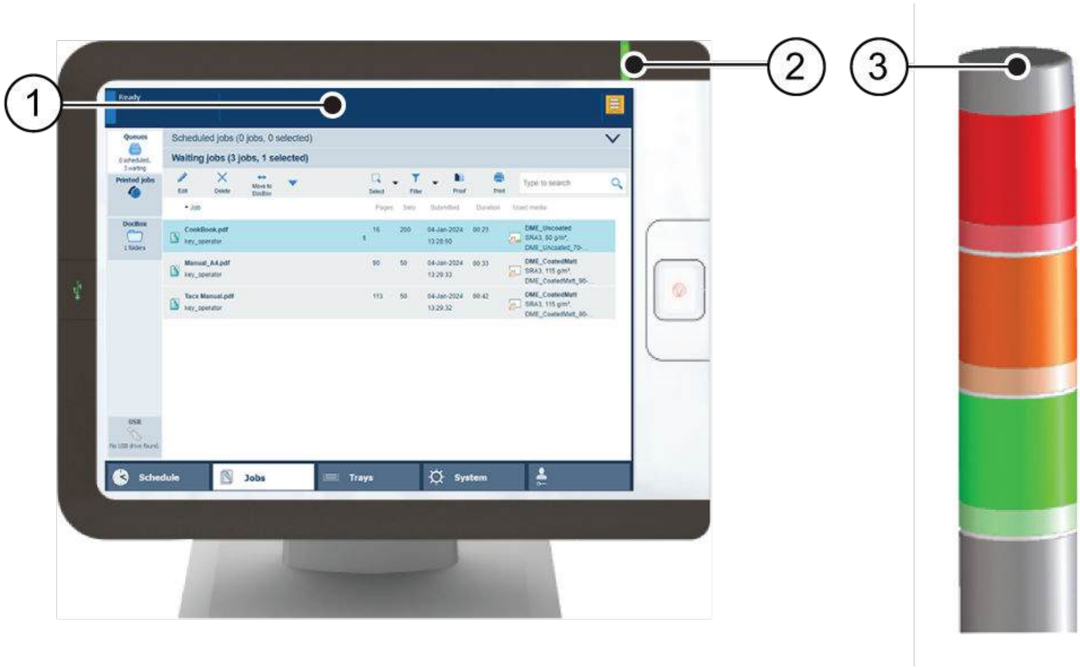
PRISMAremote Manager login window

2. Click [OK].

Learn about the printer status

The vertical status bar on the dashboard, the status LED on the control panel and the operator attention light show the same status color. PRISMAremote Manager / PRISMAsync Remote Manager Classic and PRISMAremote Monitoring display the status color remotely.

You can change the warning time or disable warnings. Choose a warning time that gives you sufficient time to prepare media or remove prints without a printer stop.



The status bar on the dashboard (1), the control panel (2) and the operator attention light (3)

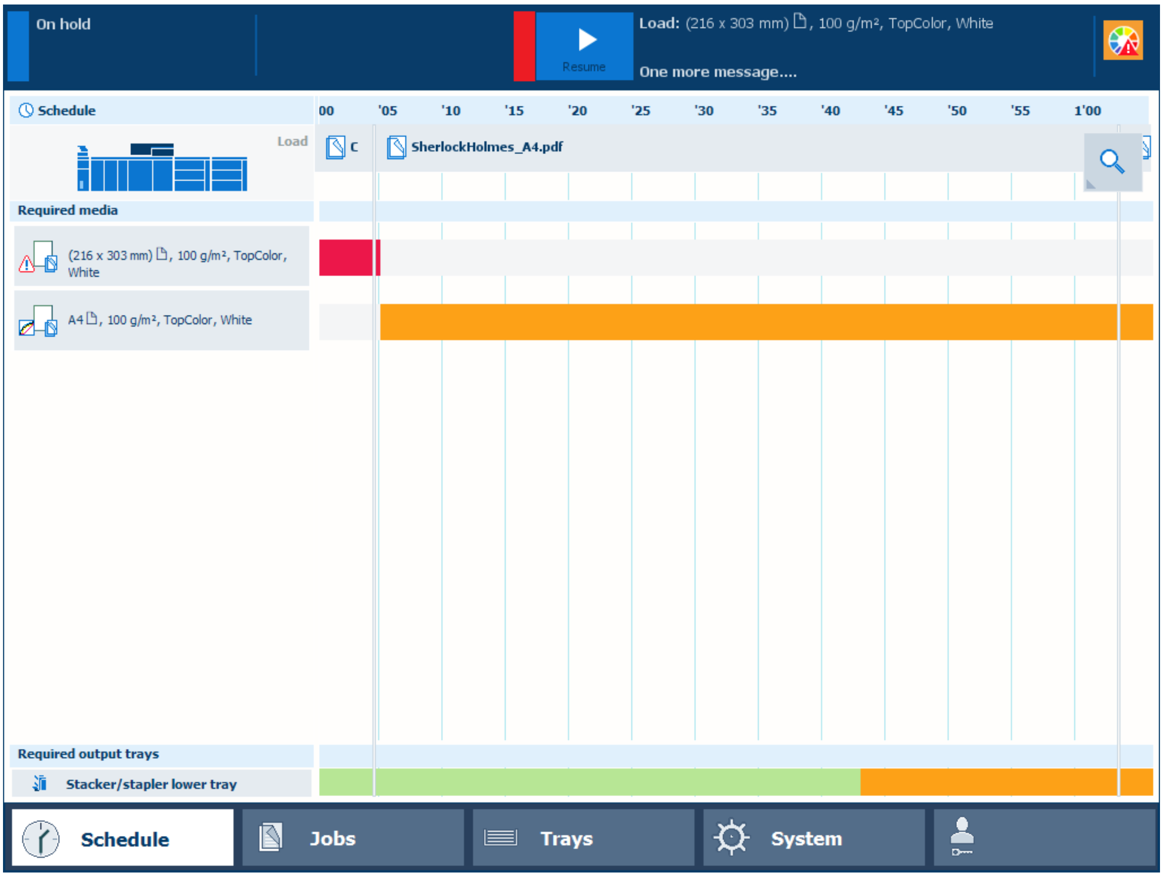
Color	Action required
	Green means that there is no action expected soon.
	Orange means that you must refill paper trays, add toner, staples or remove prints soon.
	Red means that you must perform an action immediately because the printer cannot continue printing. This can have the following reasons: 1. Media for the current job is not available. 2. One or more output trays are full. 3. One of the waste containers is absent or full. 4. A paper jam has occurred.

Keep printing with the schedule

The **schedule** is the daily up to eight-hour plan board of the control panel and PRISMAremote Manager / PRISMAsync Remote Manager Classic. It shows the list of scheduled jobs and predicts the job production time. Furthermore, you read the upcoming events: loading paper trays or removing prints from the output trays.

- **Green** means that the media and the output location are available
- **Gray** means that the availability of the media and output location is unknown. This occurs when the printer is in transaction printing mode or for streaming jobs.

- **Orange** means that the operator is needed within the specified time to load media or remove prints.
- **Yellow** means that the printer cannot determine the filling level in the paper trays or in the output location. This can occur when the thickness of the media is unknown or when prints are folded.
- **Red** means that the operator is needed to load media or to remove prints.



Schedule

When action is required

The **dashboard** shows messages and symbols when an action is expected soon or immediately.

The dashboard when the print system needs you to load media

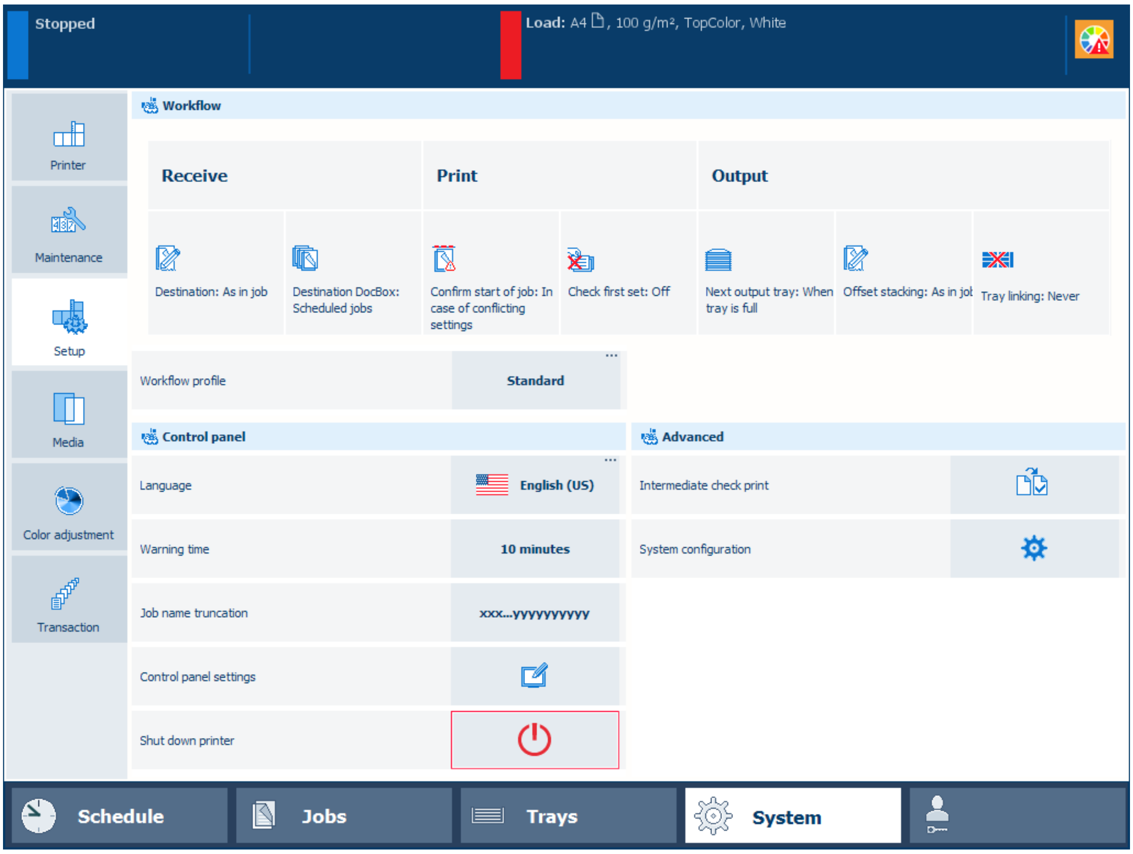
The **dashboard** shows the printer status (1), the name and progress of the active job or streaming job (2) as well as messages (3) and icons (4) of the expected actions. To prevent the printer from stopping, check the filling levels of supplies and waste on a regular basis.

Chapter 5
Define defaults

Adjust the control panel

You can adjust a number of control panel settings.

1. Touch [System]→[Setup] to go to the control panel adjustment settings.



Control panel adjustment settings

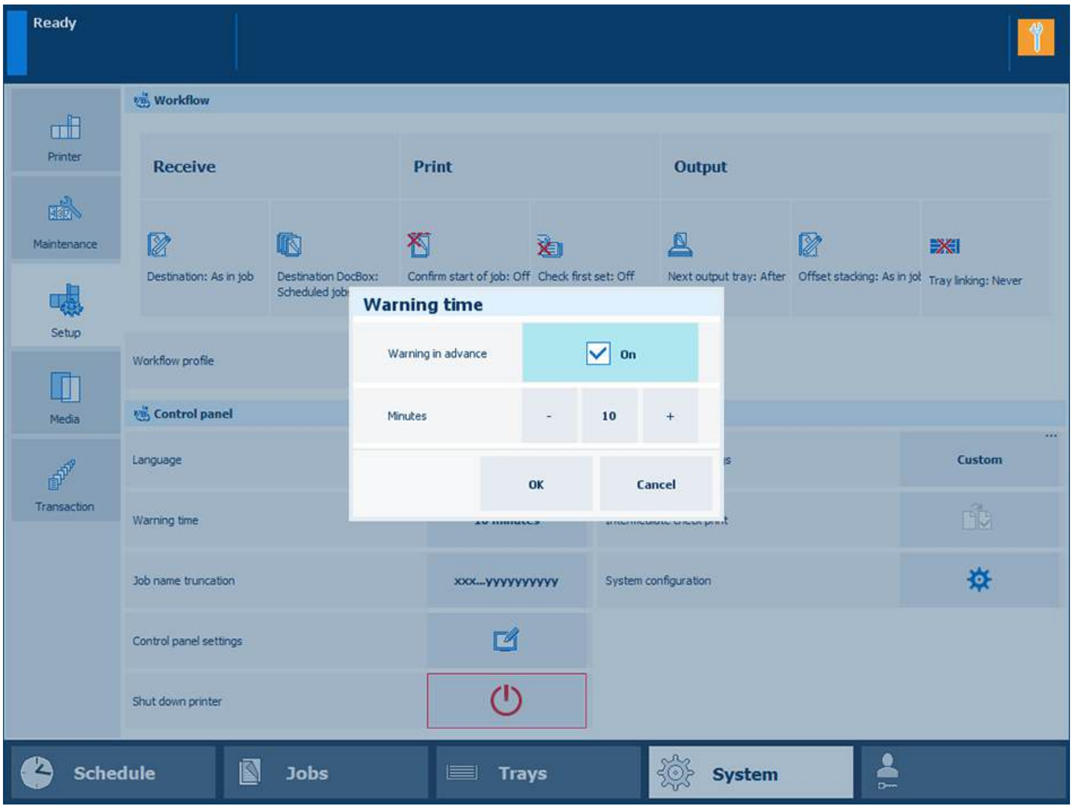
Change the display language of the control panel

1. Touch [Language].
2. Select the required language.
3. Touch [OK].

Adjust the warning time

The [Warning time] setting defines how long in advance you are notified about upcoming events.

1. Touch [Warning time].



[Warning time] setting

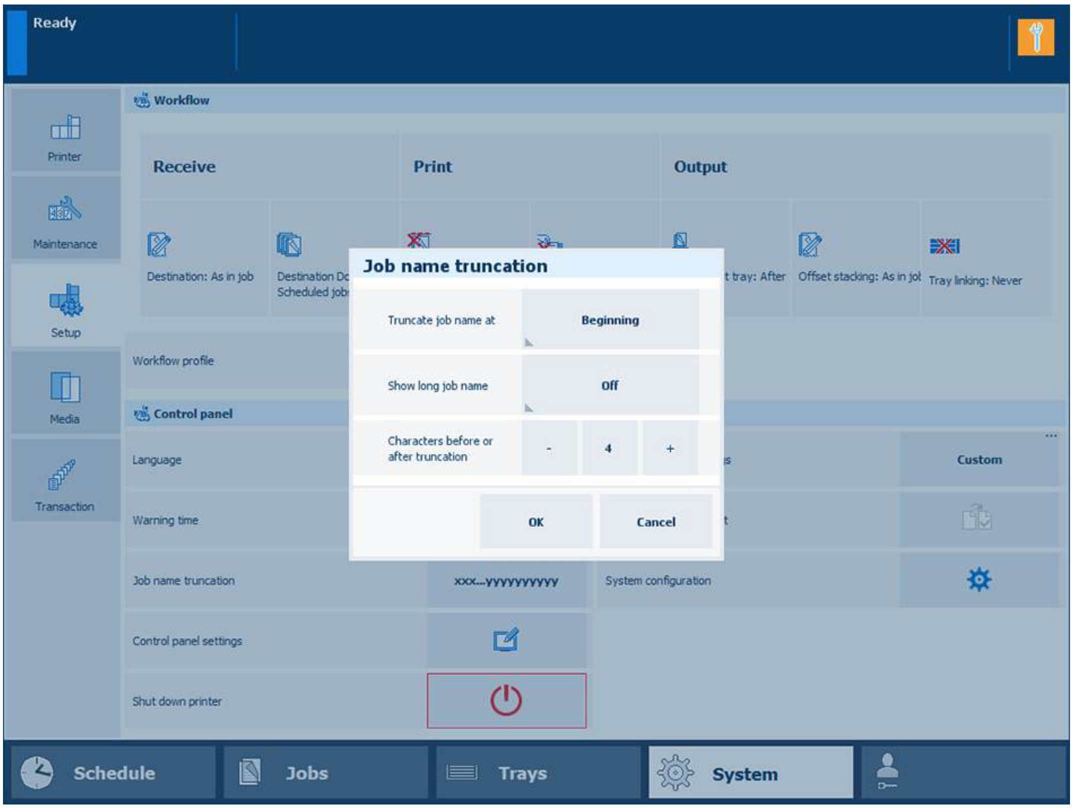
2. Indicate if you want to be notified in advance.
3. Touch the - or + button to decrease or increase the warning time in increments of 1 minute.
4. Touch [OK].

Define the display of job names

Job names can be very long. You can indicate if you want to show the name using two lines in the job list or how you want to truncate the name.

When you truncate a name at the beginning, you can indicate how many characters are removed before the truncation. When you truncate a name at the end, you can indicate how many characters are removed after the truncation.

1. Touch [Job name truncation].

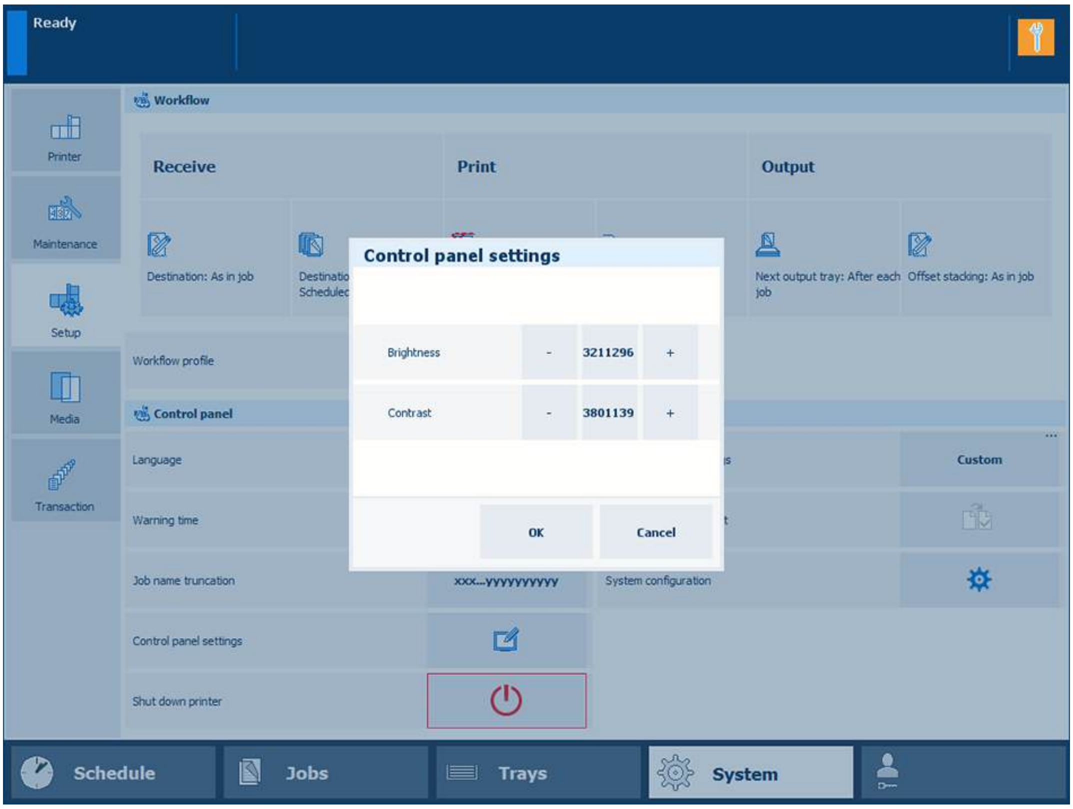


[Job name truncation] setting

2. Indicate if you want to use the long job name.
3. Indicate how you want to shorten the job name.
4. Touch [OK].

Improve the readability of the control panel

1. Touch [Control panel settings].



Readability settings

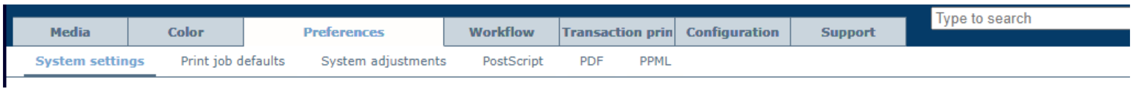
2. Choose the automatic adjustment method or touch the + and - buttons to adjust the brightness and contrast manually.
3. Touch [OK].

Define printer defaults

Define the accessibility settings

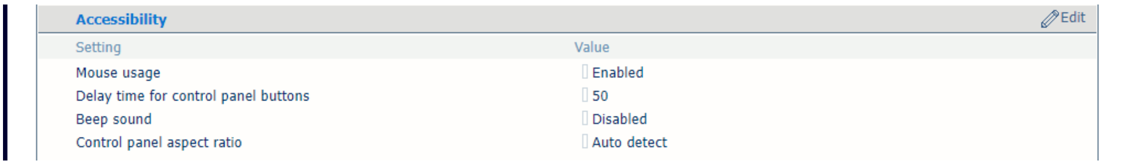
You can define the accessibility settings.

1. Open the Settings Editor and go to: [Preferences]→[System settings].



[System settings] tab

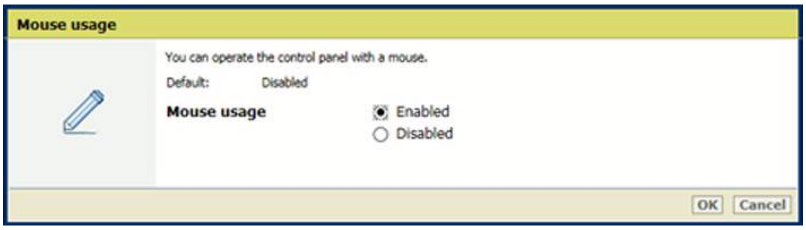
2. Go to the [Accessibility] section.



[Accessibility] section

Enable the use of the mouse to operate the control panel

1. Use the [Mouse usage] setting to enable or disable the use of the mouse.

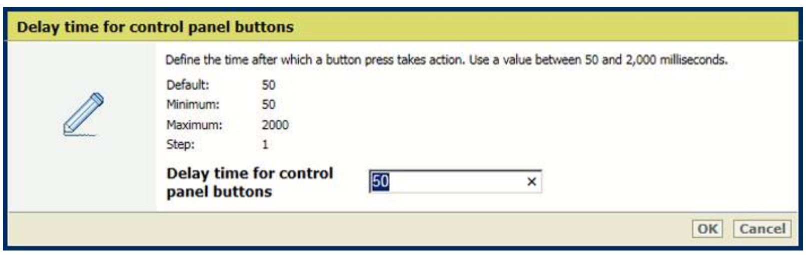


[Mouse usage] setting

2. Click [OK].

Adjust the delay time of a button press

1. Use [Delay time for control panel buttons] setting to define the time after which a button press takes action.

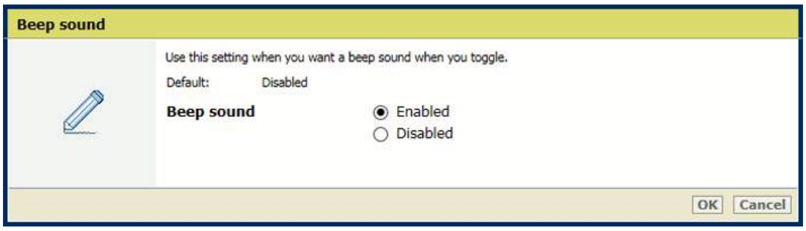


[Delay time for control panel buttons] setting

2. Click [OK].

Enable a beep sound during toggling

1. Use the [Beep sound] setting to enable or disable the beep sound.

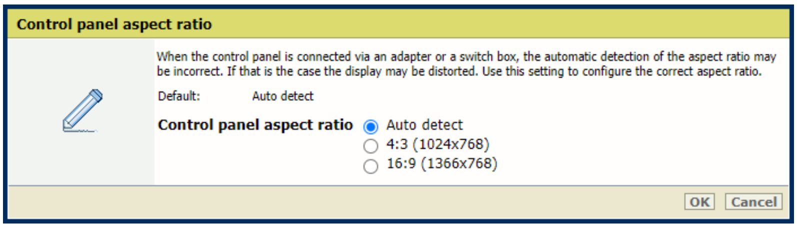


[Beep sound] setting

2. Click [OK].

Adjust the aspect ration of the control panel

1. Use the [Beep sound] setting to manually adjust the aspect ratio of the control panel. Change this setting only if the aspect ratio of the control panel is incorrect.



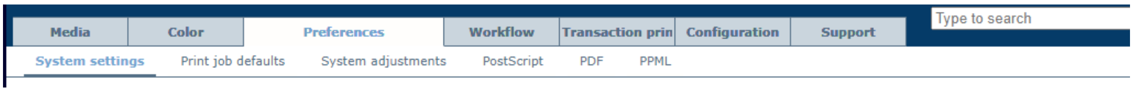
[Beep sound] setting

2. Click [OK].

Define the regional settings

You can define the regional settings. The system administrator can adjust the time and time zone.

1. Open the Settings Editor and go to: [Preferences]→[System settings].

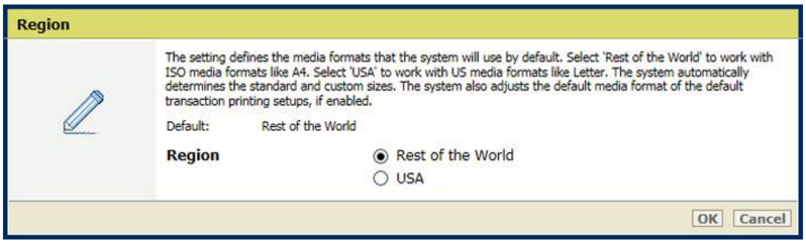


[System settings] tab

2. Go to the [Regional settings] section.

[Regional settings] section

3. Use the [Region] setting to select the media format.
   - [USA]: US media formats.
   - [Rest of the World]: ISO (European) media formats.



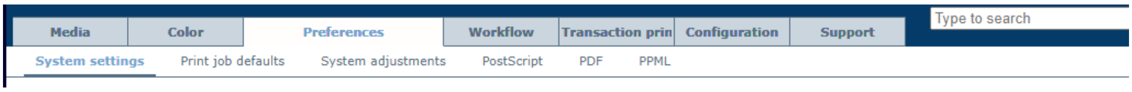
[Region] setting

4. Click [OK].

Define system of measurement

You can define whether you want to use metric or imperial system of measurement.

1. Open the Settings Editor and go to: [Preferences]→[System settings].

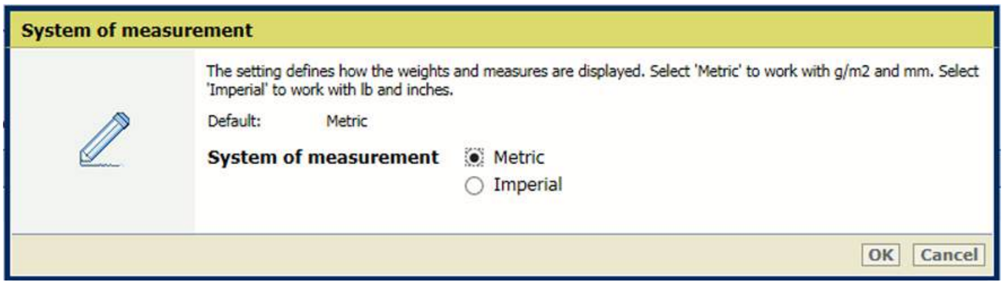


[System settings] tab

2. Go to the [Regional settings] section.

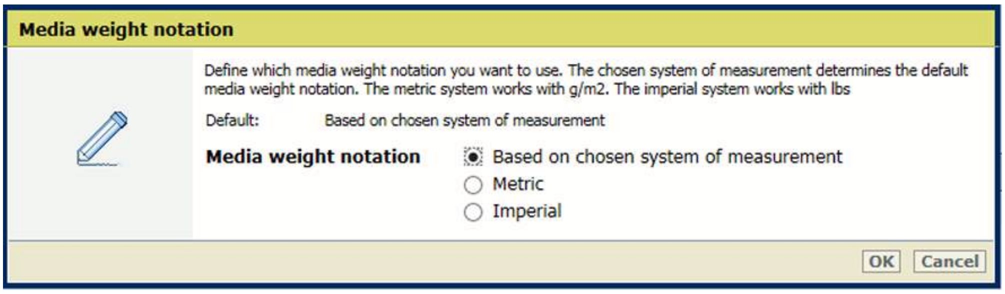
[Regional settings] section

3. Use the [System of measurement] setting to select the system of measurement.
   - [Metric]: work with g/m2 and mm.
   - [Imperial]: work with lb and inch.



[System of measurement] setting

4. Use the [Media weight notation] setting to select the media weight notation.
   - [Based on chosen system of measurement]: uses the notation as selected in the [System of measurement] setting.
   - [Metric]: work with g/m2.
   - [Imperial]: work with lb.

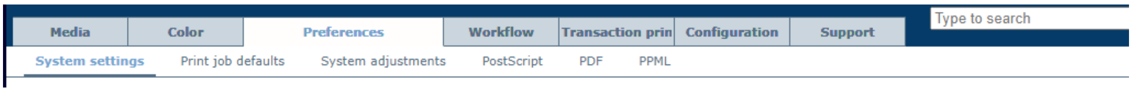


[Media weight notation] setting

Define storage of printed jobs

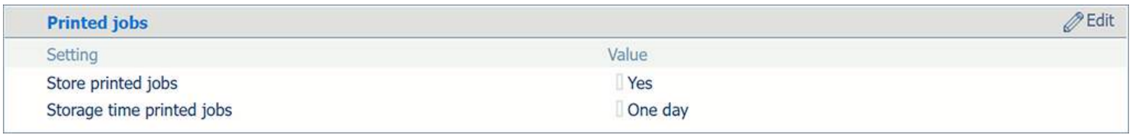
You can define if printed jobs are displayed on the control panel. This allows operators to reprint printed jobs. When printed jobs are visible, you can define their storage time.

1. Open the Settings Editor and go to: [Preferences]→[System settings].



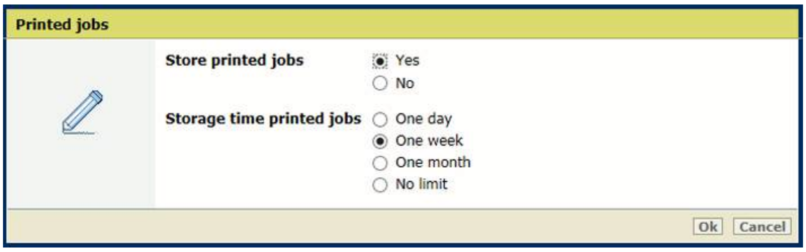
[System settings] tab

2. Go to the [Printed jobs] section.



[Printed jobs] section

3. Use the [Store printed jobs] setting to indicate if you want to display the list of printed jobs.



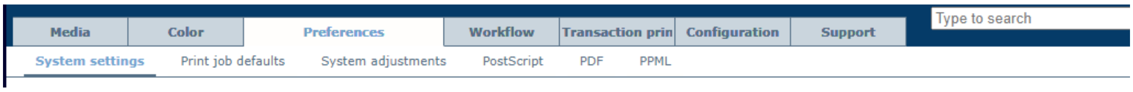
[Store printed jobs] setting

4. Use the [Storage time printed jobs] setting to select the storage time of printed jobs: [One day], [One week], [One month], or [No limit].
5. Click [OK].

Configure sleep mode, automatic wake-up, and shutdown mode

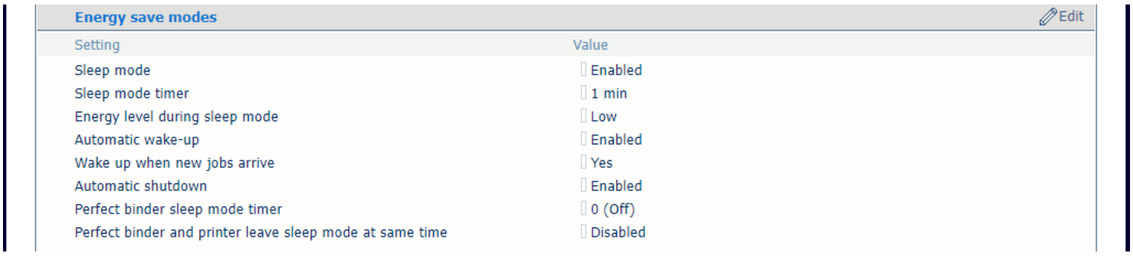
You can configure the sleep mode, automatic wake-up, as well as shutdown mode for your printer.

1. Open the Settings Editor and go to: [Preferences]→[System settings].



[System settings] tab

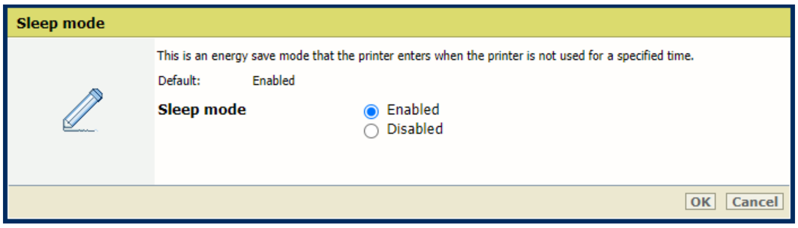
2. Go to the [Energy save modes] section.



[Energy save modes] section

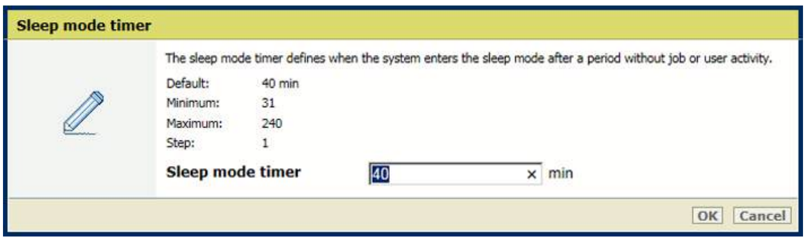
Define the sleep mode and sleep mode timer

1. Use the [Sleep mode] setting to enable the sleep mode.



[Sleep mode] setting

2. Click [OK].
3. Use the [Sleep mode timer] setting to set the sleep mode timer.

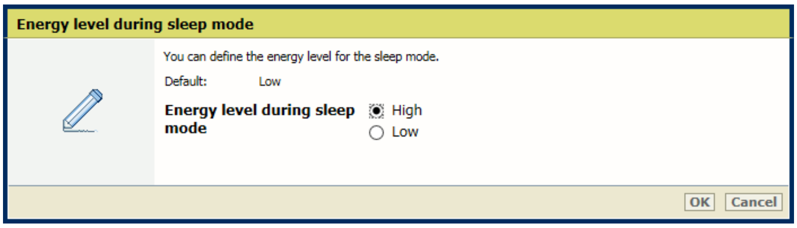


[Sleep mode timer] setting

4. Click [OK].

Define the energy level for sleep mode

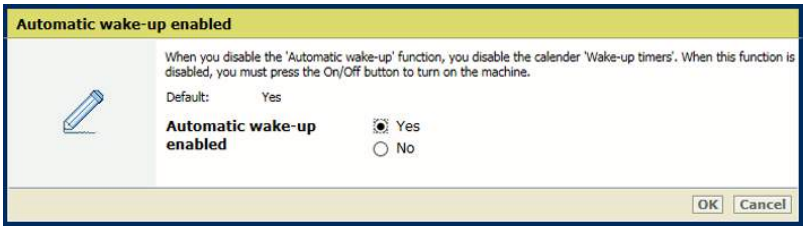
Use the [Energy level during sleep mode] setting to define the energy level. The printer leaves the sleep mode faster at a higher energy level.



[Energy level during sleep mode] setting

Define automatic wake-up

1. Go to the [Energy save modes] section.
2. Ensure the sleep mode is enabled.
3. Use the [Automatic wake-up] setting to enable the automatic awaking from the sleep mode.

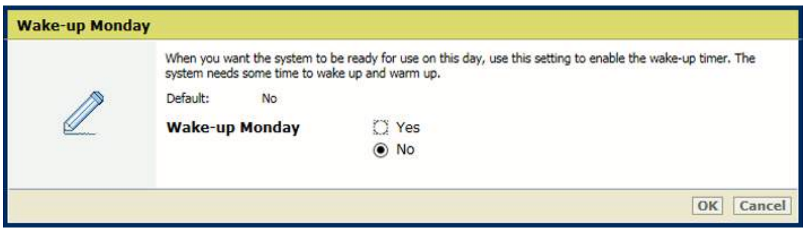


[Automatic wake-up] setting

4. Click [OK].
5. Go to the [Wake-up timers] section.

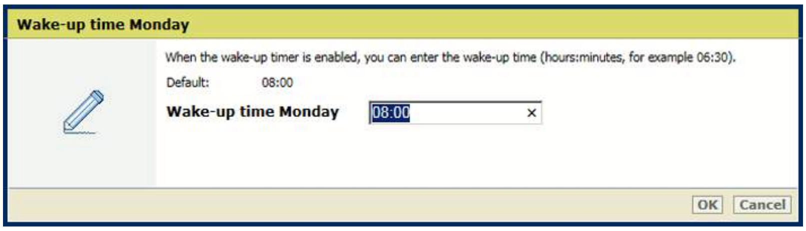
[Wake-up timers] section

6. Indicate for each day if an automatic awaking must occur.



Wake-up setting

7. Click [OK].
8. Define the wake-up time per day.

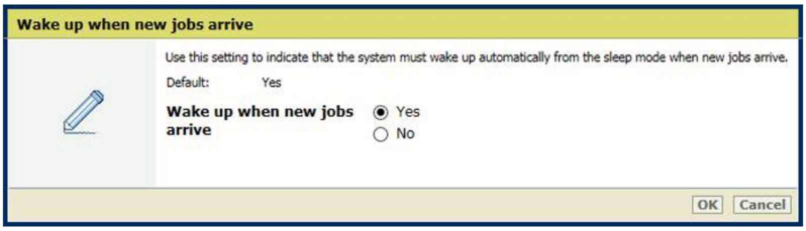


Wake-up time setting

9. Click [OK].

Define automatic wake-up for new jobs

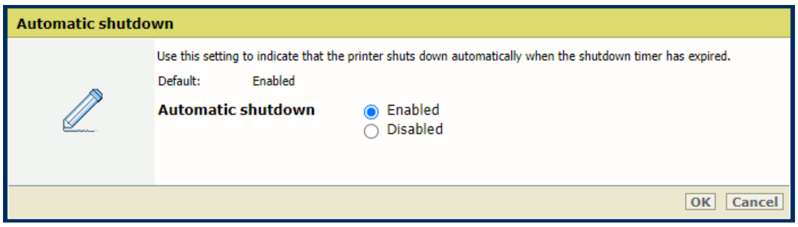
Use the [Wake up when new jobs arrive] setting to indicate whether the printer must wake up from the sleep mode for a new job.



[Wake up when new jobs arrive] setting

Define automatic shutdown

1. Go to the [Energy save modes] section.
2. Ensure the sleep mode is enabled.
3. Use the [Automatic shutdown] setting to enable the automatic shutdown.

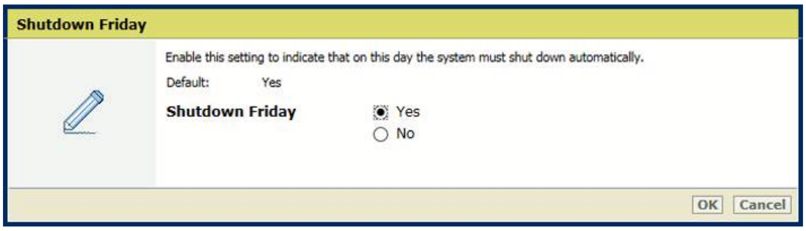


[Automatic shutdown] setting

4. Click [OK].
5. Go to the [Wake-up timers] section.

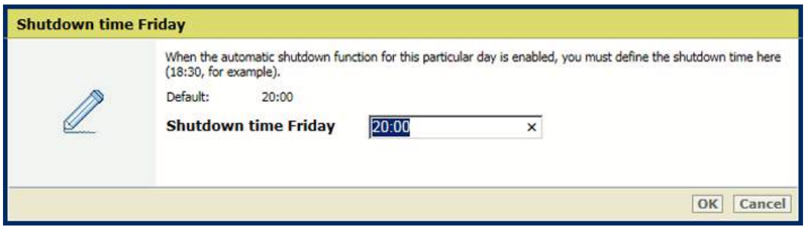
[Wake-up timers] section

6. Indicate for each day if an automatic shutdown must occur.



Shutdown setting

7. Click [OK].
8. Define the shutdown time per day.

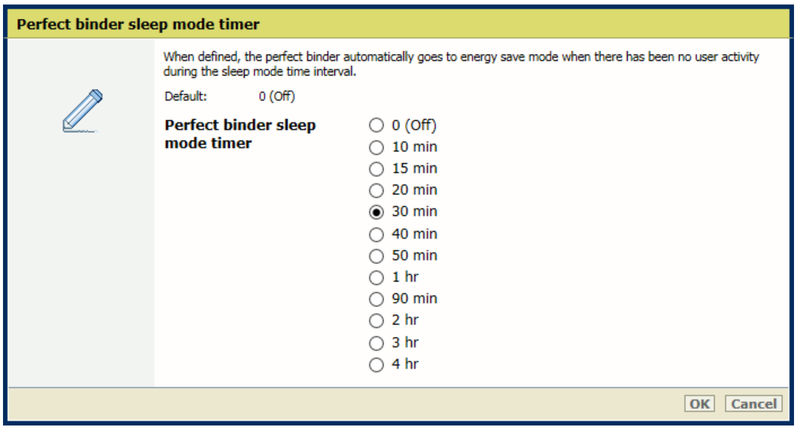


Shutdown time setting

9. Click [OK].

Define the sleep mode timer for Perfect Binder

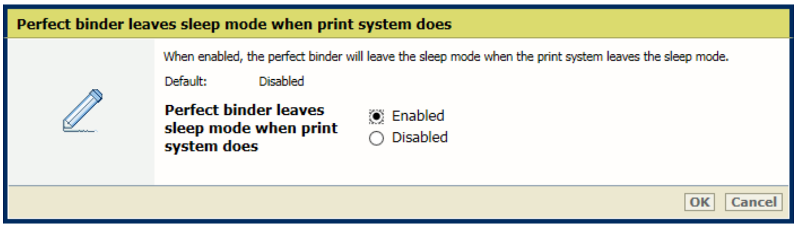
Use the [Perfect binder sleep mode timer] setting to define the time after which the Perfect Binder enters the sleep mode.



[Perfect binder sleep mode timer] setting

Define that Perfect Binder leaves sleep mode when printer does

Use the [Perfect binder and printer leave sleep mode at same time] setting to indicate whether the Perfect Binder leaves the sleep mode when the printer does.



[Perfect binder and printer leave sleep mode at same time] setting

More information

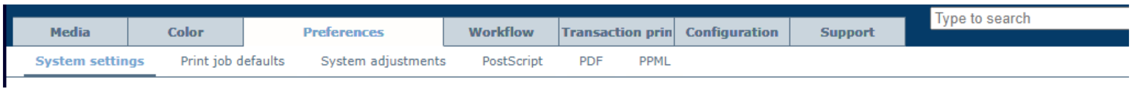
Learn about power states of the printer on page 44

Configure orange alert light for consumables and waste locations

For consumables and waste locations, you can choose whether the orange alert light is shown. The orange alert light indicates upcoming loading, refilling or replacement tasks.

The orange alert light is shown on the vertical status bar on the dashboard, the status LED on the control panel, the printer status in PRISMAsync Remote Manager and the operator attention light of the printer.

1. Open the Settings Editor and go to: [Preferences]→[System settings].

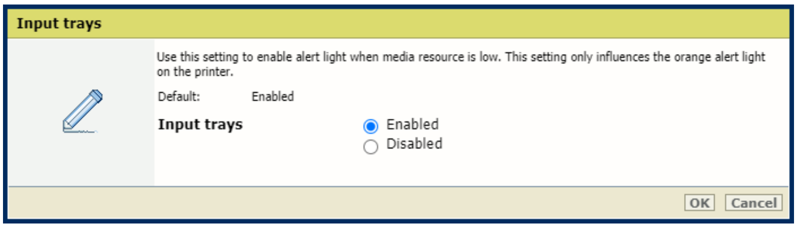


[System settings] tab

2. Go to the [Alert light] section.

[Alert light] section

3. Enable or disable the orange alert light for the following consumables and waste locations:
   - input trays;
   - output trays;
   - toner consumables;
   - toner waste;
   - finisher waste.



Orange alert light for input trays

4. Click [OK].

By default, the alert light is enabled for all consumables and waste locations.

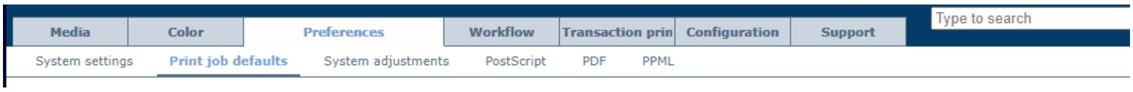
NOTE
Regardless of whether the alert light is enabled or disabled, the icons of the expected actions on the dashboard at the top of the control panel will be shown.

Define job defaults

Define default font of page numbers

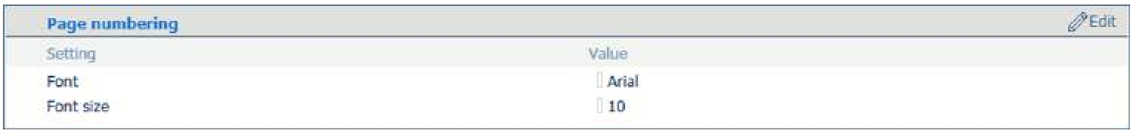
Page numbering can be enabled and defined in the job properties. You can set the default font in the Settings Editor.

1. Open the Settings Editor and go to: [Preferences]→[Print job defaults].



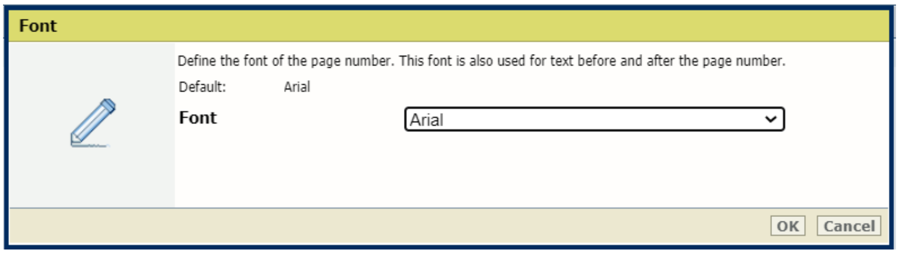
[Print job defaults] tab

2. Go to the [Page numbering] section.



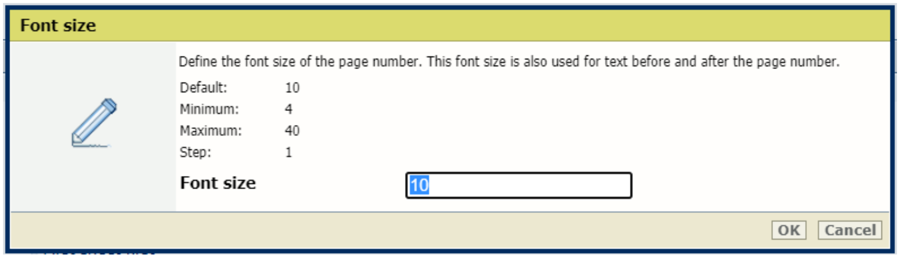
[Page numbering] section

3. Use the [Font] setting to select the font of the page number.



[Font] setting

4. Use the [Font size] setting and enter the font size.



[Font size] setting

5. Click [OK].

Define default use of special pages

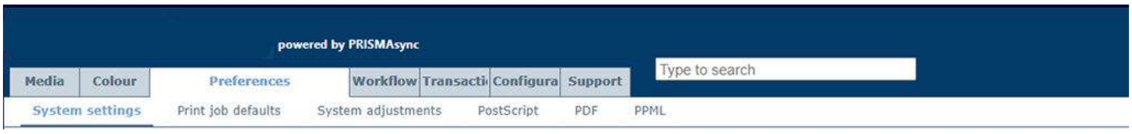
What are special pages

Banner pages, trailer pages and separator sheets are special pages that can be added to a job. The Settings Editor has settings to configure the use of special pages.

NOTE
You can import and export the media definition of special pages.

Define banner pages

1. Open the Settings Editor and go to: [Preferences]→[System settings]

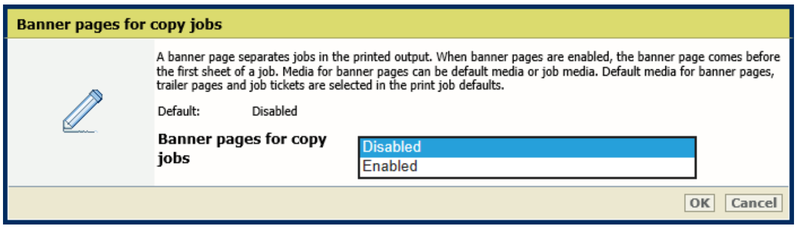


[System settings] tab

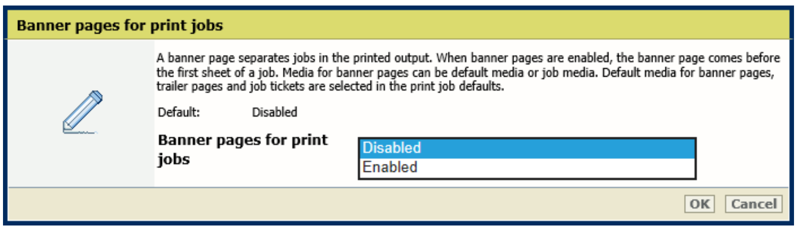
2. Go to the [Basic] section.

[Basic] section

3. Use the [Banner pages for copy jobs] and [Banner pages for print jobs] settings to indicate the use of banner pages.
   - [Enabled]: banner pages are added to all jobs.
   - [Disabled]: banner pages are never added to the jobs.

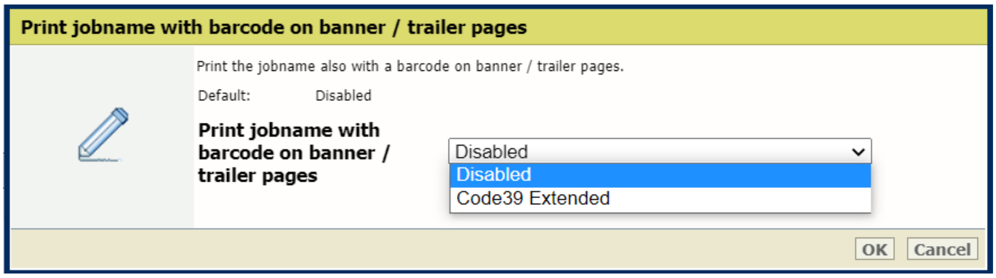


[Banner pages for copy jobs] setting



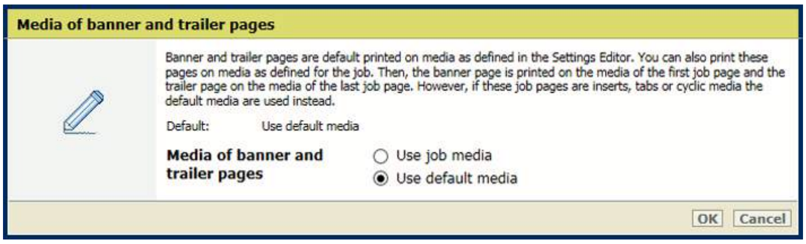
[Banner pages for print jobs] setting

4. Click [OK].
5. Use the [Print job name as barcode on banner/trailer pages] setting to add a barcode to the banner pages.



[Print job name as barcode on banner/trailer pages] setting

6. Click [OK].
7. Use the [Media of banner/trailer pages] setting to define how the media of banner pages are selected.
   - [Use job media]: the banner pages are printed on job media.
   - [Use default media]: the banner pages are printed on default media.

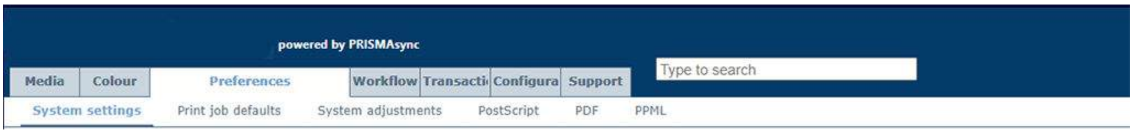


[Media of banner/trailer pages] setting

8. Click [OK].

Define trailer pages

1. Open the Settings Editor and go to: [Preferences]→[System settings].



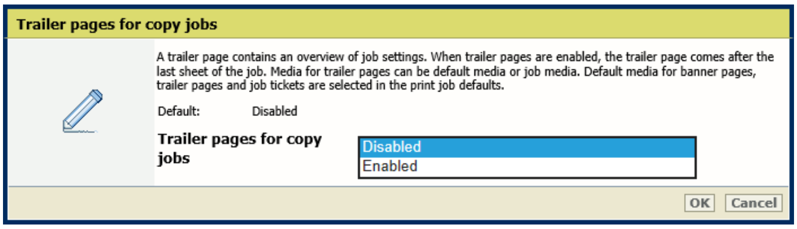
[System settings] tab

2. Go to the [Basic] section.

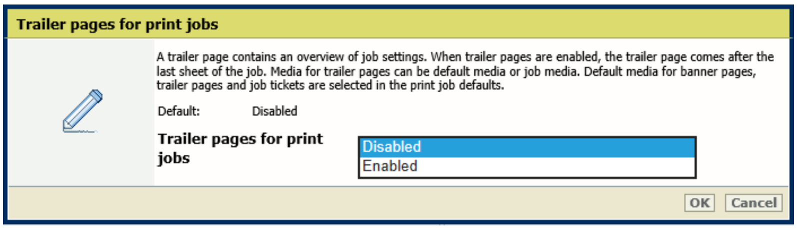
[Basic] section

3. Use the [Trailer pages for copy jobs] and [Trailer pages for print jobs] settings to indicate the use of trailer pages.

- [Enabled]: trailer pages are added to all jobs.
- [Disabled]: trailer pages are never added to the jobs.

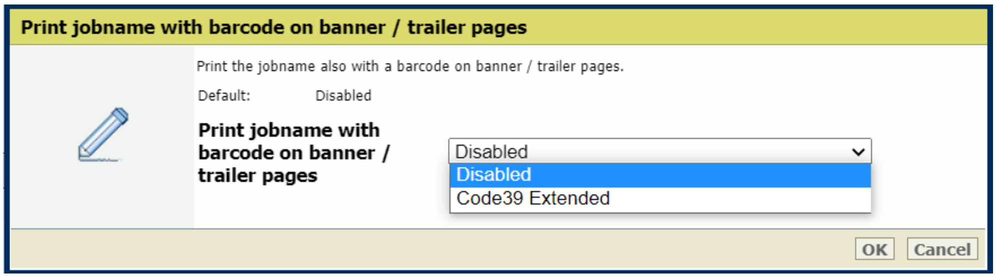


[Trailer pages for copy jobs] setting



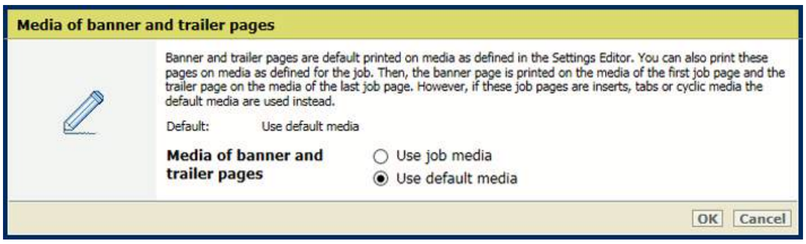
[Trailer pages for print jobs] setting

4. Click [OK].
5. Use the [Print job name as barcode on banner/trailer pages] setting to add a barcode to the trailer pages.



[Print job name as barcode on banner/trailer pages] setting

6. Click [OK].
7. Use the [Media of banner/trailer pages] setting to define how the media of trailer pages are selected.
   - [Use job media]: the trailer pages are printed on job media.
   - [Use default media]: the trailer pages are printed on default media.

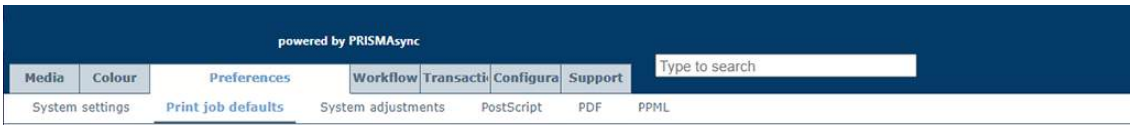


[Media of banner/trailer pages] setting

Define the default media of banner and trailer pages

Define the default media of banner and trailer pages when you have selected [Use default media] in the [Media of banner/trailer pages] setting.

1. Open the Settings Editor and go to: [Preferences]→[Print job defaults].

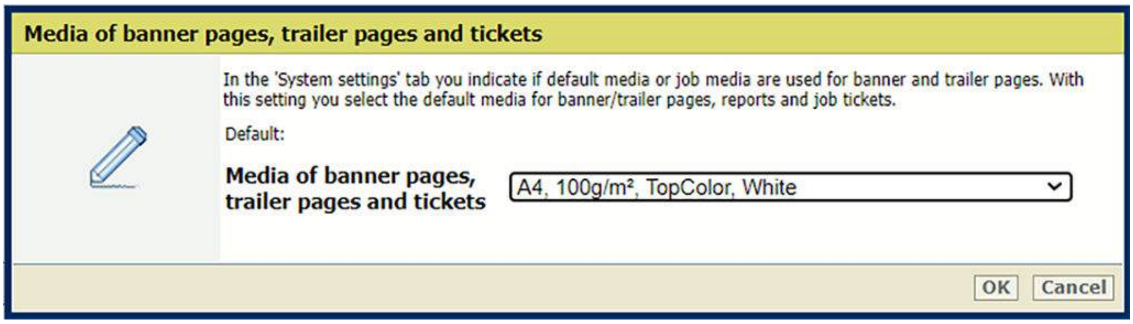


[Print job defaults] tab

2. Go to the [Banner pages, trailer pages, reports and tickets] section.

[Banner pages, trailer pages, reports and tickets] section

3. Use the [Media of banner/trailer pages, reports and tickets] setting to select the default media of banner and trailer pages.

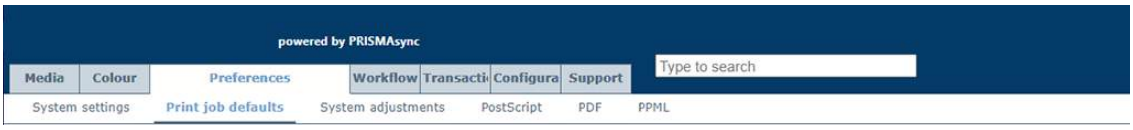


[Media of banner/trailer pages, reports and tickets] setting

4. Click [OK].

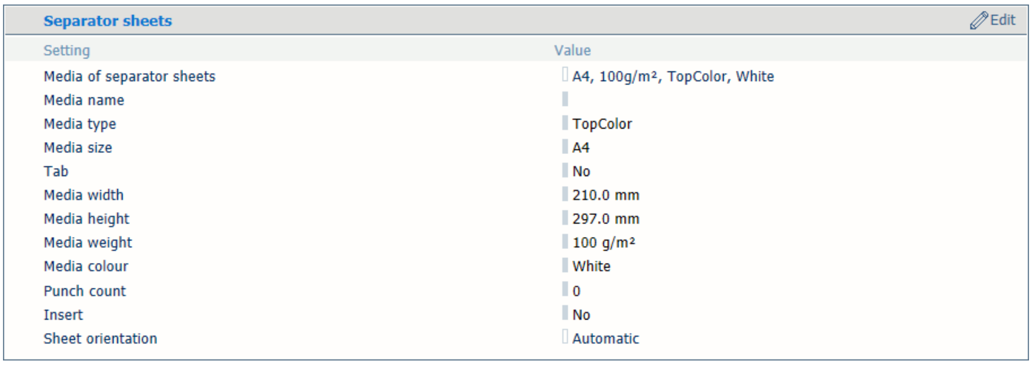
Define the default media of separator sheets

1. Open the Settings Editor and go to: [Preferences]→[Print job defaults].



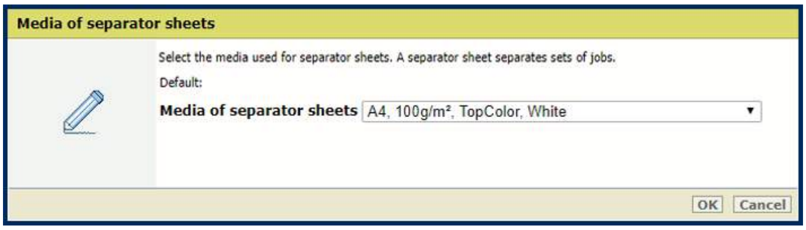
[Print job defaults] tab

2. Go to the [Separator sheets] section.



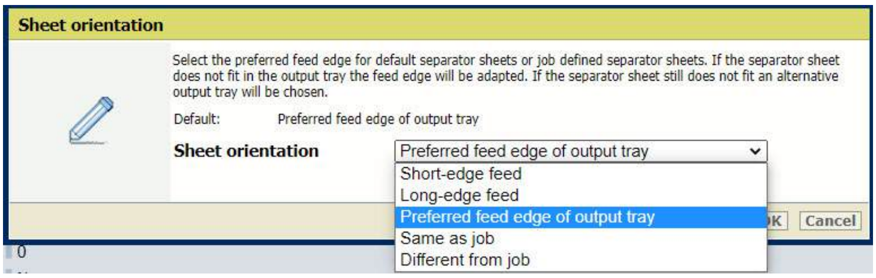
[Separator sheets] section

3. Use the [Media of separator sheets] setting to select the default media of the separator sheets.



[Media of separator sheets] setting

4. Click [OK].
5. Use the [Sheet orientation] setting to select the sheet orientation of the separator sheets.
   - [Short-edge feed]: separator sheets are delivered in the output tray with short-edge feed direction.
   - [Long-edge feed]: separator sheets are delivered in the output tray with long-edge feed direction.
   - [Preferred feed edge of output tray]: separator sheets are delivered in the output tray according to the preferred feed edge of the output tray. If the separator sheet does not fit in the output tray, the feed edge is adapted.
   - [Same as job]: separator sheets are delivered in the output tray as defined for the job.
   - [Different from job]: separator sheets are delivered in the output tray with the opposite feed direction than defined for the job.



Default orientation of separator sheets

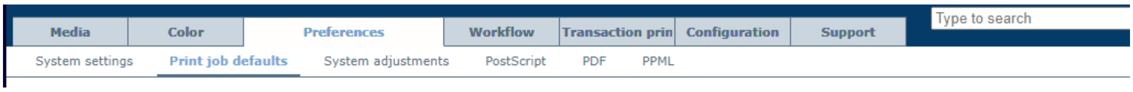
6. Click [OK].

Define default media for job tickets and covers

You can define the default media for job tickets as well as front and back covers.

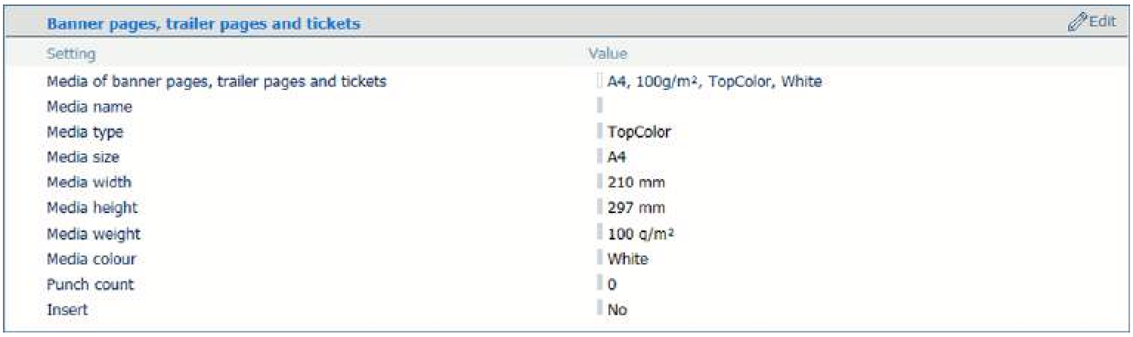
Define the default media of job tickets

1. Open the Settings Editor and go to: [Preferences]→[Print job defaults].



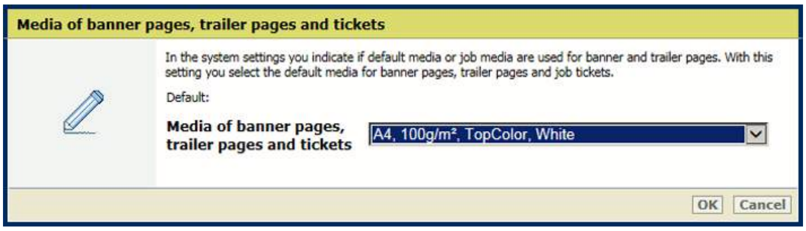
[Print job defaults] tab

2. Go to the [Banner pages, trailer pages, reports and tickets] section.



[Banner pages, trailer pages, reports and tickets] section

3. Use the [Media of banner/trailer pages, reports and tickets] setting to select the media of job tickets.



[Media of banner/trailer pages, reports and tickets] setting

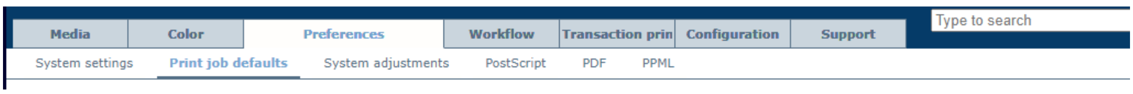
4. Click [OK].

Define the default media of covers

The Settings Editor has the setting to define the default media of the front and back covers.

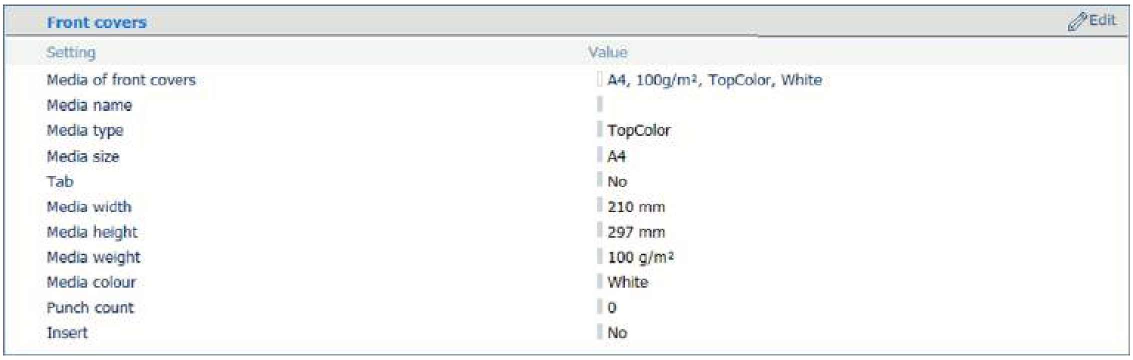
You can define the use of covers, the cover media and the media print mode of covers in the job properties.

1. Open the Settings Editor and go to: [Preferences]→[Print job defaults].



[Print job defaults] tab

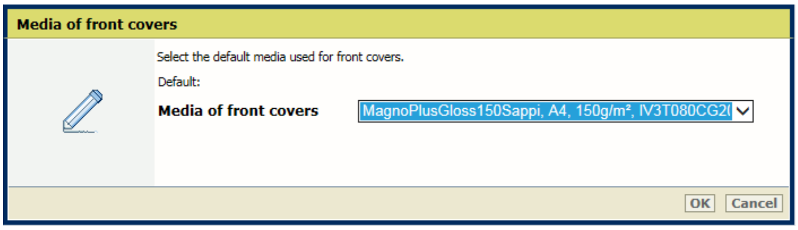
2. Go to the [Front covers] section.



Front covers	Edit
Setting	Value
Media of front covers	A4, 100g/m², TopColor, White
Media name	
Media type	TopColor
Media size	A4
Tab	No
Media width	210 mm
Media height	297 mm
Media weight	100 g/m²
Media colour	White
Punch count	0
Insert	No

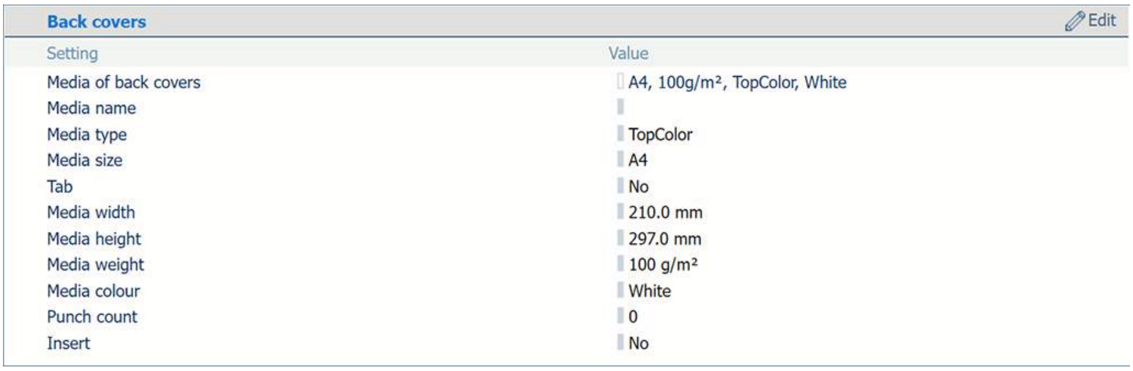
[Front covers] section

3. Use the [Media of front covers] setting to select the default media of front covers.



[Media of front covers] setting

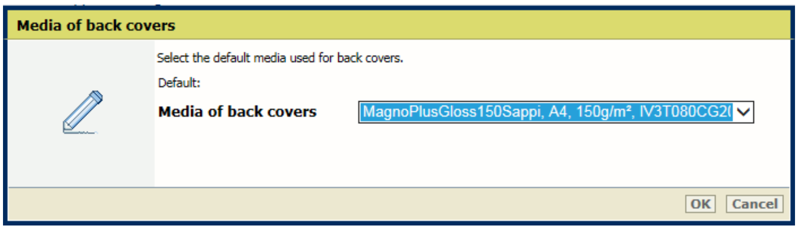
4. Click [OK].
5. Go to the [Back covers] section.



Back covers	Edit
Setting	Value
Media of back covers	A4, 100g/m², TopColor, White
Media name	
Media type	TopColor
Media size	A4
Tab	No
Media width	210.0 mm
Media height	297.0 mm
Media weight	100 g/m²
Media colour	White
Punch count	0
Insert	No

[Back covers] section

6. Use the [Media of back covers] setting to select the default media of back covers.



[Media of back covers] setting

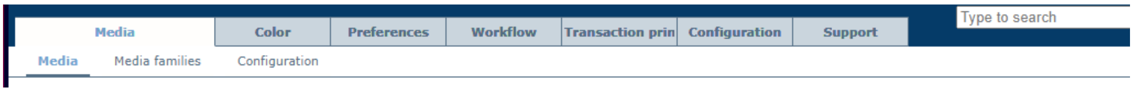
7. Click [OK].

Import and export definitions of special pages

Special pages are banner pages, trainer pages, job tickets, separator sheets and covers. You can define the default media and media print modes of special pages.

It is also possible to import or export the media definitions of special pages.

1. Open the Settings Editor and go to: [Media]→[Media].



[Media] tab

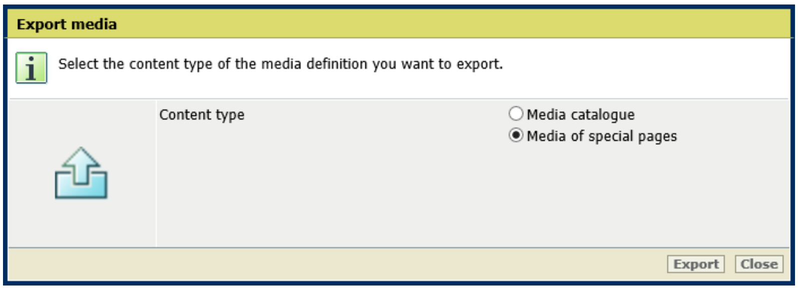
2. Go to the media catalog.

Media catalog

Export the definitions of special pages

You can export an XML file with the current media definitions of the special pages. This XML file can be imported into another printer.

1. Click [Export].



Export media definitions of special pages

2. Click [Media of special pages].
3. Click [Export].
4. Save the file: *mediausage.xml*.

Import the definitions of special pages

You can import an XML file with the media definitions of special pages.

1. Click [Import].

Import media definitions of special pages

2. Click [Media of special pages].
3. Browse to the XML file: *mediausage.xml*
4. Confirm the selection.
5. Click [Import].

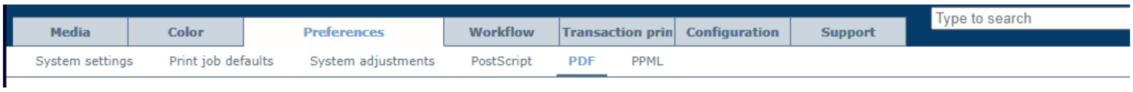
More information

- *Define default media for job tickets and covers* on page 86
- *Define default use of special pages* on page 81

Define PostScript, PDF and PPML job defaults

You can define PostScript, PDF, and PPML job defaults.

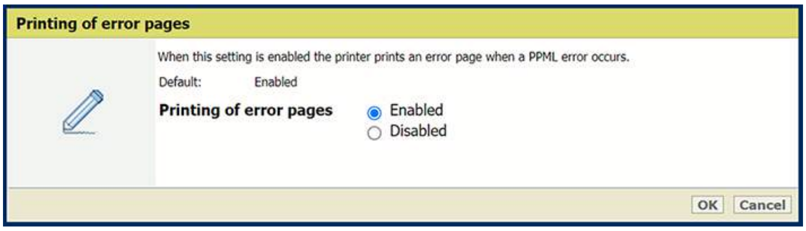
1. Open the Settings Editor and go to: [Preferences].
2. Click [PostScript], [PPML] or [PDF] tab.



[PDF] tab

Enable error pages (PPML)

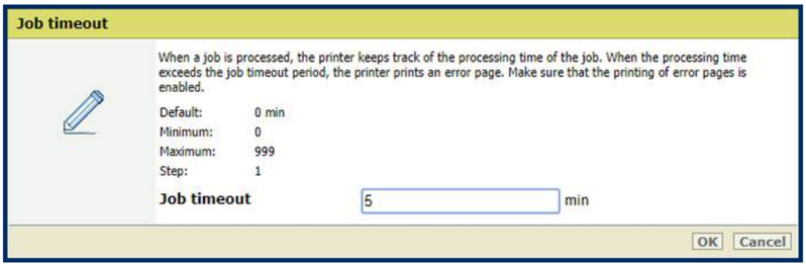
Use the [Printing of error pages] setting to enable or disable the printing of error pages.



[Printing of error pages] setting

Define job timeout (PostScript)

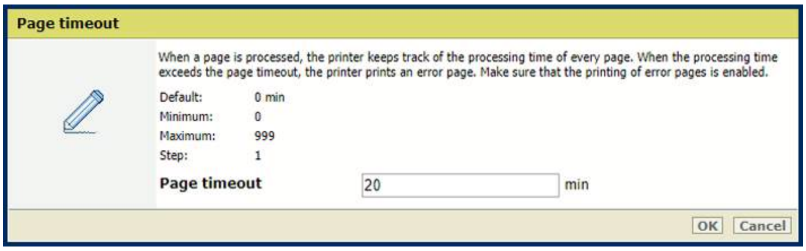
Use the [Job timeout] setting to define the job timeout. When the processing time of a PostScript job exceeds the set value, the printer prints an error page.



[Job timeout] setting

Define page timeout (PostScript)

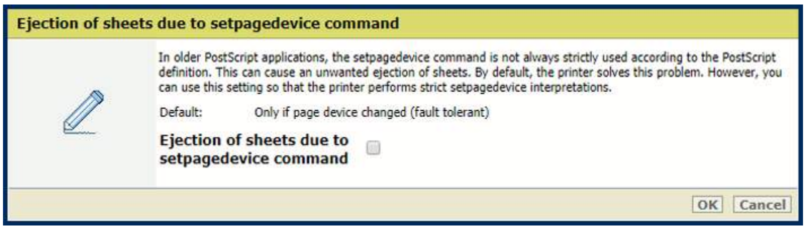
Use the [Page timeout] setting to define a page timeout. When the processing time of a PostScript page exceeds the set value, the printer prints an error page.



[Page timeout] setting

Define the interpretation of the setpagedevice command (PostScript)

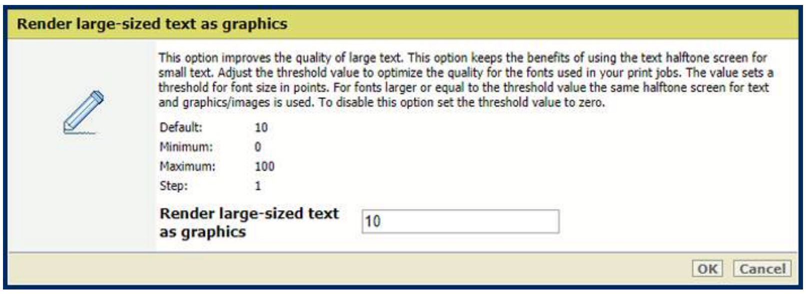
Use the [Ejection of sheets due to setpagedevice command] setting to define if the printer must perform strict setpagedevice interpretations.



[Ejection of sheets due to setpagedevice command] setting

Define the large-sized text rendering (PDF, PPML)

Use the [Render large-sized text as graphics] to define from which text size PRISMAsync must render text as graphics for optimal print quality.



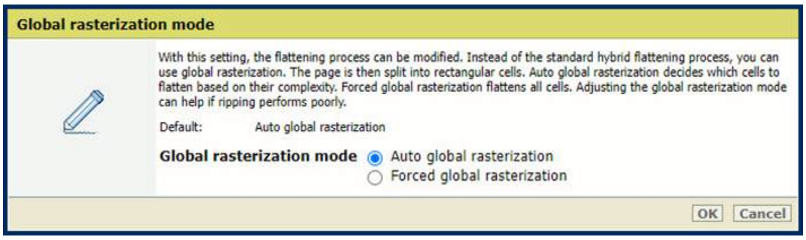
[Render large-sized text as graphics] setting

Global rasterization mode (PDF, PPML)

Use the [Global rasterization mode] setting to modify the flattening process.

- Select [Auto global rasterization] to automatically select the flattening algorithm (standard hybrid flattening or global rasterization) depending on page complexity.
- Select [Forced global rasterization] to always use global rasterization.

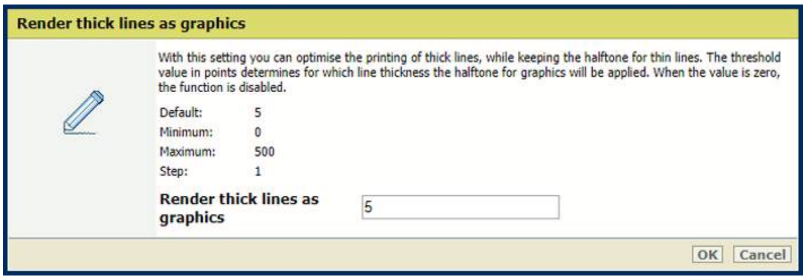
The standard hybrid flattening algorithm is fast and performs well in most cases. Each region where objects overlap is flattened in a single color-blending operation. However, standard hybrid flattening can sometimes be too memory-intensive and lead to a crash, for example, when many objects on a page are stacked on top of each other. In such cases, it is recommended to use global rasterization instead. This algorithm uses pixel-to-pixel color blending.



[Global rasterization mode] setting

Define the thick lines rendering (PDF)

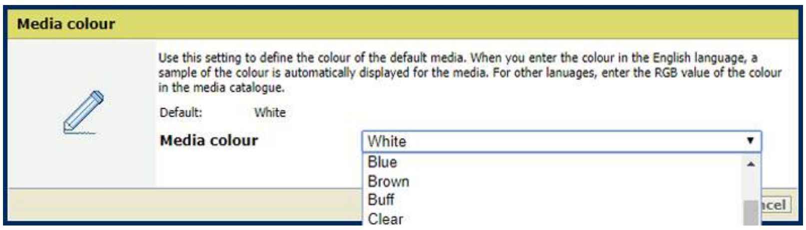
Use the [Render thick lines as graphics] setting to define from which line thickness PRISMAsync must render the lines for optimal print quality.



[Render thick lines as graphics] setting

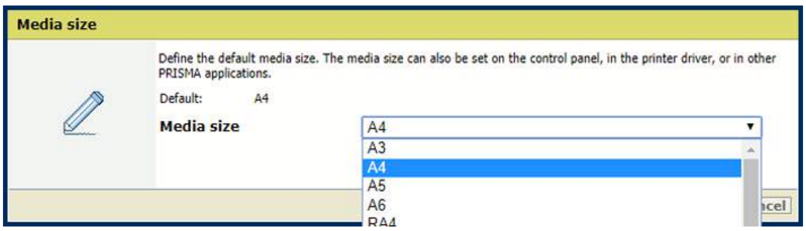
Define the default media (PostScript, PPML, PDF)

1. Use the [Media color] setting to define the color of the default media. When you enter the color name in the English language, a sample of the color is automatically displayed in the media catalog. For other languages, enter the RGB value of the color.



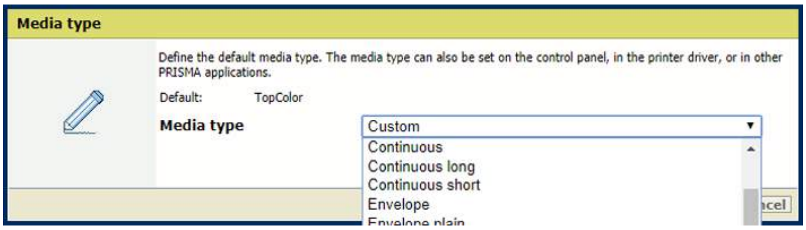
[Media color] setting

2. Use the [Media size] setting to define the default media size.



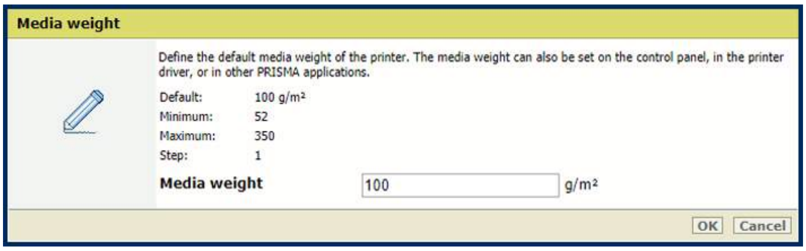
[Media size] setting

3. Use the [Media type] setting to define the default media type.



[Media type] setting

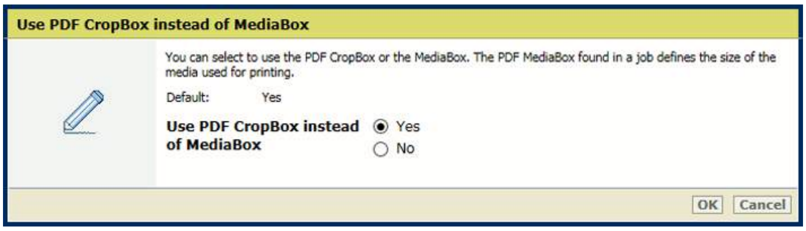
4. Use the **[Media weight]** setting to define the default media weight.



[Media weight] setting

Define the use of CropBox or MediaBox (PDF)

Use the [Use PDF CropBox instead of MediaBox] setting to use the PDF CropBox or the PDF MediaBox. The PDF MediaBox defines the size of the media used for printing.

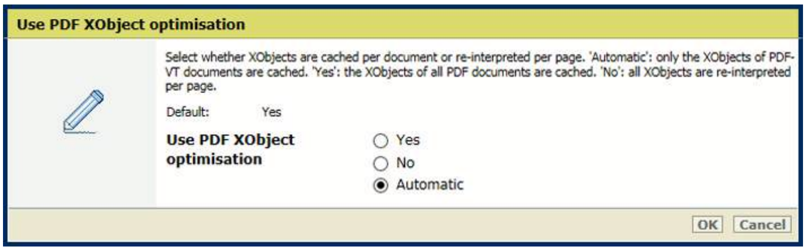


PDF CropBox

Define XObject optimization (PDF)

Use the [Use PDF XObject optimization] setting to define how XObject objects are processed. XObject objects, such as background images, are defined only once in a PDF file.

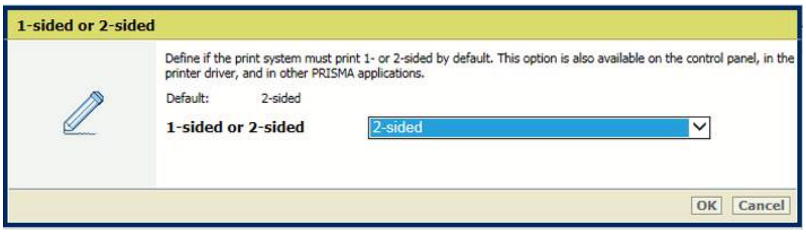
- [Automatic]: the XObject of PDF/VT jobs are cached.
- [Enabled]: the XObject of every PDF job are cached.
- [Disabled]: the XObject are reinterpreted per page.



XObjects

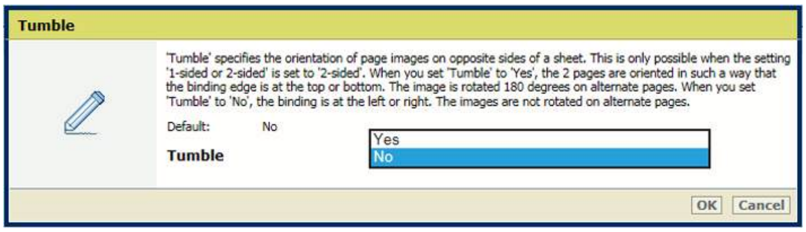
Define the default delivery of prints (PostScript, PPML, PDF)

- Use the [1-sided or 2-sided] setting to define if jobs are printed one- or two-sided.



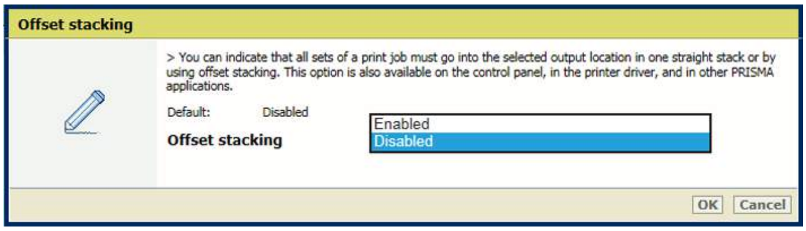
[1-sided or 2-sided] setting

- Use the [Tumble] setting to tumble two-sided documents that are bound at the top or the bottom edge. The tumble setting rotates the imposition of the back side 180 degrees.



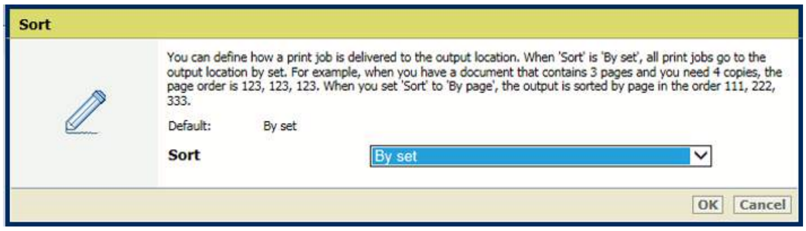
[Tumble] setting

- Use the [Offset stacking] setting to indicate if stacking occurs with or without an offset.



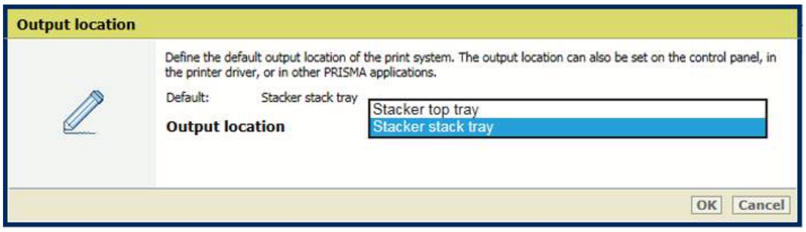
[Offset stacking] setting

- Use the [Sort] setting to define if sorting occurs by set or by page.



[Sort] setting

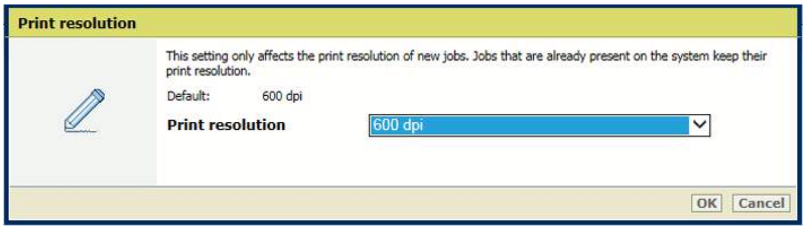
- Use the [Output tray] setting to select the output tray.



[Output tray] setting

Define the print resolution (PostScript, PPML, PDF)

Use the [Print resolution] setting to define the default print resolution.

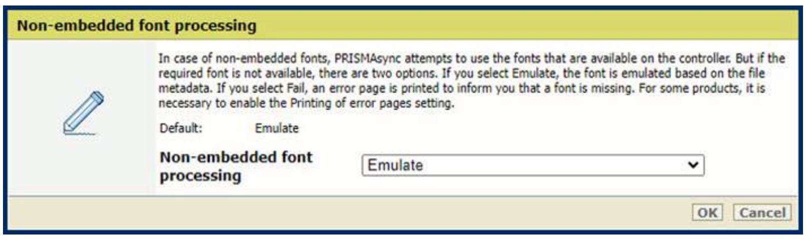


[Print resolution] setting

Define the substitution of a missing font (PDF, PPML)

In case of non-embedded fonts, PRISMAsync attempts to use the fonts that are available on the controller. If the required font is not available, there are two options.

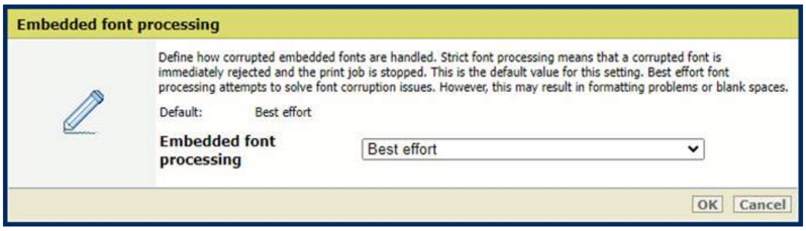
- If you select [Emulate font], the font is emulated based on the file metadata.
- If you select [Fail], an error page is printed to inform you that a font is missing. For some products, it is necessary to enable the [Printing of error pages] setting.



[Non-Embedded Font Processing] setting

To define how corrupted embedded fonts are handled, use the [Embedded Font Processing] setting.

- [Strict] font processing means that a corrupted font is immediately rejected and the print job is stopped.
- [Best effort] font processing attempts to solve font corruption issues. However, this may result in formatting problems or blank spaces.

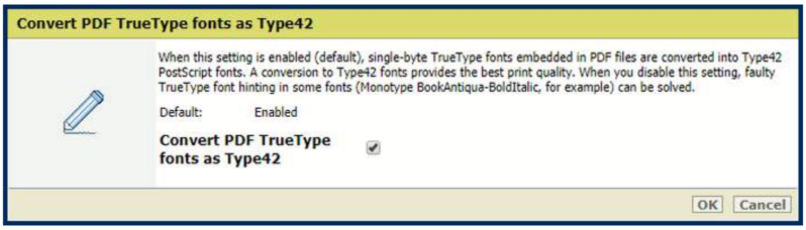


[Embedded Font Processing] setting

NOTE
You need to restart the system so that the changes can take effect. You have the option to restart now or at a later time. Keep in mind that delaying the restart results in inaccurate job previews.

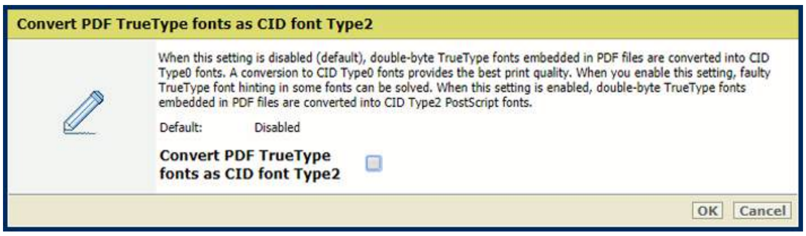
Define the font handling (PDF)

- Use the [Convert PDF TrueType fonts as Type42] setting to convert double-byte TrueType fonts embedded in PDF files into CID Type0 fonts.



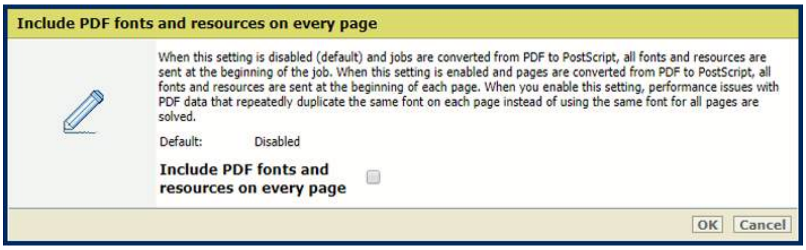
[Convert PDF TrueType fonts as Type42] setting

- Use the [Convert PDF TrueType fonts as CID font Type2] setting to convert single-byte TrueType fonts embedded in PDF files into Type42 PostScript fonts.



[Convert PDF TrueType fonts as CID font Type2] setting

- Use the [Include PDF fonts and resources on every page] setting to send all PDF fonts and resources at the beginning of each page. Be aware that this setting can cause performance issues.

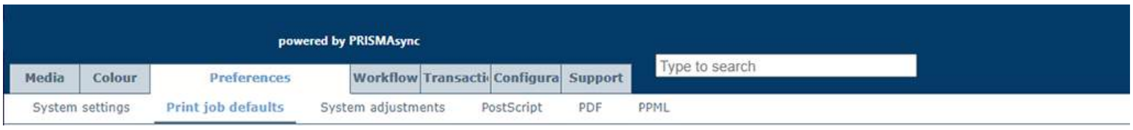


[Include PDF fonts and resources on every page] setting

Define default finisher settings

You can define finisher settings.

1. Open the Settings Editor and go to: [Preferences]→[Print job defaults].



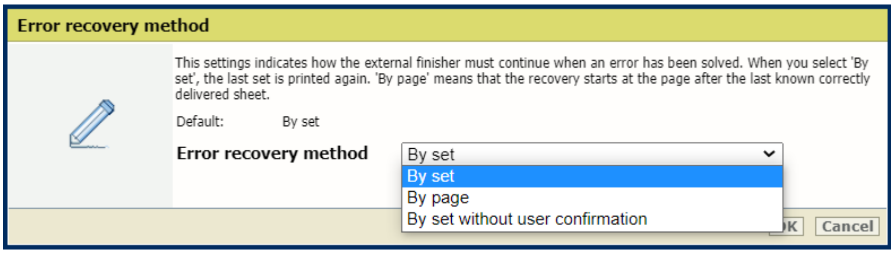
[Print job defaults] tab

Define the default external finisher settings

1. Go to the [External finisher] section.

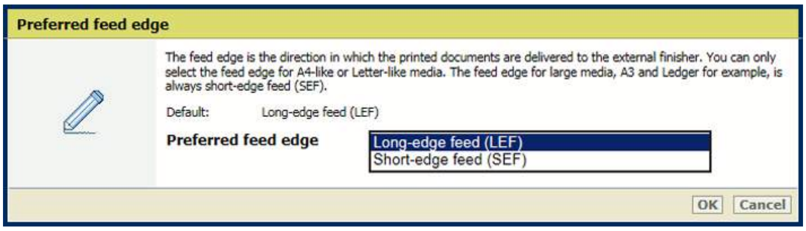
[External finisher] section

2. Use the [Error recovery method] setting to define how to continue when an error occurs.
   - [By page without user confirmation]: print the last set again.
   - [By set with user confirmation]: start at the page after the last known correctly delivered sheet. The printer restarts after user confirmation.
   - [By set without user confirmation]: print the last set again. The printer restarts automatically after error has been resolved, without user confirmation.



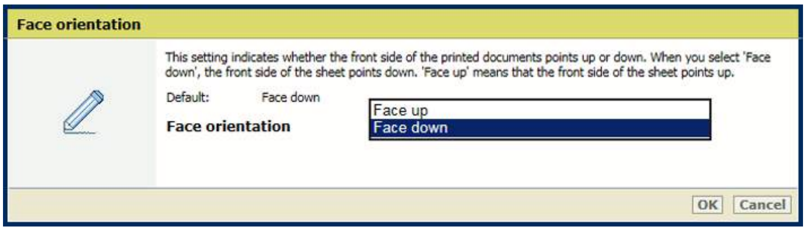
[Error recovery method] setting

3. Use the [Preferred feed edge] setting to indicate the default feed direction of the sheets to the external finisher.
   - [Long-edge feed (LEF)]: long-edge feed direction.
   - [Short-edge feed (SEF)]: short-edge feed direction.



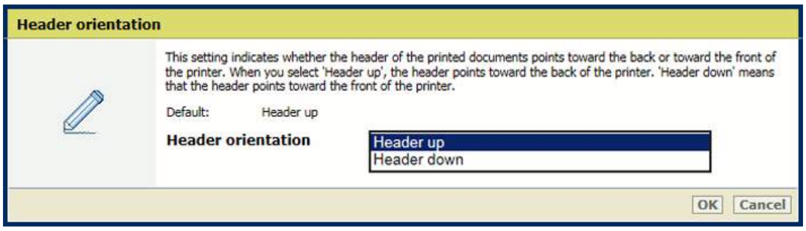
[Preferred feed edge] setting

4. Use the [Face orientation] setting to define the default face orientation.
   - [Face up]: the front side of the sheet points up.
   - [Face down]: the front side of the sheet points down.



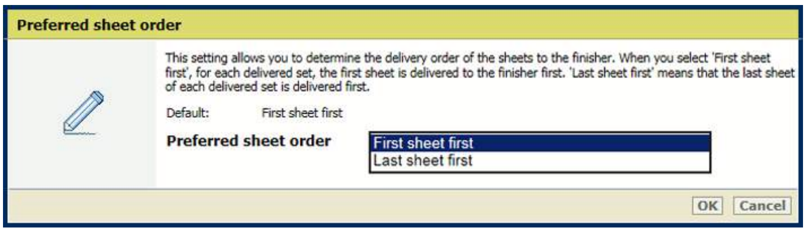
[Face orientation] setting

5. Use the [Header orientation] setting to define the default orientation of the header.
   - [Header-up]: the header points toward the back of the printer.
   - [Header-down]: the header points toward the front of the printer.



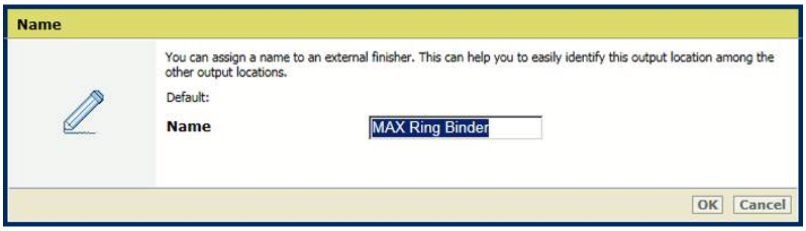
[Header orientation] setting

6. Use the [Preferred sheet order] setting to define the default order of the printed sheets within a job.
   - [First sheet first]: deliver the first sheet of each set first.
   - [Last sheet first]: deliver the last sheet of each set first.



[Preferred sheet order] setting

7. Use the [Name] setting to define a meaningful name for the external finisher.
   The name of the external finisher is visible in the job properties. Open the job properties and go to: [Print delivery]→[Output tray].



[Name] setting

Define the default stacker / stapler settings

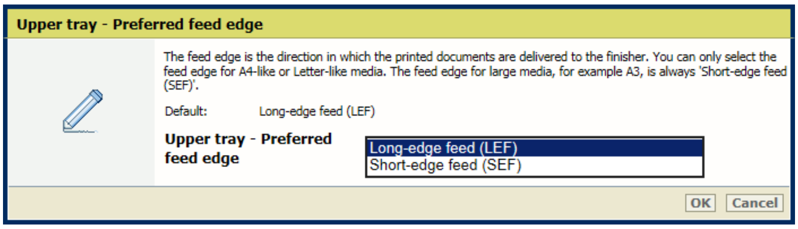
1. Go to the [Stacker/stapler] section.

Stacker/stapler

Setting	Value
Stacker/stapler	Present
Upper tray - Preferred feed edge	Long-edge feed (LEF)
Upper tray - Preferred face direction	Face down
Upper tray - Preferred header direction	Header up
Upper tray - Preferred sheet order	First sheet first
Lower tray - Preferred feed edge	Long-edge feed (LEF)
Lower tray - Preferred face direction	Face down
Lower tray - Preferred header direction	Header up
Lower tray - Preferred sheet order	First sheet first
Booklet tray - Preferred feed edge	Short-edge feed (SEF)
Booklet tray - Preferred face direction	Face up
Booklet tray - Preferred header direction	Header up
Booklet tray - Preferred sheet order	Last sheet first

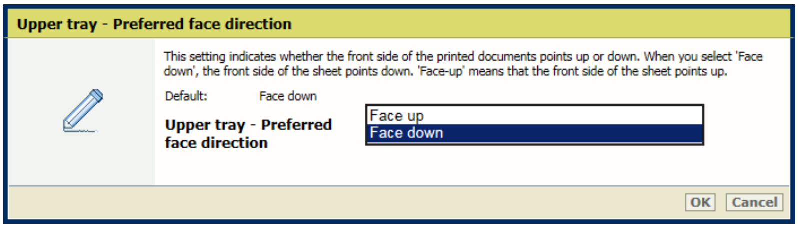
[Stacker/stapler] section

2. Use the [Upper tray - Preferred feed edge], [Lower tray - Preferred feed edge] or [Booklet tray - Preferred feed edge] setting to indicate the default feed direction of the sheets on the output tray.



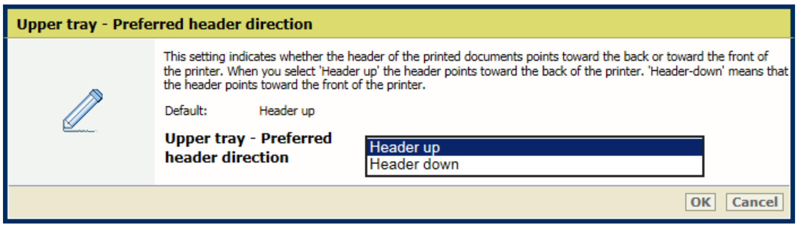
Preferred feed edge setting

3. Use the [Upper tray - Preferred face direction], [Lower tray - Preferred face direction] or [Booklet tray - Preferred face direction] setting to indicate if the front side of the sheets points up or down by default.



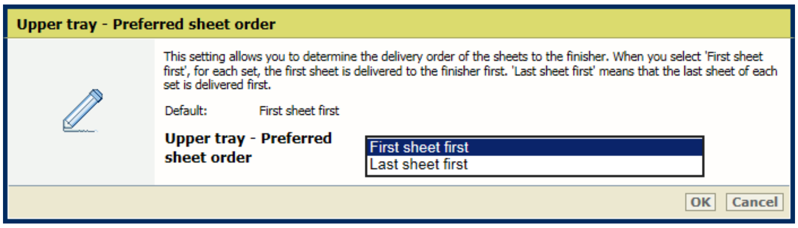
Preferred face direction setting

4. Use the [Upper tray - Preferred header direction], [Lower tray - Preferred header direction] or [Booklet tray - Preferred header direction] setting to indicate if the header points to the front side of the printer or to the back side of the printer by default.



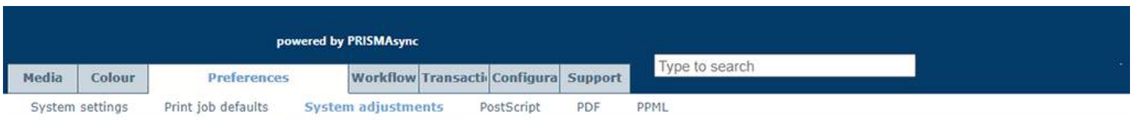
Preferred header direction setting

5. Use the [Upper tray - Preferred sheet order], [Lower tray - Preferred sheet order] or [Booklet tray - Preferred sheet order] setting to indicate the default order of the printed sheets within a job set.



Preferred sheet order setting

6. Go to [Preferences]→[System adjustments].



[System adjustments] tab

7. Go to the [Finisher adjustments] section.

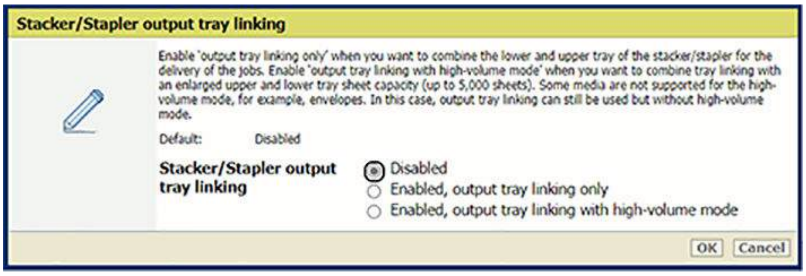
Finisher adjustments

Setting	Value
Saddle stitch fold placement adjustment	0.0 mm
Fold location for tri-fold in	0.0 mm
Fold location for Z-folding	0.0 mm
Fold location for tri-fold out	0.0 mm
Fold location for half-folding	0.0 mm
Fold location of front panel for double parallel folding	0.0 mm
Fold location of back panel for double parallel folding	0.0 mm
Stack tray - Offset stacking	0.0 mm
Minimum filling level	80 %
Maximum stack height	302 mm
Trim width adjustment in horizontal direction	2.0 mm
Fine adjustment of the Perfect Binding finishing size (horizontal)	0.0 mm
Fine adjustment of the Perfect Binding finishing size (vertical)	0.0 mm
Adjustment of the alignment guides for stapling	0.0 mm
Adjustment of the alignment of the stacker/stapler upper tray	0.0 mm
Adjustment of the alignment of the stacker/stapler lower tray	0.0 mm
Priority mode for jobs with thin paper	Focus on productivity
Priority mode for jobs with two staples	Focus on exact staple location
Stacker/Stapler output tray linking	Disabled
Professional puncher punching	Enabled
Professional puncher creasing	Enabled
Professional puncher perforating	Enabled

[Finisher adjustments] section

8. Use the [Stacker/Stapler output tray linking] setting to link the output trays of the stacker/stapler.
   - Enable [Enabled, output tray linking only] when you want to combine the lower and upper tray of the stacker/stapler for the delivery of the jobs.
   - Enable [Enabled, output tray linking with high-volume mode] when you want to combine tray linking with an enlarged upper and lower tray sheet capacity (up to 5,000 sheets).

   NOTE
   Some media are not supported for the high-volume mode, for example, envelopes. In this case, output tray linking can still be used but without high-volume mode.



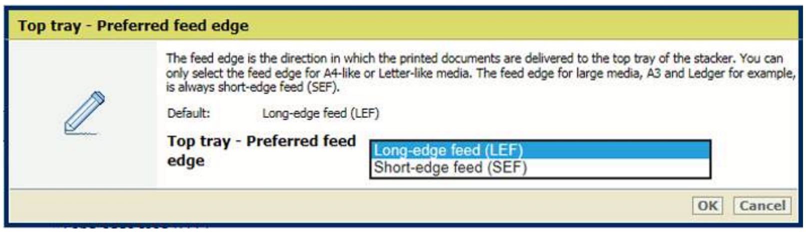
[Stacker/Stapler output tray linking] setting

Define default stacking in the high capacity stacker

1. Go to the [Stacker] section.

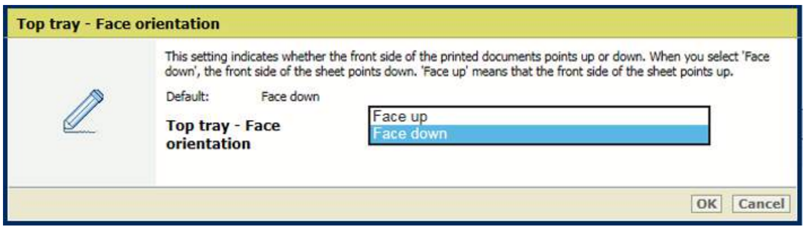
[Stacker] section

2. Use the [Top tray - Preferred feed edge] and [Stack tray - Preferred feed edge] setting to indicate if the long or short side of the sheet refers to the feed direction of the sheets by default.
   - [Long-edge feed (LEF)]: long-edge feed direction.
   - [Short-edge feed (SEF)]: short-edge feed direction.



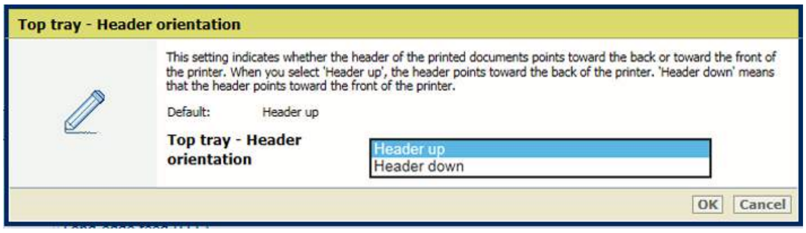
Default feed direction

3. Use the [Top tray - Face orientation] and [Stack tray - Face orientation] settings to indicate if the front side of the prints points up or down by default.
   - [Face up]: front side of first printed sheet is visible.
   - [Face down]: back side of last printed sheet is visible.



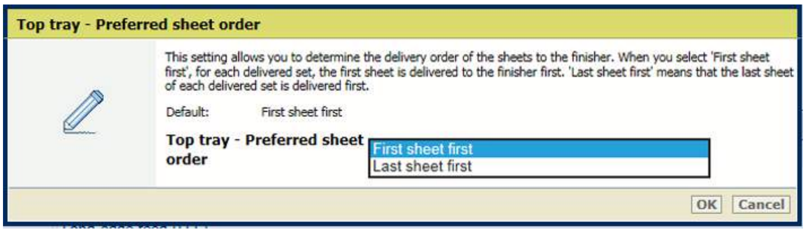
Default face orientation

4. Use the [Top tray - Header orientation] and [Stack tray - Header orientation] settings to indicate if the header of the prints points to the front side of the printer or to the back side of the printer by default.
   - [Header-up]: header of printed document pages points to the back side of printer.
   - [Header-down]: header of printed document pages points to the front side of printer



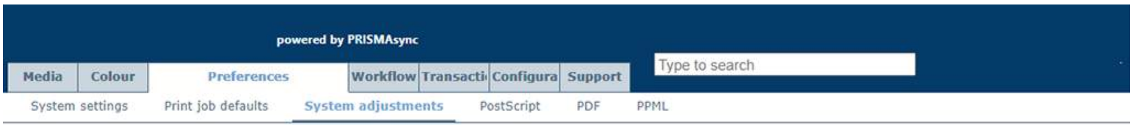
Default header orientation

5. Use the [Top tray - Preferred sheet order] and [Stack tray - Preferred sheet order] settings to define the default order of the prints.
   - [First sheet first]: the first source file page is printed first.
   - [Last sheet first]: the last source file page is printed first.



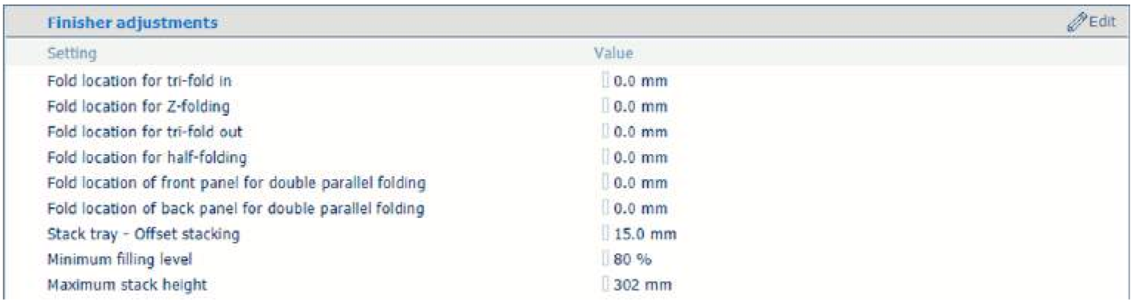
Default sheet order

6. Go to [Preferences]→[System adjustments].



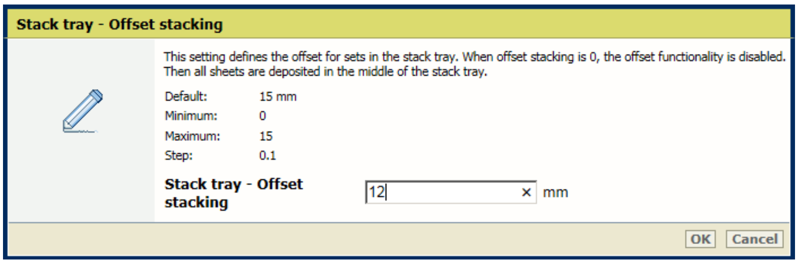
[System adjustments] tab

7. Go to the [Finisher adjustments] section.



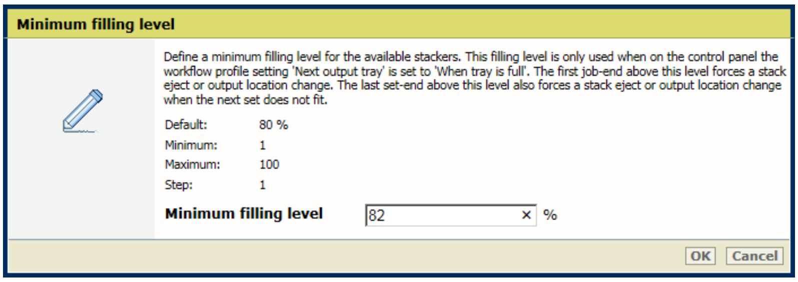
[Finisher adjustments] section

8. Use the [Stack tray - Offset stacking] setting to define the offset value. When the offset value is '0' offset stacking is disabled. The default offset value is 15 mm (0.59 inch).



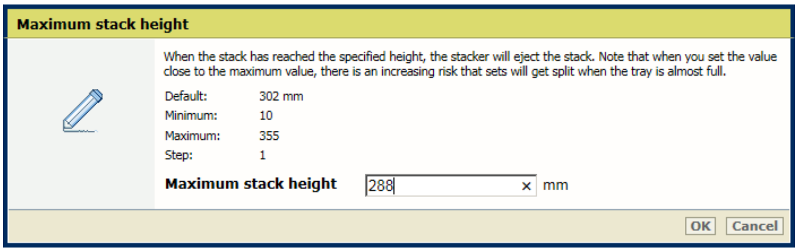
[Stack tray - Offset stacking] setting

9. Use the [Minimum filling level] setting to define the filling level in the stack tray.
   The minimum filling rate is a percentage of the maximum stack height. The default value is 80%. The high capacity stacker ejects the stack as soon as the stack height exceeds the minimum filling rate. The minimum filling rate is active in case the high capacity stacker ejects a stack when the stack tray is full.



[Minimum filling level] setting

10. Use the [Maximum stack height] setting to define the maximum stack height in the stack tray.
    The default stack height is 302 mm (11.9 inch). The minimum stack height is 10 mm (0.39 inch). The maximum stack height is 345 mm (13.58 inch). When the stack height is larger than the default stack height, there is a risk that sheets fall off the stack when the eject tray moves out.



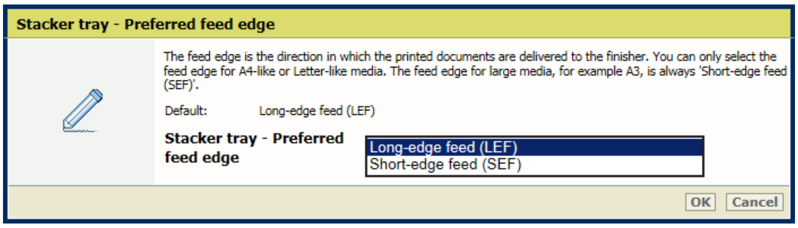
[Maximum stack height] setting

Define the default Perfect Binder settings

1. Go to: [Preferences]→[Print job defaults].
2. Go to the [Perfect Binder] section.

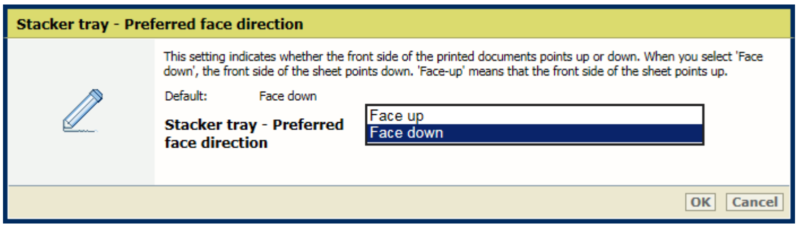
[Perfect Binder] section

3. Use the [Stacker tray - Preferred feed edge] setting to define the feed direction of the sheets on the output tray.
   - [Long-edge feed (LEF)]: long-edge feed direction.
   - [Short-edge feed (SEF)]: short-edge feed direction.



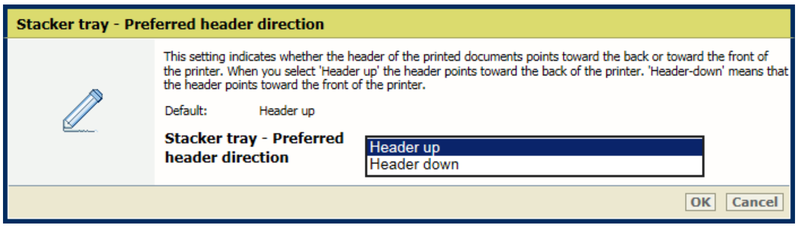
[Stacker tray - Preferred feed edge] setting

4. Use the [Stacker tray - Preferred face direction] setting to define the default face direction.
   - [Face up]: front side of first sheet points up.
   - [Face down]: back side of last printed sheet points down.



[Stacker tray - Preferred face direction] setting

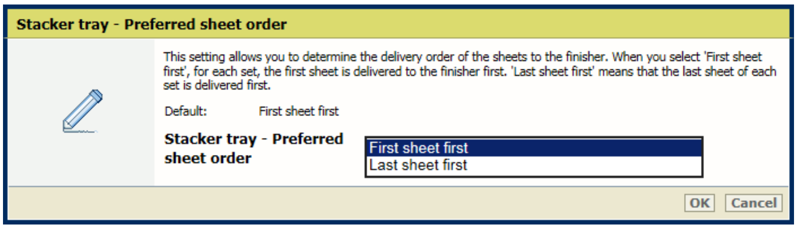
5. Use the [Stacker tray - Preferred header direction] setting to define the default header direction.
   - [Header-up]: header of printed document pages points to the back side of printer.
   - [Header-down]: header of printed document pages points to the front side of printer



[Stacker tray - Preferred header direction] setting

6. Use the [Stacker tray - Preferred sheet order] setting to define the default order of the printed sheets within a job set.
   - [First sheet first]: deliver the first sheet of each set first.

- [Last sheet first]: deliver the last sheet of each set first.

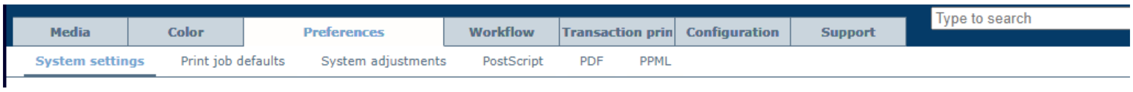


[Stacker tray - Preferred sheet order] setting

Define default job processing

You can define how jobs are processed by default.

1. Open the Settings Editor and go to: [Preferences]→[System settings].



[System settings] tab

2. Go to the [Printing workflows] and [Job management] sections.

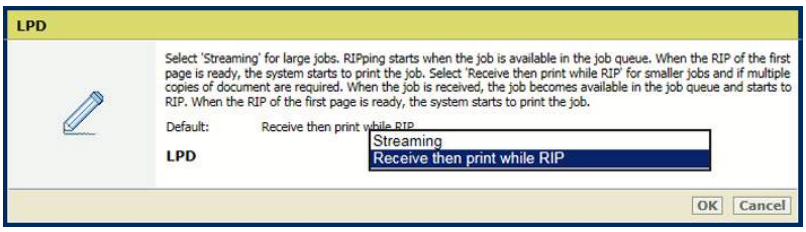
[Printing workflows] and [Job management] sections

Define the processing of jobs submitted via LPD/LPR

The printer protocol LPD can be enabled by the system administrator. The print protocol LPR must be installed on the workstation that submits the jobs.

Use the [LPD] setting to indicate when the RIP starts to process LPR jobs.

- [Streaming] is applicable to large jobs. The job is processed by the RIP and printing can start at the same time.
- [Receive, then print while RIP is in progress] is the default value and is applicable to smaller jobs that have multiple sets. The RIP starts to process the jobs after they arrive in the print queue and before the whole job has been received.



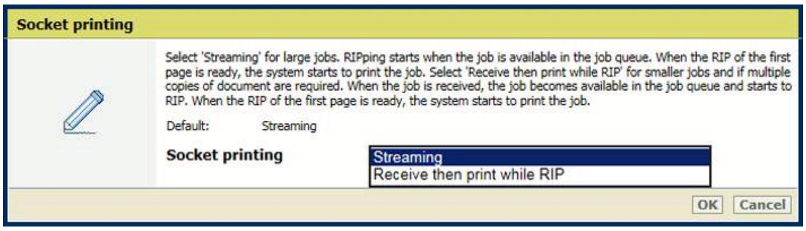
[LPD] setting

Define the processing of jobs that use the socket printing port

Socket printing can be enabled and configured by the system administrator.

Use the [Socket printing] setting to indicate when the RIP starts to process jobs that use the socket printing port.

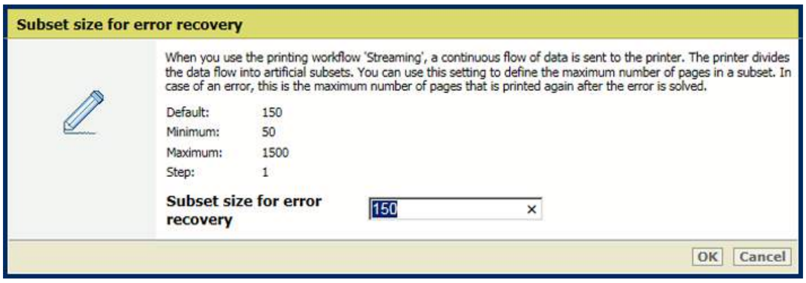
- [Streaming]: is default value and applicable to large jobs. The job is processed by the RIP and printing can start at the same time.
- [Receive, then print while RIP is in progress]: is applicable to for smaller jobs that have multiple sets. The RIP starts to process the jobs after they arrive in the print queue and before the whole job has been received.



[Socket printing] setting

Define the number of printed pages when an error occurs in a streaming job

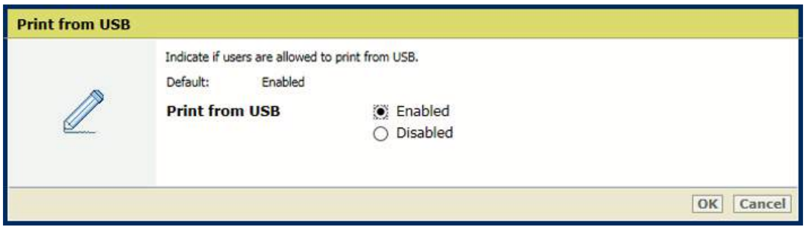
Use the [Subset size for error recovery] setting to define the number of pages that will be printed again after the error is solved.



[Subset size for error recovery] setting

Define if users are allowed to print from a USB drive

Use the [Print from USB] setting to indicate whether printing from a USB drive is allowed.



[Print from USB] setting

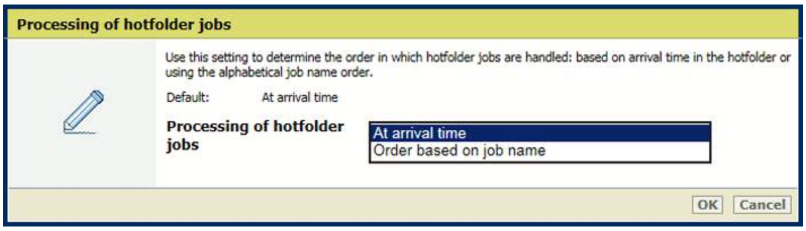
Define the processing order of hotfolder jobs

The system administrator can change the processing order of hotfolder jobs.

Use the [Processing of hotfolder jobs] setting to indicate how jobs are processed in the hotfolder.

- [At arrival time]: job processing takes place when the job arrives in the hotfolder.

- [Order based on job name]: job processing takes place according to the alphabetical order of the job names.

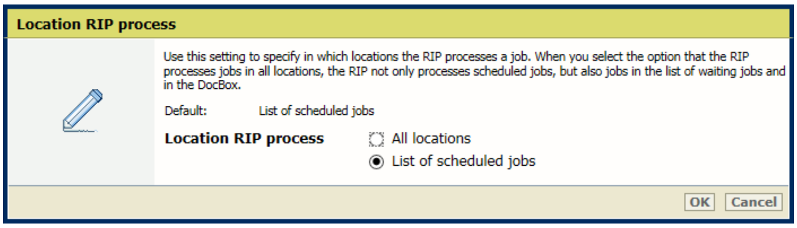


[Processing of hotfolder jobs] setting

Define the location of the RIP process

Use the [Location RIP process] setting to indicate where the RIP processes jobs when they arrive in the print queue.

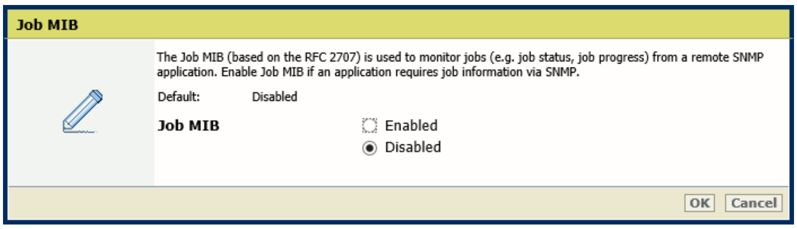
- [All locations]: the RIP can process jobs in all job destinations.
- [List of scheduled jobs]: the RIP can only process jobs when they are in the list of scheduled jobs.



[Location RIP process] setting

Define if remote monitoring of jobs via SNMP is allowed

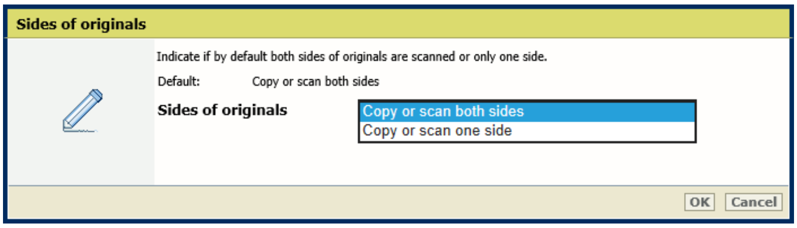
Use the [Job MIB] setting to indicate if remote monitoring of jobs via SNMP is allowed.



[Job MIB] setting

Define if default originals are 1- or 2-sided

Use the [Sides of originals] setting to indicate if originals have information on one or both sides.

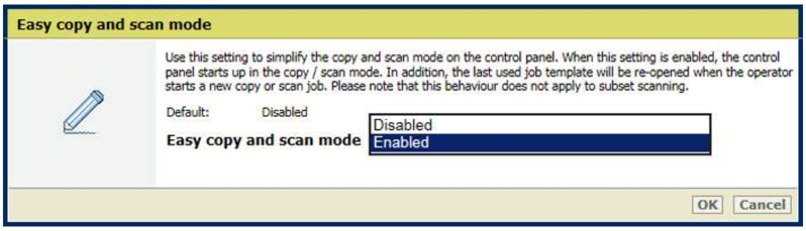


[Sides of originals] setting

Define if job templates must reopen after starting a copy or scan job

Use the [Easy copy and scan mode] setting to indicate that you want to enable the easy copy and scan mode.

- [Enabled]: the last used job template will be re-opened when the operator starts a new copy or scan job.
- [Disabled]: the last used job template will close when the operator starts a new copy or scan job.



[Easy copy and scan mode] setting

Select an accounting mode

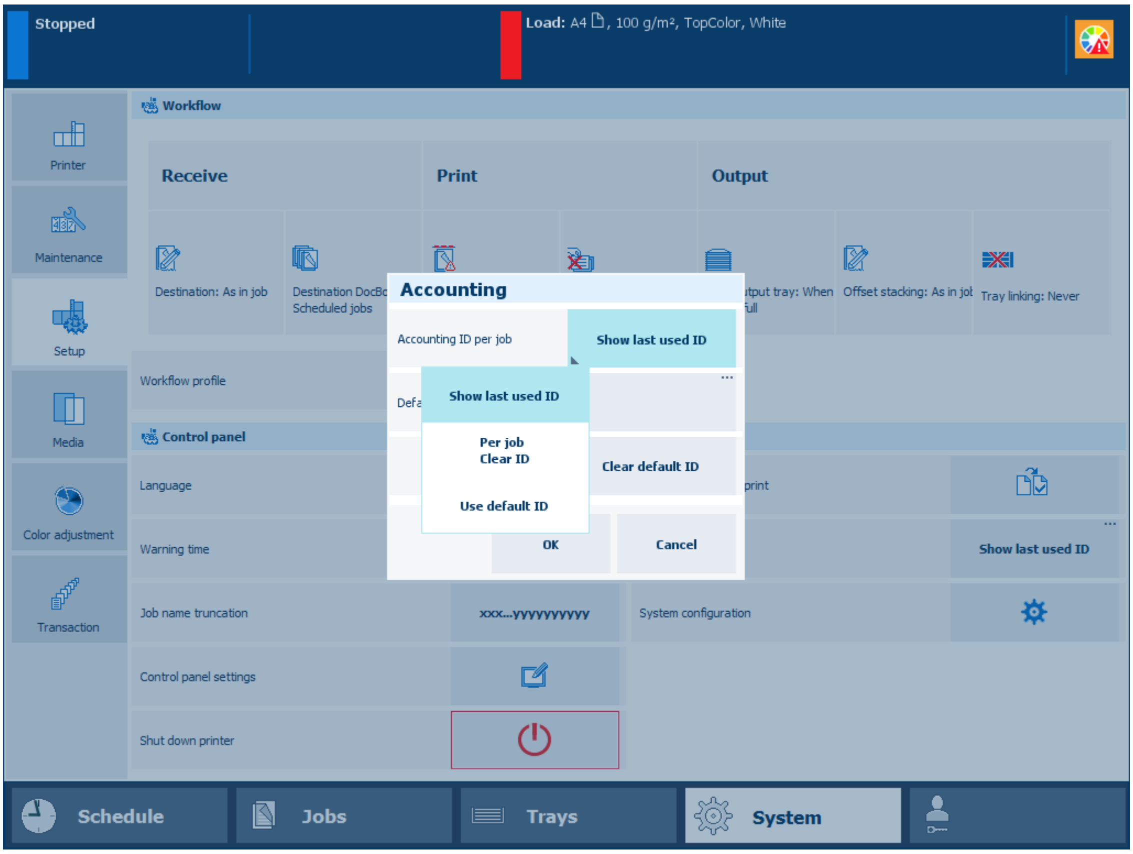
The key operator can define accounting IDs to charge job costs to departments or customers and enable the user identification in the Settings Editor ([Configuration]→[Accounting]).

PRISMAsync can check the entered accounting IDs to make sure that only valid accounting IDs are used. When a submitted job does not have a valid accounting ID, PRISMAsync sends it to the list of waiting jobs. The print system will only start to process the job when a correct accounting ID is entered.

When accounting is active, PRISMAsync Remote Printer Driver, PRISMAsync Remote Manager and the control panel ask for an accounting ID before the print system handles the job. Define the accounting mode on the control panel to determine how accounting ID's must be entered:

- [Show last used ID]
  This mode requires one accounting ID per job. You see the last used accounting ID which you can overwrite.
- [Per jobClear last used ID].
  This mode requires one accounting ID per job. You do not see the last used accounting ID.
- [Use default ID]
  This mode uses a fixed accounting ID, but you can change the accounting ID. Be aware that the fixed mode implies that the print system overrules the accounting ID that is entered in Remote Printer Driver. Use the [Default accounting ID] option to set the default accounting ID and [Clear default ID] to remove the default accounting ID

1. Touch [System]→[Setup]→[Accounting].
2. Touch [Accounting ID per job].



[Accounting ID per job] setting

3. Select an accounting mode.

4. Touch [OK].

You can download account log files and manage account IDs files in the Settings Editor: [Configuration]→[Accounting].

Download account log files

What are account log files

Account log files store details of all printed jobs. The printer generates account log files that can be downloaded for job cost calculation. Every record in the file represents a single job. The record consists of job properties and usage of inks and media.

The system administrator can define how account log file are created.

Account log files have the .CSV (comma delimited) file format. The .CSV format is supported by a large amount of business applications and is useful for transferring tabular data between programs. Account log files can be generated per day, per week, or per month. Completed account log files have the .CSV extension in the name. The current account log file (the file that can still be filled with new jobs) has the .ACL extension in the name.

Use the table below to understand how the name of the account log file is composed.

Period	Format name	Example
Completed account log file, valid for one day	*<serial number><yyyymmdd>.CSV*	93100020170709.CSV
Current account log file, valid for one day	*<serial number><yyyymmdd>.ACL*	93100020170710.ACL
Completed account log file, valid for one week	*<serial number><yyyy>W<week number>.CSV*	9310002017**W**28.CSV
Current account log file, valid for one week	*<serial number><yyyy>W<week number>.ACL*	9310002017**W**28.ACL
Completed account log file, valid for one month	*<serial number><yyyy>M<month number>.CSV*	9310002017**M**07.CSV
Current account log file, valid for one month	*<serial number><yyyy>M<month number>.ACL*	9310002017**M**07.ACL

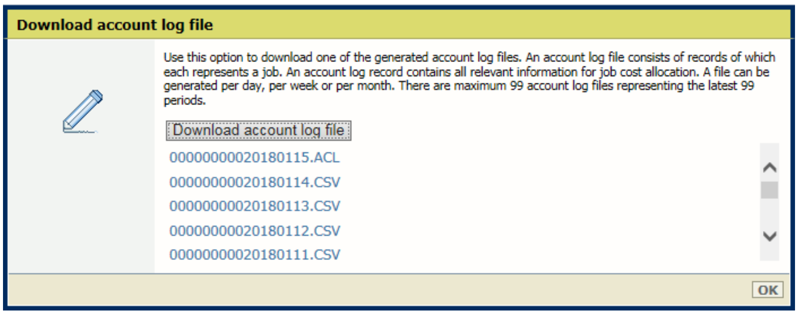
Note the following about account log files.
- Only pages that belong to the source PDF file of the job are counted and listed.
- Separator sheets are counted as printed on one side.
- Banner pages and trailer pages are not counted.
- Blank sheet sides of two-sided printed jobs are counted.
- The values of the usage of toners are expressed in milliliters (mL).

Download account log files

1. Open the Settings Editor and go to: [Configuration]→[Accounting].

[Accounting] tab

2. Click the [Download account log file] setting.



[Download account log file] setting

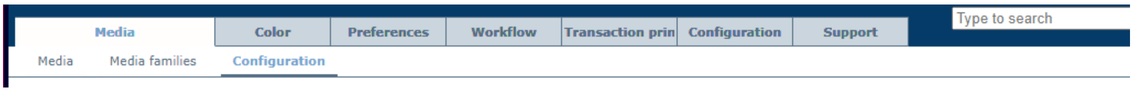
3. Select one of the available account log files.
4. Click [Download account log file].
5. Click [OK].

Define the handling of media attributes in JDF ticket

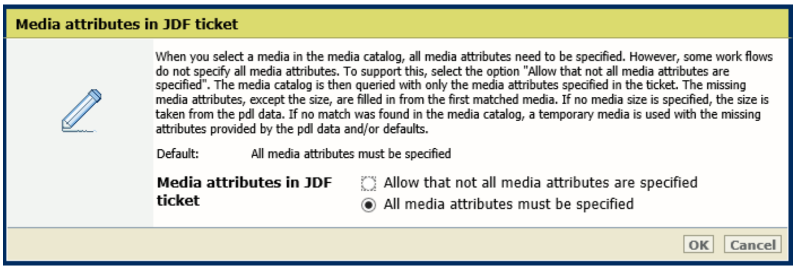
When you select a media in the media catalog, all media attributes need to be specified. However, it can occur that not all media attributes in a JDF job ticket are specified. When you allow that not all media attributes in the JDF ticket are specified, **media matching** is used to find values for these attributes. The media catalog is queried with only the media attributes specified in the ticket. If there is a single, unique match found during the media matching, all attributes of that media are used, including media size. If there is more than one match, the missing attributes, except the size, are filled in from the first matching media. If no media size is specified, the size is taken from the PDL data. If no match was found in the media catalog, a temporary media is used with the missing attributes provided by the PDL data and/or defaults.

1. Open the Settings Editor and go to: [Media]→[Configuration].

   [Media] tab

2. Use the [Media attributes in JDF ticket] setting to define how to handle media attributes in a JDF ticket.
   - [Allow that not all media attributes are specified]: media matching is used to find values for these attributes.
   - [All media attributes must be specified]: the attributes of the JDF ticket, the print document and the default media attributes are used.

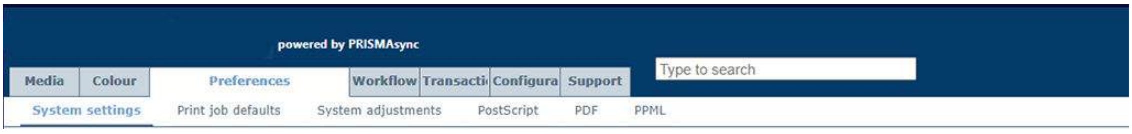
   [Media attributes in JDF ticket] setting

3. Click [OK].

Define output tray for system charts and reports

You can define a specific output tray for the delivery of system charts and reports. The default setting is [Use system default], which in most cases is the upper output tray closest to the printer.

1. Open the Settings Editor and go to: [Preferences]→[System settings].

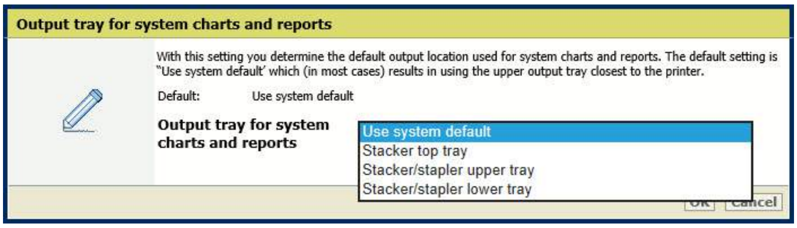


[System settings] tab

2. Go to the [Basic] section.

[Basic] section

3. Use the [Output tray for system charts and reports] setting to select the preferred output tray for system charts and reports.



[Output tray for system charts and reports] setting

4. Click [OK]

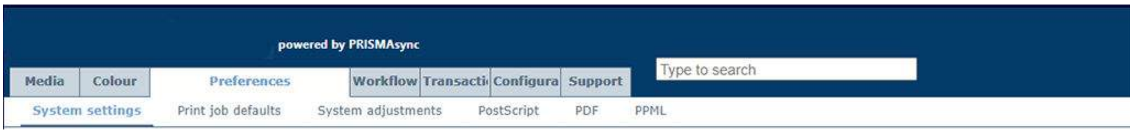
Define sensing unit adjustments

The sensing unit can automatically perform image position (media registration) and colour density adjustments during printing. This reduces registration errors and colour drift in long-run-jobs, which results in high quality prints.

Cyclic and coloured media are not supported by the sensing unit.

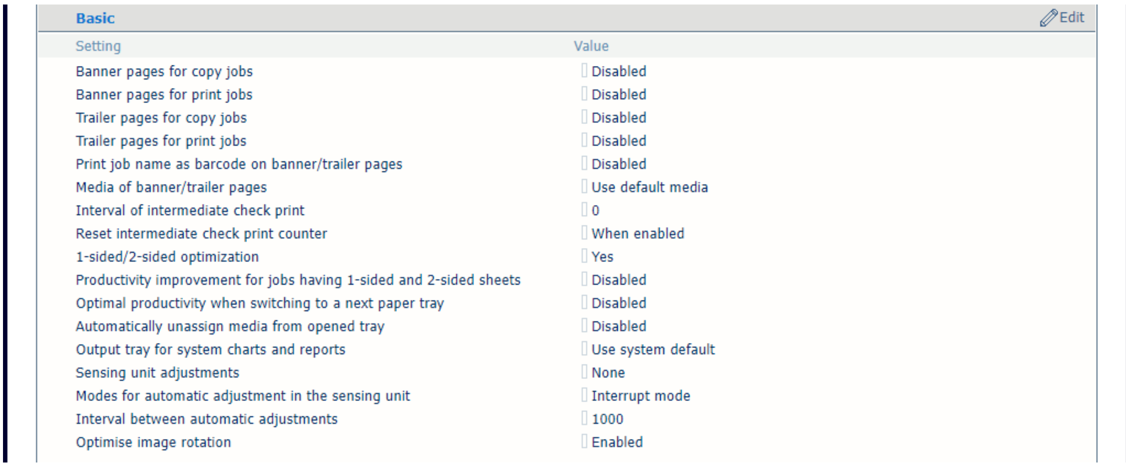
The default settings are installed in the Settings Editor. The sensing unit adjustment settings can also be changed on the control panel and in PRISMAsync Remote Manager at the job settings and Automated Workflow.

1. Open the Settings Editor and go to: [Preferences]→[System settings].



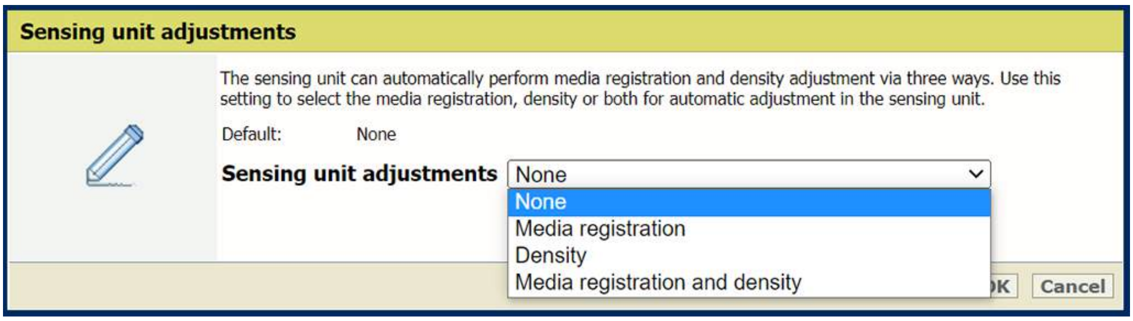
[System settings] tab

2. Go to the [Basic] section.



[Basic] section

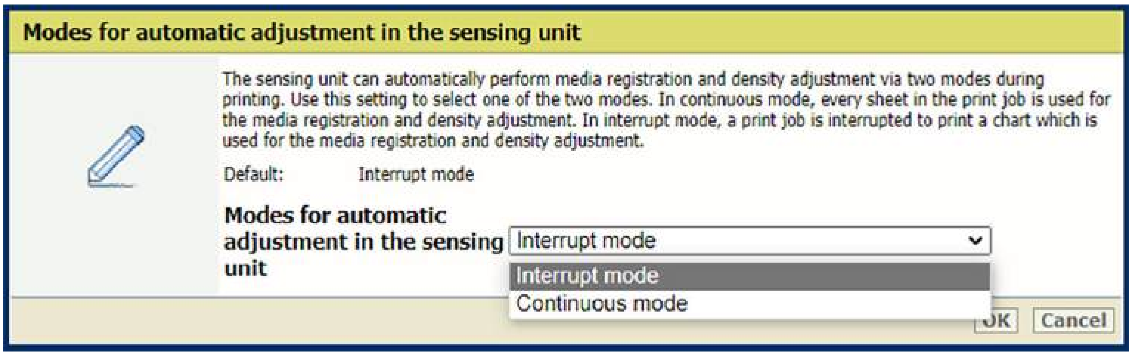
3. Use the [Sensing unit adjustments] setting to select which automatic adjustments the sensing unit performs.



[Sensing unit adjustments] setting

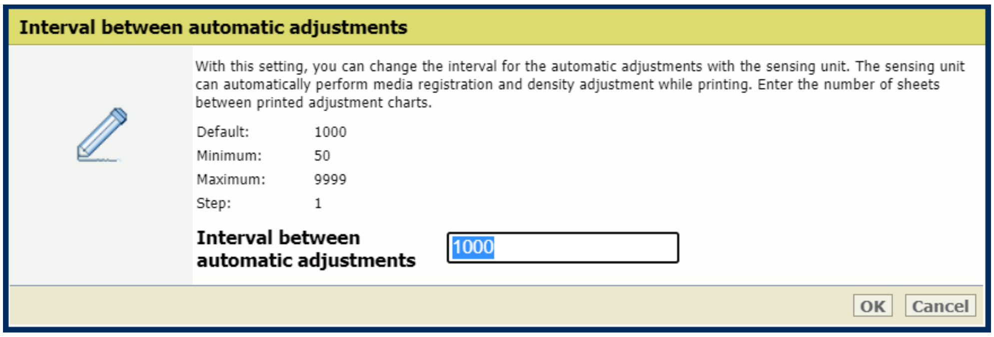
4. Click [OK].
5. Use the [Modes for automatic adjustment] setting to select the mode for the automatic adjustments. The adjustments can be done in two modes: continuous or interrupt mode.

In continuous mode, the printer prints patterns in the trim areas of every sheet of the print job. The sensing unit automatically scans and measures the patterns, any deviations are corrected automatically by the system to ensure consistency in the print quality.
In interrupt mode, measurement charts are printed at an interval. The measurement charts are purged in the sensing unit tray. The sensing unit automatically scans and measures the measurement charts, any deviations are corrected automatically by the system to ensure consistency in the print quality. When the sensing unit tray is full, the user job continues but measurement charts are not printed anymore and the operator does not get feedback.



[Modes for automatic adjustment] setting

6. Click [OK].
7. In case you use interrupt mode, indicate the interval between automatic adjustments. Use the [Interval between automatic adjustments in sensing unit] setting to set the preferred interval.



[Interval between automatic adjustments in sensing unit] setting

NOTE
This setting is only available in the Settings Editor.

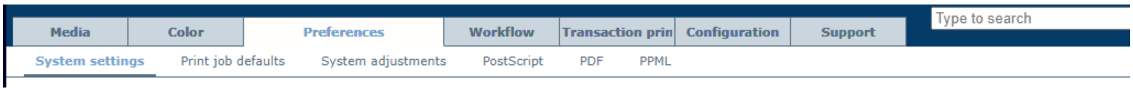
8. Click [OK].

Define print quality settings

You can define print quality settings.

NOTE
Be aware that the print quality settings affect all print jobs.

1. Open the Settings Editor and go to: [Preferences]→[System settings].



[System settings] tab

2. Go to the [Print quality settings] section.

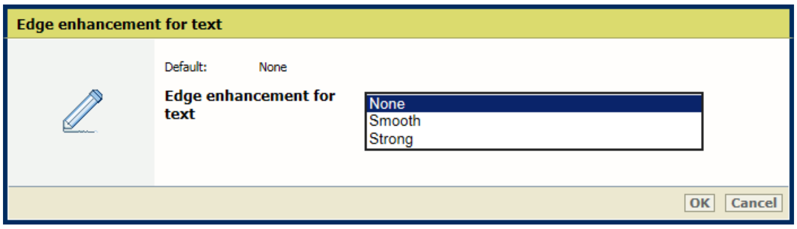
Print quality settings — Edit

Setting	Value
Edge enhancement for text	None
Edge enhancement for graphics	None
Print sharpness	Normal
Image compression optimization	Optimize for image quality
Raster normal	190 dots
Raster fine	230 dot
Stroke width adjustment	Disabled
Width adjustment in horizontal direction	Adjustment level 1
Width adjustment in vertical direction	Adjustment level 1
Condition for stroke width adjustment	Only adjustment for 1200 dpi print jobs
Sheet borders; left and right	2.5 mm
Sheet borders; top and bottom	0.4 mm

[Print quality settings] section

Define the print quality of small text

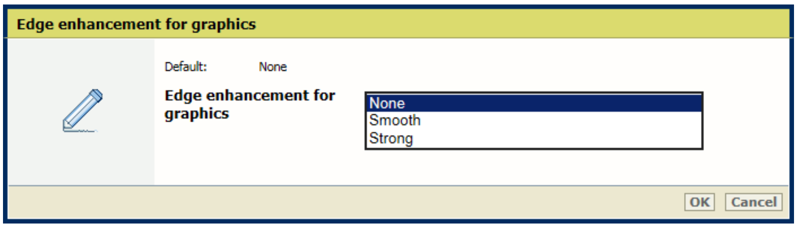
Use the [Edge enhancement for text] setting to improve the print quality of small text.



[Edge enhancement for text] setting

Define the print quality of graphics

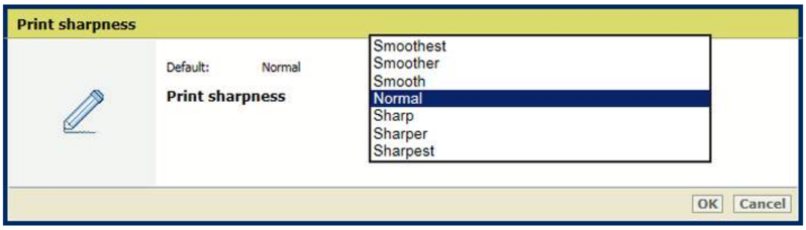
Use the [Edge enhancement for graphics] setting to improve the print quality of graphics.



[Edge enhancement for graphics] setting

Define the print sharpness

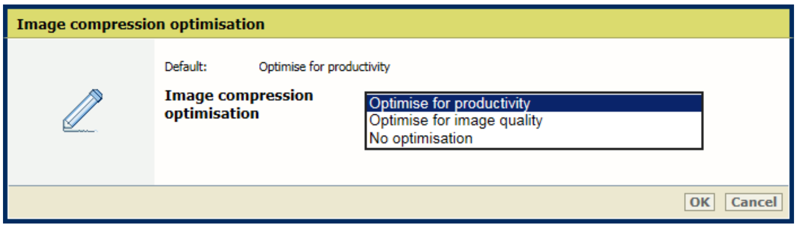
Use the [Print sharpness] setting to increase or decrease the print sharpness.



[Print sharpness] setting

Give priority to image quality or productivity

Use the [Image compression optimisation] setting to give priority to image quality or productivity.

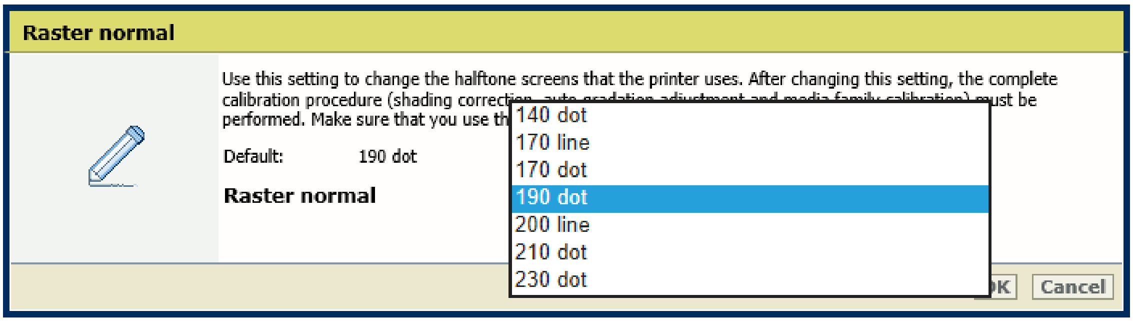


[Image compression optimisation] setting

Define the default halftone screens for the rasters

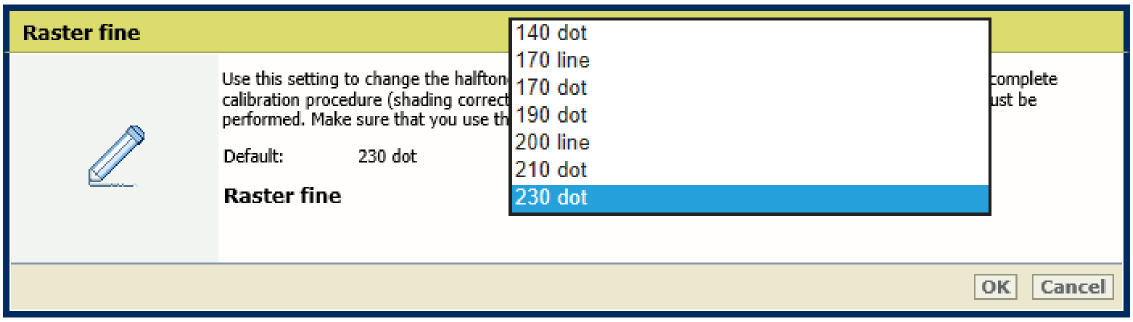
Use this setting to change the halftone screens that the printer uses. After changing this setting, the complete calibration procedure (shading correction, auto gradation adjustment and media family calibration) must be performed. Make sure that you use the matching output profiles in the media family.

1. Use the [Raster normal] setting to define the normal halftone screen.



[Raster normal] setting

2. Use the [Raster fine] setting to change the fine halftone screen.

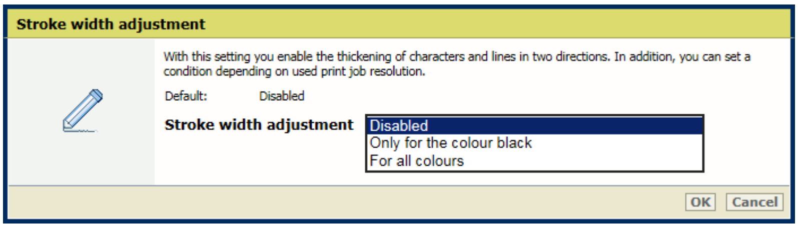


[Raster fine] setting

3. Click [OK].
4. Calibrate the printer and default media families.

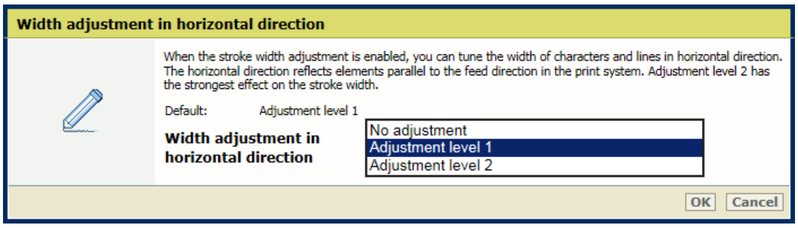
Define the print thickness for characters and lines

1. Use the [Stroke width adjustment] setting to indicate for which colours you want thicker lines.
   - [Only for the color black]
   - [For all colors]



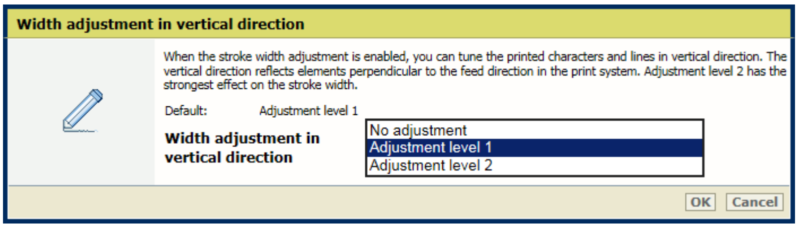
[Stroke width adjustment] setting

2. Use the [Width adjustment in horizontal direction] setting to indicate the width adjustment of characters and lines in horizontal direction.



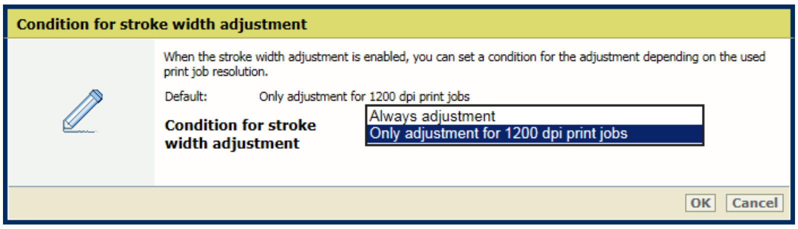
[Width adjustment in horizontal direction] setting

3. Use the [Width adjustment in vertical direction] setting to indicate the width adjustment of characters and lines in vertical direction.



[Width adjustment in vertical direction] setting

4. Use the [Condition for stroke width adjustment] setting to indicate if you want thicker characters and lines for all jobs or for 1200 dpi jobs only.



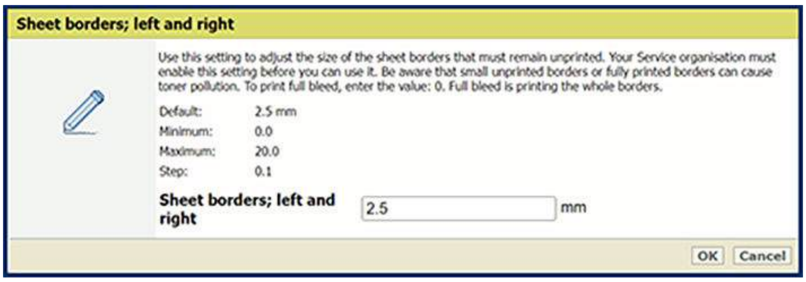
[Condition for stroke width adjustment] setting

5. Click [OK].

Adjust the size of unprinted sheet borders

Use the [Sheet border; left and right] and [Sheet border; top and bottom] settings to adjust the size of the sheet borders that must remain unprinted. Your Service organization must enable this setting before you can use it.

Be aware that small unprinted borders or fully printed borders can cause toner pollution. To print full bleed, enter the value 0. Full bleed is printing the whole borders.



[Sheet border; left and right] setting

More information

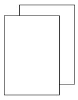
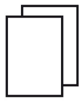
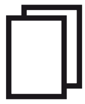
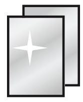
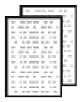
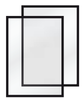
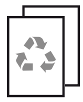
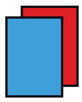
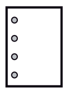
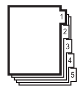
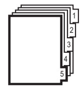
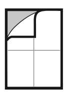
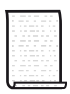
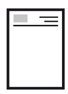
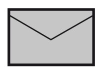
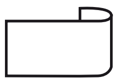
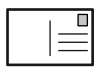
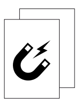
Learn about calibration on page 346

Chapter 6
Job media handling

Learn about media for your output

The paper trays of your print system can handle a variety of media types and sizes such as uncoated, offset coated, plain, recycled, thin and more.

The following table shows the types of media that the paper trays support.

 Thin	 Plain	 Heavy	 Coated
 Textured	 Vellum	 Transparencies/trans-lucents	 Recycled
 Color	 Prepunched	 Forward-order tab pa-per	 Reverse-order tab pa-per
 Labels	 Bond	 Letterhead	 Envelope type a
 Envelope type b	 Long sheets	 Postcard	 Synthetic
 Magnetic			

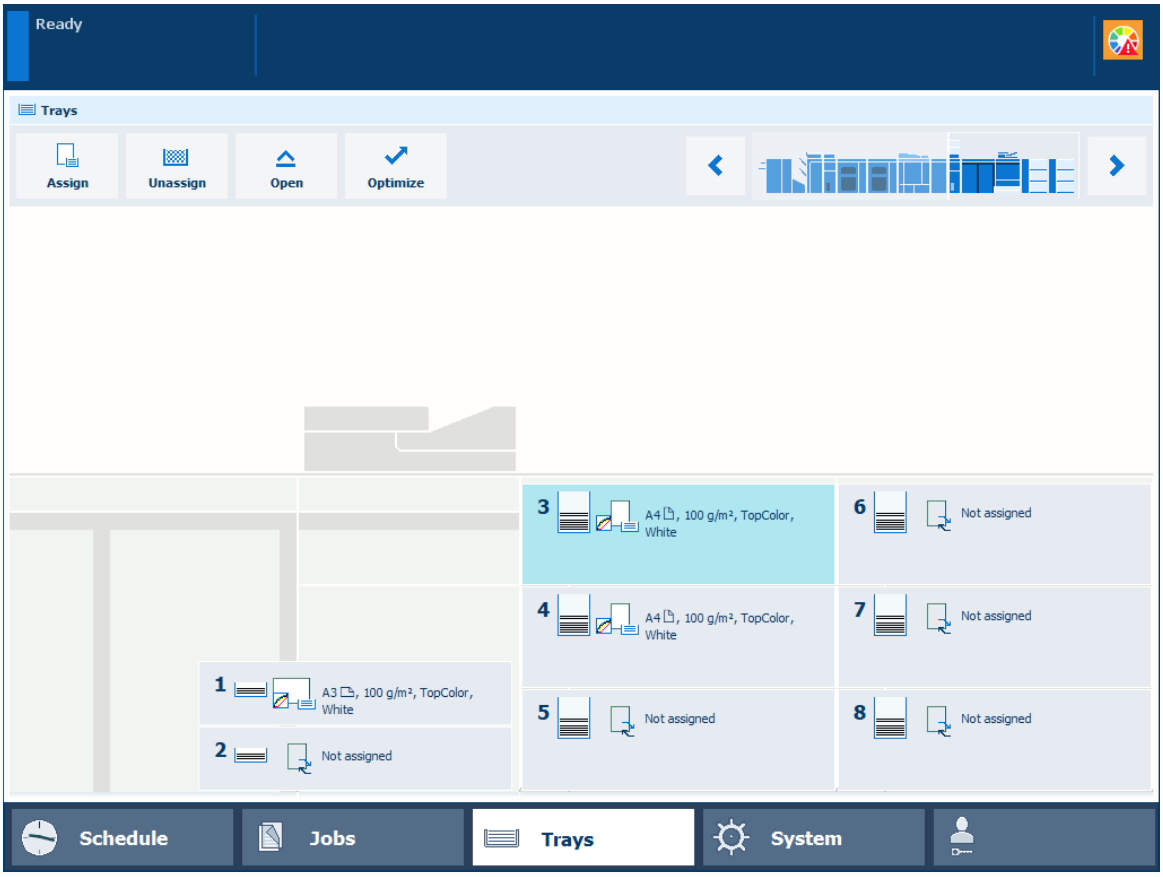
More information

Media specifications on page 613

Assign media to a paper tray

You can use the paper tray button at the right-hand side of the control panel to check the media that are loaded in the paper trays. You can assign media in the Trays view after you load the media in the paper tray. As long as the media type in the paper tray does not change, it is not necessary to assign the media after loading.

NOTE
If the [Automatically unassign media from opened tray] setting is enabled, you need to reassign the media every time you open the paper tray.



Paper trays with and without assigned media

NOTE
You can also use the [Trays] button.

Before you begin

Determine which media the jobs need.

IMPORTANT
Make sure you know how to load media into the paper trays.

Procedure:

1. Press the paper tray button .
2. Select the paper tray in which you want to load the media.
3. Use the open button of the paper tray or touch [Open] in the Trays view.

4. Load the media into the paper tray.
5. Close the paper tray.
6. Close the paper tray.
7. Select the media from the media catalog and touch [OK].

   NOTE
   Only the media that are allowed for the selected paper tray are displayed.

8. Press the paper tray button to close the Trays view.

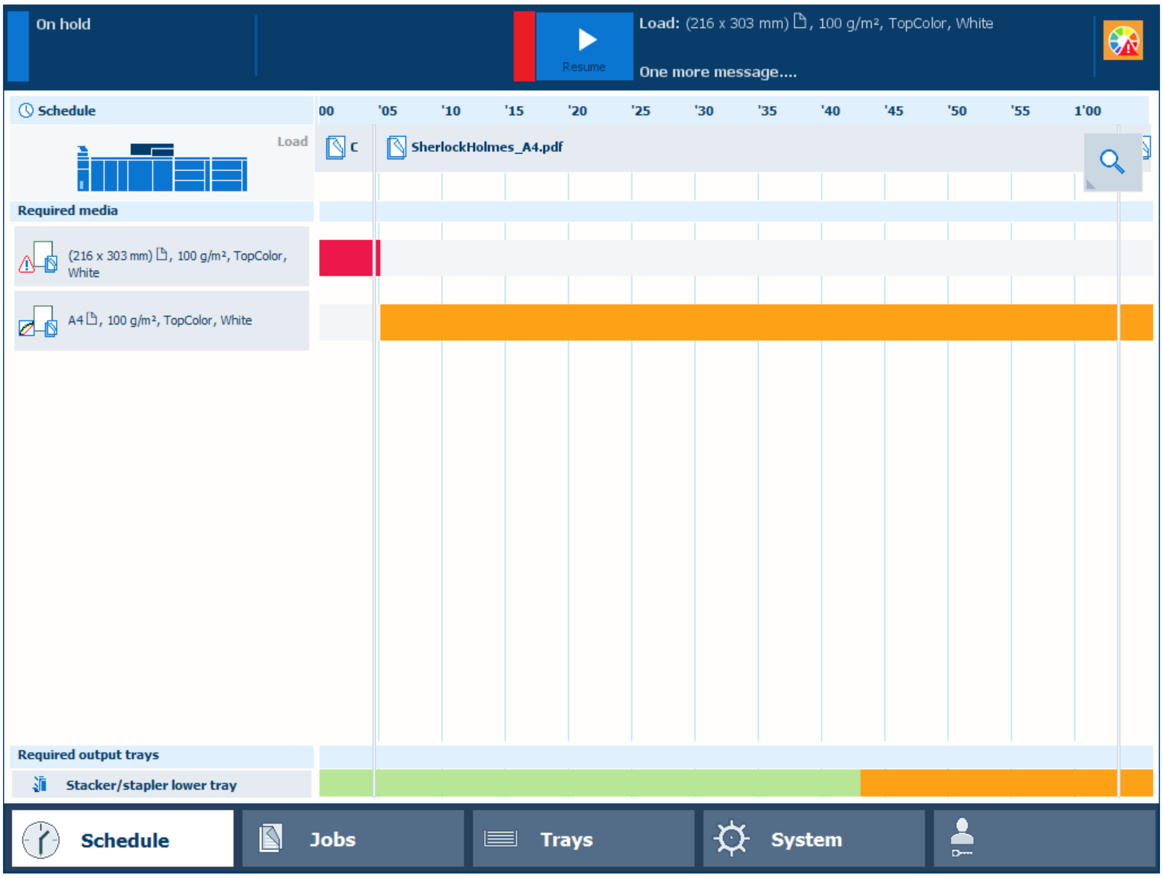
More information

Load media on page 132

Load media for the scheduled jobs

The schedule displays the required media for the scheduled jobs. You can load and assign media for jobs that are ready for printing via the Trays button on the right-hand side of the control panel or via the schedule. When you load media via the schedule, PRISMAsync automatically assigns the media to the correct paper tray.

The print system can retrieve media from any paper tray that contains the required media. You can open a paper tray when the print system is busy. The paper tray opens as soon as possible. When the print system uses the paper tray for a job and the media are also available in another paper tray, the print process continues. When there are no other paper trays with the media, the print process stops.



Schedule

Before you begin

IMPORTANT
Make sure you know how to load media into the paper trays.

Procedure

1. Touch [Schedule].
2. Select the media you want to load in the [Required media] pane.
3. Touch [Load].
4. Select a paper tray to load the media.
5. Open the paper tray and load the media into the paper tray.

6. Close the paper tray.
7. Touch [OK] to confirm.
   PRISMAsync automatically assigns the media to the correct paper tray.

More information

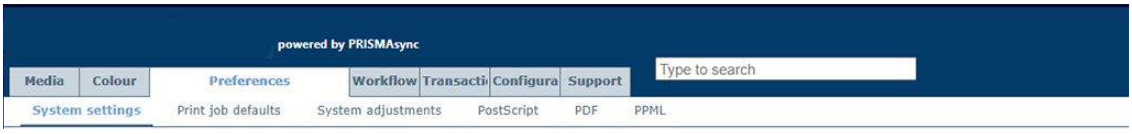
Load media on page 132

Validate media assignment

To validate media assignment and reduce the number of operator errors, there is a possibility to automatically unassign media when you open the tray. If this setting is enabled, you need to manually reassign media to the tray after you close the tray. That way, you can ensure that the correct media is assigned to the tray.

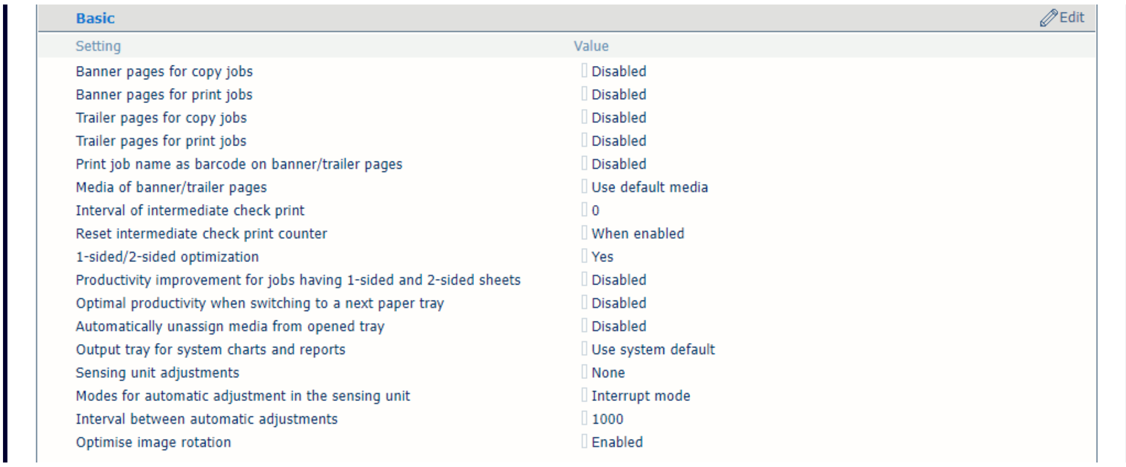
Procedure

1. Open the Settings Editor and go to: [Preferences]→[System settings] .



[System settings] tab

2. Go to the [Basic] section.



Setting	Value
Banner pages for copy jobs	Disabled
Banner pages for print jobs	Disabled
Trailer pages for copy jobs	Disabled
Trailer pages for print jobs	Disabled
Print job name as barcode on banner/trailer pages	Disabled
Media of banner/trailer pages	Use default media
Interval of intermediate check print	0
Reset intermediate check print counter	When enabled
1-sided/2-sided optimization	Yes
Productivity improvement for jobs having 1-sided and 2-sided sheets	Disabled
Optimal productivity when switching to a next paper tray	Disabled
Automatically unassign media from opened tray	Disabled
Output tray for system charts and reports	Use system default
Sensing unit adjustments	None
Modes for automatic adjustment in the sensing unit	Interrupt mode
Interval between automatic adjustments	1000
Optimise image rotation	Enabled

[Basic] section

3. Use the [Automatically unassign media from opened tray] setting to indicate whether you want media to be unassigned when the tray is opened. By default, this setting is disabled.

[Automatically unassign media from opened tray]

4. Click [OK].

Load media

Check and prepare media before loading

Before you load media into the paper trays, it is important to check and prepare the media. In this topic you read how to check and prepare the following media:

- Paper
- Long sheets
- Envelopes
- Transparencies

CAUTION

- **When you handle paper, be careful not to cut your hands on the edges of the paper.**
- **Do not store paper in places exposed to open flames. This can cause the paper to ignite which could result in burns or a fire.**

Check and prepare plain paper

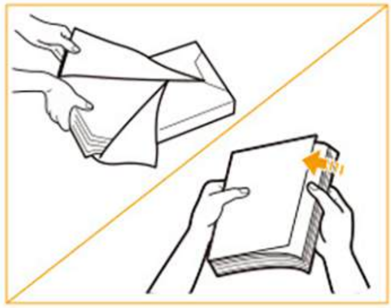
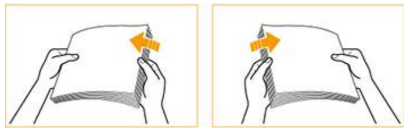
When there are load and storage instructions on the paper package, follow these instructions.

When you print on media that have absorbed moisture, steam may be released from the output area of the machine. This is due to the evaporation of moisture in the paper caused by the high temperature in the print module.

NOTE

- Use paper recommended by Canon for high-quality output.
- Some commercially available media types are not appropriate for your print system. Contact your local authorized Canon dealer to order media.

	Action	
1	Make sure the media specifications match the paper tray specifications.	
2	Check that the media is in good condition. IMPORTANT Do not load the following media. This can cause a paper jam. • Severely curled or wrinkled paper • Thin straw paper • Heavy paper (more than 220 g/m² (80 lb cover) • Paper which has been printed using a thermal transfer printer • The reverse side of paper which has been printed using a thermal transfer printer • Tracing paper	

	Action	
3	Remove the packaging and fan the sheets of paper several times. IMPORTANT You must fan thin paper, recycled paper, prepunched paper, heavy paper, transparencies, and tab papers before you load them.	
4	Align the edges of the sheets.	
5	When the paper is curled, straighten out the paper.	
6	Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.	

Check and prepare long sheets

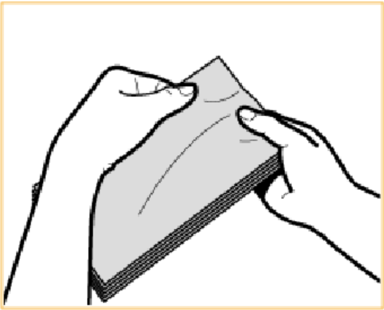
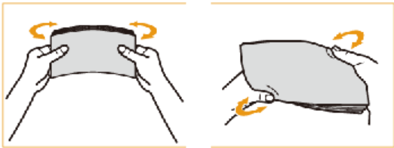
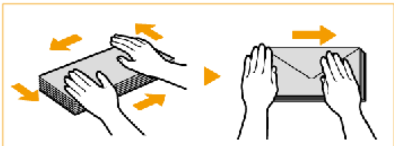
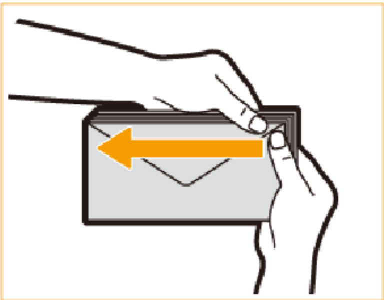
	Action	
1	Make sure the media specifications match the paper tray specifications.	
2	Check that the media is in good condition. IMPORTANT Do not load the following media. This can cause a paper jam. • Severely curled or wrinkled paper • Thin straw paper • Heavy paper • Paper which has been printed using a thermal transfer printer • The reverse side of paper which has been printed using a thermal transfer printer • Tracing paper	
3	Align the edges and fan the sheets several times. If a few sheets are stuck together, separate them carefully one by one.	
4	When the paper is curled, straighten out the paper.	
5	Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.	

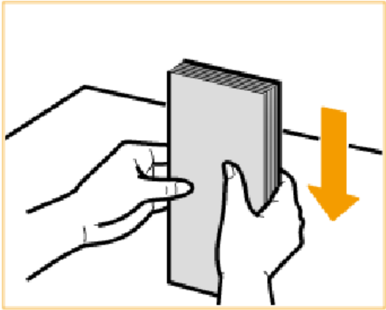
Check and prepare envelopes

You can load envelopes in all paper trays and the special feeder.

IMPORTANT

- Do not print on the back side of the envelopes (the side with the flap).
- Do not load the following types of envelopes. This can cause a paper jam, dirty printed output or a dirty machine parts.
  - Curled, creased, or folded envelopes
  - Very thick or thin envelopes
  - Damp or wet envelopes
  - Torn envelopes
  - Irregularly shaped envelopes
  - Envelopes with clasps or windows
  - Envelopes that are already sealed
  - Envelopes with holes or perforations
  - Envelopes with specially coated surfaces
  - Envelopes made of surface treated colored paper
  - Envelopes that are self-sticking and use ink, glue or other substances that can melt, burn, vaporize or emit smells by the heat of the fixing unit
- When there is a temperature difference between the storage location and the location where the envelopes are printed, take the envelopes to the print location on time.

	Action	
1	Make sure the media specifications match the paper tray specifications.	
2	Take five envelopes and fix curls or bends.	
3	Loosen and stack the five envelopes together.	
4	Place the envelopes on a clean, level surface and move your hands in the direction of the arrows to remove the air inside the envelopes..	
5	Hold down the four corners of the envelopes firmly, so that the envelopes stay flat.	

	Action	
6	Align the envelopes on a flat surface.	
7	Tightly rewrap the remaining media in the original package, and store the package in a dry place away from direct sunlight.	

Check and prepare transparencies

Transparencies have a different front and back side. Make sure you load transparencies so that the print system prints on the front side.

IMPORTANT
- Do not load transparencies while the print system is printing. This can cause a paper jam.
- Only use transparencies intended for your print system. Other transparencies can damage the print system.
- Only hold the edges of the sheets and avoid touching the print surface.
- When transparencies stick together, fan the stack and do not load more than 100 sheets. Use new transparencies when a paper jam occurs.
- Immediately take printed transparencies from the output tray to avoid folded transparencies. This can cause a paper jam.

Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

More information

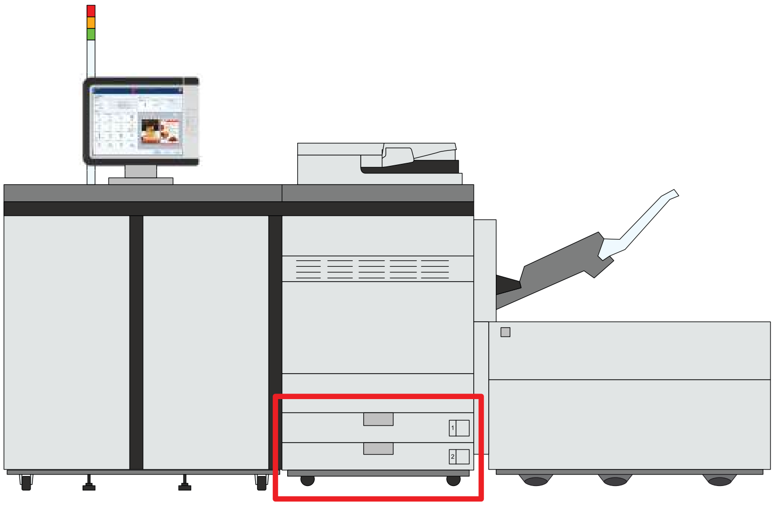
- *Media specifications* on page 613
- *Paper input specifications* on page 618

Load media into the internal paper trays

Each internal paper tray can hold up to 550 sheets, 80 g/m² (22-lb bond). The paper trays feed the media face down.

There is no sensor in the paper trays that can detect the media. After you open and close a paper tray, the printer does not change the loaded media settings.

When you load envelopes in an internal paper tray, you need the envelope guide.



Location of the internal paper trays

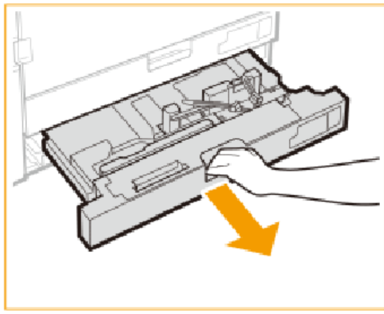
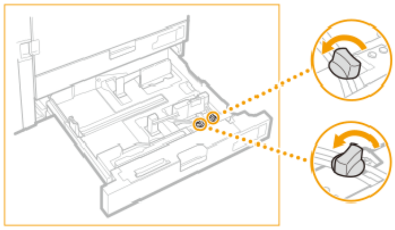
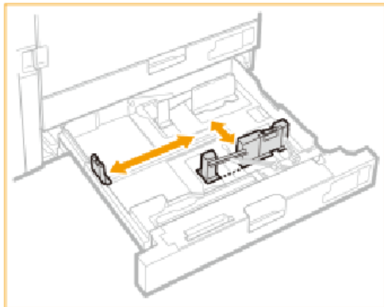
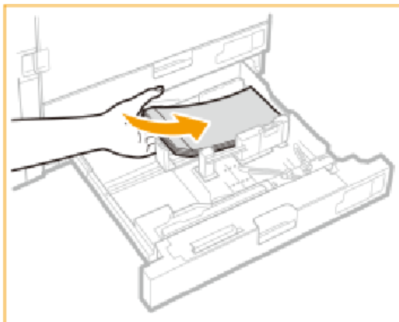
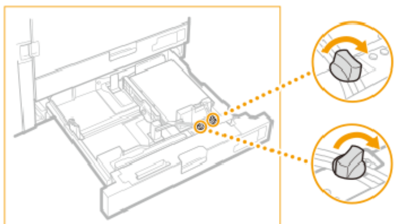
CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

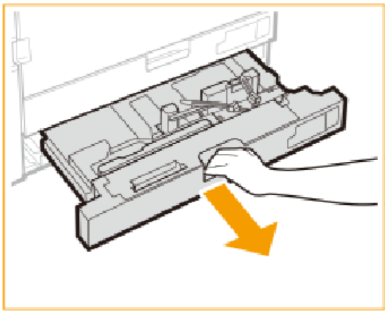
IMPORTANT
- Never place paper or other objects in the empty parts of a paper tray. This can cause a paper jam.
- Make sure you follow the instructions carefully. When you do not load the media correctly, a paper jam, dirty machine parts or poor print quality can occur.

Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

	Action	
1	Grip the handle and pull out the paper tray until it stops.	
2	Loosen the knobs.	
3	Slide the paper guides to match the media size you want to load. IMPORTANT Slide the guides until they click into place. This prevents a paper jam, poor print quality and dirty machine parts.	
4	Place the media stack against the right side of the paper tray. IMPORTANT Make sure the loaded the media stack does not exceed the loading limit mark (▼▼▼ or ✉▼▼ for envelopes).	
5	Tighten the knobs.	

	Action	
6	Gently close the paper tray and make sure it clicks into place. **CAUTION** **When you close the paper tray, be careful not to get your fingers caught. This can cause personal injury.**	

After you finish

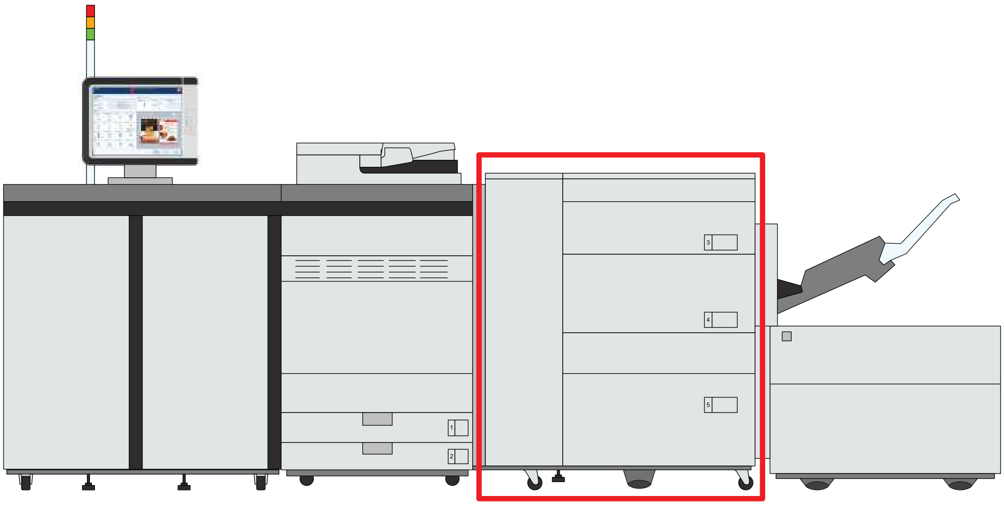
- When a paper jam occurs or you notice poor image quality, turn over the media stack and reload the media stack. Do not reload textured, single-sided coated or already printed paper to avoid a paper jam. For these media, take a new stack.
- Tightly rewrap the remaining media in the original package and store the package in a dry place, away from direct sunlight or high temperatures.

More information

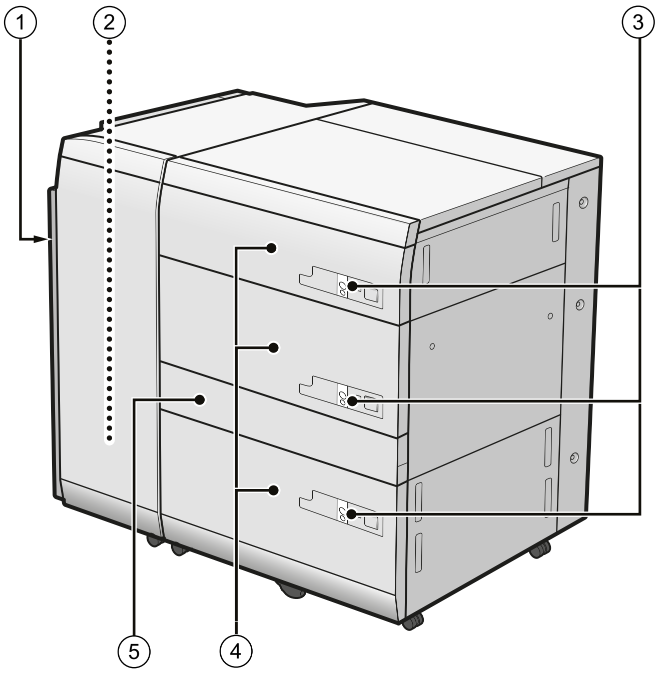
- *Feed instructions for envelopes and tab paper to the paper input optionals* on page 605
- *Load envelopes into the internal paper trays* on page 162
- *Check and prepare media before loading* on page 132

Load media into the paper module

The paper module increases the input capacity of the print system. The upper and middle paper trays can hold up to 1,500 sheets, 80 g/m² (22 lb bond); the lower deck can hold up to 2,000 sheets, 80 g/m² (22 lb bond). The paper trays feed the media face up.



Location of the paper module



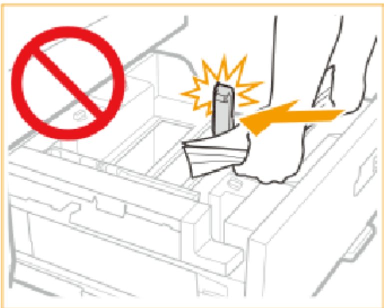
Paper module

	Description paper module
1	Front cover to access the paper path when a paper jam occurs
2	Error tray where the paper is output in case multiple sheets are fed
3	Tray buttons to open a paper tray

	Description paper module
4	Paper trays to hold the media
5	Paper path cover to access the paper path when a paper jam occurs NOTE This cover only opens if Paper Deck Path Kit-B is connected.

CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

IMPORTANT
- When you lift the inside lifter of the paper tray, for example because you dropped something, do not lift the lifter more than 50 mm (2 inch) or diagonally. This can cause a malfunction or machine damage.
- Do not hit the right-hand paper guide when you load paper. This can cause a malfunction or machine damage.



Right-hand paper guide

- Do not hit the feeding roller when you load paper. This can cause a malfunction or machine damage.

Feeding roller

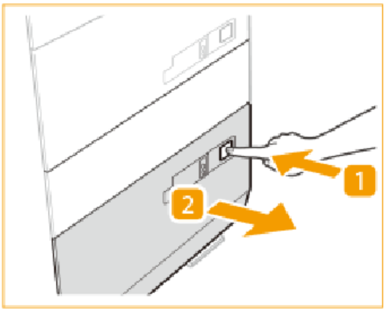
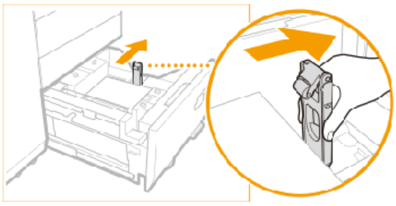
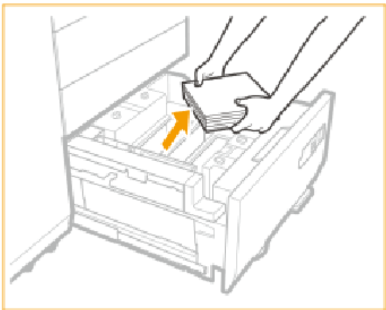
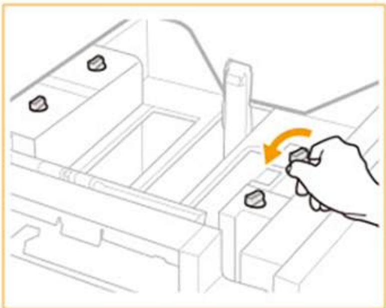
IMPORTANT
- Never place media or other objects in the empty parts of the paper tray. This can cause a paper jam.
- Make sure you follow the instructions carefully. When you do not load the media correctly a paper jam, dirty machine parts, or poor print quality can occur.

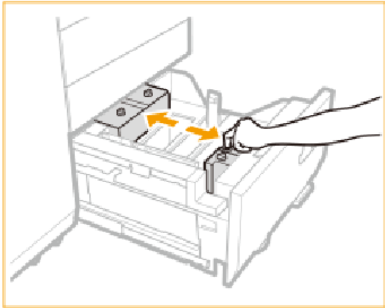
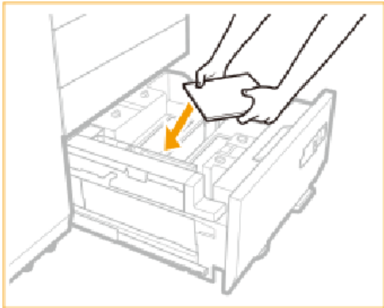
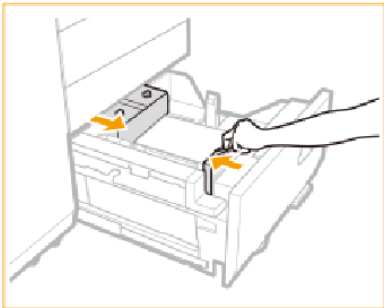
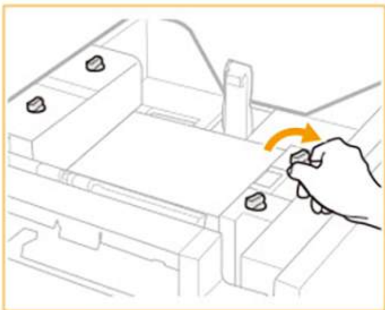
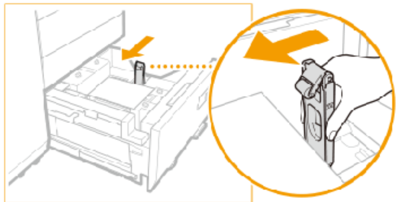
NOTE
- When you change from plain paper to coated paper, the print system fans the media first.
- Check that the height of the loaded media stack is maximum 20 mm (0.8 inch) to avoid curled and creased edges.

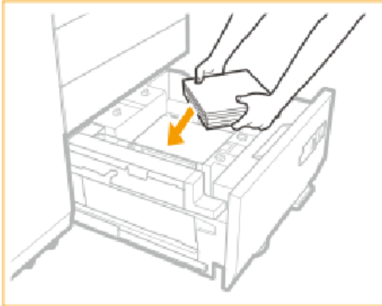
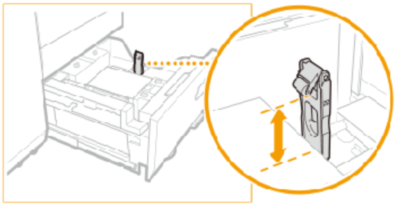
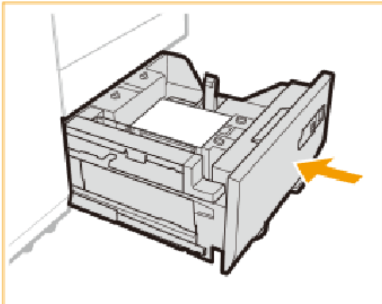
Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

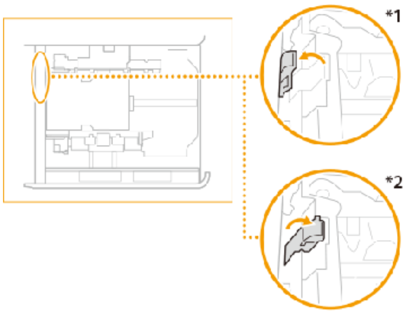
	Action	
1	Press the button (1) to open the paper tray (2).	
2	Slide the right-hand paper guide to the right-hand side of the paper tray.	
3	Lift the feeding support roller and remove all media.	
4	Loosen 4 knobs.	

	Action	
5	Squeeze the lever and slide the paper guides so that there is enough space to load the media.	
6	Load a media stack of approximately 10 mm (0.4 inch).	
7	Squeeze the lever and slide the paper guides against the media stack and check the size mark for the exact alignment.	
8	Tighten 4 knobs.	
9	Slide the right-hand paper guide towards the media stack.	

	Action	
10	Continue to load the next media stacks of approximately 10 mm (0.4 inch). Load the paper below the rollers of the right-hand paper guide. IMPORTANT • Make sure the loaded media stack does not exceed the loading limit mark (▼▼▼). • Make sure the loaded envelope stack does not exceed the envelope loading limit mark (✉▼▼). Load a maximum of approximately 30 envelopes.	 
11	Gently close the paper tray and make sure it clicks into place. **CAUTION** **When you close the paper tray, be careful not to get your fingers caught. This can cause personal injury.**	

NOTE

When you use the media paper of 351 g/m² (130 lb cover) or more, move the pressure switching lever to the 351–400 g/m² (130 lb cover - 148 lb cover) position. When you use the media that is lighter than 351 g/m² (130 lb cover), make sure to return the pressure switching lever to the original position (52–350 g/m²).



Pressure switching lever position

*1 351–400 g/m² (130 lb cover–148 lb cover) position
*2 52–350 g/m² (14 lb bond–130 lb cover) position

After you finish

- When a paper jam occurs or you notice poor image quality, turn over the media stack and reload the media stack. Do not reload textured, single-sided coated or already printed paper to avoid a next paper jam. For these media, take a new stack.

- Tightly rewrap the remaining media in the original package and store the package in a dry place, away from direct sunlight or high temperatures.

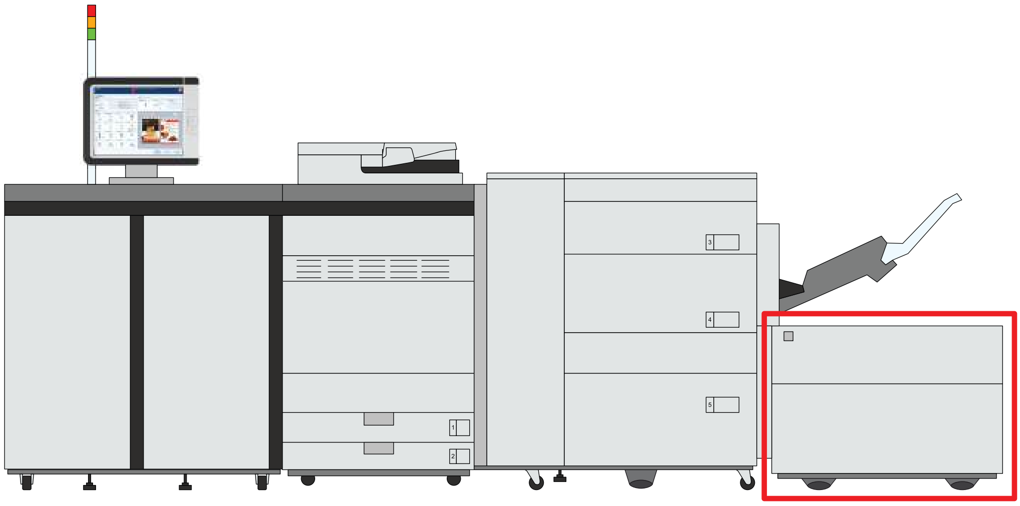
More information

- *Feed instructions for envelopes and tab paper to the paper input optionals* on page 605
- *Check and prepare media before loading* on page 132

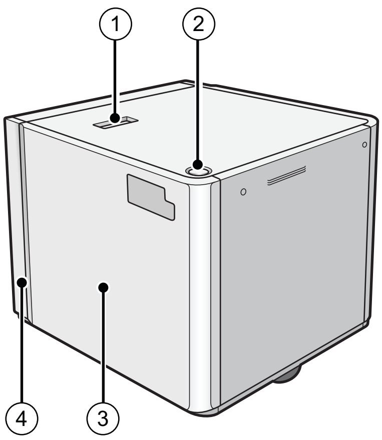
Load media into the bulk paper module

The input capacity of the print system can be extended with a bulk paper module. This bulk paper module can hold up to 3,500 sheets of paper, 80 g/m² (22 lb bond). The paper tray feeds the media face up.

When you load envelopes in the bulk paper module, you need the envelope guide.



Location of the bulk paper module



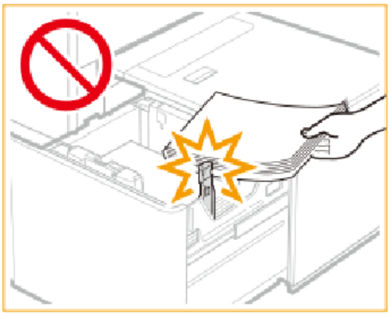
Bulk paper module

	Description bulk paper module
1	Release button to move the bulk paper module
2	Button to open the paper tray
3	Paper tray to hold the media
4	POD Deck Lite Attachment Kit-C to attach the bulk paper module to the printer

CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

IMPORTANT
- Do not lift the inside lifter of the paper tray if the printer is on. This can cause a malfunction or machine damage. If you need to lift the inside lifter of the paper tray, for example, because you dropped something, turn off the printer with the paper trays open.

IMPORTANT
- Do not hit the right-hand paper guide when you load the media. This can cause a malfunction or machine damage.

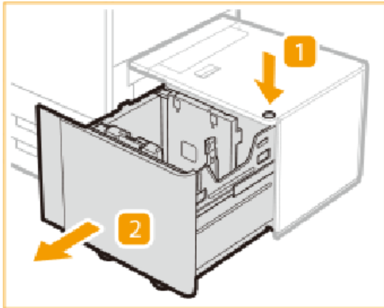
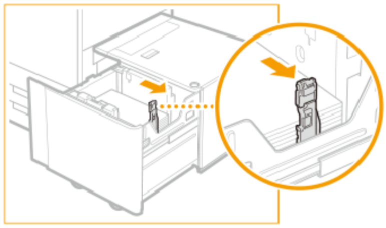
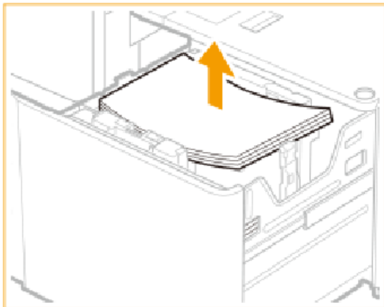
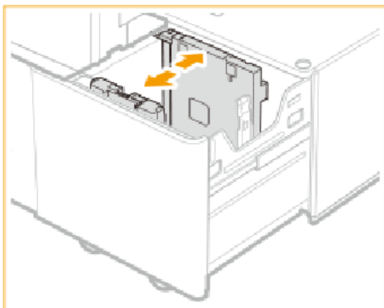


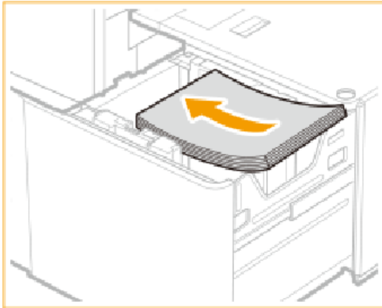
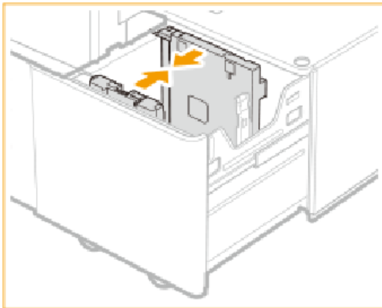
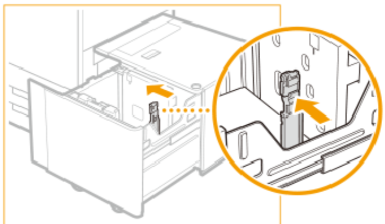
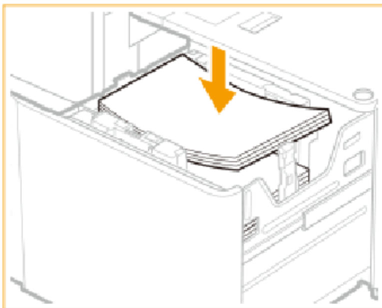
- Never place paper or other objects in the empty parts of the paper tray. This can cause a paper jam.
- Do not insert paper clips or other objects into the paper blower port on the bulk paper module.
- Make sure you follow the instructions carefully. When you do not load the media correctly, a paper jam, dirty machine parts or poor print quality can occur.

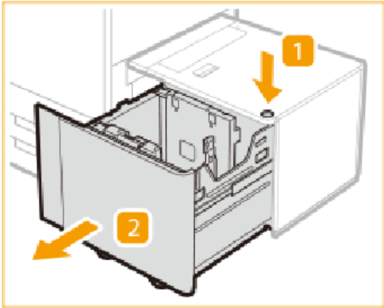
Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

	Action	
1	Press the button (1) to open the bulk paper tray (2).	
	When the media size does not change continue with step 5. Otherwise, go to step 2.	
2	Slide the right-hand paper guide to the right-hand side of the paper tray.	
3	Remove all media.	
4	Close the paper tray and open it again.	
4	Squeeze the lever and slide the paper guides so that there is enough space to load the media.	

	Action	
5	Load a media stack of approximately 20 mm (0.8 inch) against the left-hand side wall of the paper tray.	
6	Squeeze the front-side paper guide and slide the paper guides towards, and against, the media stack.	
7	Slide the right-hand paper guide to the media stack and check the size mark for the exact alignment. IMPORTANT When you do not align the right-hand paper guide properly, a paper jam, poor print quality, or dirty machine parts can occur.	
8	Load all remaining paper into the paper tray. IMPORTANT • Make sure the loaded media stack does not exceed the loading limit mark (▼▼▼). • When you load envelopes, make sure the loaded envelope stack does not exceed the envelope loading limit mark (/). Use the correct orientation for the envelopes.	

	Action	
9	Gently close the paper tray and make sure it clicks into place. **CAUTION** **When you close the bulk paper module, be careful not to get your fingers caught. This can cause personal injury.**	

After you finish

- When a paper jam occurs or you notice poor image quality, turn over the media stack and reload the media stack. Do not reload textured, single-sided coated or already printed paper to avoid a next paper jam. For these media, take a new stack.
- Tightly rewrap the remaining media in the original package and store the package in a dry place, away from direct sunlight or high temperatures.

More information

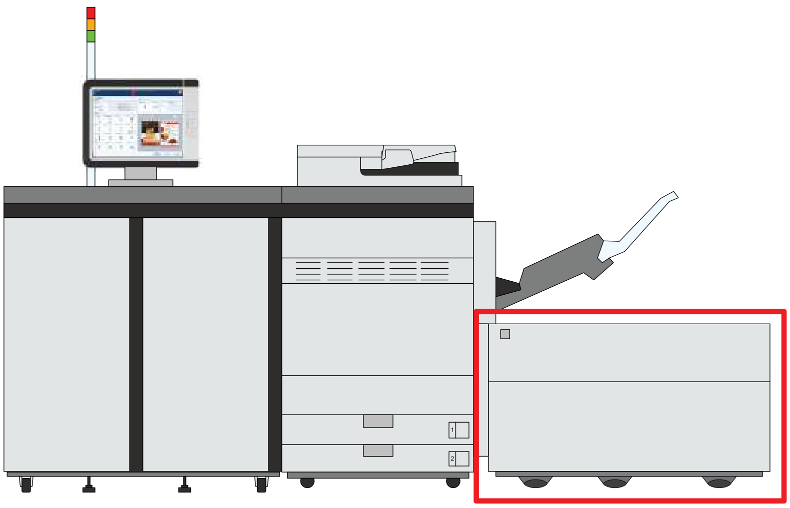
- *Feed instructions for envelopes and tab paper to the paper input optionals* on page 605
- *Check and prepare media before loading* on page 132

Load media into the bulk paper module XL

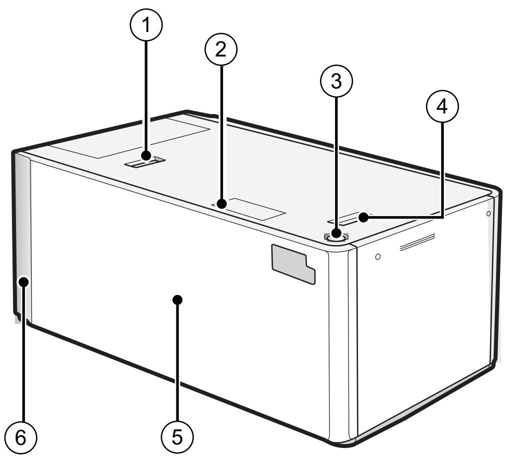
The input capacity of the print system can be extended with a bulk paper module XL. This module can hold up to 3,500 sheets of paper, 80 g/m² (22 lb bond). The paper tray feeds the media face up.

In addition to long sheets, the bulk paper module XL can also handle standard size media.

When you load envelopes in the bulk paper module XL, you need the envelope guide.



Location of the bulk paper module XL



Bulk paper module XL

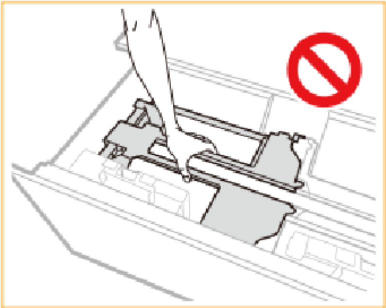
	Description bulk paper module
1	Release button to move the bulk paper module XL
2	Paper size switch indicator to indicate that the paper size is changed to use long sheets
3	Button to open the paper tray
4	Release handle to move the bulk paper module XL away from the printer

	Description bulk paper module
5	Paper tray to hold the media
6	POD Deck Lite Attachment Kit-C to attach the bulk paper module XL to the printer

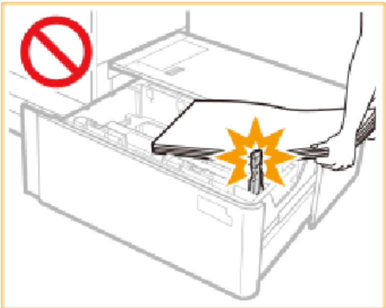
CAUTION

- **Make sure to ground the bulk paper module XL before you connect its power plug to the power source.**
- **Make sure you remove the power plug from the power source before you disconnect the ground connection of the bulk paper module XL.**
- **When you handle paper, be careful not to cut your hands on the edges of the paper.**

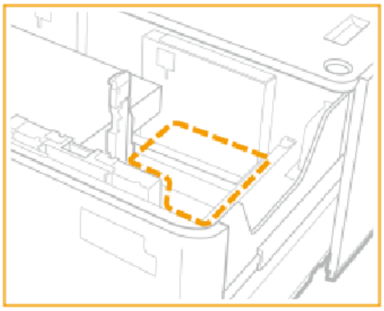
IMPORTANT
- When you lift the inside lifter of the paper tray, for example because you dropped something, do not lift the lifter more than 50 mm (2 inch) or diagonally. This can cause a malfunction or machine damage.
- Do not lift the inside lifter of the paper tray if the printer is on. This can cause a malfunction or machine damage. If you need to lift the inside lifter of the paper tray, for example, because you dropped something, turn off the printer with the paper trays open.



- Do not hit the right-hand paper guide when you load the media. This can cause a malfunction or machine damage.



- Never place paper or other objects in the empty parts of the paper tray. This can cause a paper jam.

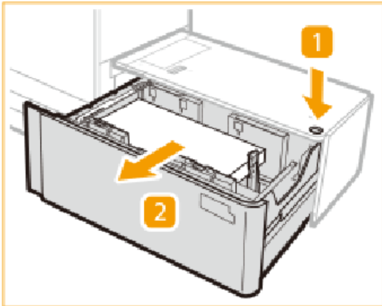
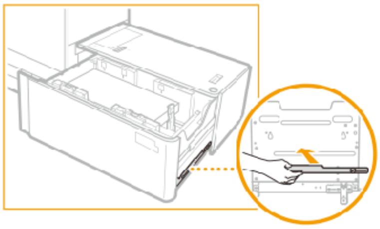
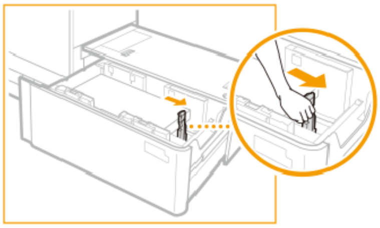
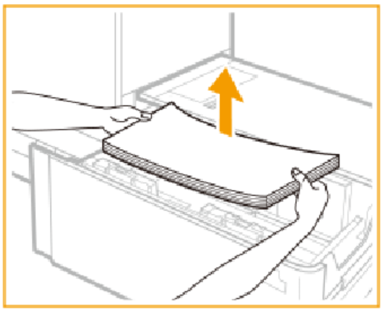


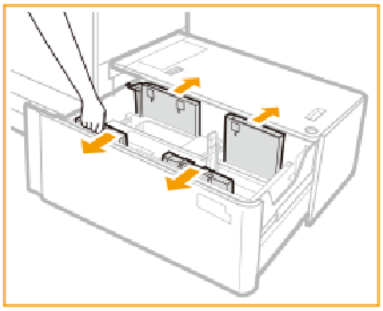
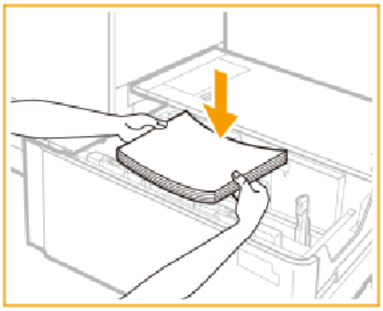
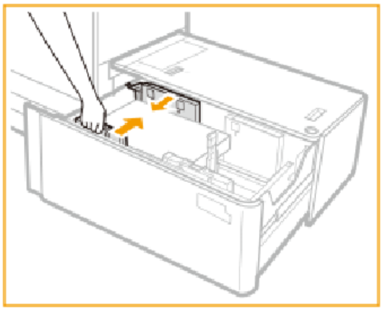
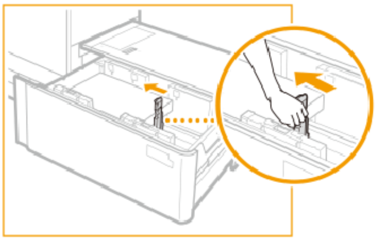
IMPORTANT
- Do not insert paper clips or other objects into the paper blower port on the bulk paper module.
- Make sure you follow the instructions carefully. When you do not load the media correctly, a paper jam, dirty machine parts, or poor print quality can occur.

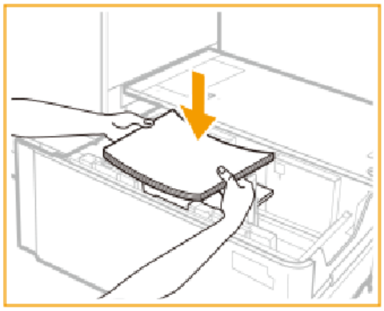
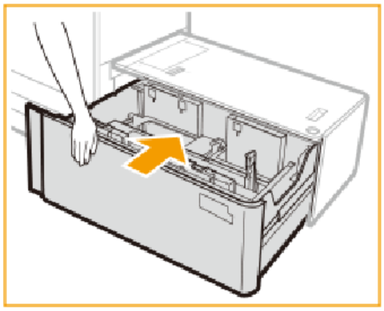
Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

	Action	
1	Press the button (1) to open the bulk paper tray (2).	
2	Store the fixing bracket for long sheets at the right-hand side of the bulk paper tray.	
3	Slide the right-hand paper guide to the right-hand side of the paper tray so that there is enough space to load the media.	
4	Remove all media. NOTE If the right-hand paper guide is in the position to hold long sheets, the paper size indicator illuminates. In this case, move the right-hand paper guide in the position for standard size media.	

	Action	
5	Squeeze the lever and slide the paper guides so that there is enough space to load the media.	
6	Load a media stack of approximately 10 mm (0.4 inch).	
7	Squeeze the lever and slide the paper guides against the media stack.	
8	Slide the right-hand paper guide against the media stack. IMPORTANT Make sure that you align the right-hand paper guide with the indicated paper size marks. When you do not align the right-hand paper guide properly, a paper jam, poor print quality, or dirty machine parts can occur.	

	Action	
9	Continue to load the next media stacks of approximately 20 mm (0.8 inch). IMPORTANT • Make sure the loaded media stack does not exceed the loading limit mark ( ). • When you load envelopes, make sure the loaded envelope stack does not exceed the envelope loading limit mark (/). Use the correct orientation for the envelopes.	
10	Gently close the paper tray and make sure it clicks into place. **CAUTION** **When you close the bulk paper module, be careful not to get your fingers caught. This can cause personal injury.**	

After you finish

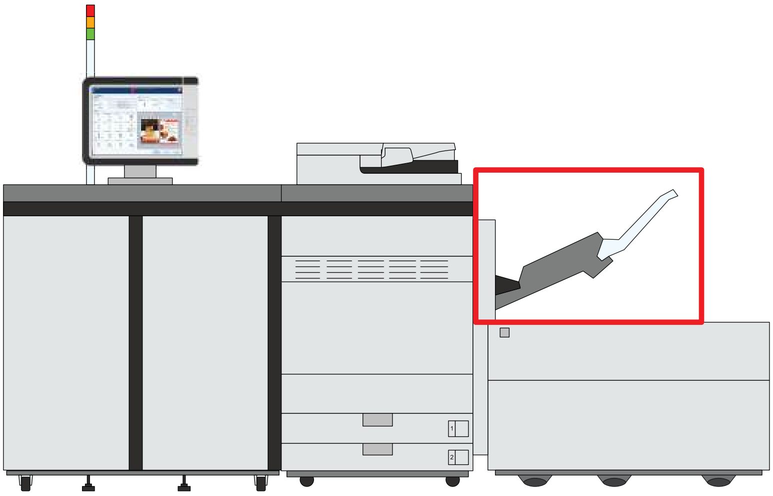
- When a paper jam occurs or you notice poor image quality, turn over the media stack and reload the media stack. Do not reload textured, single-sided coated, or already printed paper to avoid a paper jam. For these media, take a new stack.
- Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

More information

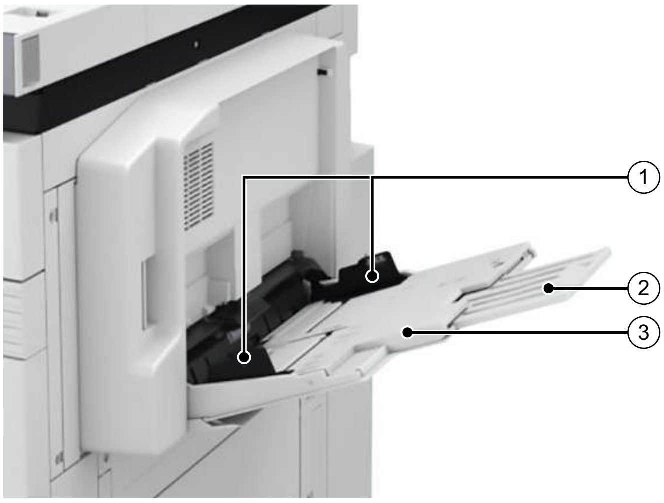
- *Feed instructions for envelopes and tab paper to the paper input optionals* on page 605
- *Load long sheets into the bulk paper module XL* on page 168
- *Check and prepare media before loading* on page 132

Load media into the special feeder

The special feeder enables you to feed media manually, for example, when you need only a few sheets of special media for your job. The special feeder can hold up to 250 sheets. The special feeder feeds the media face up.



Location of the special feeder



Special feeder

	Description special feeder
1	Paper guides to be adjusted to match the media size
2	Auxiliary tray to place large media
3	Paper tray to load the media

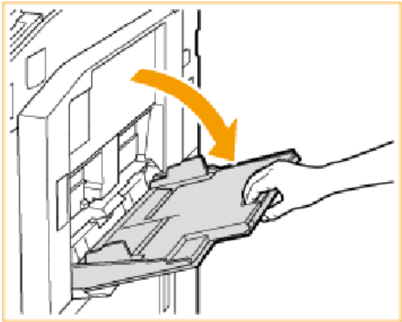
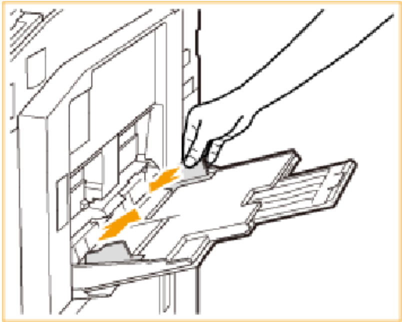
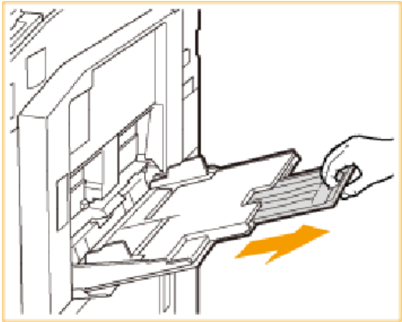
CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

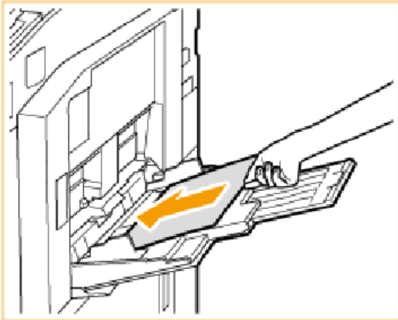
IMPORTANT
- Do not load different sizes or types of media at the same time.
- It is very important that you define the correct job settings for media, such as heavy paper or transparencies. The fixing unit can become dirty, which will require a service visit. Moreover, the image quality can become poor.
- When heavy media does not pass through properly, feed the sheets one by one.
- When envelopes do not pass through properly, feed the envelopes one by one.
- Do not collect more than ten printed envelopes in the output tray.
- Envelopes can crease during the print process.
- Make sure you follow the instructions carefully. When you do not load media correctly, a paper jam, dirty machine parts or poor print quality can occur.

Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

	Action	
1	Open the special feeder.	
2	Squeeze and slide the paper guides towards the sides of the tray.	
3	Pull out the auxiliary tray for large media sizes.	

	Action	
4	Load the media into the special feeder. IMPORTANT • Make sure the loaded media stack does not exceed the loading limit mark (▼▼▼). • Load a maximum of ten envelopes. Use the correct orientation for the envelopes.	
5	Align the media precisely between the paper guides.	

After you finish

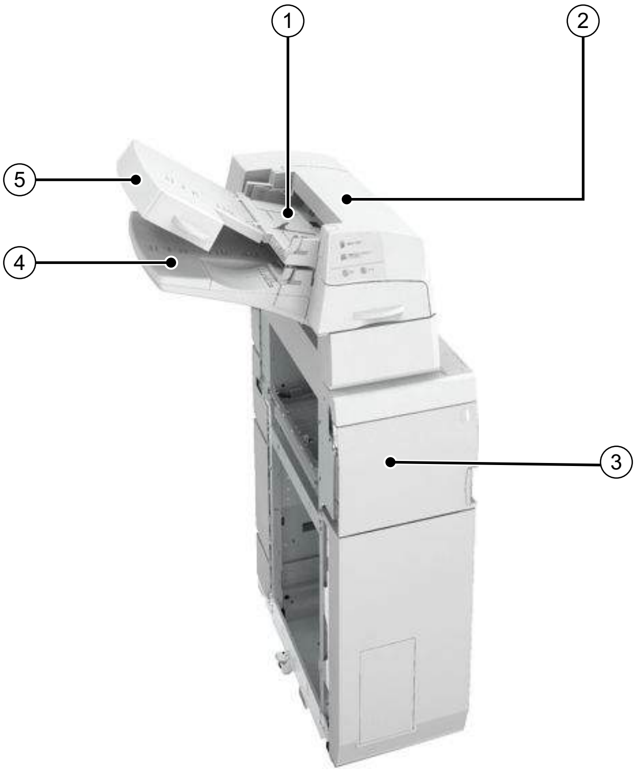
- When a paper jam occurs or you notice poor image quality, turn over the media stack and reload the media stack. Do not reload textured, single-sided coated, or already printed paper to avoid a paper jam. For these media, take a new stack.
- Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

More information

- *Check and prepare media before loading* on page 132
- *Feed instructions for envelopes and tab paper to the paper input optionals* on page 605
- *Load long sheets into the special feeder* on page 174

Load media into the inserter

The optional inserter can hold up to 200 sheets, 80 g/m² (22 lb bond), such as inserts, preprinted sheets and booklet covers. The print system cannot print sheets fed by the inserter. Use the same feed direction in the upper and lower tray.



Inserter

	Description inserter
1	Upper tray to place media
2	Upper cover to remove jammed paper
3	Front cover to access the paper path when a paper jam occurs.
4	Lower tray to place media
5	Lower tray cover to load media in the lower tray

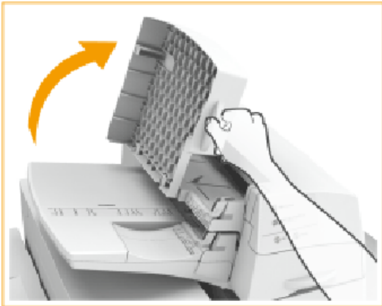
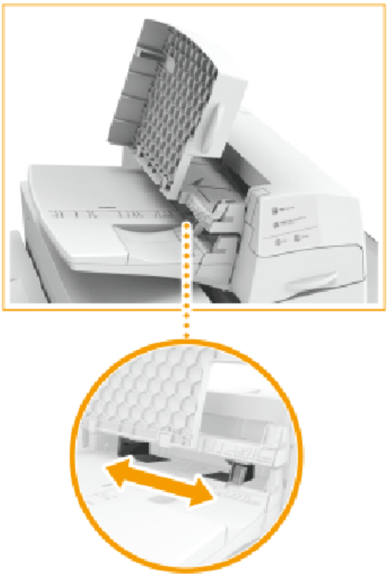
CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

IMPORTANT
Make sure you follow the instructions carefully. When you do not load the media correctly, a paper jam, dirty machine parts, or poor print quality can occur.

Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

	Action	
1	Adjust the paper guides to match the media size. When you use the lower tray, first open the lower tray cover.	 Upper tray  Lower tray cover  Lower tray

	Action	
2	Load the paper face up into the inserter. When you use the lower tray, close the lower tray cover. Use the feed instruction when you use preprinted paper. IMPORTANT Make sure the height of the media stack does not exceed the loading limit mark (▼▼▼).	 Upper tray  Lower tray

After you finish

Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

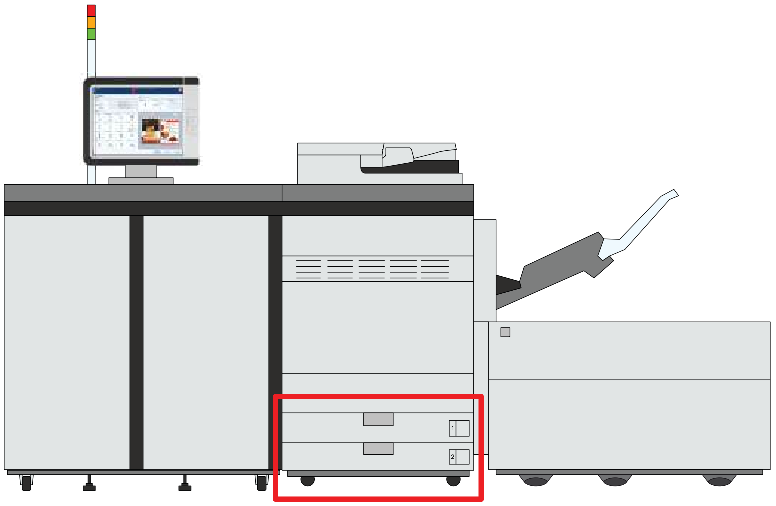
More information

- *Check and prepare media before loading* on page 132

Load envelopes, tab paper and long sheets

Load envelopes into the internal paper trays

When you load envelopes into the internal paper tray, you need an envelop guide Envelope Feeder Attachment-F.



Location of the internal paper trays

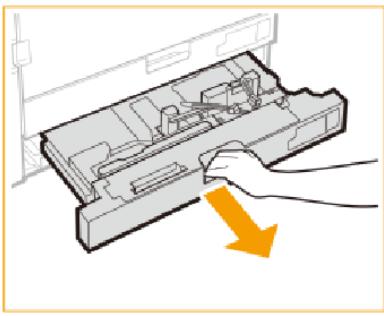
CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

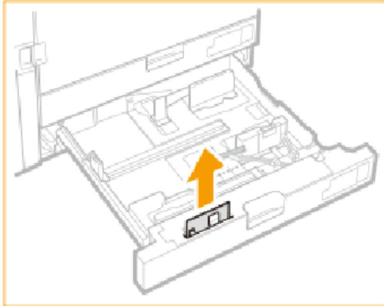
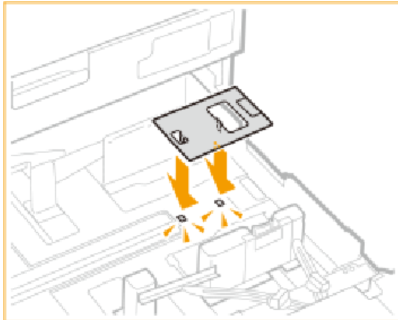
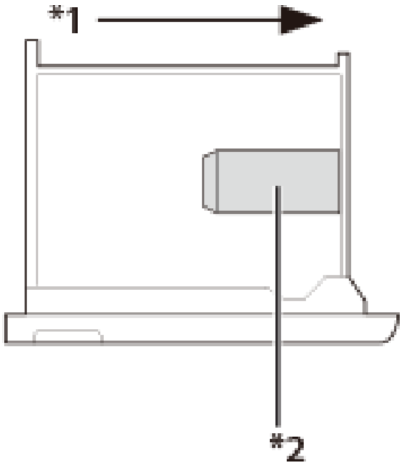
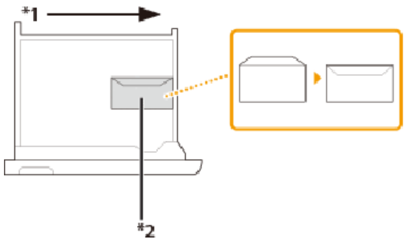
IMPORTANT
- Make sure you follow the instructions carefully. When you do not load the media correctly, a paper jam, dirty machine parts or poor print quality can occur.

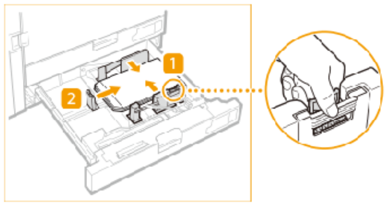
Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode, if applicable.

Procedure

	Action	
1	Grip the handle and pull out the paper tray until it stops.	

	Action	
2	Lift up the envelope guide straight from its holder.	
3	Attach the envelope guide as shown in the illustration.	
4	Load one envelope against the right-hand side of the paper tray. Envelope orientation 1: Nagagata 3, Kakugata 2 Envelope orientation 2: No. 10 (COM10), ISO-C5, DL, Monarch, Yougatanaga 3	 Envelope orientation 1  Envelope orientation 2 *1 Feeding direction *2 Print side down

	Action	
5	Squeeze the top of the front guide and slide the guides towards, and against, the envelope (1 and 2). IMPORTANT Slide the guides until they click into place to prevent a paper jam, poor print quality, or dirty machine parts.	
6	Load all envelopes. IMPORTANT • Make sure that the height of the stack does not exceed approximately 33.5 mm (1.3 inch). This is approximately 50 envelopes. • Make sure that the height of the envelopes stack does not exceed the envelope loading limit mark ( ). Use the correct orientation for the envelopes.	
7	Gently close the paper tray and make sure it clicks into place. **CAUTION** **When you close the paper tray, be careful not to get your fingers caught. This can cause personal injury.**	
8	Return the envelope guide to its holder when the job is ready.	

After you finish

IMPORTANT
Make sure you put the envelope guide into its holder to avoid a paper jam.

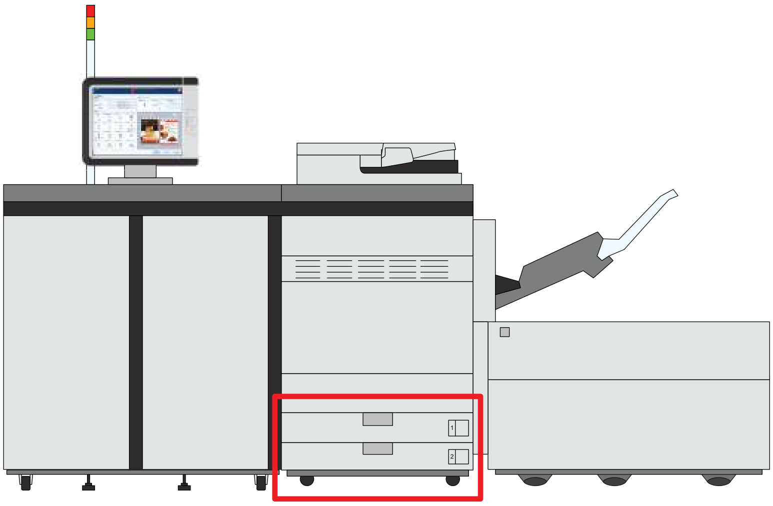
Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

More information

- *Check and prepare media before loading* on page 132
- *Feed instructions for envelopes and tab paper to the paper input optionals* on page 605

Load tab paper into the internal paper trays

When you load tab paper into the internal paper tray, you need the tab paper guide,Tab Feeding Attachment-F.



Location of the internal paper trays

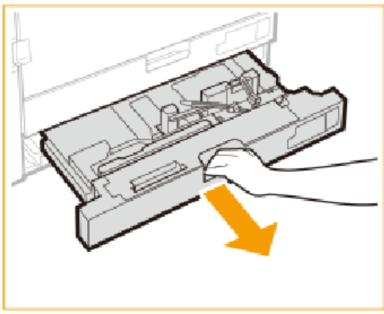
IMPORTANT
- Only use A4/LTR tab paper.
- Never place media or other objects in the empty parts of the paper trays. This can cause a paper jam.
- Make sure you follow the instructions carefully. When you do not load the media correctly, a paper jam, dirty machine parts or poor print quality can occur.

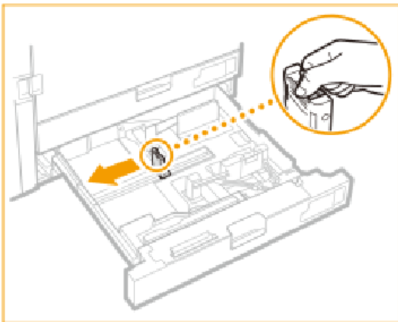
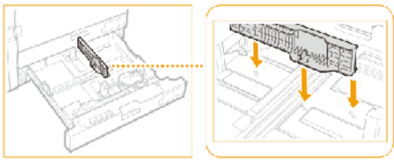
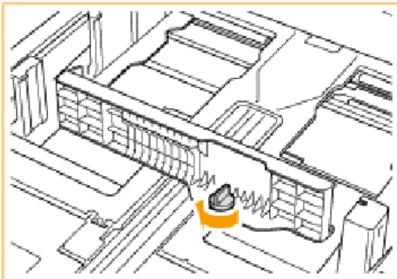
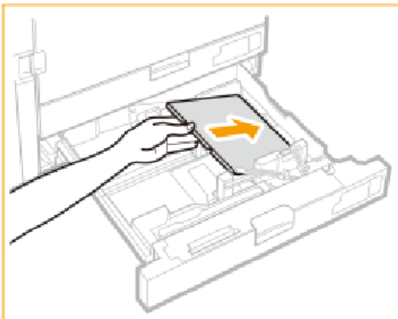
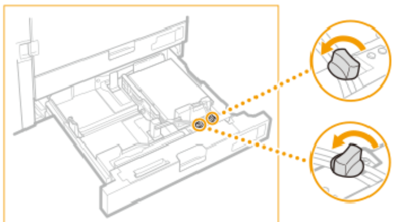
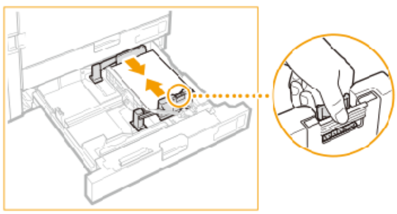
CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

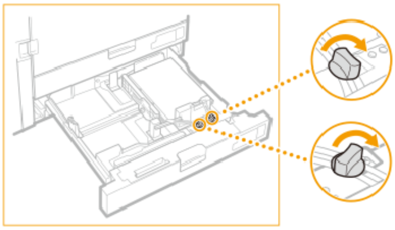
Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

	Action	
1	Grip the handle and pull out the paper tray until it stops.	

	Action	
2	Adjust the position of the paper guide.	
3	Place the tab paper guide so that it is aligned with the mark for A4.	
4	Turn the knob to secure the tab paper guide.	
5	Load the tab paper with the tabs on the left side. IMPORTANT Make sure the loaded the media stack does not exceed the loading limit mark (▼▼▼ or ✉▼▼ for envelopes).	
6	Loosen the knobs.	
7	Squeeze the top of the front guide and slide the guide against the paper.	

	Action	
8	Tighten the knobs.	
9	Gently close the paper tray and make sure it clicks into place. **CAUTION** **When you close the paper tray, be careful not to get your fingers caught. This can cause personal injury.**	
10	Return the tab paper guide back to its holder when the job is ready.	

After you finish

IMPORTANT
Make sure you put the tab paper guide into its holder to avoid a paper jam.

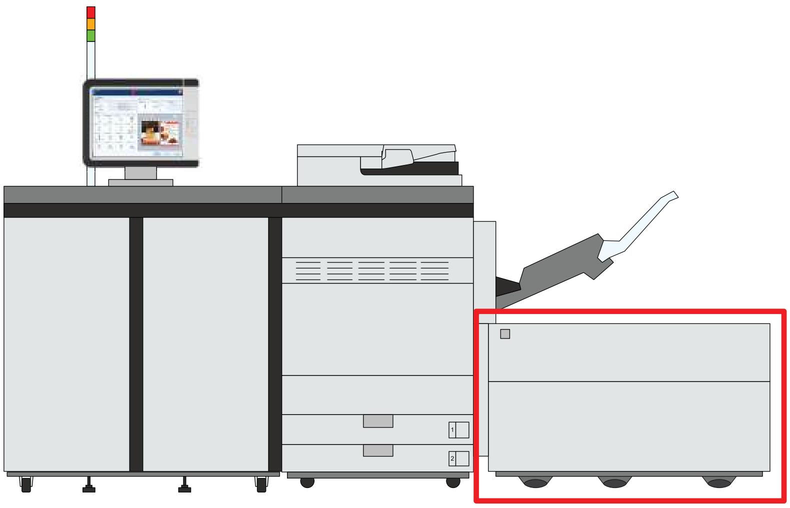
Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

More information

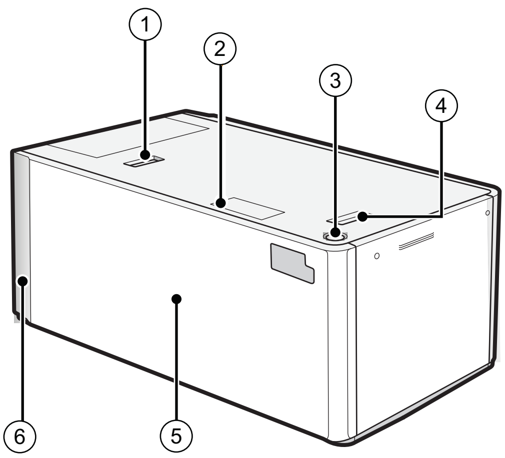
- *Check and prepare media before loading* on page 132
- *Feed instructions for envelopes and tab paper to the paper input optionals* on page 605

Load long sheets into the bulk paper module XL

The bulk paper module XL can hold up to 3,500 sheets of paper, 80 g/m² (22 lb bond). The paper tray feeds the media face up.



Location of the bulk paper module XL

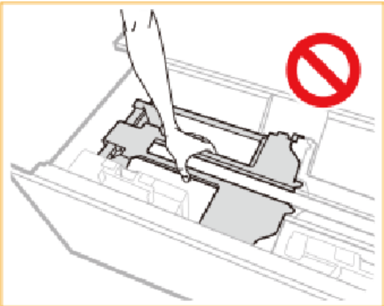


Bulk paper module XL

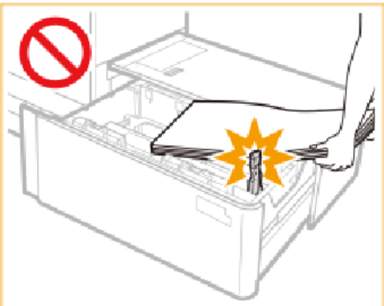
	Description bulk paper module
1	Release button to move the bulk paper module XL
2	Paper size switch indicator to indicate that the paper size is changed to use long sheets
3	Button to open the paper tray
4	Release handle to move the bulk paper module XL away from the printer
5	Paper tray to hold the media
6	POD Deck Lite Attachment Kit-C to attach the bulk paper module XL to the printer

CAUTION
- **Make sure to ground the bulk paper module XL before you connect its power plug to the power source.**
- **Make sure you remove the power plug from the power source before you disconnect the ground connection of the bulk paper module XL.**
- **When you handle paper, be careful not to cut your hands on the edges of the paper.**

IMPORTANT
- When you lift the inside lifter of the paper tray, for example because you dropped something, do not lift the lifter more than 50 mm (2 inch) or diagonally. This can cause a malfunction or machine damage.
- Do not lift the inside lifter of the paper tray if the printer is on. This can cause a malfunction or machine damage. If you need to lift the inside lifter of the paper tray, for example, because you dropped something, turn off the printer with the paper trays open.



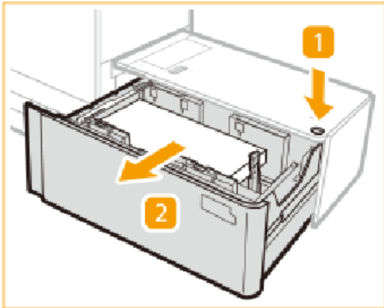
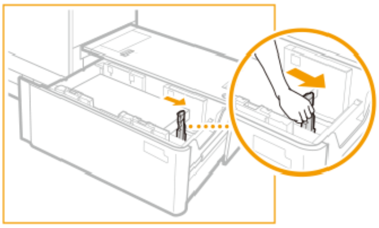
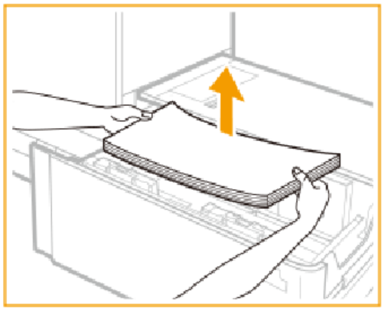
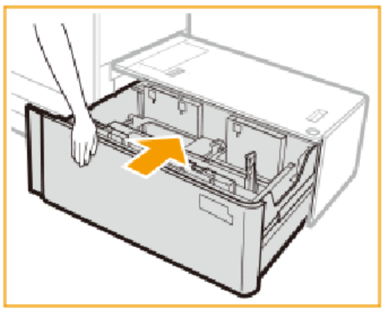
- Do not hit the right-hand paper guide when you load the media. This can cause a malfunction or machine damage.

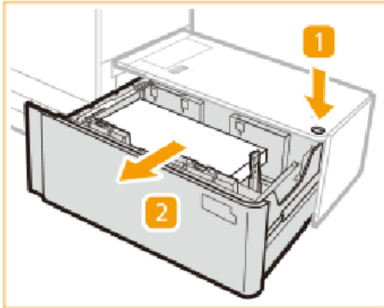
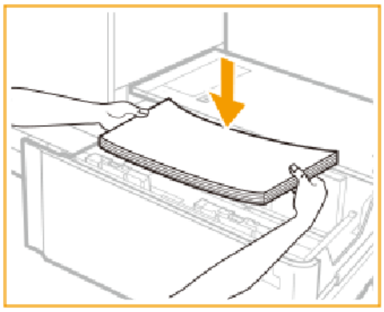
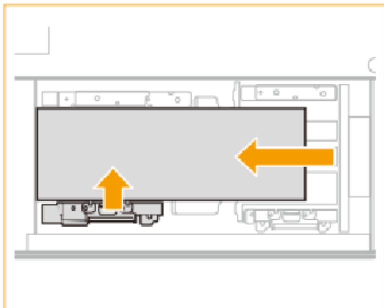
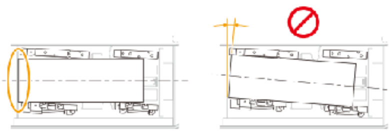
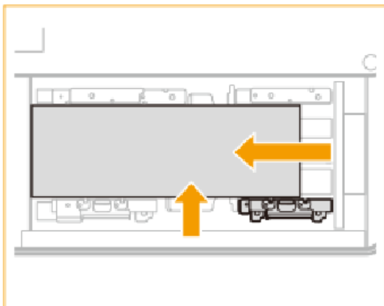


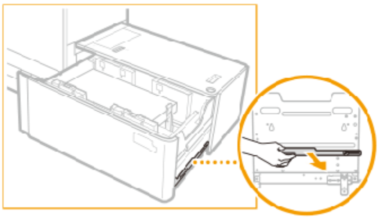
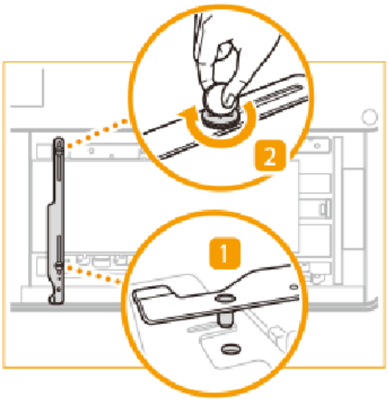
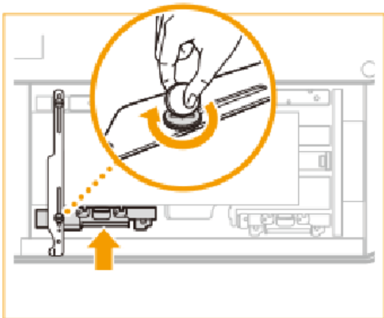
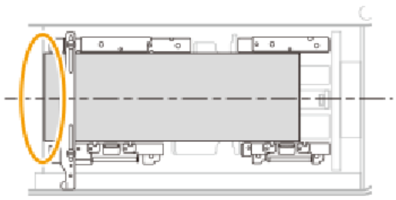
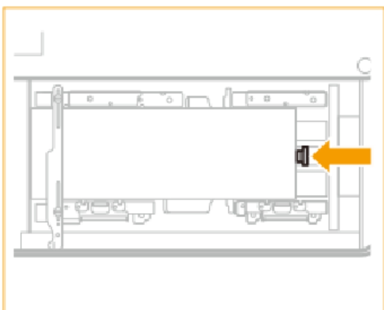
Before you begin

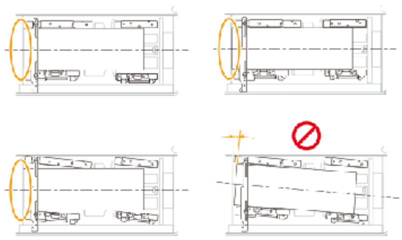
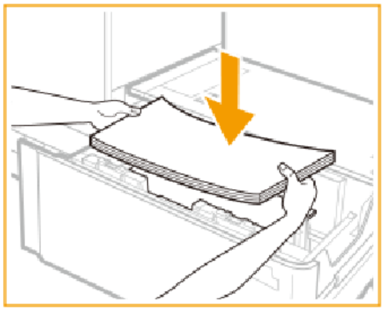
- Check and prepare the media.
- Awake the print system from sleep mode, if applicable.

Procedure

	Action	
1	Press the button (1) to open the bulk paper tray (2).	
	When the media size does not change continue with step 6, otherwise go to step 2.	
2	Slide the right-hand paper guide to the right-hand side of the paper tray so that there is enough space to load the media. NOTE The paper size indicator flashes or illuminates.	
3	Remove all media.	
	If the paper switch lamp flashes, continue with step 6. Otherwise, if the paper switch lamp illuminates, go to step 4.	
4	Close the bulk paper tray and wait until both the paper switch lamp and open button illuminate. IMPORTANT Do not open the bulk paper tray while both the paper switch lamp and open button are flashing.	

	Action	
5	Press the button (1) to open the bulk paper tray (2).	
6	Load a media stack of approximately 10 mm (0.4 inch).	
7	Keep the media against the left-hand wall and squeeze the left-hand paper guide to slide the paper guide against the media stack. IMPORTANT • Make sure the media stack is placed parallel to the left-hand wall. • Do not place the left-hand paper guide to tightly to the media stack. This can cause the media to partly rise.	 
8	Push the media stack against the left-hand wall and right-hand paper guide.	

	Action	
9	Take the fixing bracket for long sheets from the right-hand side of the bulk paper tray.	
10	Place the fixing bracket with the fixing pin in the indicated hole (1) and tighten the rear screw (2).	
11	Hold the left-hand paper guide against the media stack and tighten the front screw.	
12	Make sure that both corners on the left-hand side of the media stack touch the left-hand wall.	
13	Slide the paper guide against the media stack. IMPORTANT Use the paper size marks to place the paper guide properly.	

	Action	
14	Slide the right-hand paper guide to the media stack.	
15	Make sure that the media stack is flush against the left-hand wall.	
16	Continue to load the next media stacks of approximately 20 mm (0.8 inch). IMPORTANT • Make sure the loaded media stack does not exceed the loading limit mark (). • When you load envelopes, make sure the loaded envelope stack does not exceed the envelope loading limit mark (/). Use the correct orientation for the envelopes.	
17	Gently close the paper tray and make sure it clicks into place. **CAUTION** **When you close the bulk paper module, be careful not to get your fingers caught. This can cause personal injury.**	

After you finish

- When a paper jam occurs or you notice poor image quality, turn over the media stack and reload the media stack. Do not reload textured, single-sided coated, or already printed paper to avoid a paper jam. For these media, take a new stack.
- Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

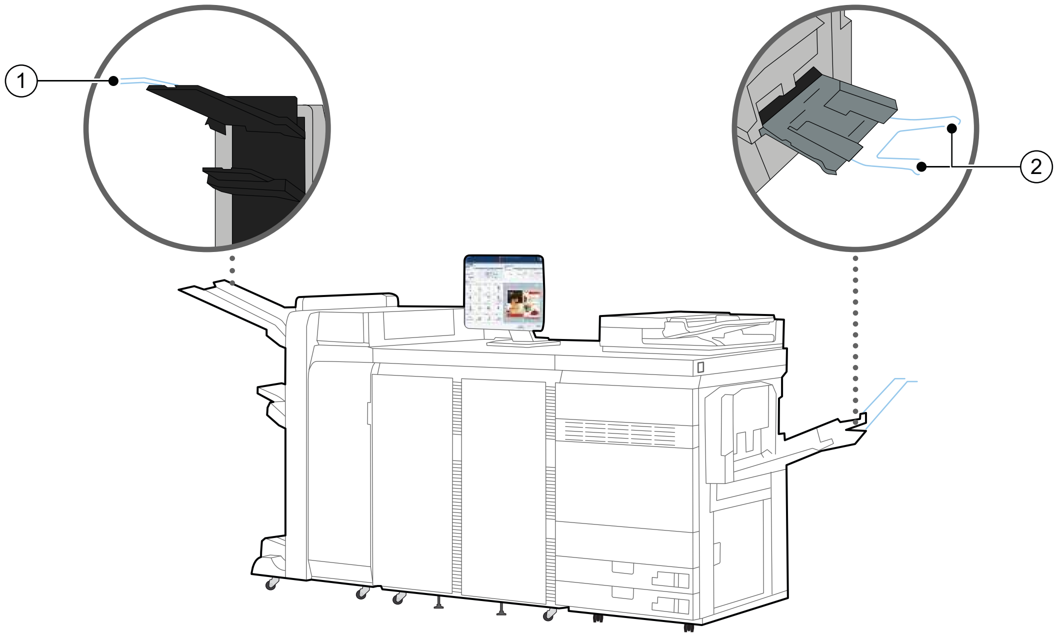
More information

- *Check and prepare media before loading* on page 132

Load long sheets into the special feeder

The extension guides of the special feeder for long sheets (Long Sheet Tray-B) enable you to feed long sheets, for example, when you need to print book jackets, banners, posters or maps. The special feeder can hold up one long sheet. The special feeder feeds the media face up.

You have to use this extension guide together with the alignment tray, Stack Bypass Alignment Tray-D.

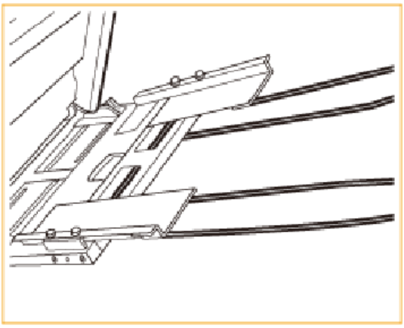


Extension guides for long sheets (Long Sheet Tray-B)

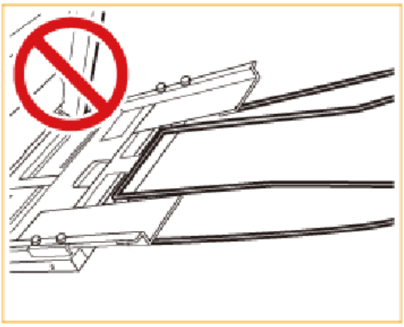
	Description extension guides
1	Extension guides to place the long sheets in the correct position in the tray
2	Extension guides to collect long sheets

CAUTION
When you handle paper, be careful not to cut your hands on the edges of the paper.

IMPORTANT
- Load only one sheet at a time.
- Do not load envelopes.
- Feed vellum media one sheet at a time and remove each sheet after delivery in the output tray. When you load several sheets together, a paper jam can occur.
- It is very important that you define the correct job settings for media, such as heavy paper or transparencies. The fixing unit can become dirty, which may require a service visit. Moreover, the image quality can become poor.
- Check that the alignment guides are attached correctly to avoid paper jams.



Correctly attached alignment guides



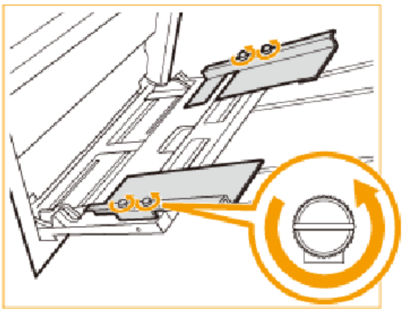
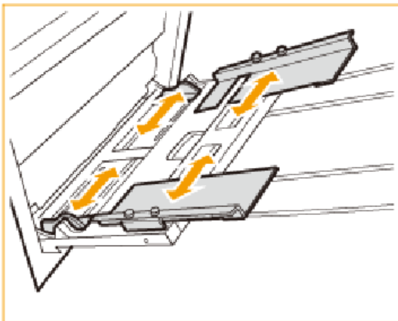
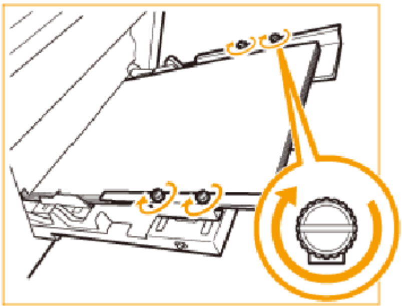
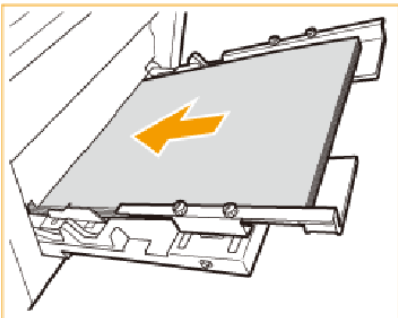
Incorrectly attached alignment guides

- Make sure you follow the instructions carefully. When you do not load media correctly, a paper jam, dirty machine parts or poor print quality can occur.

Before you begin

- Check and prepare the media.
- Awake the print system from sleep mode if applicable.

Procedure

	Action	
1	Loosen 4 knobs of the alignment guides.	
2	Adjust the position of the alignment guides and paper guides to the width of the media.	
3	Tighten the knobs.	
4	Load one sheet into the special feeder. IMPORTANT Make sure the loaded media stack does not exceed the loading limit mark (■).	

After you finish

Tightly rewrap the remaining media in the original package, and store the package in a dry place, away from direct sunlight or high temperatures.

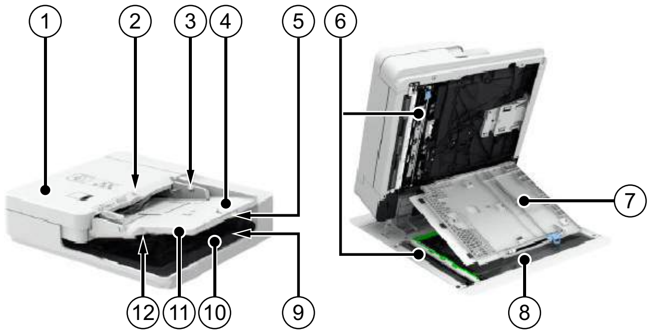
More information

Check and prepare media before loading on page 132

Chapter 7
Prepare copy and scan jobs

Use the automatic document feeder to copy or scan

The automatic document feeder enables you to copy or scan a set of originals. The automatic document feeder scans both sides of the originals at the same time.



Automatic document feeder

	Description automatic document feeder
1	Cover, to access the original feed path when a paper jam occurs.
2	Originals input indicator, to indicate that there are originals in the automatic document feeder.
3	Paper guides, to place the originals in the correct position in the tray.
4	Originals tray, to place the originals.
5	Originals tray extension, to place large originals (A3/11 inch x 17 inch).
6	Scan area, where the automatic document feeder scans the originals.
7	Cover, to access the scan area for cleaning activities or to solve paper jams.
8	Glass plate, to copy or scan an original, such as a page of a book, a heavy or delicate document or a transparency.
9	Originals receiving tray, to collect scanned originals.
10	Originals receiving tray extension, to collect scanned, large originals.
11	Original retainer
12	Originals output indicator, to indicate that the automatic document feeder is scanning.

IMPORTANT

- Thin originals can become creased in a high temperature or high humidity environment.
- When you scan long originals (432 mm - 630 mm, 17 inch - 24.8 inch), push back the extension originals tray and feed the originals by hand to prevent creased originals.
- Do not place the following originals into the automatic document feeder that are difficult to feed, such as:
  - Originals with tears or large binding holes
  - Severely curled originals or originals with sharp folds
  - Clipped or stapled originals
  - Carbon-backed paper
  - Transparencies and other highly transparent media
- Always smooth out folds in originals before you place them into the automatic document feeder.
- Do not drop clips or other objects into the gap of the originals tray.
- Do not add or remove originals during the scan process.
- Do not place objects in the original output area. This can cause damage to originals.
- Do not scan originals more than 30 times. This can cause folded and creased originals.
- When the rollers of the automatic document feeder get dirty due to originals written with pencil, perform the automatic document feeder cleaning procedure.

	Action	
1	Adjust the paper guides to fit the size of your originals. You can place a set of mixed-size originals into the automatic document feeder.	
2	Place the originals face up into the originals tray; the originals input indicator * illuminates green light. The position of the originals influences the staple, fold and punch location. **CAUTION** **Do not insert your hands in the gap of the originals tray. This can cause personal injury.**	
3	Define the copy or scan settings and touch [Start] . When the originals receiving tray detects the scanned originals, the originals output indicator * illuminates white light.	

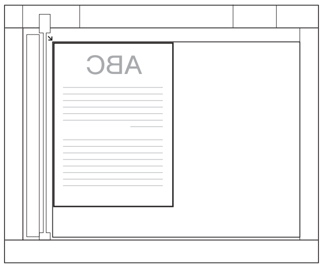
	Action	
4	Remove the scanned originals from the originals receiving tray when the originals output indicator * blinks.	

More information

- *Clean the rollers of the automatic document feeder* on page 535
- *Paper input specifications* on page 618
- *Feed instructions* on page 596

Use the glass plate to copy or scan

You must use the glass plate to scan a page of a magazine or book, a heavy or delicate document, or a transparency.

	Action	
1	Lift the automatic document feeder approximately 300 mm (11.8 inch) to allow the sensor to detect the size of the original.	
2	Place and align an original face down. The position of the original influences the staple, fold and punch location in the printed output.	 
3	Gently close the automatic document feeder. ⚠ **CAUTION** **To prevent damage to the glass plate and personal injury, carefully close the cover and do not push onto the cover.**	
4	Define the copy or scan settings and touch [Start]. ⚠ **CAUTION** **Be aware that the light of the glass plate is very bright.**	
5	Remove the original from the glass plate.	

More information

- *Paper input specifications* on page 618
- *Feed instructions* on page 596

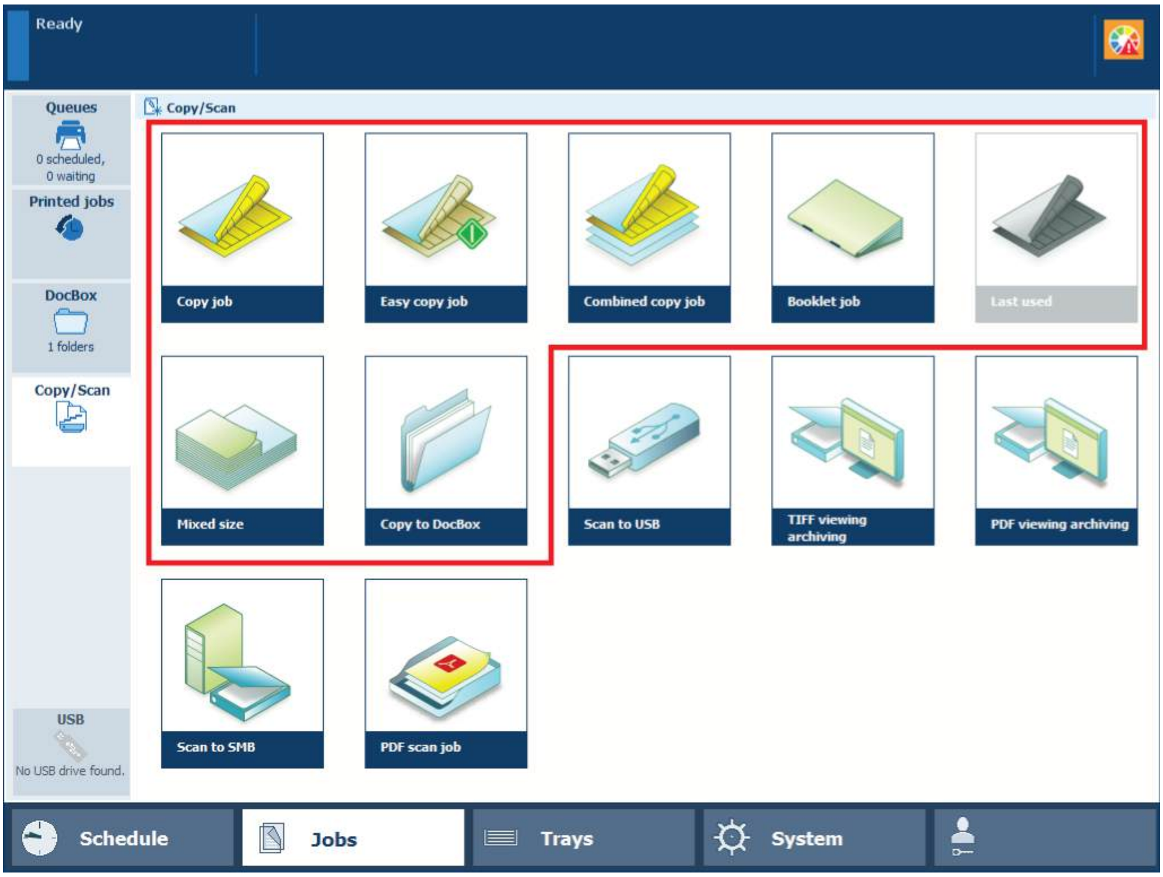
Prepare copy jobs

Learn about copy job templates and settings

The copy function shows a number of factory default templates to copy paper originals. When you select a template, PRISMAsync automatically uses copy and original settings for the job. You can change each individual setting of a template and define your own templates.

Default copy templates

The table below show the available copy job templates.



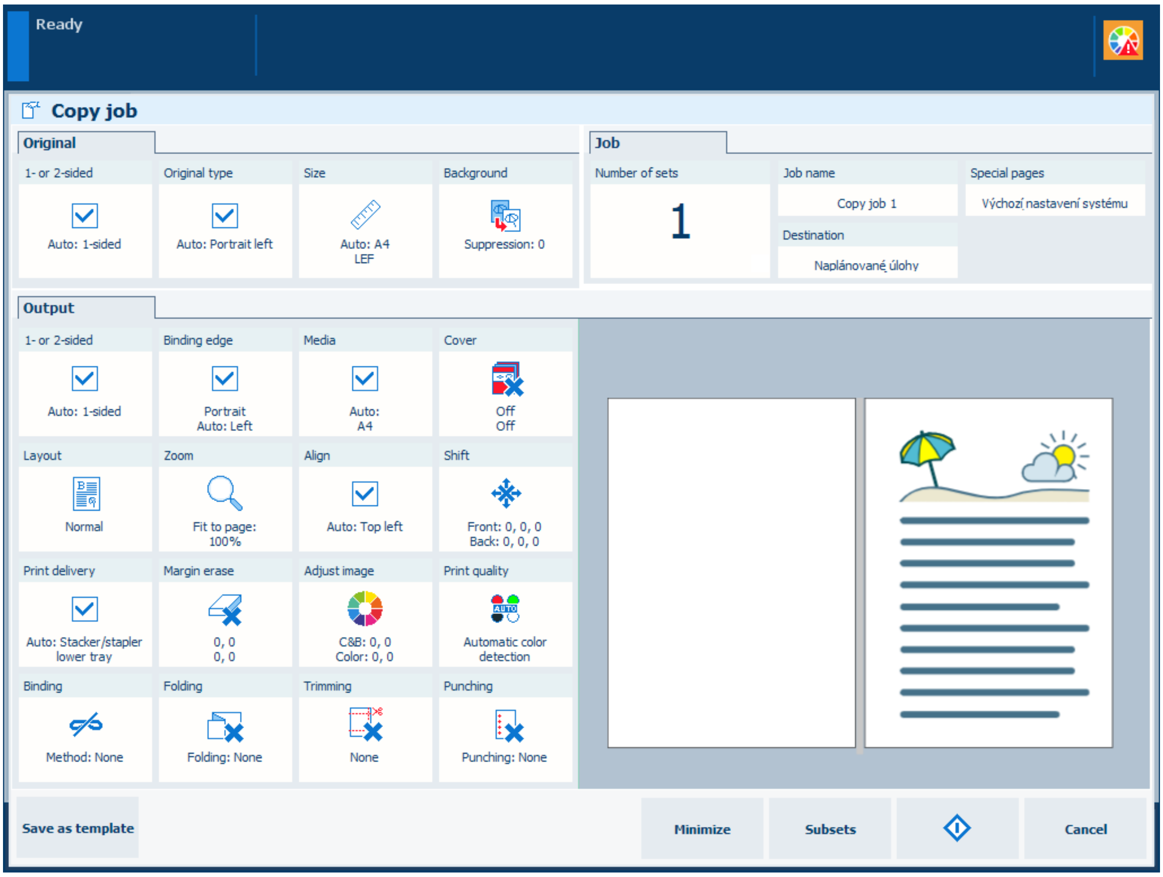
[Copy/Scan] templates

Template	When to use
[Copy job]	When you have a copy job.
[Easy copy job]	When you have a simple copy job.
[Combined copy job]	When you have a copy job with subsets of originals.
[Booklet job]	When you need copied booklets.
[Mixed size]	When you have a copy job with mixed-size originals. The copy will also have these media sizes.
[Last used]	When you want to reuse the settings of the previous copy job. You cannot use this function for a combined copy job.

Template	When to use
[Copy to DocBox]	When the destination of the copy job is the first available DocBox folder or a DocBox folder of your choice.

Copy job settings

The following tables show the copy job properties you can change. Use the [Minimize] setting to minimize the copy job definition when you want to do other tasks on the control panel.



[Copy job] settings

[Original] settings

Job setting	What you can define
[1- or 2-sided]	**Original sides to copy** Indicate if one or two sides of the original must be copied.
[Original type]	**Binding edge** Binding position along the long or short side The binding edge automatically adjusts the margin shift direction, the orientation and staple position.
[Size]	**Original size** Indicate the standard or custom size of originals.

Job setting	What you can define
[Background]	**Background of the original** Indicate if a vague or yellowish background must be suppressed.

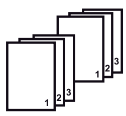
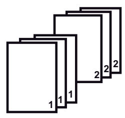
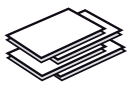
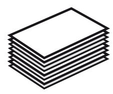
[Job] settings

Job setting	What you can define
[Number of sets] [Check first set]	**Number of sets** Indicate the number of sets: • Number of sets (maximum 65000). • Indicate if only the first set needs to be printed.
[Job name]	**Job name** Change the name of the job.
[Destination]	Indicate the location of the job: • List of [Waiting jobs] • List of [Scheduled jobs] • A [DocBox] folder
[Special pages]	**Banner pages, trailer pages and separator sheets** • **[Overrule default banner / trailer page settings]**: indicate if you want to overrule the default banner page, trailer page, and separator sheets settings. • **[Use a banner page]**: indicate if the job must include a banner page. The banner page contains: the sender name, the recipient name, the accounting ID, the job name, the printer name, the operator instruction, the cost center, and the number of sets. • **[Use a trailer page]**: indicate if the job must include a trailer page. The trailer page contains: the username, the recipient name, accounting ID, the job name, the printer name, the operator instruction, the number of pages in a set , the number of sets, the number of staples, the number of folded sheets, the number of punched sheets, the number of creased sheets, the number of inserts, job received time, job start time, job completion time, the number of sheets per job media. • **[Media of banner/trailer page]**: define the media for the banner and trailer page. • **[Use separator sheets]**: indicate the use of separator sheets to separate job sets. Separator sheets are not printed. • **[Separator sheet after N sets]**: define after how many sets you want a separator sheet. • **[Media of separator sheets]**: define the media for the separator sheet.
[Accounting ID]	**Accounting information** Enter information for billing and charging job costs: • [Accounting ID], to identify a user or a group of users. • [Cost center], to identify an organizational department, project or group. • [Note], to enter additional information required for accounting.

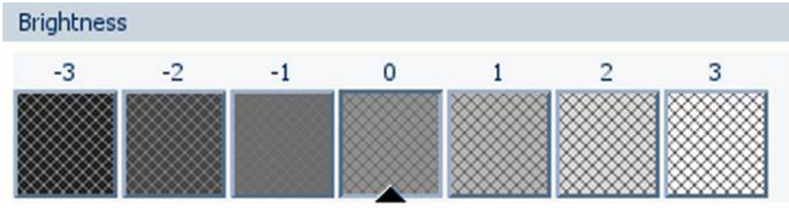
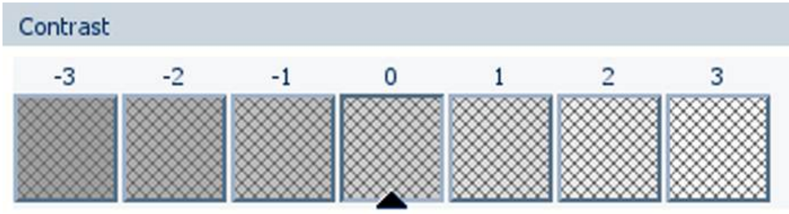
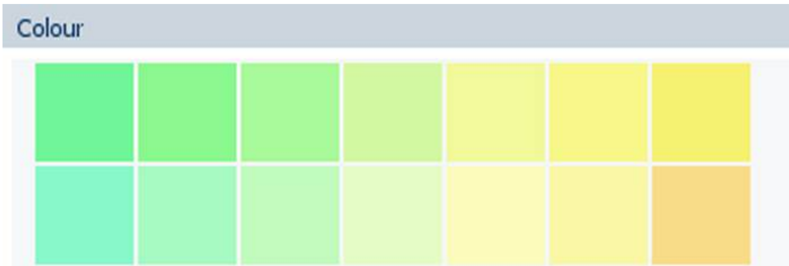
Output settings

Job setting	What you can define
[1- or 2-sided]	**Sides to print on** Indicate if the copies are printed on one or both sides.
[Binding edge]	**Orientation and binding location** The orientation of the content is part of the original settings and cannot be changed. You can change the binding edge: the sheet edge to which the previous page is bound.
[Media]	**Media** Select the media for the job.
[Cover]	**Usage of covers** Indicate if you want to use covers and select the cover media. You can also indicate if you want to copy on one or both sides of the cover. The default media that are used for covers are configured in the Settings Editor.
[Layout]	**Layout of the pages on the sheets** Indicate if and how you want to print more than one page of a document on a single sheet: • **[Multiple up]** , to print subsequent source file pages on one document page. • **[Imposition template]**, to fold a document: once to create a booklet, twice to create a quarto or three times to create an octavo. The sheets are printed in such a way that the pages of the folded document are presented in sequential order. • **[Same-up]**, to print the same page source file page on a document page.
[Zoom]	**Scaling factor** Indicate if you want to scale the image according to the page size, or set the zoom factor manually (25% - 400%).
[Align]	**Alignment of the image to a sheet location** Indicate how you want to align the image on the sheet: to a corner, edge or to the middle of a sheet.
[Shift]→[Margin shift] [Shift]→[Image shift]	**Mirrored and shifted page layout** Indicate if you want to apply a **margin shift**. When you adjust the default margin shift you create more or less space at the location of the binding edge, with a maximum shift of 100 mm (3.94 inch). Indicate if you want to use an **image shift**. When you adjust the image shift you move the image on the sheet, with a maximum shift of 100 mm (3.94 inch).
[Margin erase]	**Erase the margin of an image** Indicate if you want to erase the margins of the image. When you adjust the margin erase you erase areas of the image, with a maximum of 100 mm (3.94 inch).

Delivery settings

Job setting	What you can define
[Print delivery]→ [Output tray]	**Output tray** Indicate which output tray to use. The selected finishing method can require a specific output tray: • [Stacker/stapler booklet tray]: for booklets. • [eWire tray]: for wire-bound books with the eWire. Use this setting together with the [Punching]→[Die set dependent] setting to deliver punched sets of sheets to the eWire. • Name of external finisher: for finishing with a DFD (Document Finishing Device). You can define the name of the DFD in the Settings Editor. Location: [Preferences]→[Print job defaults]→[External finisher]→[Name].
[Print delivery]→ [Sort]	**Sorting method** Indicate how you want to sort the output, [By page] or [By set]
	 Sort by set  Sort by page
[Print delivery]→ [Offset stacking]	**Stacking method** Indicate if you want stacking with or without an offset. Offset stacking depends on the active workflow profile.
	 Offset stacked  Stacked
[Print delivery]→ [Advanced settings]	**Delivery method** Indicate the delivery method of the sheets in the output tray of your choice: • [Print order], indicate the delivery order of the sheets. • [Sheet orientation], indicate which sheet orientation is applied. • [Rotation], indicate if the images are rotated on the sheet.

Print quality settings

Job setting	What you can define
[Adjust image]	**Image adjustment** Use the preview to check the results when you change the following settings: [Brightness] 
	[Contrast] 
	[Color] 
[Color / black & white]	**Color quality** Indicate whether you want colored or black & white copies: • [Automatic color detection] • [Black and white] • [Color]

Finishing settings

Job setting	What you can define
[Binding]	**Binding method** Indicate how to bind your document.

Job setting	What you can define
[Folding]	**Folding method** Indicate how to fold the printed documents. • [None] • [Tri-fold in] • [Tri-fold out] • [Parallel fold] • [Half-fold] • [Z-fold] • [Multi half-fold]
[Creasing]	**Creasing method** Indicate how to crease the printed documents. • [None] • [Center] • [C-fold]
[Sheets]	Indicate which sheets you want to be creased. • [None] • [All sheets] • [First sheet] • [Last sheet] • [First and last sheet]
[Trimming]	**Trimming method** Define how to trim the printed document. • [Trim size]: trim the long edge and short edges of the printed document. • [Finishing size]: trim the printed document to the specified document size
[Punching]	**Punching method** Define how to punch the prints. • [None] • [2 holes] • [3 holes] • [Die set dependent] • [Die set dependent, double punch] • [Die set dependent, saddle]
[Perforating]	**Perforating method** Define how to perforate the prints. • [None] • [Single edge] • [Single center] • [Double]
[Perforation location]	When you select [Double] in [Perforating], define the perforation location.

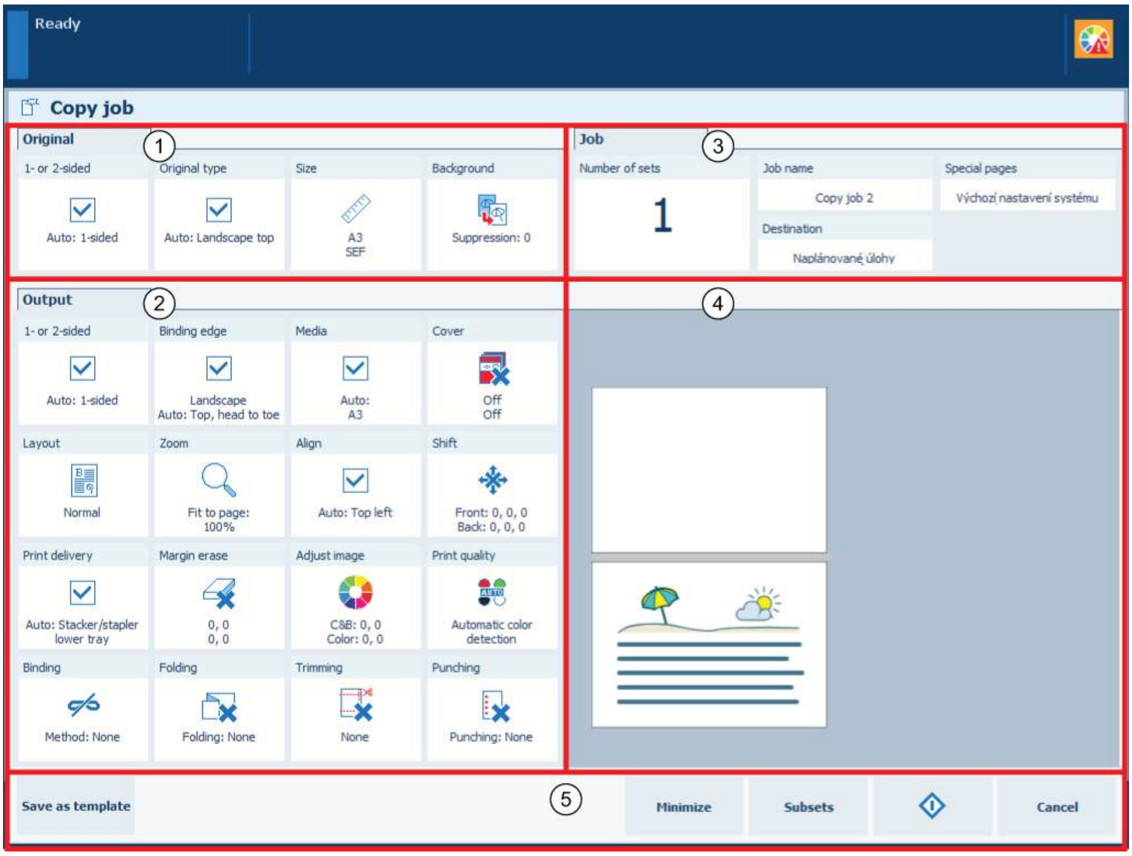
More information

- *Define default use of special pages* on page 81
- *Choose a workflow profile* on page 240

Make a copy

You start a copy job from the automatic document feeder or the glass plate. When the scan area has scanned the originals, the copy goes to the destination you indicated.

1. Place a set of originals into the automatic document feeder or place an original on the glass plate.
2. Touch [Jobs]→[Copy/Scan].
3. Select a copy template.
4. Define the original settings in the [Original] pane (1).



[Copy job] settings

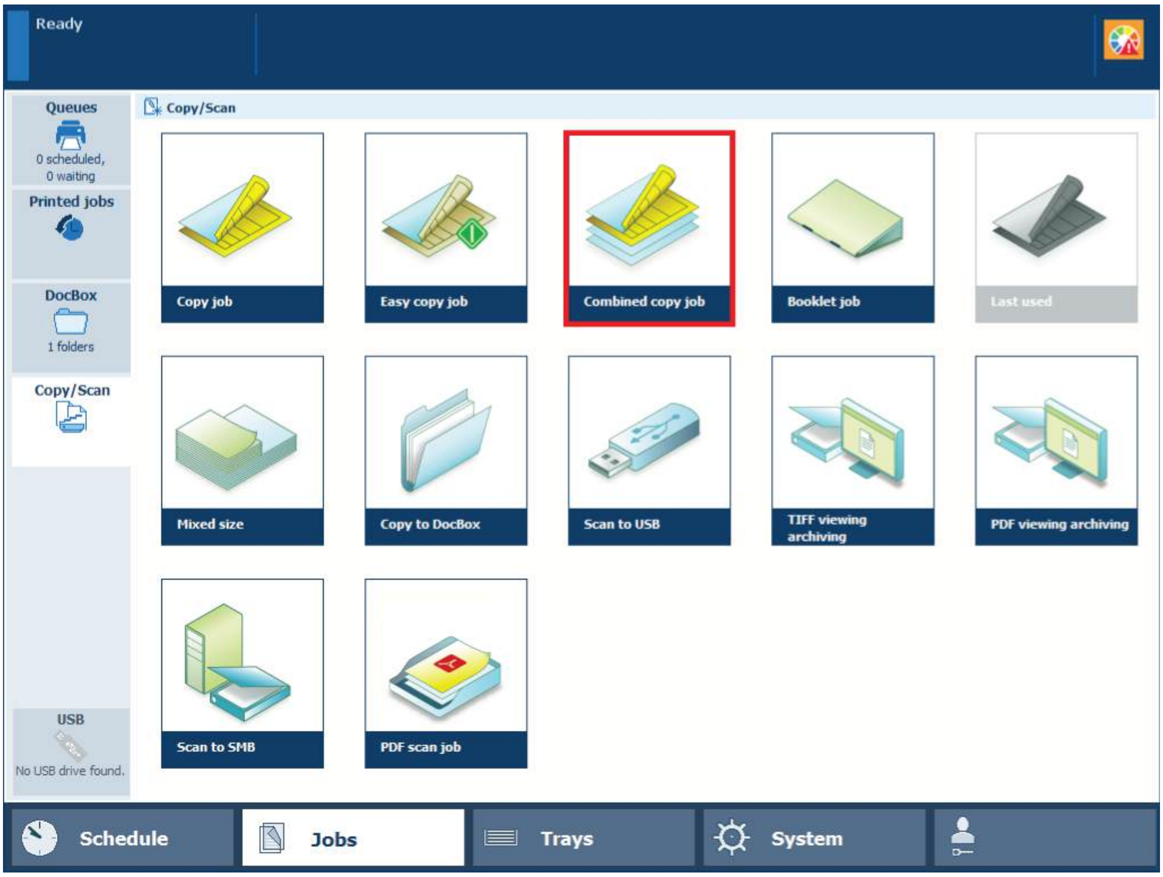
5. Define the output settings in the [Output] pane (2).
6. Enter a job name and the number of sets in the [Job] pane (3).
7. Select the destination of the copy (3).
   To scan now and print later, select the list of waiting jobs or a DocBox.
8. Check the results of your settings in the preview pane (4).
9. Touch the start button (5).

More information

- *Use the automatic document feeder to copy or scan* on page 178
- *Learn about copy job templates and settings* on page 182
- *Use templates for recurring jobs* on page 198

Copy subsets (combined copying)

You can combine different subsets of originals into a copy.



[Combined copy job] template

1. Touch [Jobs]→[Copy/Scan]→[Combined copy job].
2. Place the first subset into the automatic document feeder or on the glass plate.
3. Define the settings for the subset and the copy.
4. Touch [Scan].
5. Place the next subset into the automatic document feeder or on the glass plate.
6. Define the settings for the subset and the copy.
7. Touch [Scan].
8. Repeat step 5, 6, and 7 for each following subset.
9. Touch [Ready] when the last subset is scanned.
10. Define the destination and enter a name for the copy.
11. Touch the start button to start the copy job.

More information

- *Use the automatic document feeder to copy or scan* on page 178
- *Learn about copy job templates and settings* on page 182
- *Use templates for recurring jobs* on page 198

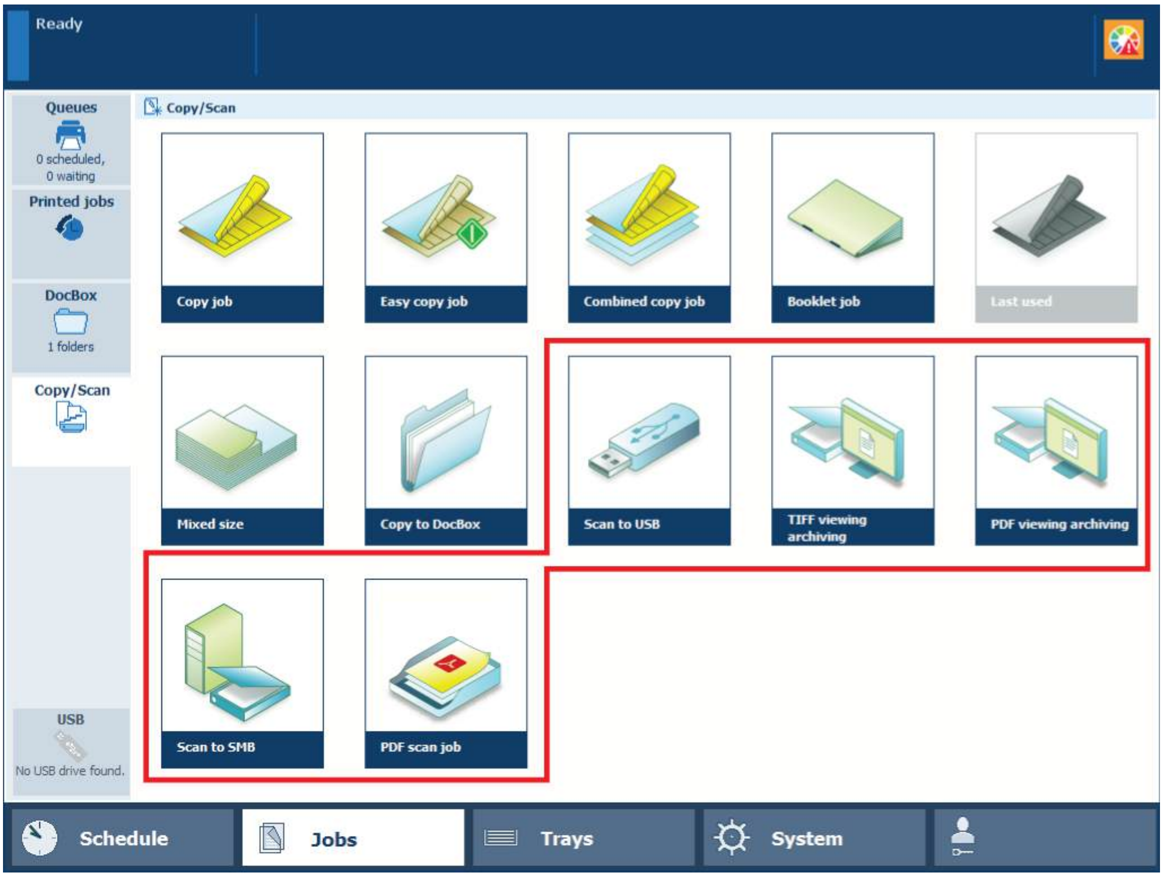
Prepare scan jobs

Learn about scan jobs templates and settings

The scan function contains a number of factory default templates to create a digital document from originals. When you select a template, PRISMAsync automatically uses scan and original settings for the job. You can change each individual setting of a template and define your own templates.

Scan job templates

The table below show the available scan job templates.

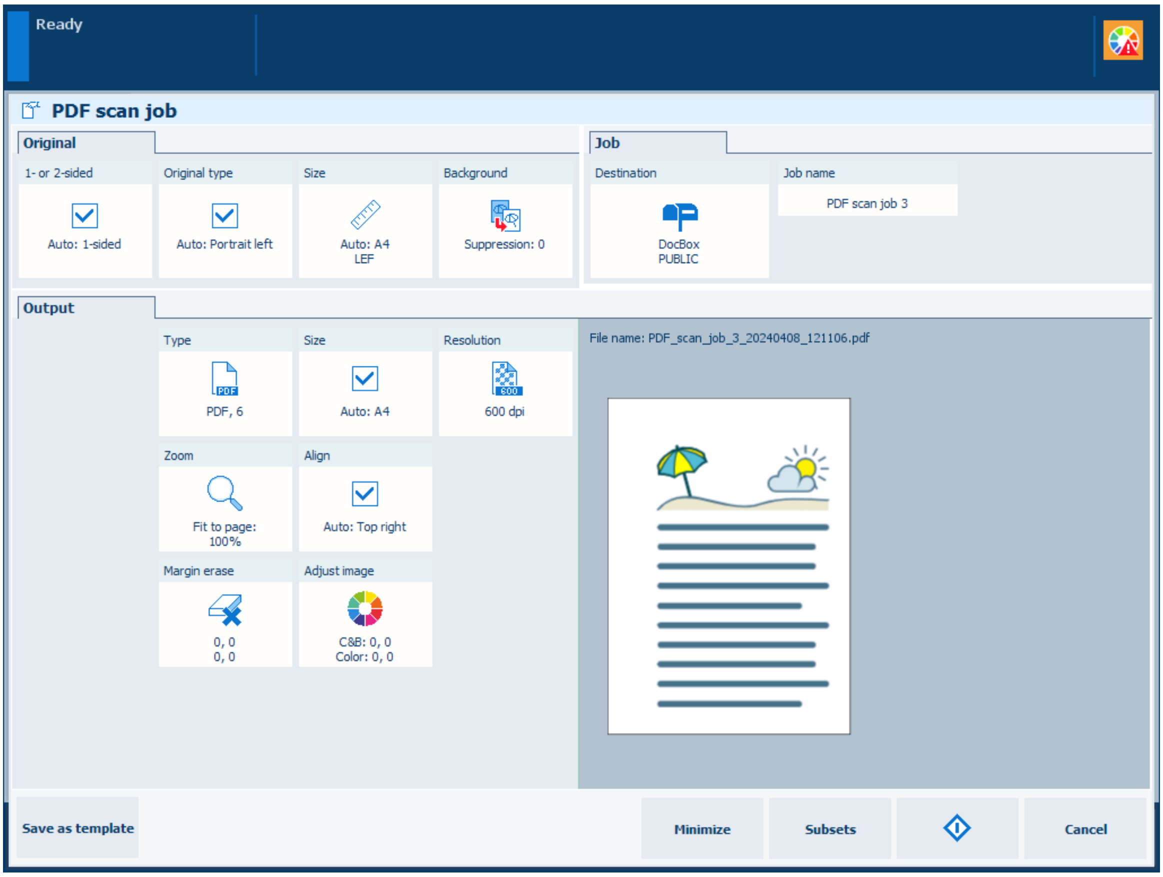


[Copy/Scan] templates

Template	When to use
[Last used]	When you want to reuse the settings of the previous scan job. You cannot use this function for a combined scan job.
[Scan to USB]	When you want to scan to your USB device.
[TIFF viewing archiving] [PDF viewing archiving]	When you want to view a lot of scanned documents. The documents are scanned at a lower resolution and stored with a smaller file size.
[Scan to SMB]	When you want to store on a server or workstation via the SMB protocol.

Template	When to use
[PDF scan job]	When you want to store your scanned document in a DoxBox,

The following tables show the scan job properties you can change. Minimize the scan job definition when you want to do other tasks on the control panel.



[PDF scan job] settings

Original settings

Job setting	What you can define
[1- or 2-sided]	**Original sides to copy** Indicate if one or two sides of the original must be copied.
[Original type]	**Binding edge** Binding position along the long or short side The binding edge automatically adjusts the margin shift direction, the orientation and staple position.
[Size]	**Original size** Indicate the a standard or custom sizes of originals.
[Background]	**Background of the original** Indicate if a vague or yellowish background must be suppressed.

Document settings

Job setting	What you can define
[Type]	**File type** Indicate how you want store the scan files: • [Format], to select the format of the scan file. • [Compression], the compression factor for TIFF scan files. The compression factor influences the size of the scan file. • [Quality], the quality factor for JPG and PDF scan files. • [PDF/A], this is the standard ISO format for archiving purpose
[Size]	**Page size** Page size of the digital document.
[Resolution]	**Scan resolution** The scan resolution influences the size of the scan file.

Layout settings

Job setting	What you can define
[Zoom]	**Scaling factor** Indicate if you want to scale the image according to the page size, or set the zoom factor manually (25% - 400%).
[Align]	**Alignment of the image to a sheet location** Indicate how you want to align the image on the sheet: to a corner, edge or to the middle of a sheet.
[Margin erase]	**Erase the margin of an image** Indicate if you want to erase the margins of the image. When you adjust the margin erase you erase areas of the image, with a maximum of 100 mm (3.94 inch).

Image quality settings

Job setting	What you can define
[Adjust image]	**Image adjustment** Use the preview to check the results when you change the following settings: • [Brightness] • [Contrast] • [Color]
[Print quality]	**[Color / black & white]** Indicate whether you want colour, black & white or automatic colour detection for your scans: • [Automatic color detection] • [Color] • [Black and white]

Destination settings

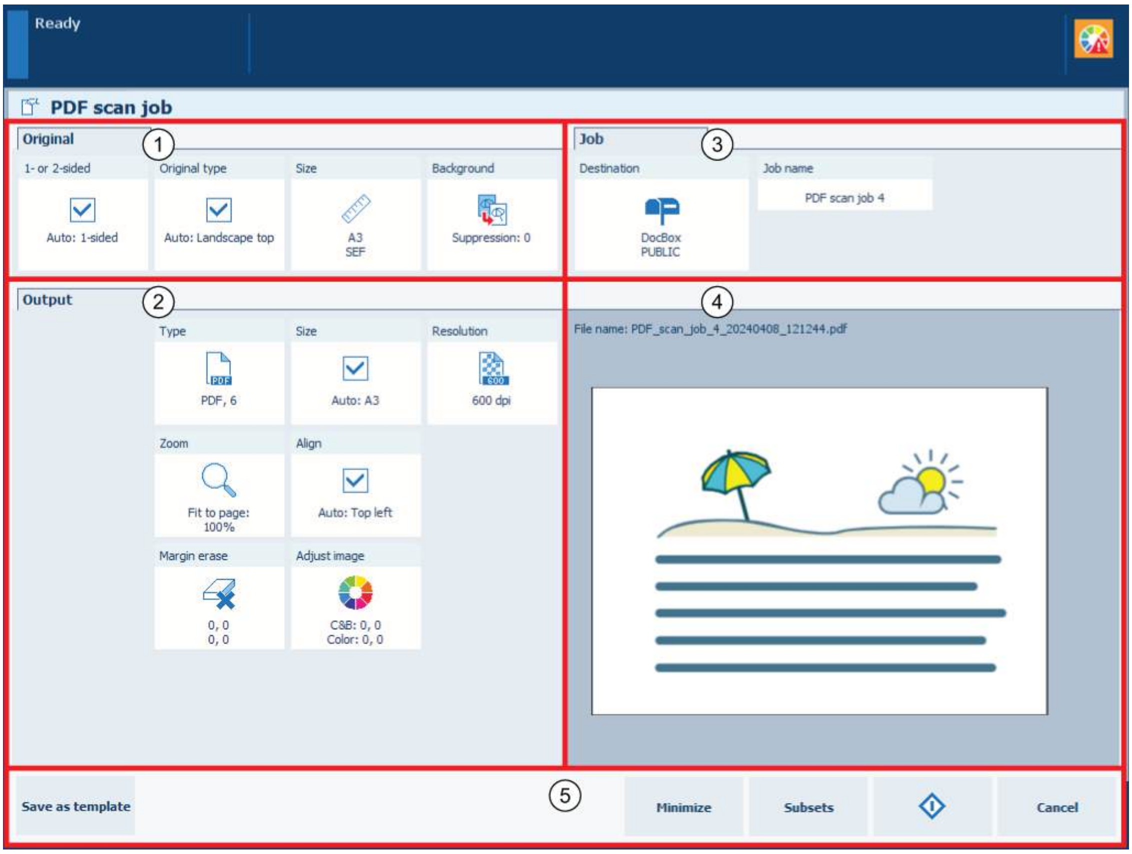
Job setting	What you can define
[Destination]	**Destination** The available destinations depend on the configuration of the scan function: • FTP • USB • Email • SMB • DocBox • List of waiting jobs • WebDAV
[Job name]	**Job name** Enter a new name for the job. This name is automatically supplemented with a date and time.

Make a scan

You can start a scan job from the automatic document feeder or the glass plate. When the originals have been scanned, the scan file goes temporarily to the list of scan files before it arrives in the indicated destination.

PRISMAsync automatically defines a number of settings for the originals and the scan file. However, you can overrule these automatic settings. The automatic settings have the prefix "Auto".

1. Place a set of originals into the automatic document feeder or place an original on the glass plate.
2. Touch [Jobs]→[Copy/Scan].
3. Select a scan template.
4. Define the original settings in the [Original] pane (1).



[PDF scan job] settings

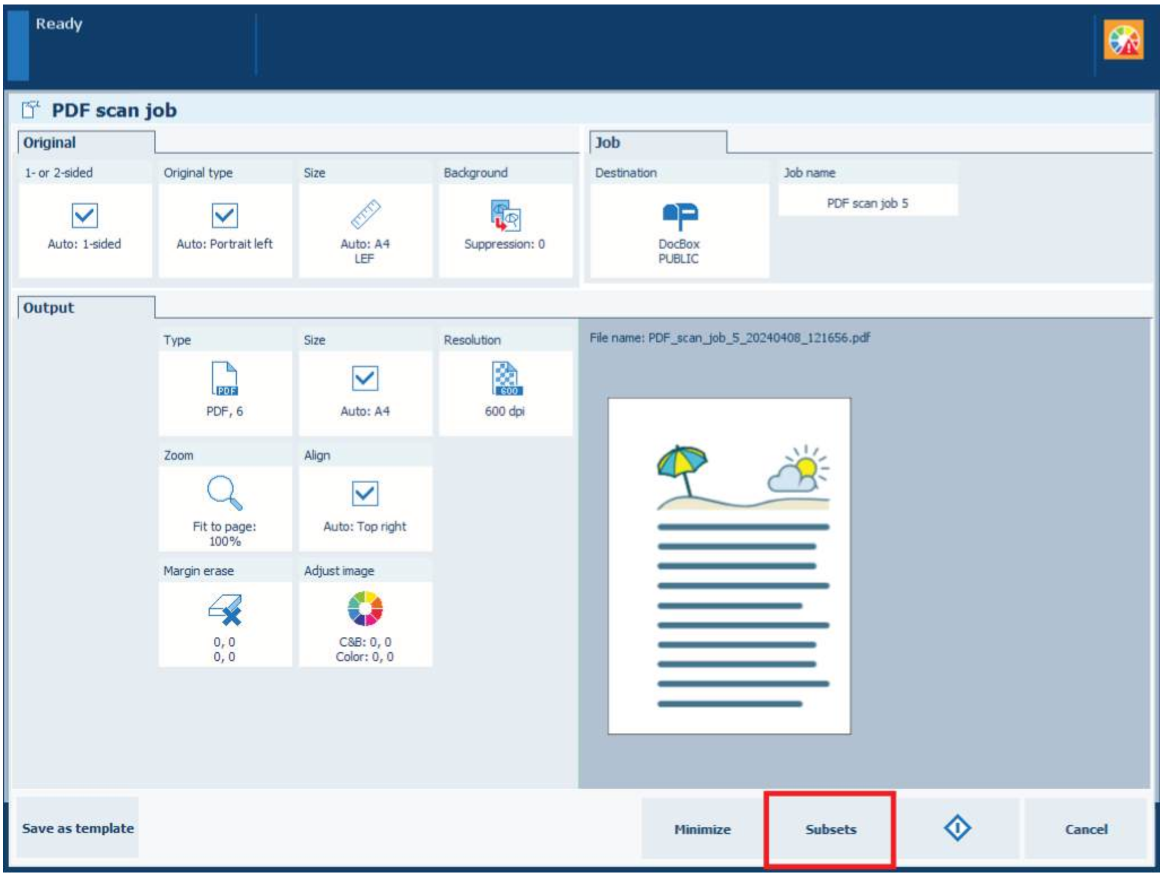
5. Define the output settings in the [Output] pane (2).
6. Select the destination of the scan file in the [Job] pane (3).
   - For scan to email define the subject of the email, the addresses, and your accounting ID.
   - For scan to USB insert the USB drive into the USB port and select the required folder on the drive.
   - For scan to SMB select the SMB share and select the SMB share directory when configured.
7. Enter a job name and a file name ID (3).
8. Check the results of your settings in the preview pane (4).
9. Touch the start button .
10. Touch the eject button and remove the USB drive, if applicable.

More information

- *Use the automatic document feeder to copy or scan* on page 178
- *Learn about scan jobs templates and settings* on page 191
- *Use templates for recurring jobs* on page 198

Scan subsets (combined scanning)

You can combine different subsets of originals into a scan file.



[Subsets] function

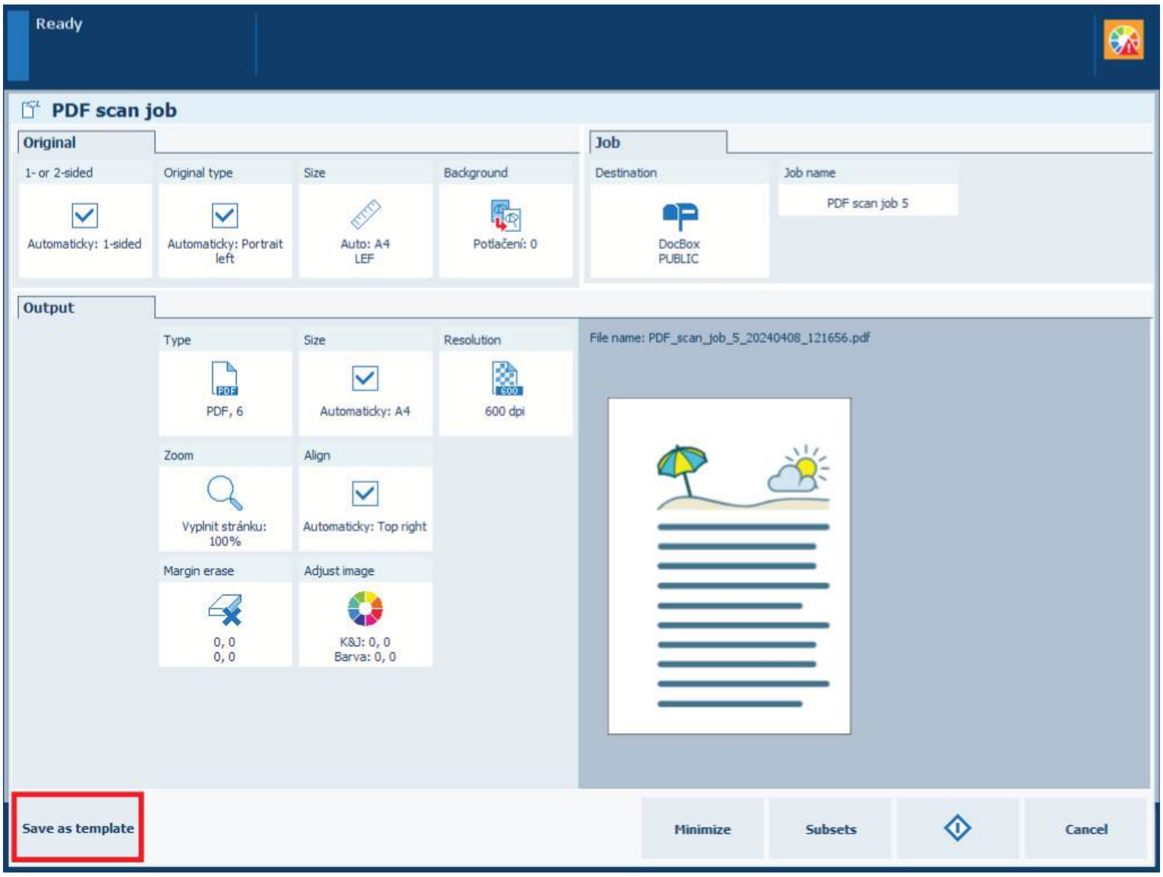
1. Touch [Jobs]→[Copy/Scan].
2. Select one the available scan templates.
3. Touch [Subsets].
4. Place the first subset into the automatic document feeder or on the glass plate.
5. Define the settings for the subset and the scan file.
6. Touch [Scan].
7. Place the next subset into the automatic document feeder or on the glass plate.
8. Define the settings for the subset and the scan file.
9. Touch [Scan].
10. Repeat steps 7, 8, and 9 for each subset.
11. Touch [Ready] when the last subset is scanned.
12. Define the destination for the scan file, enter the job name and the file name ID for the scan file.
13. Touch the start button to start the scan job.

More information

- *Use the automatic document feeder to copy or scan* on page 178
- *Learn about scan jobs templates and settings* on page 191
- *Use templates for recurring jobs* on page 198

Use templates for recurring jobs

The print system offers a number of default templates. When you carry out copy or scan jobs with the same settings on a regular basis, create custom templates to store the settings for these jobs.



Store settings of a copy or scan job in a template

1. Touch [Jobs]→[Copy/Scan].
2. Select one of the available job templates.
3. Adjust the template settings.
4. Touch [Save as template].
5. Enter a name for the custom template and touch [OK].
6. Touch [Cancel].
7. Touch [Jobs]→[Copy/Scan] to view the new template.

After you finish

To change a custom template, touch the custom template for two seconds. You can rename, move or delete the custom template.

To store the settings of the last job in a custom template, touch the [Last used] template for two seconds.

More information

Learn about copy job templates and settings on page 182

Chapter 8
Transaction printing

Learn about the transaction printing workflow

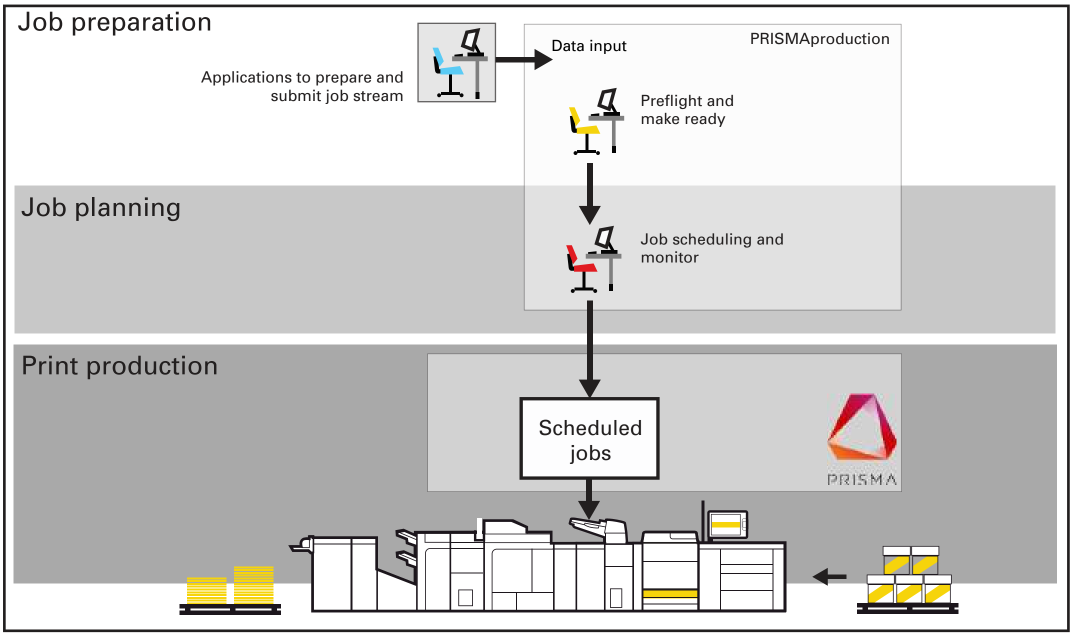
The **transaction printing mode** fits print environments that print large numbers of business critical data, such as invoices, checks, and salary slips.

The imagePRESS V1000 uses the IPDS or PJL/PCL protocol to communicate between the data submitting system and the printer. PRISMAsync Print Server uses the SRA (Scalable Raster Architecture) module to perform the communication with the data submitting system.

The communication with the data submitting system is bi-directional, which means that the printer is able to provide feedback about the job progress and the status of the printer.

Transaction print jobs

The following illustration shows the transaction print job workflow from job preparation to the delivery of the prints. The output and workflow management application PRISMAproduction provides a single point of control to spool, manage and monitor transaction print jobs.



Job workflow for transaction printing mode

1. **Job preparation**
   Jobs start with one or more applications to generate data resources and to design documents. The data resources and documents are submitted to the printer with PRISMAproduction. The prepress staff uses PRISMAproduction to check the documents (pre-flighting) and to make job adjustments (make-ready) before submitting the jobs. The IPDS data stream can have containers with PDF documents. The active transaction setup specifies the print properties.
2. **Job planning**
   One or more jobs are part of a data stream submitted to the printer during a connection session. When the data stream includes job separators, the printer can distinguish the

separate jobs. In the job monitor of PRISMAproduction you can manage the jobs and define the job order. It is not possible to change the job order or edit transaction print jobs from the control panel. However, if required, you can stop the printing of the data stream from the control panel.

3. **Print production**
The operator keeps the printer running. Correct media are loaded in the paper trays, finished output and waste are removed in time and there are sufficient supplies of consumables in the printer. In transaction printing mode, you cannot remove individual jobs. Only the entire data stream can be aborted from the control panel. The printer constantly sends information about the printer status and the job progress to PRISMAproduction.

Activate the transaction printing mode

When the transaction printing mode is active, the data stream enters the printer via a separate IPDS or PCL port. Meanwhile, all submitted **document print jobs** come in the list of waiting jobs. The printer begins to process these jobs, after you leave the transaction printing mode and return to the document printing mode.

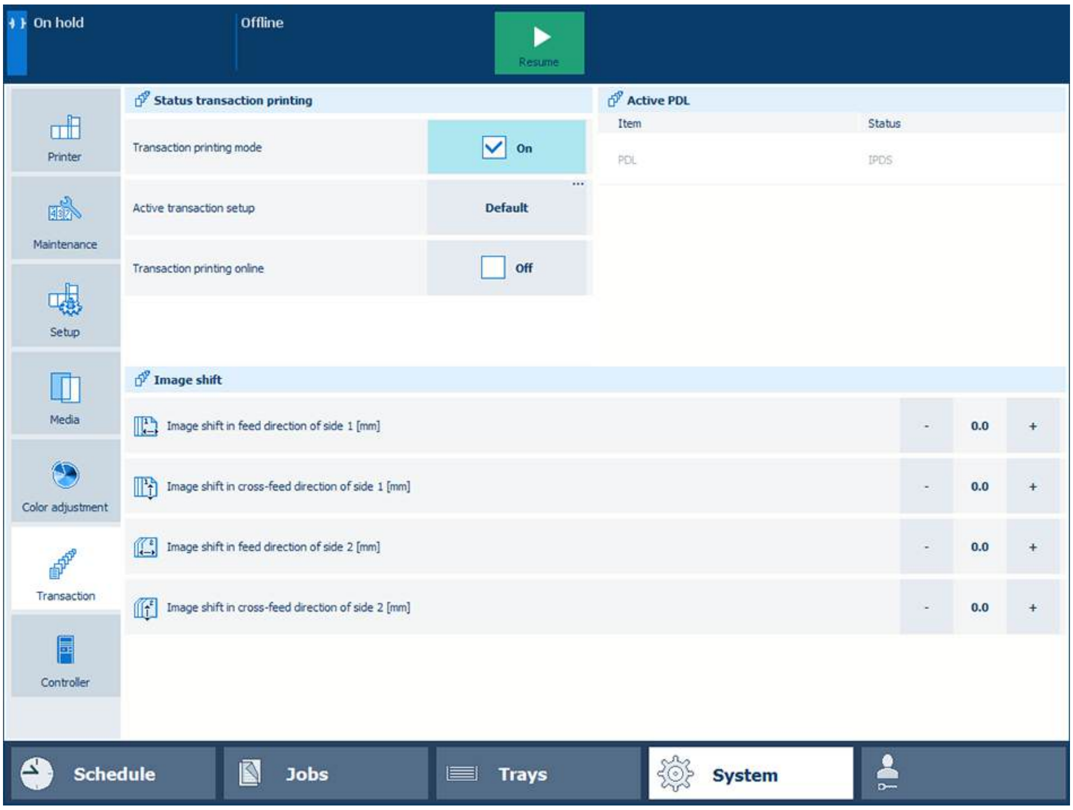
Transaction printing mode is active	

The transaction printing mode uses the workflow profile settings.

You switch to the transaction printing mode on the control panel. Next, you bring transaction printing online on the control panel or in the Settings Editor.

Activate transaction printing mode on control panel

1. Touch [Jobs] →[Queues] and ensure the list of scheduled jobs is empty.
2. Touch [System]→[Transaction].
3. Touch the [Transaction printing mode] check box.



Activate the transaction printing mode on control panel

Bring transaction printing online or offline on control panel

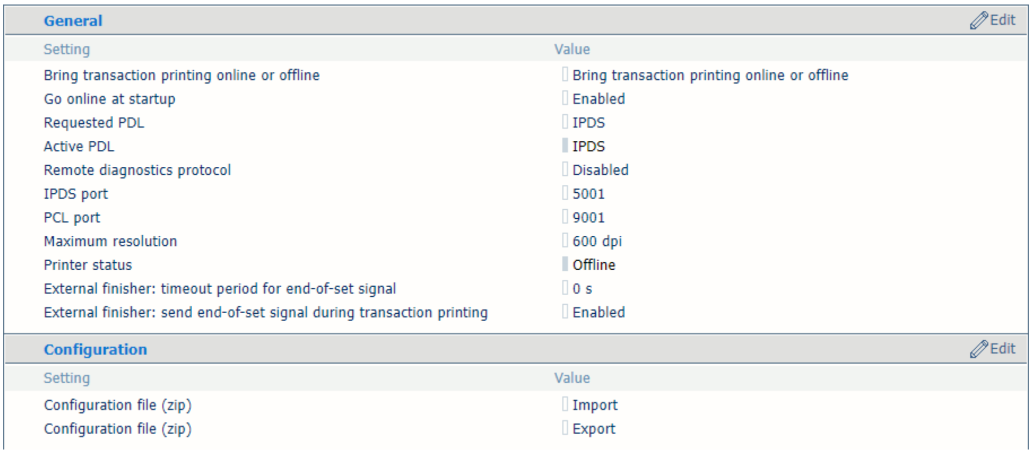
1. Touch [System]→[Transaction].
2. Use the [Transaction printing online] check box.

Bring transaction printing online or offline in Settings Editor

1. Open the Settings Editor and go to: [Transaction printing]→[Settings].

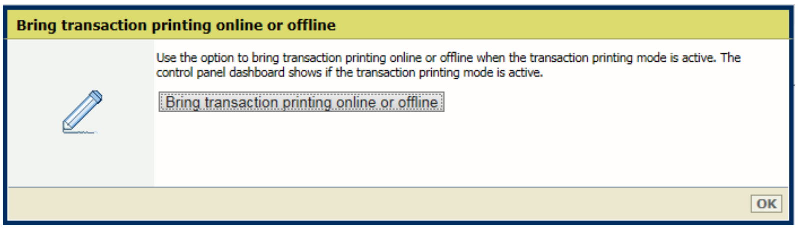
[Settings] tab

2. Go to the [General] section.



[General] section

3. Read the [Printer status] value to know the status of transaction printing.
4. Use the [Bring transaction printing online or offline] setting to bring transaction printing online or offline.



Bring transaction printing online or offline in Settings Editor

5. Click [Bring transaction printing online or offline].
6. Click [OK].

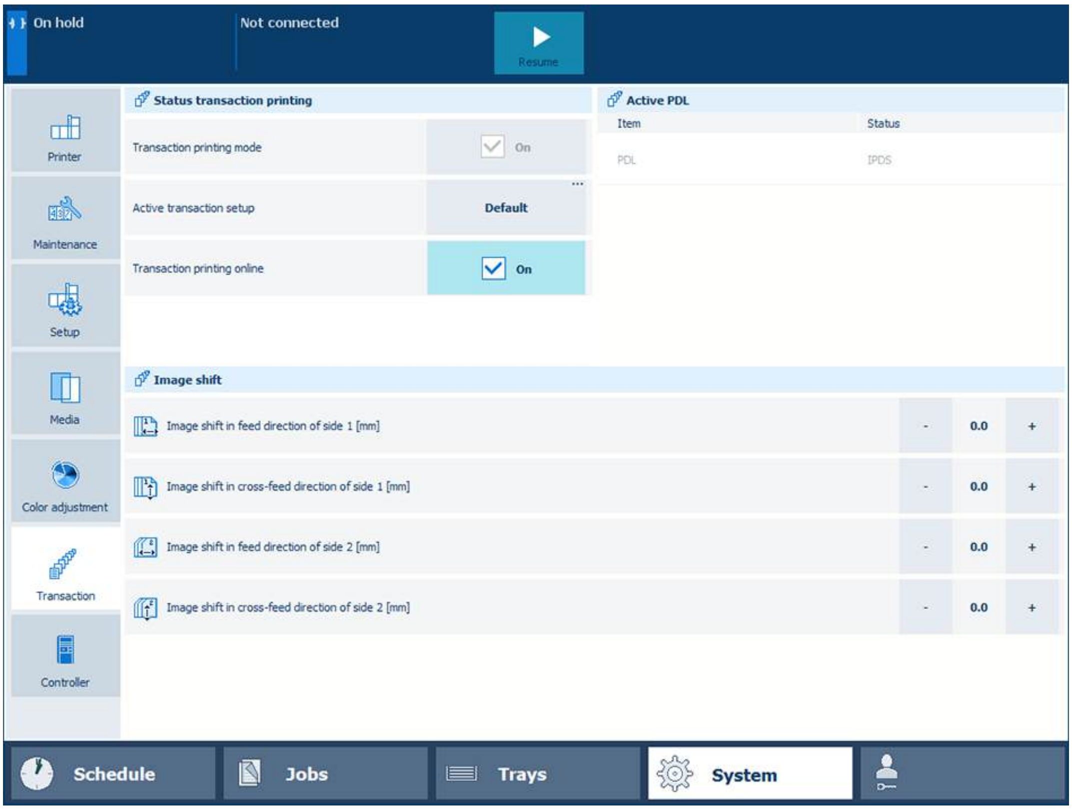
More information

- *Activate the document printing mode* on page 239
- *Learn about workflow profiles* on page 237
- *Choose a workflow profile* on page 240
- *Load a transaction setup* on page 211
- *Switch the Page Description Language (PDL)* on page 204

Switch the Page Description Language (PDL)

Only one transaction printing Page Description Language (PDL) can be active at the printer: IPDS or PCL.

The control panel shows the active PDL.



Active PDL on control panel

When your printer has the IPDS and the PCL licenses then you can change the PDL.

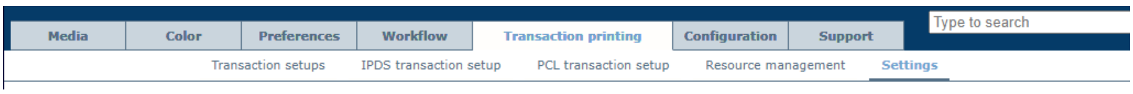
When both licenses are installed, the default PDL is IPDS.

When you change the PDL, the PDL becomes active when transaction printing is put online.

The transaction printing setup is activated by the IPDS or PCL data stream.

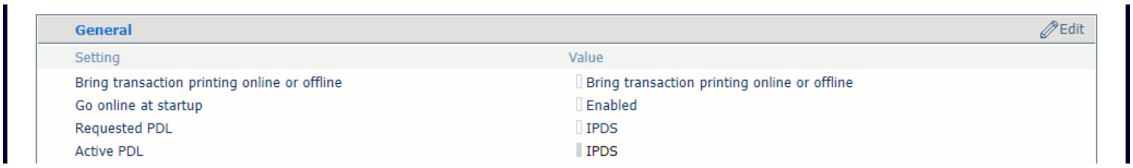
The loaded IPDS or PCL transaction setup becomes active.

1. Open the Settings Editor and go to: [Transaction printing]→[Settings].



[Settings] tab

2. Go to the [General] section.

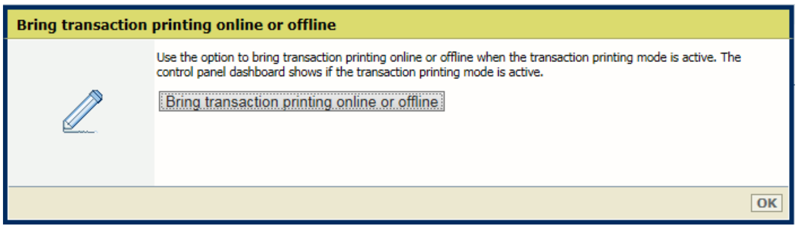


[General] section

3. The current PDL is displayed in the [Active PDL] field.

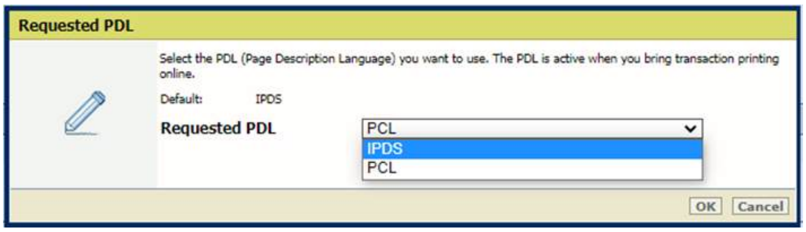
   NOTE
   You can only change the PDL if the printer is licensed for two transaction printing protocols.

4. Click [Bring transaction printing online or offline] to put transaction printing offline.



[Bring transaction printing online or offline] in Settings Editor

5. Use the [Requested PDL] setting to select the PDL you want to use.



[Requested PDL]

6. Click [Bring transaction printing online or offline] to put transaction printing is online.
7. The new PDL is displayed in the [Active PDL] field.
8. Click [OK].

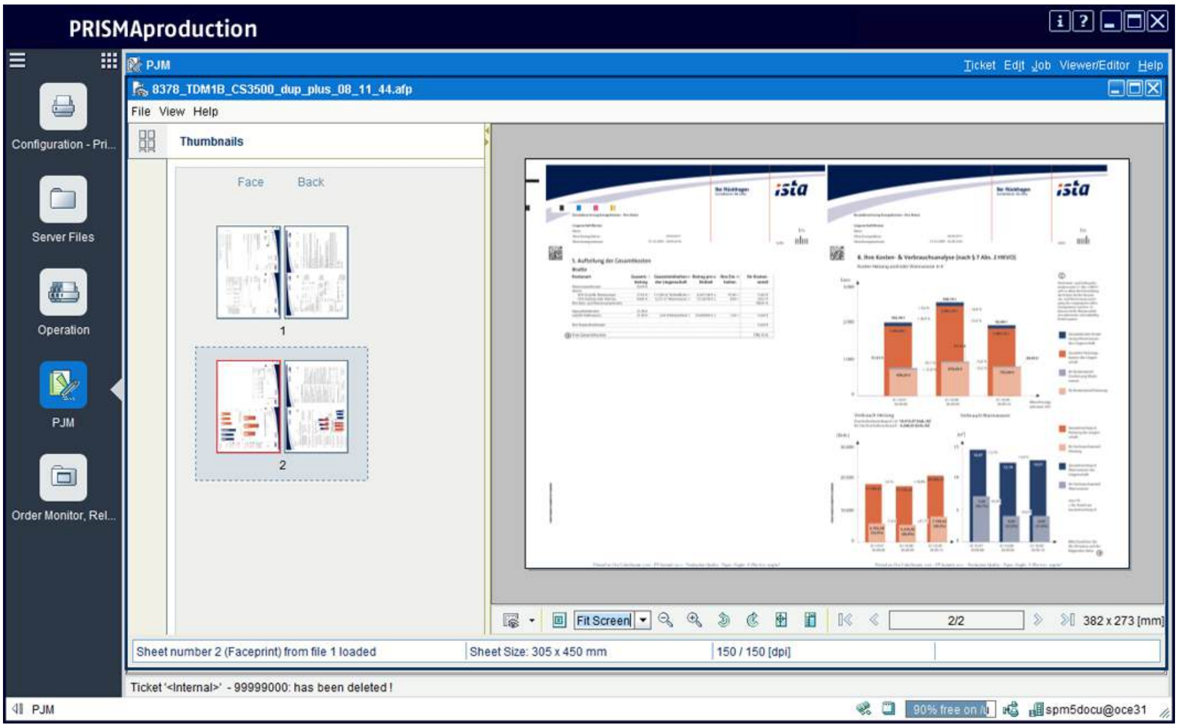
More information

- *Learn about transaction setups* on page 207
- *Define the general transaction printing settings* on page 233

Monitor or stop transaction print jobs

PRISMAproduction provides a single point of control to proof, spool, interrupt, reprint , and monitor transaction print jobs.

Use the Stop button on the control panel to stop printing and remove transaction print jobs from the list of scheduled jobs.



View a job in PRISMAproduction

When you want to interrupt printing, or abort the streaming data, open PRISMAproduction and go to the job list. Indicate the required printer action. When you delete jobs in PRISMAproduction, these jobs are no longer present in the print queue of the printer.

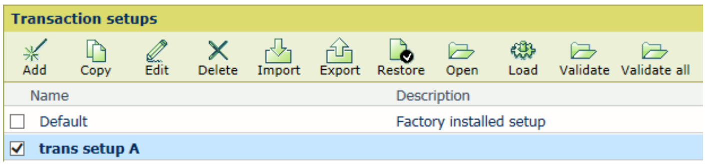
For more information on PRISMAproduction, see the manuals that belong to the application.

Work with transaction setups

Learn about transaction setups

What are transaction setups

A transaction printing setup is a set of attributes that is used for transaction printing. You can create different setups, but only one setup can be loaded, which means it is active. The attributes of the active transaction setup define the media, color, print delivery and other properties of the transaction print job.



[Transaction setups] menu

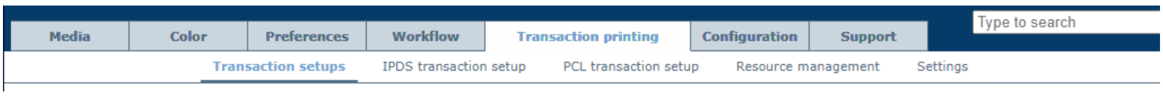
You can only load a transactions setup when transaction printing is offline.

NOTE
In PRISMAproduction you can attach a transaction setup to a job ticket. When the data stream uses this job ticket, the attached transaction setup is automatically loaded for the next data stream.

PDL and IPDS transaction setups

Only one transaction printing Printer Control Language (PDL) can be active at the printer: IPDS or PCL.

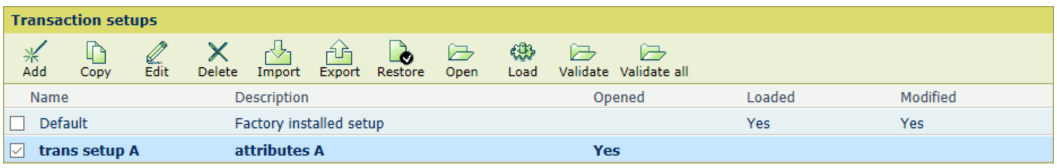
It depends on the data stream if the IPDS or the PCL attributes of the transaction setup become active.



[Transaction setups] menu

Status of transaction setups

The list of transaction setups shows the status of the transaction setups.



Status of transaction setups

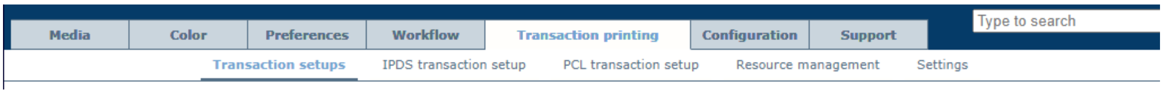
Status	
[Opened]	There is only one transaction setup with the status [Opened]. This transaction setup attributes are visible when you click the [IPDS transaction setup] and [PCL transaction setup] tab. You can also open a transaction setup with the status: [Loaded].
[Loaded]	There is only one transaction setup with the status [Loaded]. This transaction setup is active. When the PDL is IPDS, you can load a transaction setup when transaction printing is online and the printer has been stopped. When the PDL is PCL, you can only load a transaction setup when transaction printing is online.
[Modified]	You can change the attributes of a transaction setup with the status: [Loaded]. The status [Modified] indicates that the active transaction setup has been changed but the changes have not taken effect yet. Use the [Load] setting to re-load the transaction setup.

Define a new transaction setup

You can define a new transaction setup or change the name and description of a current transaction setup. To define the attributes, you first need to open the transaction setup.

Go to the transaction setups

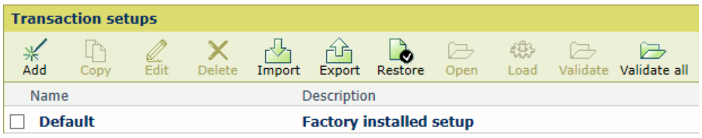
Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



[Transaction setups] tab

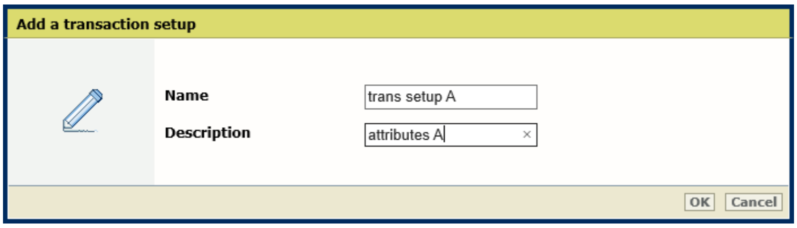
Add a new transaction setup

1. Click [Add].



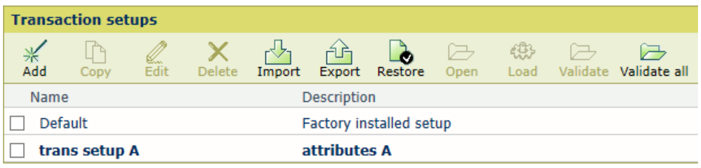
[Transaction setups] menu

2. Enter a unique name and the description.



[Add a transaction setup]

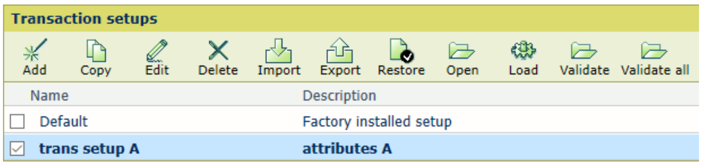
3. Click [OK].



[Transaction setups]

Copy a transaction setup

1. Select the transaction setup and click [Copy].

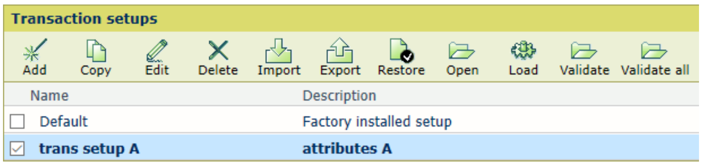


[Transaction setups] menu

2. Enter a unique name and the description.
3. Click [OK].

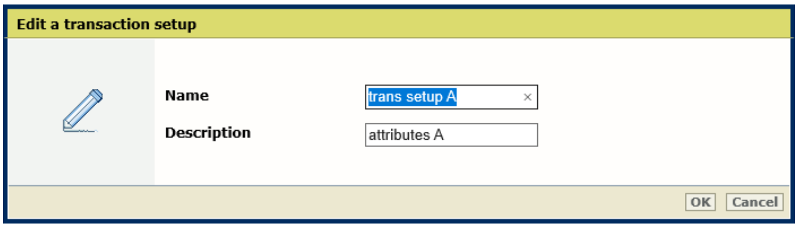
Edit a transaction setup

1. Select the transaction setup and click [Edit].



[Transaction setups] menu

2. Enter a unique name and the description.



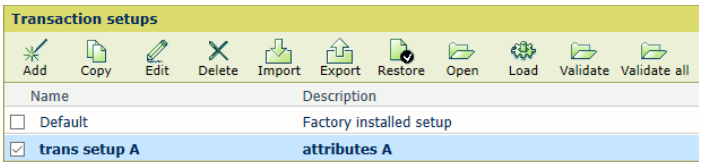
[Edit a transaction setup]

3. Click [OK].

Delete a transaction setup

You cannot delete an active transaction setup and a transaction setup that is open. At least one transaction setup must remain available in the list.

1. Select one or more transaction setups you want to delete.
2. Click [Delete].



[Transaction setups] menu

More information

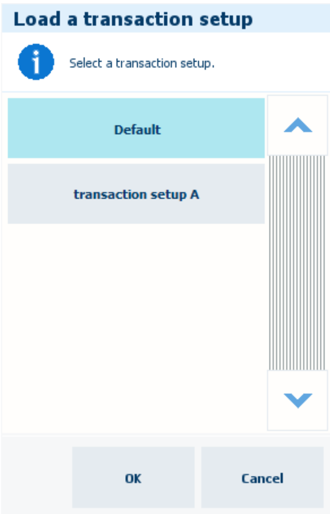
- *Learn about transaction setups* on page 207
- *Open a transaction setup to change settings* on page 213

Load a transaction setup

When you load a transaction setup, you make it active. You can only load a transaction setup when transaction printing is offline (PCL and IPDS) or when you press the Stop button on the control panel (only IPDS).

NOTE
In PRISMAproduction you can attach a transaction setup to a job ticket. When the data stream uses this job ticket, the attached transaction setup is automatically loaded for the next data stream.

Load a transaction setup on control panel



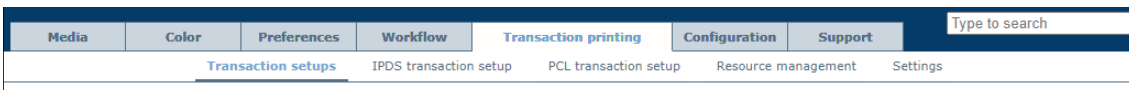
[Load a transaction setup]

1. Touch [System]→[Transaction].
2. Press the Stop button .
3. Touch [Active transaction setup] option and select one of the available transaction setups.
4. Click [OK].
5. Touch the [Resume] button .

Load a transaction setup in Settings Editor

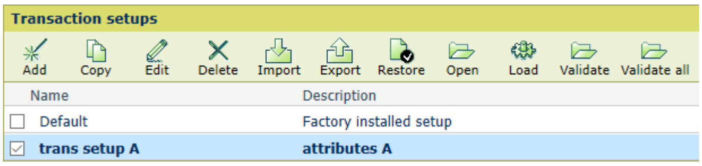
You load a transaction setup in the following situations.
- You want to apply another transaction setup than the active (loaded) transaction setup.
- You want to apply the changes of an active (loaded) transaction setup. You can read the status of the [Modified] column to see if a transaction setup must be loaded again.

1. Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



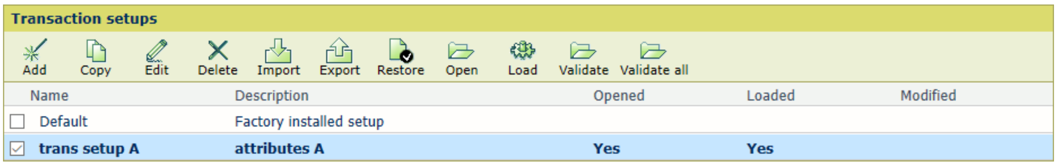
[Transaction setups] tab

2. Ensure transaction printing is offline.
3. Select one of the available transaction setups.



[Transaction setups]

4. Click [Load].
5. Bring transaction printing online.
   The status of the transaction setup is [Loaded].



[Transaction setups]

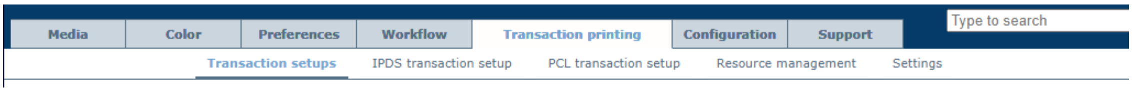
More information

- *Learn about transaction setups* on page 207

Open a transaction setup to change settings

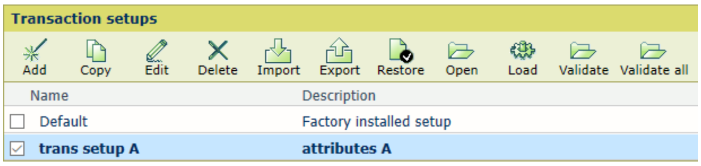
You open a transaction setup to check or change its settings.

1. Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



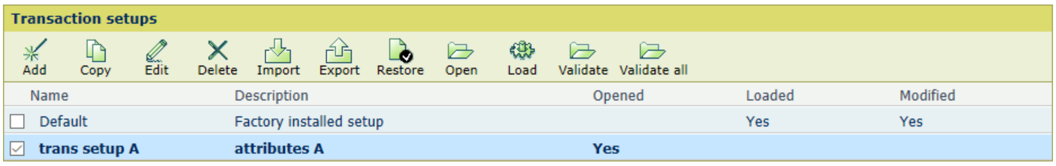
[Transaction setups] tab

2. Select the transaction printing setup.



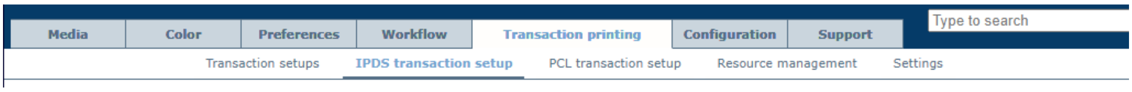
[Transaction setups]

3. Click [Open].
   The status of the transaction setup is: [Opened].



[Transaction setups]

4. Click the [IPDS transaction setup] or [PCL transaction setup] tab to define the settings.



[IPDS transaction setup] tab

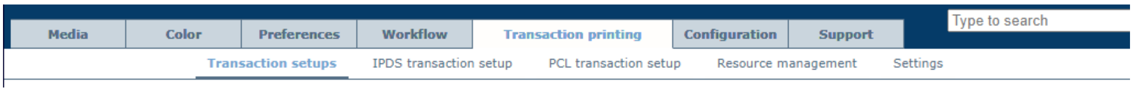
More information

- *Learn about transaction setups* on page 207
- *Define the settings of an IPDS transaction setup* on page 214
- *Switch the Page Description Language (PDL)* on page 204

Define the settings of an IPDS transaction setup

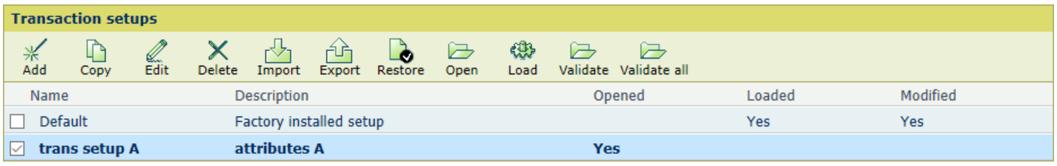
A transaction setup with the status [Opened] can be adjusted.

1. Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



[Transaction setups] tab

2. Check that the transaction setup you want to change has the status: [Opened].



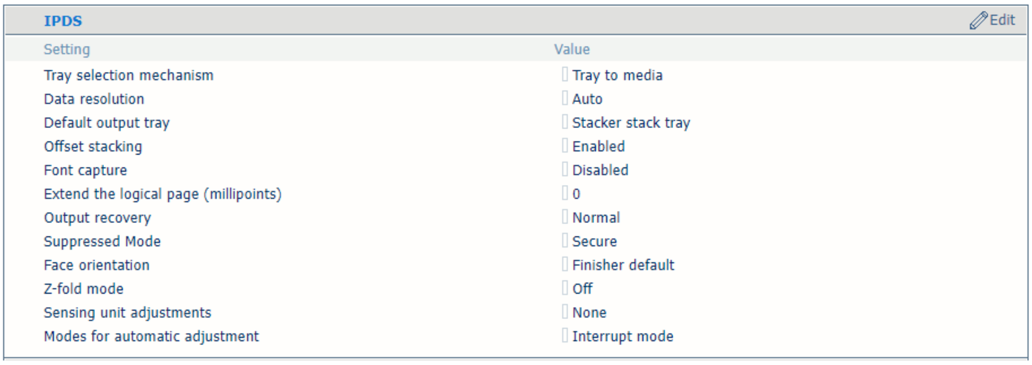
[Transaction setups]

3. Click the [IPDS transaction setup] tab.

[IPDS transaction setup] tab

4. Change the settings one by one or click [Edit] in the title bar to adjust the settings of a group in one dialog box.
5. Validate the transaction setup.
6. Make sure that you reload a transaction setup that is active when you make the changes.

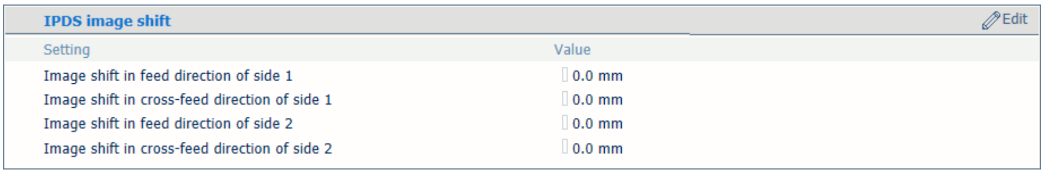
Define the IPDS settings



IPDS transaction setup settings

[IPDS] settings	Description
[Tray selection mechanism]	A transaction printing job always uses the media definitions as specified in Settings Editor. Select [Tray to tray] or [Tray to media]. The IPDS data stream contains a reference to a logical tray. There are 36 logical trays that can be assigned to a physical paper tray ([Tray to tray]) or to media of the media catalog ([Tray to media]).
[Data resolution]	Select the resolution of the bitmaps the printer accepts.
[Default output tray]	Select the default output tray.
[Offset stacking]	Select if stacking occurs with or without an offset.
[Font capture]	Indicate if fonts and other resources can be captured to the printer disk.
[Extend the logical page (millipoints)]	When you want to print with a slightly larger page size, you can increase the logical page size with the entered number of millipoints (unit of angle measurement).
[Output recovery] [Suppressed Mode]	Adjust the error recovery method to keep the printed output according to the original source. First select [Suppressed], and then use the [Suppressed Mode] mode to select [Secure]. • [Normal]: PRISMAsync Print Server recovers the print process from the point the error occurred. This may result in double prints. • [Suppressed], [Normal]: the host recovers the print process from the point the error occurred. This may result in double prints. • [Suppressed Mode], [Secure]: the host recovers errors and prevents duplicates. Intermediate check prints are disabled. The printer reports to the host which sheets are involved in errors.
[Face orientation]	Select the delivery orientation in the output tray: [Face up] or [Face down].
[Z-fold mode]	When a mixed-size set must both be Z-folded and stapled.
[Sensing unit adjustments] [Modes for automatic adjustment]	The sensing unit can automatically perform media registration and density adjustment via three ways. Use this setting to select the media registration, density or both for automatic adjustment in the sensing unit. The sensing unit can automatically perform media registration and density adjustment via two modes during printing. Use this setting to select one of the two modes. In continuous mode, every sheet in the print job is used for the media registration and density adjustment. In interrupt mode, a print job is interrupted to print a chart which is used for the media registration and density adjustment.

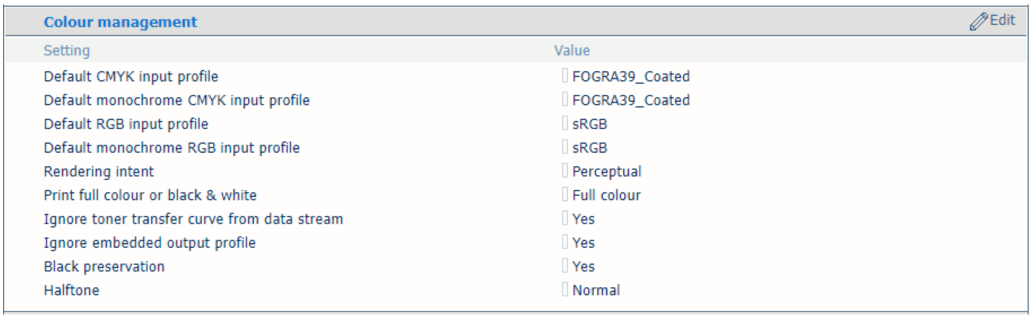
Define the default image shift settings



[IPDS image shift] settings

Image shift settings	Description
[Image shift in feed direction of side 1] [Image shift in cross-feed direction of side 1]	Use these settings to define the image shift. When you use pre-printed media with marked areas for specific text, misalignments can occur. Also use this setting to better align variable data on pages, such as names or addresses. There are settings for the feed direction and the cross-feed direction in combination with the sheet side. You can also adjust the image shift on the control panel.

Define the color management settings

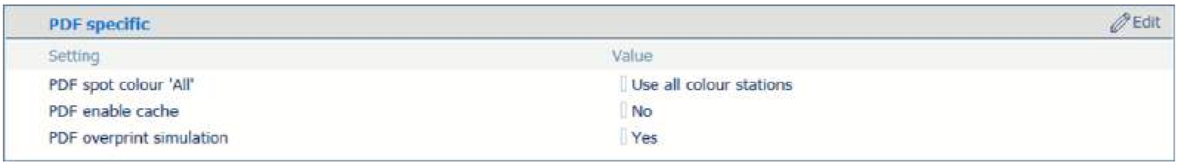


[Color management] settings

Color management settings	Description
[Default CMYK input profile], [Default monochrome CMYK input profile], [Default RGB input profile], [Default monochrome RGB input profile]	Select a default input profile for the four types of input profiles.
[Rendering intent]	Select the rendering intent.
[Print full color or black & white]	Use this setting to define how jobs are printed: in black & white or in color.
[Ignore toner transfer curve from data stream]	The printer can use or ignore toner transfer curves that are in the data stream.
[Ignore embedded output profile]	Use this setting to ignore the RGB or CMYK embedded output profiles.

Color management settings	Description
[Black preservation]	Use the setting to apply pure black preservation when possible. Pure black preservation means that the color black is composed of 100% K ink. When pure black preservation is not possible or disabled, the color black is composed of a mixture of two or more C, M, Y, and K inks.
[Halftone]	Indicate the halftone.

Define the PDF settings



PDF settings

PDF attributes	Description
[PDF spot color 'All']	PDF data have a number of pre-defined spot color names. The color name 'All' means that the RIP uses a 100% coverage of all the available colors. The color 'All' is intended for alignment marks. You can force the RIP to only use 100% black. Note that if the printer supports a separate MICR station, the color name 'All' implies that MICR pixels are used as well.
[PDF enable cache]	Indicate if you want to use the PDF cache to automatically identify and store objects that are used more than once in the job.
[PDF overprint simulation]	Indicate if simulation of overprinting of all colors must be applied. If this setting is disabled, the colors on top will knock out all underlying colors.

Define the tray-to-tray mapping

You link logical trays (indicated by a number from 1 to 15) to physical paper trays. The printer uses media that are assigned to these physical paper trays. The schedule shows the media the job uses.

IPDS tray-to-tray mapping	Edit
Setting	Value
Logical tray for paper module 1, tray 1	1
Logical tray for paper module 1, tray 2	2
Logical tray for paper module 1, tray 3	3
Logical tray for paper module 1, tray 4	4
Logical tray for paper module 2, tray 1	5
Logical tray for paper module 2, tray 2	6
Logical tray for paper module 2, tray 3	7
Logical tray for paper module 2, tray 4	8
Logical tray for paper module 3, tray 1	9
Logical tray for paper module 3, tray 2	10
Logical tray for paper module 3, tray 3	11
Logical tray for paper module 3, tray 4	12
Logical tray for roll feeder	13

[IPDS tray-to-tray mapping] settings

Define the logical tray values for all physical paper trays the jobs use.

Define the tray-to-media mapping

You link logical trays (indicated by a number from 1 to 36) to media of the media catalog. The printer maps the logical tray to the physical paper trays that hold these media. The schedule shows the media the job uses.

IPDS tray to media for tray 2	Edit
Setting	Value
Media catalogue item	Medium2, A4, 100g/m², TopColor
Media name	Medium2
Media type	TopColor
Media size	A4
Media width	210 mm
Media height	297 mm
Media weight	100 g/m²
Media colour	
Punch count	0
Insert	No
Feed direction	Long-edge feed (LEF)

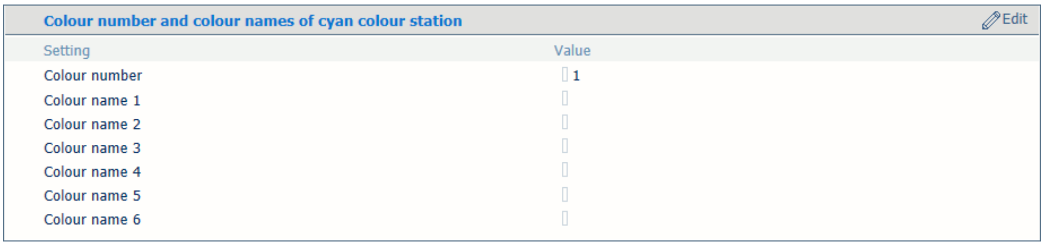
[IPDS tray to media for tray 2]

1. Use the [Media catalog item] setting to select media from the media catalog.
2. Use the [Feed direction] setting to select the feed direction of the media.

Define the paper tray linking

Input trays that contain the same media as the tray defined for the job are automatically linked. This means that when one of the paper trays is empty, printing continues from another paper tray that holds the same media.

Define the color stations



Colour number and colour names of cyan colour station Edit

Setting	Value
Colour number	1
Colour name 1	
Colour name 2	
Colour name 3	
Colour name 4	
Colour name 5	
Colour name 6	

[Color number and color names of cyan color station]

You can define custom names or numbers for the color stations.

Color station definition	Description
[Color number]	When you want to select color stations by using a number, enter a unique value (1-254). The Black color station has the value of zero and you cannot change this number. The other default color numbers are: • Cyan, 1 • Magenta, 2 • Yellow, 3
[Color name 1]	Names are encoded by UTF-16 Unicode. For every color station 6 names can be specified, to enable a color selection by a name. For the cyan, magenta, and yellow color stations, the default lists of color names are empty. The literally spelled names 'Gray', 'Cyan', 'Magenta ', 'Yellow ', 'Black ', 'All ', 'None' are reserved for PDF and cannot be used.

More information

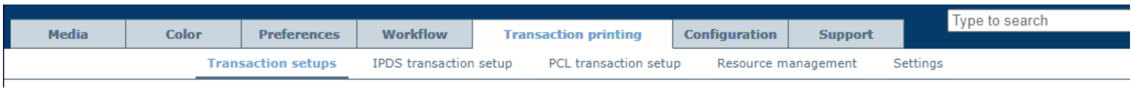
- *Learn about transaction setups* on page 207
- *Open a transaction setup to change settings* on page 213
- *Validate a transaction setup* on page 226
- *Load a transaction setup* on page 211
- *Define the resources* on page 231
- *Define an input profile* on page 448
- *Overrule the set image shift for transaction print jobs* on page 234

Define the settings of a PCL transaction setup

Define the transaction setup settings

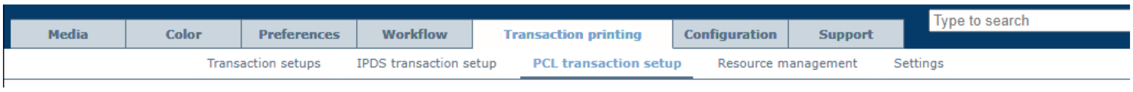
A transaction setup with the status [Opened] can be adjusted.

1. Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



[Transaction setups] tab

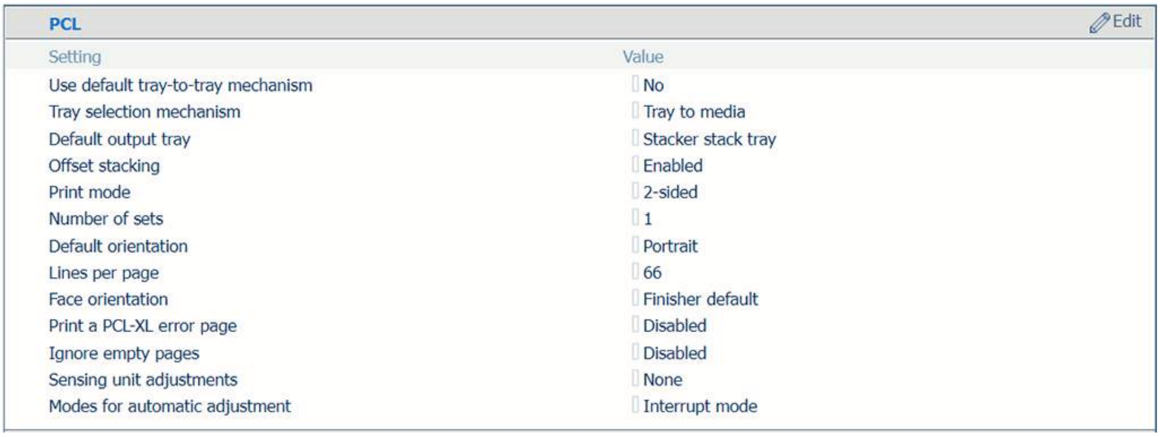
2. Check that the transaction setup you want to change has the status: [Opened].
3. Click the [PCL transaction setup] tab.



[PCL transaction setup] tab

4. Change the settings one by one or click [Edit] in the title bar to adjust the settings of a group in one dialog box.
5. Validate the transaction setup.
6. Make sure that you reload a transaction setup that is active when you made the changes.

Define the PCL settings

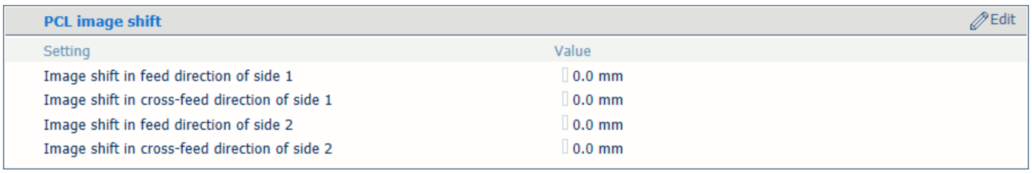


PCL settings

[PCL] settings	Description
[Use default tray-to-tray mechanism]	In [Tray to tray] mode, a default tray mapping table can be used. You cannot change the contents of this table.
[Tray selection mechanism]	A transaction printing job always uses the media definitions as specified in Settings Editor. Select [Tray to tray] or [Tray to media]. The PCL data stream contains a reference to a logical tray. There are 36 logical trays that can be assigned to a physical paper tray ([Tray to tray]) or to media of the media catalog ([Tray to media]).

[PCL] settings	Description
[Default output tray]	Select the default output tray.
[Offset stacking]	Select if stacking occurs with or without an offset.
[Print mode]	Indicate if the printer prints the jobs one-sided, two-sided or two-sided with a rotated back side ([Tumble]).
[Number of sets]	Enter the number of sets.
[Default orientation]	Select the document orientation.
[Lines per page]	Enter the number of lines on a page. More lines result in smaller line heights. The number of lines per page depends on the document orientation and the paper size.
[Face orientation]	Select the delivery orientation in the output tray: [Face up] or [Face down].
[Print a PCL-XL error page]	Indicate the printer must print an error page when a PCL-XL error occurs.
[Ignore empty pages]	Indicate if the printer must ignore empty pages.
[Sensing unit adjustments] [Modes for automatic adjustment]	The sensing unit can automatically perform media registration and density adjustment via three ways. Use this setting to select the media registration, density or both for automatic adjustment in the sensing unit. The sensing unit can automatically perform media registration and density adjustment via two modes during printing. Use this setting to select one of the two modes. In continuous mode, every sheet in the print job is used for the media registration and density adjustment. In interrupt mode, a print job is interrupted to print a chart which is used for the media registration and density adjustment.

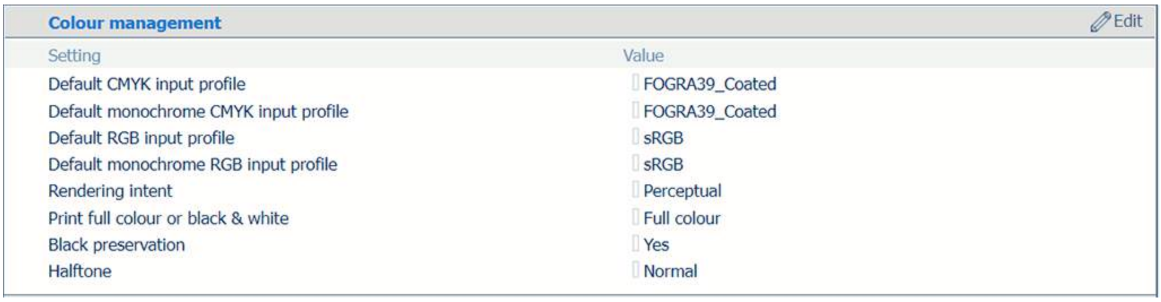
Define the [PCL image shift] settings



[PCL image shift] settings

[PCL image shift] settings	Description
[Image shift in feed direction of side 1] [Image shift in cross-feed direction of side 1] [Image shift in feed direction of side 2] [Image shift in cross-feed direction of side 2]	Use these settings to define the image shift. When you use pre-printed media with marked areas for specific text, misalignments can occur. Also use this setting to better align variable data on pages, such as names or addresses. There are settings for the feed direction and the cross-feed direction in combination with the sheet side. You can also adjust the image shift on the control panel.

Define the color management settings



Color management settings

Color management settings	Description
[Default CMYK input profile], [Default monochrome CMYK input profile], [Default RGB input profile], [Default monochrome RGB input profile]	Select a default input profile for the four types of input profiles.
[Rendering intent]	Select the rendering intent.
[Print full color or black & white]	Use this setting to define how jobs are printed: in black & white or in color.
[Black preservation]	Use the setting to apply pure black preservation when possible. Pure black preservation means that the color black is composed of 100% K ink. When pure black preservation is not possible or disabled, the color black is composed of a mixture of two or more C, M, Y, and K inks.
[Halftone]	Indicate the halftone.

Define the tray-to-tray mapping

Use the [Default media size as in tray] setting to define which paper tray uses the default media size. Therefore, you select one of the available paper trays. You link logical trays (indicated by a number from 1 to 16) to physical paper trays. The printer uses media that are assigned to these physical paper trays. The schedule shows the media the job uses.

Define the logical tray values for all physical paper trays the jobs use.

PCL tray-to-tray mapping	
Setting	Value
Default media size as in tray	Paper module 1, tray 1
Escape sequence for paper module 1, tray 1	1
Escape sequence for paper module 1, tray 2	2
Escape sequence for paper module 1, tray 3	4
Escape sequence for paper module 1, tray 4	8
Escape sequence for paper module 2, tray 1	20
Escape sequence for paper module 2, tray 2	21
Escape sequence for paper module 2, tray 3	22
Escape sequence for paper module 2, tray 4	23
Escape sequence for paper module 3, tray 1	24
Escape sequence for paper module 3, tray 2	25
Escape sequence for paper module 3, tray 3	26
Escape sequence for paper module 3, tray 4	27
Escape sequence for roll feeder	99

Tray selection for tray-to-tray mode

Define the tray-to-media mapping

You link logical trays (indicated by a number from 1 to 36) to media of the media catalog. The printer maps the logical tray to the physical paper trays that hold these media. The schedule shows the media the job uses.

PCL tray to media for media 1	
Setting	Value
Escape sequence	1
Media catalogue item	
Media name	Medium1
Media type	TopColor
Media size	A4
Media width	210 mm
Media height	297 mm
Media weight	100 g/m²
Media colour	
Punch count	0
Insert	No
Feed direction	Long-edge feed (LEF)
Media print mode	Use PCL Default

Tray-to-media mapping

1. Use the [Default media size as in tray] setting to select which media defines the media size.

PCL tray to media	
Setting	Value
Default media size as media	Media 1

[PCL tray to media]

2. Use the [Media catalog item] setting to select media from the media catalog.
3. Use the [Feed direction] setting to select the feed direction of the media.
4. Use the [Media print mode] setting to select the media print mode of the selected media. The option [Use PCL Default] refers to the [Media print mode] setting in the list of PCL settings.

Define the paper tray linking

Input trays that contain the same media as the tray defined for the job are automatically linked. This means that when one of the paper tray is empty, printing continues from another paper tray that holds the same media.

More information

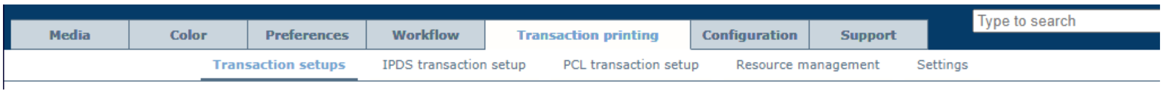
- *Learn about transaction setups* on page 207
- *Open a transaction setup to change settings* on page 213
- *Validate a transaction setup* on page 226
- *Load a transaction setup* on page 211
- *Overrule the set image shift for transaction print jobs* on page 234

Import, export, or restore transcation setups

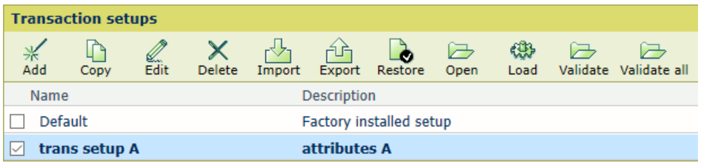
You can import, export, and restore transaction setups.

Go to the transaction setups

Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



[Transaction setups] tab



[Transaction setups] menu

Import transaction setups

1. Click [Import].
2. Browse to the XML file that contains the transaction setups.
3. Select [Replace] to replace the current transaction setups. Select [Merge] to add the transaction setups to the current transaction setups.
4. Click [OK].
5. Validate the imported transaction setups.

Export transaction setups

1. Click [Export].
2. Click [OK].
   The current transaction setup definitions are saved in an XML file.

Restore the default transaction setup

IMPORTANT
Be aware that this function removes all custom transaction setups.

1. Click [Restore].
2. Click [OK].

More information

Validate a transaction setup on page 226

Validate a transaction setup

What is transaction setup validation

When you have defined, changed, or imported a transaction setup, you can validate the transaction setup to check its media definitions. The validation procedure is performed for the active PDL.

When you use the [Validate] function, the printer checks for each tray if the media definition is valid. It depends on the tray selection mode ([Tray to media] or [Tray to tray]) how the media elements are checked.

Validation in [Tray to media] mode

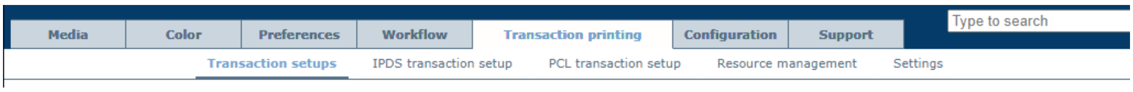
The printer checks the following:
- The media set per tray are available in the media catalog.
- The media print mode set per tray is defined for the media family of the media.

Validation in [Tray to tray] mode

The printer checks the following:
- The media families of the assigned media in the paper trays must have the IPDS media print mode.

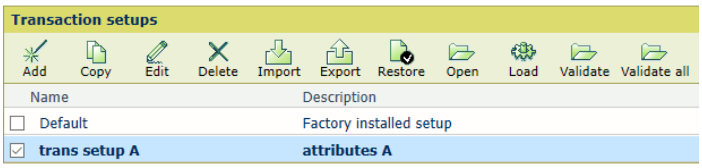
Validate a transaction setup

1. Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



[Transaction setups] tab

2. Click [Validate].



[Transaction setups] menu

The transaction setup is opened so you can adjust its settings.

3. The report appears.
4. Read the report.
5. Click [OK] to close the report.

More information

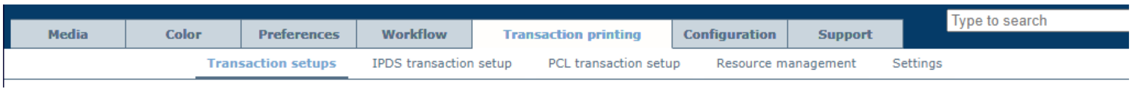
- *Configure the media* on page 420
- *Assign media to a paper tray* on page 127
- *Read the transaction setup validation report* on page 228

Validate all transaction setups

You can validate all available transaction setups. The validation procedure is performed for the active PDL. The report displays the number of valid and invalid trays per transaction setup. When the report shows invalid transaction setups, use the [Validate] function to examine the problems per tray.

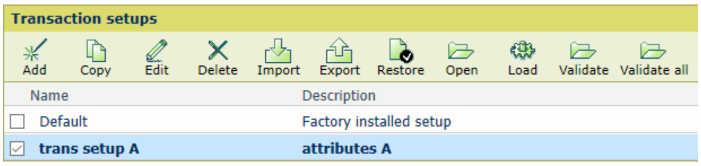
The report displays for every transaction setup the tray selection mechanism ([Tray to media] or [Tray to tray]) and the number of valid and invalid trays. For transaction setups with problems, use the [Validate] function to get more information.

1. Open the Settings Editor and go to: [Transaction printing]→[Transaction setups].



[Transaction setups] tab

2. Click [Validate all].



[Validate all] button

3. The report appears.
4. Read the report.
5. Click [OK] to close the report.

More information

- *Validate a transaction setup* on page 226
- *Read the transaction setup validation report* on page 228

Read the transaction setup validation report

Example: Validation in [Tray to media] mode for IPDS

A transaction setup has the following settings.

IPDS

Setting	Value
Tray selection mechanism	Tray to media
Data resolution	Auto
Default output tray	Stacker stack tray
Offset stacking	Enabled
Font capture	Disabled
Extend the logical page (millipoints)	0
MICR mode (if licensed)	Disabled
Output recovery	Normal
Suppressed Mode	Secure
Face orientation	Finisher default
Media print mode	LowCost

IPDS settings

IPDS tray to media for tray 1

Setting	Value
Media catalogue item	
Media name	RedLabelSuperior80CanonG1000
Media type	IV2T150CG00PG1000
Media size	A4
Media width	210.0 mm
Media height	297.0 mm
Media weight	80 g/m²
Media colour	White
Punch count	0
Insert	No
Feed direction	Long-edge feed (LEF)
Media print mode	Use IPDS Default

Tray 1 media settings

IPDS tray to media for tray 2

Setting	Value
Media catalogue item	
Media name	RedLabelSuperior80CanonG800
Media type	IV2T150CG00PG0800
Media size	A4
Media width	210.0 mm
Media height	297.0 mm
Media weight	80 g/m²
Media colour	White
Punch count	0
Insert	No
Feed direction	Long-edge feed (LEF)
Media print mode	Use IPDS Default

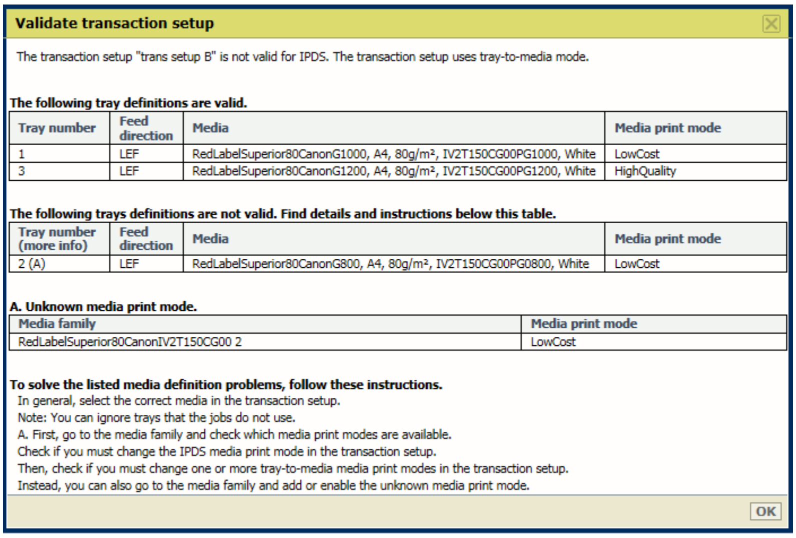
Tray 2 media settings

IPDS tray to media for tray 3

Setting	Value
Media catalogue item	
Media name	RedLabelSuperior80CanonG1200
Media type	IV2T150CG00PG1200
Media size	A4
Media width	210.0 mm
Media height	297.0 mm
Media weight	80 g/m²
Media colour	White
Punch count	0
Insert	No
Feed direction	Long-edge feed (LEF)
Media print mode	HighQuality

Tray 3 media settings

The validation report found that the transaction setup is not valid.



Validate transaction setup

The transaction setup "trans setup B" is not valid for IPDS. The transaction setup uses tray-to-media mode.

The following tray definitions are valid.

Tray number	Feed direction	Media	Media print mode
1	LEF	RedLabelSuperior80CanonG1000, A4, 80g/m², IV2T150CG00PG1000, White	LowCost
3	LEF	RedLabelSuperior80CanonG1200, A4, 80g/m², IV2T150CG00PG1200, White	HighQuality

The following trays definitions are not valid. Find details and instructions below this table.

Tray number (more info)	Feed direction	Media	Media print mode
2 (A)	LEF	RedLabelSuperior80CanonG800, A4, 80g/m², IV2T150CG00PG0800, White	LowCost

A. Unknown media print mode.

Media family	Media print mode
RedLabelSuperior80CanonIV2T150CG00 2	LowCost

To solve the listed media definition problems, follow these instructions.
In general, select the correct media in the transaction setup.
Note: You can ignore trays that the jobs do not use.
A. First, go to the media family and check which media print modes are available.
Check if you must change the IPDS media print mode in the transaction setup.
Then, check if you must change one or more tray-to-media media print modes in the transaction setup.
Instead, you can also go to the media family and add or enable the unknown media print mode.

OK

Example of validation report in tray-to-media mode

Valid tray: 1 and 3

- Tray 1. The media family of the media (*RedLabelSuperior80CanonG1000*) has the IPDS media print mode: *LowCost*.
- Tray 3. The media family of the media (*RedLabelSuperior80CanonG1200*) has the media print mode: *HighQuality*.

Invalid trays: 2

- Tray 2. The media family of the media (*RedLabelSuperior80CanonG1200*) does not have the IPDS media print mode: *LowCost*.

Example: Validation in [Tray to tray] mode for IPDS

The transaction setup has the following settings.

IPDS Edit

Setting	Value
Tray selection mechanism	Tray to media
Data resolution	Auto
Print resolution	600 dpi
Default output tray	Stacker stack tray
Offset stacking	Enabled
Font capture	Disabled
Extend the logical page (millipoints)	0
MICR mode (if licensed)	Disabled
Output recovery	Normal
Suppressed Mode	Secure
Face orientation	Finisher default
Media print mode	Use Media Family Default

IPDS settings

The validation report found that the transaction setup is not valid.

Validate transaction setup

The transaction setup "trans setup B" is not valid for IPDS. The transaction setup uses tray-to-tray mode.

The following tray definitions are valid.

Tray number	Feed direction	Media	Media print mode
1	LEF	RedLabelSuperior80CanonG1000, A4, 80g/m², IV2T150CG00PG1000, White	LowCost

The following trays definitions are not valid. Find details and instructions below this table.

Tray number (more info)	Feed direction	Media	Media print mode
2 (A)	LEF	RedLabelSuperior80CanonG800, A4, 80g/m², IV2T150CG00PG0800, White	LowCost
3 (A)	LEF	RedLabelSuperior80CanonG1200, A4, 80g/m², IV2T150CG00PG1200, White	LowCost

A. Unknown media print mode.

Media family	Media print mode
RedLabelSuperior80CanonIV2T150CG00 2	LowCost
RedLabelSuperior80CanonIV2T150CG00 3	LowCost

To solve the listed media definition problems, follow these instructions.
In general, go to the control panel and assign the correct media to the required paper trays.
Note: You can ignore paper trays that the jobs do not use.
A. First, go to the media family and check which media print modes are available.
Then, change the IPDS media print mode in the transaction setup.
Instead, you can also go to the media family and add or enable the unknown media print mode.

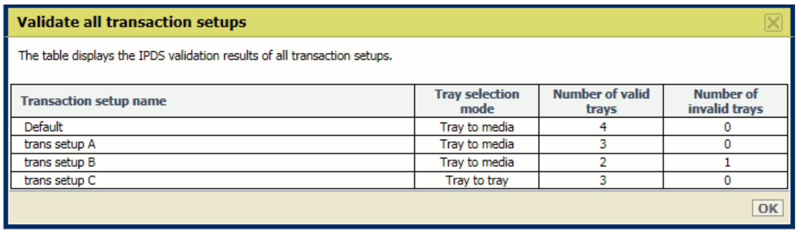
OK

Example of validation report in tray-to-tray mode

Valid paper tray: 1
- The media family of the media (RedLabelSuperior80CanonG1000) assigned to paper tray 1 has the IPDS media print mode: *LowCost*.

Invalid paper trays: 2 and 3
- The media family of the media (RedLabelSuperior80CanonG800) assigned to paper tray 2 does not have the IPDS media print mode: *LowCost*.
- The media family of the media (RedLabelSuperior80CanonG1200) assigned to paper tray 3 does not have the IPDS media print mode: *LowCost*.

Example: Validation report of all transaction setups for IPDS



Transaction setup name	Tray selection mode	Number of valid trays	Number of invalid trays
Default	Tray to media	4	0
trans setup A	Tray to media	3	0
trans setup B	Tray to media	2	1
trans setup C	Tray to tray	3	0

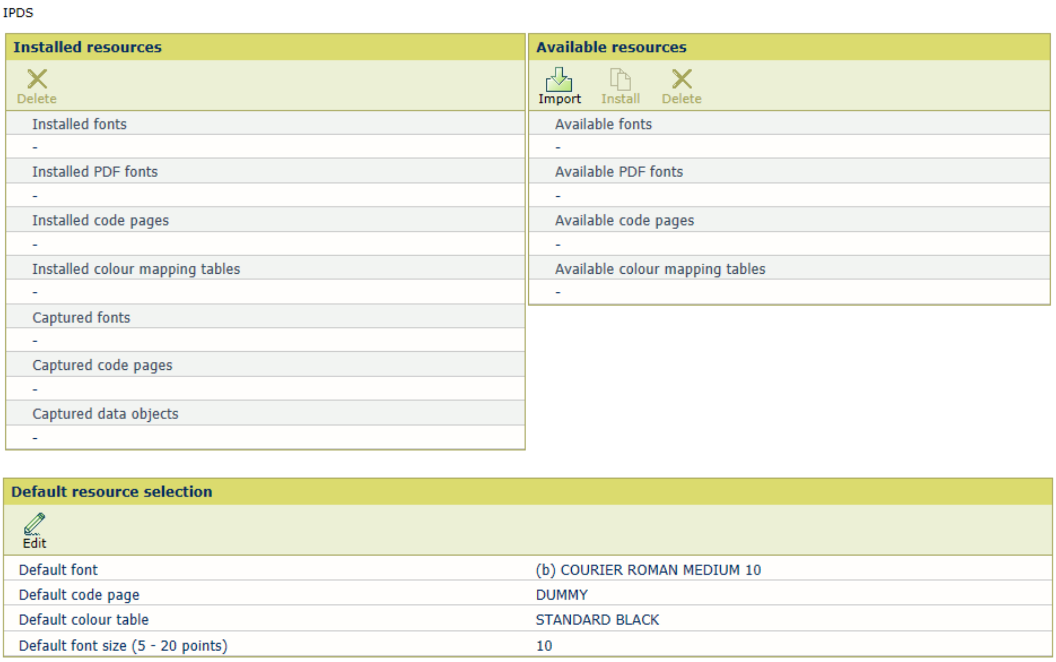
Example of validation report for all transaction setups

Define the resources

Type of resources

There are several types of resources that you can manage and use for transaction printing:
1. Fonts
2. Code pages
3. Color mapping tables
4. Data objects, such as font files, image types or color management resources

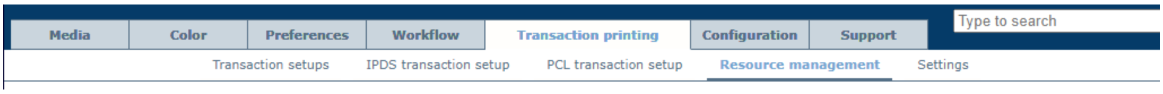
The imagePRESS V1000 uses the following types of resources.
1. **Captured resources**: The IPDS host can download a resource via the IPDS data stream if a required resource is not available on the printer. To increase the performance, the downloaded resources can be captured on the printer disk for later reuse. After capturing, the IPDS host gets informed about available captured resources but still must activate the captured resources. Captured resources are visible in the Settings Editor. Only when the [Font capture] setting in the loaded transaction setup is enabled the capturing of resources can take place. This setting also controls whether captured or installed resources are selectable.
2. **Imported resources**: Resources can be imported one by one or in a batch. Imported resources are visible in the Settings Editor.
3. **Installed resources**: Imported resources can be installed. The installed resources are visible in the Settings Editor.
4. **Permanent resources**: Permanent resources are factory installed and are not visible in the Settings Editor.



Installed resources

Go to transaction printing resource management

Open the Settings Editor and go to: [Transaction printing]→[Resource management].



[Resource management] tab

Resource management

Use the table below to see what can be done for each type of resource.

Tasks	Fonts	Fonts for embedded PDF	Code pages	Color mapping table	Data objects
Import resource file **	Yes	Yes	Yes	Yes	No
Install imported resource **	Yes	Yes	Yes	Yes	No
Delete imported resource **	Yes	Yes	Yes	Yes	Yes
Delete installed resource **	Yes	Yes	Yes	Yes	Yes
Delete permanent resource **	No	No	No	No	No
Delete captured resource **	Yes	No	Yes	No	Yes
Capture resources ***	Yes*	No	Yes*	No	Yes
Set default resources **	Yes	No	Yes	Yes	No

* A captured bitmap font is related to a specific captured code page. When one of these captured resources is deleted, the other related captured resource is also deleted.

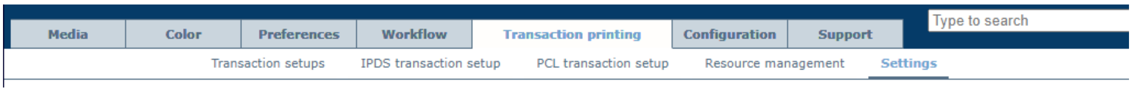
** You can only perform this task when transaction printing is offline.

*** Capturing resources can only take place when transaction printing is online.

Define the general transaction printing settings

You can set a number of transaction printing settings.

1. Open the Settings Editor and go to: [Transaction printing]→[Settings].



[Settings] tab

2. Go to the [General] and [Configuration] sections.

[General] and [Configuration] sections

3. Use the [Go online at startup] setting if you want the printer to stay online after a reboot or restart.

   NOTE
   If the printer is offline before a reboot or restart, the printer status will not change.

4. Use the [Remote diagnostics protocol] setting to enable the remote diagnostics protocol.
5. Use the [IPDS port] setting to set the IPDS port if applicable.
6. Use the [PCL port] setting to set the PCL port if applicable.
7. You can import or export SRA configuration files on request of your Service organisation. Click [Import] or [Export] in the [Configuration file (zip)] fields.

   IMPORTANT
   Only use these settings on request of your Service organization. Otherwise, you can damage the printer.

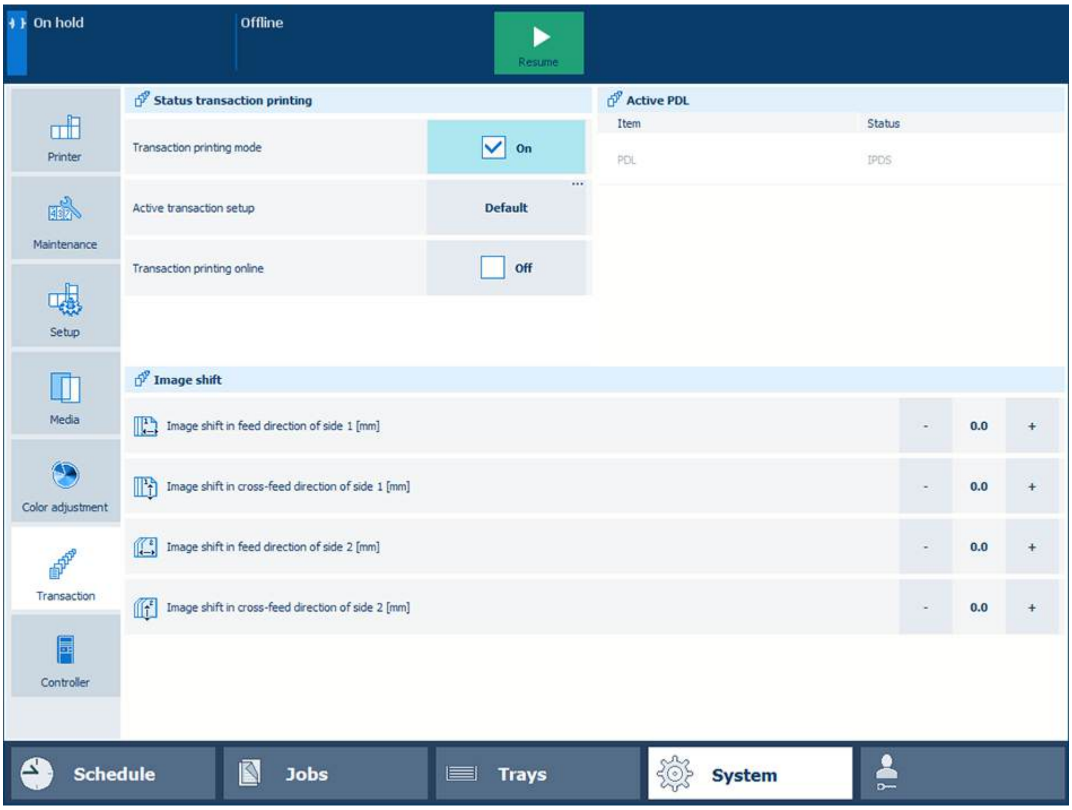
More information

- *Switch the Page Description Language (PDL)* on page 204
- *Activate the transaction printing mode* on page 202
- *Learn about transaction setups* on page 207

Overrule the set image shift for transaction print jobs

The transaction setup has settings to define the image shift. From the control panel you can overrule the image shift definition of a transaction setup. This is useful when you notice that variable data, such as names or addresses, must be better aligned on the page.

Be aware that the image shift adjustments you do on the control panel are lost when you leave the transaction printing mode.



Change image shift of transaction print jobs on control panel

1. Touch [System]→[Transaction].
2. Bring transaction printing offline.
   The [Image shift] section shows the image shift as defined in the active transaction setup.
3. Make the changes in the [Image shift] section.
4. Bring transaction printing online.

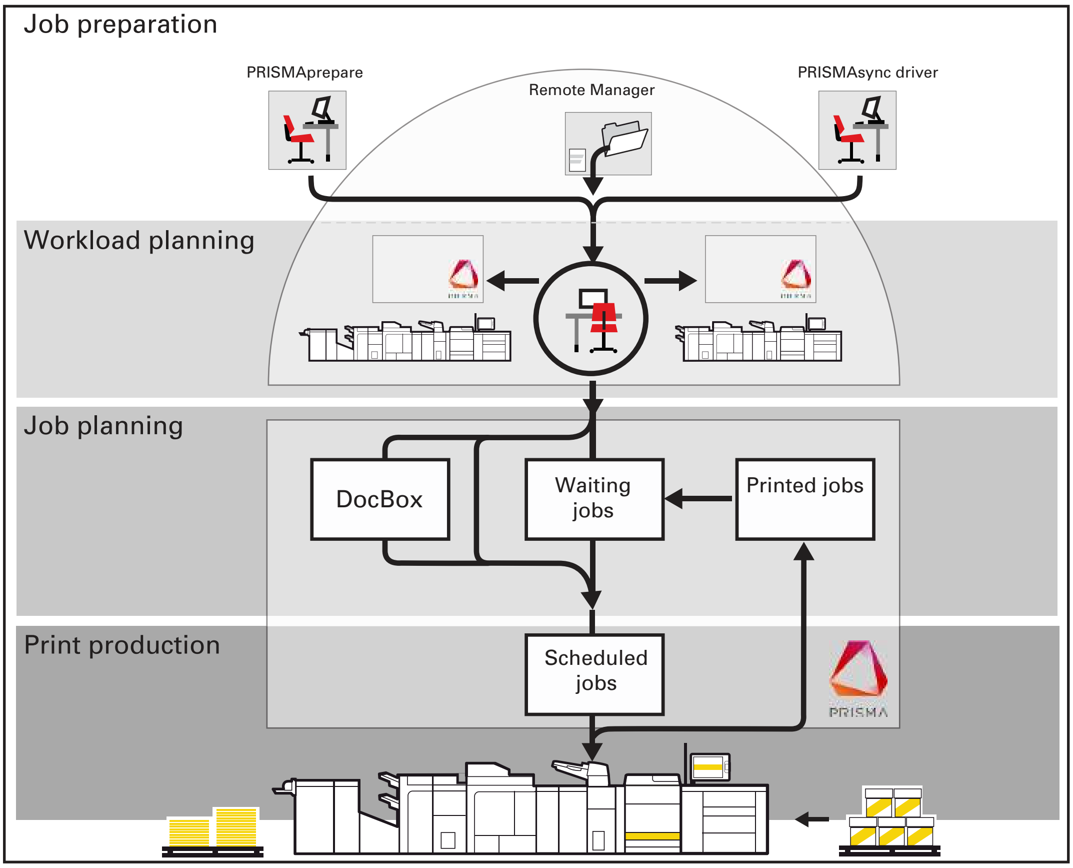
More information

- *Define the settings of an IPDS transaction setup* on page 214

Chapter 9
Document printing

Learn about the document printing workflow

The following illustration shows the document print job workflow from the job preparation to the delivery of the prints.



Job workflow for document printing mode

1. **Job preparation**
   You can use PRISMAprepare, the remote printer driver, LPR, or hotfolders to submit jobs. The PDF documents can be created with specific graphical or office applications.
2. **Workload planning**
   PRISMAremote Manager / PRISMAsync Remote Manager Classic is used to monitor the connected print systems, paper trays, scheduled jobs, and upcoming actions to manage the workload across all connected print systems. With the PRISMAremote Monitoring app and PRISMAremote Manager / PRISMAsync Remote Manager Classic you monitor the print production of the PRISMAsync Print Server printers remotely.
3. **Job planning**
   Print jobs are visible on the control panel: in the list of waiting jobs, the list of scheduled jobs or in a DocBox folder. The job properties determine the destination of the job. Job properties can be changed when the job arrives at the destination.
4. **Print production**
   The printer prints the jobs according to the job sequence in the list of scheduled jobs. Job media are loaded in the paper trays, prints and waste are removed in time and there are sufficient supplies of consumables in the printer.

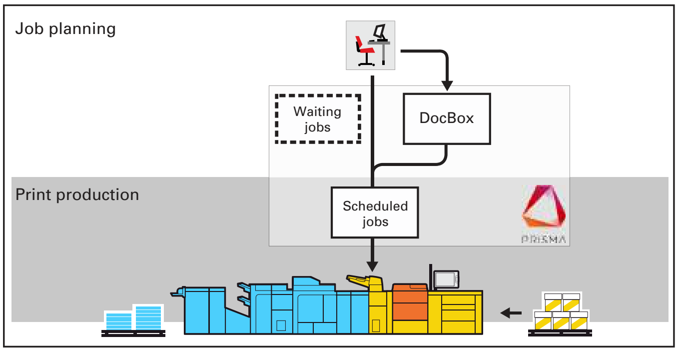
Learn about workflow profiles

The printer provides workflow profiles to manage your workload from the control panel.

A workflow profile determines the following:
- Destination of print jobs: the list of scheduled jobs or the list of waiting jobs.
- Destination of DocBox jobs that are printed: the list of scheduled jobs or the list of waiting jobs.
- Confirm the start of job.
- Job printing: only first set or all sets.
- Delivery of the prints in the trays.
- Offset stacking and the use of banner pages.

Below you find a description of the standard workflow profiles.

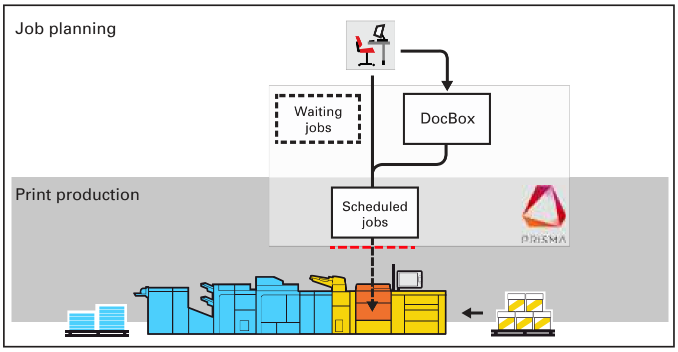
Standard workflow profile



Standard workflow profile

- The standard workflow profile fits a productive workflow with sufficient control over jobs.
- Jobs go to the list of scheduled jobs and the print system prints the jobs.
- The print system selects another output tray for each next job.

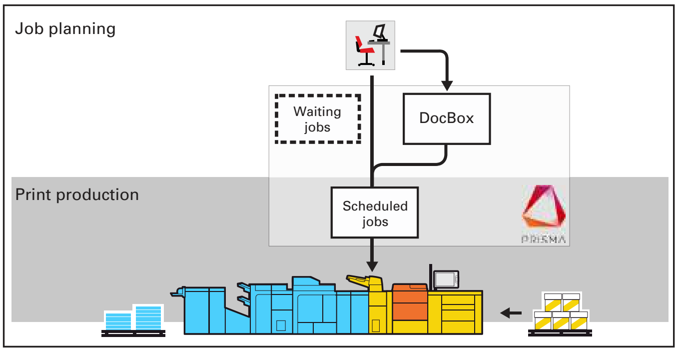
Job by job workflow profile



Job by job workflow profile

- The job-by-job workflow profile fits a workflow in which every job needs attention.
- All jobs come in the list of scheduled jobs and you start the jobs one by one from the list of scheduled jobs.
- The print system selects another output tray for each next job.

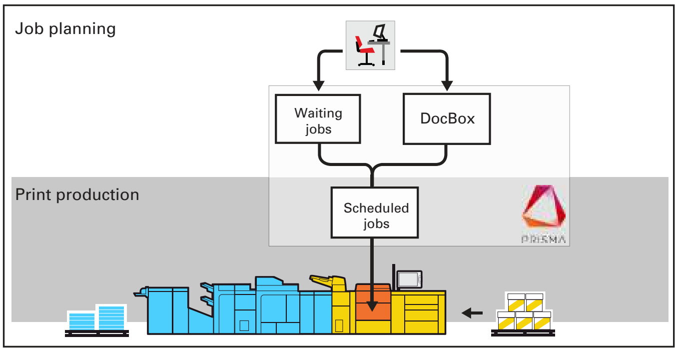
Check and print workflow profile



Check and print workflow profile

- The check-and-print workflow profile is suitable for a workflow in which every job requires attention. You check the print quality and layout setting of the first set.
- All jobs are received in the list of scheduled jobs and only the first set of the job is printed. After approval of the first set, you give the print command to print the other sets in one run.
- The print system selects another output tray for each next job.

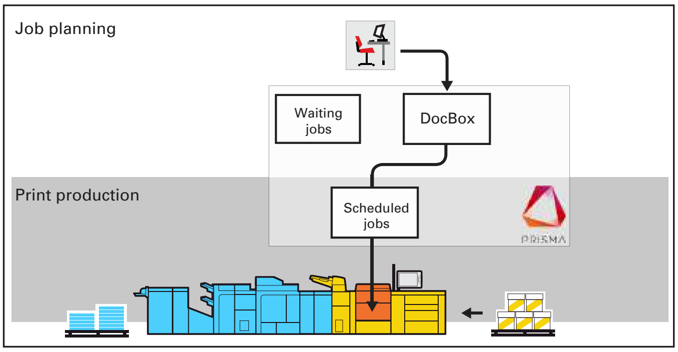
Manual planning workflow profile



Manual planning workflow profile

- The manual planning workflow fits a workflow in which you want to determine the print priority of the jobs.
- All jobs are received in the list of waiting jobs.
- The print system selects another output tray for each next job.

Unattended workflow profile



Unattended workflow profile

- The unattended workflow profile fits a workflow in which productivity is very important.
- All jobs go to the list of scheduled jobs and are printed.
- To keep the system running, you should make sure consumables remain available and you remove printed output and waste on time.

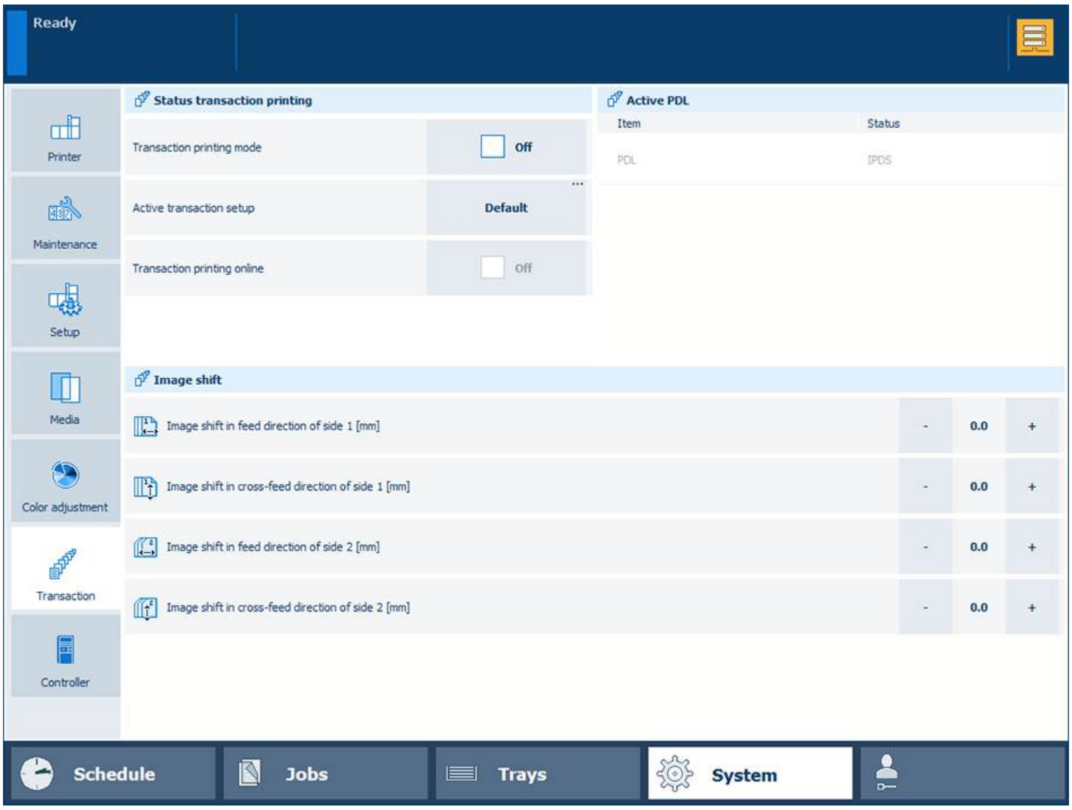
More information

Choose a workflow profile on page 240

Activate the document printing mode

In case the transaction printing mode is active and you want to go to the document printing mode, follow this instruction. As long as the document printing mode is active, the printer cannot accept transaction print jobs.

Document printing symbol on the dashboard of the control panel	



Activate document printing mode from control panel

1. Touch [System]→[Transaction].
2. Touch [Transaction printing online]→[Off].
3. Touch [Transaction printing mode]→[Off].

More information

- *Learn about the document printing workflow* on page 236
- *Learn about the transaction printing workflow* on page 200

Choose a workflow profile

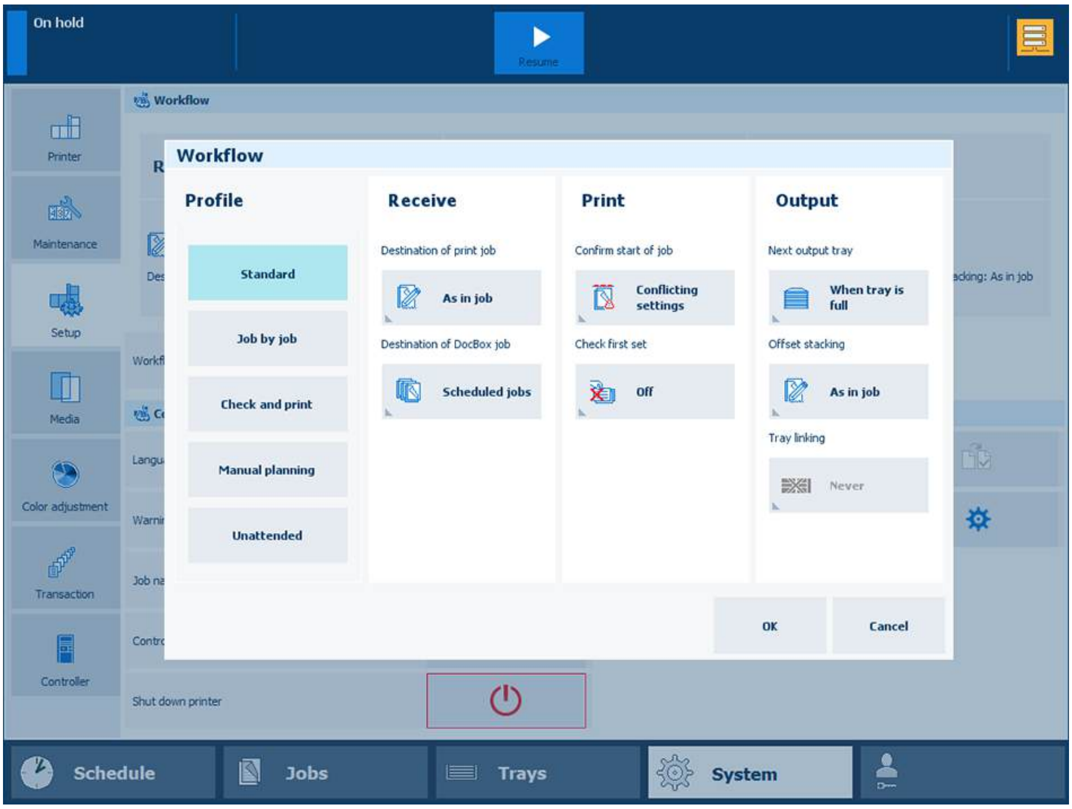
The printer provides workflow profiles to manage your workload from the control panel.

You select one of the available standard workflow profiles and, if required, change one or more attributes so it fits the print production requirements. Only one custom workflow profile can be active at the same time.

Go to the workflow profiles

1. Touch [System]→[Setup].
2. Touch the name of the [Workflow profile].

Use one of the available workflow profiles



Attributes of a standard workflow profile

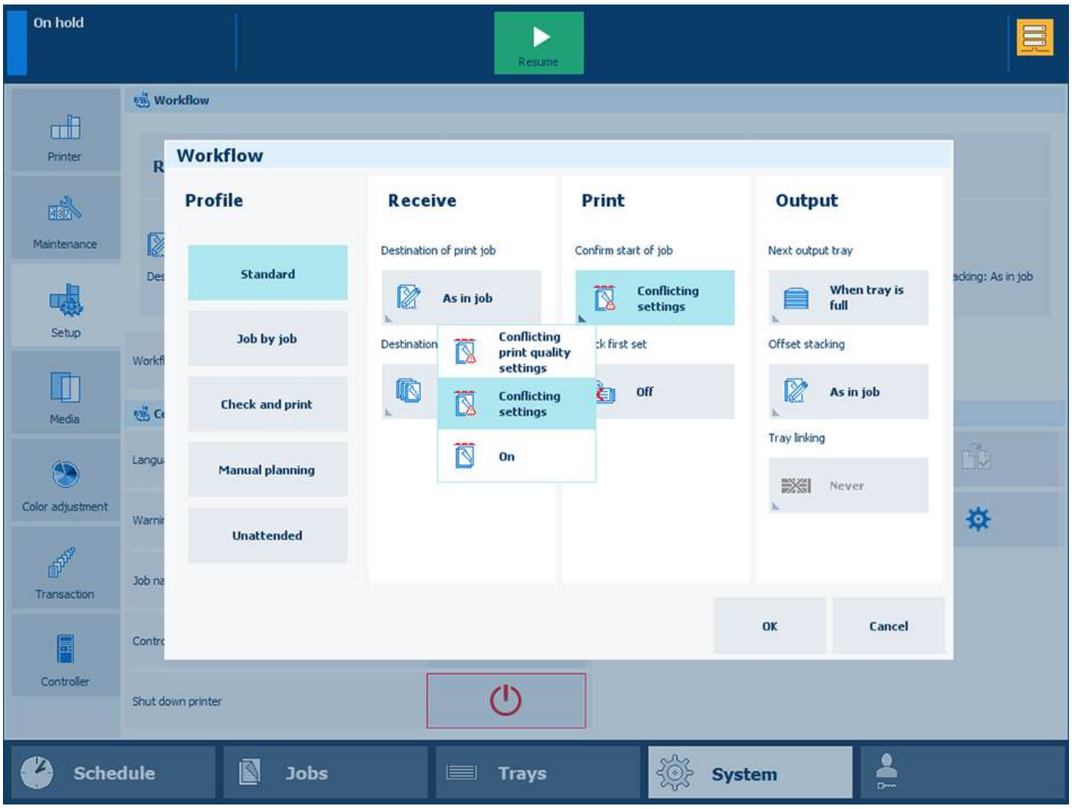
1. Touch one of the workflow profiles.
2. Touch [OK].

Adjust one of the standard workflow profiles

When you change an attribute of an automated workflow the name changes into [Custom].
1. Select the workflow profile.
2. Adjust one or more attributes.
3. Touch [OK].

Indicate when to confirm the start of a job

You can indicate when the printer shows a dialogue to confirm the start of a job.



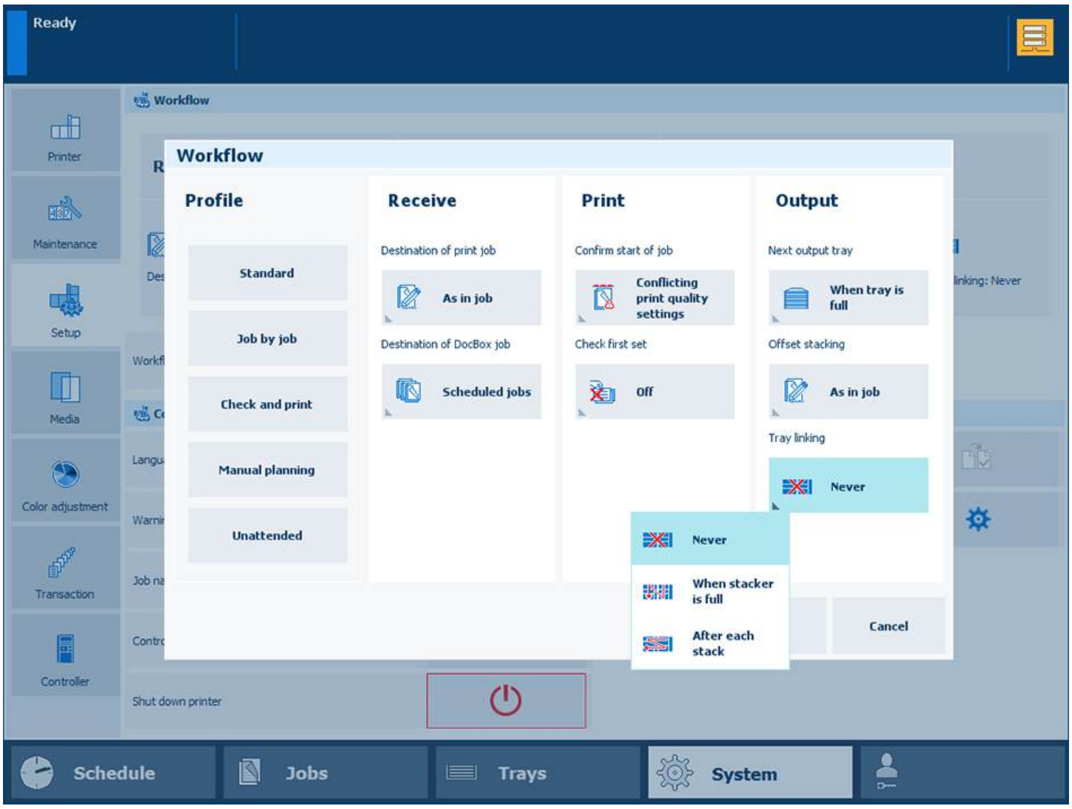
Confirm start of job

1. Select the workflow profile.
2. Use the [Confirm start of job] function to select the one of the following values.
   - [On]: Confirm that a job can start.
   - [Confirm start of job: In case of conflicting settings]: Confirm that a job can start with default settings in case the job has conflicting settings.
   - [Confirm start of job: In case of conflicting print quality settings]: Confirm that a job can start with default settings in case the job has conflicting print quality settings. The requested media print mode is unknown. The job will be printed with the default media print mode instead of the requested media print mode.
     You can decide to cancel the job and first create the required media print mode in the Settings Editor.

Define tray linking

You can enable or disable the linking of output trays, when there are two high capacity stackers.

Tray linking means that the prints delivery automatically switches to the other high capacity stacker.



Attributes of a workflow profile

1. Select the workflow profile.
2. Use the [Tray linking] function to indicate if tray linking can occur.
   - [Never]: there is no tray linking.
   - [When stacker is full]: tray linking occurs when the first high capacity stacker tray is full.
   - [After each stack]: tray linking occurs by switching to the other high capacity after the delivery of a stack.
3. Touch [OK].

More information

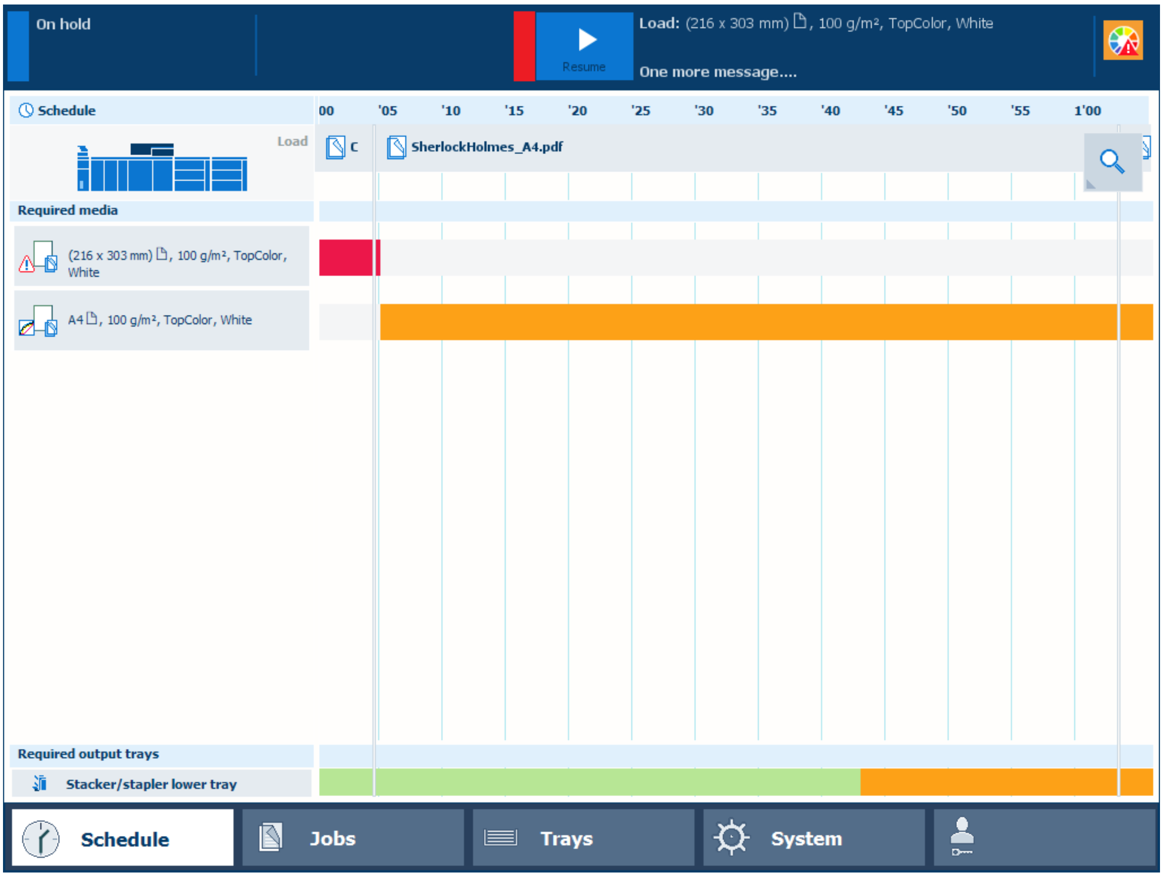
Learn about workflow profiles on page 237

Work with the schedule in document printing mode

When you work in document printing mode, the schedule predicts the job production time and offers a daily up to eight-hour plan board. The schedule provides all information of the scheduled jobs on a moving timeline, so you are able to intervene in time and keep your printer running.

Essential information about the current print process and upcoming actions are displayed in order to achieve maximum productivity. From the schedule, you simply start to load new media or media of almost empty trays.

The status color indicates the printer status and informs when you must prepare or perform an action. You can set the warning time according to your wishes.



Schedule in document printing mode

The screen with the schedule provides the following information in document printing mode:

1. The **timeline** shows the scheduled jobs and gives a prediction of the job completion times.
2. The **required media** pane shows the media of all scheduled jobs.
3. The [Load] area shows in which paper trays media you select are loaded.
   You can load media for the scheduled jobs. Therefore, select the media and touch [Load].
4. The **output trays** area shows which output trays the printer will use for the scheduled jobs.

	The vertical red-white bar indicates when a stack eject or full tray is expected.

5. The **zoom** option changes the scale of the timeline.
6. The **message area** of the dashboard shows the upcoming events and other information. The dashboard shows one message at a time. When there are more messages, the control panel

displays the first required or the most important message. To see all messages, touch the message area.

7. The **maintenance area** of the dashboard shows symbols of the upcoming or immediate maintenance actions.

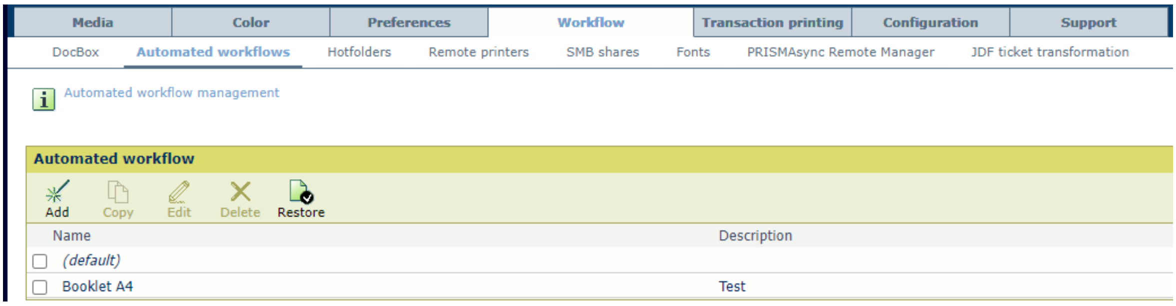
More information

- *Learn about the printer status* on page 61
- *Load media for the scheduled jobs* on page 129

Work with automated workflows

Learn about automated workflows

Automated workflows are managed in the Settings Editor. An automated workflow bundles a series of pre-set attributes to define job and workflow properties. You can address an automated workflow when you submit one or more PDF or PostScript jobs.



Automated workflows

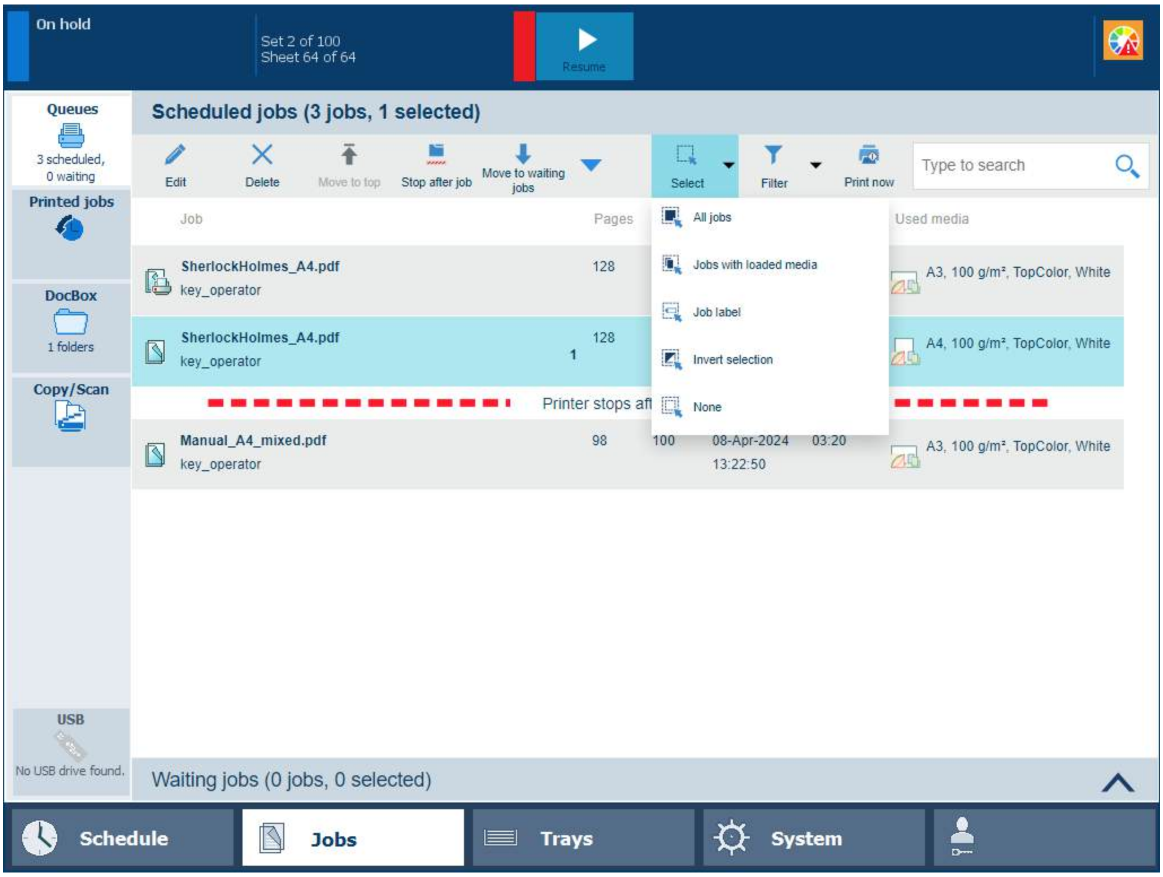
You can address an automated workflow at several locations.

- **Hotfolder**: The hotfolder definition has a link to one of the available automated workflows. PDF files submitted via a hotfolder shortcut on the desktop retrieve the properties of an automated workflow or the properties of a JDF ticket. The system administrator can create new hotfolders.

  NOTE
  You can use the ticket editor to create or modify a JDF ticket.

- **LPR printing**: PDF files can be submitted to the printer with the name of an automated workflow in the LPR command.
- **PRISMAsync Remote printer driver**: You can submit jobs with one of the available automated workflows.
- **PRISMAremote Manager / PRISMAsync Remote Manager Classic**: You can submit jobs with one of the available automated workflows.
- **Control panel**: When you use the [Apply automated workflow] option, you can apply an automated workflow. The job is re-submitted again.

The [Job label] field in the print queue refers to the name of the automated workflow. You can select or filter print jobs according to this job label.



Select jobs with a certain label

NOTE
When you have not selected one of the available automated workflows during the job submission, the printer automatically uses the *(default)* automated workflow.

Factory default automated workflow

The printer has one factory defined automated workflow: ***(default)***. The name of this automatic workflow is an empty string and cannot be changed. It is not possible to delete this automated workflow.

The printer uses this automated workflow when there is no other automated workflow selected, for example when the jobs properties are in the job ticket.

IMPORTANT
Do not change the attributes of the factory default automated workflow: ***(default)***. This automated workflow is used in case no other automated workflow is addressed and can change the job properties you defined at other locations.

More information

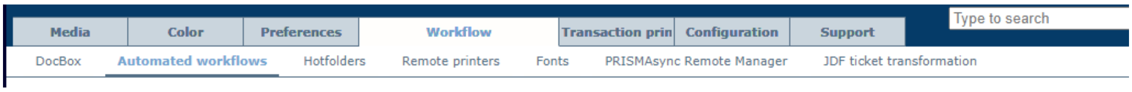
- *Define a new automated workflow* on page 247
- *Submit jobs via LPR* on page 257
- *Submit jobs via hotfolders* on page 258
- *Use the job ticket editor* on page 260
- *Edit with automated workflow in job destination* on page 263

Define a new automated workflow

An automated workflow bundles a series of pre-determined settings to define jobs.

Go to the automated workflows

1. Open the Settings Editor and go to: [Workflow]→[Automated workflows].



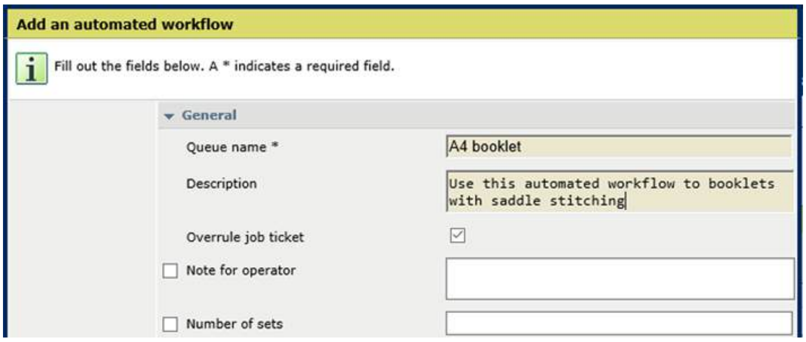
[Automated workflows] tab

Add a new automated workflow

1. Click [Add].

[Automated workflows] menu

2. Define the settings.



[Add an automated workflow]

3. Click [OK].

Copy an automated workflow

1. Select one of the available automated workflows.
2. Click [Copy].
3. Define the settings.
4. Click [OK].

Edit an automated workflow

You cannot rename the automated workflow: *(default)*, but you can change its settings.

IMPORTANT
Do not change the settings of the factory default automated workflow: ***(default)***. This automated workflow is used in case no other automated workflow is addressed and changes the job settings you defined at other locations.

1. Select one of the available automated workflows.
2. Click [Copy].
3. Define the settings.
4. Click [OK].

Delete an automated workflow

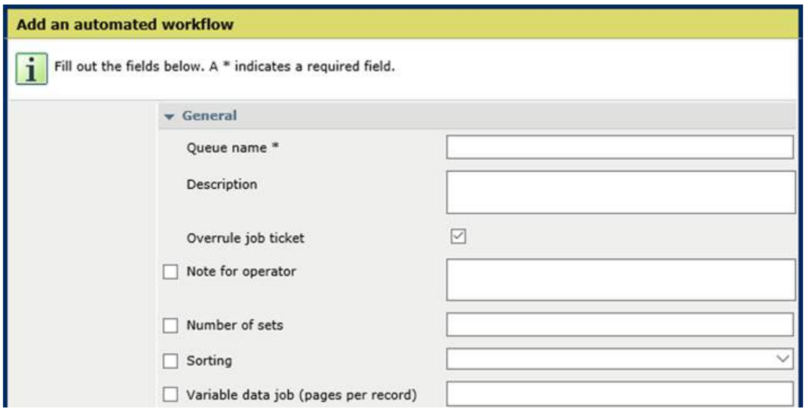
You cannot delete the automated workflow: *(default)*.
1. Select one or more automated workflows.
2. Click [Delete].

Restore the default automated workflow

IMPORTANT
Be aware that this option removes all custom automated workflows.

1. Click [Restore].
2. Click [OK].

[General] settings



[Add an automated workflow]

Setting	Description
[Queue name]	Name of the automated workflow. For LPD/LPR printing this name is the name of the queue. The label name of a job that has used an automated workflow, also refers to this name.
[Description]	Description of the automated workflow.
[Overrule job ticket]	Job settings that are defined on other locations, are by default ignored to ensure that only the settings of the automated workflow are used.
[Note for operator]	Create a print or finishing instruction for the operator.
[Number of sets]	Indicate the number of sets.
[Sorting]	Indicate if you want to sort the output per set or per page.

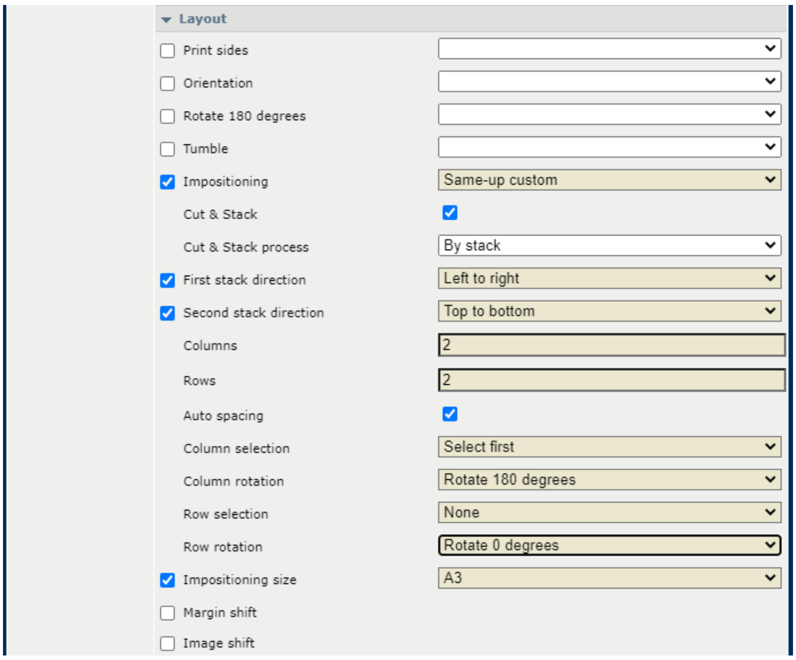
Setting	Description
[Variable data job (pages per record)]	Indicate if the PDF job is handled as variable data print job. Then, enter the pages per record.
[Variable data job (pages per record)]	Use this setting to indicate that the job is a variable data job. Enter the number of pages that composes a record.

[Media] settings

[Media]

Setting	Description
[Media]	Select the media from the media catalogue.
[Fit to media size]	Indicate if the page is scaled or clipped according to the media size.

[Layout] settings



[Layout]

Setting	Description
[Print sides]	Indicate if the output is printed 1- or 2-sided.
[Orientation]	Indicate if the reading direction of the document is parallel to the short edge or long edge of the page.

Setting	Description
[Rotate 180 degrees]	Indicate if pages are rotated 180 degrees.
[Tumble]	Indicate if you bind two-sided documents at the top or the bottom edge. The tumble option rotates the image on the back side of the sheet 180 degrees.
[Impositioning]	Indicate if the multiple pages are printed on a sheet, for example, to create a booklet.
[Impositioning size]	Select the impositioning size in case media are not selected.
[Margin Shift]	Define the margin shift of the front side and the back side in two directions. Enter: *0* to indicate that there is not shift required.
[Image shift]	Define the image shift of the front side and the back side in two directions. Enter: *0* to indicate that there is not shift required.
If [Impositioning] is set to [Same-up custom], the following settings become available:	
[Cut & Stack]	Indicate if you want to cut and stack the sheets.
[Cut & Stack process]	Select how you want to stack the sheets.
[First stacking direction]	Select the first stacking direction.
[Second stacking direction]	Select the second stacking direction.
[Columns]	Select the number of columns.
[Rows]	Select the number of rows.
[Auto spacing]	Indicate if you want to use the automatic gutter spacing option. The automatic gutter spacing option sizes gutters so that the document's pages fill the entire available space in the layout. The [Auto spacing] setting becomes available only if [Print trim marks] is selected.
[Column spacing ({0})]	Define the distance between columns. The [Column spacing ({0})] setting becomes available only if [Print trim marks] is selected.
[Row spacing ({0})]	Define the distance between rows. The [Row spacing ({0})] setting becomes available only if [Print trim marks] is selected.
[Column selection]	Select specific columns.
[Column rotation]	Select how you want to rotate the column(s) that you selected.
[Row selection]	Select specific rows.
[Row rotation]	Select how you want to rotate the row(s) that you selected.

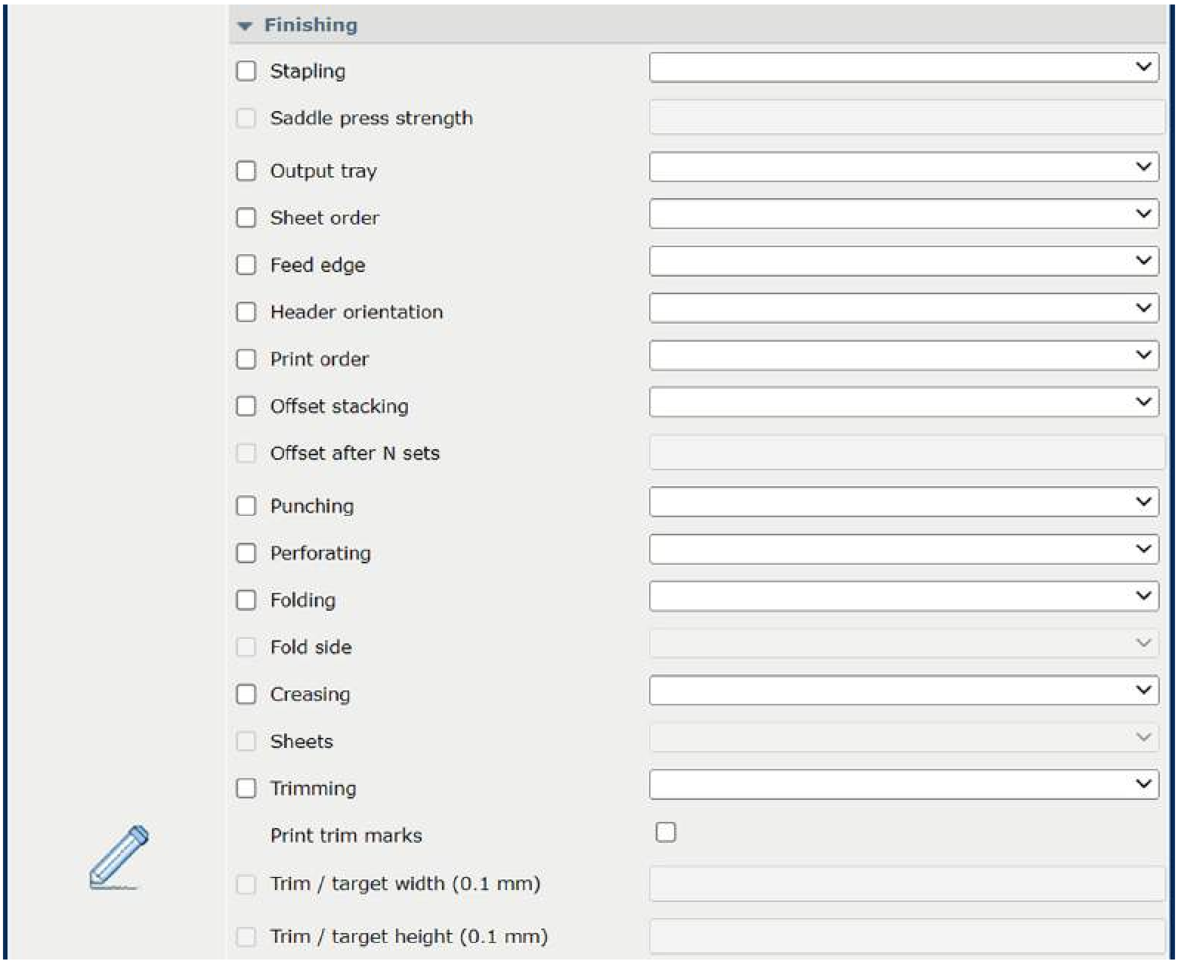
NOTE
You can set [Impositioning] to [Same-up custom] only when your printer has the Advanced Impose licence installed.

[Special pages] settings

[Special pages]

Setting	Description
[Front or booklet cover]	Select the media from the media catalogue.
[Printing on front cover]	Indicate if the front cover is printed.
[Back cover]	Select the back cover media from the media catalogue.
[Printing on back cover]	Indicate if the back cover is printed.
[Perfect binding]	Indicate if and where the job is bound.

[Finishing] settings



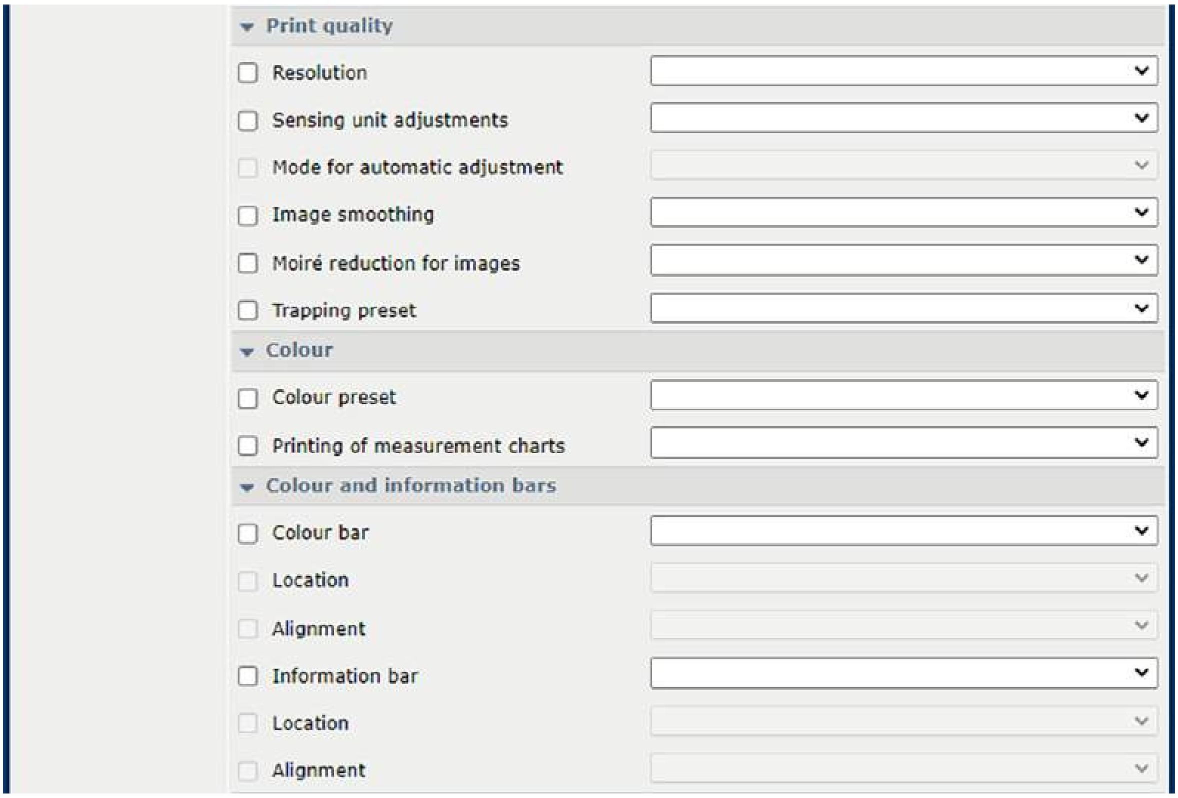
[Finishing]

Setting	Description
[Stapling]	Select the staple method and location.
[Saddle press strength]	Define the value for saddle press stapling: between -10 and +10.
[Output tray]	Select the output tray.
[Sheet order]	Select the order of the sheets in the output tray.
[Feed edge]	Select the orientation of the sheets in the output tray.
[Header orientation]	Select the orientation of the sheets in the output tray.
[Print order]	Indicate the delivery order of the sheets.
[Offset stacking]	Indicate if the sets are delivered with an offset.
[Offset after N sets]	Indicate if a series of sets are delivered with an offset.
[Punching]	Select the number of punch holes and their location.
[Perforating]	Select the perforation type and location.
[Folding]	Select the folding method.
[Fold side]	Indicate the fold side: [Inside] or [Outside].

Setting	Description
[Creasing]	Select the creasing method.
[Sheets]	Indicate the sheets you want to crease.
[Trimming]	Select the trimming method.
[Print trim marks]	Indicate if you want to print trim marks.
[Width ({0})]	Indicate the width that must be trimmed.
[Height: ({0})]	Indicate the height that must be trimmed.
[Horizontal trim shift: ({0})]	Shift the trim marks horizontally.
[Vertical trim shift: ({0})]	Shift the trim marks vertically.
[Horizontal alignment]	Indicate if the trim marks need to be aligned horizontally.
[Vertical alignment]	Indicate if the trim marks need to be aligned vertically.

NOTE
[Horizontal trim shift: ({0})], [Vertical trim shift: ({0})], [Horizontal alignment] and [Vertical alignment] settings become available only if:
- You have the Advanced Impose licence installed on your printer;
- [Impositioning] is set to [Same-up custom];
- [Print trim marks] is selected.

[Print quality], [Color], and [Color and information bars] settings



[Print quality], [Color] and [Color and information bars]

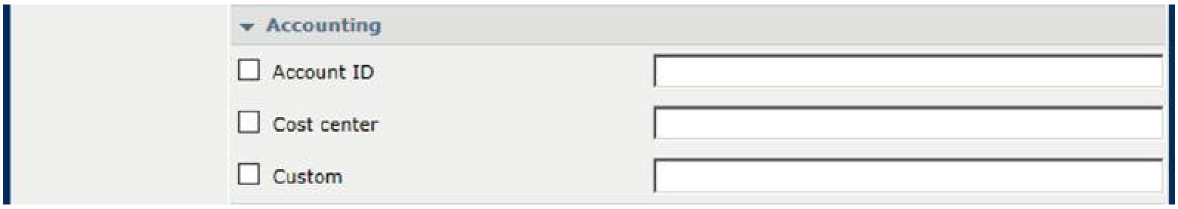
Setting	Description
[Resolution]	Use this setting to indicate the print resolution.
[Sensing unit adjustments]	Use this setting to select an automatic adjustment to be performed by the sensing unit, either: media registration, density, media registration and density or none.
[Mode for automatic adjustment]	Use this setting to select the mode for the automatic adjustment in the sensing unit: continuous or interrupt mode.
[Image smoothing]	Use this setting to apply an anti-aliasing algorithm to smooth images with a low resolution.
[Moiré reduction for images]	Apply a Moiré reduction algorithm to enhance photographic images. When images have a resolution below 300 dpi, the Moiré reduction only takes effect when image smoothing is enabled.
[Trapping preset]	Select a trapping preset.
[Color preset]	Select a colour preset.
[Printing of measurement charts]	Use this setting only for printing measurement charts for G7 calibration or external profiling. IMPORTANT Improper use of this setting may cause pollution of the printer.
[Color bar], [Location], [Alignment]	Select a colour bar and its location.
[Information bar], [Location], [Alignment]	Select an information bar and its location.

[Workflow] settings

[Workflow]

Setting	Description
[Job destination]	Select the destination of the jobs: scheduled jobs, waiting jobs or DocBox.
[DocBox name]	Select the DocBox.
[Printing workflow]	Select the job processing on the PRISMAsync Print Server.
[Banner pages for print jobs]	Indicate if a banner page is used.
[Trailer pages for print jobs]	Indicate if a trailer pages is used.
[Media of banner/trailer page]	Select the media of the banner and trailer page.
[Separator sheets for print jobs]	Indicate if separator sheets are used. You can use separator sets per a series of sets, if required.
[PDF XObject optimization]	Select whether PDF/X objects are cached per document or per page.
[PDF to PostScript conversion]	Indicate if you want to convert PDF to PostScript instead of only using the native PDF RIP, which is the default setting value.

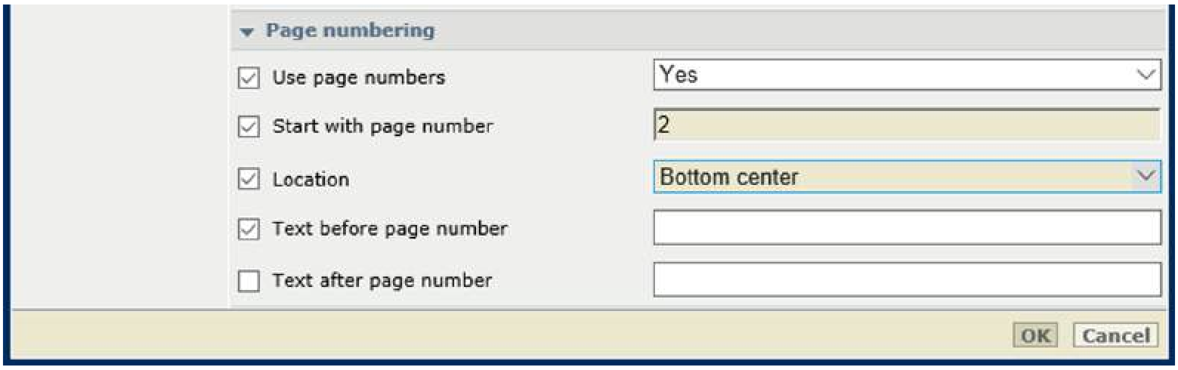
[Accounting] settings



[Accounting]

Setting	Description
[Accounting ID]	Enter accounting ID.
[Cost center]	Enter cost centre.
[Custom]	Enter extra account information.

[Page numbering] settings



[Page numbering]

Setting	Description
[Use page numbers], [Start with page number], [Location], [Text before page number], [Text after page number]	Indicate if you want to use page numbers. The font and the font size are pre-defined. Indicate if and where the page numbers are printed. Enter the text that is additionally printed before or behind the page number.

More information

- *Learn about automated workflows* on page 245
- *Advanced impositioning* on page 316
- *Submit jobs via LPR* on page 257
- *Submit jobs via hotfolders* on page 258
- *Use the job ticket editor* on page 260
- *Edit with automated workflow in job destination* on page 263

Submit jobs via LPR

PDF files can be submitted to the printer with an LPR command. When you print via LPR the queue name is the name of an automated workflow. Jobs submitted via LPR take over the attributes of the automated workflow.

An automated workflow bundles a series of pre-set attributes to define jobs.

[Automated workflow] menu

1. On your workstation, make sure the Windows LPR command has been installed.
2. Use the LPR command line and the following command format:
   lpr -S<printer name> -P<automated workflow name> <pdf name>

More information

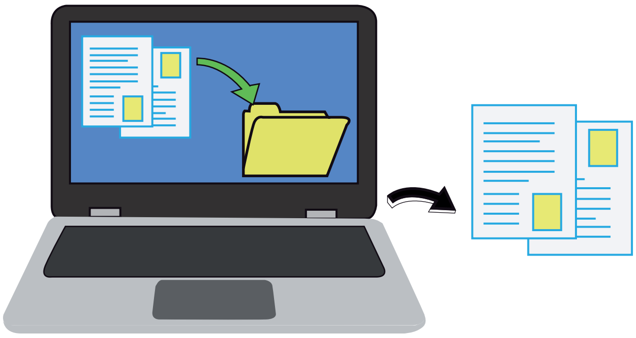
- *Learn about automated workflows* on page 245
- *Define a new automated workflow* on page 247
- *Submit jobs via hotfolders* on page 258
- *Use the job ticket editor* on page 260
- *Edit with automated workflow in job destination* on page 263

Submit jobs via hotfolders

What are hotfolders

Hotfolders enable you to submit print jobs by simply drag & drop ready-to-print PDF files onto a folder shortcut on your desktop. The job properties are determined by the automated workflow connected to that hotfolder.

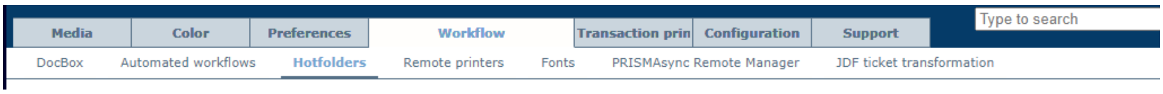
When there is a JDF ticket in the hotfolder, the print properties of the JDF ticket and the connected automated workflow are used. If a property is defined in the JDF ticket and the automated workflow, then the property of the automated workflow is applied. Unless the automated workflow setting [Overrule job ticket] is disabled.



Drag & drop print files

It depends on the [Overrule job ticket] setting of the automated workflow whether the settings of the automated workflow or the job ticket are used.

The system administrator creates the hotfolders in the Settings Editor and defines the user authentication for the hotfolders.



[Hotfolders] tab

Create a shortcut to a hotfolder on a Windows workstation

1. Start the standard Windows wizard to map a network drive.
2. In the dialogue, use the link to connect to the PRISMAsync hotfolder to share documents or files.
3. Enter the path to the hotfolder that is shown in the Settings Editor at the [Workflow]→ [Hotfolders] tab .
   Use the following path.
   - **WebDAV**: *http(s)://<hostname>/dav/<hotfolder name>* or *http(s):||<IPaddress>| dav| <hotfolder name>*.
   - **SMB**: *<hostname>**<hotfolder name>* or *<IPaddress>**<hotfolder name>*.
4. Enter your username and password.
5. Enter a name for the network location.

Create a shortcut to a hotfolder on a Mac workstation

NOTE
Use a third party WebDAV client, for example CyberDuck. Use always the latest version of CyberDuck.

1. Open CyberDuck.
2. Select + and choose WebDAV (HTTP).
3. Click [More Options].
4. Enter a Nickname if desired.
5. Enter de IP address of the PRISMAsync Print Server in the Server name field.
6. Enter the username and password of the PRISMsync Print Server.
7. Enter the path to the hotfolder.
   Use the following format to define the path: `/dav/hotfolder name`
8. Click [Close].

Use a JDF ticket

1. Go the hotfolder shortcut.
2. Drag & drop the Default_ticket.jdf file on the hotfolder shortcut.

Drag & drop PDF files on a hotfolder

1. Double-click the hotfolder shortcut to display the print properties that are defined for the hotfolder.
2. Select the PDF files and drag & drop these on the hotfolder shortcut.
   After printing, the PDF files are removed from the hotfolder.

More information

- *Learn about automated workflows* on page 245
- *Define a new automated workflow* on page 247
- *Submit jobs via LPR* on page 257
- *Use the job ticket editor* on page 260
- *Edit with automated workflow in job destination* on page 263

Use the job ticket editor

What is the job ticket editor

The **JDF ticket** is a file that contains the instructions to handle and print a job. The Settings Editor has a ticket editor to create, load and save JDF tickets. Job Definition Format (JDF) is a standard format to automate the print production between different applications. It is an XML format that describes the job ticket, the message description, and the message interchange.

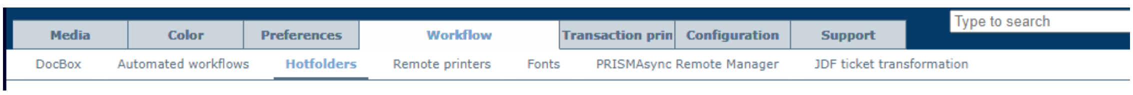
The JDF ticket that has the name **default_ticket.jdf** can be dropped in the hotfolder shortcut to define the properties of the PDF jobs.

Some settings are only available in the automated workflow or only in the ticket editor:

Settings available in automated workflow only	Settings available in ticket editor only
[Variable data job (pages per record)]	Preview feature with [Sheet view] and [Document view] options
[Clip pages to media size]	[Recipient name]
[Ignore the media size]	Media catalog via [Media] option
[Margin shift]	[Print trim marks]
[Offset after N sets]	Color preset editor via [Print quality]→[Color preset]→[Edit]
[Moiré reduction for images]	
[Trapping preset]	
[Printing of measurement charts]	
[Printing workflow][Streaming], or [Receive then print while RIP]	
[PDF XObject optimization]	
[PDF to PostScript conversion]	

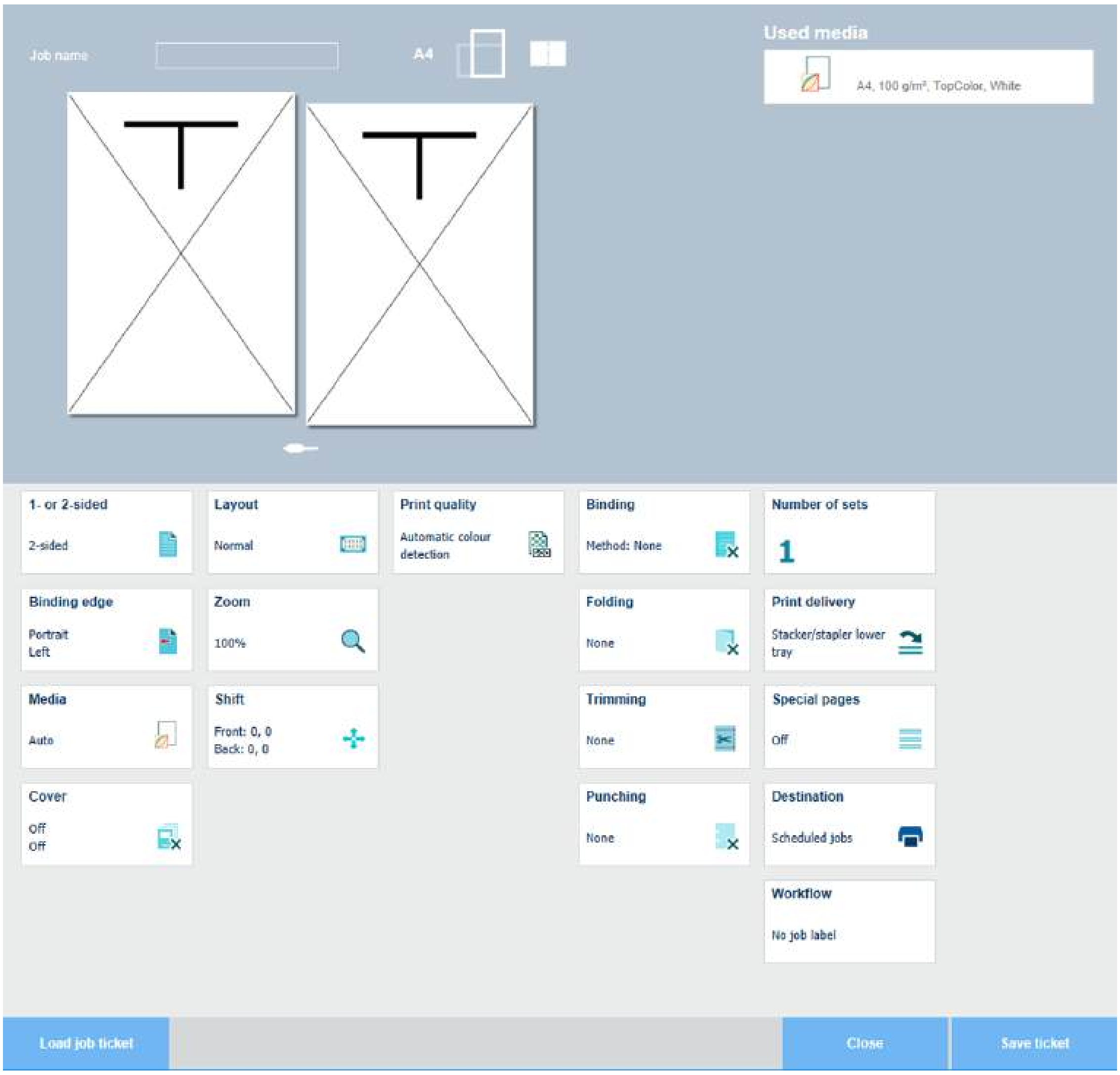
Go to the job ticket editor

1. Open the Settings Editor and go to: [Workflow]→[Hotfolders].



[Hotfolders] tab

2. Click [Ticket editor].



Job ticket editor

Create a JDF ticket

1. Define the job properties you want to include into the JDF ticket.
2. Click [Save ticket].
3. Save the JDF job ticket.

Load a JDF ticket

1. Click [Load job ticket].
2. Browse to the JDF ticket.
3. Click [Open].

More information

- *Learn about automated workflows* on page 245
- *Define a new automated workflow* on page 247
- *Submit jobs via LPR* on page 257
- *Submit jobs via hotfolders* on page 258

• *Edit with automated workflow in job destination* on page 263

Edit with automated workflow in job destination

With the function [Edit with automated workflow] you can apply an automated workflow or edit with automated workflows on PDF or PS jobs only on the control panel or with PRISMAsync Remote Manager. You can use the function in the following situations:
- You want to change multiple job settings of a job that is stored in one of the queues. You define an automated workflow that covers these job settings.
- You want to change multiple jobs in the same way. You define an automated workflow that covers the job settings.
- You want to adjust the settings of an automated workflow on the spot. You enable [Show settings] and adjust the settings to your preference.

NOTE
The job destination that is defined in the automated workflow will be ignored. The job remains in its location.

IMPORTANT
Please note that the [Edit with automated workflow] function will reset all job settings to the initial values.

The [Edit with automated workflow] function is available in the following locations.

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Edit with automated workflow]	✔	✔	✔	

The [Show settings] function offers edit possibilities for all selected jobs. Previous job settings are not shown, all the fields in the dialogue are empty. When a value is set in this dialogue, it will be applied to all selected jobs. The settings are similar to the settings of the Automated Workflow Editor, but [Edit with automated workflow] does not contain a job preview, CMYK curves, pixel precise preview, and page programming.

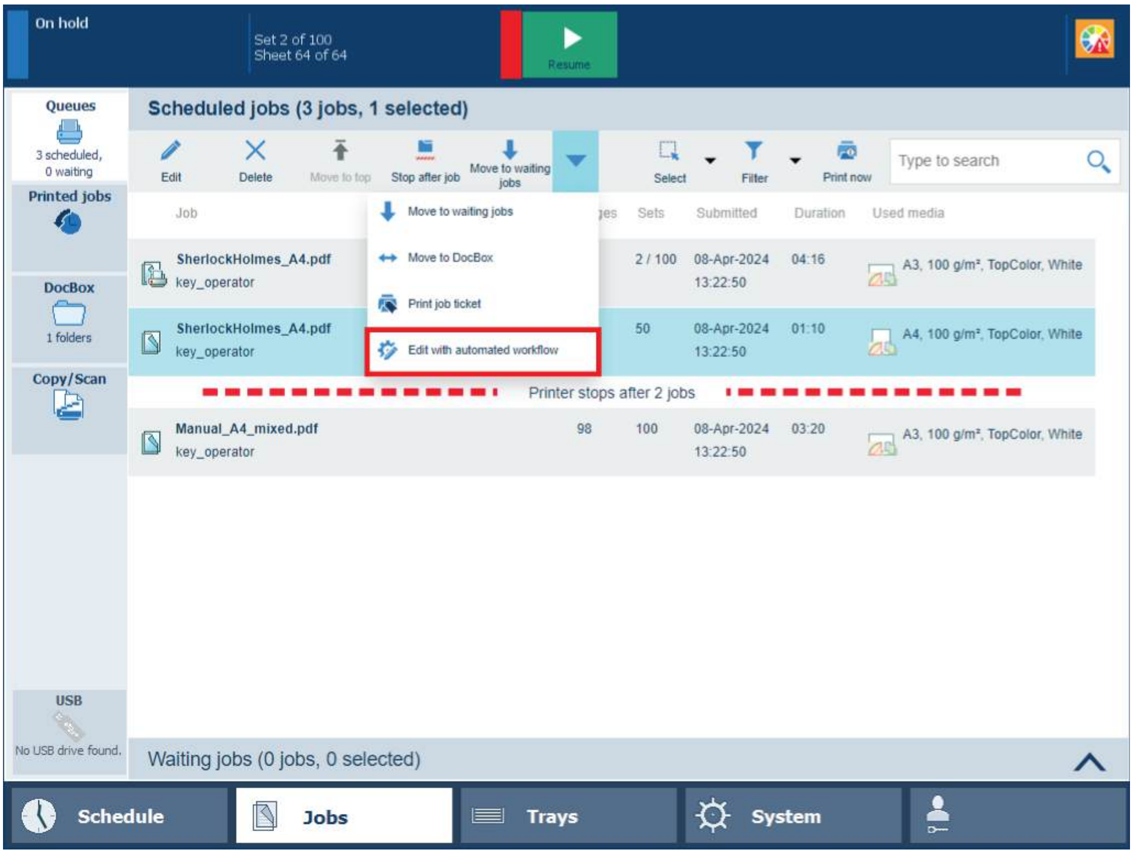
After the changed settings are confirmed, the new updated settings will be applied. Be aware of the following:
- Settings of an automated workflow or setting that you had previously changed via the [Edit] function on the control panel will be lost. Settings that are assigned with [Edit with automated workflow] have priority. Settings that are left empty in [Edit with automated workflow] will receive the original setting (the setting it had before it was changed with the [Edit] function).
- Conflicts will be resolved. The jobs will be printed except if you have indicated that the printer stops when jobs with conflicting settings occur. Then you have to confirm the change to continue printing.
- PDF and PS jobs will be re-ripped.

A setting to overrule job ticket is present. By default, this setting is enabled. This means that the settings of the [Edit with automated workflow] function will have priority. For example:
- When a predefined automated workflow is chosen and overrule job ticket = **disabled**, the ticket settings have more priority. Example: ticket has media A3, automated workflow has media A4 => result media A3
- When a predefined automated workflow is chosen and overrule job ticket = **enabled**, the automated workflow settings have more priority. Example: ticket has media A3, automated workflow has media A4 => result Media A4

[Edit with automated workflow] on control panel

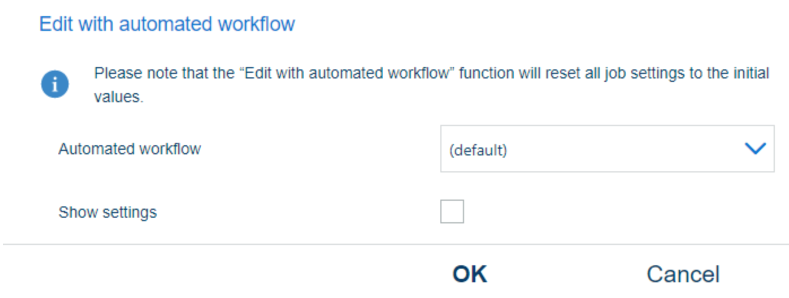
1. Go to the control panel.

2. Select one or more jobs.
3. Touch [Edit with automated workflow] from the drop-down list.



[Edit with automated workflow] button

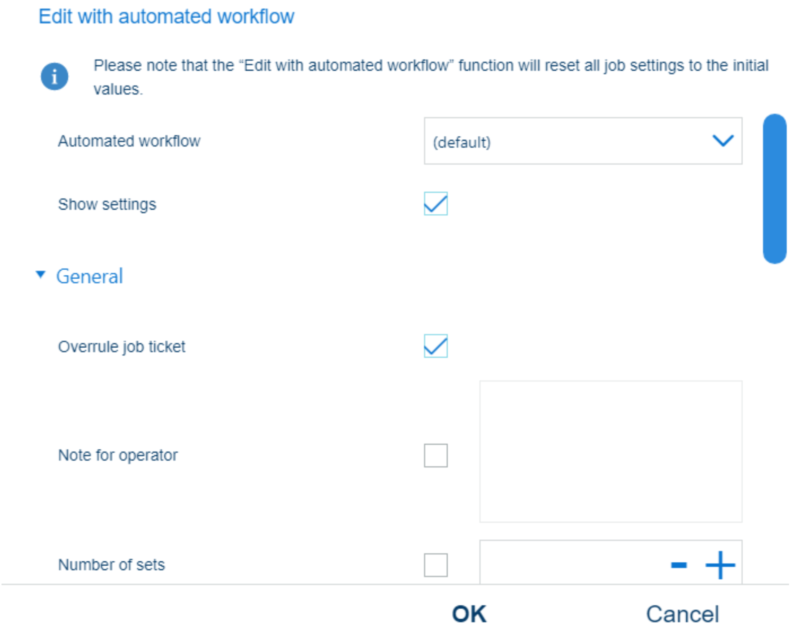
4. Apply an automated workflow:
   - Select the required automated workflow in the [Automated workflow] drop-down list.



NOTE

To restore original jobs settings, leave the [Automated workflow] drop-down list *empty* and touch [OK].

   - Touch [OK].
5. Change the settings of the automated workflow:
   - Select the [Show settings] check box to view all settings.



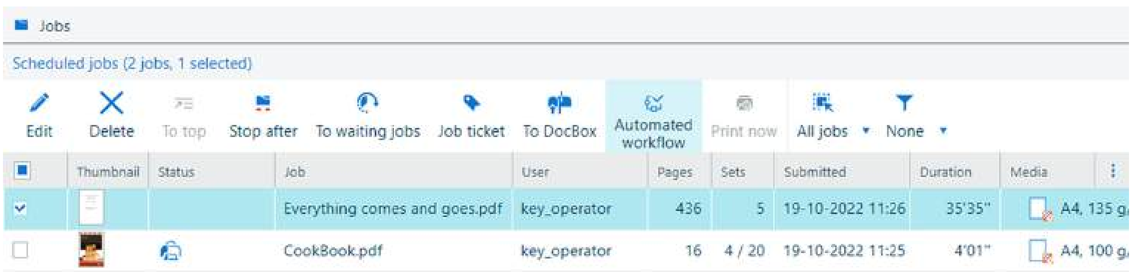
- Touch [OK].

NOTE
A description of all settings can be found here: *The settings of an automated workflow* on page 267.

NOTE
You can filter jobs based on job label. The [Job label] field is the name of the automated workflow that has been applied.

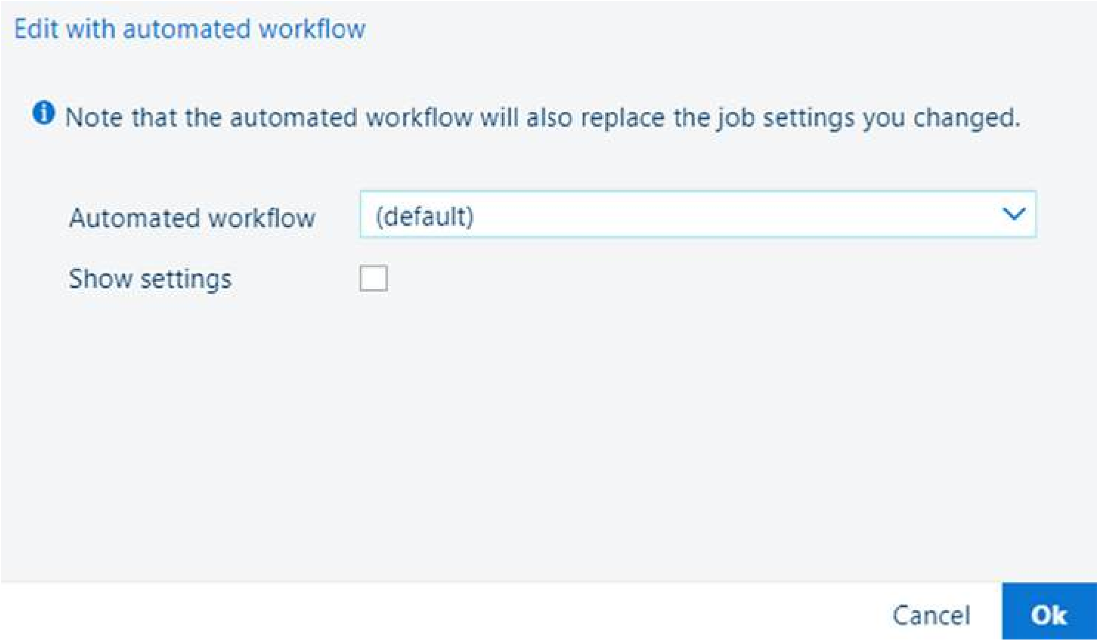
[Edit with automated workflow] in PRISMAremote Manager

1. Go to the PRISMAremote Managerr.
2. Select one or more jobs.
3. In the [Jobs] app, click the [Automated workflow] button.



[Automated workflow] in PRISMAremote Manager

4. You can apply an automated workflow:
   - Select the required automated workflow in the [Automated workflow] drop-down list.

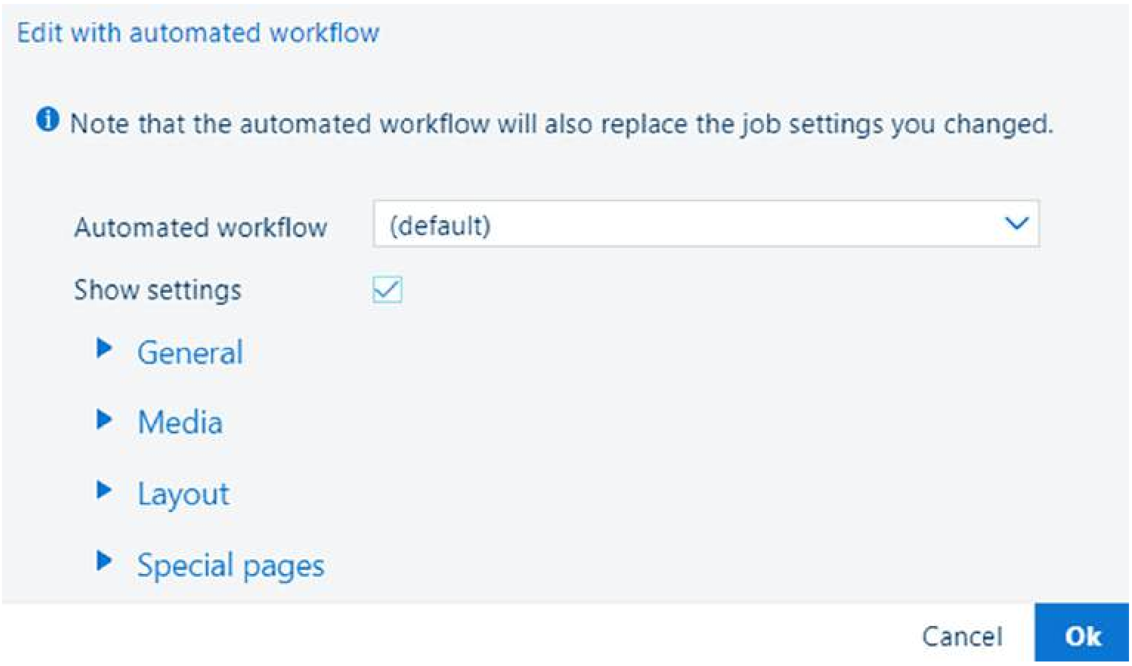


[Automated workflow] drop-down list

NOTE
To restore original jobs settings, leave the [Automated workflow] drop-down list*empty* and click [OK].

- Click [OK].

5. You can change the settings of the automated workflow:
   - Select the [Show settings] checkbox to view all settings.



[Show settings] checkbox

6. Click [OK].

NOTE
A description of all settings can be found here: *The settings of an automated workflow* on page 267 .

The settings of an automated workflow

[General] and [Media] settings

[General] settings	Description
[Overrule job ticket]	The properties of the automated workflow can overrule the properties defined in the job ticket. This is the default behaviour of an automated workflow. When you want to apply a JDF ticket in a hotfolder, ensure the check box is not checked.
[Note for operator]	Create a print or finishing instruction for the operator.
[Number of sets]	Enter the number of sets.
[Sorting]	Select if sorting occurs by set or by page.

[Media]settings	Description
[Media]	Select the media from the media catalogue.
[Fit to media size]	Use this setting to indicate how the document pages are resized according to the size of the selected media: [Scale every page to media size], [Clip pages to media size], or [Ignore the media size]. When you select the attribute [Ignore the media size], the media size defined in the job is applied.

[Layout] settings

[Layout] settings	Description
[Print sides]	Select if jobs are printed one- or two-sided.
[Orientation]	Select the reading direction of the prints: [Portrait] or [Landscape].
[Rotate 180 degrees]	Use this setting to rotate the imposition 180 degrees.
[Tumble]	Select the binding edge of the output. The [Tumble] setting rotates the imposition of the back side 180 degrees.
[Impositioning]	Select the required imposition.
[Impositioning size]	Select the size of the imposition in case the media are unknown.
[Margin Shift]	Select this setting to define the margin shift of the front side and the back side in two directions. Enter: '0' to indicate that there is not shift required.
[Image shift]	Select this attribute to define the image shift of the front side and the back side in two directions. Enter: '0' to indicate that there is no shift required.
If [Impositioning] is set to [Same-up custom], the following settings become available:	
[Cut and stack]	Use this setting to indicate if you want to cut and stack the sheets.
[Cut and stack process]	Select how you want to stack the sheets.

[Layout] settings	Description
[First stacking direction]	Select the first stacking direction.
[Second stacking direction]	Select the second stacking direction.
[Columns]	Select the number of columns.
[Rows]	Select the number of rows.
[Auto spacing]	Indicate if you want to use the automatic gutter spacing option. The automatic gutter spacing option sizes gutters so that the document's pages fill the entire available space in the layout. The [Auto spacing] setting becomes available only if [Print trim marks] is selected.
[Column spacing (mm)]	Define the distance between columns. This setting only becomes available if [Print trim marks] setting is enabled.
[Row spacing (mm)]	Define the distance between rows. This setting only becomes available if [Print trim marks] setting is enabled.
[Column selection]	Select specific columns.
[Column rotation]	Select how you want to rotate the column(s) that you selected.
[Row selection]	Select specific rows.
[Row rotation]	Select how you want to rotate the row(s) that you selected.

[Special pages] and [Finishing] settings

[Special pages] settings	Description
[Front or booklet cover]	Select the media for the front cover.
[Printing on front cover]	Indicate if the cover is printed or not.
[Media print mode of front cover]	Select the required media print mode.
[Back cover]	Select the media for the back cover.
[Printing on back cover]	Indicate if the cover is printed or not.
[Perfect binding]	Indicate if and where the job is bound.

[Finishing] settings	Description
[Stapling]	Select the staple method and location.
[Saddle press strength]	Define the value for saddle press stapling: between -10 and +10.
[Output tray]	Select the paper tray of the prints.
[Sheet order]	Select the order of the prints in the output tray.
[Feed edge]	Select the orientation of the prints in the output tray.
[Header orientation]	Select the header orientation of the prints in the output tray.
[Print order]	Indicate the delivery order of the sheets.
[Offset stacking]	Indicate if the stacking occurs with or without an offset.

[Finishing] settings	Description
[Offset after N sets]	Indicate if a group of sets is delivered with an offset. N means the number of sets in a group.
[Punching]	Select the number of punch holes and their location.
[Perforating]	Select the perforation type and location.
[Folding]	Select the folding method.
[Fold side]	Indicate the fold side: [Inside] or [Outside].
[Creasing]	Select the creasing method.
[Sheets]	Indicate the pages you want to crease.
[Trimming]	Select the trimming method.
[Print trim marks]	Indicate if you want to print trim marks.
[Width ({0})]	Enter the width of the trim marks.
[Height: ({0})]	Enter the height of the trim marks.
[Horizontal trim shift: ({0})]	Use this setting to shift the trim marks horizontally.
[Vertical trim shift: ({0})]	Use this setting to shift the trim marks vertically.
[Horizontal alignment]	Indicate if the trim marks need to be aligned horizontally.
[Vertical alignment]	Indicate if the trim marks need to be aligned vertically.

[Print quality], [Color] and [Color and information bars] settings

[Print quality] settings	Description
[Resolution]	Use this setting to indicate the print resolution.
[Sensing unit adjustments]	Use this setting to select an automatic adjustment to be performed by the sensing unit: media registration, density, media registration and density or none.
[Mode for automatic adjustment]	Use this setting to select the mode for the automatic adjustment in the sensing unit: continuous or interrupt mode.
[Image smoothing]	Use this setting to apply an anti-aliasing algorithm to smooth images with a low resolution.
[Moiré reduction for images]	Apply a Moiré reduction algorithm to enhance photographic images. When images have a resolution below 300 dpi, the Moiré reduction only takes effect when image smoothing is enabled.
[Trapping preset]	Select a trapping preset.

Colour settings	Description
[Color preset]	Select a colour preset.

[Add bars] settings	Description
[Color bar], [Location], [Alignment]	When you want to print a colour bar, select one of the available colour bars. You can also select the location and the alignment.

[Add bars] settings	Description
[Information bar], [Location], [Alignment]	When you want to print an information bar, select one of the available information bars. You can also select the location and the alignment.

[Workflow] settings

[Workflow] settings	Description
[Banner pages for print jobs], [Trailer pages for print jobs], [Media of banner/trailer page]	Indicate if individual sets in the job output are separated by a banner page and / or a trailer page.
[Separator sheets for print jobs], [Separator sheet after N sets], [Media of separator sheets]	Indicate if sets or a group of sets are separated by a separator sheet. N means the number of sets in a group.
[PDF XObject optimization]	Use this section to define how XObjects are processed. [Automatic]: the XObjects of a PDF/VT job are cached. [Enabled]: the XObjects of every PDF job are cached. [Disabled]: the XObjects are re-interpreted per page.
[PDF to PostScript conversion]	Indicate if you want to convert PDF to PostScript instead of only using the native PDF RIP, which is the default setting value.

[Print range], [Accounting] and [Page numbering] settings

[Print range] settings	Description
[First page of print range]	When you want to print a part of the document, enter the first page number of the print range. When you do not enter the last page, the range ends at the last document page.
[Last page of the print range]	When you want to print a part of the document, enter the last page number of the print range. When you do not enter the fist page, the range begins at the first document page.
[First record of the print range]	When you want to print a part of the document of a variable data job, enter the last page number of the print range. When you do not enter the last record, the range ends at the last record.
[Last record of print range]	When you want to print a part of the document a variable data job, enter the last page number of the print range. When you do not enter the fist record, the range begins at the first record.

[Accounting] settings	Description
[Accounting ID]	Enter the accounting ID.
[Cost center]	Enter the cost centre.
[Custom]	Enter extra accounting information.

[Page numbering] settings	Description
[Page numbering], [Start with page number], [Location], [Text before page number], [Text after page number]	Indicate if you want to use page numbers. The default font and the font size are pre-defined. Indicate if and where the page numbers are printed. Enter the text that is additionally printed before or behind the page number.

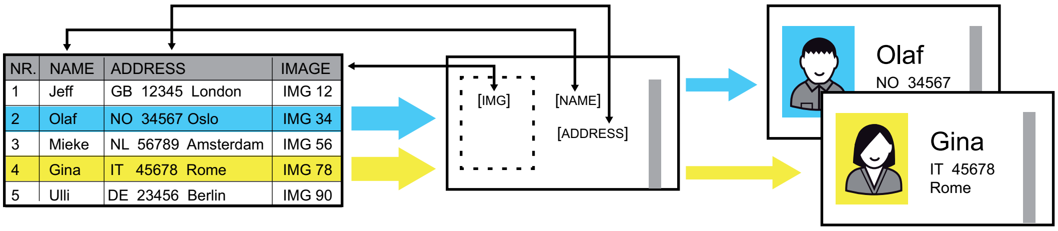
More information

Learn about automated workflows on page 245

Work with variable data printing in document printing mode

Learn about variable data printing in document printing mode

Variable data printing is used for direct marketing, advertising, and personalized letters. PRISMAprepare and document authoring tools allow to build personalized documents. Variable data, for example names and addresses, are merged with fixed elements and saved as multi-record documents.

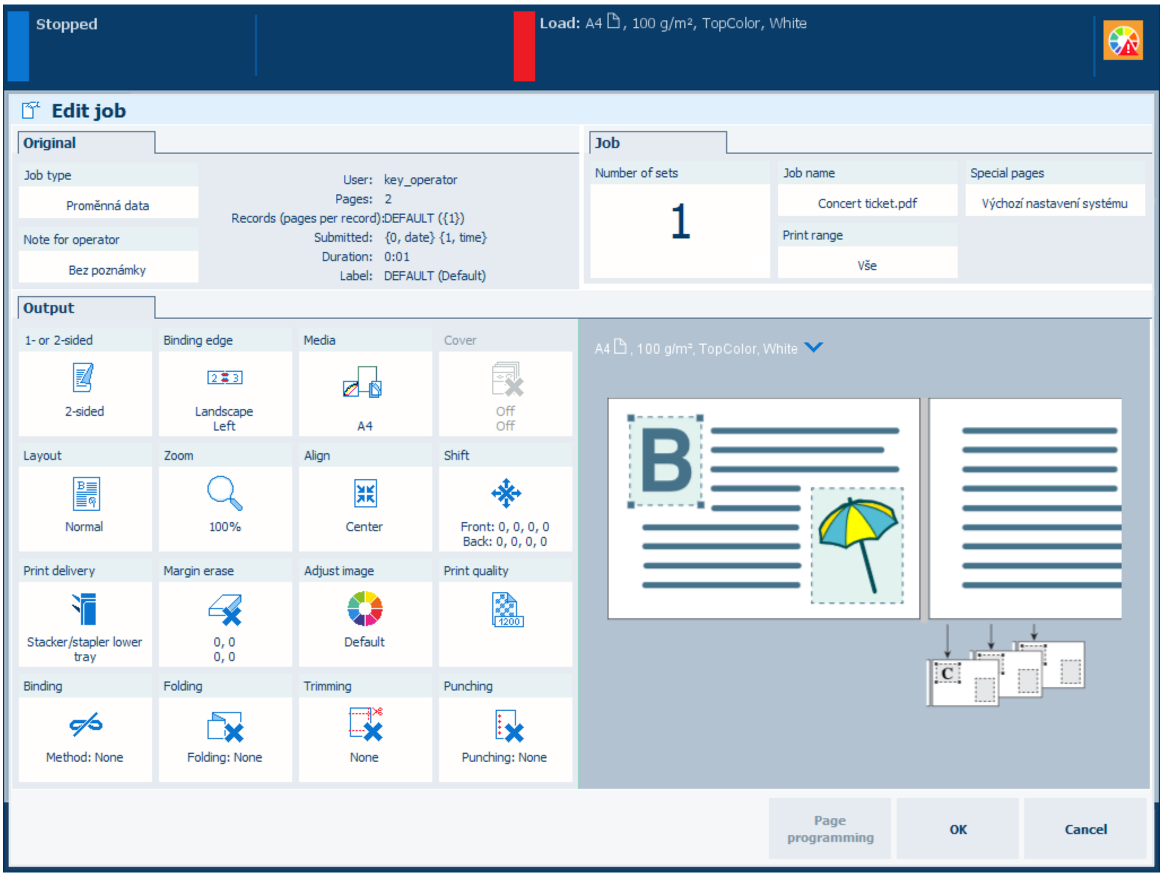


Creation of personalized documents

PRISMAsync supports several PPML formats and the PDF/VT format for variable data printing.

Also manually created VDP jobs are supported. When PRISMAsync receives a variable data print job, it distinguishes the record structure. Records are treated as print sets. Submit variable data jobs to PRISMAsync with PRISMAprepare or an automated workflow.

Large jobs can be converted into VDP jobs.



Properties of a variable data job

The icon indicates a variable data job in the job list.

More information

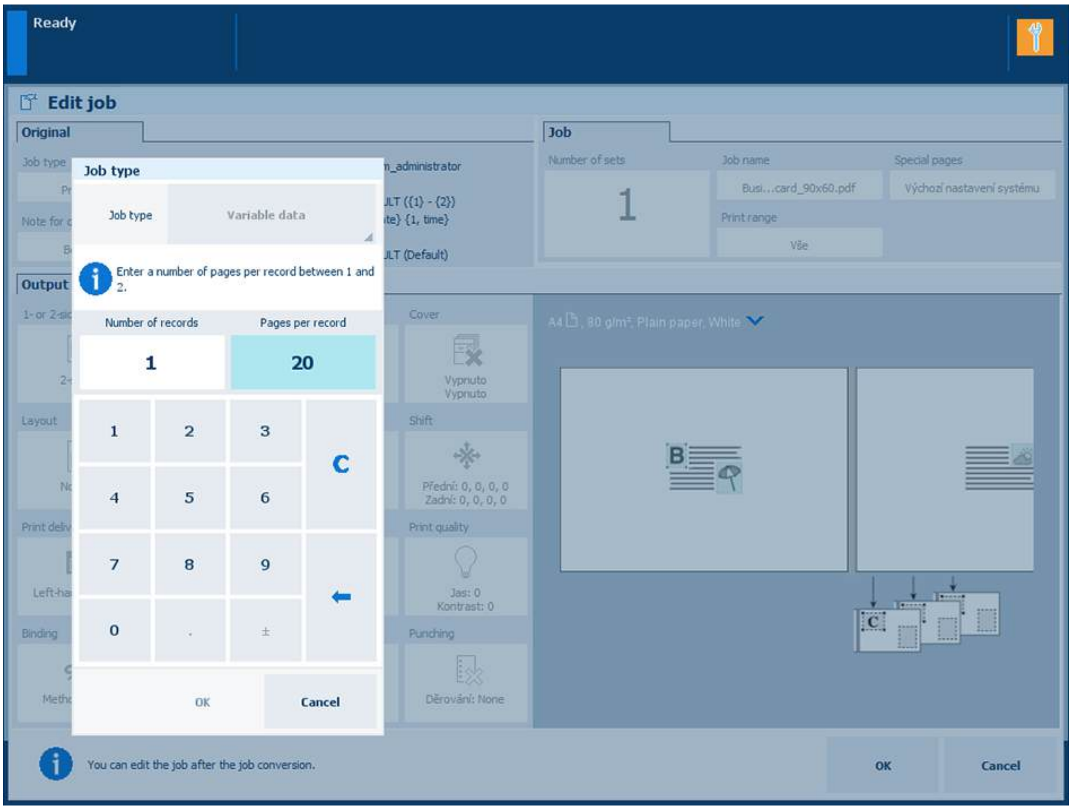
- *Printer specifications* on page 611
- *Learn about automated workflows* on page 245
- *Convert a job to a variable data job* on page 274

Convert a job to a variable data job

If you print a large document that has the structure of a variable data document, you can instruct the print system to handle this job as a variable data job. This can be useful if you only want a proof print or only want to print part of the job.

To indicate how the job is composed, you enter either the number of records or the pages per record.

You can also convert jobs to variable data jobs with an automated workflow.



Indicate the structure of the job

Procedure

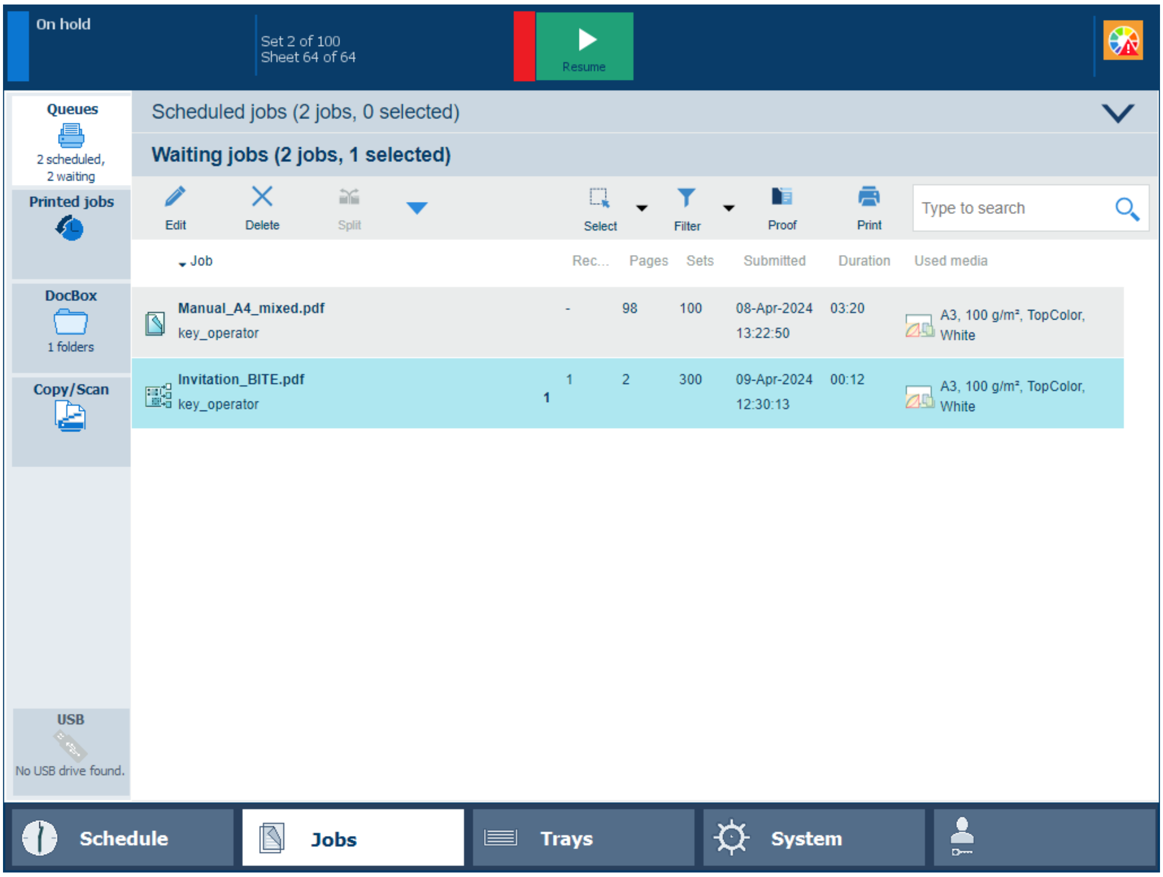
1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to convert.
4. Touch [Edit], or double-tap the job.
5. Touch [Job type].
6. Touch [Normal] -> [Variable data].
7. Define how often the master document occurs in the job at [Number of records] or enter the number of pages in the master document at [Pages per record].
8. Touch [OK] to store the new job type.
9. Touch [OK] to start the conversion.

After you finish

After conversion you can make the job settings and print the variable data job.

Print a variable data job

When PRISMAsync receives a job with variable data (PDF/VT job), it distinguishes the record structure. PRISMAsync treats records as print sets with fixed contents.



A PDF/VT job in the list of scheduled jobs

Procedure

When the PDF/VT job arrives in the print queue, you can do the following:

1. Change the settings of the job.
   Be aware that when you change settings of a PDF/VT job, these settings are applicable to all records of the job. Page programming is not available for PDF/VT jobs.
2. Print a proof of the first record or print a range of records.
3. Print a range of records.
4. Interrupt the print process after a certain record, and resume the print process of the job later.

Plan the jobs

Learn about job management in the queues

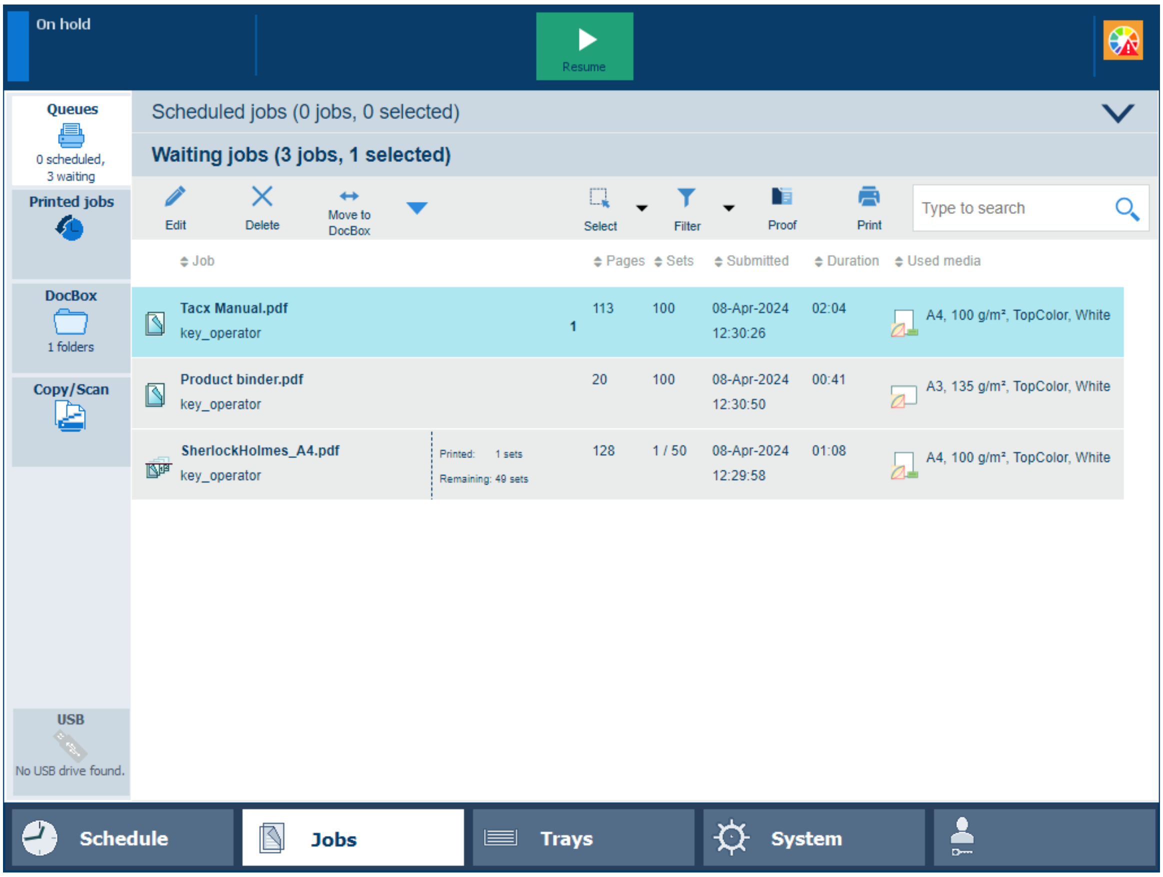
The destination of a print job is determined when a job is submitted. However, the active workflow profile and the automated workflow can overrule the destination. Completed print jobs are stored in the list of printed jobs, if enabled.

When the print system is in the transaction printing mode, PRISMAsync routes the streaming jobs directly to the list of scheduled jobs.

Waiting jobs

When jobs arrive in the list of waiting jobs, you determine the print order of the jobs. You are also able to change print job settings or perform page programming. You can select one or more jobs by touching them or with one of the options in the [Select] menu. Give the [Print] command and the jobs go to the list of scheduled jobs.

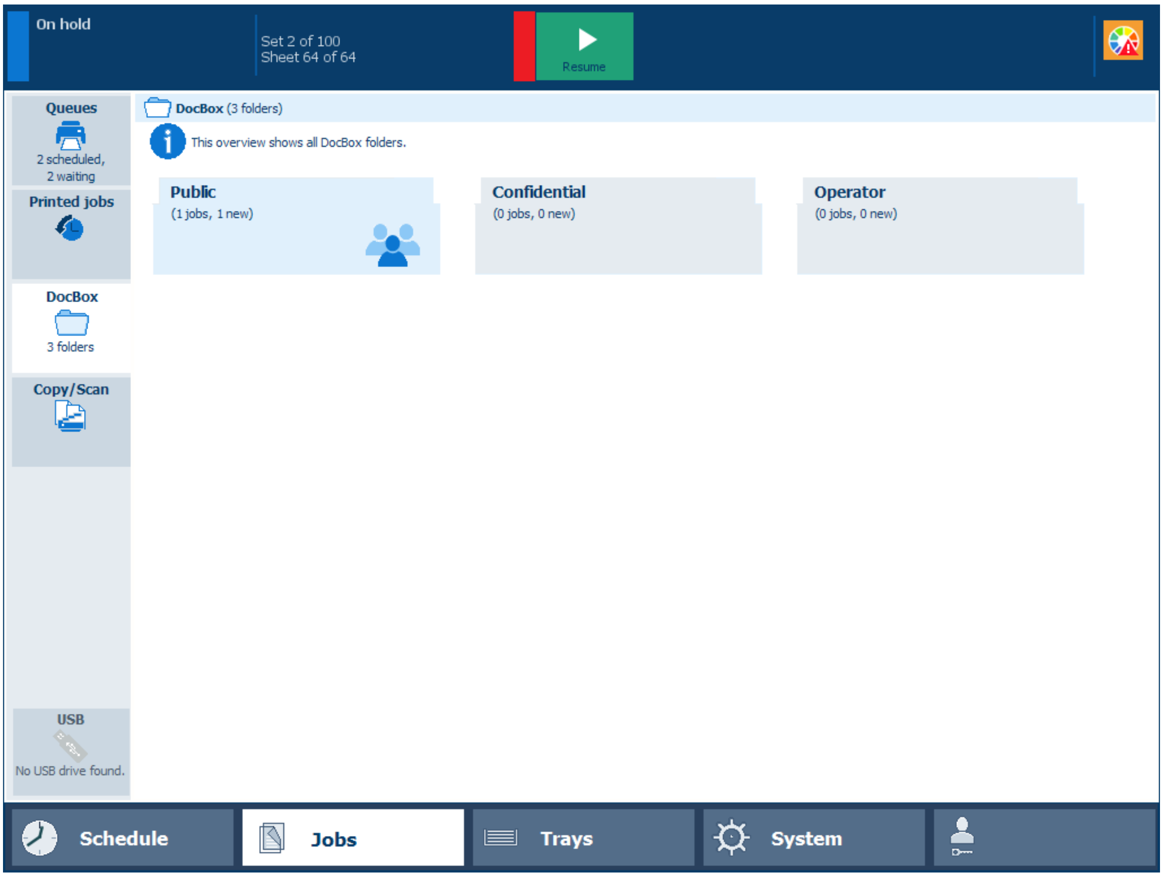
To increase the print productivity, you can use the job bundle function to combine several jobs into a single job. The original jobs are no longer visible, but you can recover the original jobs with the split function. You can change some settings and the print order of the individual jobs in the bundled job.



List of waiting jobs

DocBox

The DocBox is a job destination to collect jobs before printing. DocBox folders make it easy to group jobs according to a job owner or a department. There can also be DocBox folders for jobs that must be printed according specific requirements. The DocBox folder [Public] can be accessed by all operators.



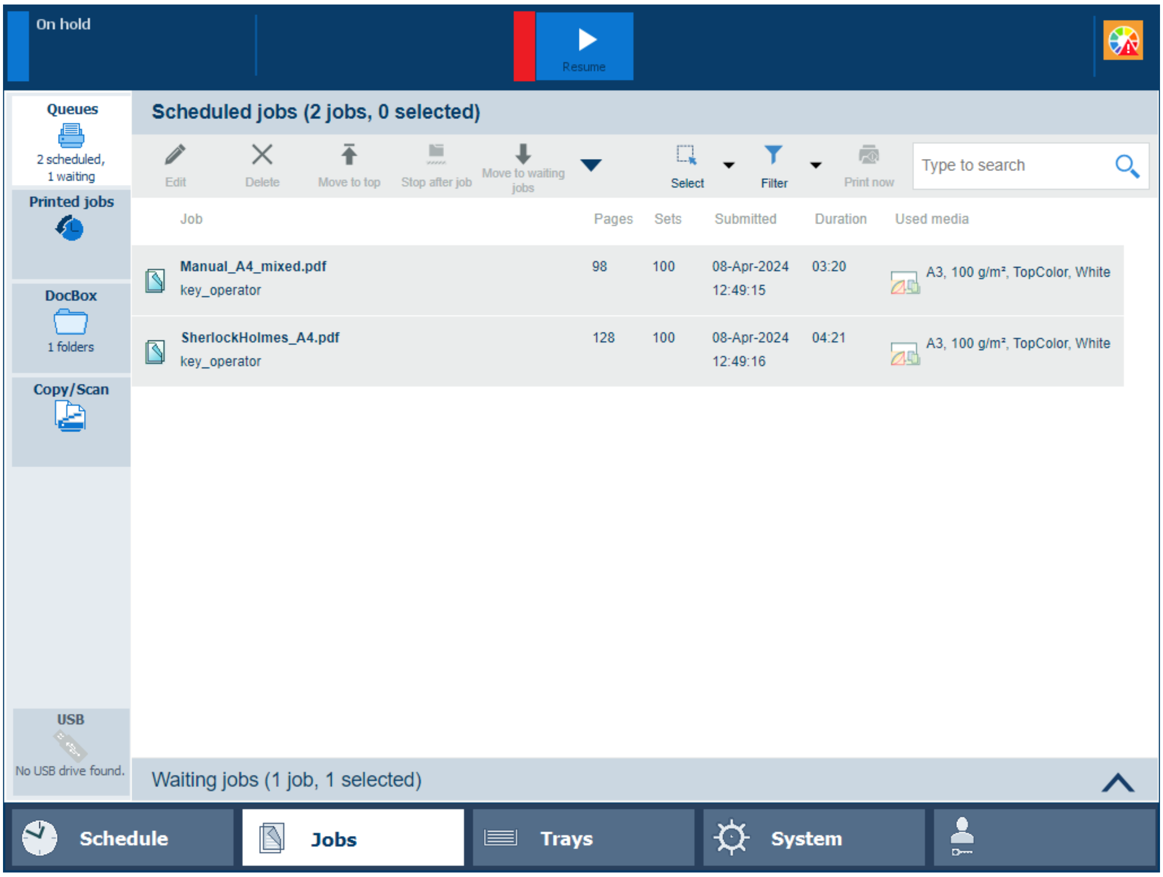
The DocBox folders

The system administrator creates the DocBox folders. The system administrator can protect DocBox folders with a PIN. The DocBox folder [Public] can always be accessed without PIN. You use the [Lock] option to protect DocBox job information.

Scheduled jobs

The print system prints the jobs in the sequence of the list of scheduled jobs. However, you are able to change the print sequence when one or more jobs require priority. In addition, you can decide to postpone or stop a print job.

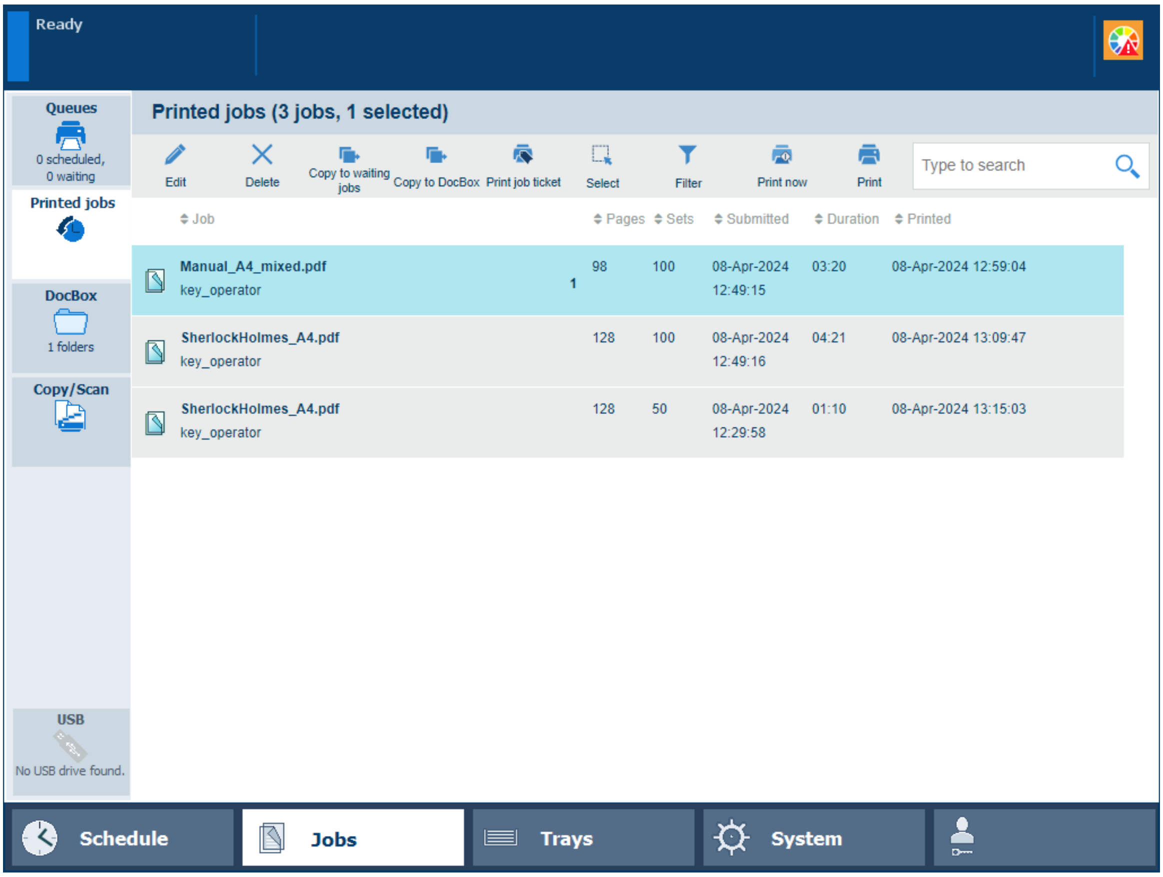
The schedule gives you up to eight hours of plan-ahead predictability into the print production. It tells you everything you need to know to avoid an idle print system.



List of scheduled jobs

Printed jobs

When a job is done, it is visible in the list of printed jobs. To re-print a printed job, you must select a printed job from the printed job list and touch the [Print] option. Via this option printed jobs can easily be re-directed to the scheduled job list. Printed jobs remain available after the print system is shut down. Proof prints or stopped jobs are not visible in the list of printed jobs.



List of printed jobs

In the following situations, you are not able to reprint jobs:
- The list of printed jobs is disabled in the Settings Editor.
- The print system is in the transaction printing mode.
- PRISMAsync already removed the print jobs from the list.

More information

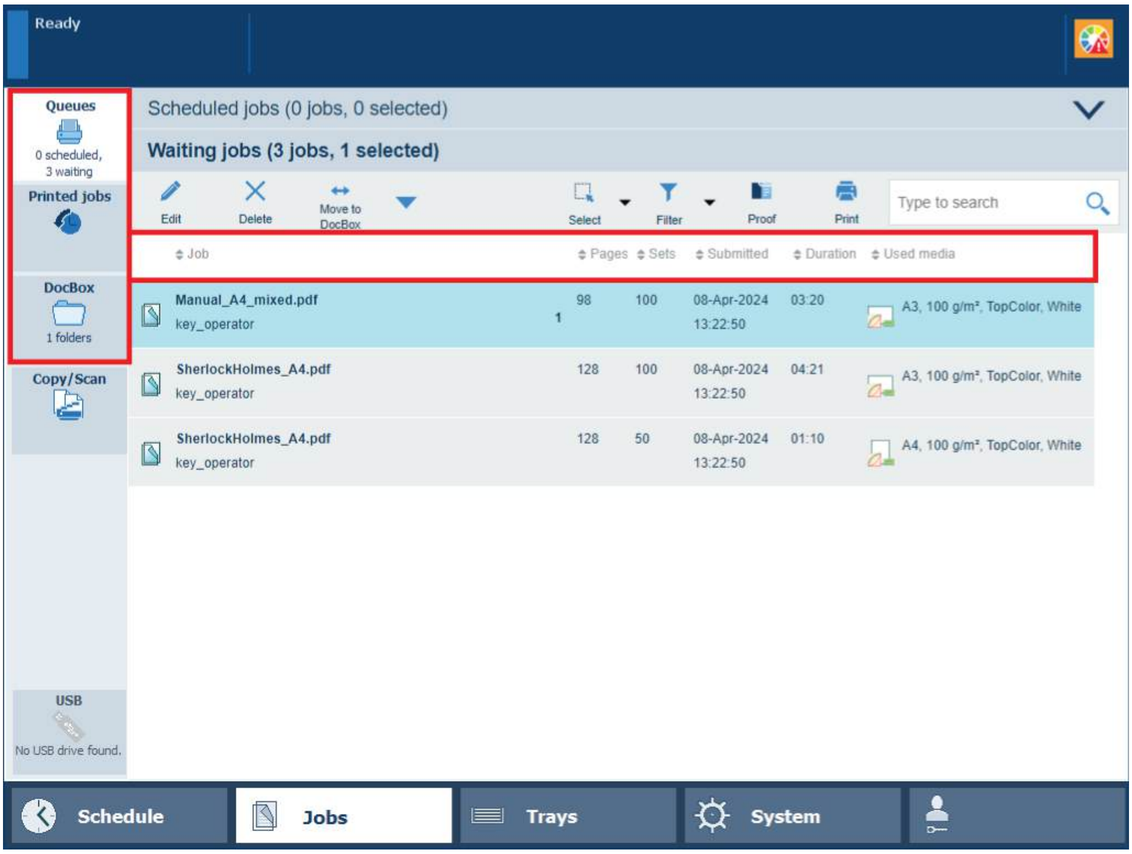
- *Choose a workflow profile* on page 240
- *Work with the schedule in document printing mode* on page 243
- *Learn about the document printing workflow* on page 236

Find, filter, and select jobs

To organize the print workflow to your preferences, you can find, filter and select print jobs.

Go to the job locations

1. Touch [Jobs].
2. Select the required job location:
   - [Queues]: the list of scheduled and the list waiting jobs.
     The waiting and scheduled jobs cannot be displayed in one view. Touch or to switch between the two lists.
   - [DocBox]: the DocBox folders.
   - [Printed jobs]: the list of printed jobs, when enabled.



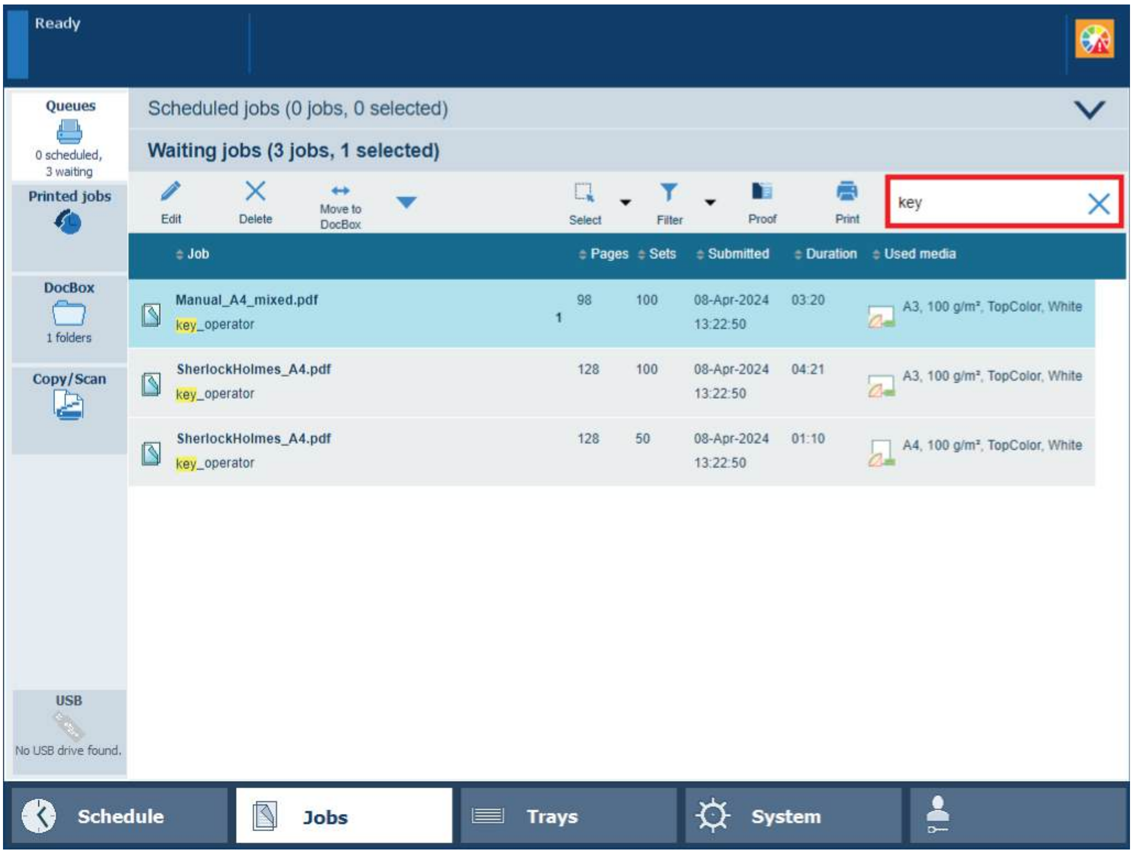
Job locations and information about jobs

The jobs show information about:
- [Job]: job name and sender name.
- [Pages]: number of source file pages.
- [Sets]: number of sets.
- [Submitted]: date and time of job submission.
- [Duration]: estimated print duration, based on media size, number of sheets sides to be printed, and the number of sets.
- [Used media]: job media.
- The bold number indicate how many jobs are currently selected.
- The job symbols explain the status of the job. The media symbols explain the status of the media.

Find jobs

The search function makes it possible to find jobs with a specific attribute, for example job name, sender's name, or media.

1. Go to the job location.
2. Enter the search text in the search field

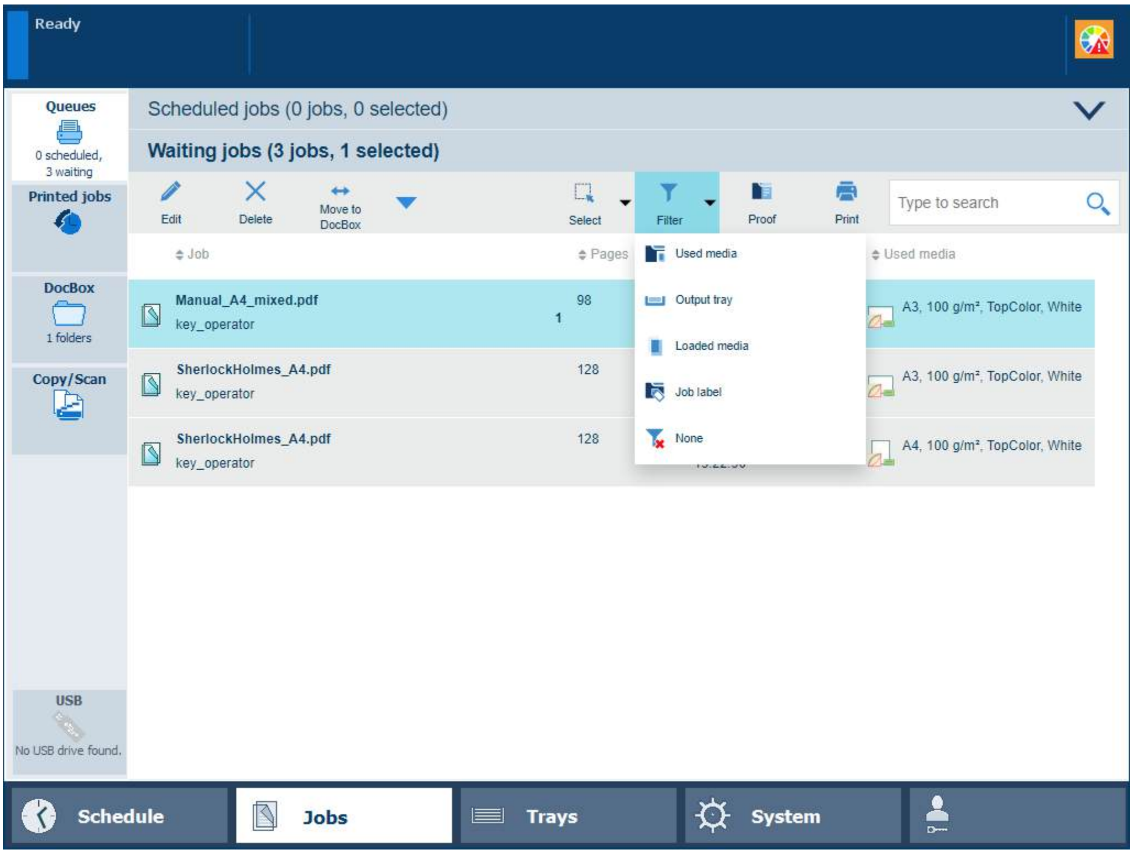


All jobs of a specific user in the list of waiting jobs

Filter jobs

Job filtering is useful when you want to limit the amount of jobs to select from.

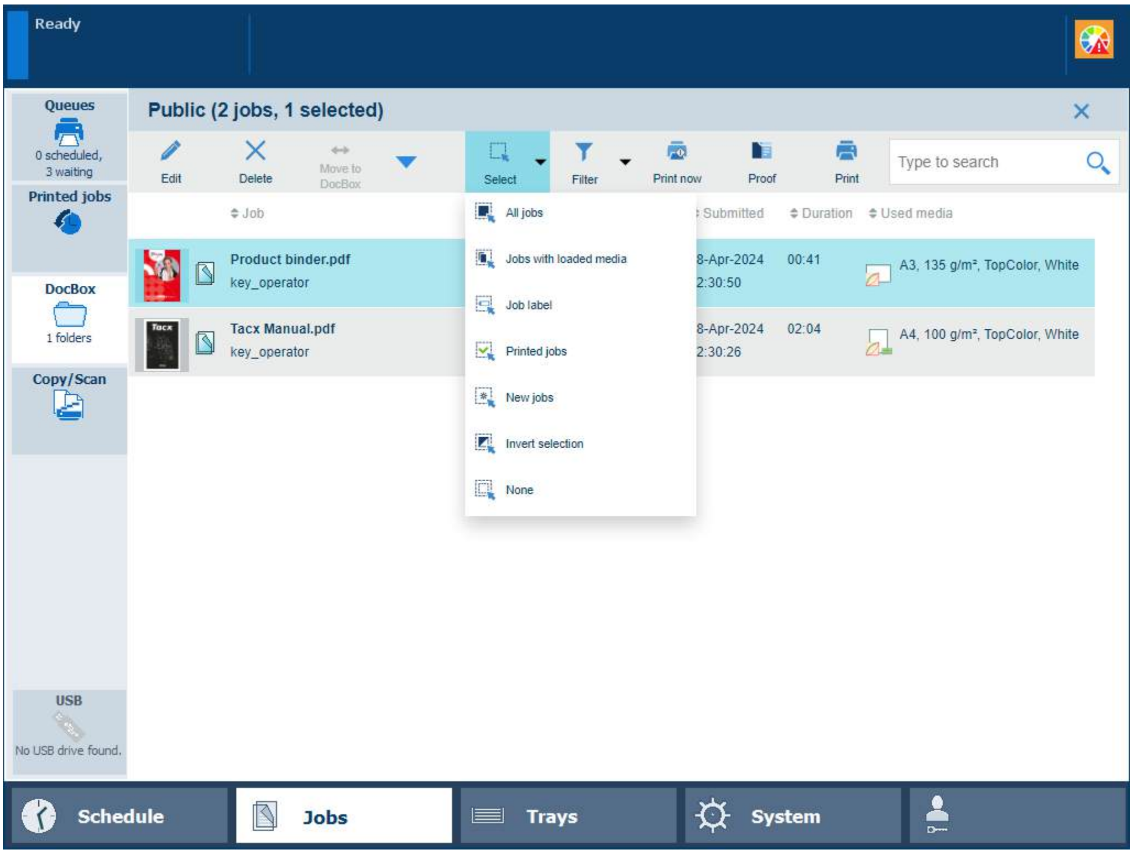
1. Go to the job location.
2. Select the required filter option:
   - [Used media]: jobs that use certain media.
   - [Output tray]: jobs that use a certain output tray.
   - [Loaded media]: jobs that use media that are in paper trays.
   - [Job label]: jobs that use a specific job label.
     The job label refers to the name of the applied automated workflow.
   - [None]: remove the filter and show all jobs in the job location.



The filter options in the list of waiting jobs

Select jobs

1. Go to the job location.
2. Select and deselect jobs by tapping the job entries or use the [Select] menu to select the required selection option:
   - [All jobs]: all jobs in the job destination.
   - [Jobs with loaded media]: jobs that use media that are in the paper trays.
   - [Job label]: jobs with a specific job label.
     The job label refers to the name of the applied automated workflow.
   - [Invert selection]: select the unselected jobs and unselect the selected jobs.
   - [None]: no jobs.
   - [New jobs]: new jobs in the DocBox folder.
   - [Printed jobs]: printed jobs in the DocBox folder.
   - Touch and hold a job to bring a multiple job selection back to one job.



The select options in the DocBox folder

More information

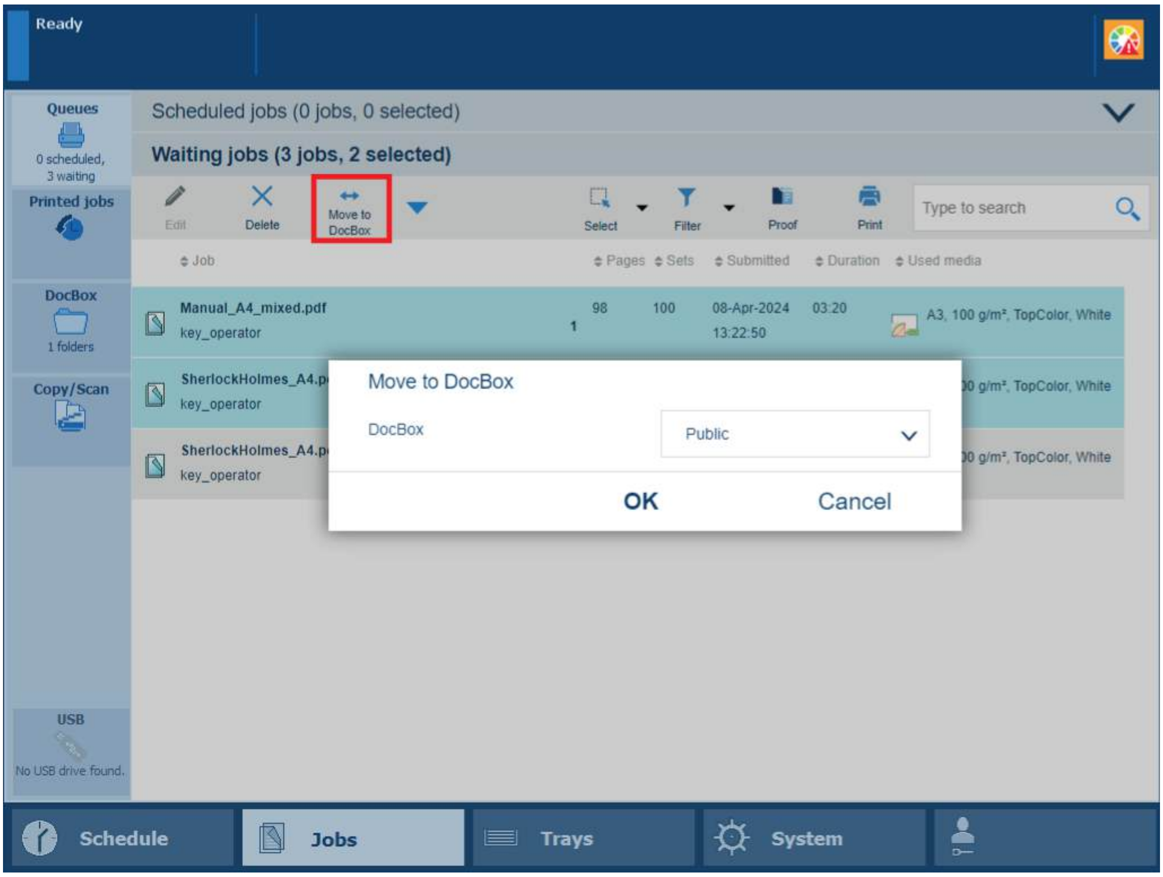
Learn about job management in the queues on page 277

Move jobs to another destination

The destination of a print job is defined during the job preparation. After arrival on the printer, you can decide to move one or more jobs to another location. For example, because you want to print a job later or want to store a job. You can also decide to store or reprint an already printed job.

Move a job to a DocBox folder

Move the selected jobs to a (different) DocBox folder.



Move a waiting job to a DocBox folder

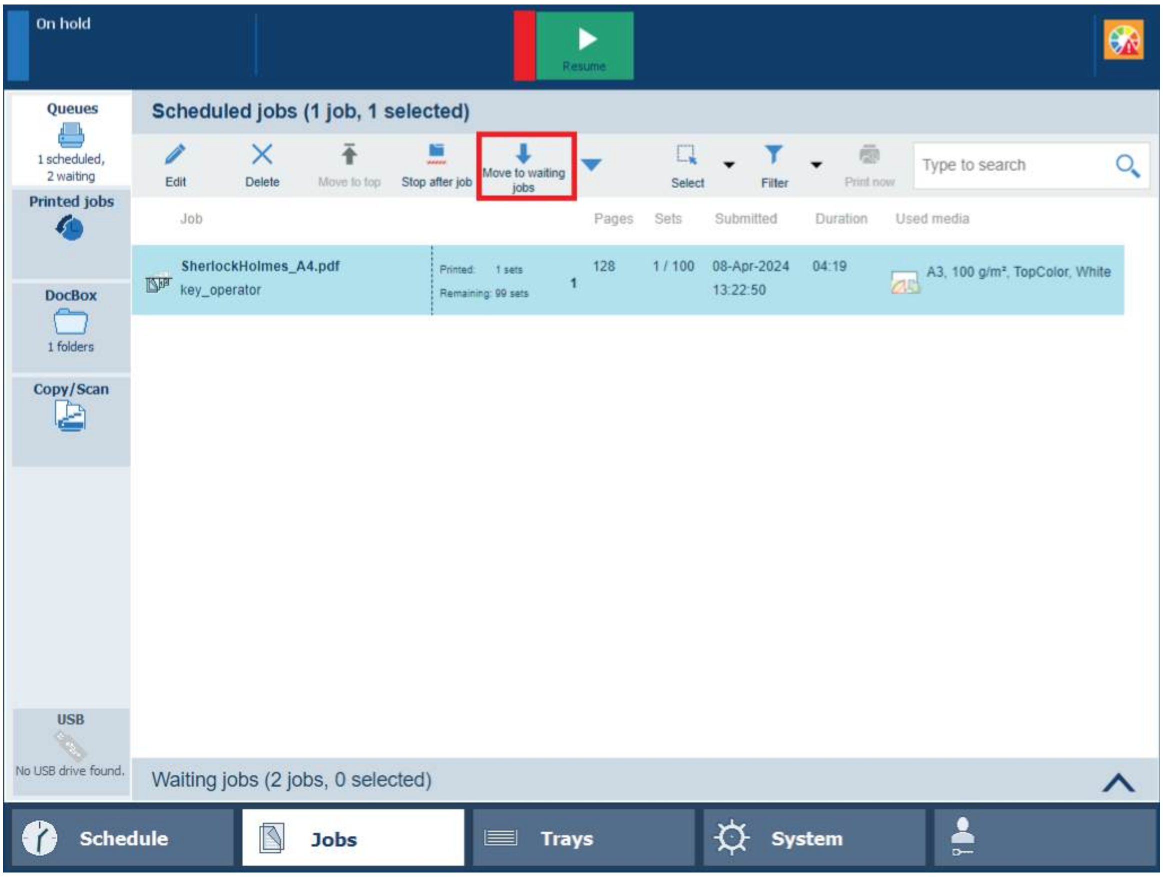
You can move jobs to a DocBox folder from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Move to DocBox]	✔	✔	✔	

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the jobs you want to move.
4. Touch [Move to DocBox].
5. Select the required DocBox folder
6. Touch [OK].

Move a job to the waiting jobs

Move the selected jobs to the list of waiting jobs



Move scheduled job to waiting jobs

You can move jobs to the list of waiting jobs from the following locations:

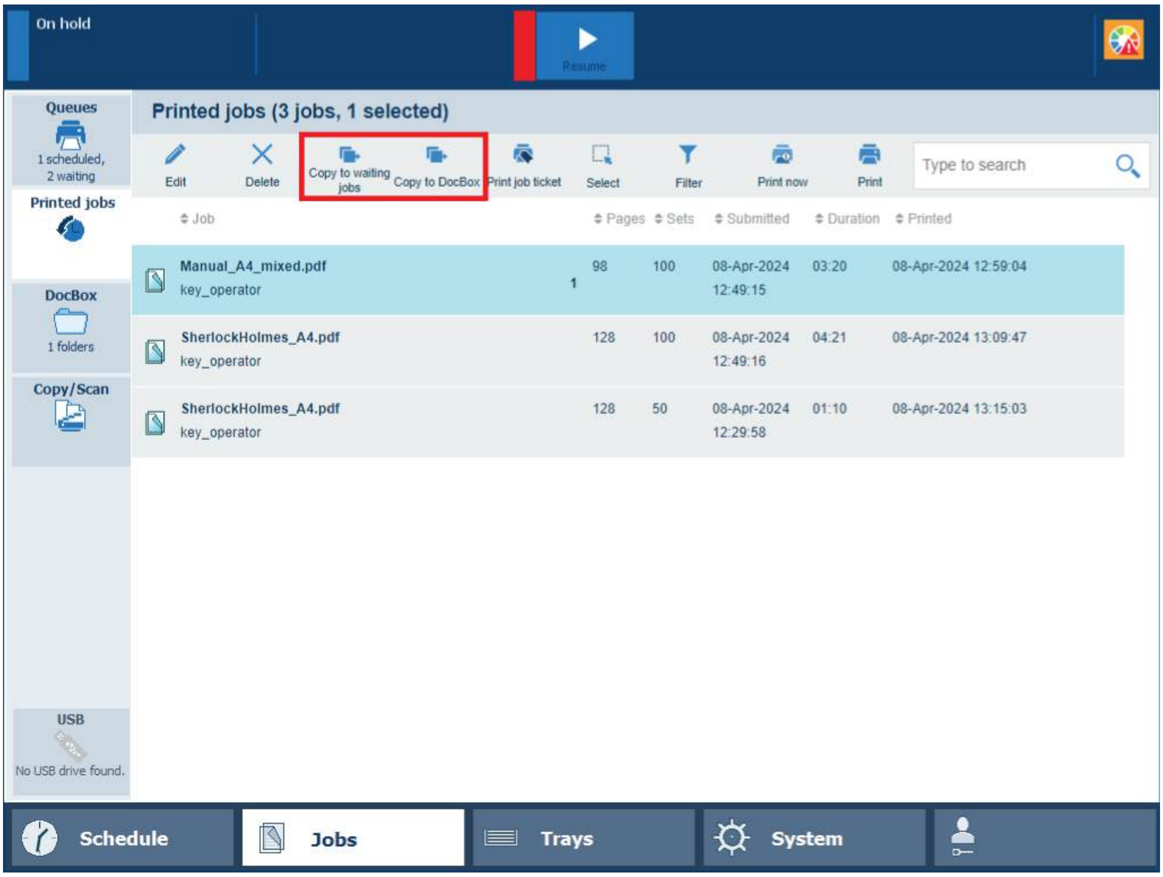
Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Move to waiting jobs]	✓			

NOTE
When you want to move the active job to the list of waiting jobs, you must first press the [Stop] button twice to stop the job.

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the jobs you want to move.
4. Touch [Move to waiting jobs].

Make a copy and move a printed job

Make a copy of a selected job and move the copy to a DocBox folder or the list of waiting jobs.



Copy printed job to DocBox or waiting jobs

You can copy and move printed jobs from the following location:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Copy to waiting jobs]				✓
[Copy to DocBox]				✓

1. Touch [Jobs]→[Printed jobs].
2. Select the jobs you want to copy and move.
3. Select the location for the selected job:
   - Touch [Copy to waiting jobs] to make a copy and move the job to the list of waiting jobs.
   - Touch [Copy to DocBox] to make a copy and move the job to a DocBox folder.
     Select the DocBox folder and touch [OK].

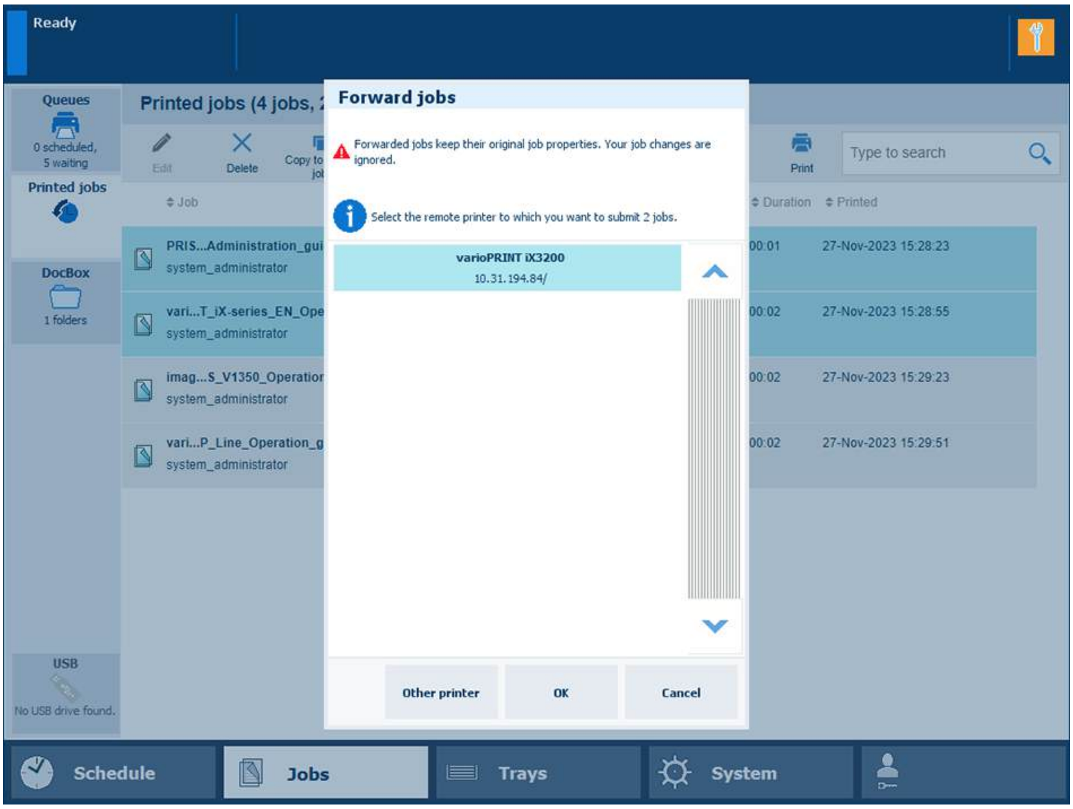
Forward a job to another printer

When there are remote printers configured in the Settings Editor, you can forward jobs to these printers. Forwarded jobs keep the properties that were defined during the job preparation.

You can forward jobs from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Forward]	✓	✓	✓	✓

1. Touch [Jobs]→[Printed jobs].
2. Go to the location of the job.
3. Select the jobs you want to forward.
4. Touch [Forward].
5. Select a printer from the list or enter another printer name.



[Forward jobs] to other printer

6. Touch [OK].

More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281

Combine jobs

When do you combine jobs

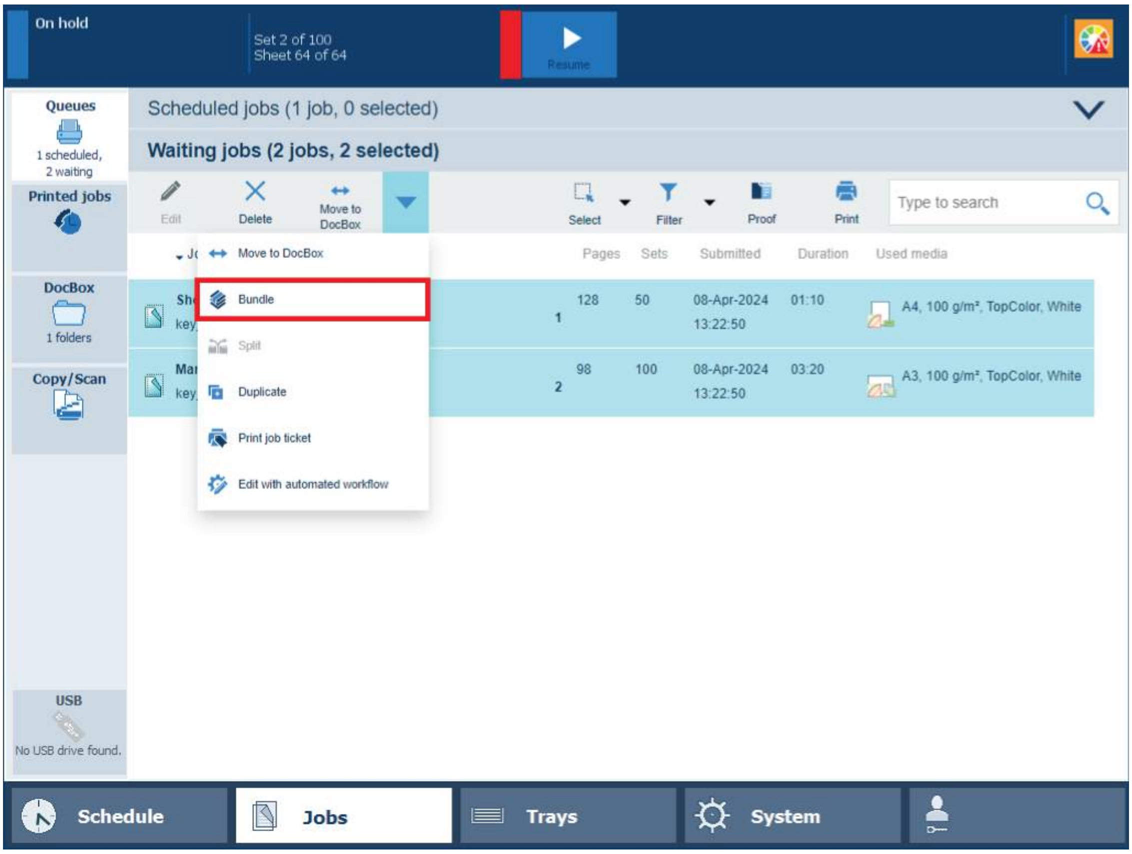
To increase the print productivity, you can combine several waiting jobs or DocBox jobs into a single job. You can change some properties of the combined job and the job order.

The original jobs are no longer visible in the location. However, you can always recover the original jobs with the [Split] option.

The combined job gets the name of the first individual job.

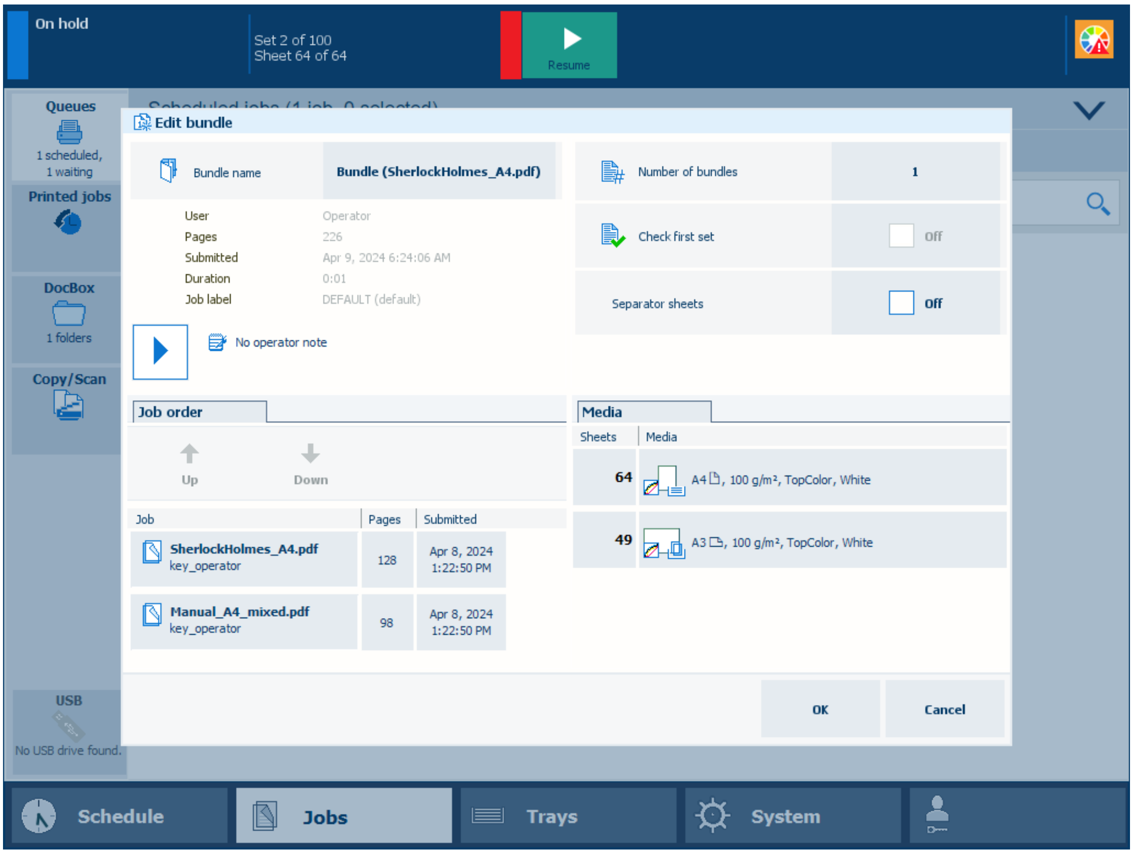
Combine waiting jobs

1. Go to the list of waiting jobs.
2. Select the jobs you want to combine.
3. Touch [Bundle].



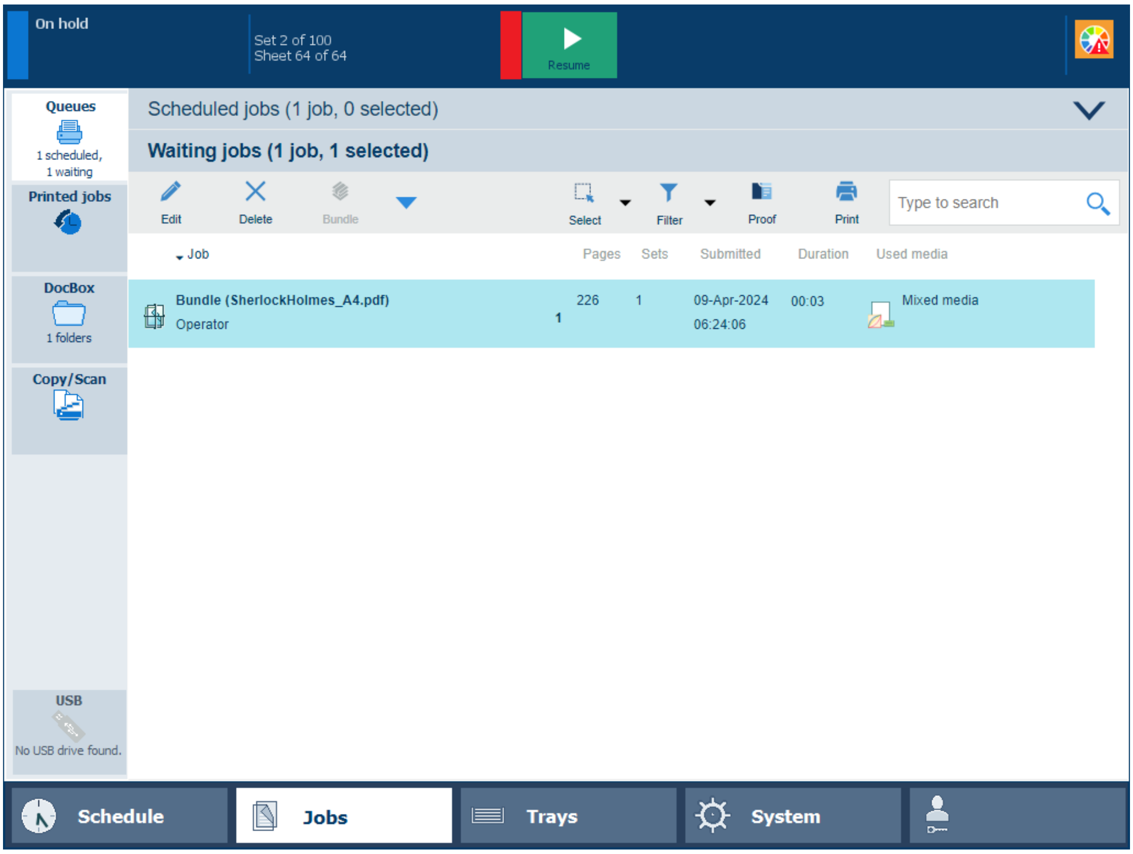
Bundle jobs

4. If required, change the job name, the job order, and other properties.



Properties of a bundle job

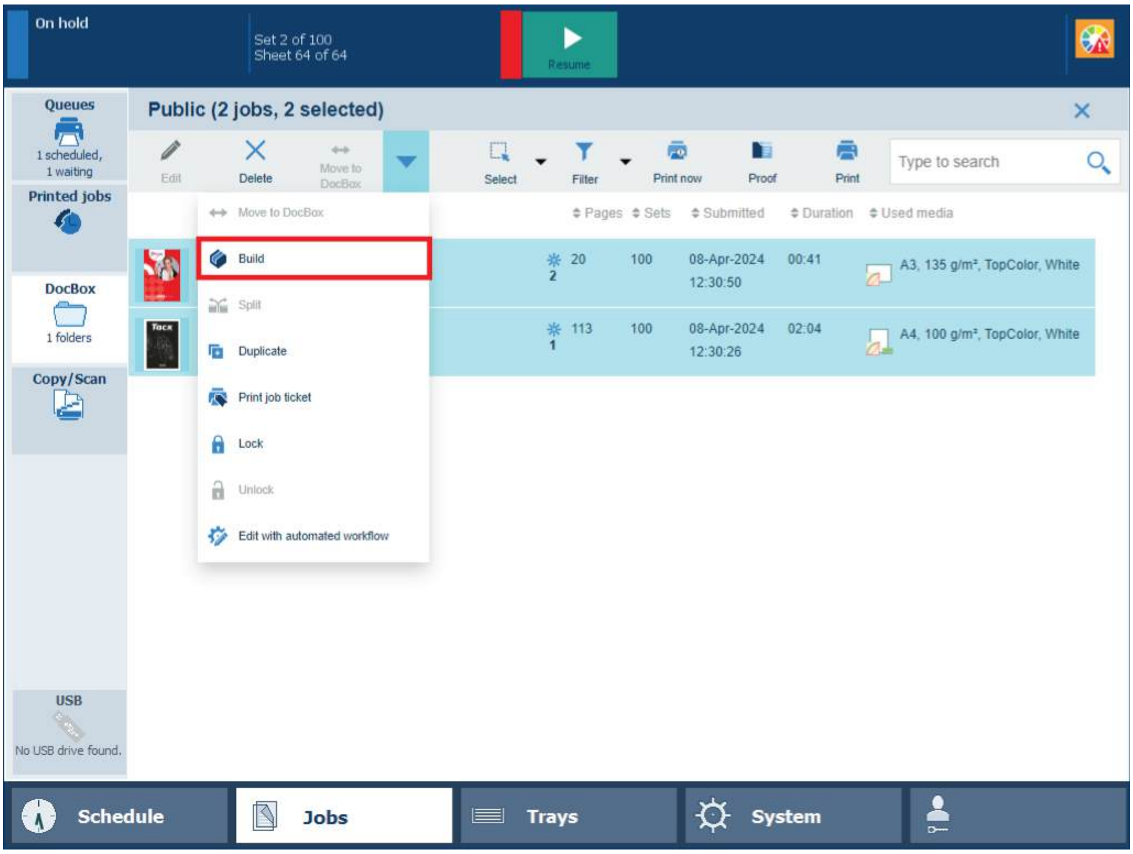
5. Touch [OK].
6. The job is in the list.



Bundle job

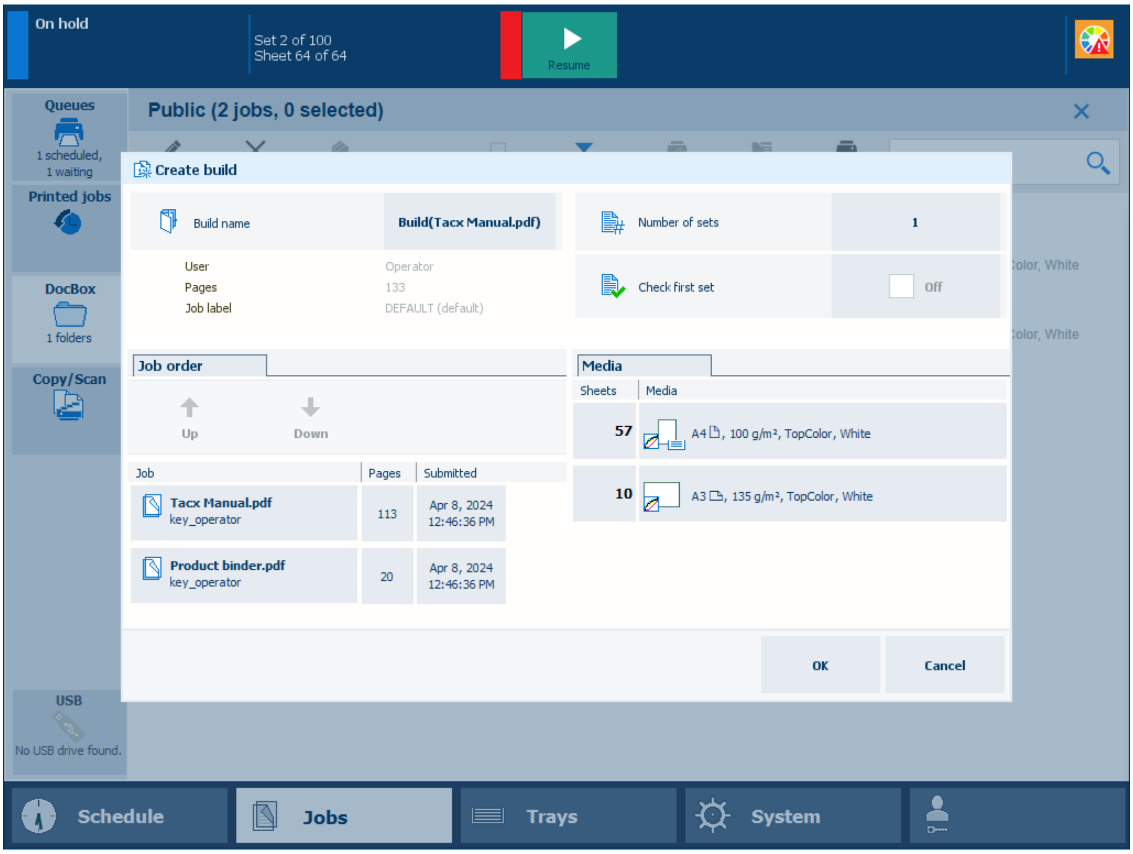
Combine DocBox jobs

1. Go to a DocBox folder.
2. Select the jobs you want to combine.
3. Touch [Build].



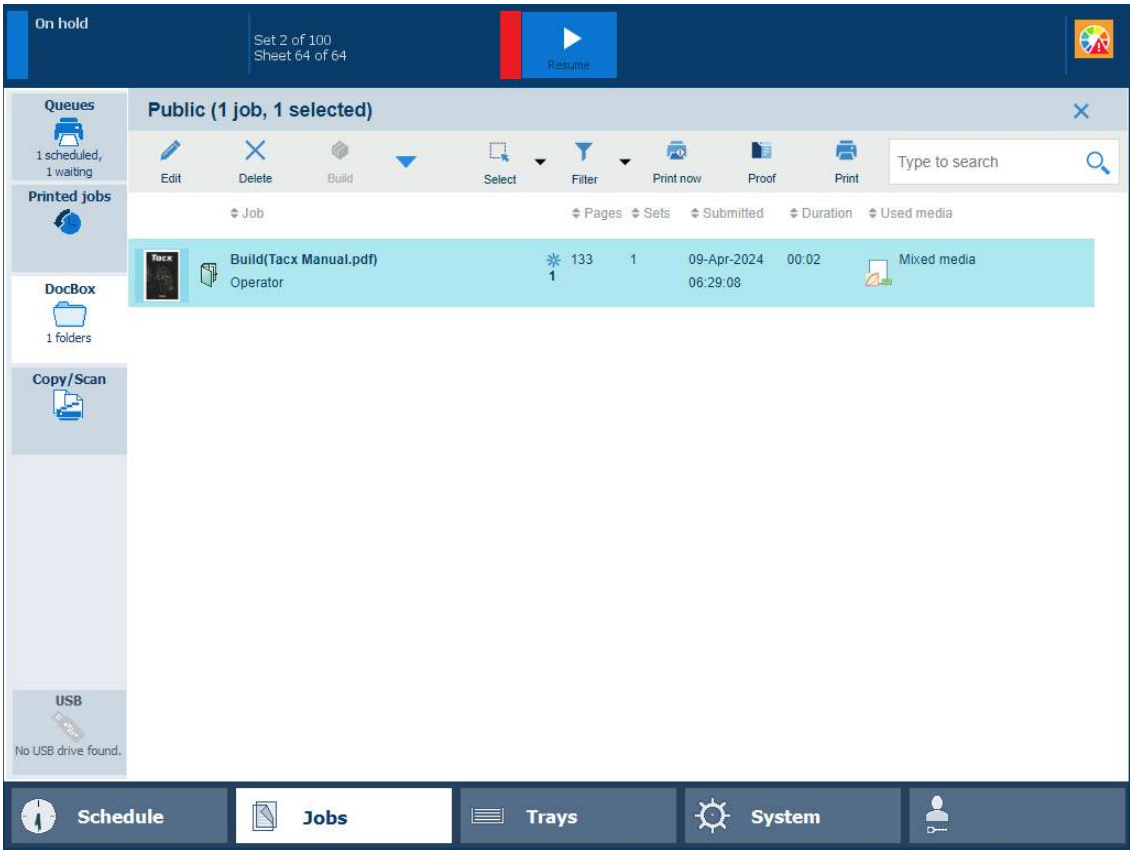
Create build

4. If required, change the job name, the job order, and other properties.



Properties of a build job

5. Touch [OK].
6. The job is in the list.



Build job

Split combined jobs into the original jobs

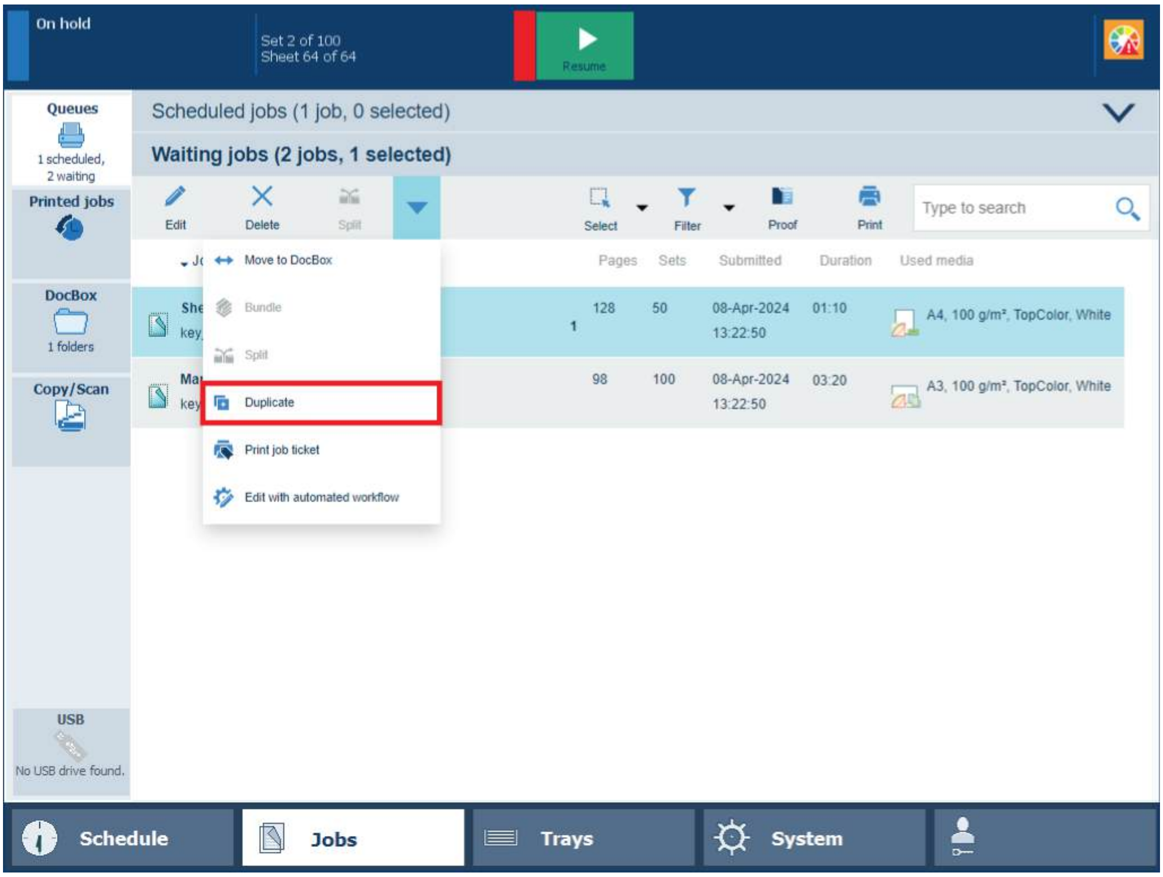
1. Select the combined job.
2. Touch [Split].

More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281

Duplicate jobs

When you want to use different print settings for the same source file, you can duplicate a job.



Duplicate jobs

You can duplicate jobs at the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Duplicate]		✔	✔	

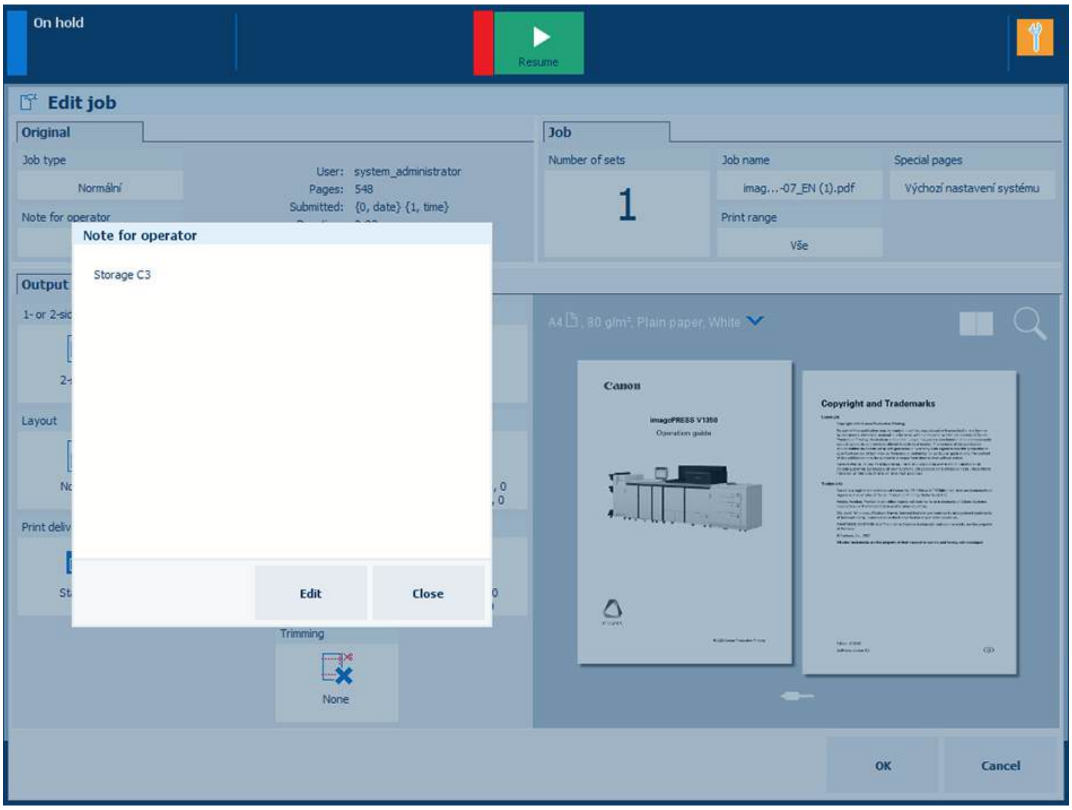
1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to duplicate.
4. Touch [Duplicate].

More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281

Create a note for the operator

You can create a note for the operator on several locations.



[Note for operator] on the control panel

A job that has a note for the operator has the symbol.

1. Touch the job.
2. In the [Original] pane, touch [Note for operator].
3. Touch [Edit].
4. Enter a message for the operator.
5. Touch [OK].
6. Touch [Close].
7. Touch [OK].

More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281

Delete print jobs

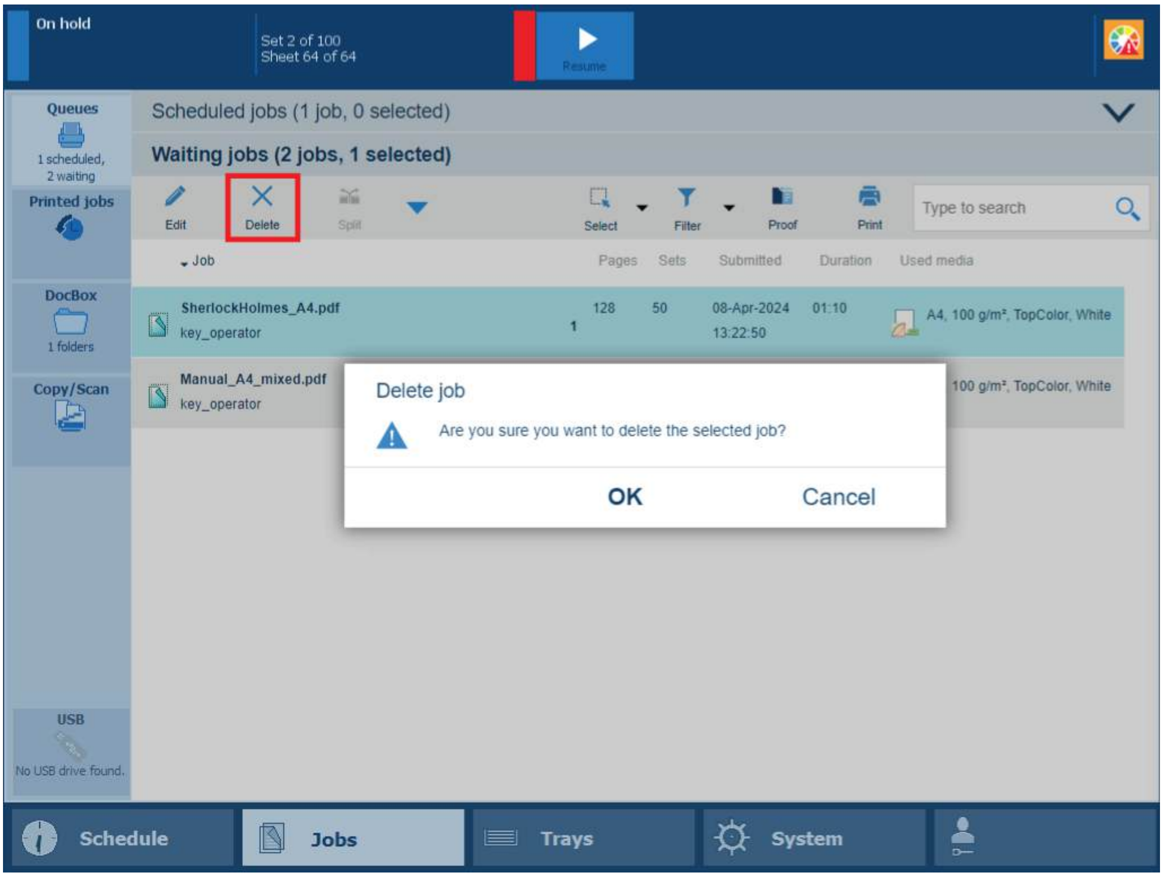
When do you delete jobs

You can remove print jobs before or after they are printed.

For security reasons, the system administrator can enable the E-shredding option. E-shredding overwrites the deleted job data and prevents the data recovery of a removed job.

When the control panel shows the printed jobs, you can reprint jobs. Printed jobs can automatically be removed by the printer after a certain storage period. When the printer does not remove printed jobs automatically, remove these jobs manually to prevent a full system disk.

Jobs that have not been printed completely and proof prints do not come in the list of printed jobs.



Delete function and confirmation window in list of waiting jobs

You can delete jobs from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Delete]	✓	✓	✓	✓

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the jobs you want to delete.
4. Touch [Delete].

5. Confirm the deletion action.

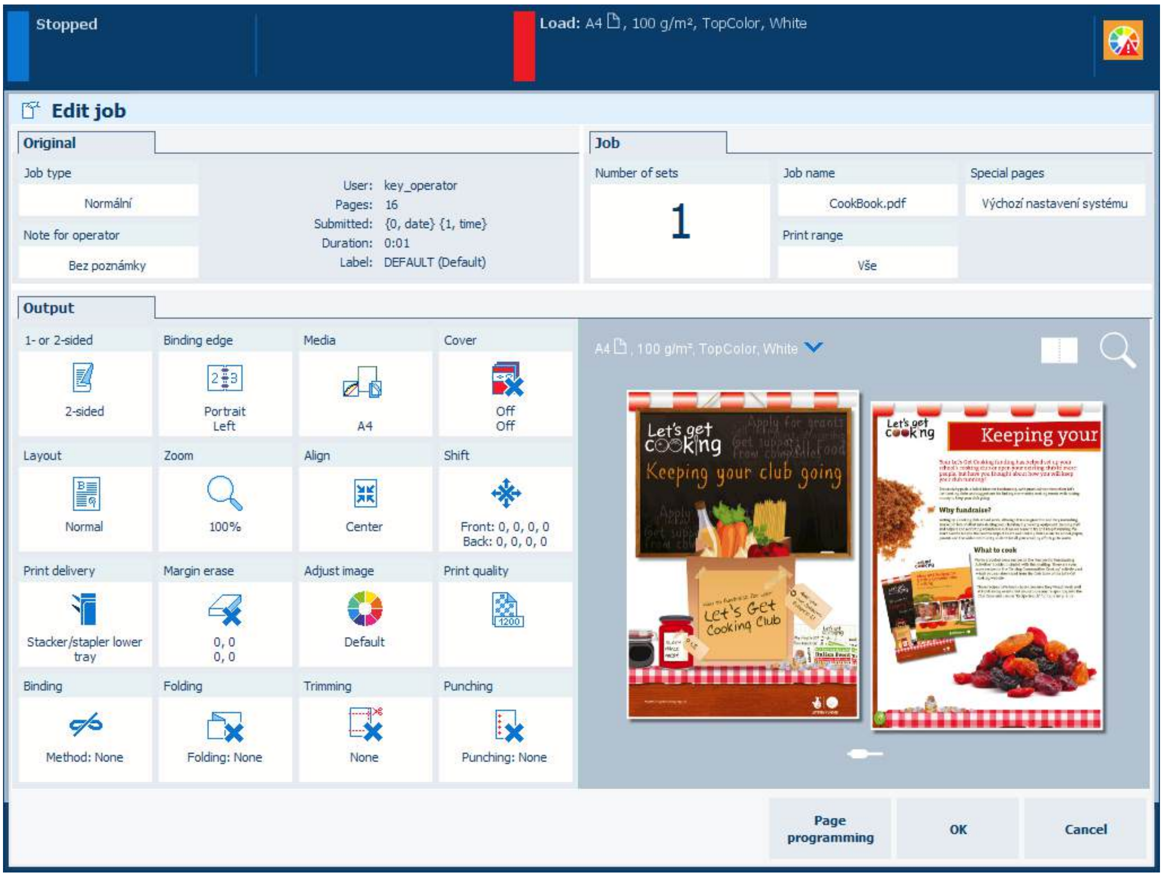
More information

- *Learn about job management in the queues* on page 277.
- *Find, filter, and select jobs* on page 281.

Change jobs

Learn about the print job settings

Print job settings are available in PRISMAsync Remote Printer Driver, PRISMAremote Manager and on the control panel.



Print job settings on the control panel

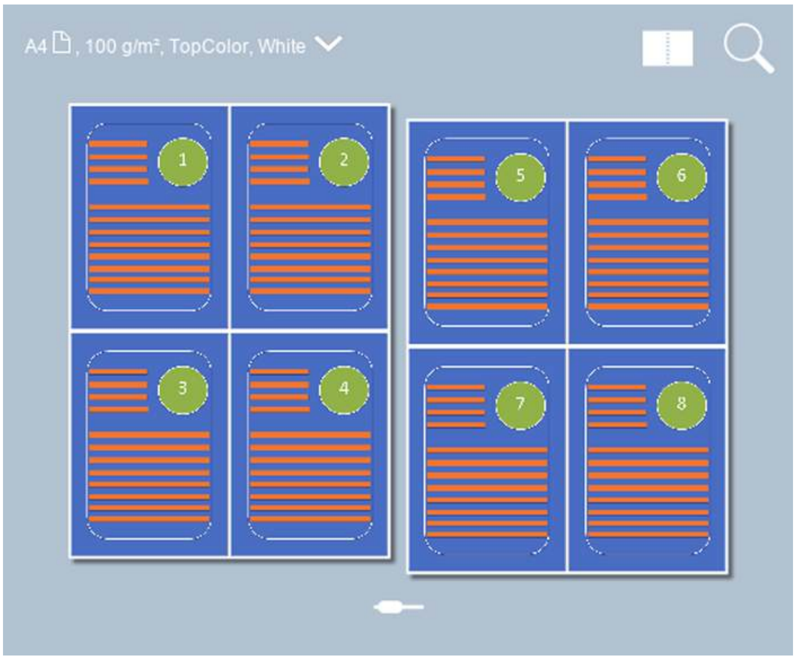
[Original] and [Job] settings

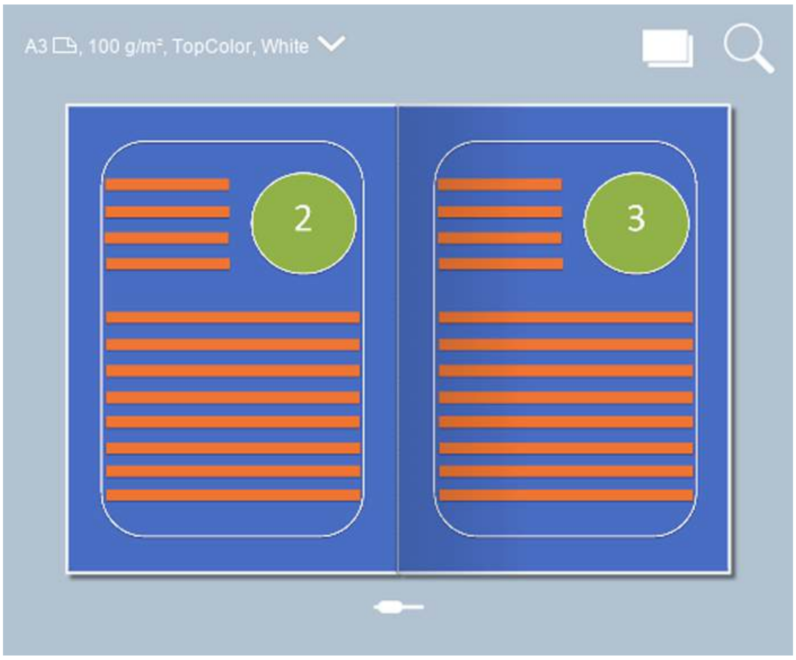
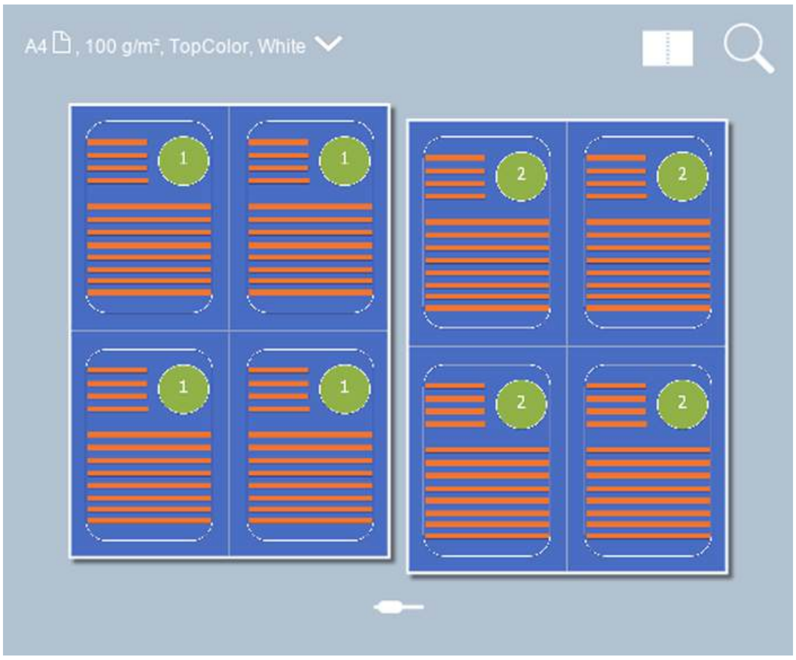
Job setting	What you can define
[Job type]	**Job type** Indicate the type of job. • [Normal] • [Variable data]
[Note for operator]	**Job instruction** Print or finishing instruction for the operator.
[Number of sets] [Check first set]	**Number of sets** Indicate the number of sets: • Number of sets (maximum 65,000). • Indicate if only the first set needs to be printed.

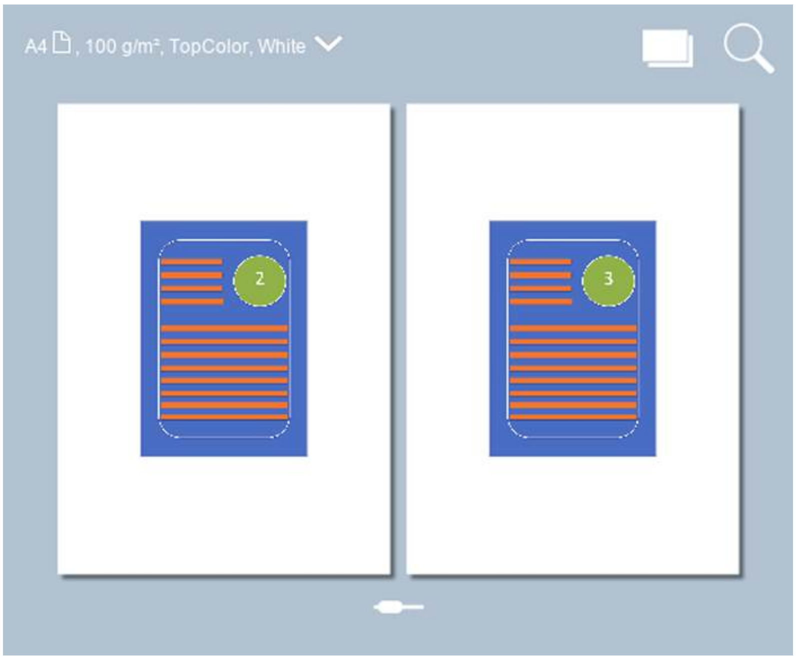
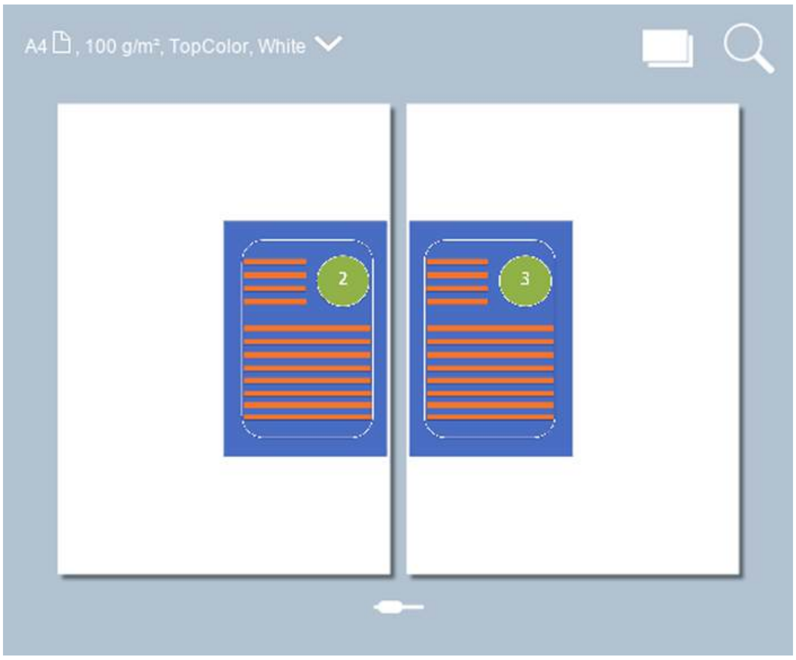
Job setting	What you can define
[Job name]	**Job name** Enter a new name for the job.
[Print range]	**Print range** Indicate if you want to print all pages or records, or want to select a print range.
[Special pages]	• **[Overrule default banner / trailer page settings]**: indicate if you want to overrule the default banner page, trailer page and separator sheets settings. • **[Use a banner page]**: indicate if the job must include a banner page. The banner page contains: the sender name, the recipient name, the accounting ID, the job name, the printer name, the operator instruction, the cost centre and the number of sets. • **[Use a trailer page]**: indicate if the job must include a trailer page. The trailer page contains: the username, the recipient name, accounting ID, the job name, the printer name, the operator instruction, the number of pages in a set , the number of sets, the number of staples, the number of folded sheets, the number of punched sheets, the number of creased sheets, the number of inserts, job received time, job start time, job completion time, the number of sheets per job media. • **[Media of banner/trailer page]**: define the media for the banner and trailer page. • **[Use separator sheets]**: indicate the use of separator sheets to separate job sets. Separator sheets are not printed. • **[Separator sheet after N sets]**: define after how many sets you want a separator sheet. • **[Media of separator sheets]**: define the media for the separator sheet.
[Accounting ID]	**Accounting information** Enter information for billing and charging job costs: • [Accounting ID], to identify a user or a group of users. • [Cost center], to identify an organizational department, project, or group. • [Note], to enter additional information required for accounting.

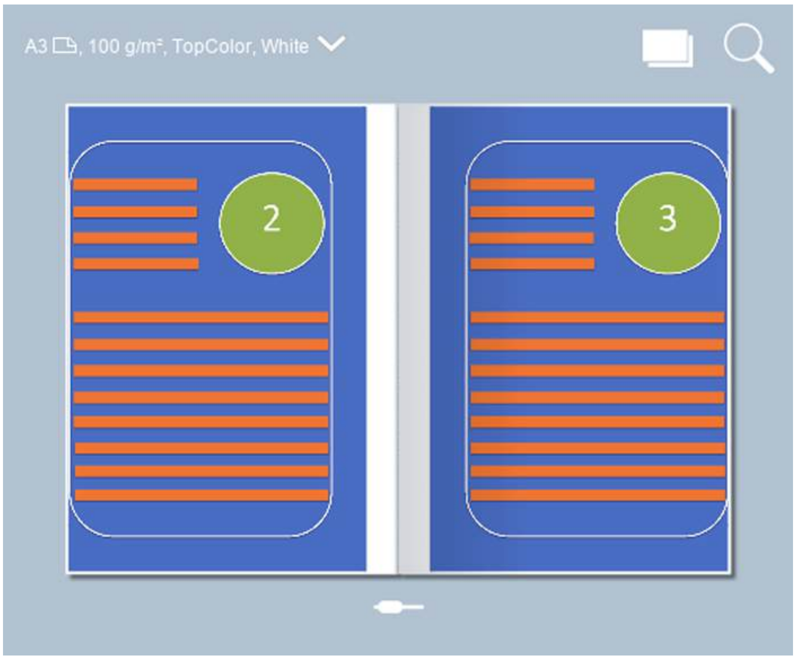
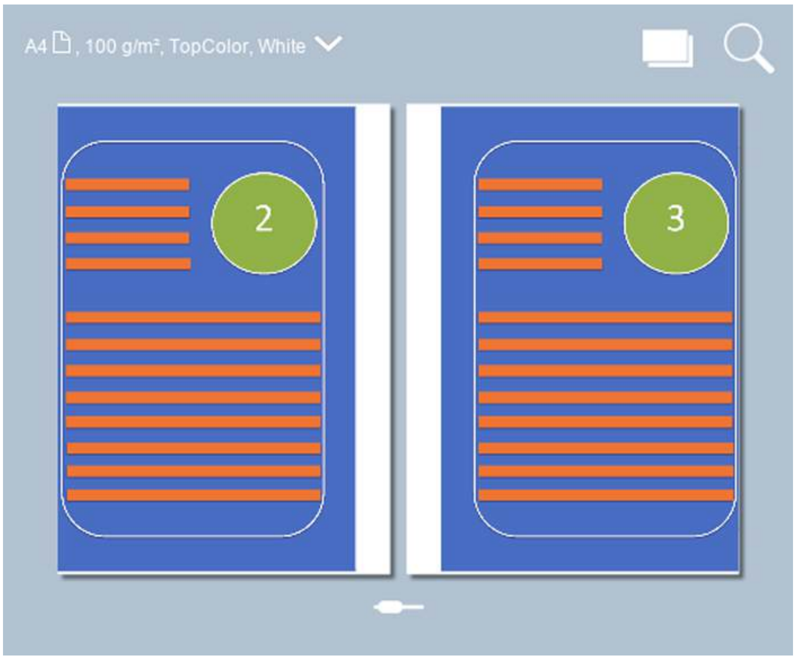
Output settings

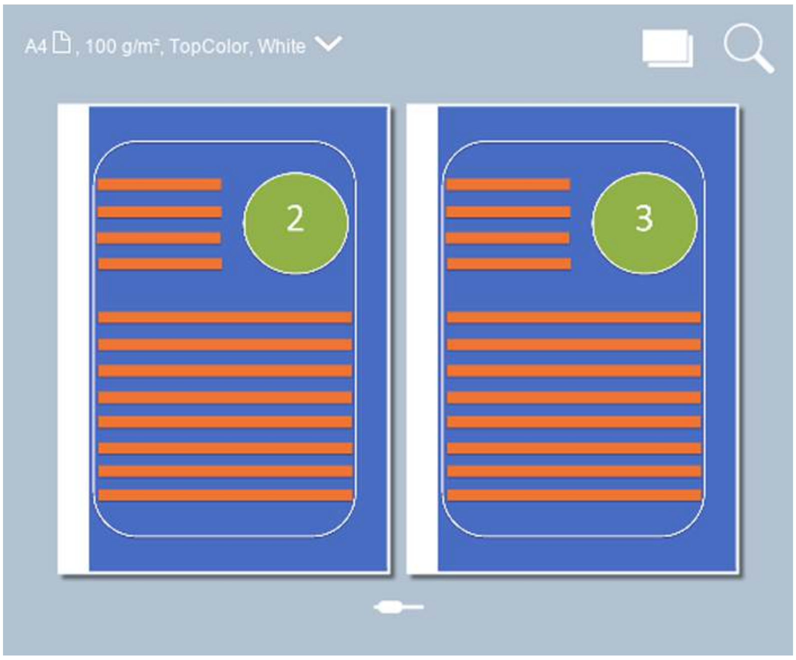
Job setting	What you can define
[1- or 2-sided]	**Sides to print on** Indicate if the prints are printed on one or both sides.
[Binding edge]	**Orientation and binding location** The orientation of the content is part of the original settings and cannot be changed. You can change the binding edge: the sheet edge to which the previous page is bound.
[Media]	**Media** Select the media for the job.

Job setting	What you can define
[Cover]	**Usage of covers** Indicate if you want to use covers and select the cover media. You can also indicate if you want to print on one or both sides of the front and back cover. The default media that are used for covers are configured in the Settings Editor.
[Layout]→[Multiple up]	**Multiple pages on a sheet** Indicate if you want to print subsequent source file pages on one document page.  Imposition example of [Layout]→[Multiple up] and [Zoom]→[Fit to page] option (sheet view)

Job setting	What you can define
[Layout]→[Imposition template]	Indicate if you want to fold a document: once to create a booklet, twice to create a quarto or three times to create an octavo. The sheets are printed in such a way that the pages of the folded document are presented in sequential order.  Imposition example of A4 document on A3 media with [Layout]→[Booklet] option (document view)
[Layout] →[Same-up]	**Same page multiple times on a sheet** Indicate if you want to print the same source file page multiple times on a document page  Imposition example of [Layout] →[Same-up] and [Zoom]→[Fit to page] option (sheet view)

Job setting	What you can define
[Zoom]	**Scaling factor** Indicate if you want to scale the image according to the page size, or set the zoom factor manually (25% - 400%).  Imposition example of [Zoom]→[50%] option (document view)
[Align]	**Alignment of the image to a sheet location** Indicate how you want to align the image on the sheet: to a corner, edge or to the middle of a sheet.  Imposition example of the [Align]→[Center left] and [Zoom]→[50%] option (document view)

Job setting	What you can define
[Shift]→[Margin shift]	**Margin shift in the source document** Indicate if you want to shift the margin of the source document to create more space at binding edges, with a maximum shift of 100 mm (3.94 inch).  Imposition example of A4 document on A3 media with [Booklet] and [Margin shift] option (document view)
[Shift]→[Image shift]	**Image shift on the sheet** Indicate if you want to shift the image on the sheet, with a maximum shift of 100 mm (3.94 inch).  Imposition example with [Image shift]→[Front side] and [Image shift]→[Back side] option (document view)

Job setting	What you can define
[Margin erase]	**Erase the margin of the document** Indicate if you want to erase the margins of the image. When you adjust the margin erase you erase areas of the image, with a maximum of 100 mm (3.94 inch).  Imposition example with [Margin erase] option (document view)

[Print delivery] settings

Job setting	What you can define
[Print delivery]→ [Output tray]	**Output tray** Indicate which output tray to use. The selected finishing method can require a specific output tray: • [Stacker/stapler booklet tray]: for booklets. • [eWire tray]: for wire-bound books with the eWire. Use this option together with the [Punching]→[Die set dependent] option to deliver punched sets of sheets to the eWire. • Name of external finisher: for finishing with a DFD (Document Finishing Device). You can define the name of the DFD in the Settings Editor. Location: [Preferences]→[Print job defaults]→[External finisher]→[Name]. • [Perfect Binder tray]: only visible when selected by PRISMAprepare or PRISMAsync Remote Printer Driver.
[Print delivery]→ [Sort]	**Sorting method** Indicate how you want to sort the output, [By page] or [By set]
[Print delivery]→ [Offset stacking]	**Stacking method** Indicate if you want stacking with or without an offset. Offset stacking depends on the active workflow profile.

Job setting	What you can define
[Print delivery]→[Advanced settings]	**Delivery method** Indicate the delivery method of the sheets in the output tray of your choice: • [Print order], indicate the delivery order of the sheets. • [Sheet orientation], indicate which sheet orientation is applied. • [Rotation], indicate if the sheets are rotated in the output tray.

Print quality settings

Job setting	What you can define
[Adjust image]	**Adjustment of CMYK values on job level** Adjust the CMYK curves on job level when needed. The PRISMAsync Print Server colour management and print engine ensure optimal colour reproduction on the media you use. However, sometimes you want to edit CMYK values on job level for a better result.
[Print quality]→[Color preset]	**Selection of the colour preset** If required, select a default colour preset for the job: • [Office documents], predefined colour settings for the reproduction of text and graphical lines in office documents. • [Photographic content], predefined colour settings for the reproduction of photographs and images
[Print quality]→[Edit color settings]	**Definition and adjustment of the colour preset** Adjust the colour settings of the selected colour preset and save the colour preset if required. The colour presets are also created in the Settings Editor.
[Print quality]→[Edit color settings]: • [Input profile] • [Rendering intent] • [Standard rules CMYK saturation intent]	**Input profile and rendering intent** Indicate which input profiles and rendering intents you want to use. Furthermore, you can indicate if embedded profiles and embedded rendering intents must be overruled. For CMYK you can indicate if the colour management system must preserve pure process colours or comply to standard colour management rules.
[Print quality]→[Edit color settings]→[Spot colors]	**Spot colour matching** Indicate if you want to use the spot colour table that stores standard and custom spot colours. Spot colours are managed in the Settings Editor, but they can also be added with the spot colour editor.
[Print quality]→[Edit color settings]→[Halftones]	**Halftone selection** Indicate if you want to change the default selection of the halftones: • Halftone [Fine] for sharper texts or improvement of the colour gradient in colour areas. • Halftone [Normal] for text, images and graphics.

Job setting	What you can define
[Print quality]→[Edit color settings]→[Other settings]	**Other colour presets** There are several other colour settings, each for a specific situation: • **[Print black & white]**. Indicate if all colours are printed using the black channel only. • **[Use PDF overprint simulation]**. Indicate if you want that opaque objects look transparent. Then, the underlying objects get visible. • **[Use PDF/X output intent]**. Indicate if you want to print according to the embedded PDF/X output intent. Then, the Device CMYK definitions are ignored. • **[Overprinting black]**. Use this setting to force black objects to print over background colour. This option is used by the native PDF RIP to prevent registration artifacts. • **[Preserve pure black]**. Indicate if pure black preservation is applied when possible. Pure black preservation means that the colour black is composed of 100% K ink or toner. When pure black preservation is not possible or disabled, the colour black is composed of a mixture of C, M, Y, and K inks. • **[Black Point Compensation (BPC)]**. Black Point Compensation (BPC) is used in combination with the relative colourimetric rendering intent. Black Point Compensation (BPC) scales input colours relative to the output black in order to preserve details in dark areas. When the output black is rather light, [Enhanced BPC] is preferred over [Adobe BPC] because it has a better performance.
[Print quality]→[Resolution]	**Print resolution** Indicate the print resolution.
[Print quality]→[Color bar]	**Colour patches on the printed output** Indicate if you want to print an extra colour bar of colour patches on the printed output to check the ink density, dot grain and contrast. Colour bars are created in the Settings Editor.
[Print quality]→[Information bar]	**Values of used colour settings on the printed output** Indicate if you want to print the values of the used colour settings on the printed output. Information bars are created in the Settings Editor.

Job setting	What you can define
[Print quality]→[Quality adjustments]	**Print quality adjustments** There are several settings available to adjust the print quality: • **[Sensing unit adjustments]** (only if sensing unit is installed) Use this setting to select an automatic adjustment to be performed. • **[Mode for automatic adjustment]** (only if sensing unit is installed) Use this setting to select the mode for the automatic adjustment in the sensing unit. • **[Image smoothing]** Use this function when you want to apply an anti-aliasing algorithm to smooth images with a low resolution. • **[Moiré reduction for images]** Use this function when you want to apply a Moiré reduction algorithm to enhance photographic images. Please note that when images have a resolution below 300 dpi, the Moiré reduction only takes effect when the image smoothing function is enabled. • **[Trapping]** When you want to improve the object registration, you can select one of the available trapping presets or create a new trapping preset. • **[Color / resolution priority]** Use this setting for sharp edges when graphic objects with a high toner density may appear blurry at the edges of the graphic. The graphic objects may become less saturated. In general, a larger resolution improves the print quality but decreases the job productivity. • **[Print sharpness]** Use this setting to change the print resolution. In general, a larger resolution improves the print quality but influences the job productivity. • **[Edge enhancement for text]** Use this setting to improve the print quality for text. • **[Render large-sized text as graphics]** Use this setting to improve the print quality of large-sized text. This option keeps the benefits of using the text halftone screen for small text.
[Print quality]→[Page numbering]	**Page numbering** Indicate if you want to have page numbers on the printed documents.

Finishing settings

Job setting	What you can define
[Binding]	**Binding method** Indicate how to bind your document.
[Folding] [Creasing]	**Folding or creasing method** Indicate how to fold or crease the printed documents.

Job setting	What you can define
[Trimming]	**Trimming method** Indicate how to trim the printed documents.
[Punching] [Perforating]	**Punching or perforating method** Depending on the installed die set, select the method for punching or perforating.

More information

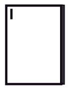
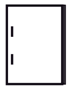
- *Define default use of special pages* on page 81
- *Choose a workflow profile* on page 240
- *Define a color bar* on page 491
- *Define a color preset* on page 440
- *Adjust CMYK curves of a job* on page 481
- *Learn about the color-based workflow* on page 434

Learn about finishing for your output

When the print system has finishing options, you select staple, punch, trim, booklet and book options in Remote Printer Driver, the PRISMA software, and the control panel.

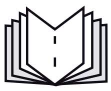
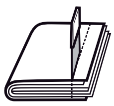
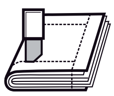
Below you find an overview of the finishing options the print system supports.

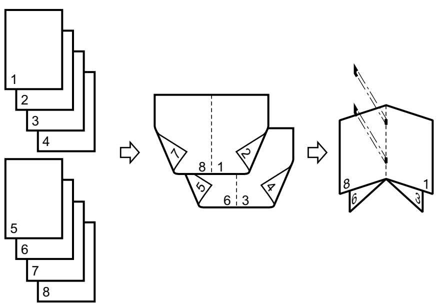
Stapling

Stapling options		Finishing optional
 Corner stapling	 2-side stapling	Stacker / stapler

The feed direction of originals is important for stapled copies.

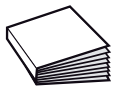
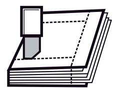
Booklet making

Booklet options	Finishing options
 Saddle-stitched booklet	Stacker / stapler Booklet maker
 Saddle-stitched booklet, leading-edge trimmed	Stacker / stapler Booklet maker Booklet trimmer
 Saddle-stitched booklet, trimmed at three sides	Stacker / stapler Booklet maker Booklet trimmer Two-knife booklet trimmer

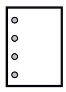
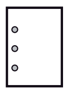
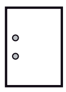


Print and page order of booklets

Perfect binding

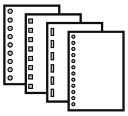
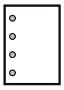
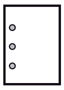
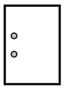
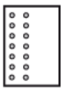
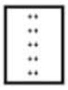
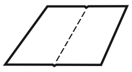
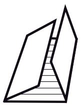
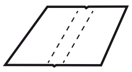
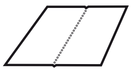
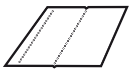
Perfect binding options		Finishing options
 Perfect bound book	 Perfect bound book, trimmed at three sides	Perfect binder

Punching

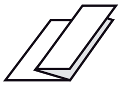
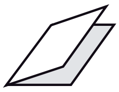
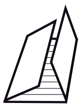
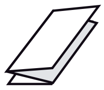
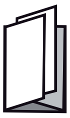
Punching options		Finishing options
 4-holes punching	 3-holes punching	Stacker / stapler with punch unit
 2-holes punching		

Professional punching, creasing, and perforation

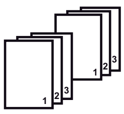
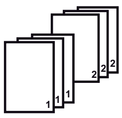
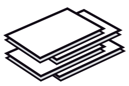
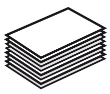
NOTE
If you are using the Multi Function Professional Puncher-C, select [Punching], [Perforating], or [Creasing] from the operator panel and change the required setting on the operation panel of the external puncher.

Professional punching, creasing, and perforation		Finishing options
 Punching with various punch patterns	 4-holes punching	Professional puncher
 3-holes punching	 2-holes punching	
Double punch 	Saddle punch 	
 Single crease, in the center	 C-fold crease	
 Book cover crease	 Single perforation, in the center	
 Double perforation	 Single perforation, at the edge	

Folding

Folding options		Finishing options
 Z-fold	 Half-fold	Folder
 Tri-fold in	 Tri-fold out	
 Parallel fold		

Stacking and sorting

Stacking and sorting		Finishing options
 Sort by set	 Sort by page	Stacker / stapler High capacity stacker
 Offset stacked	 Stacked	Stacker / stapler High capacity stacker

More information

Finishers specifications on page 624

Preview job settings

The job properties area of PRISMAsync includes a preview of the document to print. The preview exactly shows the results of current impositioning and print settings. You immediately see the effect of changing values. You can preview the settings remotely with PRISMAsync Remote Manager or locally on the control panel.

Realistic preview

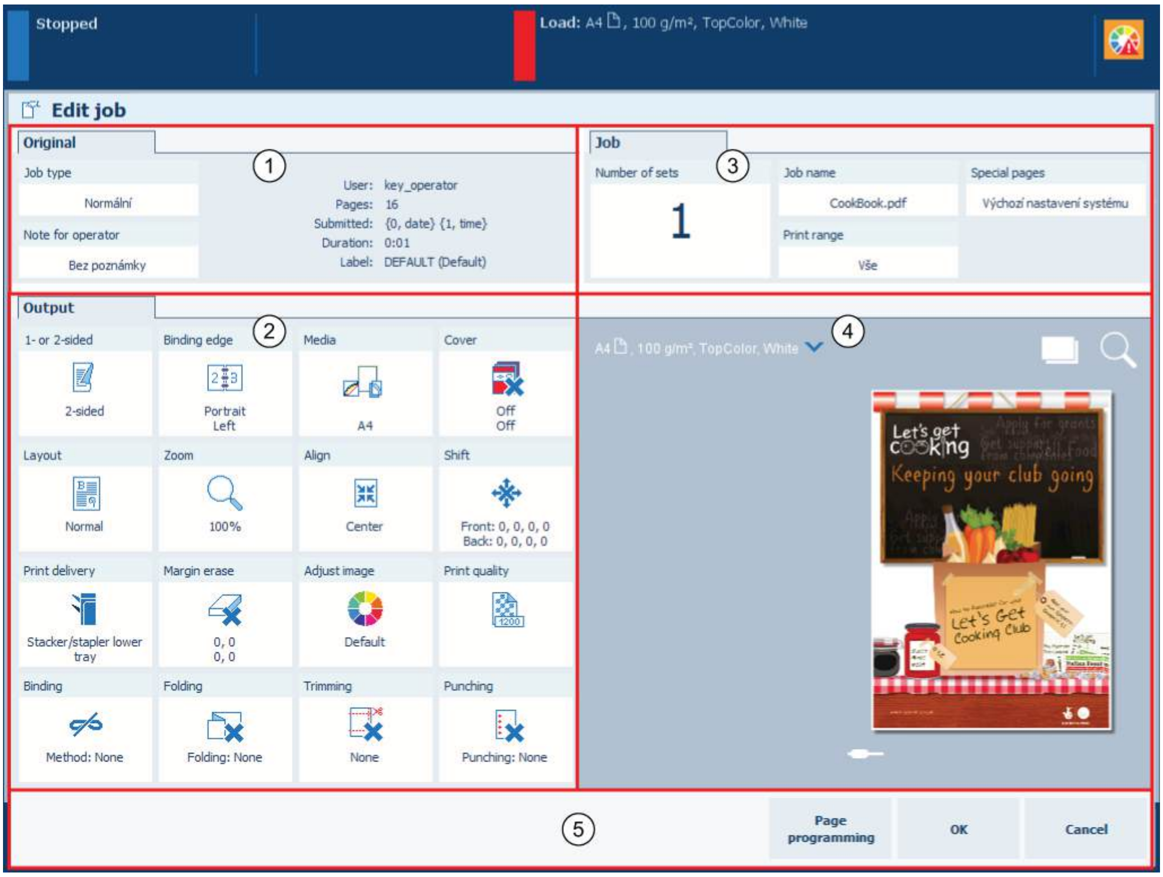
1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to preview.
4. Touch [Edit], or double-tap the job to open the [Edit] window.
5. Touch to view the result of the job and imposition settings in the sheet view .
6. Touch the icon and browse the document.
7. Touch to view how the document will look after printing in the document view.
8. Use the zoom function to view specific details.

More information

- *Learn about job management in the queues* on page 277
- *Learn about the print job settings* on page 299
- *Change job settings* on page 315

Change job settings

Job settings are usually made during the job preparation. You can decide to change the print job settings after you checked the first set of a print job, after a proof, or after you print a job ticket. You can change the settings remotely with PRISMAsync Remote Manager or locally on the control panel.



Print job settings

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to change.
4. Touch [Edit], or double-tap the job to open the [Edit] window.
5. Check the job information in the [Original] pane (1).
6. Change the settings in the [Output] pane (2).
7. Change the settings in the [Job] pane (3).
8. Use the optional [Page programming] function (5), if required.
9. Check the results of the changed settings in the realistic preview (4).
10. Touch [OK] (5).

More information

- *Learn about job management in the queues* on page 277
- *Learn about the print job settings* on page 299
- *Advanced impositioning* on page 316
- *Preview job settings* on page 314
- *Use page programming* on page 325

Advanced impositioning

The Advanced impose license enables you to make use of advanced layout and finishing settings and have more control over how the job is printed.

If you have the Advanced impose license, the following advanced print job settings become available on the control panel and in the Settings Editor:

1. Additional imposition templates: added booklet impositions as well as quarto and octavo (see more information below);
2. Custom same-up imposition, which enables you to edit the gutter, change the number/ orientation of rows and columns, use the cut and stack functionality;
3. Trim shift and trim alignment which make it possible to define the custom sheet size after trimming.

Imposition templates

Introduction

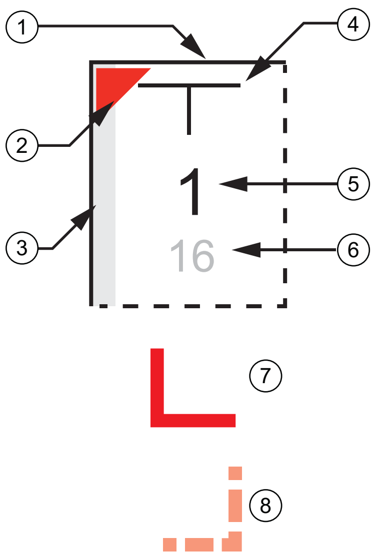
Imposition consists of an arrangement of pages on the sheet in order to obtain faster printing, simplify binding and reduce paper waste. Correct imposition minimizes printing time by maximizing the number of pages per impression.

You can configure an imposition template for your product on the control panel and via automated workflow settings in the Settings Editor.

The table below describes the structure and arrangement of pages into default imposition templates.

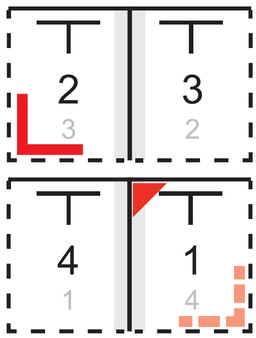
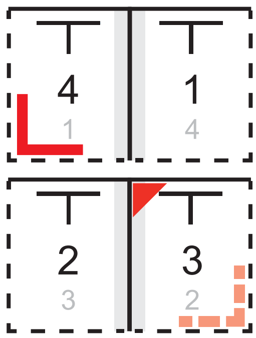
NOTE
The type of finisher and the finishing settings that you select directly influence your choice of the imposition template.

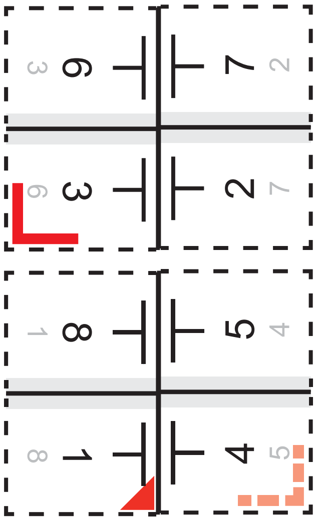
How to interpret the imposition template

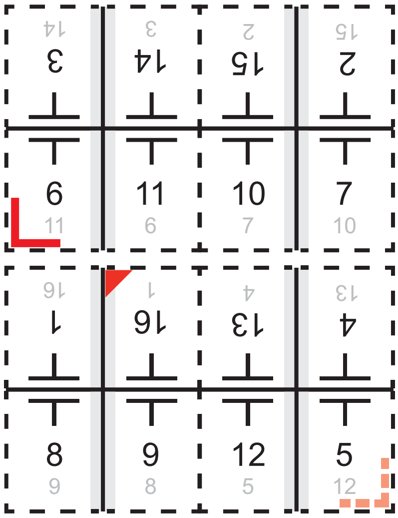
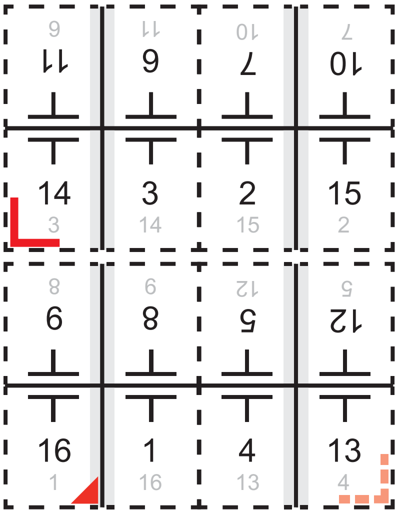
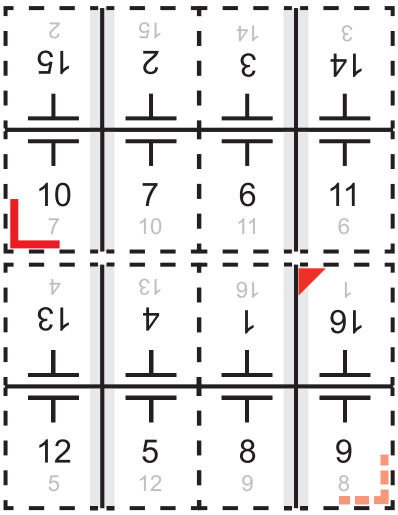


Imposition template

Number	Description
1	*Solid line*: the top side of the product after the sheet is folded
2	The top-left corner on the front side of a folded sheet
3	*Solid line with gray bar*: binding edge
4	Top of the page
5	Index of the page to be imposed
6	Index of the page to be imposed when pages are printed in reverse order (because of different binding options)
7	Reference corner on the front side of an unfolded sheet
8	Reference corner on the front side of an unfolded sheet (when located on the opposite side)

Type	Columns x Rows	Folding sequence	Arrangement of pages on the sheet
[Booklet (F4-1)]	2x1	1. • Left-hand part goes over the right-hand part • The size of the part being folded: 1/2 of the unfolded sheet's size	
[Booklet (F4-2)]		1. • Left-hand part goes under the right-hand part • The size of the part being folded: 1/2 of the unfolded sheet's size	

Type	Columns x Rows	Folding sequence	Arrangement of pages on the sheet
[Quarto (F8-7)]	2x2	1. • Left-hand part goes over the right-hand part • The size of the part being folded: 1/2 of the unfolded sheet's size 2. • Bottom part goes over the top part • The size of the part being folded: 1/2 of the unfolded sheet's size	

Type	Columns x Rows	Folding sequence	Arrangement of pages on the sheet
[Octavo (F16-6)]	4x2	1. • Left-hand part goes over the right-hand part • The size of the part being folded: 1/2 of the unfolded sheet's size 2. • Bottom part goes over the top part • The size of the part being folded: 1/2 of the unfolded sheet's size 3. • Left-hand part goes over the right-hand part • The size of the part being folded: 1/4 of the unfolded sheet's size	
[Octavo (F16-7)]		1. • Left-hand part goes over the right-hand part • The size of the part being folded: 1/2 of the unfolded sheet's size 2. • Bottom part goes over the top part • The size of the part being folded: 1/2 of the unfolded sheet's size 3. • Left-hand part goes under the right-hand part • The size of the part being folded: 1/4 of the unfolded sheet's size	
[Octavo (F16-8)]		1. • Left-hand part goes over the right-hand part • The size of the part being folded: 1/2 of the unfolded sheet's size 2. • Bottom part goes under the top part • The size of the part being folded: 1/2 of the unfolded sheet's size 3. • Left-hand part goes under the right-hand part • The size of the part being folded: 1/4 of the unfolded sheet's size	

Impose business cards

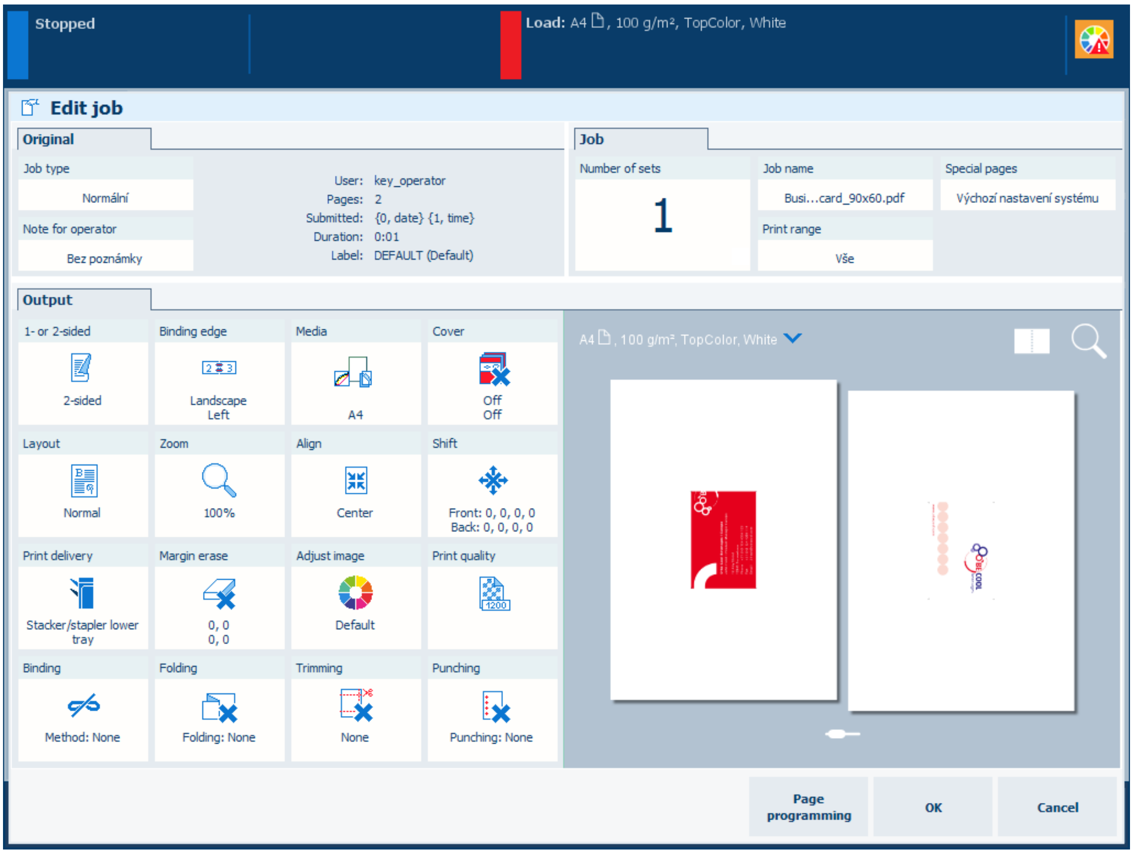
Introduction

The advanced layout and finishing settings can be useful for creating various product types, for instance, business cards. The procedure below demonstrates how the advanced settings can make the imposition easily customizable.

Procedure

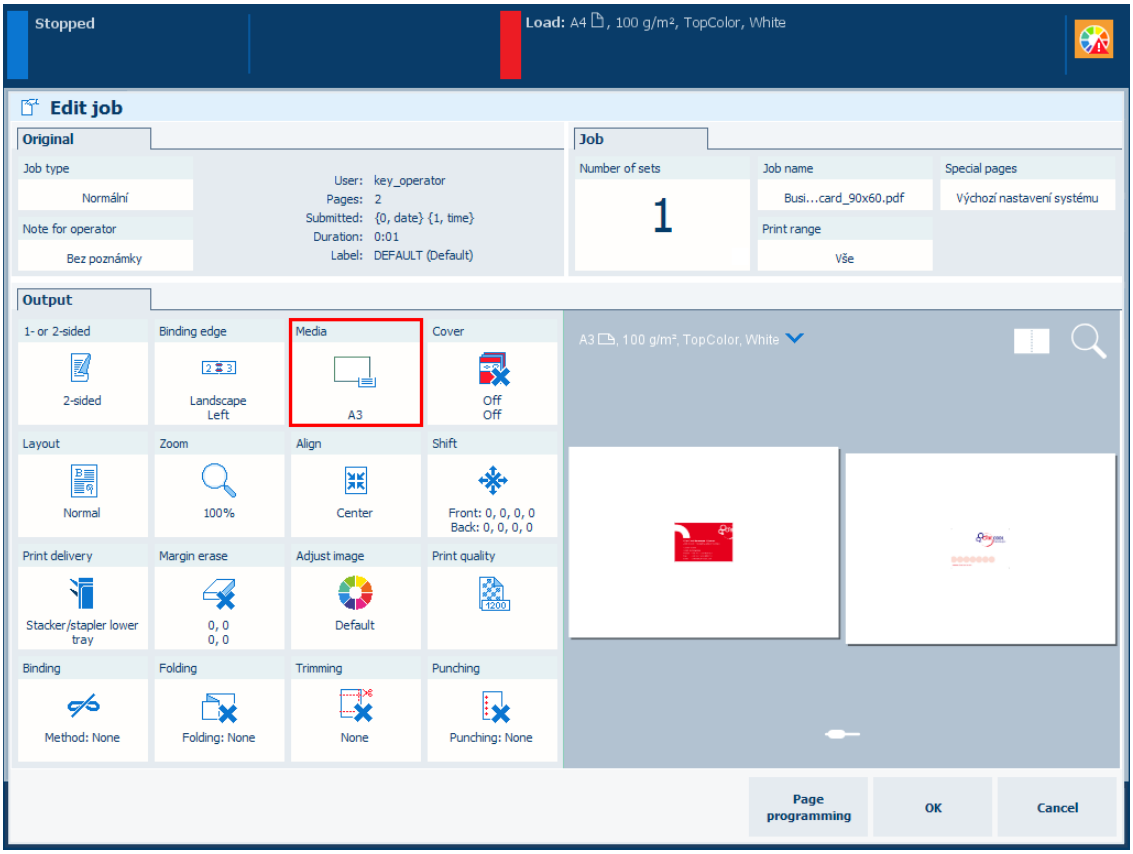
This procedure shows an example of imposition of business cards. Business cards are printed on a double-sided sheet. After trimming, different stacks are created. In the end, each stack contains the same number of the business cards.

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to edit.
4. Touch [Edit] or double-tap the job to open the [Edit] window.



[Edit] window

5. Touch [Media].
6. Select a media of size A3.
7. Touch [OK].

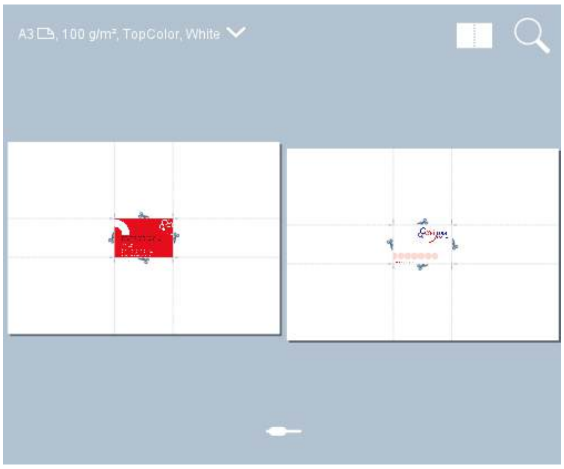
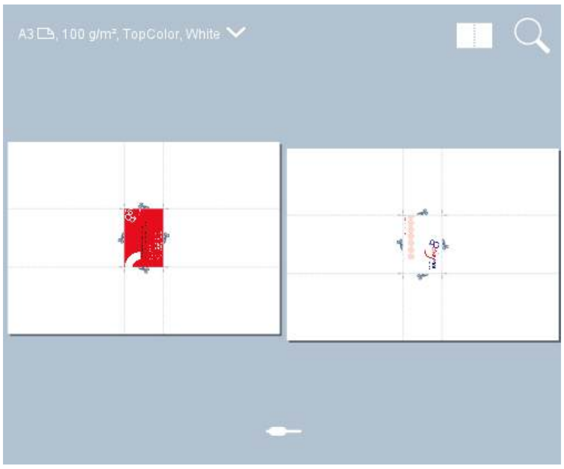


[Media] setting

8. Define the required size of your business card with the help of trimming settings. Do the following:
   1. Touch [Trimming]→[Print trim marks].
   2. Touch [Size]→[Finishing size] to define the sheet size after trimming. The drop-down list contains predefined sheet sizes, e.g. A4.
      You can also select [Custom] to define the size yourself. You can enter the custom measurements in [Height: ({0})] and [Width: ({0})].

      NOTE
      Width is parallel to the short edge of the sheet. Height is parallel to the long edge of the sheet. If you swap the measurements for [Height: ({0})] and [Width: ({0})], the orientation of your business card on the printed sheet will change.

   3. Touch [OK].

<table>
<tr><th>Settings</th><th>Realistic preview</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><td>
</td><td>
</td></tr>
</table>

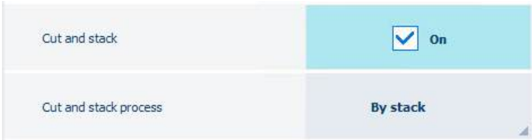
NOTE

You are required to define the trimming settings to be able to customize the end size of your product. If you do not select [Print trim marks], you will **not** be able to define the needed imposition.

9. Define the layout settings as follows:
   1. Touch [Layout]→[Same-up].
   2. Select [Custom].

[Custom] setting

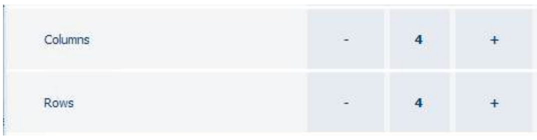
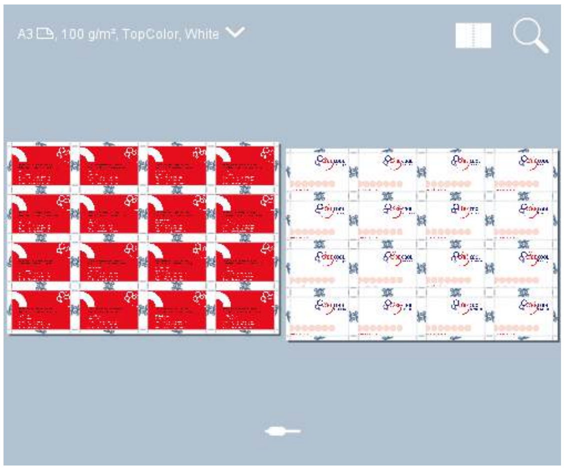
   3. Enable [Cut and stack].
   4. Touch [Cut and stack process] and select [By stack].



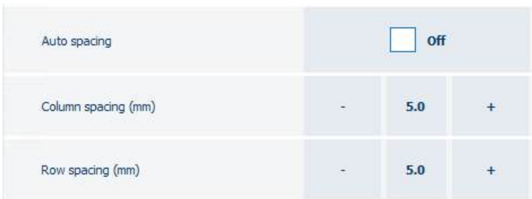
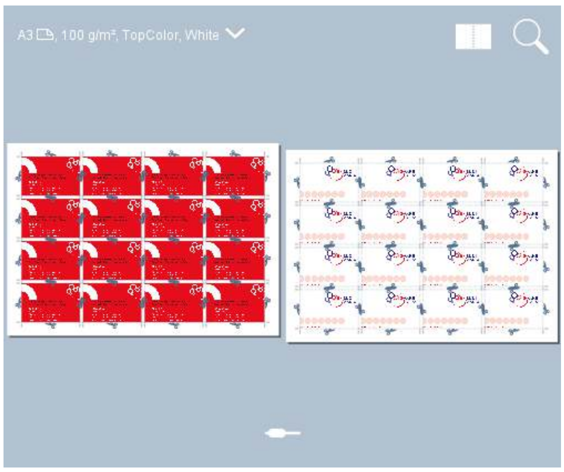
[Cut and stack process] setting

5. Change the number of rows and columns in order to have the maximum repetition of pages per one sheet.

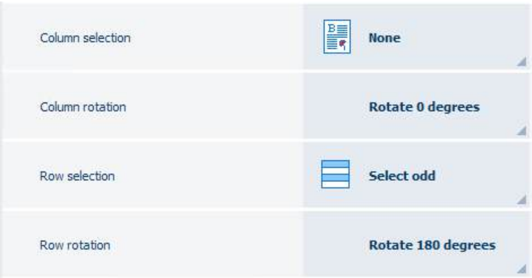
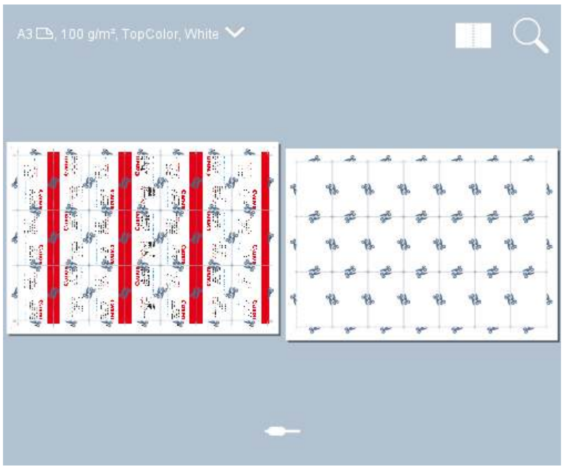
   NOTE
   Regardless of feed direction, rows are parallel to the short edge of the sheet and columns are parallel to the long edge of the sheet.

Settings	Realistic preview
	

6. • If you **do not** want PRISMAsync to automatically calculate column and row spacing, disable [Auto spacing]. Define[Column spacing (mm)] and [Row spacing (mm)] manually, for example, 5 mm.
   • If you **do not** want PRISMAsync to automatically define the orientation of your business card,[Auto spacing] also needs to be disabled.

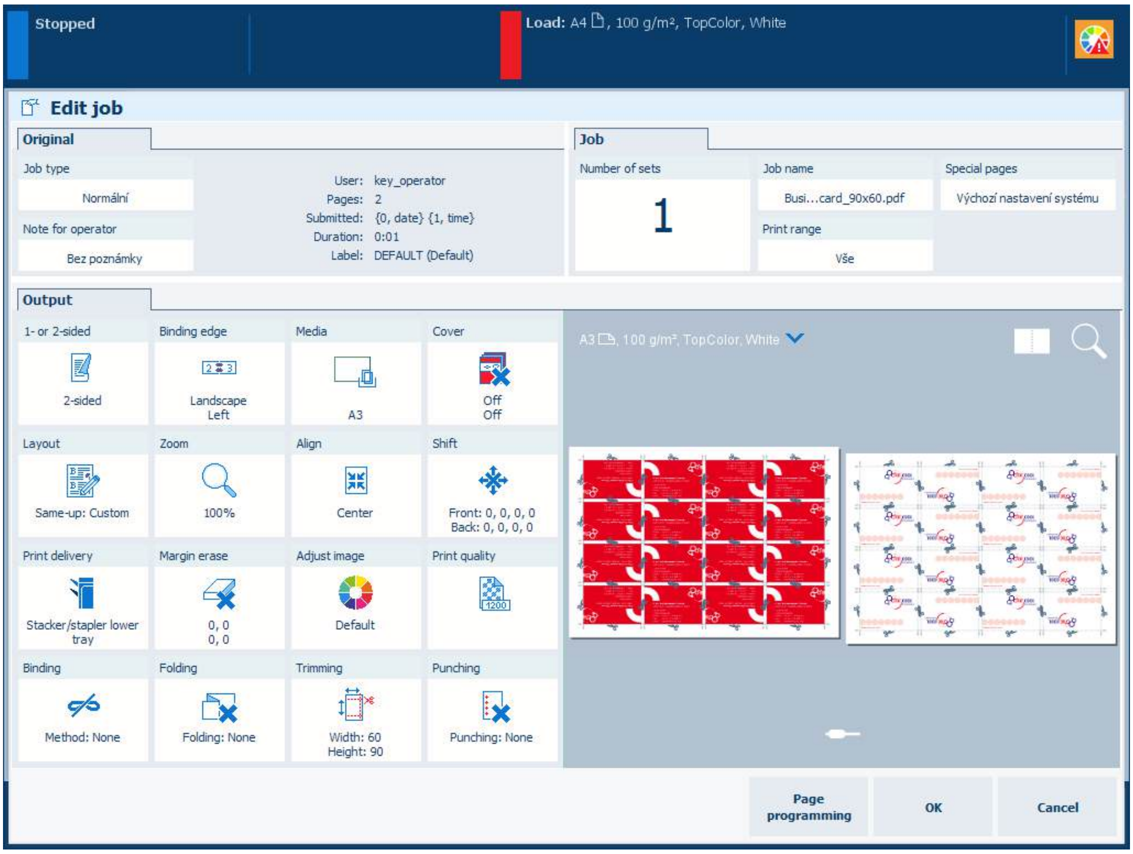
Settings	Realistic preview
	

7. It is also possible to select, flip and rotate rows and/or columns. For instance, these layout settings can be useful when one side of your business cards has the same color. Placing the sides of the same color next to one another makes trimming less prone to imprecision.

Settings	Realistic preview
	

8. Touch [OK].

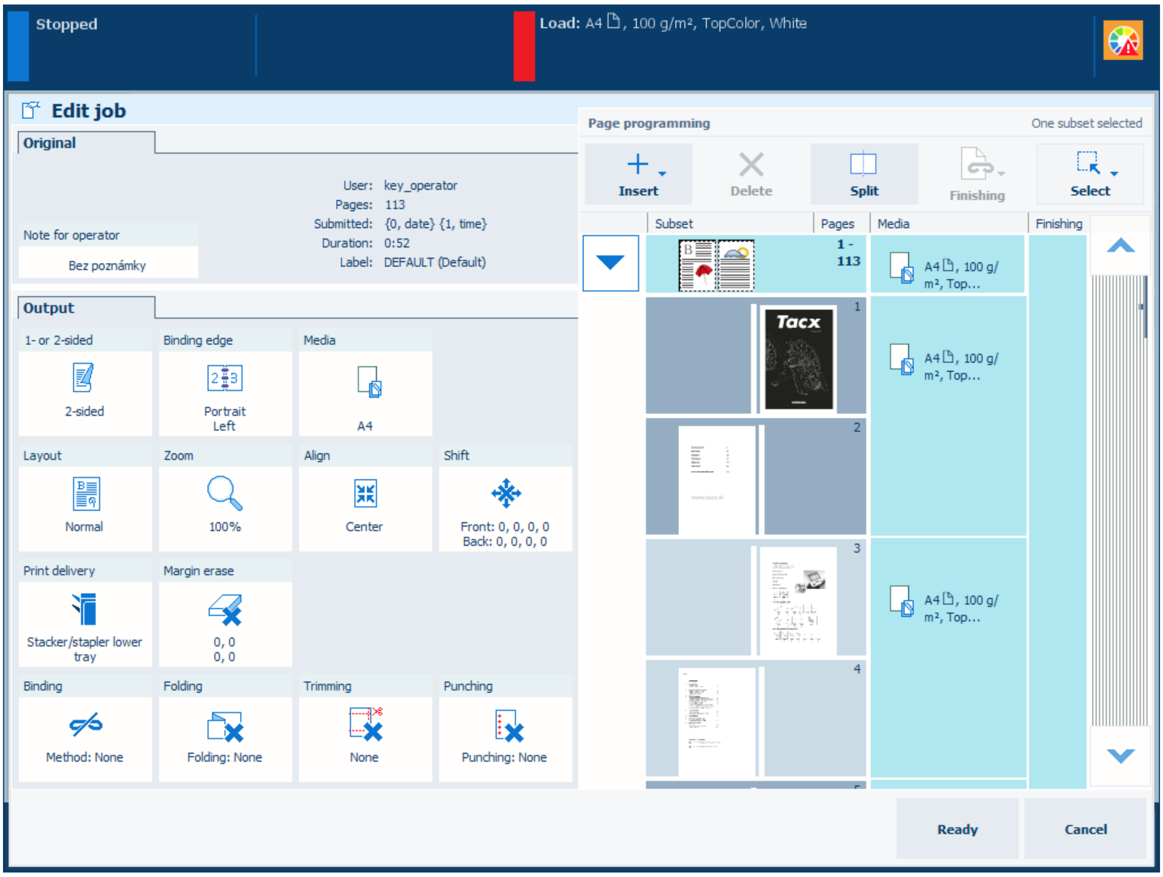
10. The business cards are ready to be printed. Touch [OK].



Imposed business cards

Use page programming

With the optional page programming function, you can create subsets and page ranges to apply different layout, media and finishing settings within the job. The page programming window alerts you when you create settings that do not match job properties.



Page programming

Before you begin

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to edit.
4. Touch [Edit], or double-tap the job to open the [Edit] window.
5. Touch [Page programming].

Create a new subset

1. Select a subset from the list of subsets.
2. Touch [Split].
3. Select the first page of the new subset.

Combine subsets

1. Select the subsets to combine.
2. Touch [Merge] to create the new subset.
3. Define the settings in the [Output], if required.

Change the settings of a subset

1. Select the subset, sheet or page you want to change.
   - Touch one or more subsets from the list of subsets.
   - Use the drop-down icon to select pages or sheets of a subset.
   - Use the [Select] menu to select a page range.
2. Define the settings in the [Output].
3. Repeat step 1 and 2, if required.
4. Touch [Ready] to apply the settings.

Merge finishing settings

1. Select the subsets that must have the same finishing settings.
2. Touch [Merge finishing] and select the subset to use the finishing settings.
   Use [Split finishing] to get the original finishing settings.

Add a page to a page range or subset

1. Select the subset or page range.
   - Use the subset drop-down icon to select pages of a subset.
   - Use the [Select] menu to select a page range.
2. Indicate where you want add a page.
   - Use [Insert] →[Insert page before] to add a page before a page range or subset.
   - Use [Insert]→[Insert page after] to add a page after a page range or subset.

Delete a page, sheet, or subset

1. Select the part you want to delete.
   - Use the subset drop-down icon to select sheets and pages of a subset.
   - Use the [Select] option to select a page range.
2. Touch [Delete] to delete the subset or page range.

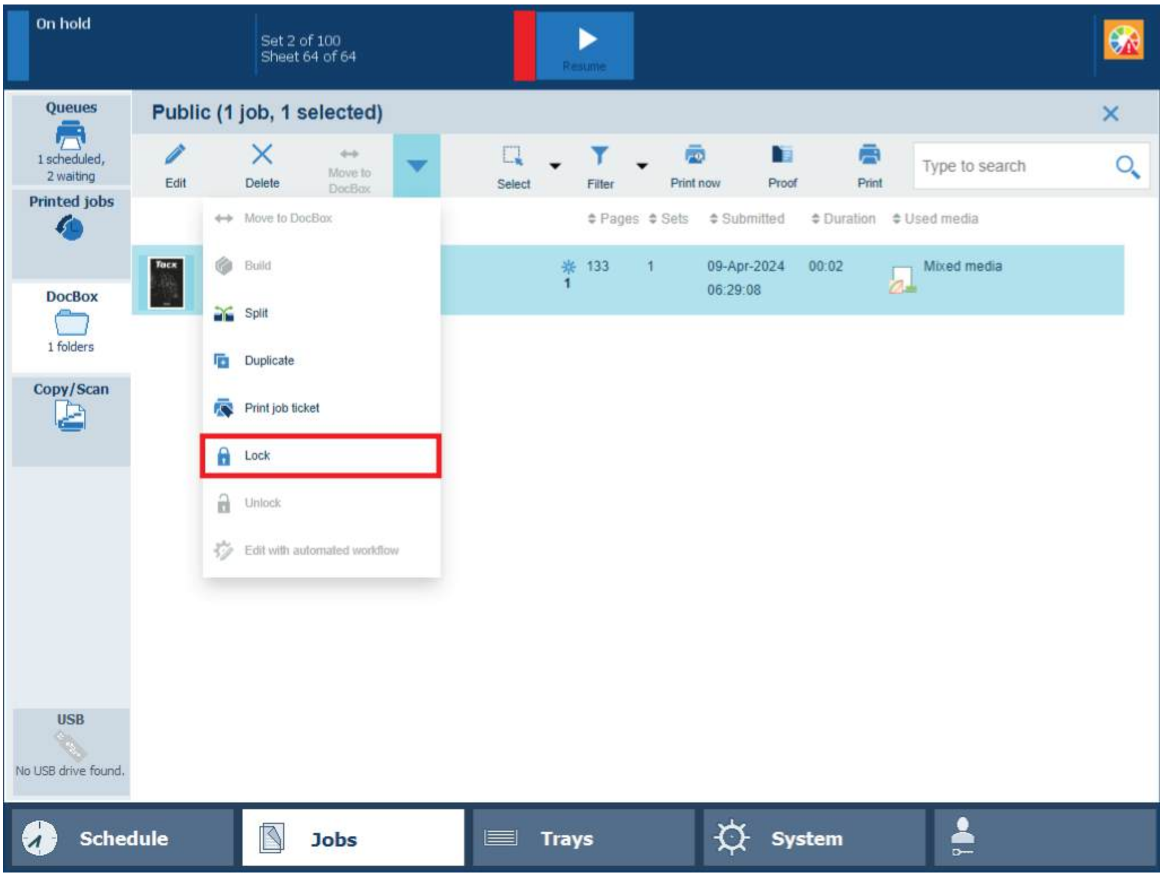
More information

- *Learn about the print job settings* on page 299

Lock DocBox job settings

You can lock a DocBox job to protect its settings.

The lock icon indicates a locked job.



The lock function in DocBox

1. Touch [Jobs]→[DocBox].
2. Select the DocBox folder that contains the job you want to lock.
3. Enter a PIN, if requested.
4. Select the job you want to lock.
5. Select [Lock] from the drop-down menu.
   If required, unlock the job again with the [Unlock] function.

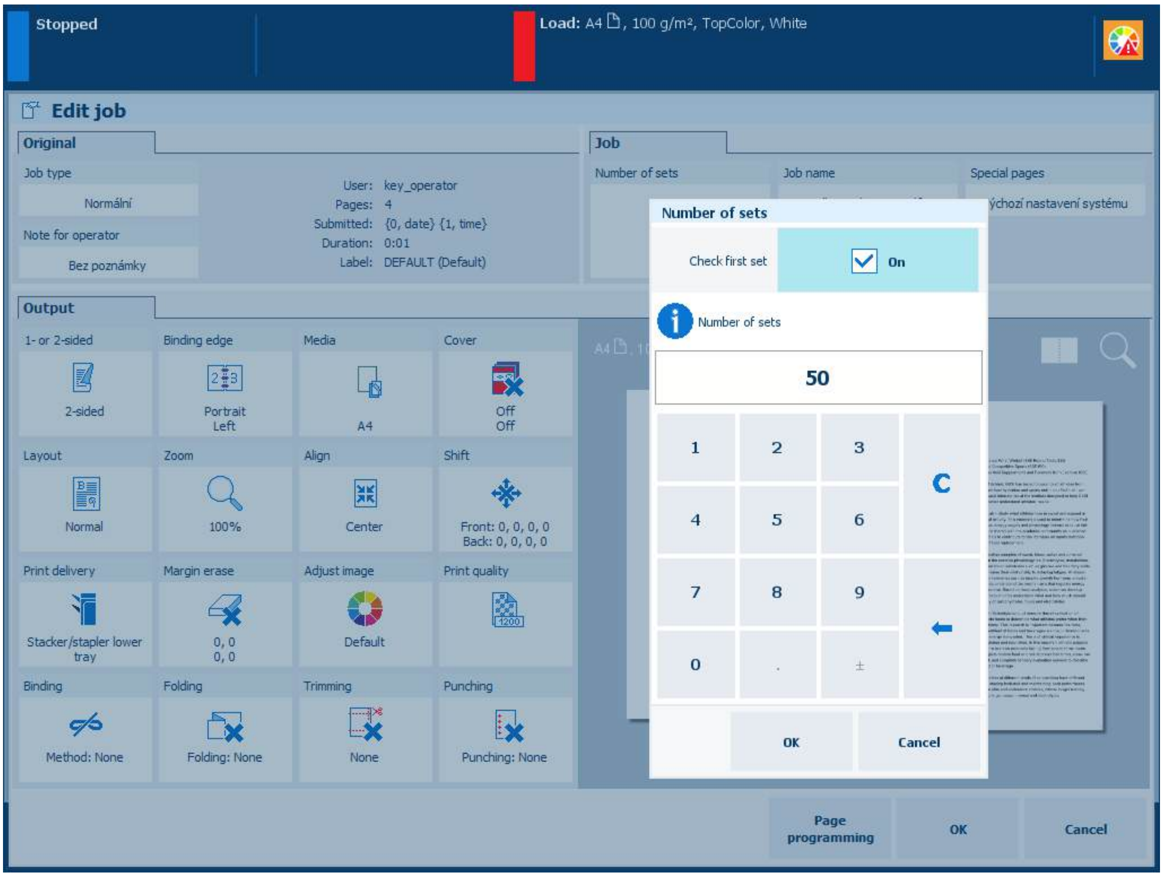
More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281

Check prints

Check the first set

Especially for jobs with many sets, you can first print and check one set before you print the remaining sets with the check first set function.



Check first set function

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to print.
4. Touch [Edit].
5. Touch [Number of sets].
6. Touch [Check first set].
7. Check the print quality of the first set.
   - If the printed output is not as expected:
     1. Touch [Edit] to change the job settings.
     2. Touch the [Resume] button ▷ to print the remaining sets.
   - If the printed output is as expected, touch the [Resume] button ▷ to print the remaining sets.

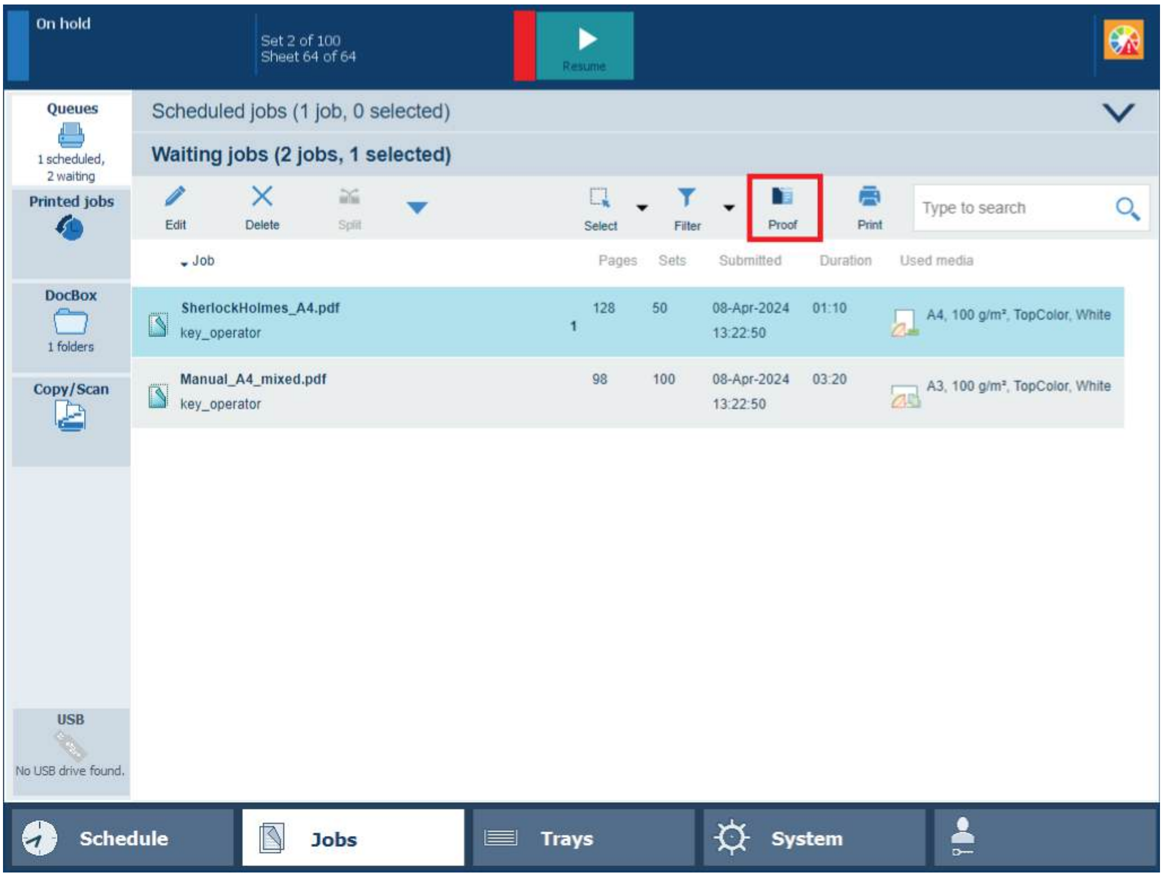
More information

- *Choose a workflow profile* on page 240
- *Change job settings* on page 315
- *Make a proof* on page 330

- *Make intermediate check prints* on page 331
- *Print a job ticket* on page 333

Make a proof

When you make a proof, the printer prints an extra set or record of the job. A copy of the job goes to the last position of the list of scheduled jobs. The original job remains in the list of waiting jobs or in the DocBox. The magnifying glass icon indicates a proof print.



[Proof] button in the list of waiting jobs

You can make a proof from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Proof]		✓	✓	

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the jobs you want to proof.
4. Touch [Proof].

More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281
- *Check the first set* on page 328
- *Make intermediate check prints* on page 331
- *Print a job ticket* on page 333

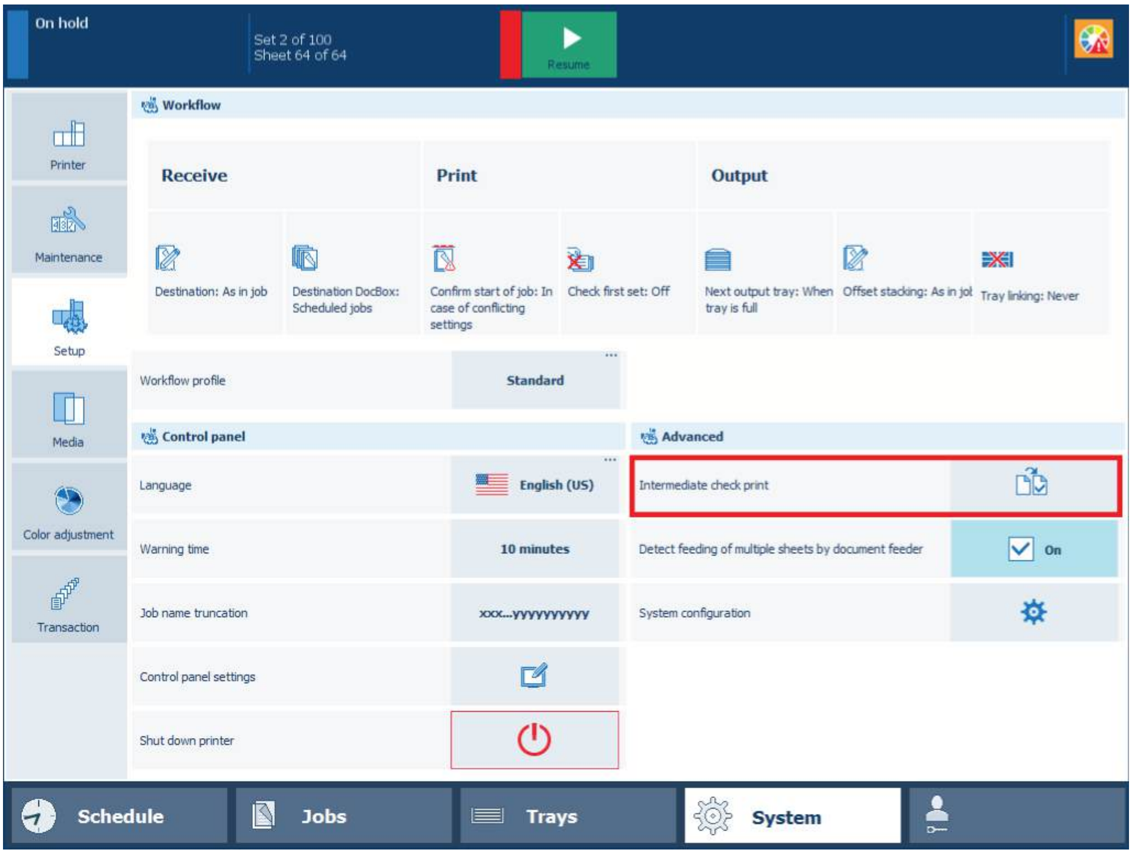
Make intermediate check prints

An intermediate check print is a print sample of a job sheet. You can use these sheets to check the print quality. Intermediate check prints are printed according to a set interval, but can also be printed on demand.

Intermediate check prints arrive in the output tray that is closest to the print module.

Make an intermediate check print from the control panel

1. Touch [System] →[Setup].

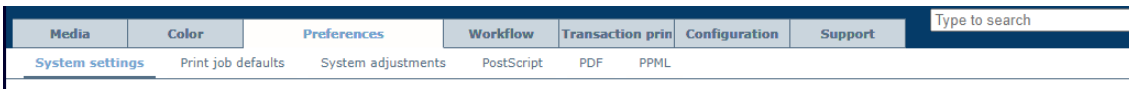


Intermediate check print function

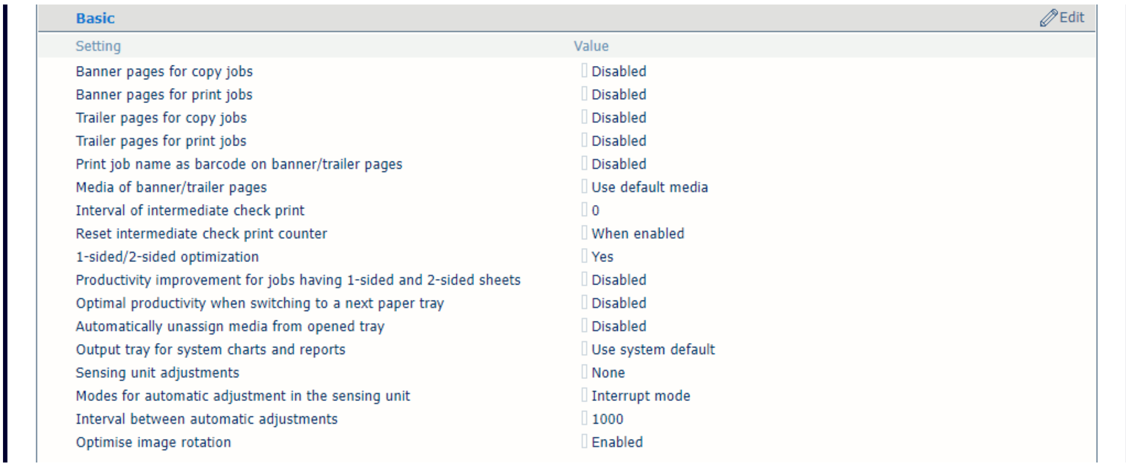
2. Touch [Intermediate check print].

Print intermediate check prints automatically

1. Open the Settings Editor and go to: [Preferences]→[System settings].

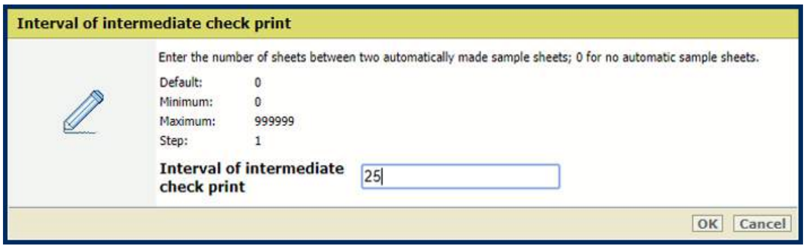


2. Go to the [Basic] section.



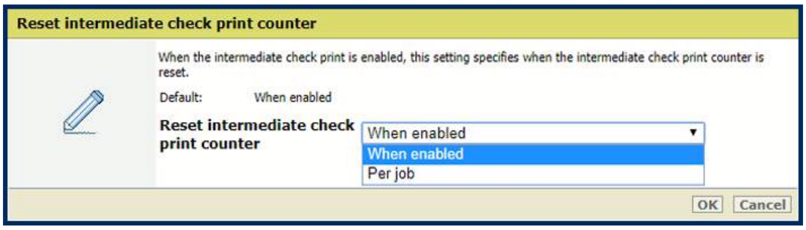
[Basic] settings

3. Use the [Interval of intermediate check print] setting to define the interval of intermediate check prints. The printing of intermediate check prints is disabled when the value is: 0.



Enable intermediate check prints

4. Use the [Reset intermediate check print counter] setting to define when the counter is reset.
   - [When enabled]: reset when the [Interval of intermediate check print] setting is set to enabled.
   - [Per job]: reset for each new job.



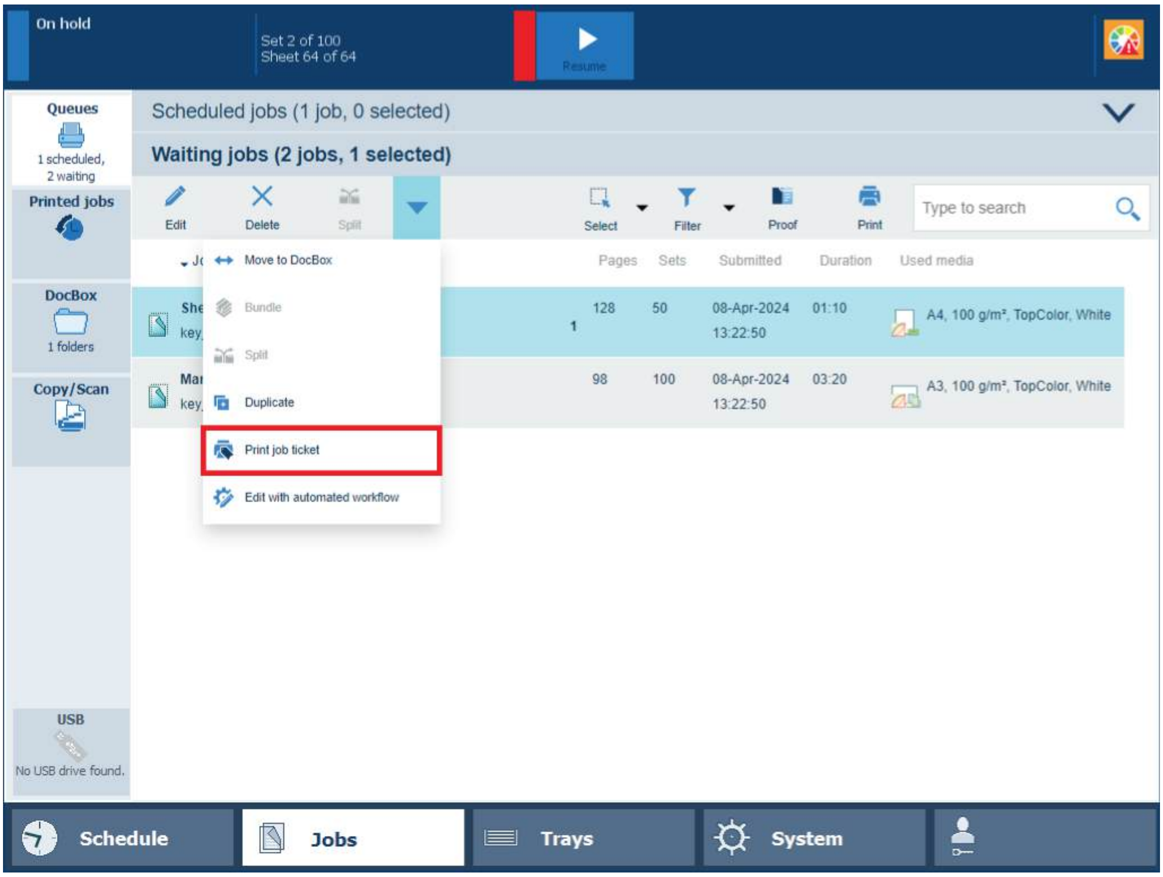
Reset the counter of intermediate check prints

More information

- *Check the first set* on page 328
- *Make a proof* on page 330
- *Print a job ticket* on page 333

Print a job ticket

You can check the job settings by printing a job ticket. The job ticket print goes to the last position of the list of scheduled jobs. The job ticket is printed in the printer language and shows the job properties.



Print a job ticket

You can print a job ticket from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Print job ticket]	✔	✔	✔	✔

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the jobs for which you want to print a job ticket.
4. Touch [Print job ticket].

More information

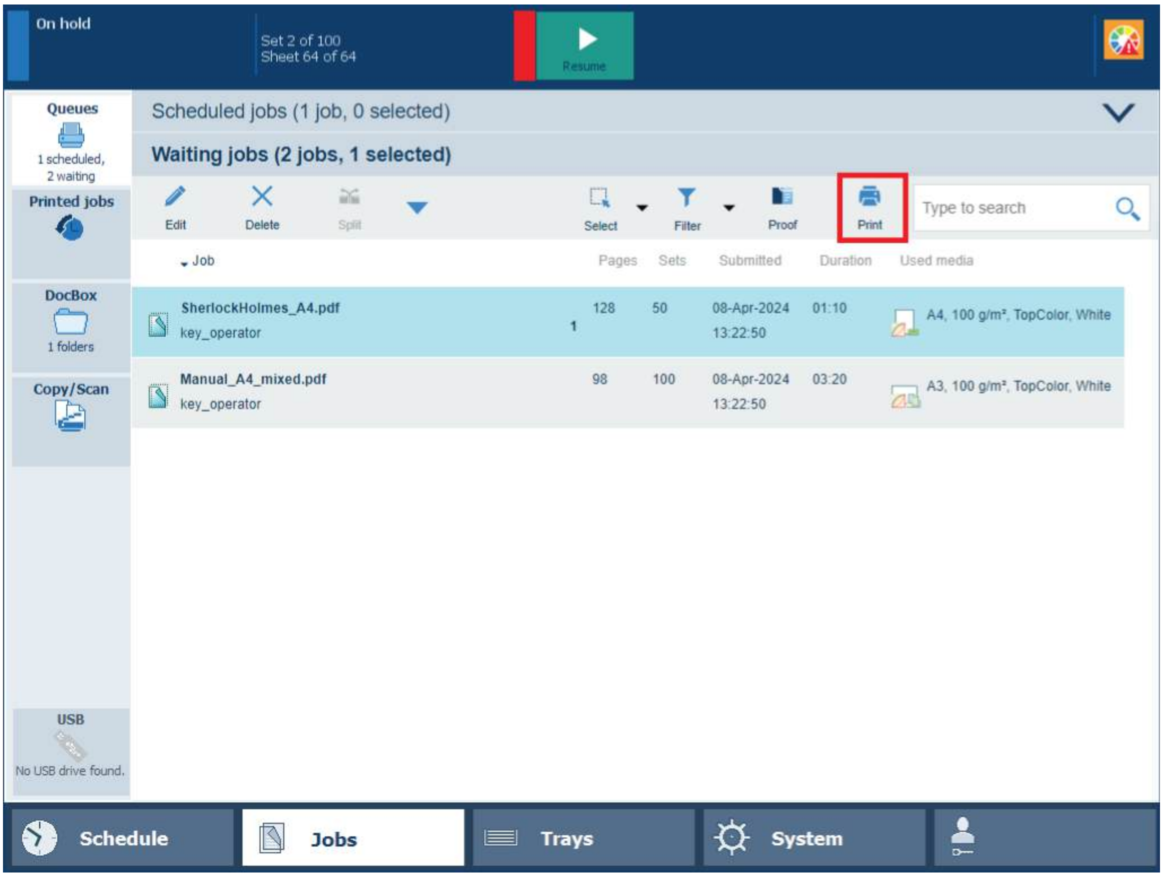
- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281
- *Check the first set* on page 328
- *Make a proof* on page 330
- *Make intermediate check prints* on page 331

Print and monitor jobs

Print a job

Print a job

The **Print** function sends one or more jobs to the **end of the list of scheduled jobs**. There they are printed according to their position in the list. Printed scheduled jobs go to the list of printed job. Printed DocBox jobs remain in the DocBox folder.



The print function in the list of waiting jobs

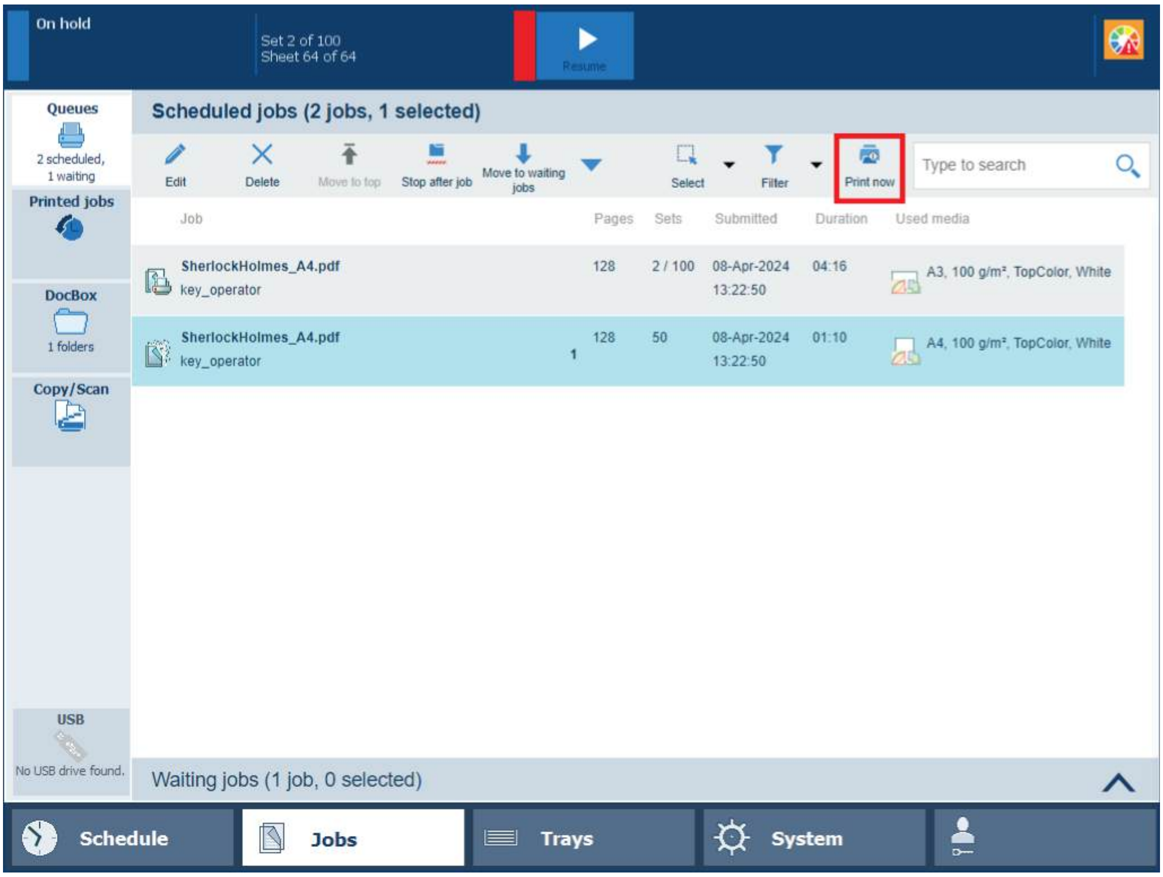
You can print a job from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Print]		✔	✔	

1. Touch [Jobs].
2. Go to the location of the job.
   - To print a waiting job, touch [Queues]→[Waiting jobs].
   - To print a DocBox job, touch [DocBox] and select the DocBox folder. Enter a PIN, if requested.
3. Select the jobs you want to print.
4. Touch [Print].

Print a job immediately

The [Print now] function sends one or more jobs to the **first position in the list of scheduled jobs**. The printer stops the active print job after a set is finished.



The print now function in the list of scheduled jobs

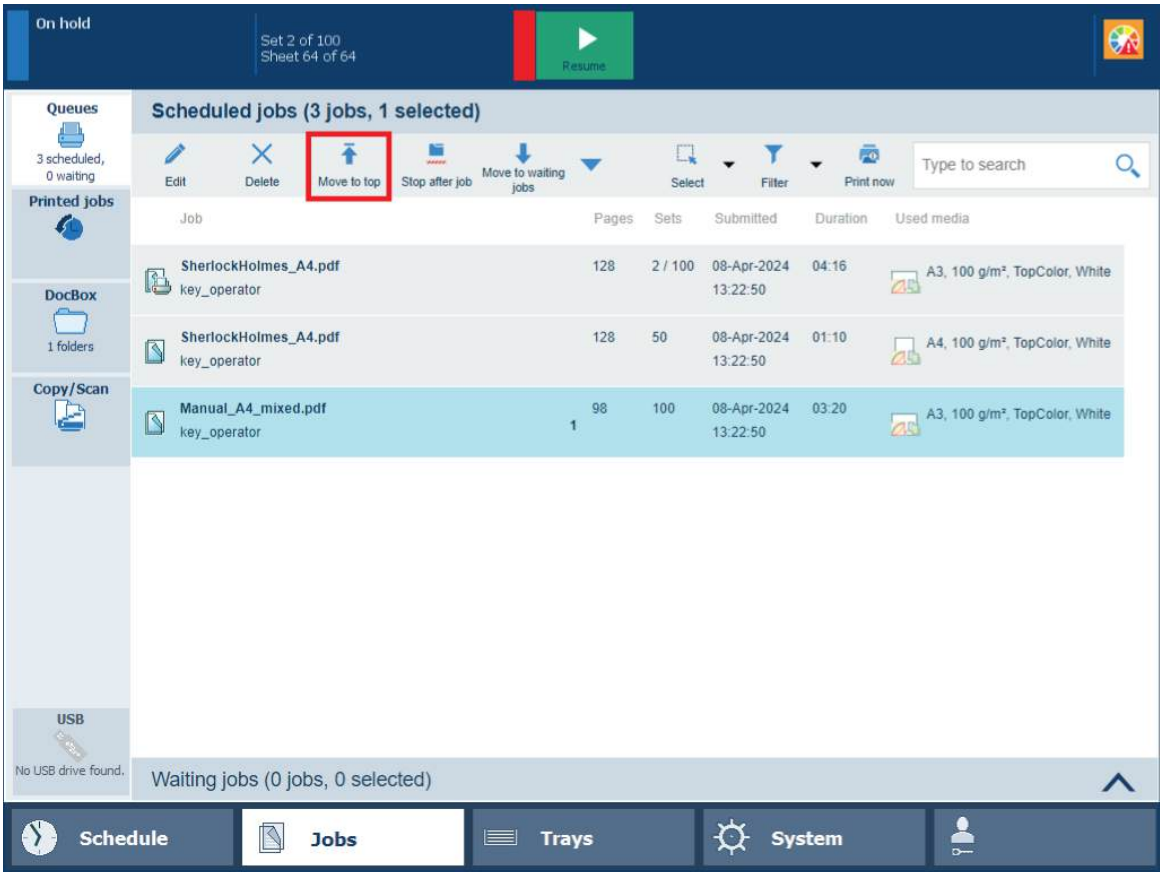
You can print a job immediately from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Print now]	✔		✔	

1. Touch [Jobs].
2. Go to the location of the job.
   - To print a scheduled job immediately, touch [Queues]→[Scheduled jobs].
   - To print a DocBox job immediately, touch [DocBox] and select the DocBox folder. Enter a PIN, if requested.
3. Select the jobs you want to print.
4. Touch [Print now].

Print a job after the active print job

The [Move to top] function sends one or more jobs to the **second position in the list of scheduled jobs**. The printer finishes the active print job .



The To Top function in the list of scheduled jobs

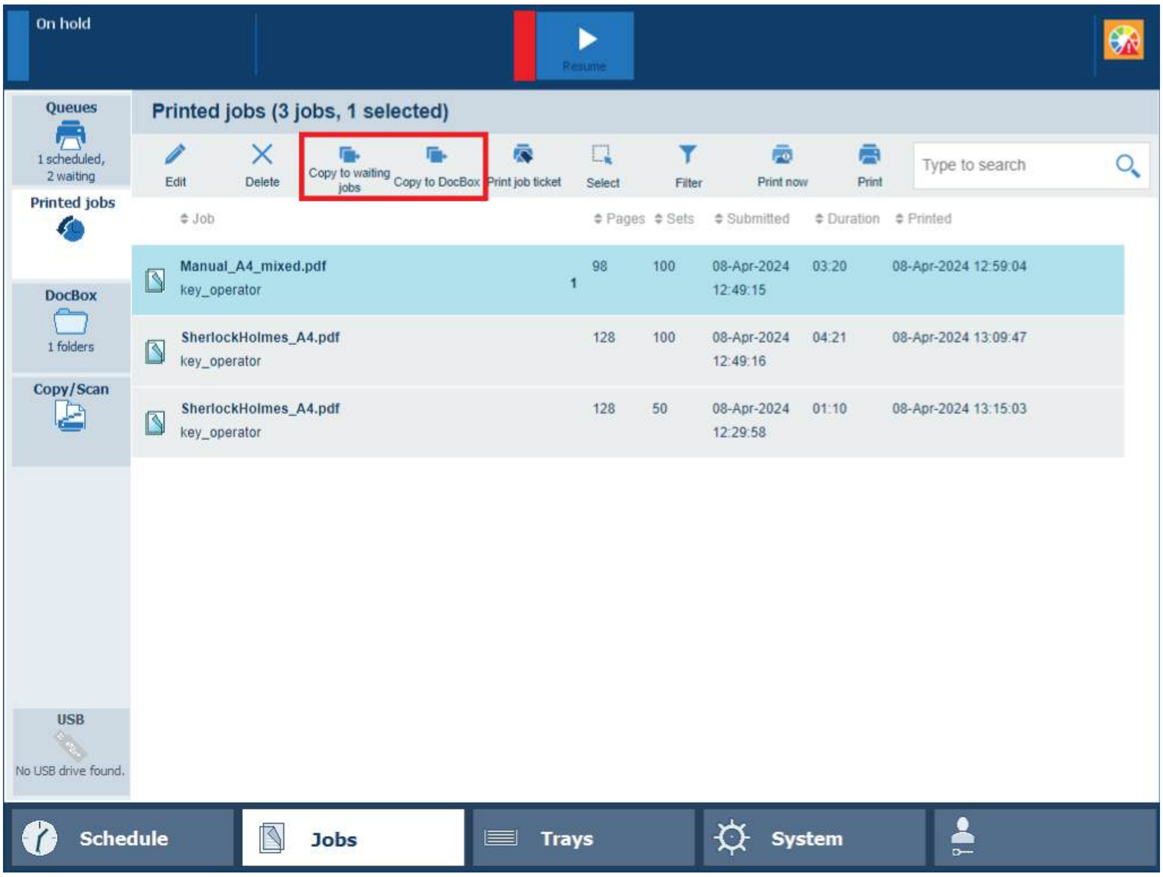
You can print a job after the active print job from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Move to top]	✓			

1. Touch [Jobs].
2. Touch [Queues]→[Scheduled jobs].
3. Select the jobs you want to print after the active print job.
4. Touch [Move to top].

Reprint a printed job

To reprint a printed job, you must first copy the printed job to waiting jobs or to DocBox. Then, you can change the jobs settings (if required) and print them.



The copy functions in the list of printed jobs

You can reprint a printed job from the following locations:

Function	Scheduled jobs	Waiting jobs	DocBox	Printed jobs
[Copy to waiting jobs]				✓
[Copy to DocBox]				✓

1. Touch [Jobs]→[Printed jobs].
2. Select the jobs you want to reprint.
3. Touch [Copy to waiting jobs] or [Copy to DocBox].
4. Go to the location of the copied jobs.
5. Select the job you want to reprint.
6. Change the job settings, if required.
7. Touch [Print].

More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281
- *Make a proof* on page 330
- *Change job settings* on page 315

Print a job from a USB device

When the USB port is available for printing jobs, you can copy print files from a USB device to the list of waiting jobs. It is also possible to print the files directly.

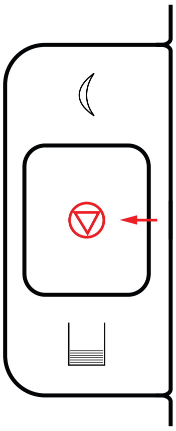
1. Insert the USB device into the USB port at the left-hand side of the control panel.
2. Touch [Jobs]→[USB].
3. Navigate to the print jobs and select one or more.
4. Select the required option:
   - Touch [Print] to copy the jobs to the list of [Scheduled jobs].
     If accounting is enabled, the jobs are sent to the list of waiting jobs. Here you must open the job and first enter the accounting ID before you can print.
   - Touch [Save] to copy the jobs to the list of [Waiting jobs] or to a DocBox folder.
5. Touch the eject button and remove the USB device.

More information

- *Learn about job management in the queues* on page 277
- *Find, filter, and select jobs* on page 281

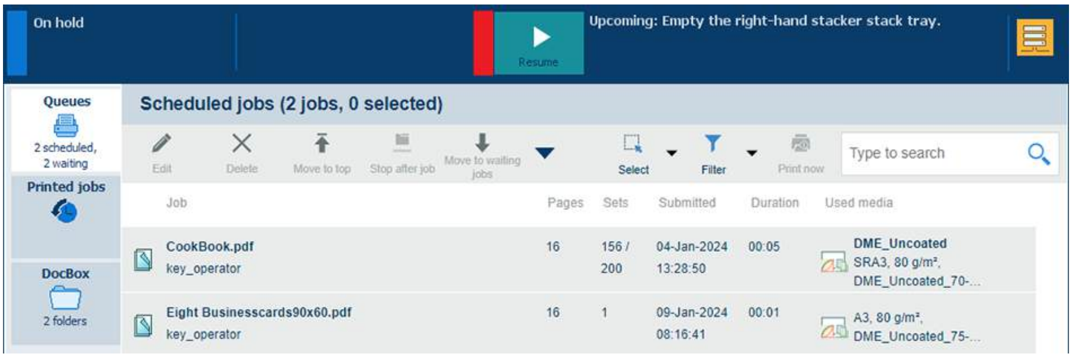
Stop printing

There are different ways to stop or delay the print production with the stop button or the stop after job function.



Stop button

In all cases you continue printing with the resume function on the control panel.



Resume function that appears after a print stop

Stop after a completed set or record

Stop printing when the print buffer is empty and a complete set or record has been printed.

1. Press the [Stop] button **once** to stop printing after the current set or record has been completed.
2. Touch the [Resume] button ▹ to resume printing.

Stop as soon as possible

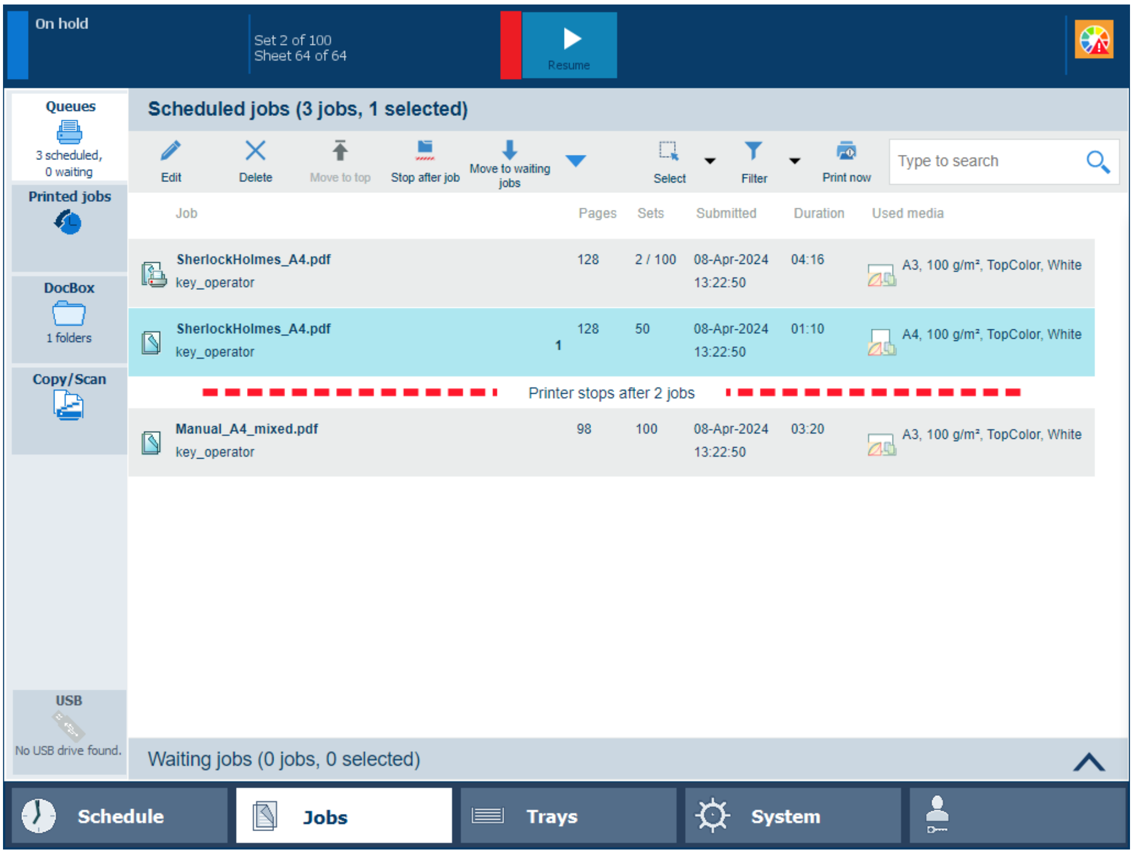
The printer stops when the print buffer is empty, which means as soon as possible.

1. Press the [Stop] button **twice** to stop the print process as soon as possible.
2. Touch the [Resume] button ▹ to resume printing.

Stop after job

Stop printing after a certain job has been completed.

1. Select the job in the list of scheduled jobs.
2. Touch [Stop after job].
   A red-white bar indicates when printing will stop.



The red-white bar

3. Touch the [Resume] button ▷ to resume printing.

Handle printed output

Remove printed output from the stacker / stapler

The output tray moves downwards during the print process to give space to the growing stack of output.

The workflow profile determines to which output trays PRISMAsync sends printed output. When an output tray has reached its limit, the printed output automatically goes to the next available tray. If all available output trays have reached their stacking limits, the printing process stops. Remove all printed output from the output trays. The output trays move upwards, and print process resumes.

Use the Settings Editor to turn high-volume stacking for the stacker / stapler on or off.

CAUTION

- **Do not place your fingers in the stacker / stapler when it is in use. This can cause personal injury or stacker / stapler damage.**

- **When you remove paper from the output tray, do not place your hands on the output tray of the stacker / stapler. The output tray can move upwards and your hands can get caught.**
- **Do not place your hands in the part of the output tray near the rollers where stapling takes place. This can cause personal injury.**

IMPORTANT
- Do not place objects onto the output trays of the stacker / stapler. This can damage the output trays.
- Do not place objects under the paper trays of the stacker / stapler. This can damage the output trays.

IMPORTANT
For **stacking** with the **stacker / stapler**:
- When the media weight is 326 g/m² - 350 g/m² (120 lb cover - 130 lb cover), use the lower tray to avoid that sheets stick together. In this case the maximum stack height is 40 mm (1.6 inch).
- When rigid media with a weight of 326 g/m² - 350 g/m² (120 lb cover - 130 lb cover) is delivered in the lower tray, paper jams and scratches in high density images can occur.

IMPORTANT
For **stapling** with the **stacker / stapler**:
- When the width of the media is small, the stapled sets can have uneven edges.
- When stapled sets have coated covers, the staples can dirty the covers of sets on the output tray.
- You cannot staple vellums, transparencies and labels.

IMPORTANT
For **booklet making** with the **stacker /stapler**:
- You cannot make booklets of vellums, transparencies, prepunched paper, tab paper and labels.
- When the cover weight is less than 64 g/m² (17 lb bond), the media can crease when saddle-stitched.
- When the cover weight is less than 64 g/m² (17 lb bond), the booklets can get uneven edges when trimmed.
- When the media size is small, the booklets can get uneven edges when saddle-stitched.
- When the media size is small, the booklets can get uneven edges when trimmed.
- When the covers use coated paper or insert sheets, the first page after the cover can stick to the back of the cover.
- When the covers use coated paper or insert sheets, toner streaks can appear on the covers.
- When the covers use coated paper or insert sheets, cracks can appear around the folds of the cover.

More information

- *Finishers specifications* on page 624
- *Define finisher adjustments* on page 560

Store printed output

To keep maximum print quality, you must store prints in an optimal storage environment before the transfer to the final destination. Use the following storage recommendations:

- Store prints on a flat surface.
- Store prints so that the sheets cannot fold or crease.
- Use a binder for storage during a long period of time (more than two years).
- Do not wrap prints in PVC material.
- Do not store prints in a location with high temperatures.
- Prints can discolor after a long period of time.
- Adhesive can stick prints; only use insoluble adhesive that is completely dry.

Chapter 10

Keep the print quality high

Calibrate the system

Learn about calibration

Importance of calibration

Calibration is very important to keeping the quality of the color reproduction high. The environment of your print system influences the print quality. A constant temperature and humidity are essential for consistent color output.

The print system automatically keeps the print quality as high as possible. However, additional calibration is essential to keep the color reproduction level stable for all media types you use. PRISMAsync Print Server offers an automated and structured workflow and feedback mechanism to integrate calibration into your daily work.

There are two types of calibrations: **printer calibration** and **media family calibration**.

Automated color tasks

It is important to perform the appropriate calibration procedures regularly. You can perform each calibration procedure separately. To ease the process of calibration PRISMAsync Print Server offers an automated procedure. When you perform the Automated color tasks the calibration procedures will be processed automatically in the correct order and with limited operator interaction.

The Automated color tasks can perform:
- Auto Gradation Adjustment
- Shading correction
- Media family calibration
- Idealliance® G7® Grayscale test.

Printer calibration

There are two procedures, which calibrate the printer:
- **Automatic gradation adjustment**
  The automatic gradation adjustment includes precise adjustments of the gradation, density and color quality of the primary colors.
- **Shading correction**
  The shading correction includes precise adjustments of slightly uneven color densities in color planes.

There are two modes available for the automatic gradation adjustment:
- **Standard, for normal media**
  This recommended mode guarantees a high print quality for most print environments.
- **Professional, for normal, heavy and extra heavy media**
  This extended mode is advised when you frequently use heavy and extra heavy uncoated media or when you frequently use coated media.

The printer calibration settings on the control panel show which printer calibration procedures are configured for your print system. The dashboard of the control panel will show the printer calibration indicator when printer calibration is required.

Configuration of printer calibration

If required, consult a color expert to decide what you need to configure for printer calibration to achieve the required color quality:
- The standard or professional mode for automatic gradation adjustment.

- How often to perform the shading correction.

IMPORTANT
It is strongly advised to perform the shading correction at least once after the installation of the print system.

IMPORTANT
It is strongly advised to configure at least the automatic gradation adjustment for **daily** printer calibration.

You can configure the required printer calibration procedures in the Settings Editor. Then the operator receives a daily reminder to perform the printer calibration.

Media for printer calibration

You define and see the media for the printer calibration in the Settings Editor. You can find it as follows: [Color]→[Color calibration].

The following media is recommended for printer calibration:

Normal media for **standard** and **professional printer calibration**:
- Top Colour Zero FSC (100 g/m² / 27 lb bond)
- Canon CS-814 (81.4 g/m² / 22 lb bond)
- Mondi Neusiedler (100 g/m² / 27 lb bond)
- International Paper Hammermill Color Copy Digital (105 g/m² / 28 lb bond)

Heavy media for **professional printer calibration**:
- Top Colour Zero FSC (250 g/m² / 92 lb cover)
- Mondi (250 g/m² / 92 lb cover)
- Mohawk Navajo Cover 98 bright (243 g/m² / 90 lb cover)

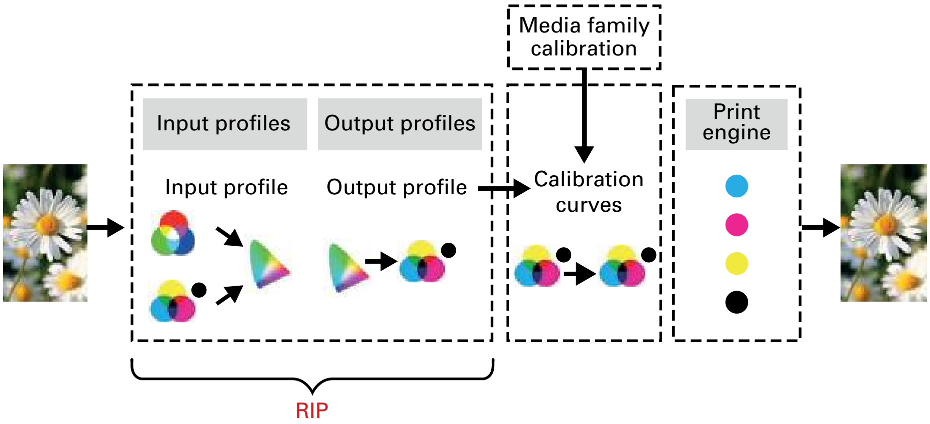
Extra heavy media for **professional printer calibration**:
- Top Colour Zero FSC (300 g/m² / 110 lb cover)
- Mondi (300 g/m² / 110 lb cover)
- International Paper Hammermill Color Copy Digital (271 g/m² / 100 lb cover)

When the recommended media is not available, you can use custom media for automatic gradation adjustment.

Media family calibration

PRISMAsync provides default **coated** and **uncoated** media families. A media family is a set of media with the same **output profiles** (one per halftone) and calibration curves. A media family calibration applies to all media of the media family. You only have to perform a media family calibration for one media of the media family.

Output profiles describe how source colors (in Lab) are mapped to the color space (in CMYK) of the printer. They contain information about the color boundaries a printer can reproduce.



Calibration curves are the result of a media family calibration to ensure that the color quality remains stable over time.

Media for the media family calibration

You define the default calibration media for a media family the first time you perform a media family calibration for that media family. Make sure that you use the same calibration media to represent the media family every time you perform a media family calibration.

Additional feedback on media family calibration

When media family calibration is completed, the control panel will show the following feedback to inform you how well colors are reproduced:

- **Comparison with previous calibration**
  The comparison with the previous calibration shows you how urgently this media family calibration was needed. This feedback helps you to determine how often media family calibration is required.
- **Measurement of paper white and primary colors**
  The feedback on the measured paper white and primary colors window shows how close the measured paper white and primary colors approach the reference values. The reference values are taken during the first media family calibration or when you create a new profile with the embedded profiler.
- **Measurement accuracy in numbers and heatmaps**
  The measurement accuracy information shows one or more heatmaps that indicate how accurately the patches were measured. PRISMAsync Print Server uses tolerance levels for patch measurements. Based on the configured tolerance levels, heatmap patches turn green, orange-like, or red. The heatmap can indicate a possible problem. When you want to change the tolerance values, contact your Service organization.

Recommended calibration scenario

You are advised to follow the next calibration scenario to keep the image quality high.

- Perform the printer calibration daily.
- Perform the shading correction once.
- Perform the media family calibration regularly.

IMPORTANT
Additional procedures are not necessary to keep the print quality high.

More information

- *Learn about calibration curves* on page 350
- *Learn about color differences and tolerances* on page 352
- *Calibrate the printer* on page 368
- *Perform shading correction* on page 370
- *Calibrate a media family* on page 373
- *Register the custom media for automatic gradation adjustment* on page 413

Learn about calibration curves

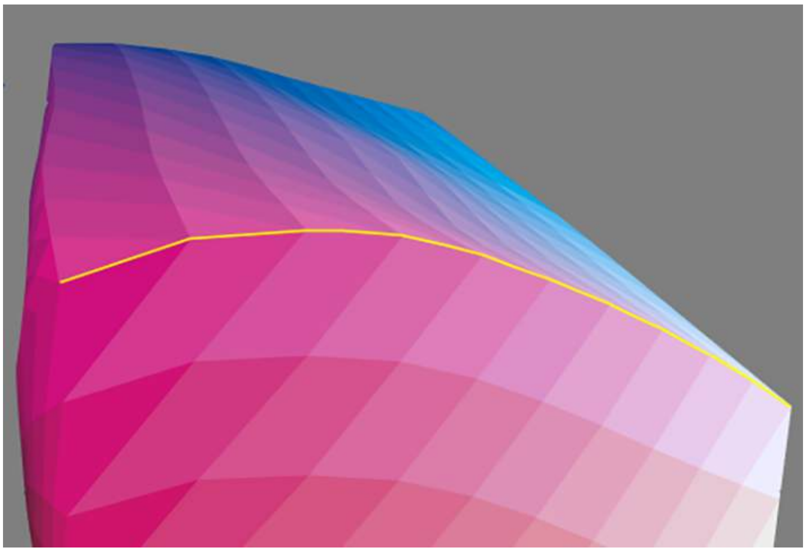
What is the purpose of calibration

The calibration process ensures that for each tone value from 0% to 100% the exact amount of toner or ink is deposited on the media. This exact amount per tone value is determined by the calibration curve.

What is Distance to White (DTW)

The amount of ink or toner on the media is difficult to measure in, for example mg / m^2. Therefore, PRISMAsync Print Server uses an indirect method to measure this amount on the media: **Distance to White (DTW)**.

The DTW for a given tone value of a process color, for example cyan, is a distance in the Lab color space. It is the distance between the measured Lab value of paper white and, the measured Lab value of the process color at that given tone value. The DTW for tone value 0 is always zero. The DTW for tone value 100% is the maximum amount of toner or ink the printer can deposit on the media: **maximum Distance to White (Max DTW)**



DTW tone values for cyan in the Lab color space

Note that the application of the DTW concept is similar to the traditional concept of optical density.

What is a calibration target

The calibration target specifies per color channel (CMYK), the required DTW value for each tone value.

The calibration must ensure that the amount of toner or ink specified in the output profile is deposited on the media. Therefore, the calibration target generally is derived from the output profile. When the output profile is installed on another printer of the same type, the calibration will automatically adjust its target to this output profile.

However, there are situations that the output profile is invalid, for example because no output profile is available for specific media. In that case, PRISMAsync Print Server allows to derive the calibration target from the calibration measurements. Then, the calibration is done in such a way that the DTW increases linearly per tone value, until it reaches the maximum measured DTW value.

This method is also used in case of a composite output profile. It is not possible to derive a calibration target from the composite output profile.

What is a calibration curve

The result of the calibration process are four calibration curves, one per color channel (CMYK). Each curve maps an input tone value to an output tone value, so that the correct amounts of toner or ink are deposited on the media.

Calibration curves can be restored to the default curves. Then, the default curves map input tone values to the same output tone values. This can be useful for testing purposes.

More information

- *Learn about calibration* on page 346
- *Learn about color differences and tolerances* on page 352
- *Calibrate a media family* on page 373
- *Restore calibration curves* on page 378
- *Define the media families* on page 423

Learn about color differences and tolerances

What are color differences

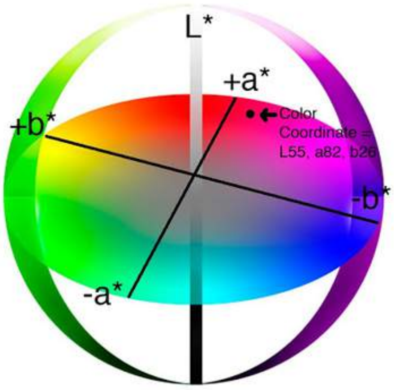
Color differences are the differences between the requested colors and reproduced colors. These color differences can be caused by factors such as printer environment, printer condition, printer gamut, media, and output profile. This topic explains how PRISMAsync compares and represents color differences.

CIELAB color values

Requested colors, printed colors and measured colors can all be defined with three numerical values. These three numerical values form together a three-dimensional color space. This makes it possible to calculate color differences by calculating the distance between two colors in a color space. For example between a requested color and the reproduced color.

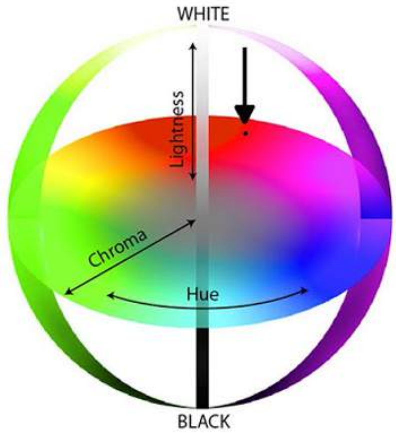
The Commission Internationale d'Eclairage (CIE) offers the following CIELAB variants to define a color space:

- **CIE L*a*b* (CIELAB)**, in which:
  L* represents the lightness (from 0 to 100)
  a* represents the green to red axis (from -128 to +127)
  b* represents the yellow to blue axis (from -128 to +127)

  CIE L*a*b*

- **CIE L*C*h* (CIELCh)**, in which:
  L* represents the lightness (from 0 to 100)
  C* represents the chroma or color saturation, represented by the distance from the lightness axis (from 0 to > 100 at the edge)
  h* represents the hue or color, represented by the angle with respect to the a-axis (from -180° to +180°)
  C* and h* are calculated from the a* and b* coordinates of the CIE L*a*b* model. The same color is thus defined in two different notations: CIE L*a*b* and CIE L*C*h*.

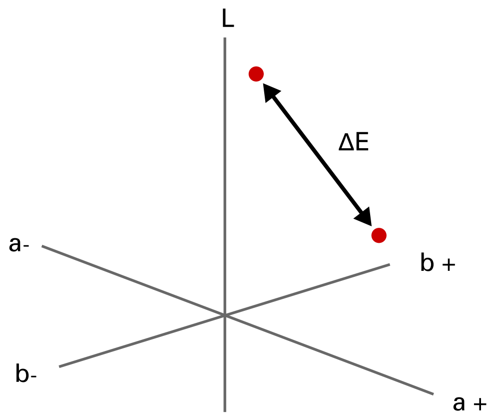


CIE L*C*h*

The CIELAB color space is device independent and based on how humans perceive color differences. These are the main reasons why PRISMAsync uses the CIELAB color space to indicate color differences using the color metrics described below.

△E (Delta E)

The most general CIE metric to express the difference between two colors in CIELAB is Delta E. PRISMAsync Print Server uses two variants of Delta E:
- Delta E 1976, ΔE_{76}, based on the calculated distance between two colors.
- Delta E 2000, ΔE_{00}, based on the calculated distance between two colors and the human perception of color differences.



Difference between two colors in CIELAB

The following Delta E_{00} range illustrates how the values of Delta E may be interpreted for a specific color evaluation.
- Delta E_{00} smaller than 1 is not visible by humans.
- Delta E_{00} between 1 and 2 is visible through close observation.
- Delta E_{00} between 2 and 10 is visible at a glance.

△H (Delta H)

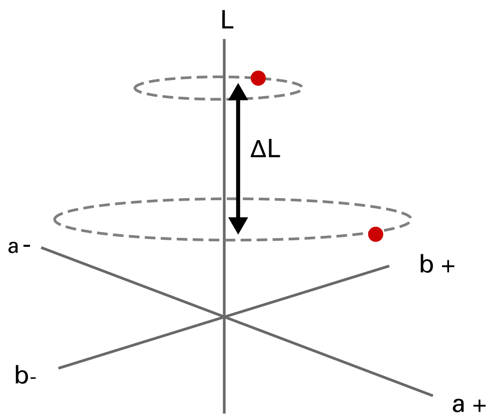
To indicate more color difference information than the total color difference Delta E, the hue error △**H (Delta H)** is used.

You can imagine Delta H as which color difference is left over when the lightness and the chroma differences are ignored.

△L (Delta L)

△**L (Delta L)** is the lightness difference between the two colors. **IΔLI** is the absolute value of △L (Delta L).

The **weighted** |ΔL| puts more emphasis on the color differences for light grays, where the K value is lower than 50%. The goal of weighting is to reduce the importance of the lightness factor for very dark grays. Humans usually do not identify errors in the very dark gray color area. This weighting definition is in accordance with the IDEAlliance G7 specification.



△Ch (Delta Ch)

Close to the gray axis, in other words for near neutral colors, the usage of Delta H is not appropriate enough. Instead, the usage of the △**Ch (Delta Ch)** metric is proposed.

The metric that is called **chromaticness** can reveal gray balance errors.

The **weighted** ΔCh puts more emphasis on the color differences between composite grays, where C is lower than 50%. The goal of weighting is to reduce the importance of the gray balance for very dark grays. These dark grays errors are difficult to handle and are usually printed with black toner. This weighting definition is in accordance with the IDEAlliance G7 specification.

Define printer calibration defaults

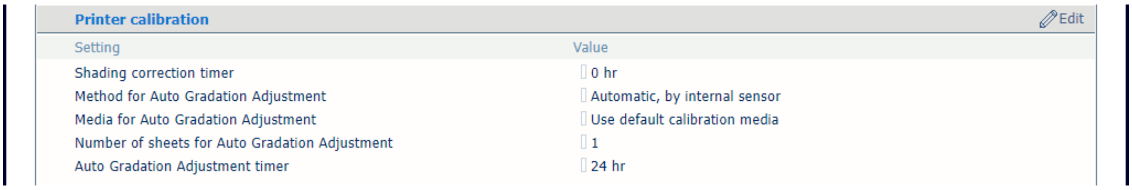
You can define printer calibration defaults.

1. Open the Settings Editor and go to: [Color]→[Color calibration].

[Color calibration] tab

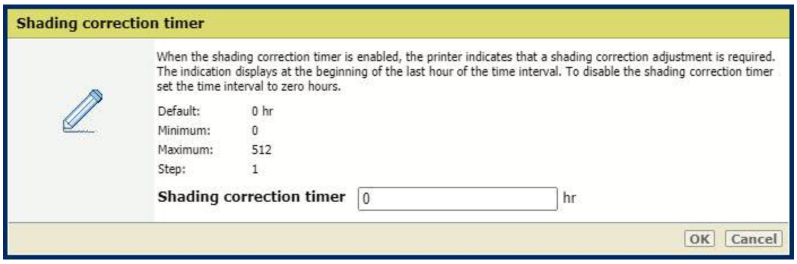
Set the shading correction timer

1. Go to the [Printer calibration] section.



[Printer calibration] section

2. Use the [Shading correction timer] setting to set the required time interval.

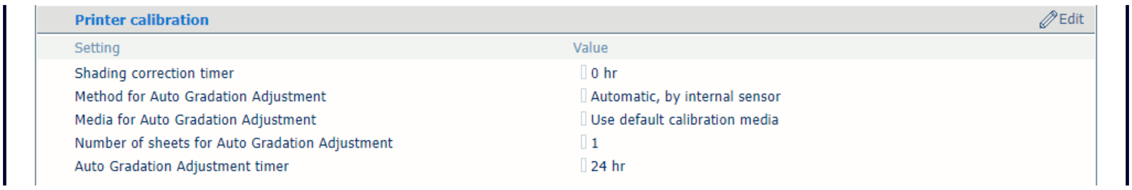


[Shading correction timer] setting

3. Click [OK].

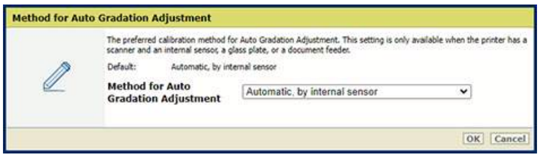
Define the method for automatic gradation adjustment

1. Go to the [Printer calibration] section.



[Printer calibration] section

2. Use the [Method for Auto Gradation Adjustment] setting to define the preferred calibration method for automatic gradation adjustment.

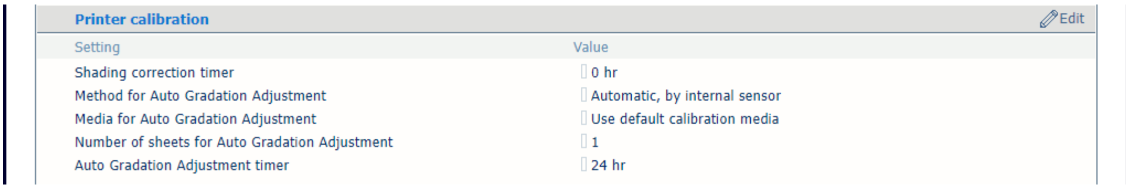


[Method for Auto Gradation Adjustment] setting

3. Click [OK].

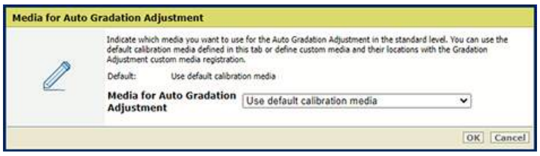
Define the media for automatic gradation adjustment

1. Go to the [Printer calibration] section.



[Printer calibration] section

2. Use the [Media for Auto Gradation Adjustment] setting to define which media must be used for automatic gradation adjustment.



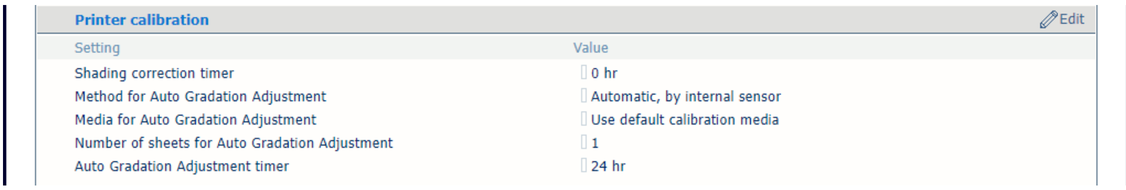
[Media for Auto Gradation Adjustment] setting

3. Click [OK].

Define the number of sheets for automatic gradation adjustment

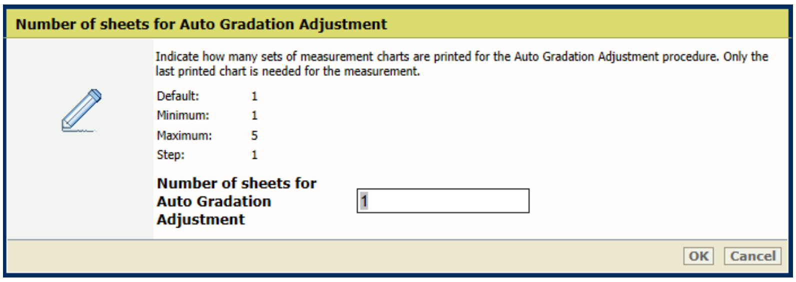
The higher number of sheets increases the print quality. However, the duration of the procedure is extended.

1. Go to the [Printer calibration] section.



[Printer calibration] section

2. Use the [Number of sheets for Auto Gradation Adjustment] setting to indicate the number of sheets for auto gradation adjustment.
   The total number of charts printed is the indicated number of sheets plus three.

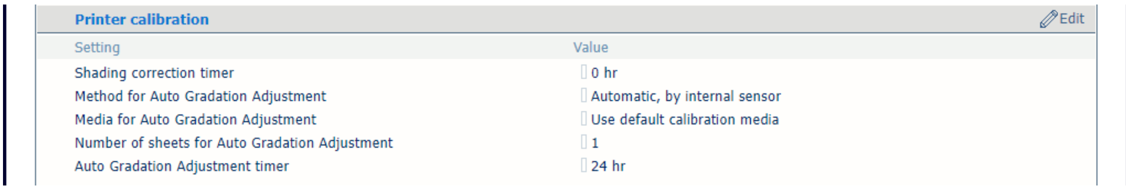


[Number of sheets for Auto Gradation Adjustment] setting

3. Click [OK].

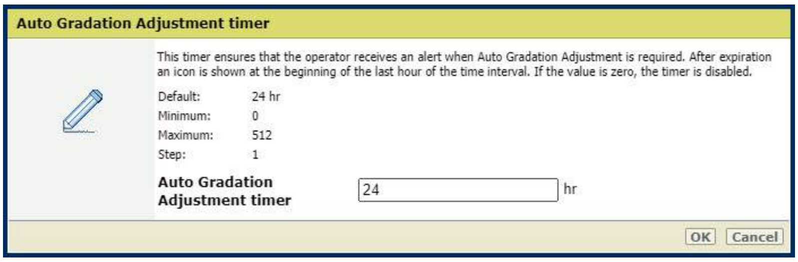
Set the automatic gradation adjustment timer

1. Go to the [Printer calibration] section.



[Printer calibration] section

2. Use the [Auto Gradation Adjustment timer] setting to define the required time interval between the procedures. It is recommended to set the time interval so that you get a reminder to perform the automatic gradation adjustment every day.

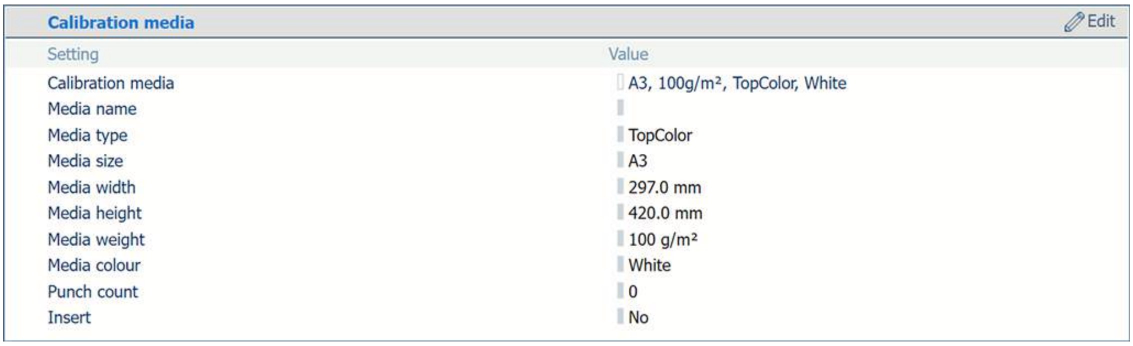


[Auto Gradation Adjustment timer] setting

3. Click [OK].

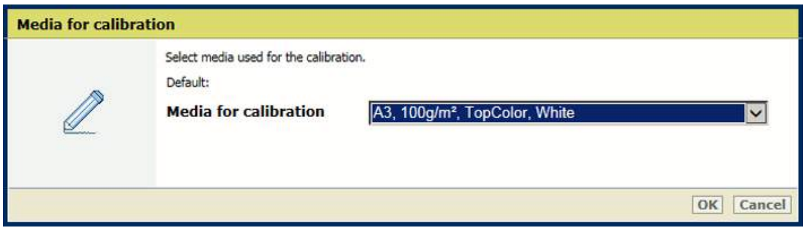
Define the media for printer calibration

1. Go to the [Default calibration media] section.



[Default calibration media] section

2. Use the [Default calibration media] setting to select the media for printer calibration.



[Default calibration media] setting

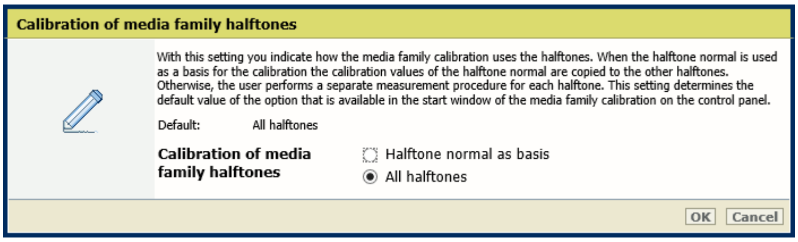
3. Click [OK].

Define how the media family uses the halftones

1. Go to the [Media family calibration] section.

[Media family calibration] section

2. Use the [Calibration of media family halftones] setting to indicate how the media family calibration uses the halftones.



[Calibration of media family halftones] setting

3. Click [OK].

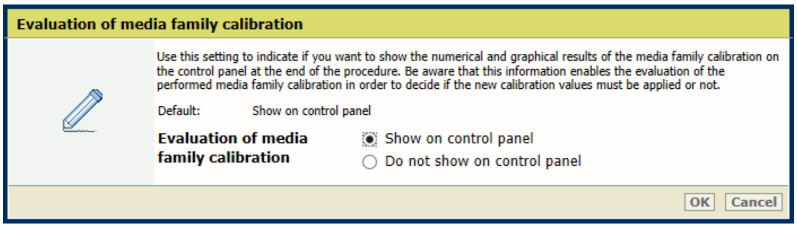
Enable the evaluation of the media family calibration on the control panel

NOTE
Only the service operator can change this setting.

1. Go to the [Media family calibration] section.

[Media family calibration] section

2. Use the [Evaluation of media family calibration] setting to indicate if you want to show the results of the media family calibration on the control panel at the end of the procedure.



[Evaluation of media family calibration] setting

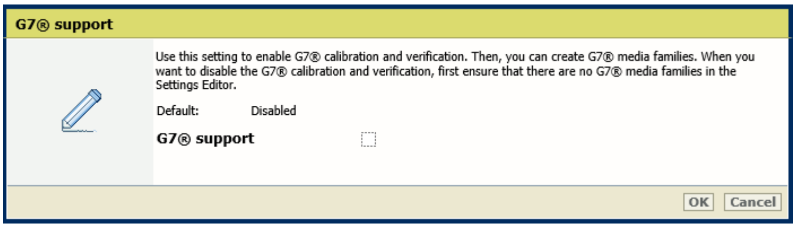
3. Click [OK].

Enable G7 calibration

1. Go to the [G7®] section.

[G7®] section

2. Use the [G7® support] setting to enable G7 calibration.



3. Click [OK].

You can only disable G7 support when all G7 media families are removed.

Measure charts with the i1Pro3 spectrophotometer

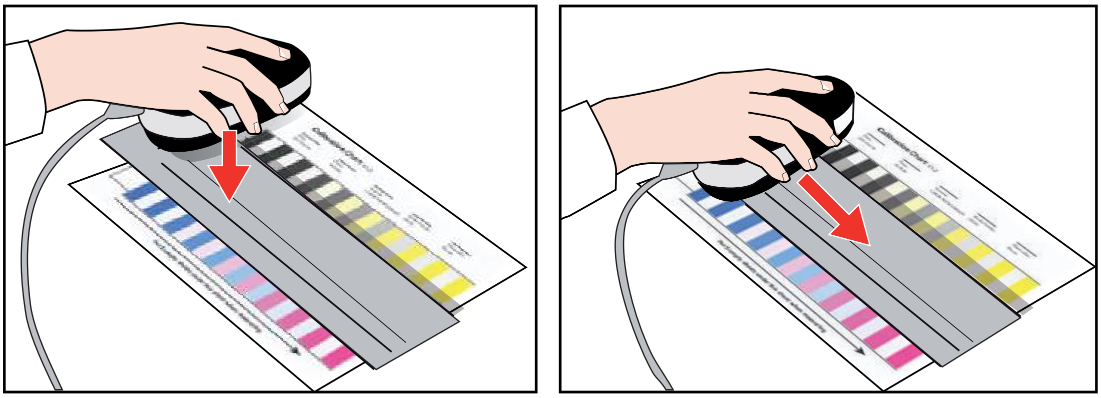
For Auto Gradation Adjustment, media family calibration, shading correction, and Idealliance® G7® Grayscale test, you can use the i1Pro3 spectrophotometer (i1Pro2 also supported) to measure charts.

NOTE
Except for shading correction, you can also use the inline spectrophotometer, which is the preferred method.

You can use the ruler that belongs to the equipment to prevent that you touch the toner layer on the chart with the i1Pro3 spectrophotometer.

The i1Pro3 supports M0 and M1 measurement modes. Read the user manual of the i1Pro3 spectrophotometer for more information.

Prepare the measurement



i1Pro3 spectrophotometer

1. Ensure you know which target media must be used to perform the measurement.
2. Prepare the target media to print the charts.
3. Connect the i1Pro3 spectrophotometer to the USB port of the control panel.
4. To use the M1 measurement mode enable the setting. Open the Settings Editor and go to: [Color]→[Color calibration], section [M1 measurement mode].
5. Make sure the protective slider shows the white calibration tile of the calibration plate and place the i1Pro3 spectrophotometer on the calibration plate.
6. Press the measurement button. The status indicator lights pulsate white when the i1Pro3 spectrophotometer is ready for use.

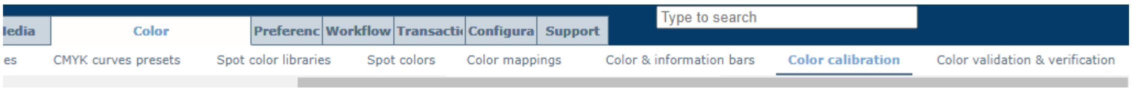
Measure charts

1. Place five blank sheets under the calibration chart.
2. Position the i1Pro3 spectrophotometer on the chart or in the carriage on the ruler.
3. Press the measurement button for 1 second and then move the i1Pro3 spectrophotometer across the first row of patches.
4. Repeat step 3 for each other row.

Automated color tasks

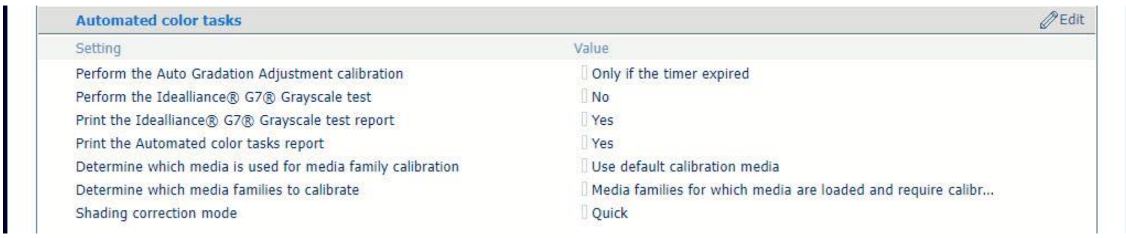
It is important to perform the appropriate calibration procedures regularly. It is possible to perform each calibration procedure separately. However, to ease the process of calibration, PRISMAsync Print Server offers an automated procedure. When you perform the automated color tasks, the calibration procedures will be carried out automatically in the correct order and with limited operator interaction. The automated color tasks include:
- automatic gradation adjustment;
- shading correction;
- media family calibration;
- Idealliance® G7® Grayscale test.

It is important to define the settings for the automated color tasks in the Settings Editor.
1. Open the Settings Editor and go to: [Color]→[Color calibration].



[Color calibration] tab

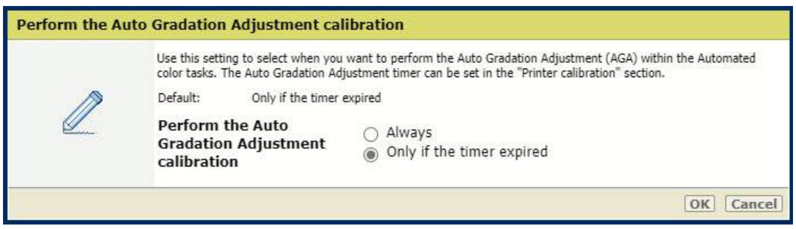
2. Go to the [Automated color tasks] section.



[Automated color tasks] section

Define when to perform the automatic gradation adjustment calibration

With the [Perform the Auto Gradation Adjustment calibration] setting, you can determine when the automatic gradation adjustment is performed.



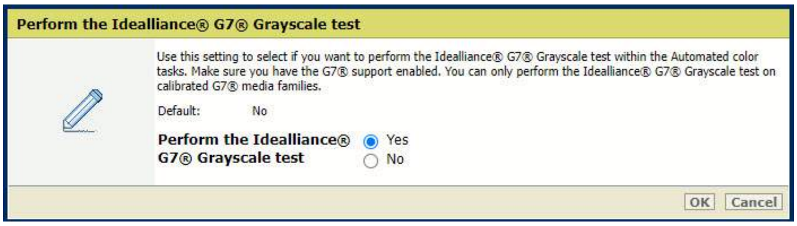
[Perform the Auto Gradation Adjustment calibration] setting

- If you want to perform the automatic gradation adjustment every time you perform the automated color tasks, select [Always].
- If you want to perform the automatic gradation adjustment after a set time, select [Only if the timer expired]. Go to [Color]→[Color calibration]→[Printer calibration] to set the timer value. It is recommended to set the timer value so that the automatic gradation adjustment is performed once a day.

The advice is to always perform the automatic gradation adjustment calibration within the automated color tasks.

Define if you want to perform the Idealliance® G7® Grayscale test

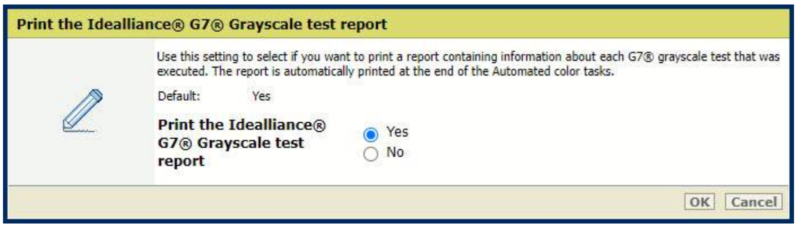
Use the [Perform the Idealliance® G7® Grayscale test] setting to define whether the Idealliance® G7® Grayscale test is performed after a G7® media family calibration.



[Perform the Idealliance® G7® Grayscale test] setting

Define whether you want to print the Idealliance® G7® Grayscale test report

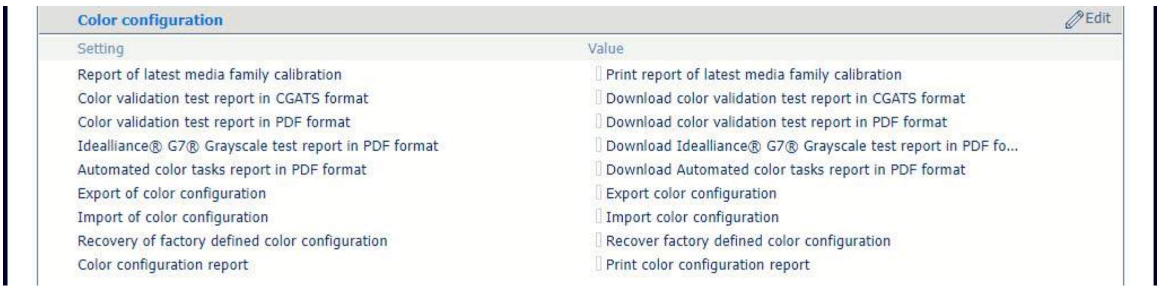
Use the [Print the Idealliance® G7® Grayscale test report] setting to indicate if you want to print the Idealliance® G7® Grayscale test report after the Idealliance® G7® Grayscale test is performed.



[Print the Idealliance® G7® Grayscale test report] setting

You can also download a detailed report of the performed Idealliance® G7® Grayscale test in the Settings Editor: [Support]→[Troubleshooting]→[Download Idealliance® G7® Targeted test report in PDF format].

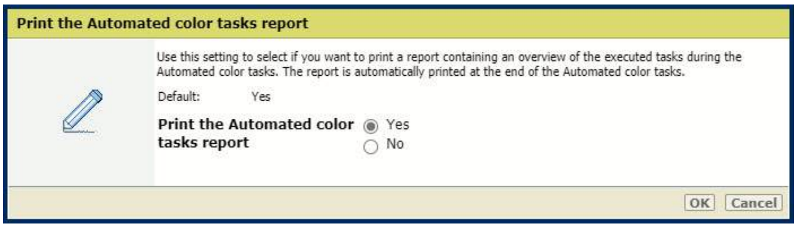
The result of every Idealliance® G7® Grayscale test is retained for four weeks. The last 100 reports can be downloaded.



[Color configuration] section

Define whether the automated color tasks report is printed

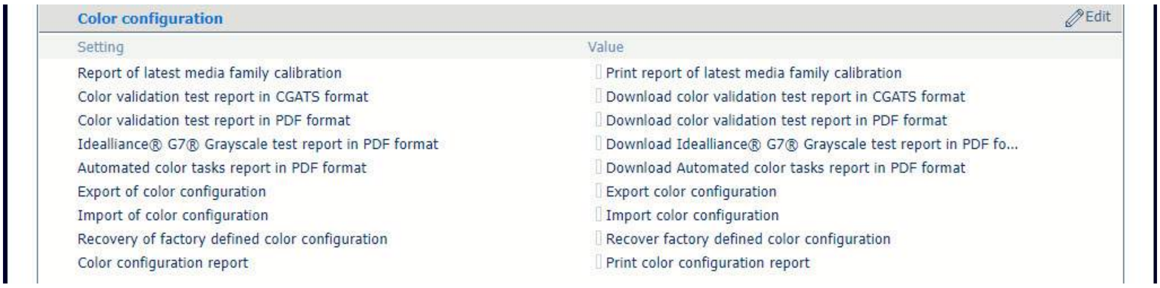
Use the [Print the Automated color tasks report] setting to indicate if you want to print the automated color tasks report after the automated color tasks are performed.



[Print the Automated color tasks report] setting

You can also download a detailed report of the performed automated color tasks in the Settings Editor: [Support]→[Troubleshooting]→[Download Automated color tasks report in PDF format].

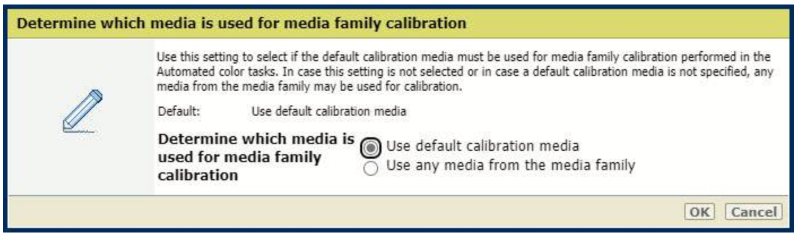
The result of every automated color task is retained for four weeks. The last 100 reports can be downloaded.



[Color configuration] section

Define which media you want to use for the media family calibration

Use the [Determine which media is used for media family calibration] setting to define which media you want to use for the media family calibration performed in the automated color tasks.

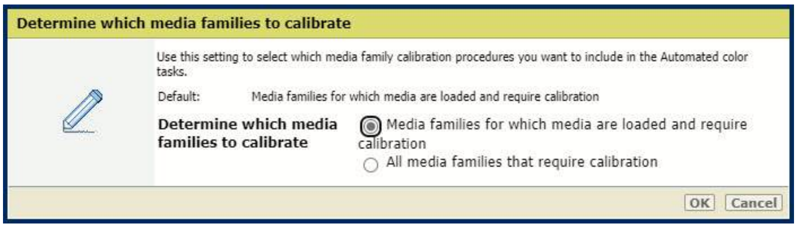


[Determine which media is used for media family calibration] setting

- If you use a specific media most of the time, select [Use default calibration media].
  Go to [Color]→[Color calibration]→[Default calibration media] to set this specific media as the default calibration media.
- If you use multiple media, [Use any media from the media family] is more convenient for calibration. In this case, you do not have to replace media in the paper trays.

Define which media families you want to calibrate

Use the [Determine which media families to calibrate] setting to define which media family calibration procedures are included in the automated color tasks.



[Determine which media families to calibrate] setting

- If you have many media families, calibration takes a lot of time and effort. To make the calibration process more efficient, you can select [Media families for which media are loaded and require calibration].
  Then you only have to calibrate media that you are going to use and that require calibration.

  NOTE
  It is advised to put all the media you need for the coming jobs in the trays before printing. This way, you can be sure that all necessary calibrations are carried out.

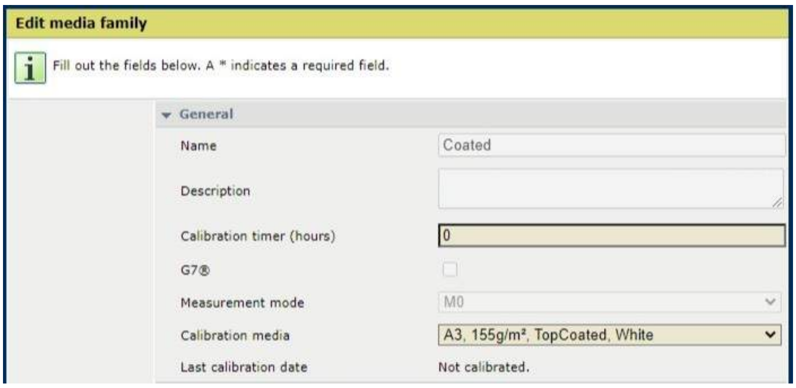
- If you want to calibrate all media families at once, select [All media families that require calibration].
  It is the easiest way to ensure that all media families are calibrated. If you have more media families than paper trays, you need to perform the automated color tasks more than once.

Define the media family calibration timer

In you want media family calibration to be performed within the automated color tasks, you need to adjust the media family calibration timer.

By default, the calibration timer for a media family is disabled (set to 0). If the calibration timer for a media family is disabled, the media family calibration will not be performed as part of the automated color tasks.

To change the media family calibration timer, go to: [Media]→[Media families]→[Edit]. It is recommended to set the timer value so that the media family calibration is performed once a day.

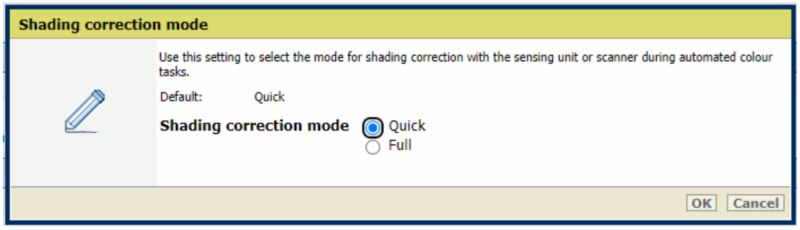


Edit media family

Define the shading correction mode

Use the [Shading correction mode] setting to select the mode for shading correction during automated color tasks.

NOTE
This setting is applicable if the sensing unit is present in your printer configuration.



[Shading correction mode] setting

- If you use the [Quick] mode, the shading correction procedure with the sensing unit takes two minutes.
- If you use the [Full] mode, the shading correction procedure with the sensing unit takes four minutes.

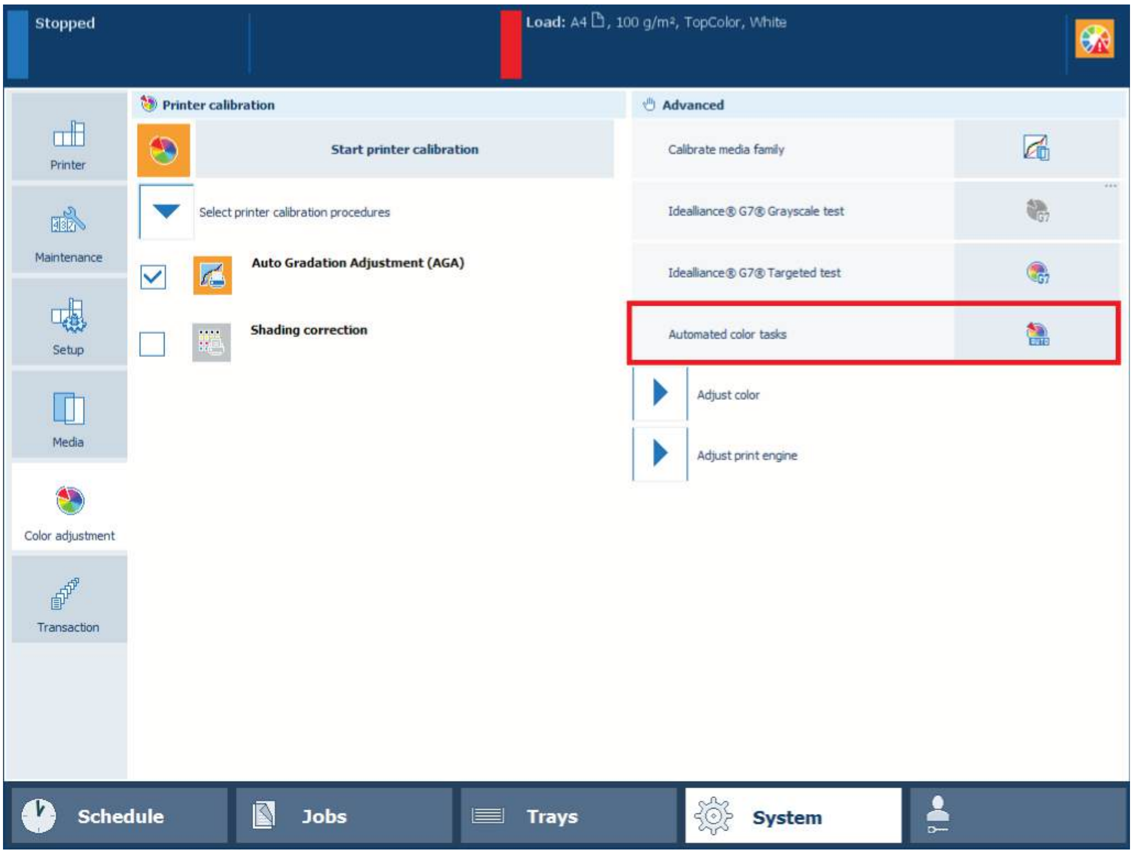
Media specification

For all automated color tasks (except for shading correction), the inline spectrophotometer is used. Use the following media for the automated color tasks:
- Media sizes: A3/11 inch x 17 inch, SRA3/12 inch x 18 inch, 330 mm x 483 mm/13 inch x 19 inch
- Media weights for media type uncoated: 64 g/m² - 300 g/m² (17 lb bond - 110 lb cover)
- Media weights for media type coated: 70 g/m² - 300 g/m² (19 lb bond - 110 lb cover)

Tools

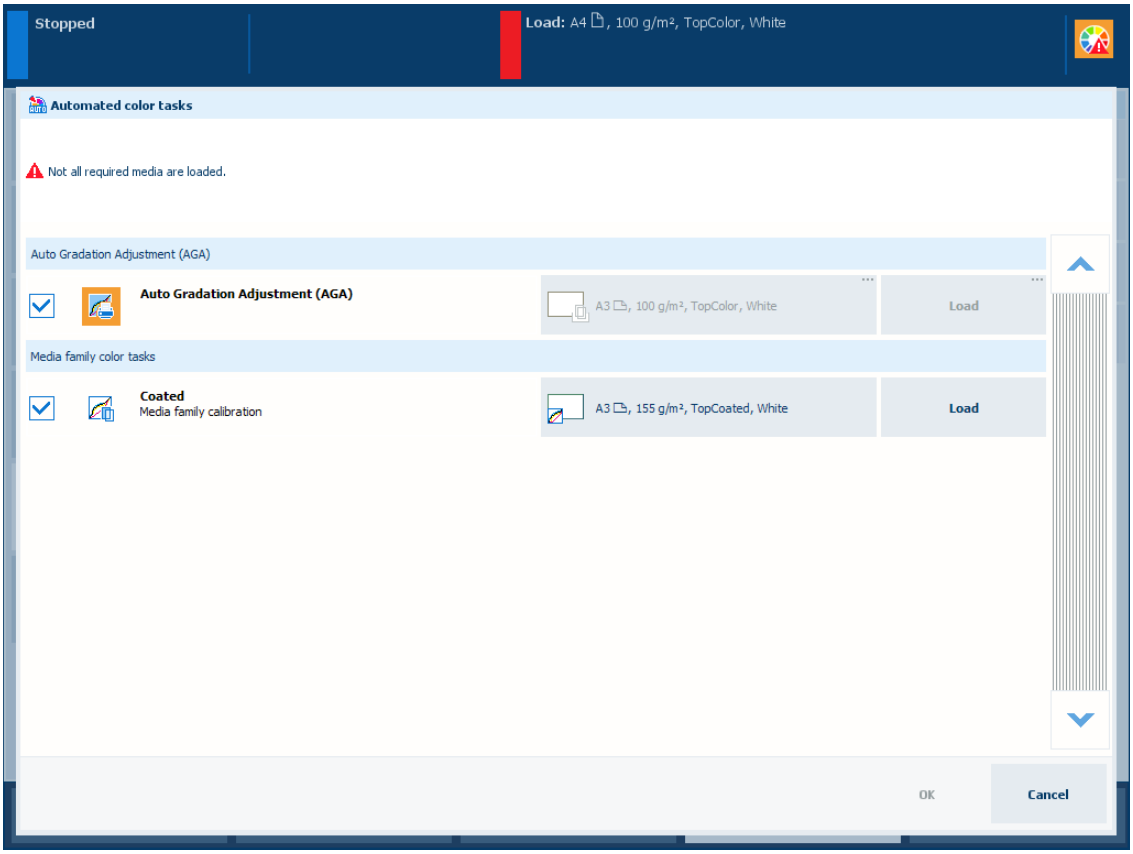
i1Pro3 spectrophotometer (i1Pro2 also supported) when you want to perform the shading correction too.

1. Check if you have defined the required settings for the automated color tasks in the Settings Editor.
2. If necessary, load the calibration media.
3. Go to the control panel and touch [System]→[Color adjustment].



The automated color tasks

4. Touch [Automated color tasks].
   You see the calibration tasks which will be performed according to the settings you defined in the Settings Editor.



5. If preferred, deselect the task you do not want to perform.
6. Touch [OK] to start the first procedure.
7. Follow the instructions on the control panel.

After the automated color tasks have been finished, an overview of the performed tasks are presented on the operator panel. When a task has failed, it is marked with a warning sign. Dependent on the setting in the Settings Editor, a report with the overview of the tasks will be printed.

You can perform the failed task again via the separate procedure to obtain more information about the cause of the failure.

More information

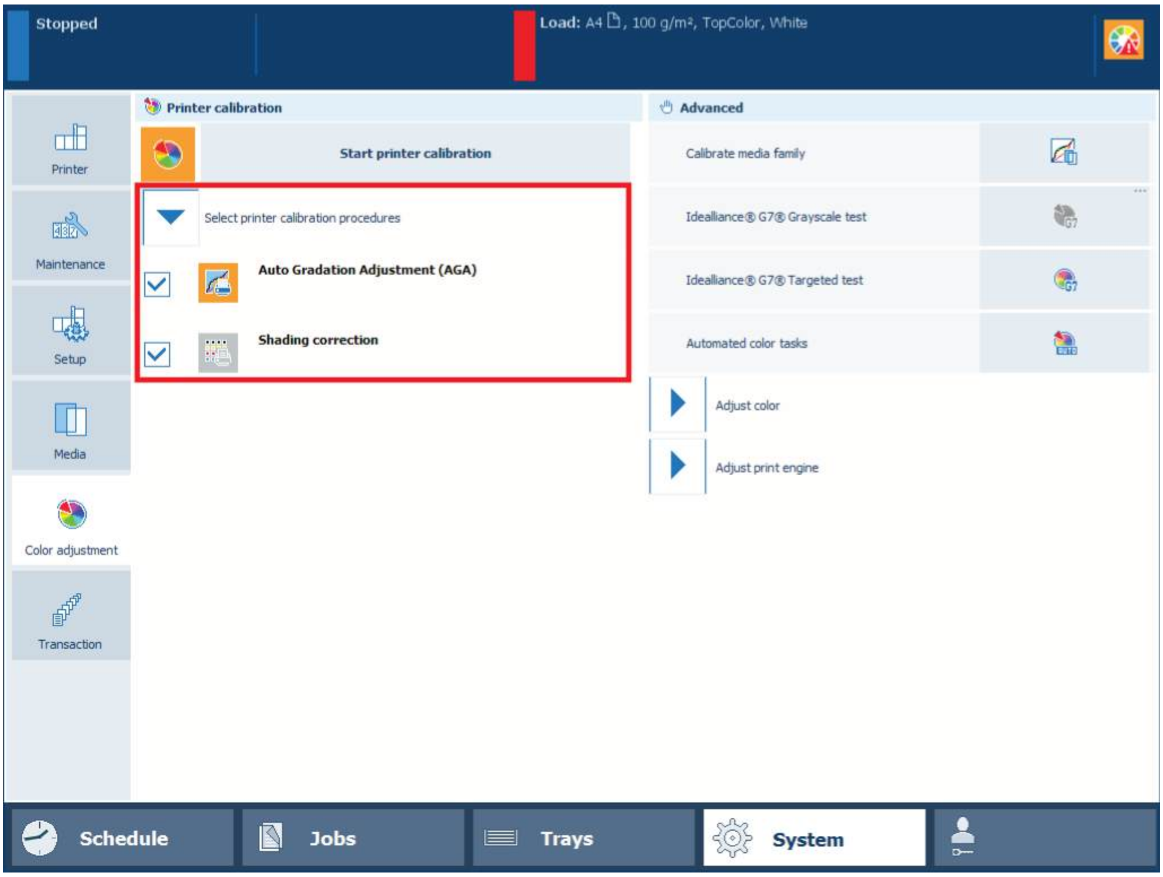
- *Learn about calibration* on page 346
- *Define printer calibration defaults* on page 355
- *Define the media families* on page 423
- *Perform a G7® media family calibration* on page 389

Calibrate the printer

It is important that the printer is calibrated daily. The orange printer calibration indicator on the dashboard helps to remind you when printer calibration is required.

	Printer calibration is required.

As part of the procedure, the system will print calibration charts. For an automatic measurement of these charts, the printer can use the internal scanner as well as the optional sensing unit.



Printer calibration settings

Before you begin

- Prepare calibration media according to the calibration media specifications. The default calibration media is defined in the Settings Editor.
- Print a job of about 100 sheets to ensure that the printer is warmed up.

Procedure

1. Load the calibration media.
2. Touch the calibration indicator on the dashboard or go to [System]→[Color adjustment].
3. If required, touch [Select printer calibration procedures] to check which calibration procedures are configured.
4. Touch [Start printer calibration].

5. Follow the instructions on the control panel.

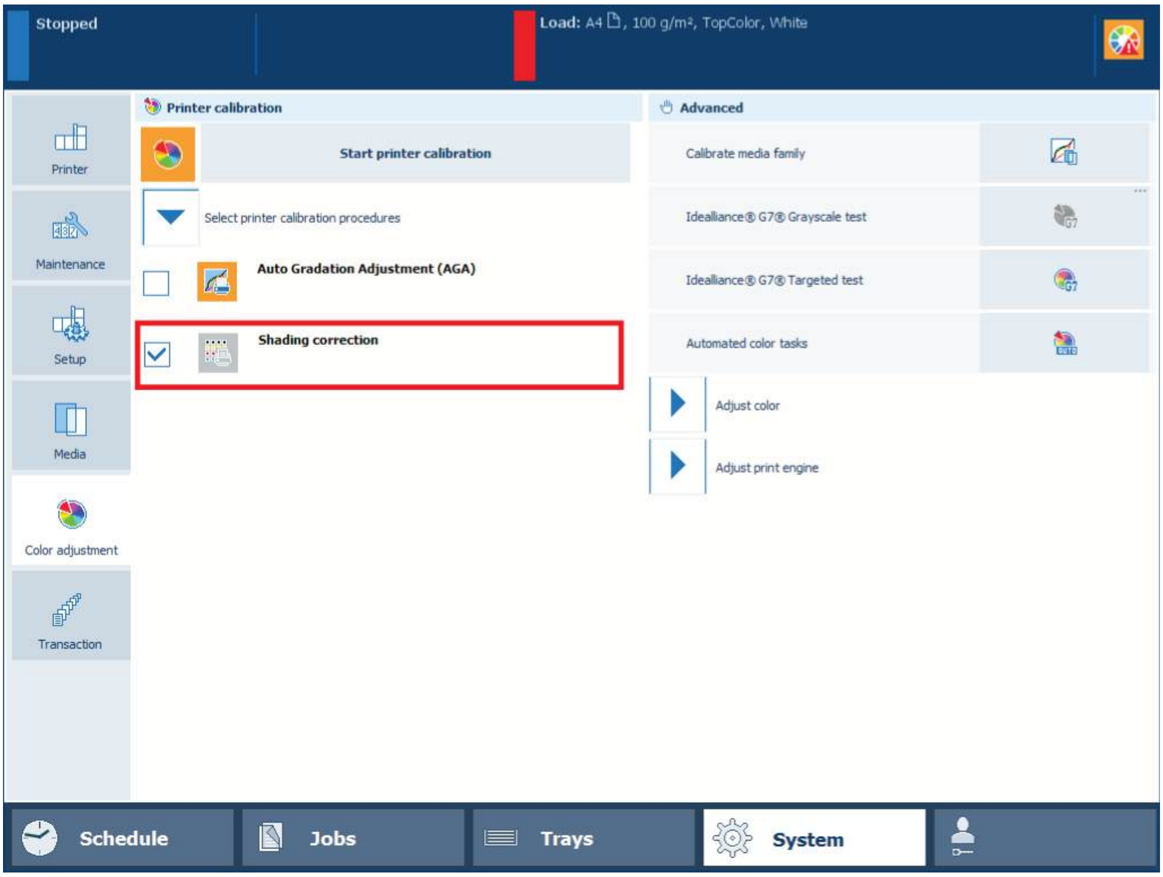
More information

Learn about calibration on page 346

Perform shading correction

In most cases, an automatic gradation adjustment is sufficient to deliver the required color quality. However, in the following situations it is necessary to perform also the shading correction:
- After the installation of the print system.
- When the printed output shows uneven color densities in the color panes.
- When your service organization replaces system parts.

If different supported calibration media are loaded and assigned, the correction procedure will select the media with the largest size. Both coated and uncoated media are suitable for this procedure.



Shading correction

Tools

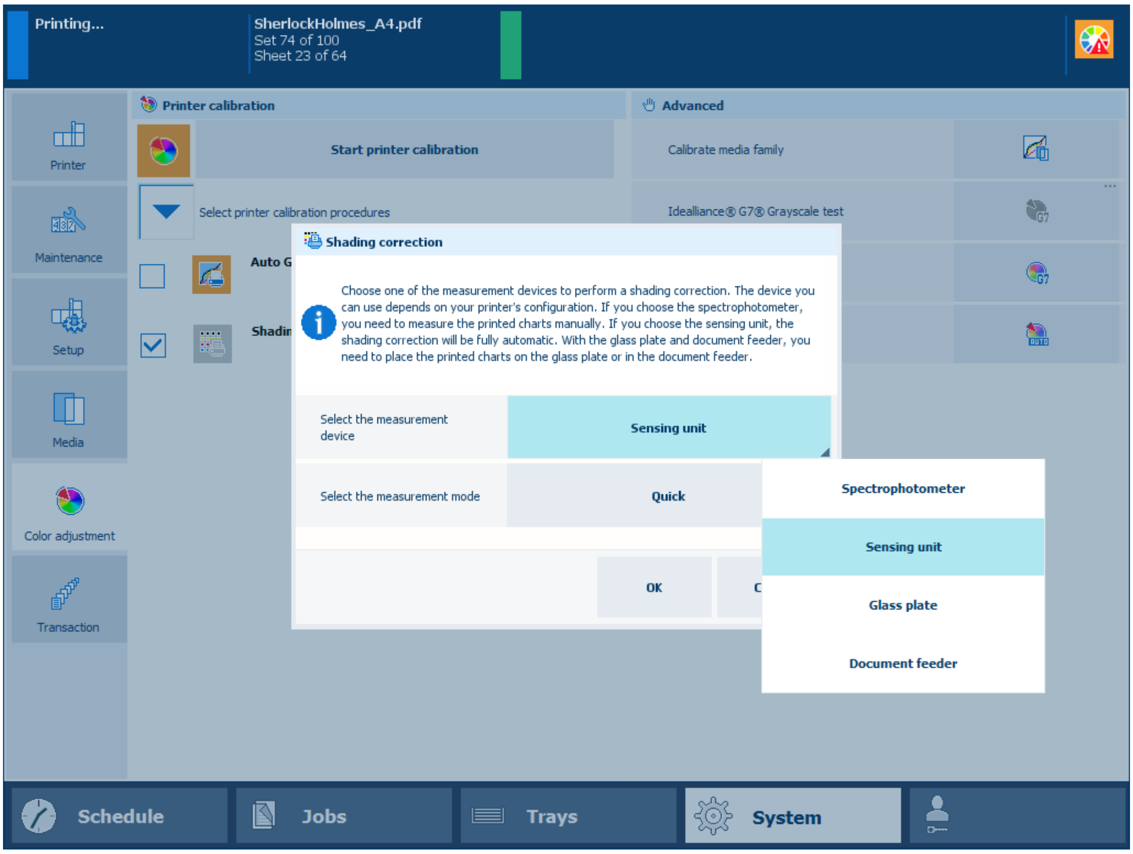
- i1Pro3 spectrophotometer (i1Pro2 also supported).
- Sensing unit (optional)
- Glass plate (optional)
- Automatic document feeder (optional)

Procedure

1. Calibrate the printer.

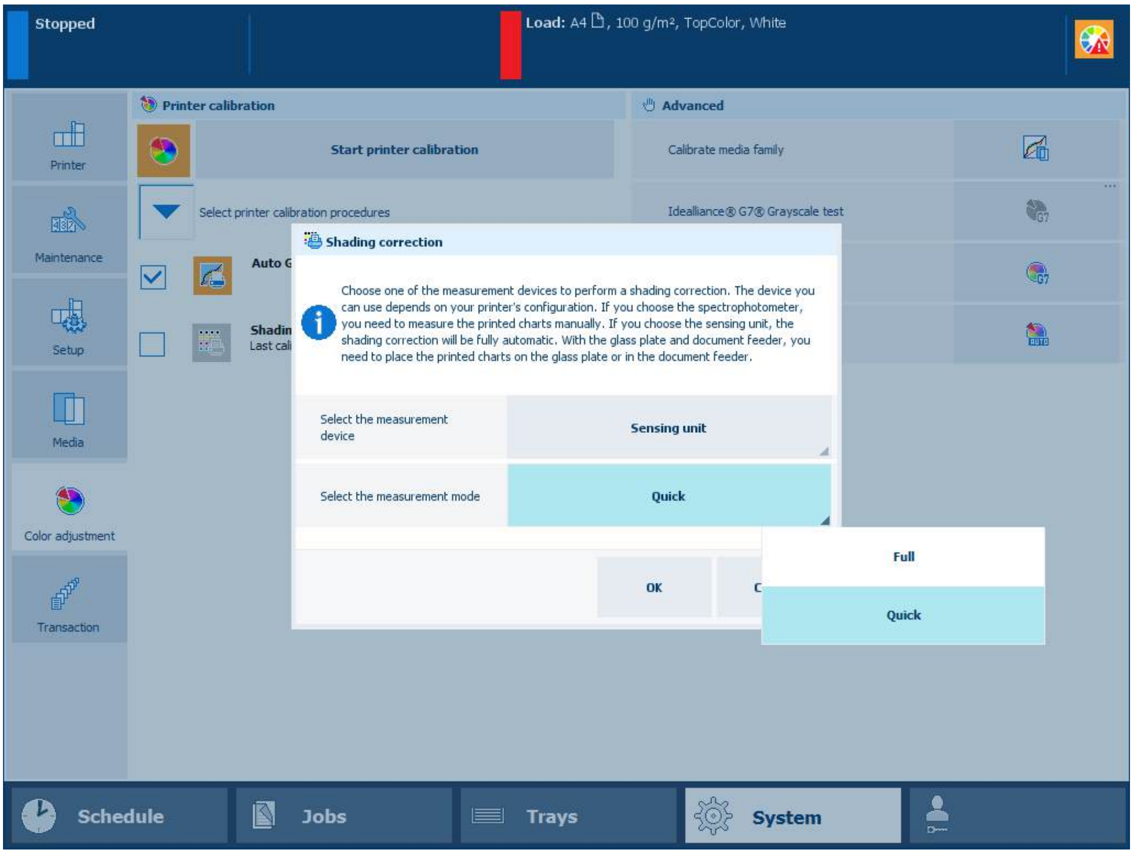
IMPORTANT
You must always perform a printer calibration before you continue with this procedure.

2. Connect the i1Pro3 spectrophotometer to the USB port of the control panel.
3. Touch [System]→[Color adjustment]→[Select printer calibration procedures].
4. Select [Shading correction].
5. Touch [Start printer calibration].
6. Select the measurement device.



Shading correction measurement device

7. Select the measurement mode: quick or full.



Shading correction measurement mode

If you use the [Quick] mode,
- the shading correction procedure with the sensing unit takes two minutes.
- the shading correction procedure with the glass plate or automatic document feeder involves printing and scanning of one chart.

If you use the [Full] mode,
- the shading correction procedure with the sensing unit takes four minutes.
- the shading correction procedure with the glass plate or automatic document feeder involves printing and scanning of two charts.

8. Touch [OK]
9. Follow the instructions on the control panel.

After you finish

When the shading correction produces a "density not within required range" error, perform a correction procedure for this error.

More information

- *Calibrate the printer* on page 368
- *Fix "density not within required range" error* on page 559

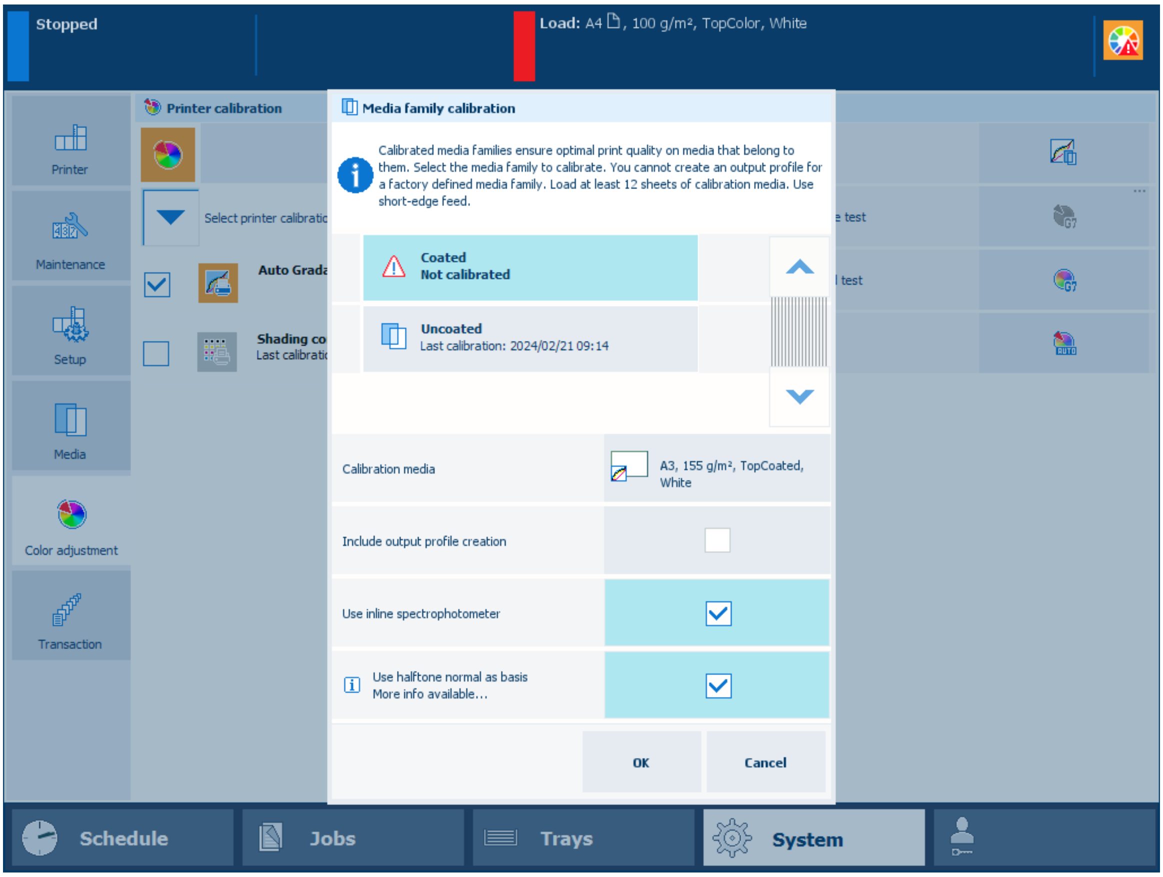
Calibrate a media family

The goal of the media family calibration is to achieve consistent color quality over time for media of the same media family.

Calibrate a media family in the following situations:
- After the installation of the print system, for the default coated and uncoated media families.
- When you create a new media family.
- When your Service organization replaces system parts.
- When you want to keep the print quality high for frequently used media families.

A curve icon indicates that a media family is not calibrated. This icon appears when a new media family was created without calibration or when the timer for media family calibration expires.

You are advised to use the inline spectrophotometer for a faster calibration. The print quality is at least as high as with the i1Pro3 spectrophotometer. You can use the inline spectrophotometer for the following media, for all other media you must use the i1Pro3 spectrophotometer:
- Media sizes: A3/11 inch x 17 inch, SRA3/12 inch x 18 inch, 330 mm x 483 mm (13 inch x 19 inch)
- Media weights for media type uncoated: 64 g/m² - 300 g/m² (17 lb bond - 110 lb cover)
- Media weights for media type coated: 70 g/m² - 300 g/m² (19 lb bond - 110 lb cover)



Media family calibration options

Tools

i1Pro3 spectrophotometer.

Before you begin

For more information about:
- Calibration, see *Learn about calibration* on page 346.
- Media family configuration, see *Define the media families* on page 423.
- Tolerance levels, see *View tolerance levels for color evaluations* on page 379.
- Metrics, see *Color validation metrics* on page 592.

Procedure

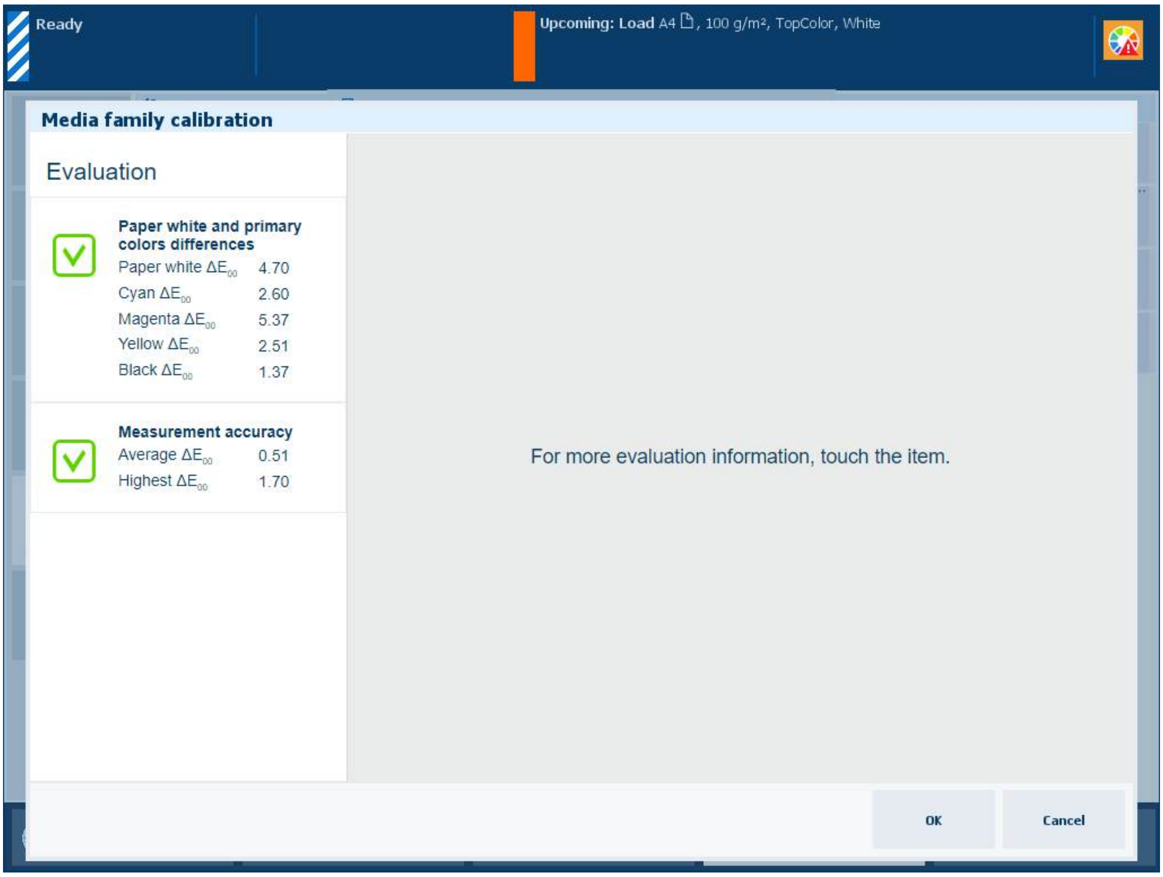
1. Calibrate the printer. (*Calibrate the printer* on page 368)

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Connect the i1Pro3 spectrophotometer to the USB port of the control panel, if required.
3. Touch [System]→[Color adjustment]→[Calibrate media family].
4. Select the media family to calibrate.
5. Select the media for calibration, if required.
   The system displays the default selected media for calibration.
6. Load the calibration media.
7. Indicate if you want to use the inline spectrophotometer or i1Pro3 spectrophotometer.
8. Touch [OK].
9. Follow the instructions on the control panel.

Result

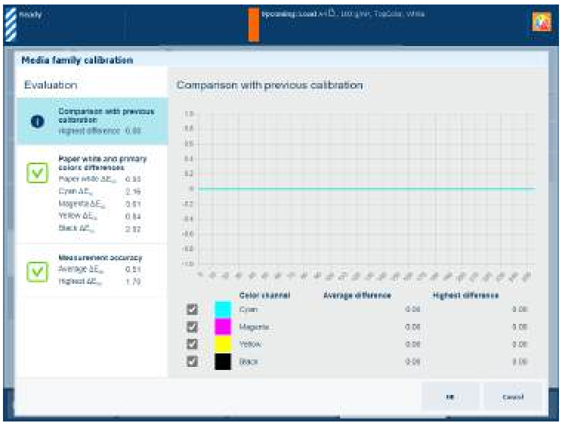
After media family calibration, the feedback window shows the results of the performed procedure.

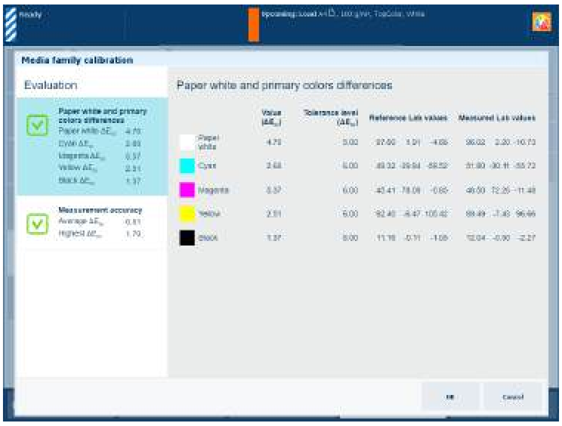


Feedback window of media family calibration procedure

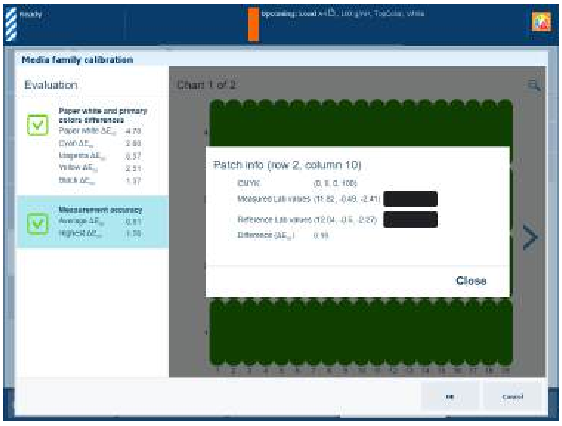
When you see two green checkmark icons, touch [OK] to save the new calibration curves.

When you see a red cross icon, use the table below to evaluate the result. Then, click [OK] to save the new calibration curves or [Cancel] to discard the results.

<table>
<tr><th>Feedback screen</th><th>Remarks</th></tr>
<tr><td rowspan="2">
</td>
<td>Description
The CMYK lines in the graph show how many percent the colour differs compared to the previous media family calibration.
The [Highest difference] and [Average difference] values of the [Comparison with previous calibration] information are indicators for the frequency of the media family calibration.

NOTE
For the first media family calibration there are no reference values and thus no feedback information.</td></tr>
<tr><td>Evaluation
When the [Highest difference] value color is less than about 2 for each color, you can calibrate less often without noticeable loss of color quality.
When the [Highest difference] value is between about 2 and about 4 for each color, you are advised to continue calibrating with the current calibration frequency.
When the [Highest difference] value is greater than about 4 for one or more colors, you are advised to calibrate more often to achieve a more stable color stability.</td></tr>
<tr><td rowspan="2">
</td>
<td>Description
The ΔE values of the [Paper white and primary colors differences] information are indicators for the difference between the reference values and measured values.
The [Reference Lab values] are taken during the first media family calibration.</td></tr>
<tr><td>Evaluation
When each ΔE value is less than the tolerance level, the color quality is fine.
When a ΔE value is greater than the tolerance level, first check that the media used is the same as the media used in the first media family calibration. Then, if required, check the printer condition.</td></tr>
</table>

<table>
<tr><th>Feedback screen</th><th>Remarks</th></tr>
<tr><td rowspan="2">
</td>
<td>Description
The ΔE values of the [Measurement accuracy] information are indicators for the accuracy by which the patches are printed and measured. The control panel displays a heatmap whose patch locations correspond with the printed chart.</td></tr>
<tr><td>Evaluation

When the average ΔE value is less than the [Value where heatmap turns green] value the patches on the control panel turn green and the colour quality is fine.
When the average ΔE value is between the [Value where heatmap turns green] value and the [Value where heatmap turns red] value, the patches on the control panel turn orange-like and the colour quality can still be fine. However, you might want to try the options in the list below.
When the average ΔE value is greater than the [Value where heatmap turns red] value the patches on the control panel turn red. You are advised to try one or more options in the list below.
</td></tr>
<tr><td>
</td>
<td>Ways to improve colour quality

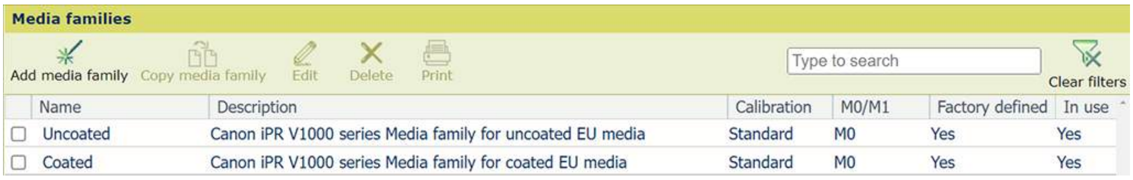
Check the patch on the printed chart
On the control panel, use the > and < symbols to scroll through the calibration charts. Use the zoom symbol to zoom in or out. Then, touch the red patch to find the row and column number to check the corresponding patch on the printed chart to solve the problem. Then, calibrate the media family again.
Calibrate the media family again
Something may have gone wrong during the use of the i1Pro3 spectrophotometer.
Calibrate the printer
See Calibrate the printer on page 368.
</td></tr>
</table>

Restore calibration curves

About restoring calibration curves

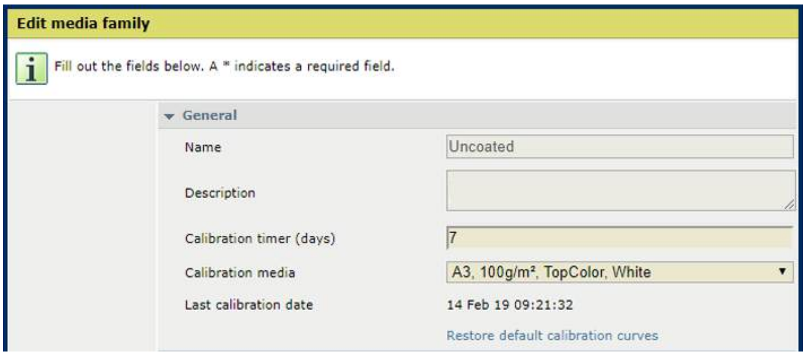
After the calibration curves have been restored, a new media family calibration is required.

1. Open the Settings Editor and go to: [Media]→[Media families].



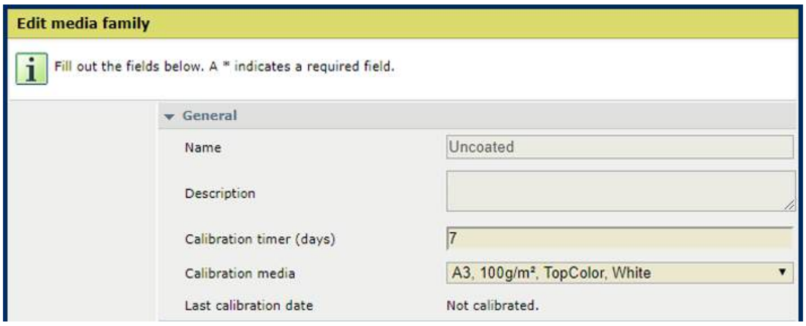
[Media families] menu

2. Go to the media family.
3. Click [Edit].
4. Click [Restore default calibration curves].



Edit media family

The [Last calibration date] setting now shows: [Not calibrated.].



Restored calibration curves

5. Click [OK].

More information

- *Learn about calibration* on page 346
- *Learn about calibration curves* on page 350
- *Calibrate a media family* on page 373

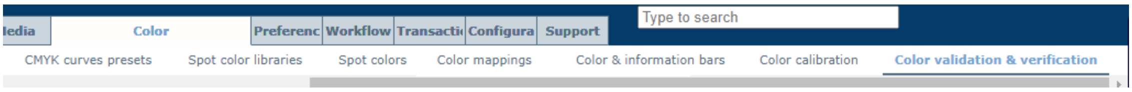
View tolerance levels for color evaluations

The evaluation of the results for media family calibration and G7® Grayscale are based on set tolerance levels.

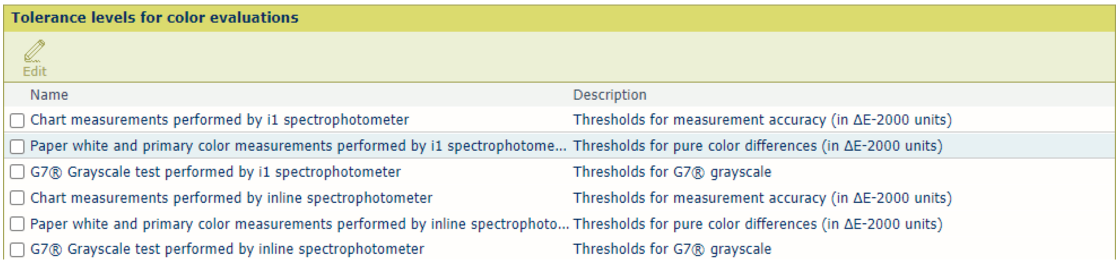
Color evaluation methods use metrics to decide if printed colors are close enough to their reference colors or not. A tolerance level or threshold level is set before the evaluation takes place. The level specifies where a calculated difference is still acceptable and where not. A tolerance level is also expressed by a △(Delta) value.

When you want to change the tolerance values, contact your Service organization.

1. Open the Settings Editor and go to: [Color]→[Color validation & verification].

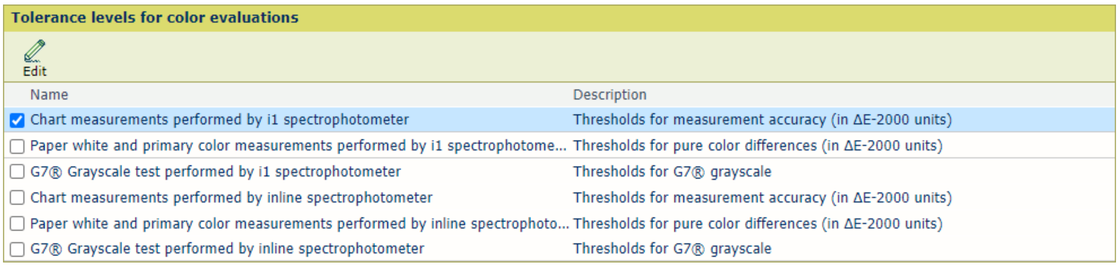


2. Go to the [Tolerance levels for color evaluations] section.



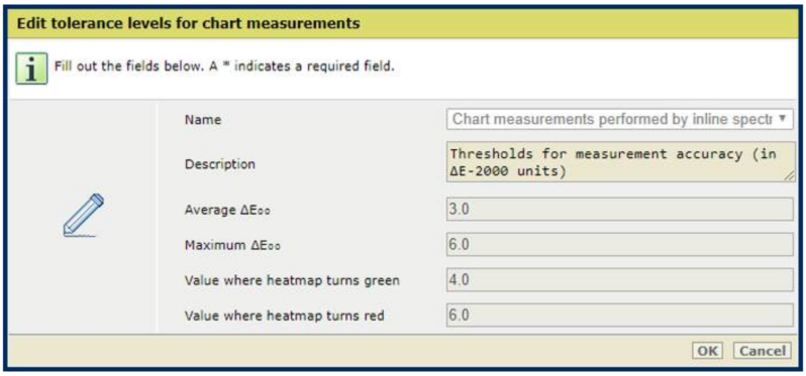
[Tolerance levels for color evaluations] section

3. To view specific tolerance levels, select one of the color evaluations.



[Chart measurements performed by inline spectrophotometer] selected

4. Click [Edit].
5. Click [OK].


Edit tolerance levels for chart measurements
Fill out the fields below. A * indicates a required field.
Name
Chart measurements performed by inline spectr
Description
Thresholds for measurement accuracy (in ΔE-2000 units)
Average ΔE00
3.0
Maximum ΔE00
6.0
Value where heatmap turns green
4.0
Value where heatmap turns red
6.0
OK
Cancel

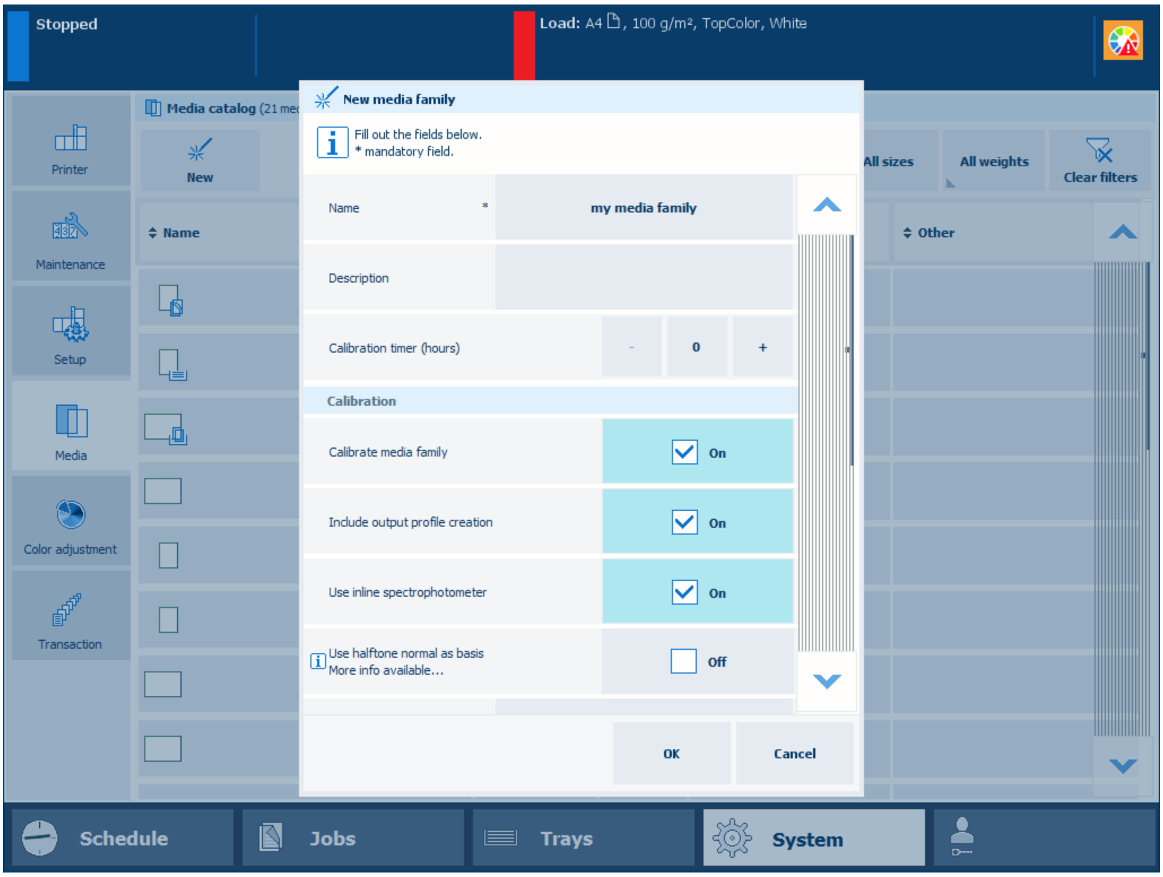
Manage media families and output profiles

Create output profiles with the embedded profiler

Output profiles determine how the system will print color. PRISMAsync uses media families to indicate which output profiles will be used to reproduce color on these media.

When the default output profiles do not achieve your color reproduction standards on specific media, you can create new output profiles. The profile accuracy test can help to determine if you need to create new output profiles for the specific media.

The creation of new output profiles can be done during the media family calibration. The embedded color profiler creates and installs an output profile and calibration curve for every halftone. As part of the procedure, you measure the calibration charts printed on the specific media.



Indicate media family calibration options and output profile creation

Before you begin

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

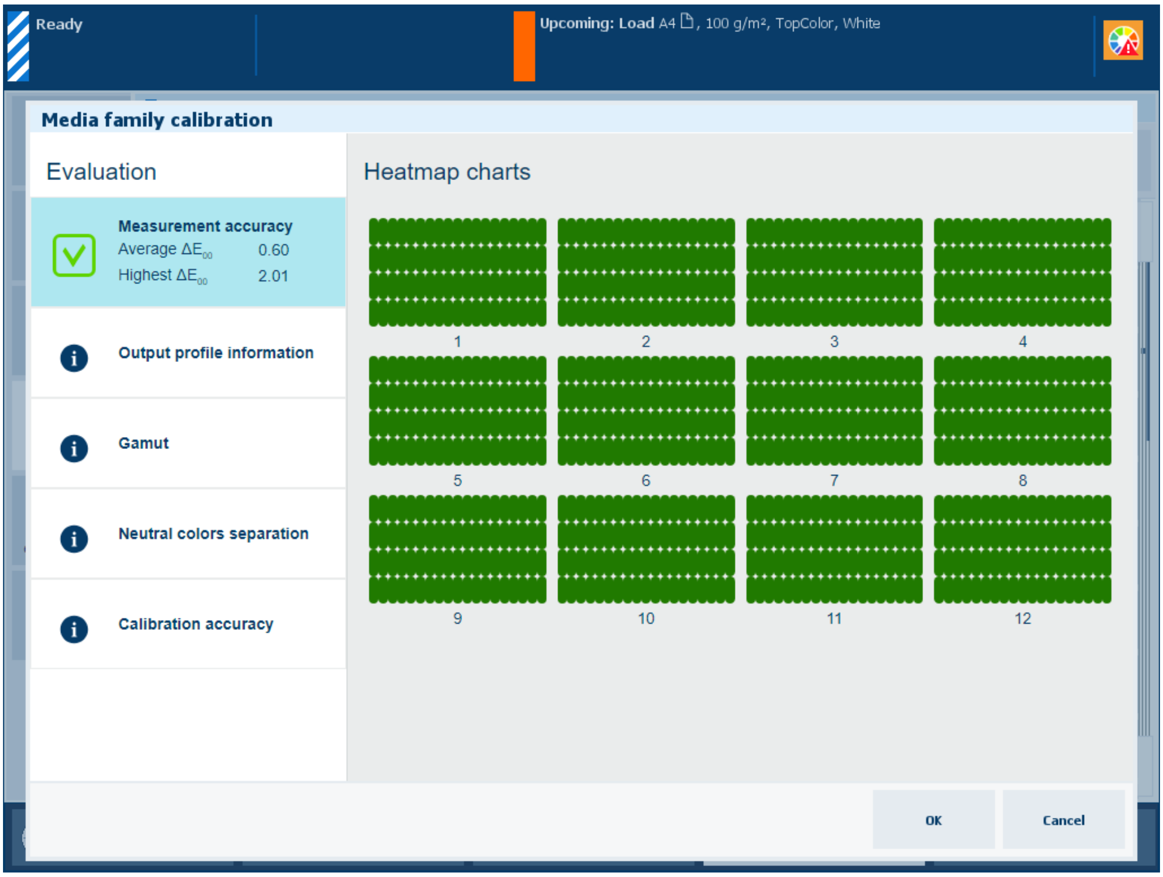
2. Perform a shading correction.

1. Touch [System]→[Media]→[New].
2. Fill out the general media attributes.

   NOTE
   For optimal print quality, you must define the correct values for the following media attributes:
   - Size
   - Weight
   - Surface type
3. Select [Create new media family].
4. Touch [OK].
5. Select [Calibrate media family].
6. Select [Include output profile creation].
7. Select [Use inline spectrophotometer] if applicable.
   You can use the online spectrophotometer with the following media sizes: A3/11 inch x 17 inch, SRA3, 305 mm x 457 mm (12 inch x18 inch), and 330 mm x 483 mm (13 inch x 19 inch).
8. Select [Use halftone normal as basis] to shorten the media family calibration.
9. Select the image adjustment, if required.
10. Load at least 12 sheets of the media for which you create the media family.
11. Touch [OK].
12. Follow the instructions on the control panel.

Result

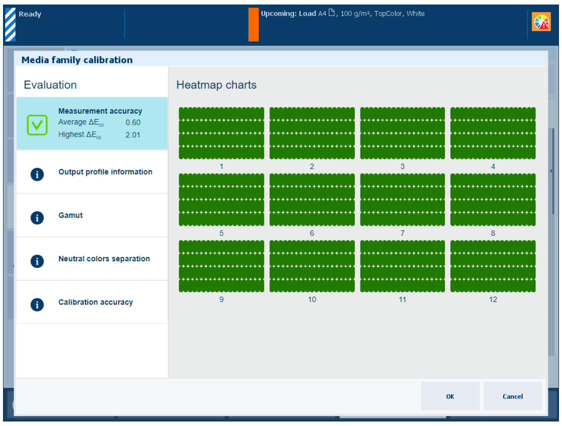
After you created the output profiles, the feedback window shows the results of the performed procedure.

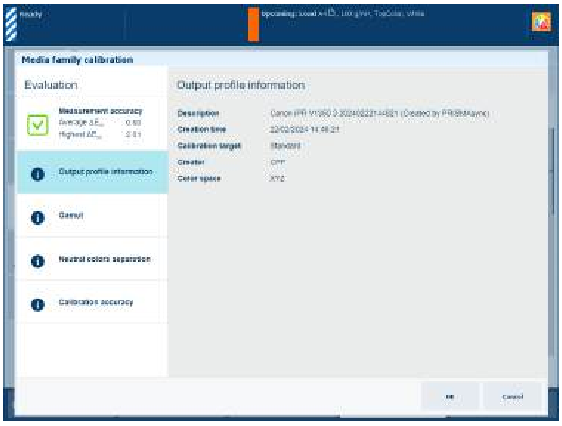
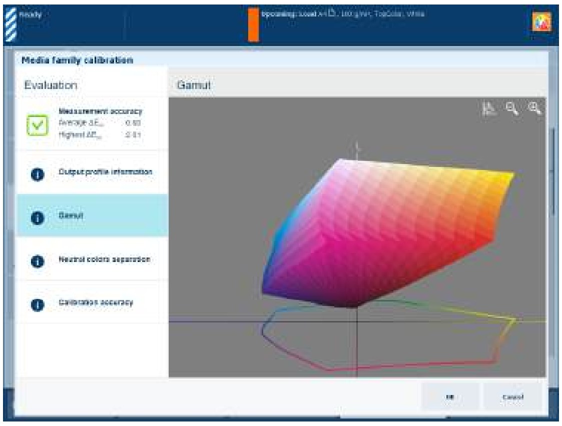


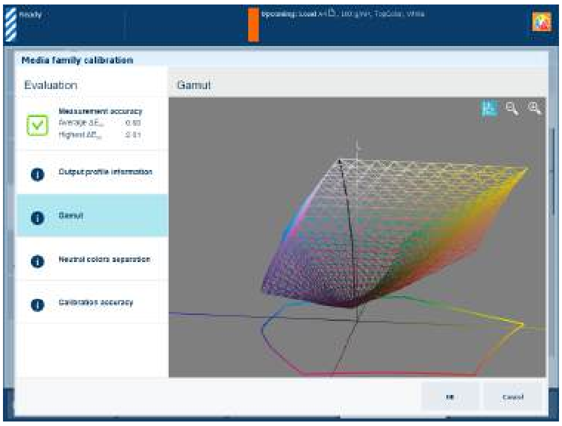
Feedback window of profile creation procedure

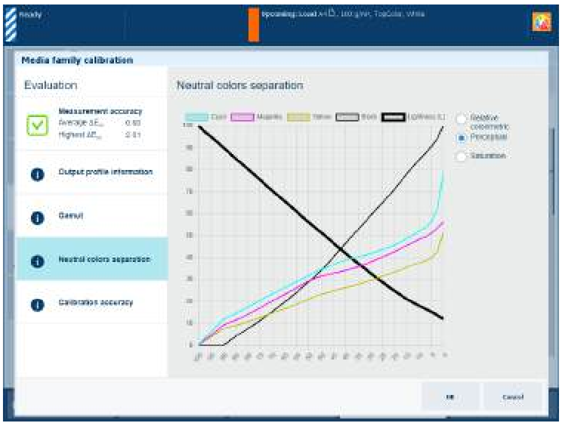
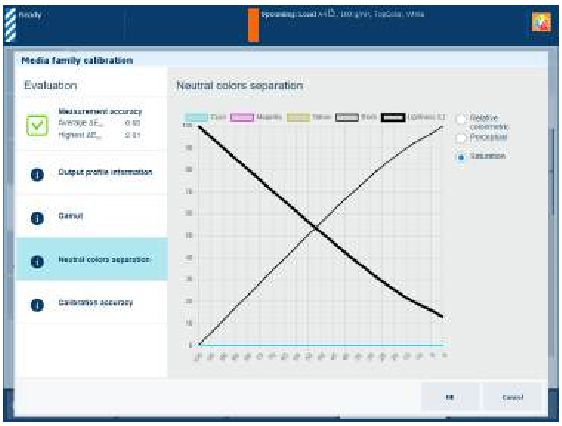
When you see a green checkmark icon, touch [OK] to save the new calibration curves.

When you see a red cross icon, use the table below to evaluate the result. Then click [OK] to save the new calibration curves or [Cancel] to discard the results.

Feedback screen	Remarks
	**Description** The ΔE values of the [Measurement accuracy] information are indicators for the accuracy by which the patches are printed and measured. The control panel displays a heatmap whose patch locations correspond with the printed chart.
	Evaluation • When the average ΔE value is **less than the [Value where heatmap turns green] value** the patches on the control panel turn **green** and the color quality is fine. • When the average ΔE value is **between the [Value where heatmap turns green] value and the [Value where heatmap turns red] value**, the patches on the control panel turn **orange-like** and the color quality can still be fine. However, you might want to try the options in the list below. • When the average ΔE value is **greater than the [Value where heatmap turns red] value** the patches on the control panel turn **red**. You are advised to try one or more options in the list below.
	**Options to improve color quality** • **Check the patch on the printed chart** On the control panel, use the > and < symbols to scroll through the calibration charts. Use the zoom symbol to zoom in or out. Touch the red patch to find the row and column number to check the corresponding patch on the printed chart to solve the problem. Then, perform this profile creation procedure again. • **Check the media** There may be something wrong with the media used. • **Check the printer condition** There may be a printer calibration or new toner required.

<table>
<tr><th>Feedback screen</th><th>Remarks</th></tr>
<tr><td>
</td><td>Description
The [Output profile information] information shows the following characteristics about the output profile: description, creation time, calibration target, creator and color space.</td></tr>
<tr><td rowspan="2">


</td><td>Description
The [Gamut] information is an indication of the quality of the output profile. It is a visual representation of the color gamut that the printer can produce with the created output profile on the used media.
Touch the wire frame icon to view the black curve that shows the gradient of the neutral colors. This curve goes from paper white to the darkest color black that the printer can produce.</td></tr>
<tr><td>Evaluation
When you see an evenly shaped model with smooth color transitions, the output profile is fine.
When you see a smooth black curve, the output profile is fine.
When you notice deviations in the color gamut like dents, imperfections or deformities, examine the printed charts carefully. There can be something wrong with the measurement, the media or the printer condition. After this is resolved, perform the above procedure to create output profiles again.</td></tr>
</table>

Feedback screen	Remarks
	**Description** The **thick black line** of the [Neutral colors separation] graph shows how gray values are reproduced with the separate colors cyan, magenta, yellow and black (CMYK). Note that the darkest lightness value is on the left-hand side and the lightest value on the right-hand side.
	Explanation for [Relative colorimetric] selection The rendering intent [Relative colorimetric] reproduces true colors, for example for proofing applications and spot colors. • With this rendering intent the **thick black line** shows a kink at a lightness of about 12. This is the darkest color black that the printer can reproduce. • The **color black** is only used at a lightness of about 92 (black starting point) and lower.
	**Explanation for [Perceptual] selection** The rendering intent [Perceptual] reproduces contrasts in dark gray areas, for example for photos. • This rendering intent uses black point compensation. Hence, the **thick black line** gradually approaches the darkest color black that the printer can reproduce. • The **color black** is only used at a lightness of about 92 (black starting point) and lower.
	**Explanation for [Saturation] selection** The rendering intent [Saturation] reproduces saturated colors for example for business graphics. • This rendering intent uses only black to create saturated colors. The **thick black line** gradually approaches the lightness of 0. • Only the **color black** is used at a lightness from 100 to 0.

Feedback screen	Remarks
	**Description** The percentage values of the [Calibration accuracy] information show how well the calibration procedure and output profile fit together. The graph shows per color channel how close the measured values meet the target values.

More information

- *Calibrate the printer* on page 368
- *Perform shading correction* on page 370
- *Learn about the color-based workflow* on page 434
- *Learn about calibration* on page 346
- *View tolerance levels for color evaluations* on page 379
- *Color validation metrics* on page 592
- *Perform a color validation test* on page 405

Learn about G7® calibration and verification

G7® is both a definition of grayscale appearance and a calibration method. The G7® calibration method uses the G7® definition to gray balance the output colors of the printer. A good gray balance is important because the human eye is very sensitive to subtle changes in grays.

G7® in PRISMAsync

PRISMAsync supports the use of G7® calibration curves created by external tooling. However, PRISMsync also offers a very easy and quick way to create G7® output profiles and calibration curves with feedback about the output profiles and calibration process.

In addition, you can perform a **Idealliance® G7® Grayscale test**. The feedback of this procedure shows immediately that your printed output meets the G7® standard.

G7® Legacy CMY versus G7® Native CMY

G7® Legacy CMY requires that 300% CMY must be neutral (a* = 0 and b* = 0). This requirement does not always provide optimal gray values when using a printer. Hence, printers use G7® Native CMY to achieve G7® compliance. With G7® Native CMY the target a* and b* values for CMY gray levels are gradually adjusted along a curve that ends at the native a* and b* values of the printer's 300% CMY level.

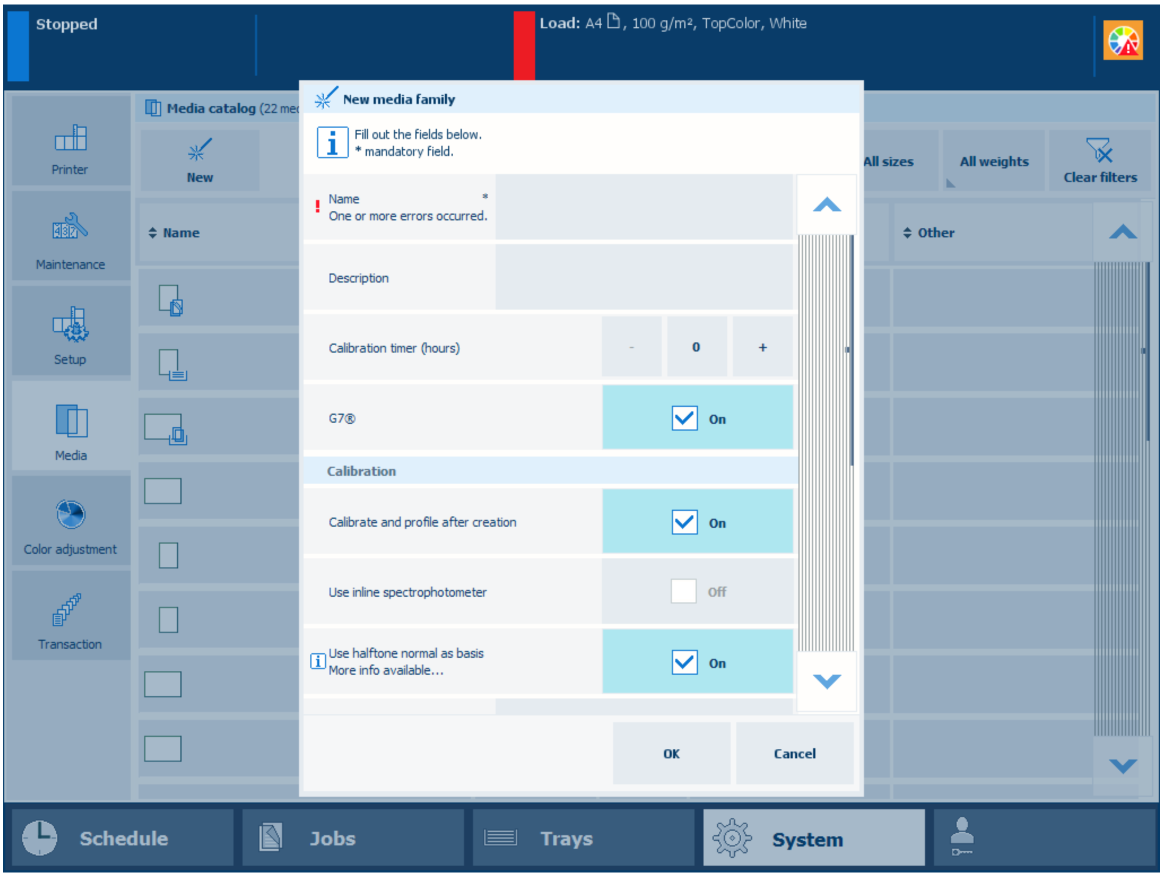
More information

- *Perform a G7® media family calibration* on page 389
- *Perform the G7® Grayscale test* on page 395
- *Learn about color differences and tolerances* on page 352

Perform a G7® media family calibration

The calibration of a G7® media family calibration includes the very easy and quick creation of G7® output profiles with the PRISMAsync embedded profiler. PRISMAsync uses media families to indicate which output profiles and calibration curves must be applied to a media. Hence, the creation of a new media family with new output profiles and calibration curves are part of this procedure. PRISMAsync has regular media families and G7® media families.

Procedure



G7® calibration options

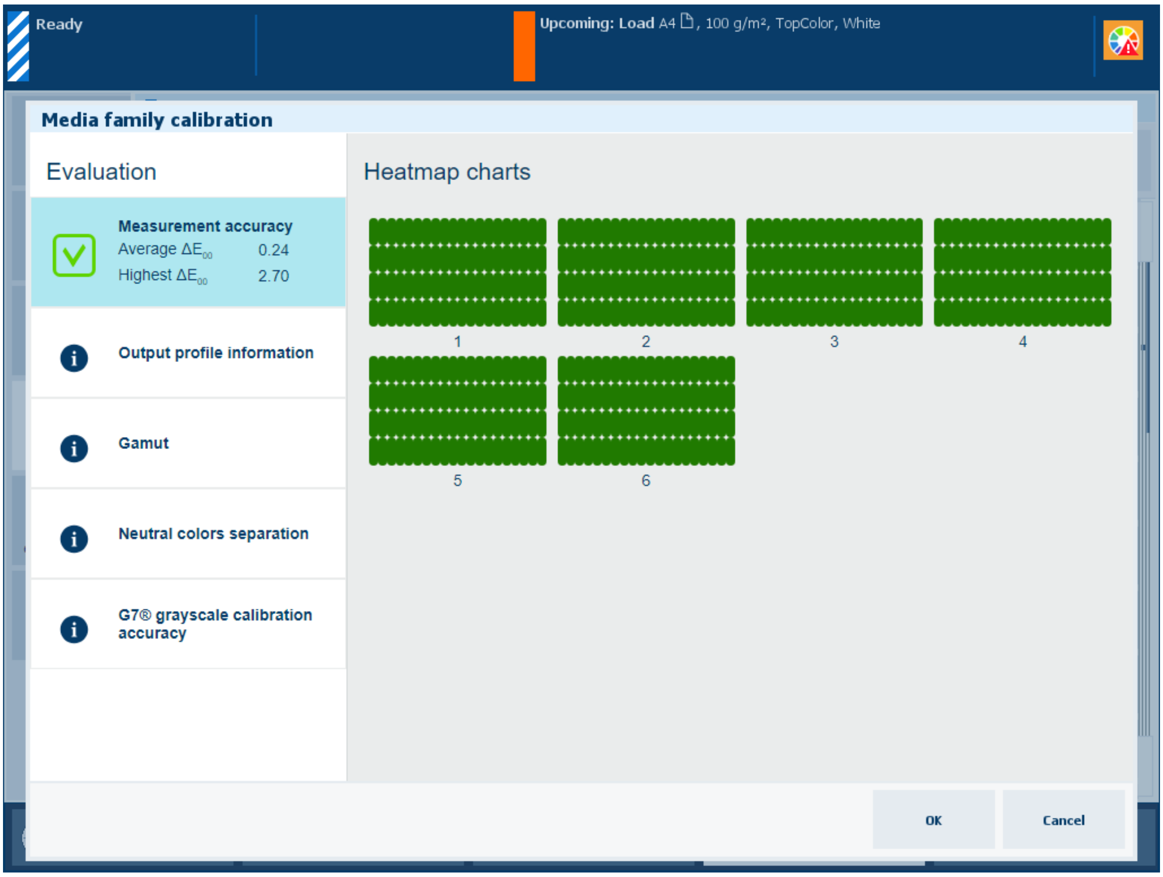
1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Touch [Create new media family] when you add new media.
3. Touch [G7®] and [Calibrate and profile after creation] when you create the new media family.
4. Select [Use inline spectrophotometer] if applicable.
   You can use the online spectrophotometer with the following media sizes: A3/11 inch x 17 inch, SRA3, 305 mm x 457 mm (12 inch x18 inch), and 330 mm x 483 mm (13 inch x 19 inch).
5. Select [Use halftone normal as basis] if required.
6. Load at least 12 sheets of the media for which you create the media family.
7. Touch [OK].
8. Follow the instructions on the control panel.

Result

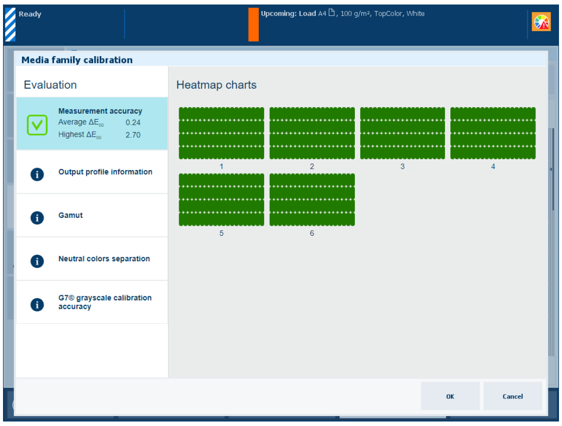
After G7® media family calibration, the feedback window shows the results of the performed procedure.

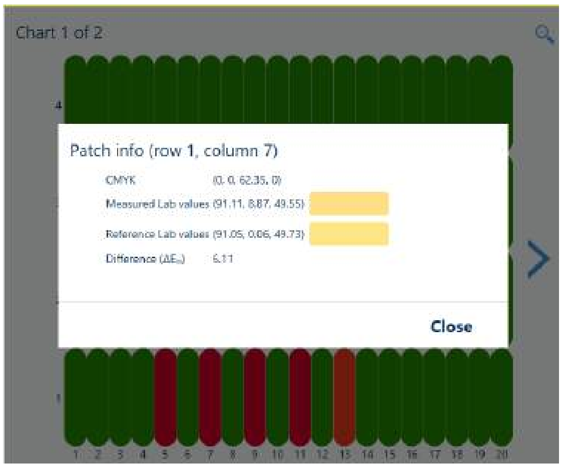


Feedback window of G7® media family calibration

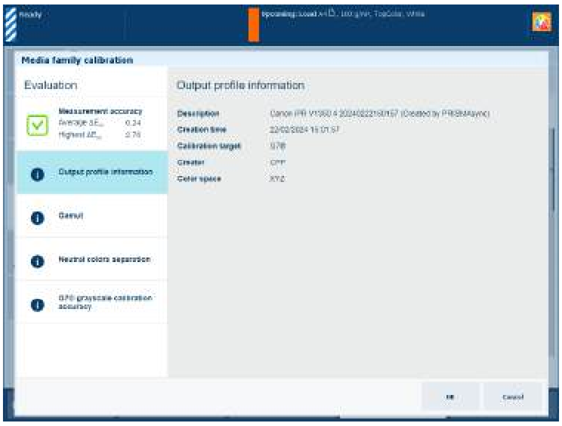
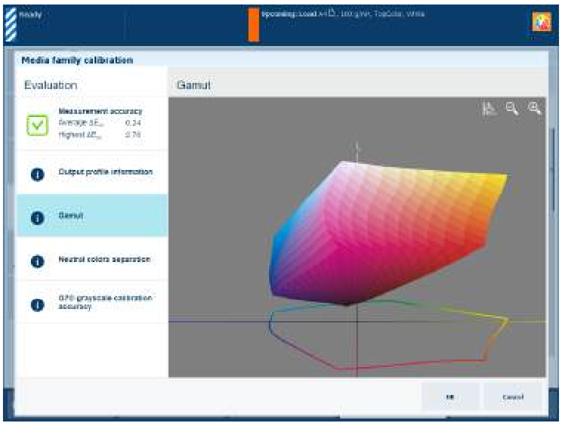
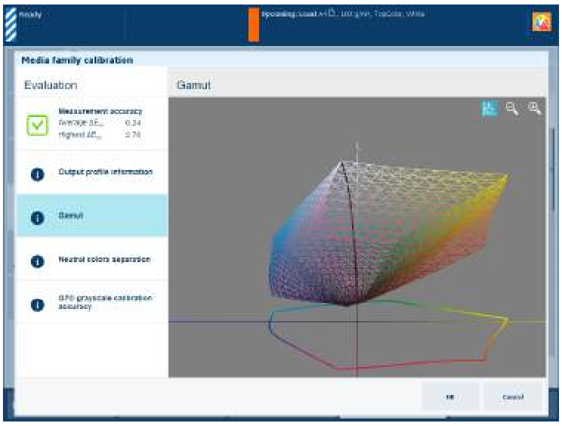
When you see a green checkmark icon, touch [OK] to save the new calibration curves.

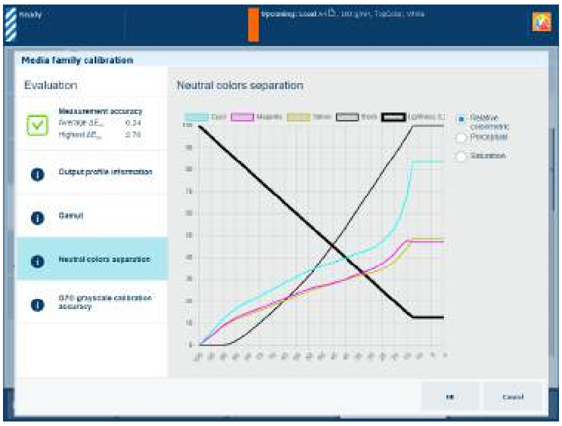
When you see a red cross icon, use the table below to evaluate the result. Then, touch [OK] to save the new calibration curves or [Cancel] to discard the results.

<table>
<tr><th>Feedback screen</th><th>Remarks</th></tr>
<tr><td rowspan="2">
</td>
<td>Description
The ΔE values of the [Measurement accuracy] information are indicators for the accuracy by which the patches are printed and measured. The control panel displays a heatmap whose patch locations correspond with the printed chart.</td></tr>
<tr><td>Evaluation

When the average ΔE value is less than the [Value where heatmap turns green] value the patches on the control panel turn green and the color quality is fine.
When the average ΔE value is between the [Value where heatmap turns green] value and the [Value where heatmap turns red] value, the patches on the control panel turn orange-like and the color quality can still be fine. However, you might want to try the options in the list below.
When the average ΔE value is greater than the [Value where heatmap turns red] value the patches on the control panel turn red. You are advised to try one or more options in the list below.
</td></tr>
<tr><td>
</td>
<td>Options to improve color quality

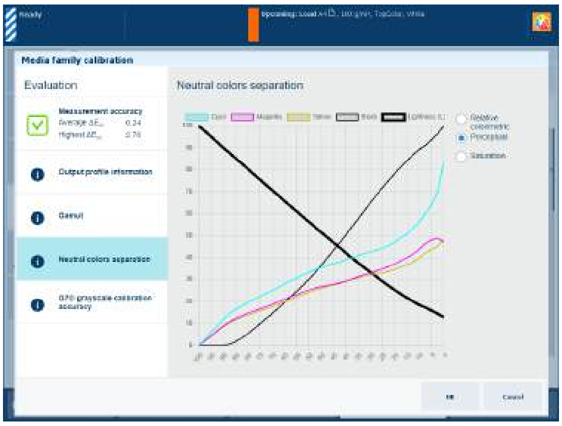
Check the patch on the printed chart
On the control panel, use the > and < symbols to scroll through the calibration charts. Use the zoom symbol to zoom in or out. Touch the red patch to find the row and column number to check the corresponding patch on the printed chart to solve the problem. Then, calibrate the media family again.
Check the media
There may be something wrong with the media used
Check the printer condition
There may be a printer calibration or new toner required.
</td></tr>
</table>

Feedback screen	Remarks
	**Description** The [Output profile information] information shows the following characteristics about the output profile: description, creation time, calibration target, creator and color space.
	**Description** The [Gamut] information is an indication of the quality of the output profile. It is a visual representation of the color gamut that the printer can produce with the created output profile on the used media. Touch the wire frame icon to view the black curve that shows the gradient of the neutral colors. This curve goes from paper white to the darkest color black that the printer can produce.
	**Evaluation** • When you see an evenly shaped model with smooth color transitions, the output profile is fine. • When you see a smooth black curve, the output profile is fine. • When you notice deviations in the color gamut like dents, imperfections or deformities, examine the printed charts carefully. There can be something wrong with the measurement, the media or the printer condition. After this is resolved, perform the above procedure to create output profiles again.

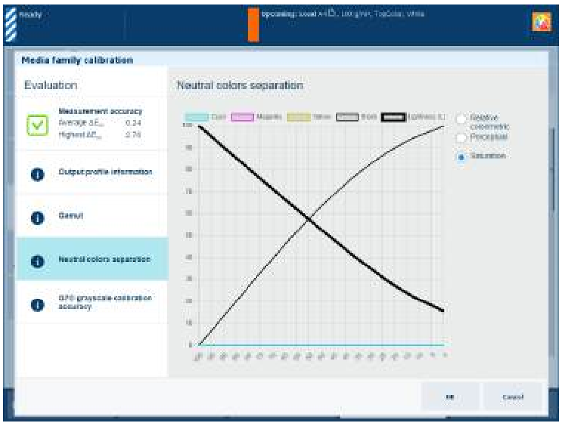
<table>
<tr><th>Feedback screen</th><th>Remarks</th></tr>
<tr><td>
</td><td>

Description
The **thick black line** of the [Neutral colors separation] graph shows how gray values are reproduced with the separate colors cyan, magenta, yellow and black (CMYK).
Note that the darkest lightness value is on the left-hand side and the lightest value on the right-hand side.
</td></tr>
<tr><td></td><td>

Explanation for [Relative colorimetric] selection
The rendering intent [Relative colorimetric] reproduces true colors, for example for proofing applications and spot colors.

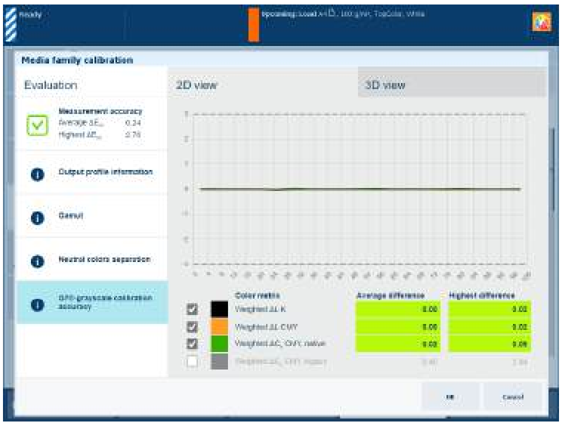
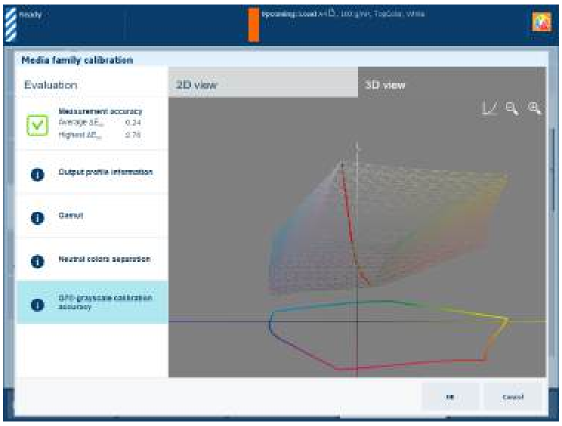
- With this rendering intent the **thick black line** shows a kink at a lightness of about 12. This is the darkest color black that the printer can reproduce.
- The **color black** is only used at a lightness of about 92 (black starting point) and lower.
</td></tr>
<tr><td>
</td><td>

Explanation for [Perceptual] selection
The rendering intent [Perceptual] reproduces contrasts in dark gray areas, for example for photos.

- This rendering intent uses black point compensation. Hence, the **thick black line** gradually approaches the lightness of 0.
- The **color black** is only used at a lightness of about 92 (black starting point) and lower.
</td></tr>
<tr><td>
</td><td>

Explanation for [Saturation] selection
The rendering intent [Saturation] reproduces saturated colors for example for business graphics.

- This rendering intent uses only black to create saturated colors. The **thick black line** gradually approaches the lightness of 0.
- Only the **color black** is used at a lightness from 100 to 0.
</td></tr>
</table>

Feedback screen	Remarks
	**Description** The percentage values of the [Calibration accuracy]→[2D view] information show how well the calibration procedure and output profile fit together. The graph shows per color channel how close the measured values meet the G7® target values. **Evaluation** • When the average percentage value is **less than the set threshold** the calibration accuracy is fine. • When the average percentage value is **greater than the set threshold** check the printer condition. You can perform the G7® verification procedure to verify if the printed output meets the G7® grayscale target.
	**Description** The [Calibration accuracy]→[3D view] information shows how well the calibration procedure and output profile fit together. The graph shows how close the measured grayscale curve (red) meets the G7® target grayscale curve (green). Touch the wire frame icon to toggle between black curve view and CMY curve view.

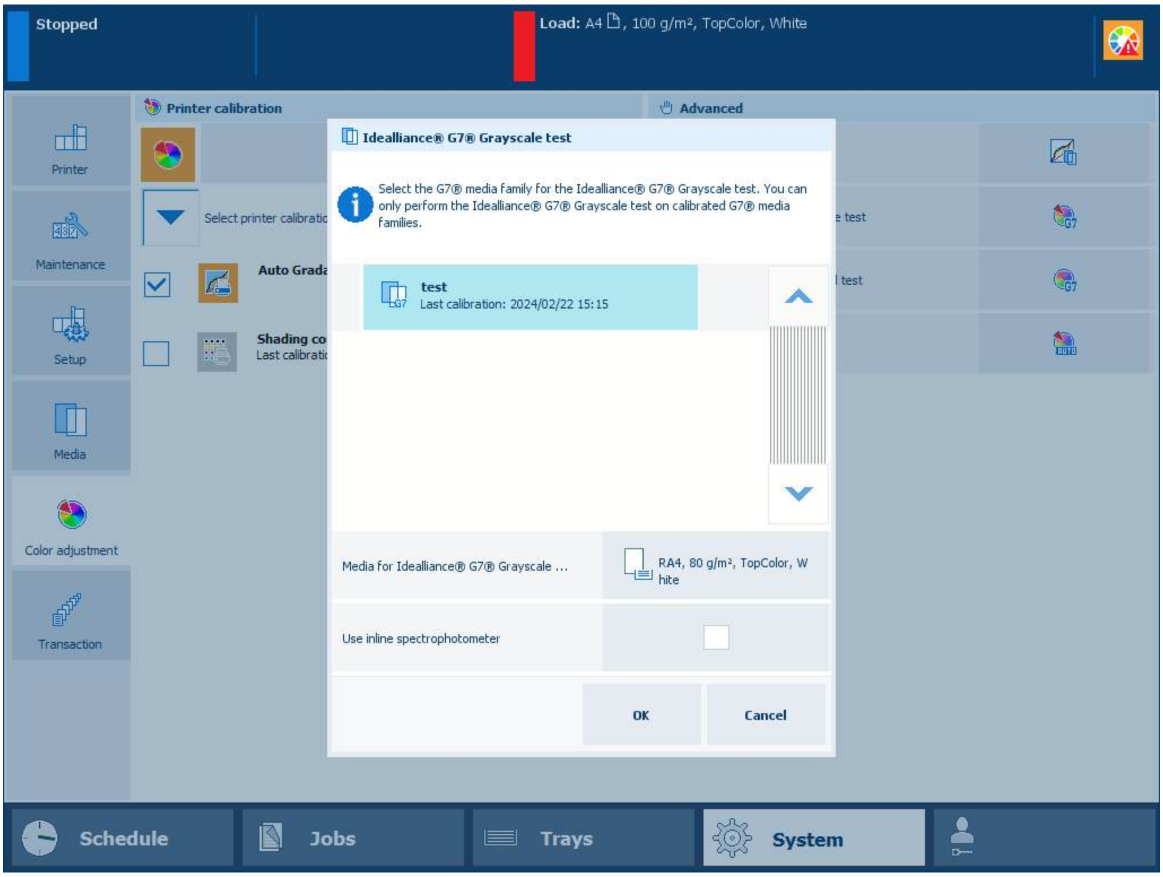
More information

- *Calibrate the printer* on page 368
- *Learn about the color-based workflow* on page 434
- *Learn about calibration* on page 346
- *View tolerance levels for color evaluations* on page 379
- *Color validation metrics* on page 592
- *Define printer calibration defaults* on page 355
- *Perform the G7® Grayscale test* on page 395

Perform the G7® Grayscale test

With PRISMAsync Print Server, you can verify if the printed output meets the G7® Grayscale target.

The G7® Grayscale test applies to all media of the media family. You only have to perform a G7® Grayscale test for one media of the media family. This is the media that is also used for media family calibration. Make sure you use that same media each time you perform a G7® Grayscale test.



Tools

i1Pro3 spectrophotometer (i1Pro2 also supported)

Before you begin

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Calibrate the media family of the media you want to validate.

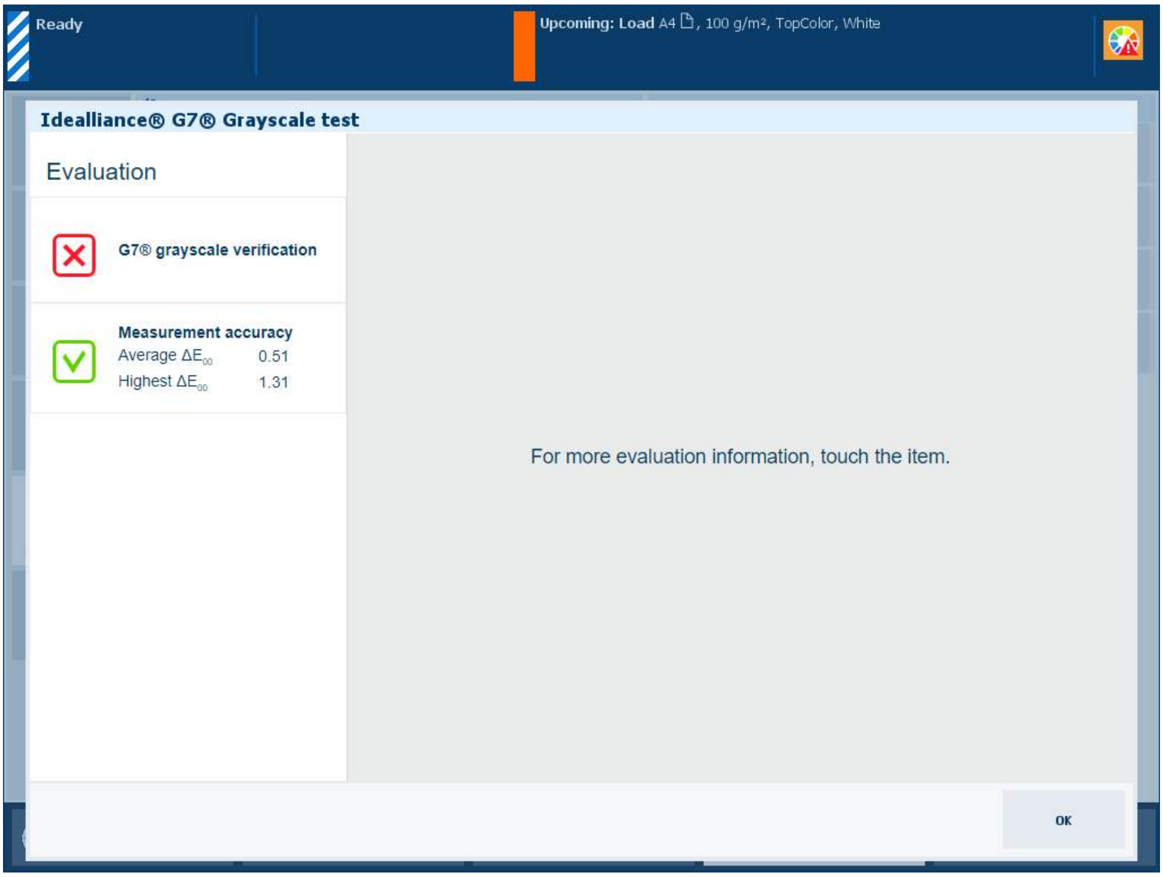
   IMPORTANT
   You must always perform a media family calibration before you continue with this procedure.

Procedure

1. If applicable, connect the i1Pro3 spectrophotometer to the USB port of the control panel.
2. Touch [System]→[Color adjustment]→[Idealliance® G7® Grayscale test].
3. Select a G7® media family.
4. Select the media for the test.
   The system displays the default media.
5. Load the media you selected.
6. Touch [OK].
7. Follow the instructions on the control panel.

Result

After you performed the G7® Grayscale test, the feedback window shows the results of the performed procedure.



Feedback window of G7® verification procedure

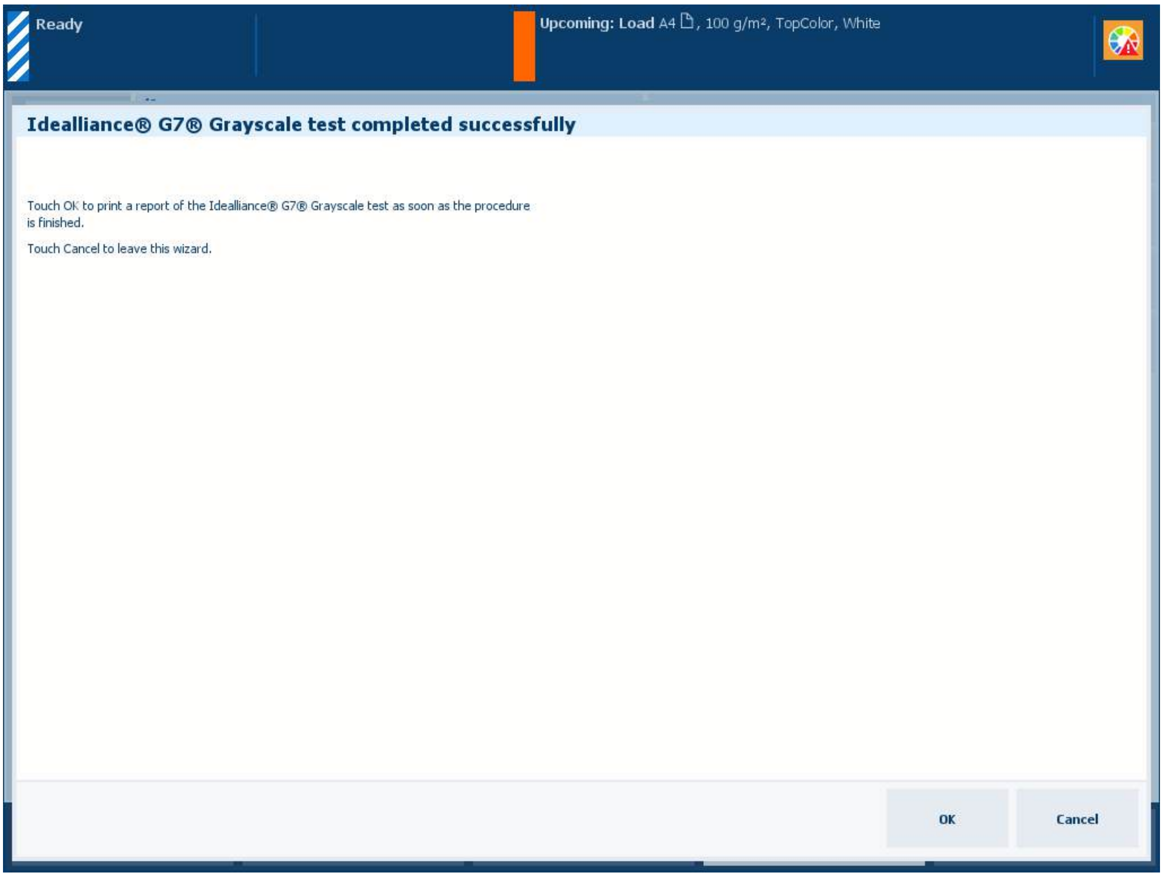
When you see two green checkmark icons, touch [OK] to continue.

When you see a red cross icon, use the table below to evaluate the result before you click [OK] to continue.

	**Description** The [G7® grayscale verification] information is an indication for the color accuracy achieved with the current printer condition, used media and corresponding output profile. The graph shows per color channel how close the measured values meet the G7® target values. The tolerances are displayed by dashed lines.
	Evaluation • When the average ΔL or ΔCh value is **less than the set threshold** the color quality is fine. • When the average ΔL or ΔCh value is **greater than the set threshold**, examine the printed output carefully. There can be something wrong with the measurement, the media or the printer condition.
	**Description** The ΔE values of the [Measurement accuracy] information are indicators for the accuracy by which the patches are printed and measured. The control panel displays a heatmap whose patch locations correspond with the printed chart.
	Evaluation • When the average ΔE value is **less than the [Value where heatmap turns green] value** the patches on the control panel turn **green** and the color quality is fine. • When the average ΔE value is **between the [Value where heatmap turns green] value and the [Value where heatmap turns red] value**, the patches on the control panel turn **orange-like** and the color quality can still be fine. However, you might want to try the options in the list below. • When the average ΔE value is **greater than the [Value where heatmap turns red] value** the patches on the control panel turn **red**. You are advised to try one or more options in the list below.

	**Options to improve color quality** • **Check the patch on the printed chart** On the control panel, use the > and < symbols to scroll through the calibration charts. Use the zoom symbol to zoom in or out. Then, touch the red patch to find the row and column number to check the corresponding patch on the printed chart to solve the problem. Then, perform this verification procedure again. • **Calibrate the media family again** Something may have gone wrong during the use of the i1Pro3 spectrophotometer. • **Calibrate the printer**

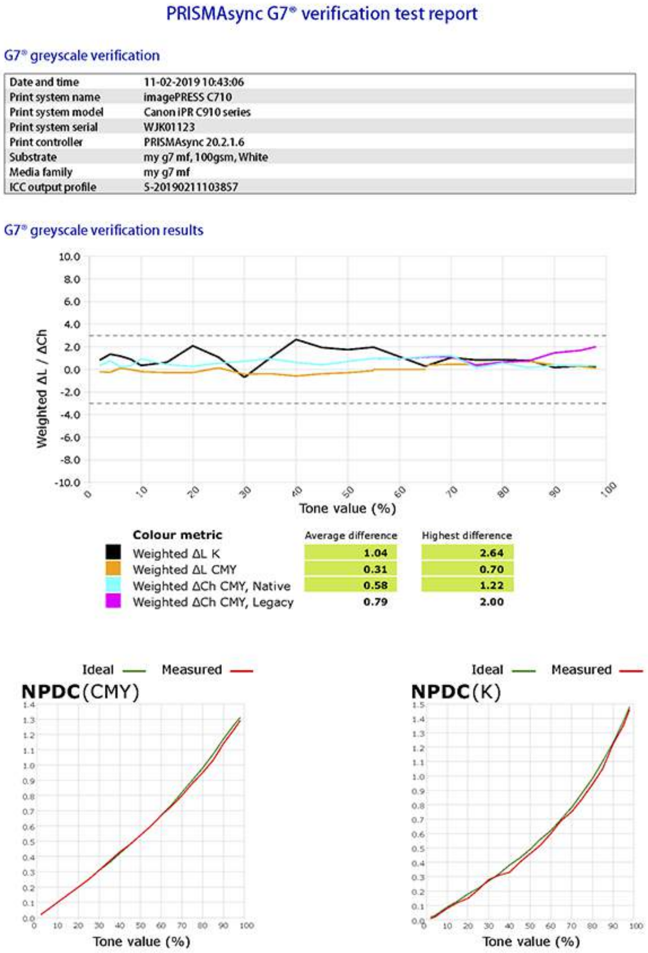
When you want to print a detailed report of the performed G7® Grayscale test, touch [OK].



Option to print a detailed report

You can also download a detailed report of the performed G7® Grayscale in the Settings Editor.

Location: [Support]→[Troubleshooting]→[Download Idealliance® G7® Targeted test report in PDF format] .



Example of G7® Grayscale test report

More information

- *Calibrate the printer* on page 368
- *Calibrate a media family* on page 373
- *Learn about the color-based workflow* on page 434
- *Learn about calibration* on page 346
- *View tolerance levels for color evaluations* on page 379
- *Color validation metrics* on page 592
- *Define printer calibration defaults* on page 355

Create G7® calibration curves with an external tool

PRISMAsync supports the use of calibration curves created by an external tool, for example CHROMiX Curve3 ™. PRISMAsync uses media families to indicate which output profiles and calibration curves must be applied to a media. Hence, the creation of a new media family with the factory default G7® output profiles is part of this procedure. PRISMAsync has regular media families and G7® media families.

Tools

A file with a chart intended for G7® calibration, for example the P2P25 or P2P51 calibration chart.

Procedure

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Create a G7® media family in the Settings Editor:
   1. Open the Settings Editor.
   2. Create a new G7® media family.
   3. Create a new media for the G7® media family.
3. Print the G7® measurement chart via an automated workflow.
   1. Create a new automated workflow.
   2. Select the media on which the measurement chart will be printed.
   3. At [Printing of measurement charts] select [G7® calibration] to print without color management.
   4. Print the G7® measurement chart via the G7® automated workflow.
4. Measure the calibration chart:
   1. Measure the chart, for example with an X-Rite ™ device.
   2. Calculate the calibration curves with the external tool.
5. Import the calculated G7® calibration curves to the G7® media family in the Settings Editor.
   1. Open the Settings Editor.
   2. Open the new G7® media family.
   3. Touch [Import CMYK curves] to import the calibration curves.

After you finish

You can only disable G7® support in the Settings Editor when all G7® media families are removed.

More information

- *Calibrate the printer* on page 368
- *Define the media families* on page 423
- *Configure the media* on page 420
- *Learn about automated workflows* on page 245
- *Learn about the color-based workflow* on page 434
- *Learn about calibration* on page 346
- *Define printer calibration defaults* on page 355

Validate colors

Learn about color validation

Color validation tests and the output profile accuracy test provide you an objective method to evaluate the color reproduction on the media you select. The reference printing conditions of the default color validation tests are already defined, as well as their quality levels and tolerance levels.

- The **output profile accuracy** test verifies how accurate the output profile fits the color reproduction properties of the media.
- The pre-defined **FOGRA** and **Idealliance® color validation tests** verify against digital print certifications. When you want to check the color reproduction of the printer against other printing conditions, you can define your own color validations tests.

Benefits of color validation tests

You can perform a color validation test for several reasons:

1. You want to check how accurate the output profile fits the color reproduction of the media family.
2. You want to know if the printer is able to print the colors on the selected media according to a reference printing condition.
3. You want to know if the printer is able to print the same colors on the selected media over time.
4. You want to check the color reproduction against: FOGRA Print Standard Digital Certification, FOGRA Validation Printing System Certification, or Idealliance® Digital Press Certification Program.
5. You want to check the color reproduction on the selected media against your own reference printing condition. Therefore, you create your own test and define your own printing condition, quality levels, tolerance levels and metrics.

Evaluation methods

The traditional **side-by-side** evaluation method checks if printed colors exactly match the colors of the reference printing condition. The **media relative** evaluation takes into account that the paper white values of the printed media and the reference media can be different. Therefore, the media relative method uses a non-linear algorithm to compensate for the different white points. The **SCCA** (Substrate-Corrected Colorimetric Aims) evaluation method, used by Idealliance® tests, is also based on the media relative evaluation.

The evaluation of the results for media family calibration and G7® grayscale test are based on set tolerance levels.

Color evaluation methods use metrics to decide if printed colors are close enough to their reference colors or not. A tolerance level or threshold level is set before the evaluation takes place. The level specifies where a calculated difference is still acceptable and where not. A tolerance level is also expressed by a Δ(Delta) value.

If you want to change the tolerance values, contact your Service organization.

More information

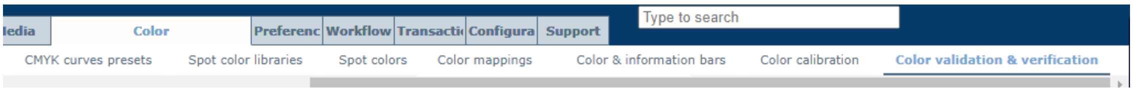
- *Color validation metrics* on page 592
- *Define the use of color validation tests* on page 402
- *Color validation metrics* on page 592

Define the use of color validation tests

Evaluation tests can have different quality levels based on different tolerance levels for the same metric.

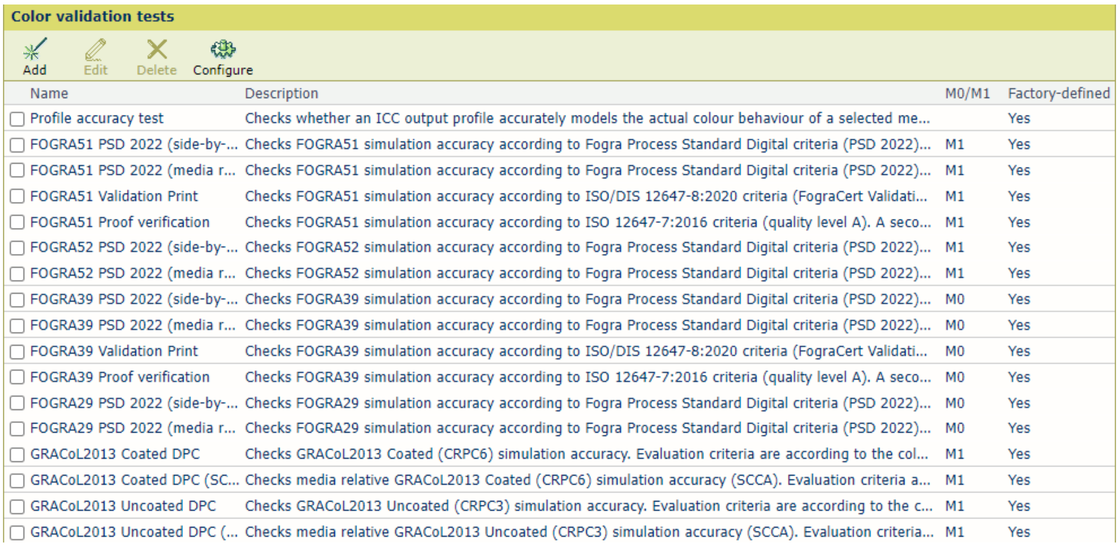
You cannot change the metrics and tolerance levels for the pre-defined FOGRA, GRACoL and SWOP color validation tests. The profile accuracy test also has metrics and tolerance levels that are not editable.

1. Open the Settings Editor and go to: [Color]→[Color validation & verification].



[Color validation & verification] tab

2. Go to the [Color validation tests] section

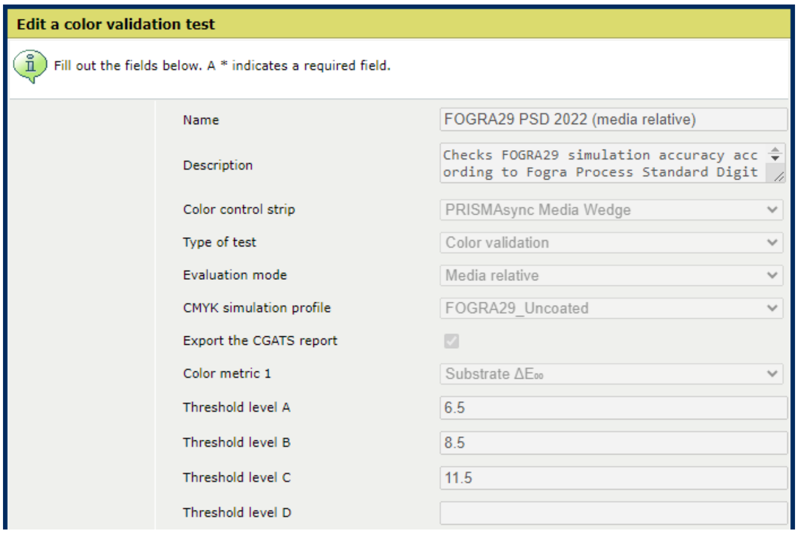


Name	Description	M0/M1	Factory-defined
Profile accuracy test	Checks whether an ICC output profile accurately models the actual colour behaviour of a selected me...		Yes
FOGRA51 PSD 2022 (side-by-...	Checks FOGRA51 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M1	Yes
FOGRA51 PSD 2022 (media r...	Checks FOGRA51 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M1	Yes
FOGRA51 Validation Print	Checks FOGRA51 simulation accuracy according to ISO/DIS 12647-8:2020 criteria (FograCert Validati...	M1	Yes
FOGRA51 Proof verification	Checks FOGRA51 simulation accuracy according to ISO 12647-7:2016 criteria (quality level A). A seco...	M1	Yes
FOGRA52 PSD 2022 (side-by-...	Checks FOGRA52 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M1	Yes
FOGRA52 PSD 2022 (media r...	Checks FOGRA52 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M1	Yes
FOGRA39 PSD 2022 (side-by-...	Checks FOGRA39 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M0	Yes
FOGRA39 PSD 2022 (media r...	Checks FOGRA39 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M0	Yes
FOGRA39 Validation Print	Checks FOGRA39 simulation accuracy according to ISO/DIS 12647-8:2020 criteria (FograCert Validati...	M0	Yes
FOGRA39 Proof verification	Checks FOGRA39 simulation accuracy according to ISO 12647-7:2016 criteria (quality level A). A seco...	M0	Yes
FOGRA29 PSD 2022 (side-by-...	Checks FOGRA29 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M0	Yes
FOGRA29 PSD 2022 (media r...	Checks FOGRA29 simulation accuracy according to Fogra Process Standard Digital criteria (PSD 2022)...	M0	Yes
GRACoL2013 Coated DPC	Checks GRACoL2013 Coated (CRPC6) simulation accuracy. Evaluation criteria are according to the col...	M1	Yes
GRACoL2013 Coated DPC (SC...	Checks media relative GRACoL2013 Coated (CRPC6) simulation accuracy (SCCA). Evaluation criteria a...	M1	Yes
GRACoL2013 Uncoated DPC	Checks GRACoL2013 Uncoated (CRPC3) simulation accuracy. Evaluation criteria are according to the c...	M1	Yes
GRACoL2013 Uncoated DPC (...	Checks media relative GRACoL2013 Uncoated (CRPC3) simulation accuracy (SCCA). Evaluation criteria...	M1	Yes

Available predefined color validation tests

View the tolerance levels of a predefined color validation test

1. Select a color validation test you are interested in.
2. Click [Edit].

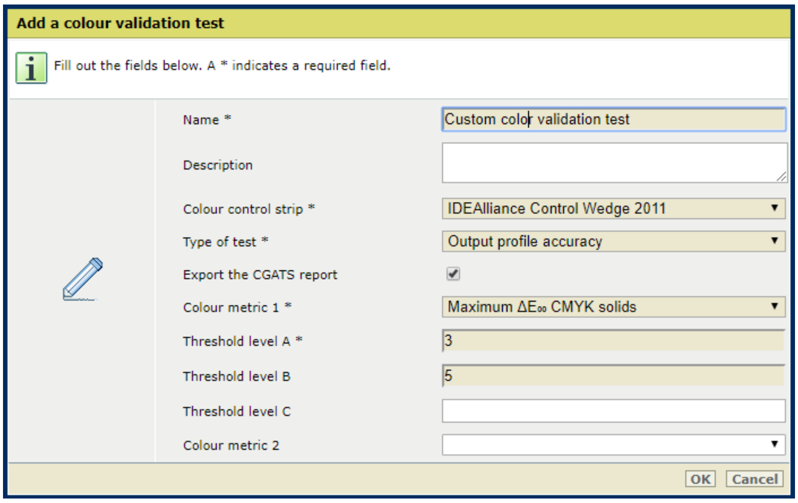


Edit a color validation test

3. After you check the tolerance levels, click [OK] to close the window.

Create a custom color validation test

1. Click [Add].



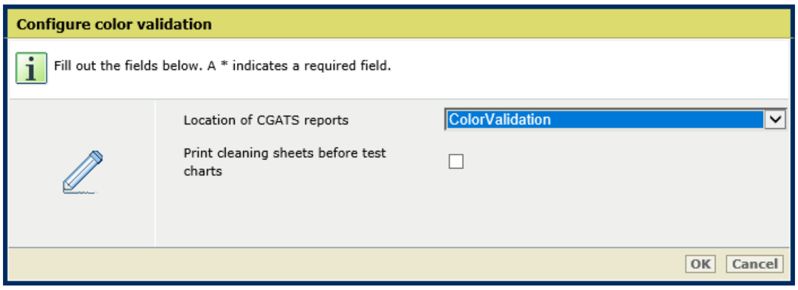
Add a color validation test

2. Define a name and description.
3. Select the target chart form the [Color control strip] drop-down list.
4. Indicate if you want to save the CGATS report.
5. Select the first metric from the [Color metric 1] drop-down list.
6. Enter the tolerance level (in ΔE) for quality level A in the [Threshold level A] field.
7. Define the other metrics and quality levels you want to add.
8. Click [OK].

Enable the storage of CGATS reports after a color validation test

1. Click [Configure].
2. Use the [Location of CGATS reports] setting to select the location of the CGATS reports.

CGATS reports are stored on an SMB share. The system administrator can configure SMB shares.

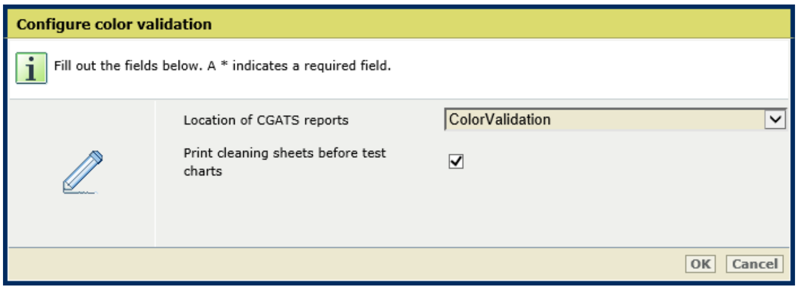


[Location of CGATS reports] setting

3. Click [OK].

Enable the printing of cleaning sheets before the test charts

1. Click [Configure].
2. Use the [Print cleaning sheets before test charts] setting to enable the printing of cleaning sheets before the test charts.



[Print cleaning sheets before test charts] setting

3. Click [OK].

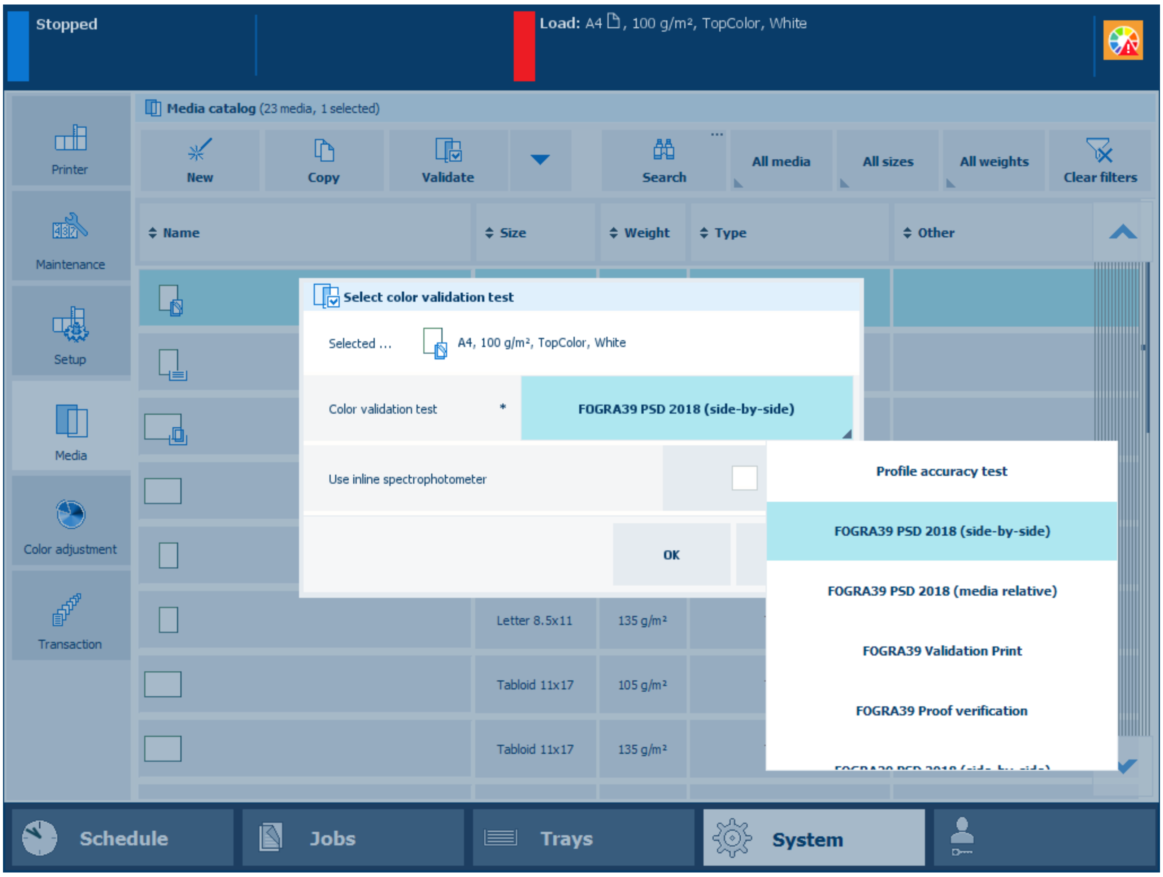
More information

- *Learn about color validation* on page 401
- *Perform a color validation test* on page 405
- *Color validation metrics* on page 592

Perform a color validation test

The color validation test gives an indication of the color quality for the used media and the linked output profile.

Evaluation tests can have different quality levels based on different tolerance levels for the same metric. You can define custom color validation tests and a storage location for a detailed CGATS report.



Available color validation tests

Tools

i1Pro3 spectrophotometer (i1Pro2 also supported).

Before you begin

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Calibrate the media family of the media you want to validate.

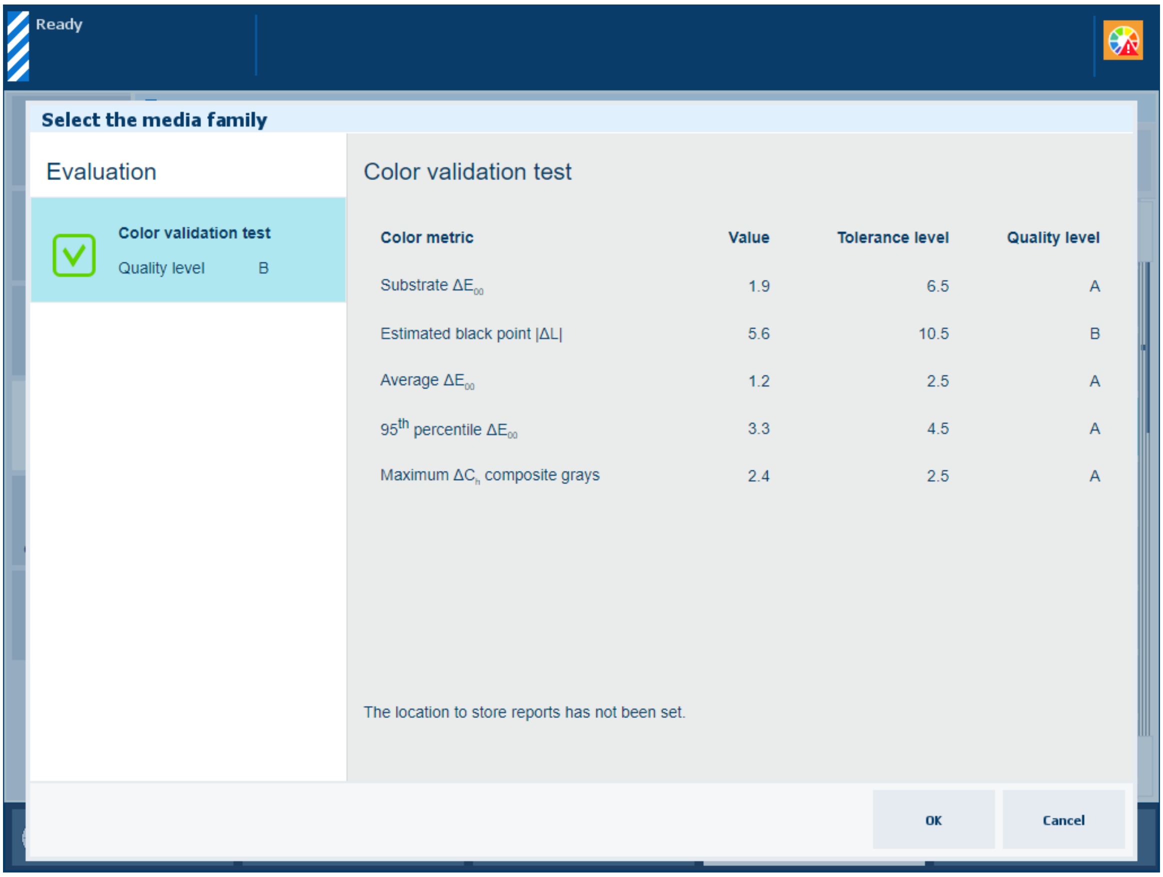
   IMPORTANT
   You must always perform a media family calibration before you continue with this procedure.

Procedure

1. If applicable, connect the i1Pro3 spectrophotometer to the USB port of the control panel.
2. Touch [System]→[Media].
3. Select the target media.
4. Touch [Validate].
5. Load and assign approximately 10 sheets of the target media in one of the paper trays.
6. Select a color validation test.
7. Select [Use inline spectrophotometer] if applicable.
   You can use the online spectrophotometer with the following media sizes: A3/11 inch x 17 inch, SRA3, 305 mm x 457 mm (12 inch x18 inch), and 330 mm x 483 mm (13 inch x 19 inch).
8. Touch [OK].
9. Follow the instructions on the control panel.

Result

After you performed the color validation, the feedback window shows the results of the performed procedure.

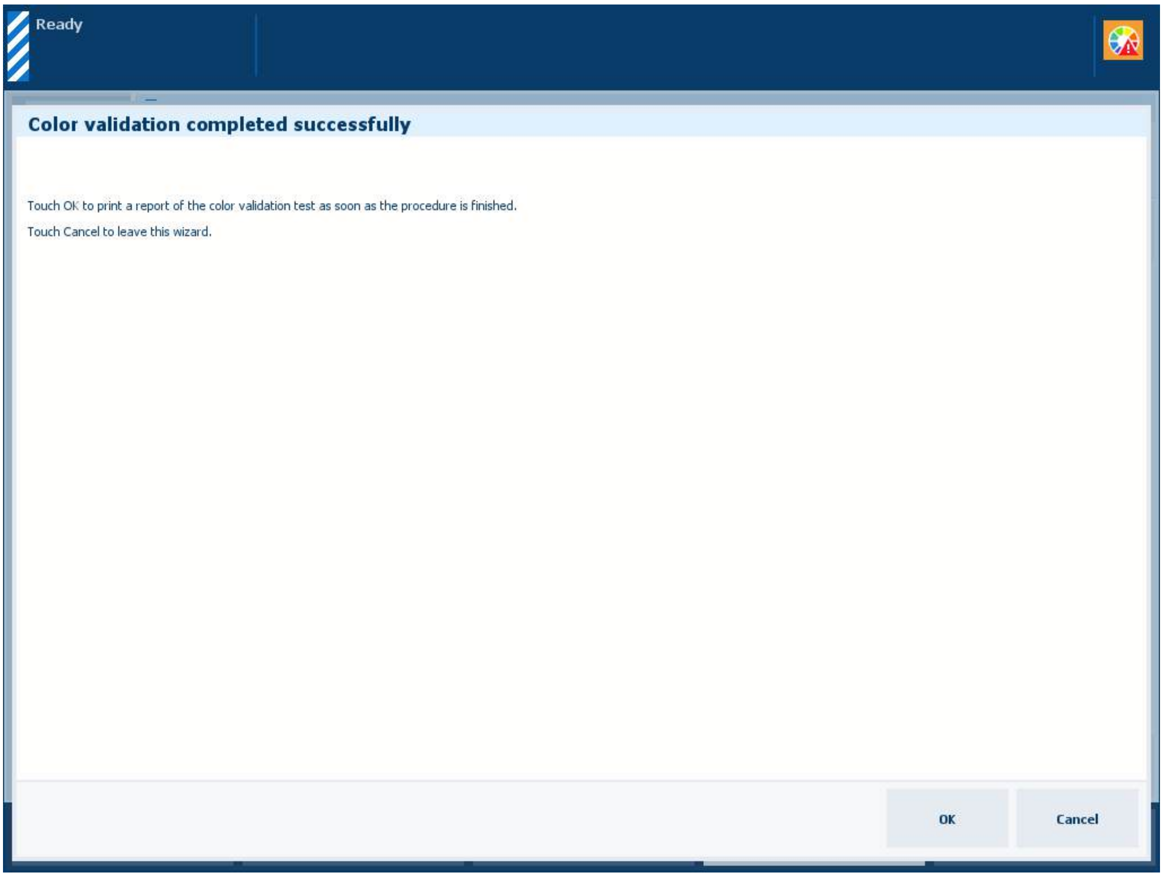


Feedback window of color validation test

Use the table below to evaluate the result and touch [OK] to close the window.

Profile accuracy test	Color validation test
Description of profile accuracy test The [Color validation test] information of the **pre-defined profile accuracy test** indicates if a media fits the media family. The test also indicates how accurate colors are reproduced on the used media with corresponding output profile.	**Description of color validation test** The [Color validation test] information of the **pre-defined color validation tes**t indicates if the printed output meets an external color standard with the current printer condition, used media and corresponding output profile.
Evaluation of profile accuracy test The test shows the measured value, the tolerance level, and the assigned quality level. • When the **quality level is A** the highest quality level is achieved and the color accuracy is fine. • When the **quality level is B** the quality level is still within specifications and the color accuracy is fine. • When the **quality level is not shown** then the quality level is not achieved. You are advised to create a new media family for this medium.	**Evaluation of color validation test** The test shows the measured value, the tolerance level, and the assigned quality level. • When the **quality level is A** the highest quality level is achieved and the color standard is met. • When the **quality level is B or lower** the quality level is still within specifications and the color standard is met. • When the **quality level is not shown** then the quality level is not achieved and the color standard is not met. You are advised to use a medium that is more suitable for this standard.

When you want to print a detailed report of the performed color validation test, touch [OK].



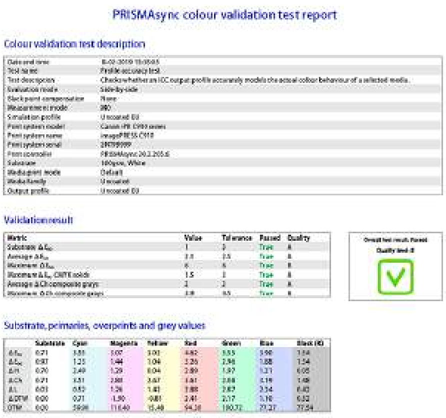
Print a detailed report

You can also download a detailed report of the performed color validation test in the Settings Editor.

Location:

[Support]→[Troubleshooting]→[Download color validation test report in CGATS format]

[Support]→[Troubleshooting]→[Download color validation test report in PDF format]



PRISMAsync colour validation test report

Colour validation test description

Date and time	[illegible]
Test name	Profile accuracy test
Test description	Checks whether an ICC output profile accurately models the actual colour behaviour of a selected media.
Evaluation mode	Side-by-side
Black point compensation	None
Measurement mode	[illegible]
Simulation profile	Uncoated [illegible]
Print system model	Canon iPR C910 series
Print system name	imagePRESS C910
Print system serial	[illegible]
Print controller	PRISMAsync [illegible]
Substrate	100gsm, White
Media print mode	Default
Media family	Uncoated
Output profile	Uncoated [illegible]

Validation result

Metric	Value	Tolerance	Passed	Quality
Substrate ΔE_{00}	1	3	True	A
Average ΔE_{00}	[illegible]	[illegible]	True	A
Maximum ΔE_{00}	6	6	True	A
Maximum ΔE_{00} CMYK solids	1.5	3	True	A
Average ΔCh composite grays	2	2	True	A
Maximum ΔCh composite grays	[illegible]	[illegible]	True	A

Overall test result: Passed

Quality level: A

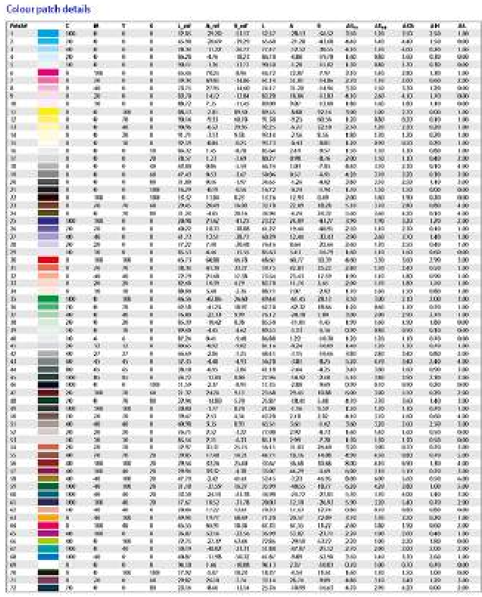
Substrate, primaries, overprints and grey values

	Substrate	Cyan	Magenta	Yellow	Red	Green	Blue	Black (K)
ΔE_{00}	0.71	3.55	3.07	3.03	4.82	3.55	3.90	1.14
ΔE_{00}	0.97	1.23	1.44	1.04	2.26	2.98	1.88	1.54
ΔH	0.70	2.49	1.29	0.04	2.89	1.97	1.21	0.05
ΔCh	0.71	3.51	2.88	2.67	3.61	2.88	3.19	1.48
ΔL	0.03	0.52	1.36	1.43	3.88	2.87	2.34	0.42
ΔDTW	0.00	0.21	-1.90	-0.81	2.41	2.17	3.10	0.52
DTW	0.00	59.98	116.40	15.48	94.38	100.72	77.27	77.54

Example of color validation report (front side)

The front side of the test report shows the following information:

1. Description of the test
2. Result of the test
3. Summary of the metrics shown on the back side for: substrate, cyan, magenta, yellow, red, green, blue and black.



Colour patch details

Example of validation report (back side)

The back side of the test report shows the metrics per measured patch.

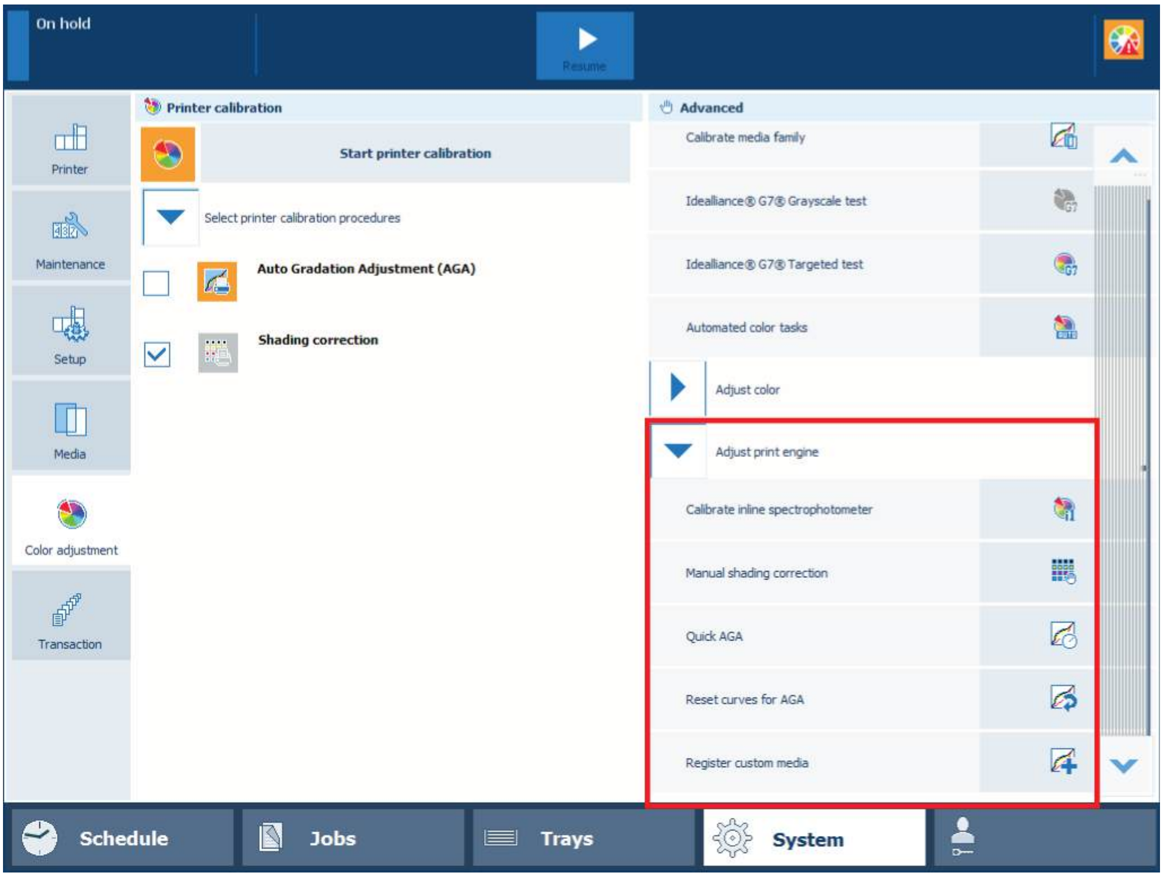
More information

- *Calibrate the printer* on page 368
- *Perform a color validation test* on page 405
- *Learn about calibration* on page 346
- *Learn about color validation* on page 401
- *Learn about color differences and tolerances* on page 352
- *Color validation metrics* on page 592
- *Define the use of color validation tests* on page 402

Perform advanced color adjustments

Learn about engine adjustments

In most cases, a printer calibration is sufficient to deliver the required color quality. However, there may be situations when you still want to make a color adjustment.



Engine adjustment procedures

More information

- *Perform manual shading correction* on page 411
- *Register the custom media for automatic gradation adjustment* on page 413

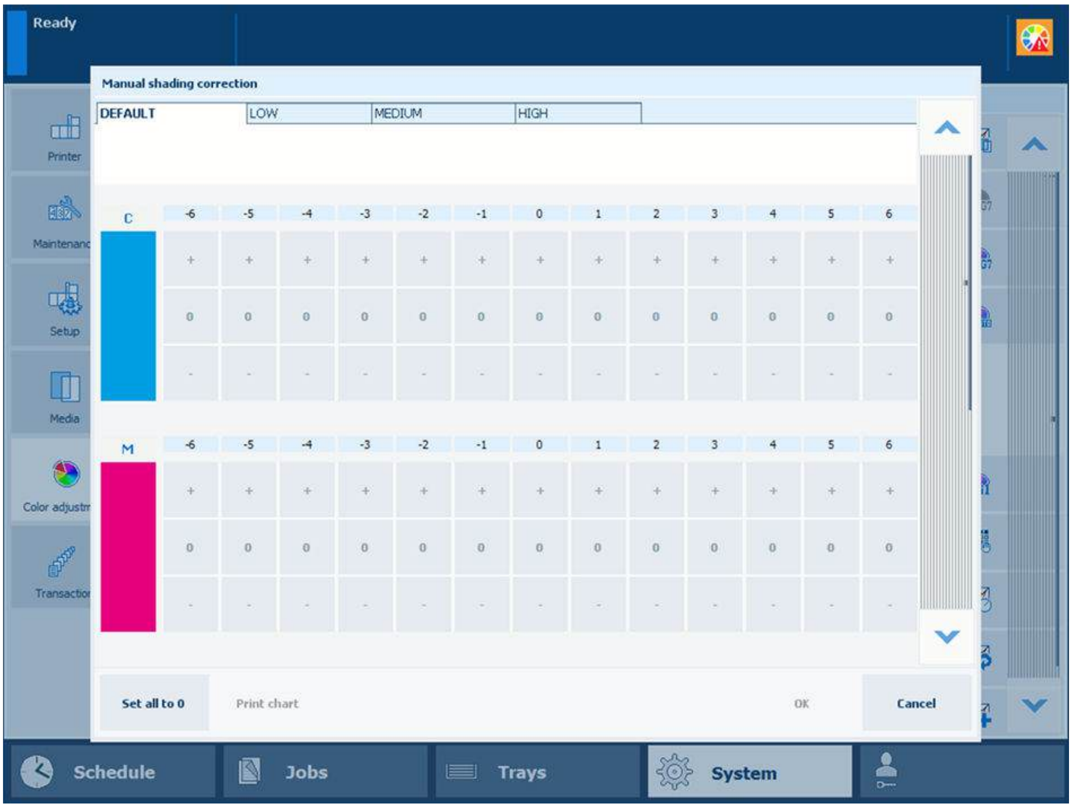
Perform manual shading correction

Perform manual shading correction when you want to correct the uniformity of the color panes in the direction perpendicular to the feed direction or the color panes close to the edges of the sheet.

Tools

i1Pro3 spectrophotometer (i1Pro2 also supported)

Perform manual shading correction



Manual shading correction window

1. Calibrate the printer.

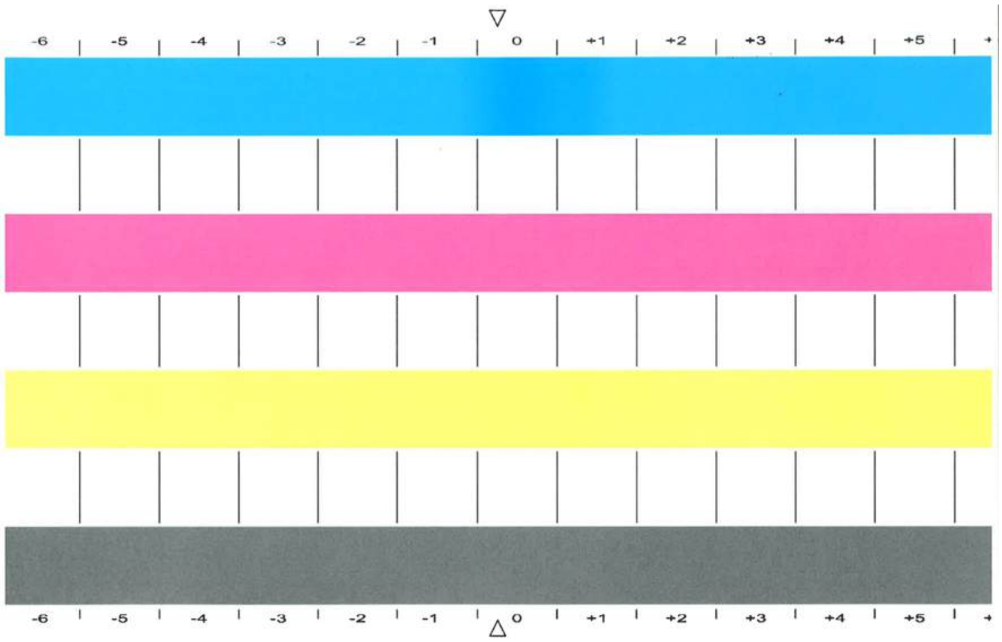
   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Check the print quality to decide if you want to proceed.
3. Perform automatic shading correction.
4. Check the print quality to decide if you want to proceed.
5. Load the largest size uncoated media you use, for example, SRA3.
6. Touch [System]→[Color adjustment]→[Adjust print engine]→[Manual shading correction].
7. Select a density range: low, medium, or high.

   NOTE
   The values in the [Default] tab can be viewed but not edited.

8. Touch [Print chart] to print the gradation chart.

NOTE
The gradation chart for the currently selected density range is printed.



Gradation chart

9. Check each color bar on uneven color densities.
10. Make the corrections on the locations where you see deviations.
    The locations are indicated per color with a number from -6 to 6.
    The initial values are displayed between brackets.
11. Repeat step 7 - 9 until each color bar has a uniform color.
12. Touch [OK].

After you finish

In the exceptional case where the manual shading correction does not deliver the required color quality, do the following:

1. Touch [System]→[Color adjustment]→[Adjust print engine]→[Manual shading correction].
2. Touch [Set all to 0].
3. Perform automatic shading correction.
4. Continue with manual shading correction.

More information

- *Calibrate the printer* on page 368
- *Perform shading correction* on page 370

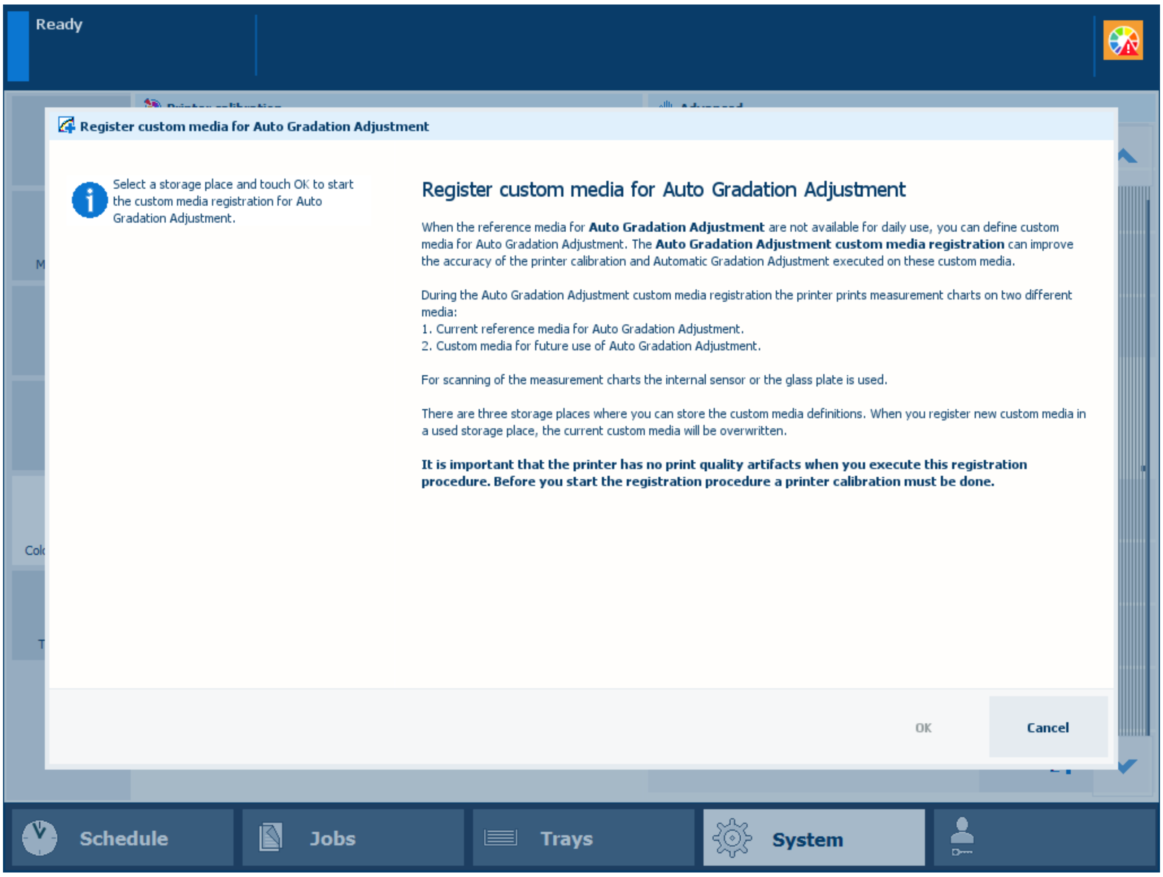
Register the custom media for automatic gradation adjustment

Normally, you use the recommended media for the automatic gradation adjustment for optimal print quality.

When the recommended media is not sufficiently available, you can use custom media for automatic gradation adjustment. Automatic gradation adjustment uses an internal sensor to register the custom media.

IMPORTANT
For optimal print quality on custom media, you are advised to register the custom media you regularly use.

For the registration of custom media, you also need some sheets of the recommended calibration media. At the end of the registration procedure you can decide if you want to use the registered custom media for automatic gradation adjustment.



Storage place for custom media at the left-hand side

Before you begin

- Make sure that [High gloss mode] setting is disabled. To check whether the setting is disabled, open the Settings Editor and go to: [Preferences]→[System settings]→[Basic].
- Prepare approximately 20 sheets of the recommended calibration media. These sheets are used as reference media during the procedure.
- Prepare approximately 20 sheets of the custom media.

Procedure

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Check the print quality to decide if you want to proceed.

   IMPORTANT
   Make sure the print system has no print quality artifacts when you execute this registration procedure.

3. Touch [System]→[Color adjustment]→[Adjust print engine].
4. Touch [Register custom media].
5. Select a storage place to store the custom media.
6. Touch [OK].
7. Follow the instructions on the control panel.
8. Indicate if you want to apply the custom media for the automatic gradation adjustment.
9. Close the menu.

IMPORTANT
When the custom media is added to the media catalog, make sure that the color setting of the custom media is **white**. If the color setting is not set to white, the media will not appear in the automated color tasks schedule.

After you finish

You can find the media for standard automatic gradation adjustment in the Settings Editor. Location: [Color]→[Color calibration].

More information

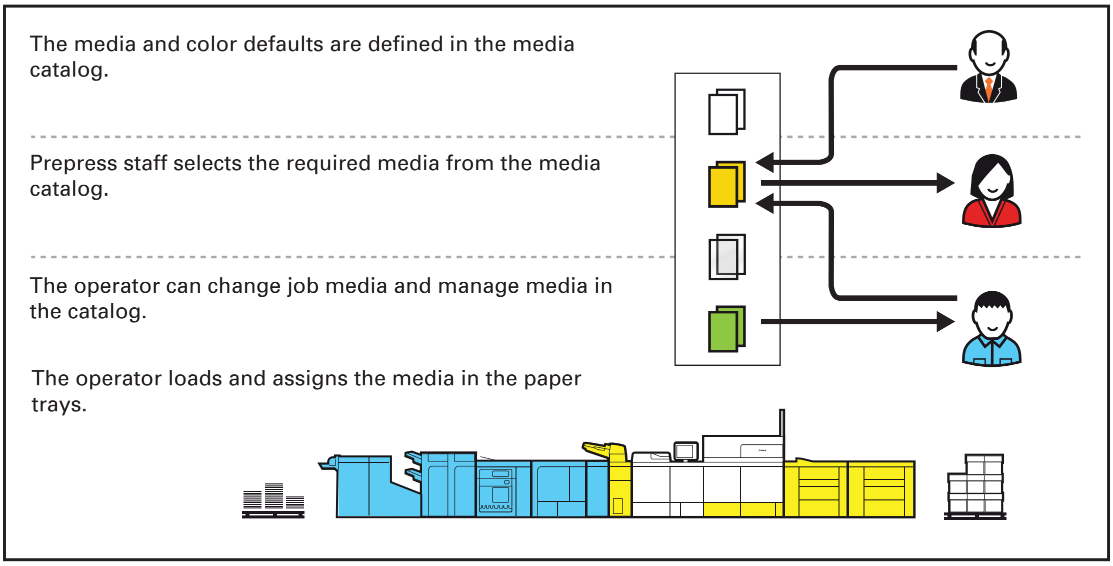
- *Learn about calibration* on page 346
- *Check and prepare media before loading* on page 132
- *Calibrate the printer* on page 368

Chapter 11
Manage media definitions

Learn about the media-based workflow

The **media catalog** is the backbone of the media-based workflow. The media catalog contains the media that the print system supports.

The prepress staff select the media for the job from the media catalog. The control panel shows the operator the media the job needs.



Media-based workflow

During the initial configuration of the print system, the media catalog is filled with media and media attributes. Media belong to a media family, which ensures optimal color output settings. (*Learn about the color-based workflow* on page 434)

When you work with the media catalog, you can take advantage of the following benefits:
- The media in the media catalog store print quality attributes that the entire print system uses. Each time you select the media from the media catalog, the print system automatically applies the same print quality settings.
- The control panel shows the operator which media to load.
- The control panel, Remote Printer Driver and the PRISMA software access the same media catalog.
- You can also use the media catalog for copy jobs.

Define the media attributes on the control panel or with the Settings Editor. (*Interfaces to access the printer* on page 38)

The system administrator determines if operators are allowed to manage the media catalogue from the control panel.

Media and image quality

Media has a big influence on the image quality and the productivity of a print system. The performance of a print system, the print quality and the consumption of consumables depend on media factors, such as:
- Media type
- Media weight
- Surface
- Moisture content
- Smoothness

To ensure optimal output quality and performance, media must comply with the paper specifications of your print system. Store the media in a dry environment away from direct sunlight. Before you load the media into the paper trays, it is important to check the media sheets.

Temporary media

You can assign media that are not included in the media catalog. When the job on these temporary media is printed, you find these temporary media in the media section of the System view on the control panel. For an optimal print quality you are advised to add the temporary media with all its important media attributes to the media catalog.

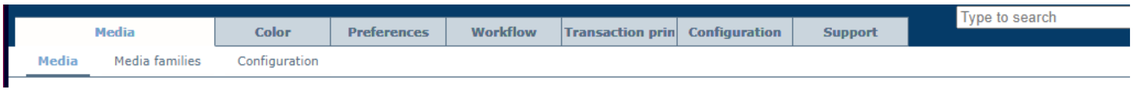
The temporary media remains available in the System view as long as there are jobs in the queue that use these temporary media, or these temporary media are assigned to a paper tray.

Define the media catalog

The PRISMAsync Print Server media catalog lists all media that can be selected for jobs. You can change several media settings.

Go to the media catalog on the Settings Editor

1. Open the Settings Editor and go to: [Media]→[Media].



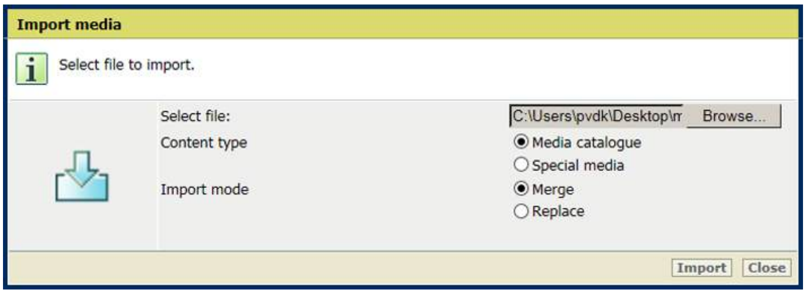
[Media] tab

Media catalog

Import a media catalog or special media definitions

Special media is the media for banner and trailer pages, separator sheets, tickets, covers, and auto gradation adjustment.

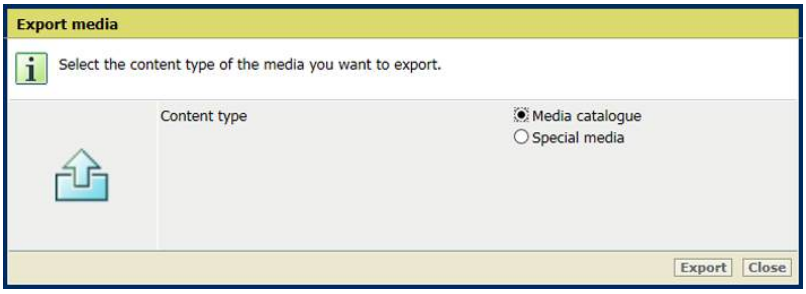
1. Click [Import].
2. Indicate if you want to import the media catalog or the special media definitions.



3. Indicate if you want to replace or expand the media catalog or special media definitions.
4. Browse to the file.
5. Confirm the selection.
6. Click [Import].
7. Check the media family and surface type when you import a media catalog from a printer that runs a software version lower than 4.1.

Export the media catalog and special media definitions

1. Click [Export].
2. Indicate if you want to export the media catalog or the special media definitions.



3. Click [Export].

Restore the media catalog

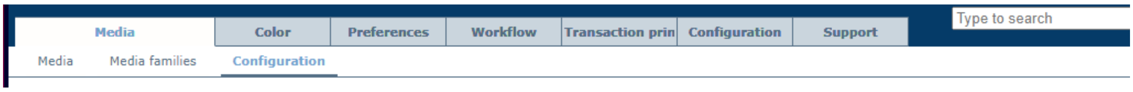
1. Click [Restore].

NOTE
You cannot undo this action.

Show warnings when media miss attributes

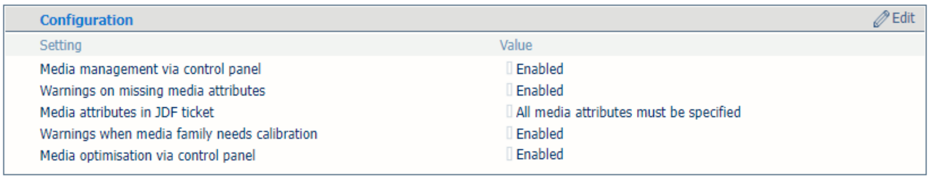
The system administrator can change this setting.

1. Open the Settings Editor and go to: [Media]→[Configuration].



[Media]>[Configuration]

2. Go to the [Configuration] section.



[Configuration] settings

3. Enable the setting [Warnings on missing media attributes].

Do media management or optimization via the control panel

The system administrator can change this setting.

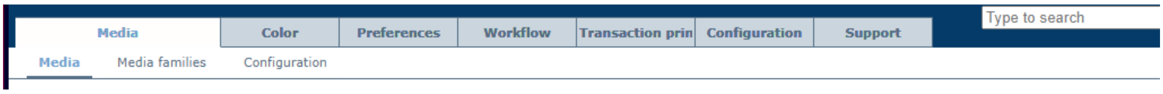
1. Enable the setting [Media management via control panel].
2. Enable the setting [Media optimization via control panel].

Configure the media

The PRISMAsync Print Server media catalog lists all media that can be selected for jobs. You can change several media settings.

Go to the media catalog

Open the Settings Editor and go to: [Media]→[Media].



The media catalog in the Settings Editor

Add media

1. Click [Add].
2. Use the table below for information on the settings.
3. Click [OK].

Copy media

1. Select the media.
2. Click [Copy].
3. Use the table below for information on the settings.
4. Click [OK].

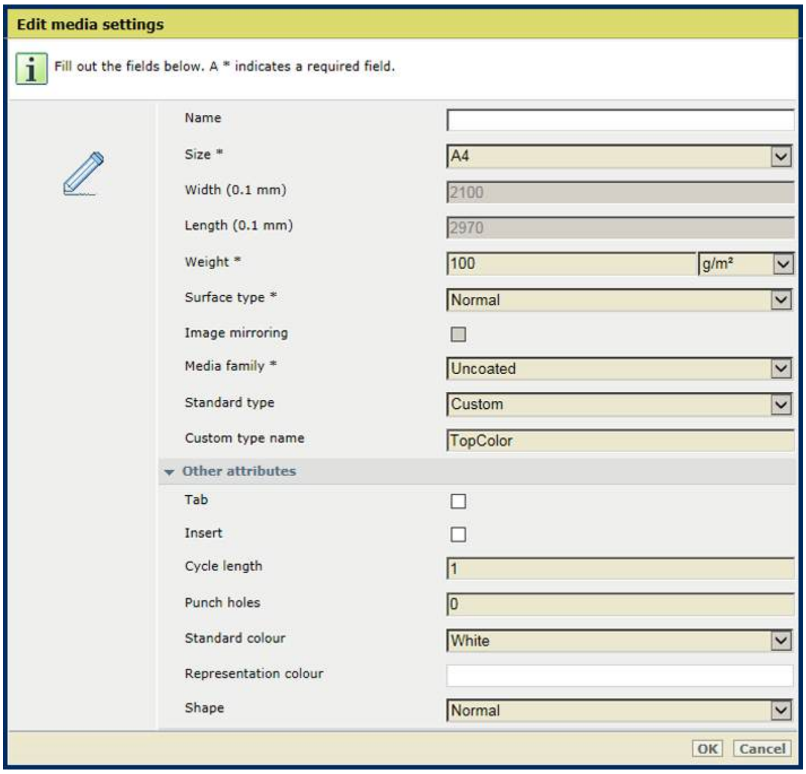
Edit media

1. Select the media.
2. Click [Edit].
3. Use the table below for information on the settings.
4. Click [OK].

Delete media

1. Select the media.
2. Click [Delete].

Configure the media settings



Media settings

Media setting	Description
[Name]	Name according to naming conventions.
[Size]	The size of the media.
[Width]	The width of the media.
[Length]	The length of the media.
[Weight]	The media weight of the media.
[Surface type]	The surface type of the media.
[Image mirroring]	Indicates if images must be mirrored.
[Media family]	The media family.
[Standard type]	The custom media type.
[Custom type name]	The custom type name according to naming conventions.
[Tab]	Indicates if the media are tab paper.
[Insert]	Indicates if the media are inserts.
[Cycle length]	Indicates the cycle length in case the media are tab paper.
[Punch holes]	Indicates the number of holes in case the media are punched.
[Standard color]	The color of the media.
[Representation color]	Indicates a custom color name.

Media setting	Description
[Grain direction]	Use this setting when the paper curl cannot be corrected or when paper jams occur frequently.

Define the media families

You can add, copy, edit and delete media families. You can also change several media family settings.

Go to the media families

1. Open the Settings Editor and go to: [Media]→[Media families].

Media Families

Add a media family

1. Click [Add].
2. Use the table below for information on the settings.
3. Click [OK].

Copy a media family

1. Select the media family.
2. Click [Copy media family].
3. Click [OK].

Edit a media family

1. Select the media family.
2. Click [Edit].
3. Use the table below for information on the settings.
4. Click [OK].

Delete a media family

1. Select the media family.
2. Click [Delete].

Print a color reference chart

1. Select the media family.
2. Click [Print].
3. Select the media.
4. Click [OK].

Define the default media family

1. Open the Settings Editor and go to: [Color]→[Color defaults].

2. Go to [Default media family].
3. Select the default media family.
4. Click [OK].

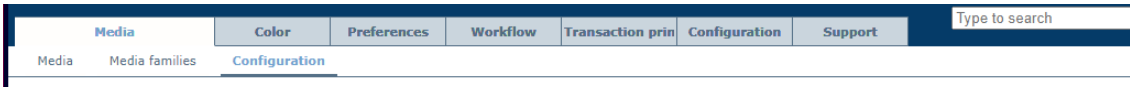
Print a media family calibration report

1. Open the Settings Editor and go to: [Support]→[Troubleshooting].

2. Click [Print report of latest media family calibration].
3. Click [OK].

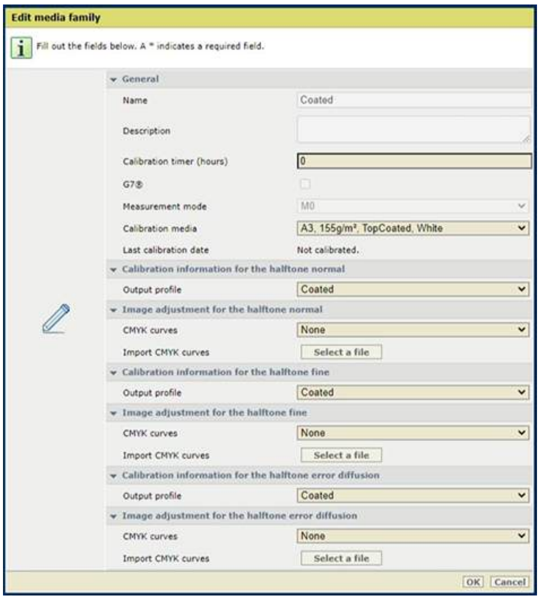
Show a warning symbol when a media family needs calibration

1. Open the Settings Editor and go to: [Media]→[Configuration].



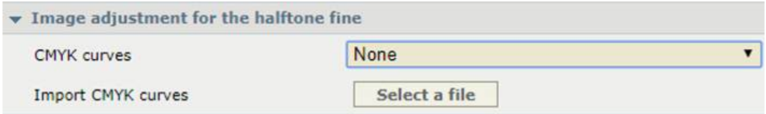
2. Enable the setting [Warnings when media family needs calibration].
3. Click [OK].

Media family settings



Edit media family

Media family settings	Description
[Name]	Name according to naming conventions.
[Description]	Indicates a description of the media family.
[Calibration timer (hours)]	Indicates the media family calibration interval, which can be set between 0 and 512 hours, in steps of 1 hour. Enter 0 to disable the media family calibration timer.
[G7®]	Indicates a media family used for G7 calibration.
[Calibration media]	Indicates the default media used for printing the calibration charts.
[Last calibration date]	Indicates when the last calibration occurred.
[Origin of calibration target]	Select the calibration target that must be applied to generate the calibration curves. • [Output profile] When the calibration target comes from the output profile, the most optimal gray balance can be achieved. • [Calibration measurements] The classic method to calculate calibration curves When you change the origin of the calibration target, new calibration curves are generated based on the last stored calibration measurements.
[Output profile]	Indicates the output profile of the media family.

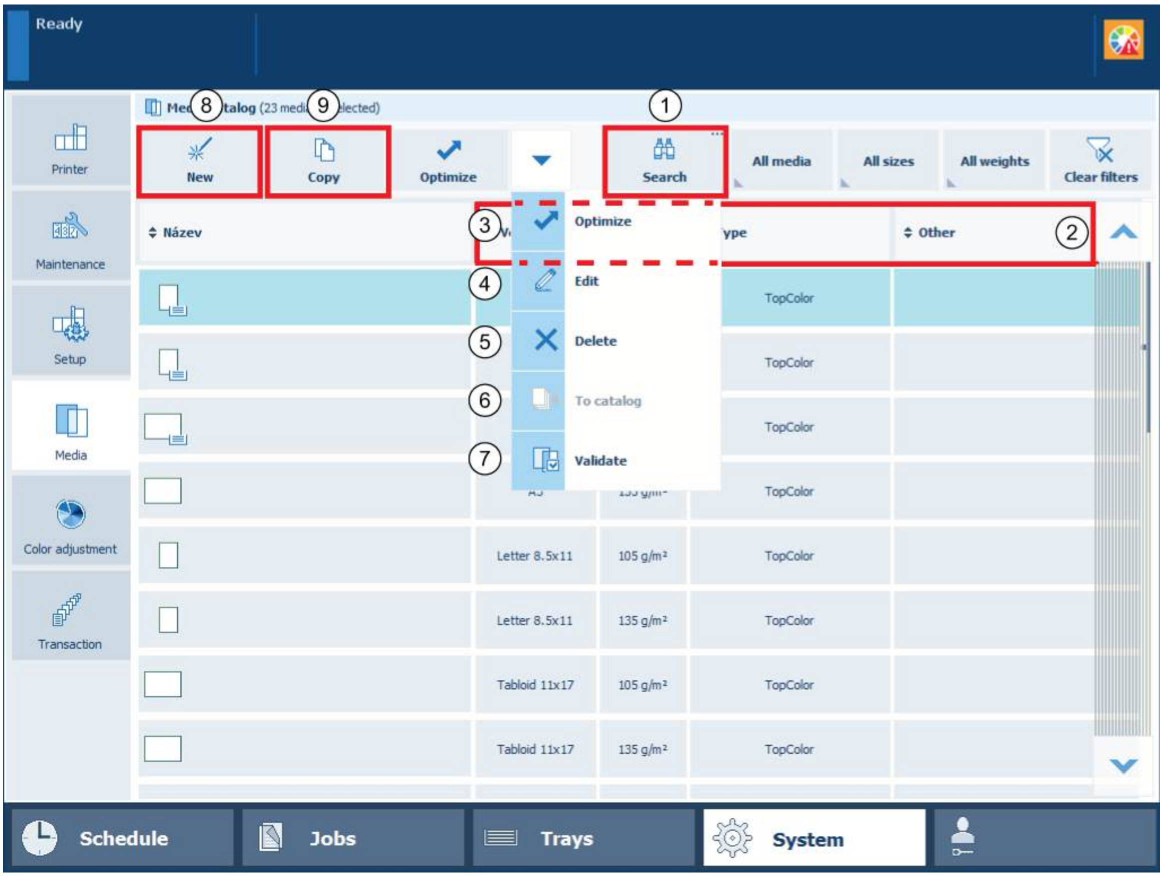
Media family settings	Description
[Import CMYK curves] [CMYK curves]	Displays if a CMYK curve is applied to a media family. • [None]: no CMYK curve is applied  • CMYK curve preset: a custom or factory defined CMYK curves preset. Use the [CMYK curves] drop-down button to select a custom or factory defined CMYK curves preset: • Cool • Lively • Saturated • Sharp • Warm  • [Anonymous]: an imported CMYK curve is applied. Use the [CMYK curves] option to import a CMYK curve for this media family - halftone combination. 

More information

- *Create output profiles with the embedded profiler* on page 381
- *Learn about calibration* on page 346
- *Learn about CMYK curves* on page 476

Manage the media from the control panel

The print system uses a central media catalog from where you can select media for jobs. ()
The media catalog can be managed with the Settings Editor. When media management via the control panel is enabled, you can also manage the media catalog on the control panel.



Media management on the control panel

1. Go to the control panel and touch: [System]→[Media].
2. Use the search button (**1**) to filter media with a specific size, type, weight, or name.
3. Touch one of the media attributes (**2**) to change the order of the media in the media catalog.
4. Use the following options to manage the media catalog:
   - [Optimize] (**3**): to start a media family calibration or perform media adjustments.
   - [Edit] (**4**): to change the name, size, type or media family of the media
   - [Delete] (**5**): to remove media that is no longer needed.
   - [To catalog] (**6**): to add temporary media to the media catalog.
   - [Validate] (**7**): to start a color validation test.
   - [New] (**8**): to add new media to the media catalog.
   - [Copy] (**9**): to add new media, based on existing media.

More information

- *Calibrate a media family* on page 373
- *Correct curled output media* on page 573
- *Adjust media registration* on page 575
- *Add temporary media to the media catalog* on page 431
- *Perform a color validation test* on page 405
- *Add media to the media catalog* on page 429

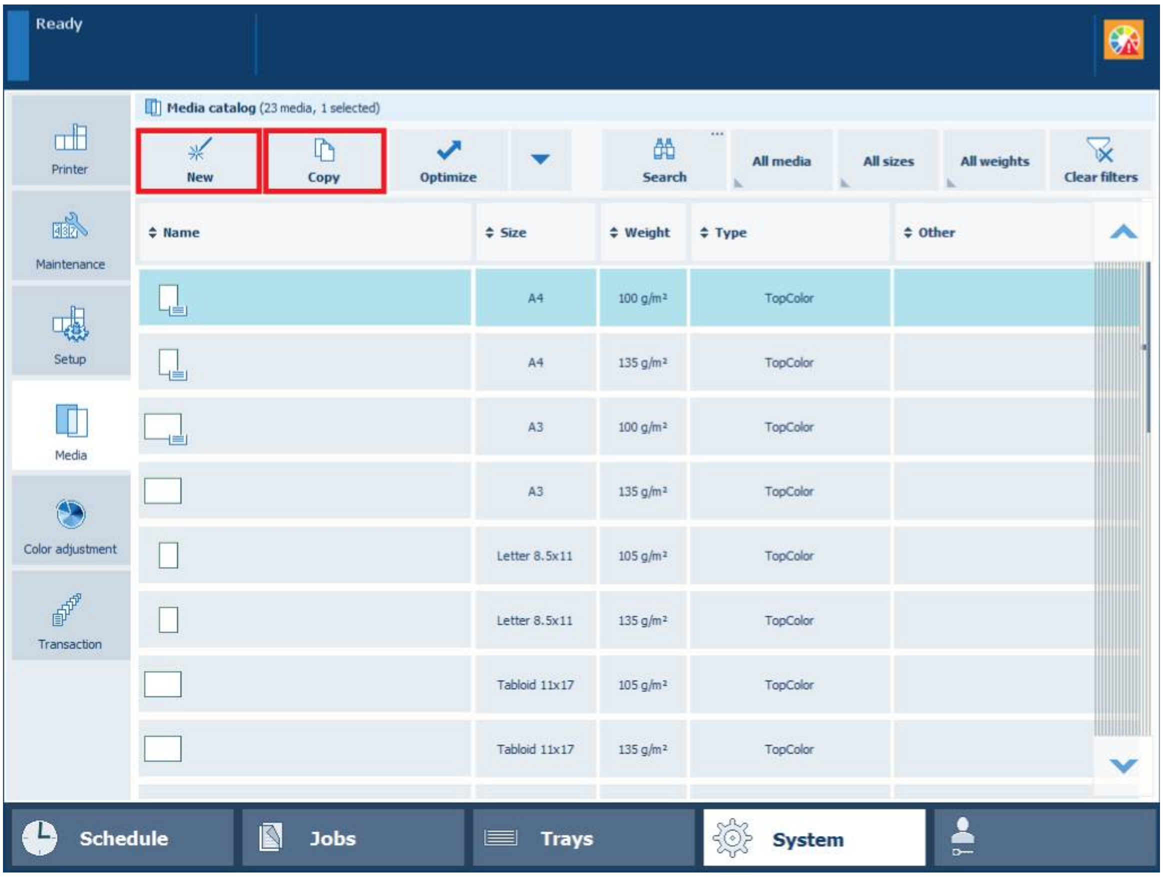
• *Manage media definitions* on page 415

Add media to the media catalog

When enabled in the Settings Editor, you can add new media to the media catalog via the control panel. You can also define whether you want media family calibration warnings and warnings on missing media attributes on the control panel.

When you use a specific media that differs significantly from the media in the available media families, the print system may not deliver the required color quality. Then, you need to create a new media family followed by a media family calibration with the specific media to improve the color quality for this media.

You can use the Canon media library to add new media to the media catalog. This media library provides a comprehensive list of recommended media with the correct media attributes. You can change the media name if required.



Media management via the control panel

1. Touch [System]→[Media].
2. Use one of the following methods to add new media.
   - Touch [Copy] to use one of the current media definitions as starting point and copy its attributes to the new media definitions.
   - Touch [New] to define all attributes from scratch.
   - Touch [New]→[Import from media library] to import a media with correct media attributes from the media library.
3. Define the name and attributes of the new media.

NOTE
For convenient retrieval of media in the media catalog, define the following media attributes:
- Name
- Standard type
- Custom type name, when the standard type is custom

IMPORTANT
For optimal print quality, define the correct values for the following media attributes. Refer to the packaging of the media for this information.
- Size
- Weight
- Surface type
- Media family

Only use the [Create new media family] option when you use a specific media that differs significantly from the media in the available media families.

4. Touch [OK].

After you finish

If you created a new media family, perform a media family calibration.

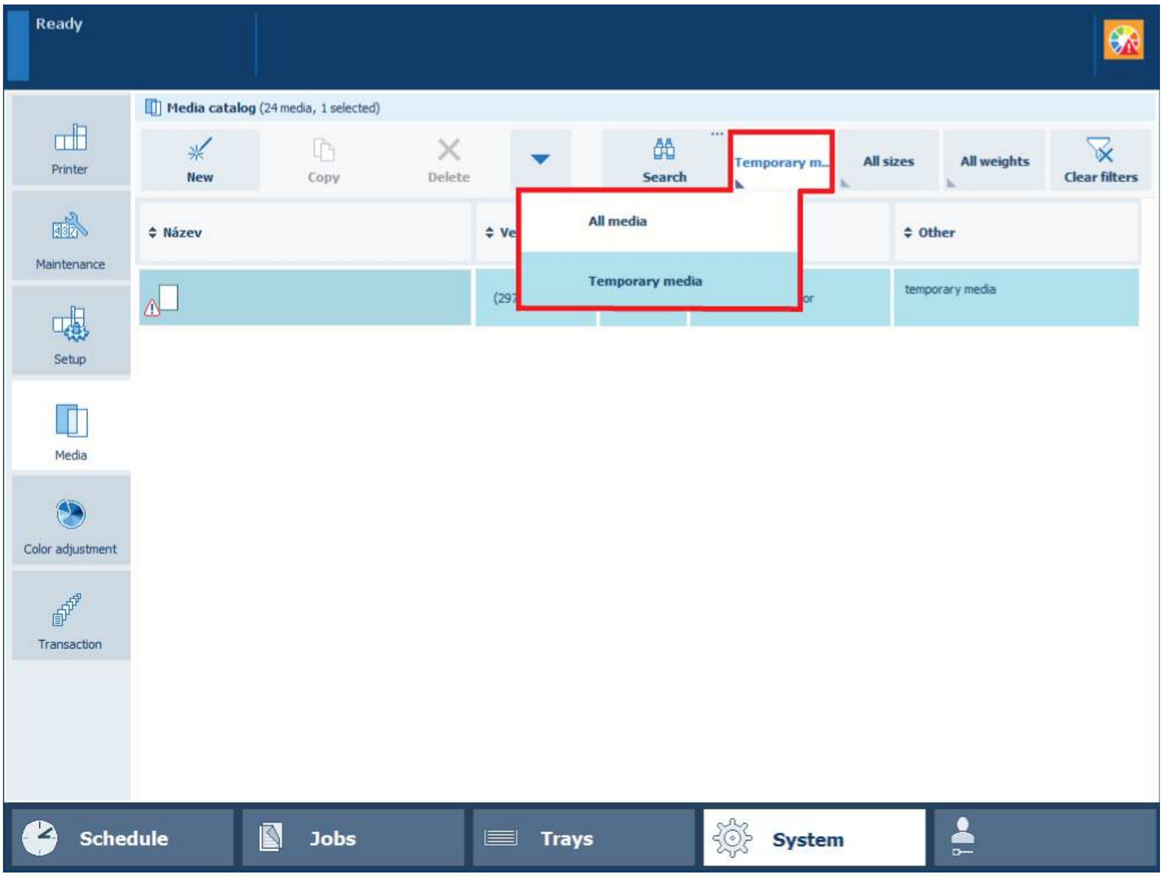
If the color quality does not meet the expectations after media family calibration, create a new media family with new output profiles.

More information

- *Define the media catalog* on page 418
- *Calibrate a media family* on page 373
- *Create output profiles with the embedded profiler* on page 381

Add temporary media to the media catalog

You can print on media that are not part of the media catalog. In a job ticket, forwarded job or PRISMAprepare job, a media can be specified that is not in the media catalogue. You can add this temporary media to the media catalog on the control panel. This is useful if you are using the temporary media more often.



Temporary media

Procedure

1. Touch [System]→[Media].
2. Select one or more temporary media.
3. Touch the drop-down icon (▼).
4. Select [To catalog] from the drop-down menu.
5. Fill out the media attributes.
6. Touch [OK].

Chapter 12

Manage color definitions

Learn about the color-based workflow

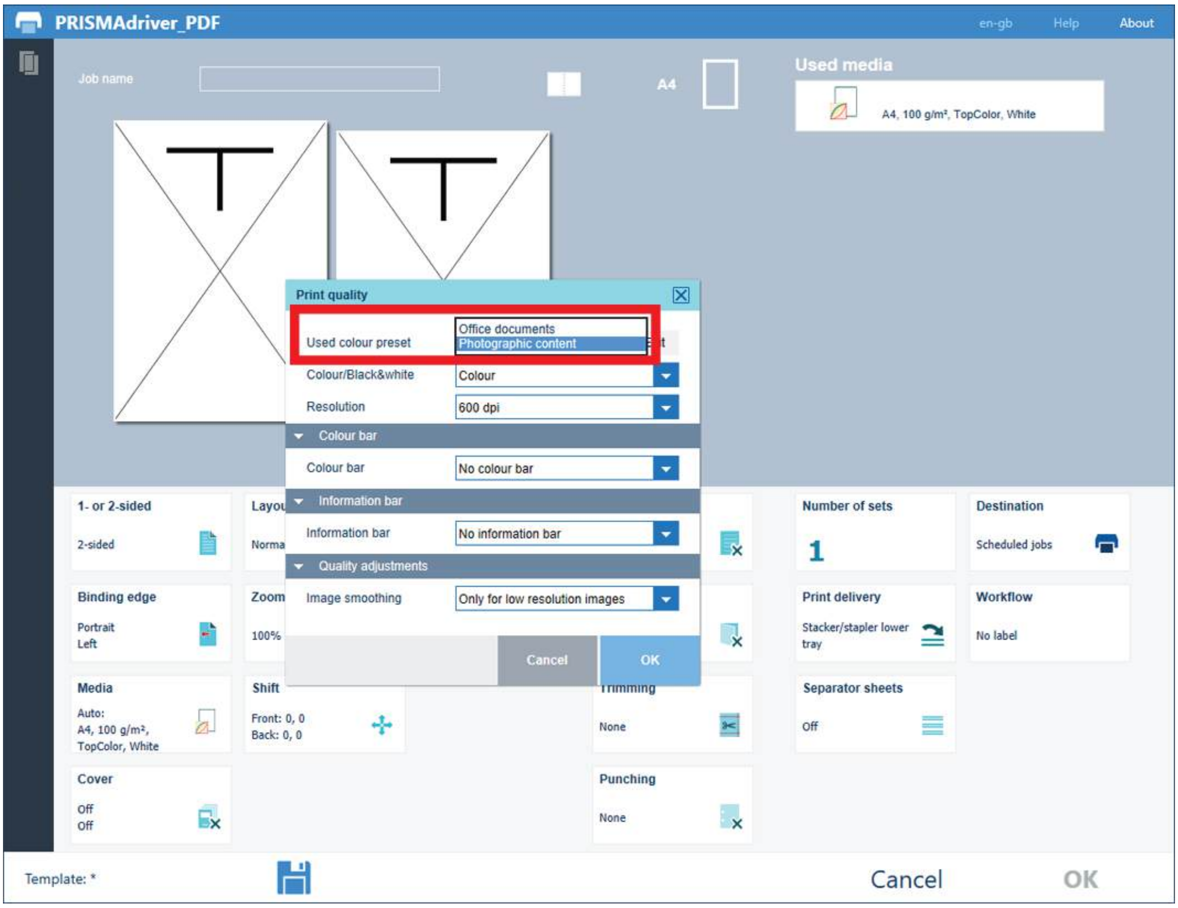
The color management system of PRISMAsync controls the reproduction of colors and handles the conversion of input colors (RGB, CMYK, spot colors) to printed colors on paper. PRISMAsync has two instruments for easy color management across the entire workflow: **color presets** and **media families.**

Furthermore, PRISMAsync offers the **advanced color management** options to increase the color features of your printer.

This topic describes how to:
- Color Presets
- Media Families
- Input and output profiles
- Advanced color management features

Color Presets

The PRISMAsync color presets simplify the complexity of color management. You select an appropriate color preset when you prepare a print job with Remote Printer Driver or PRISMA software. The definition of an automated workflow and a hotfolder also include a color preset selection. The use of color presets saves time and does not require an in-depth knowledge on color.



The color presets in PRISMAsync Remote Printer Driver

Default color presets

PRISMAsync includes the following default color presets:
- **Office documents**
  This color preset is optimal for color reproduction of text and graphical lines in office documents. PRISMAsync converts the colors to more saturated colors for the printed output.
- **Photographic content**
  This color preset is optimal for the reproduction of photographs, pictures, and images.

Rendering intents

A color preset contains among other settings a rendering intent. Input colors do not always fit in the color gamut of an output device. The rendering intent tells the color management system how to handle these colors. The following rendering intents are available on PRISMAsync:
- **Relative colorimetric**
  Frequently used and suitable for proofing applications on **reference paper**. White point compensation is applied. The white point of the output color space is the color of the used media.
- **Absolute colorimetric**
  Mainly suitable for **proofing applications on bright white paper**. No black and white point compensation is applied, so the precise input paper color is simulated on the used media.
- **Perceptual**
  Mainly suitable for **photographic input.** The colors are mapped in a continuous way and color nuances are recognized. White and black point compensation is applied.
- **Saturation**
  Mainly suitable to obtain **highly saturated colors** and **business graphics**. Out-of-gamut colors are mapped to more saturated colors. White and black point compensation is applied.

For more information about the creation of color presets:

Media Families

A media family is a set of media that has comparable print quality specifications, such as surface type, color and substrate type. A media family defines the color output profiles for the media. Each media family has three different **output profiles**, for the halftone screens Normal, Fine, and Error diffusion. When you have more than one halftone screen in a single job or page, PRISMAsync automatically applies the related output profiles for optimal color management in the job.

Input and output profiles

Input and output profiles define the transition from input colors, for example, the RGB color values of a photo, towards output colors or, for example, the CMYK color values of your printer. The color management of PRISMAsync transforms the color input space to the color output space via the standards of the ICC, the International Color Consortium.

PRISMAsync comes with high quality color output profiles for both coated and uncoated media. PRISMAsync links output profiles to media via media families. Each media family has three different output profiles, for the halftone screens Normal, Fine, and Error diffusion.

Advanced color management features

The Advanced color management license offers the following color features:
- **Device simulation**, to simulate or proof other print engines or offset presses with your print system.
- **Device Link support**, to import Device Link profiles for special CMYK conversions.
- **Named color profile import and export**, to extend the functionality of spot color libraries.

- **Information bars**, to print extra metadata on printed output. There is one default information bar, but you can compose other information bars. Information bar selection is possible in Remote Printer Driver, PRISMAprepare, and via automated workflows or hotfolders.
- **Color control bars**, to print extra color patches for control of color output via FOGRA verification. Using a color bar you can check the ink density, dot grain and contrast. Use default or compose custom color bars. The color bar selection is available in the Remote Printer Driver, PRISMAprepare, and via automated workflows or hotfolders.
- **Color mappings**, to convert RGB, CMYK to a specific color, for example, your corporate color. Color mappings are useful to map colors in the input file to a fixed color of a spot color library.
- **Leave out separations**, to exclude specific spot colors from printing. You can exclude layers for printing when you map the colors of these layers to a spot color with the value “None”. You can use leave out separations for white separations, cutting marks and spot layers to indicate the use of foil or a gold print.
- **Embedded profiling**, to create your own ICC or G7 output profiles with the embedded color profiler.

More information

- *Define color defaults* on page 437
- *Add media to the media catalog* on page 429
- *Create output profiles with the embedded profiler* on page 381
- *Define the media families* on page 423

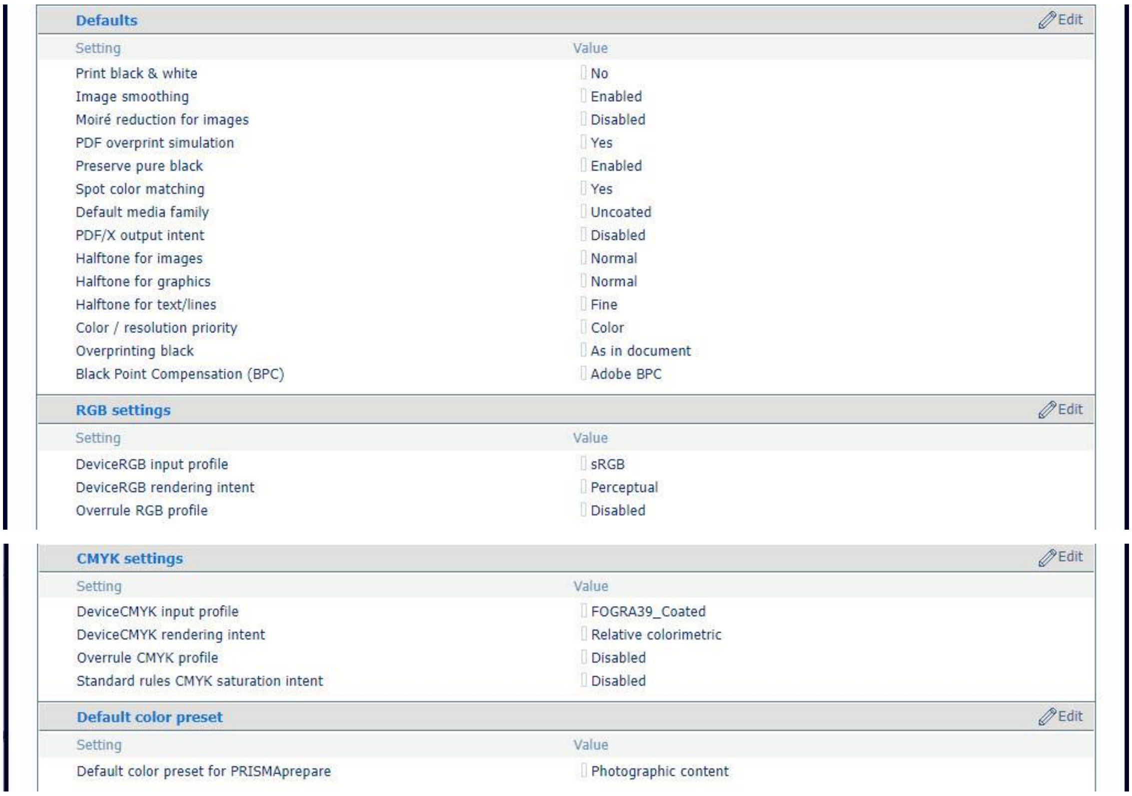
Define color defaults

You can change several color, RGB, and CMYK settings.

Go to the color defaults

1. Open the Settings Editor and go to: [Color]→[Color defaults].

[Color defaults] tab



[Color defaults] settings

Define the color defaults

1. Use the [Print black & white] setting to print in black & white or in color.
2. Use the [Image smoothing] setting to prevent unsmooth lines and image blocks in the document. This occurs when source objects have a lower resolution than the printer. The [Image smoothing] interpolation method only affects images below 300 dpi.
3. Use the [Moiré reduction for images] setting to apply a Moiré reduction algorithm to enhance photographic images.

NOTE
When images have a resolution below 300 dpi, the Moiré reduction only takes effect when the [Image smoothing] setting is enabled.

4. Use the **[PDF overprint simulation]** setting to make opaque objects look transparent. Underlying objects become visible. If this setting is disabled, the colors on top will knock out all underlying colors.
5. Use the **[Preserve pure black]** setting to apply pure black preservation when possible. Pure black preservation means that the color black is composed of 100% K ink. When pure black preservation is not possible or disabled, the color black is composed of a mixture of two or more C, M, Y, and K inks.
6. Use the **[Spot color matching]** setting to enable or disable spot color matching. If a source file contains a spot color, the printer must know the spot color definition to exactly print the required color.
7. Use the [Default media family] setting to define the default media family for print jobs that arrive without media family information.
8. Use the **[Use PDF/X output intent]** setting to indicate if PDF source files are printed according to their embedded output intent. PDF/X specifies the printing conditions for which a PDF/X file is created. These printing conditions are called output intents. The printer can handle PDF/X compliant PDF source files. Then, the output intent overrules the DeviceCMYK rendering intent and DeviceCMYK input profile.
9. Use the **[Halftone for images]** setting to change the default halftone for images.
   - [Normal]: for sharper texts or improvement of the color gradient in color areas.
   - [Fine]: for text, images and graphics.
   - [Error diffusion]
10. Use the **[Halftone for graphics]** setting to change the default halftone for graphics.
11. Use the **[Halftone for text/lines]** setting to change the default halftone for text and lines.
12. Use the **[Color / resolution priority]** setting if graphic objects with a high toner density may appear blurry at the edges of the graphic. Use this setting for sharp edges. The graphic objects may become less saturated.
13. Use the **[Overprinting black]** setting to force the printer to print pure black text and graphics over the background color. The [Overprinting black] prevents that white lines appear around black characters and graphics.
    - [As in document]: overprinting black is not applied.
    - [Enabled for text]: overprinting black is applied to texts.
    - [Enabled for text and graphics]: overprinting black is applied to texts and graphics.
14. The **[Black Point Compensation (BPC)]** setting applies to the relative colorimetric rendering intent. Details in dark regions of the document can be lost with the standard color conversion. The Black Point Compensation aligns the darkest level of black achievable (black point) of the source to the darkest level of black achievable on the printer.
    - [Disabled]: Black Point Compensation is not applied.
    - [Adobe BPC]: the Adobe implementation of [Black Point Compensation (BPC)].
    - [Enhanced BPC]: when the black point in the document is rather light, select [Enhanced BPC].

Define the DeviceRGB settings

The Settings Editor stores the DeviceRGB input profiles from which you can select a default DeviceRGB input profile.

1. Use the **[DeviceRGB input profile]** setting to select the default DeviceRGB input profile.
2. Use the **[DeviceRGB rendering intent]** setting to select the default DeviceRGB rendering intent that defines the color conversion strategy for out-of-gamut colors.
3. Use the **[Overrule RGB profile]** setting to indicate if the embedded DeviceRGB input profile is overruled or not.

Define the DeviceCMYK settings

The Settings Editor stores the DeviceCMYK input profiles from which you can select a default DeviceCMYK input profile.

1. Use the **[DeviceCMYK input profile]** setting to select the default DeviceCMYK input profile.
2. Use the **[DeviceCMYK rendering intent]** setting to select the default DeviceCMYK rendering intent that defines the color conversion strategy for out-of-gamut colors.
3. Use the **[Overrule CMYK profile]** setting to indicate if the embedded DeviceCMYK input profile is overruled or not.
4. Use the **[Standard rules CMYK saturation intent]** setting to indicate if the standard ICC color management rules of the CMYK saturation intent must be applied. For the CMYK saturation intent, the printer by default maps pure (100%) C, M, Y and K input colors into pure C, M, Y and K output colors. However, it can be required that the standard ICC color management rules must be used without an improvement of the color management of the printer.

More information

Define an input profile on page 448

Define a color preset

A color preset consists of a series of pre-defined color settings. Color presets help to easily select the correct color settings during the job definition. The list of color presets is available in PRISMAprepare, automated workflows, PRISMAsync Remote Manager, PRISMAsync Remote Printer Driver, and on the control panel.

The job properties on the control panel and in PRISMAsync Remote Manager allow you to change one or more attributes of the job color preset. The changed color preset can be stored with another name.

There are two factory defined color presets.

- **Office documents**
  This color preset is optimal for the color reproduction of text and graphical lines in office documents. The printer converts the source file colors to more saturated output colors.
- **Photographic content**
  This color preset is optimal for the reproduction of photographs and illustrations.

Go to the color presets

1. Open the Settings Editor and go to: [Color]→[Color presets].

[Color presets] tab

[Color presets]

Add a color preset

1. Click [Add].
2. Define the attributes of the color preset.
3. Use the table below for information on the attributes.

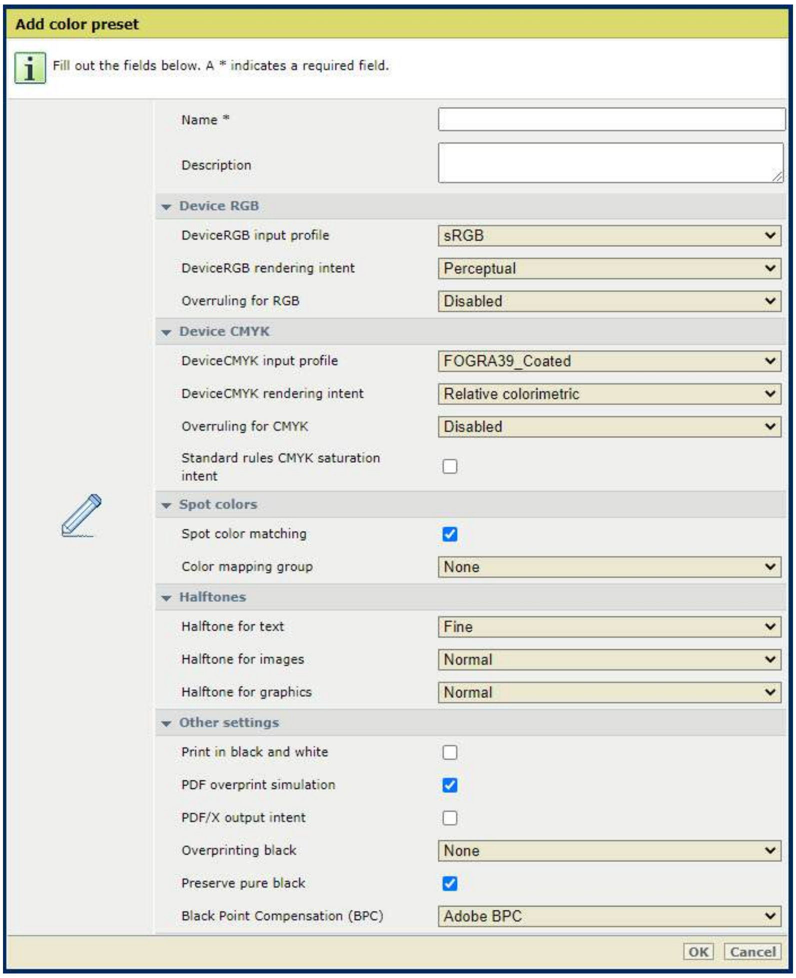
Edit a color preset

1. Select a color preset.
2. Click [Edit].
3. Define the required attributes of the color preset.
4. Use the table below for information on the options.

Delete a color preset

1. Select the color preset.
2. Click [Delete].

Define the color preset attributes



[Add color preset]

Color preset attributes	Description
[Name]	Enter a unique name of the color preset.
[Description]	Enter a description of the color preset.
[DeviceRGB input profile]	Use this setting to select the DeviceRGB input profile.
[DeviceRGB rendering intent]	Use this setting to select the DeviceRGB rendering intent.
[Overruling for RGB]	Use this setting to indicate if the embedded DeviceRGB input profile is overruled or not.
[DeviceCMYK input profile]	Use this setting to select the DeviceCMYK input profile.
[DeviceCMYK rendering intent]	Use this setting to select the DeviceCMYK rendering intent.
[Overruling for CMYK]	Use this setting to indicate if the embedded CMYK input profile is overruled or not.

Color preset attributes	Description
[Standard rules CMYK saturation intent]	Use this setting to indicate if the standard ICC color management rules of the CMYK saturation intent must be applied. For the CMYK saturation intent, the printer by default maps pure (100%) C, M, Y and K input colors into pure C, M, Y and K output colors. However, it can be required that the standard ICC color management rules must be used without an improvement of the color management of the printer.
[Spot color matching]	Use this setting to enable or disable spot color matching. If a source file contains a spot color, the printer must know the spot color definition to exactly print the required color.
[Color mapping group]	If the [Spot color matching] setting is enabled, you can select a color mapping group. With color mapping you can redefine the color values of a source color (RGB, CMYK, or spot color).
[Halftone for text]	Use this setting to define the halftone for text.
[Halftone for images]	Use this setting to define the halftone for images.
[Halftone for graphics]	Use this setting to define the halftone for graphics.
[Print in black and white]	Use this setting to indicate how jobs are printed: in black & white or in color.
[PDF overprint simulation]	Use this setting to make opaque objects look transparent. Underlying objects become visible. If this setting is disabled, the colors on top will knock out all underlying colors.
[PDF/X output intent]	Use this setting to define that PDF files are printed according to their embedded output intent. PDF/X specifies the printing condition for which the PDF/X file was created. The Device CMYK input profile and the job ticket are ignored.
[Overprinting black]	Use this setting to force the printer to print pure black text and graphics over the background color. The [Overprinting black] prevents that white lines appear around black characters and graphics. • [As in document]: overprinting black is not applied. • [Enabled for text]: overprinting black is applied to texts. • [Enabled for text and graphics]: overprinting black is applied to texts and graphics.
[Preserve pure black]	Use the setting to apply pure black preservation when possible. Pure black preservation means that the color black is composed of 100% K ink. When pure black preservation is not possible or disabled, the color black is composed of a mixture of two or more C, M, Y, and K inks.

Color preset attributes	Description
[Black Point Compensation (BPC)]	This setting applies to the relative colorimetric rendering intent. Details in dark regions of the document can be lost with the standard color conversion. The Black Point Compensation aligns the darkest level of black achievable (black point) of the source to the darkest level of black achievable on the printer. • [Disabled]: Black Point Compensation is not applied. • [Adobe BPC]: the Adobe implementation of [Black Point Compensation (BPC)]. • [Enhanced BPC]: when the black point in the document is rather light, select [Enhanced BPC].

Define the default color preset for PRISMAprepare

1. Open the Settings Editor and go to: [Color]→[Color defaults].

[Color defaults]

2. Go to the [Default color preset] section.

[Default color preset]

3. Use the [Default color preset for PRISMAprepare] setting to define the default color preset for PRISMAprepare.

[Default color preset for PRISMAprepare]

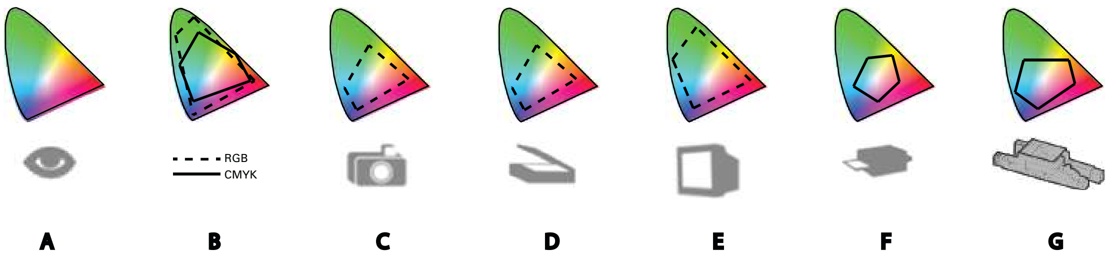
Manage input profiles and output profiles

Learn about input and output profiles

Color spaces

Devices as monitors and printers create colors in a different way. Because of these differences colors are described differently for each device.

The different methods of describing colors are called color spaces. The color space specifies how color information is represented in an image. Each device has its own individual color space and range of colors that it can display or print. In addition, each device can have one or more ICC (International Color Consortium) profiles. ICC profiles are used when an image or a object is converted to the color space of a different device.



Color spaces of human eye (A), RGB and CMYK (B), and other devices (C, D, E, F, and G)

RGB

Devices such as monitors, digital cameras, and scanners generally use RGB color spaces to describe colors. Two implementations of RGB color spaces are sRGB and Adobe RGB.

CMYK

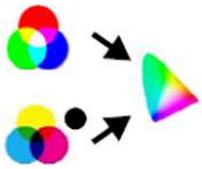
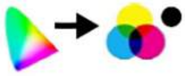
Color printers use the CMYK color space. Implementations of the CMYK color space vary from printer to printer and from medium to medium. FOGRA has several implementations of the CMYK color spaces.

Input profiles and output profiles

An **input ICC profile** can translate color data created on one device (such as a monitor) into the Profile Connection Space (PCS). The Profile Connection Space (PCS) is necessary to connect the input and output color space by describing the colors in a device-independent way. The Profile Connection Space (PCS) color space, such as the CIELAB color space, defines colors as true representations of what the human eye can perceive.

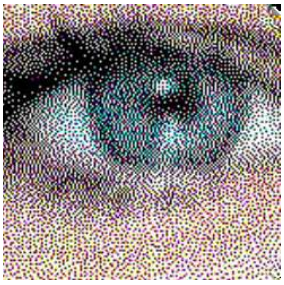
There are input profiles to translate RGB color data and input profiles to translate CMYK color data.

The output ICC profile translates the color information from the Profile Connection Space (PCS) to the CMYK color space of the printer.

Input profile	Output profile
 RGB / CMYK -> Lab	 Lab -> CMYK

Separation

The process where the output profile converts the colors from the Profile Connection Space (PCS) color space into the CMYK color space is called **separation**. The result of the separation is the creation of the CMYK separation planes.



CMYK dots in the document

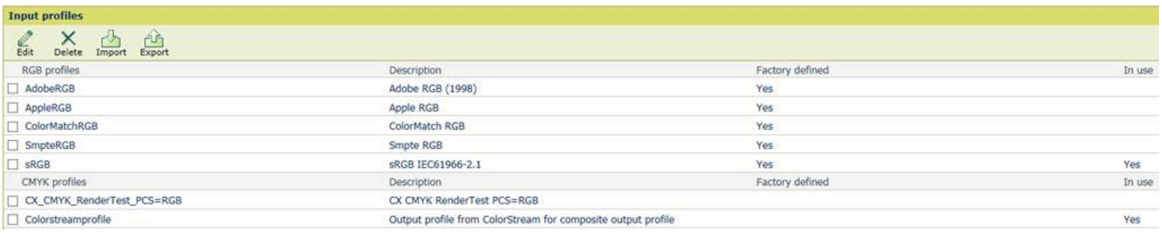
The outcome of the separation is important for the quality of the printed image, since one target color in the device-independent color space can be reproduced by different ink combinations on the printer. To calculate the CMYK values, the output profile takes into account which limits are set for the media. For example, the total ink coverage limit and the maximum drop size of the inks. Moreover, the output profile knows where CMY inks must be replaced by K ink and how much ink must be used for the darkest black.

A CMYK color value is often expressed as four percentages. Each C, M, Y, and K color channel has the potential of printing at 100%.

Input profiles on the printer

The color values of an image can be described in the RGB or CMYK color space.

The Settings Editor has a list of input profiles that can be selected. Color presets and transaction setups have a reference to one of the input profiles. You import the input profiles from the Settings Editor.



Input profiles

Edit Delete Import Export

RGB profiles	Description	Factory defined	In use
AdobeRGB	Adobe RGB (1998)	Yes	
AppleRGB	Apple RGB	Yes	
ColorMatchRGB	ColorMatch RGB	Yes	
SmpteRGB	Smpte RGB	Yes	
sRGB	sRGB IEC61966-2.1	Yes	Yes
CMYK profiles	**Description**	**Factory defined**	**In use**
CX_CMYK_RenderTest_PCS=RGB	CX CMYK RenderTest PCS=RGB		
Colorstreamprofile	Output profile from ColorStream for composite output profile		Yes

Input profiles

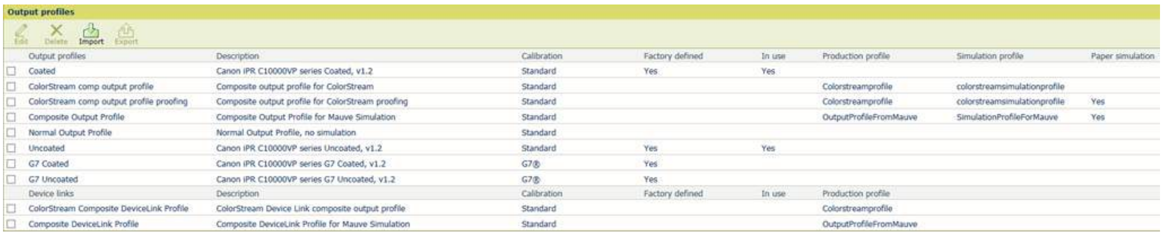
Source files can also contain input profiles. If required, you can decide to overrule these embedded profiles or not.

The list of input profiles can contain simulation profiles and production profiles to create composite output profiles for device simulation.

Output profiles on the printer

The color values in a document can be described in the CMYK color space or in the CIELAB color space.

The Settings Editor has a list of output profiles that can be linked to a media family. When media definitions change, a new output profile can be needed. You import output profiles in the Settings Editor.



Output profiles

Output profiles	Description	Calibration	Factory defined	In use	Production profile	Simulation profile	Paper simulation
Coated	Canon iPR C10000VP series Coated, v1.2	Standard	Yes	Yes			
ColorStream comp output profile	Composite output profile for ColorStream	Standard			Colorstreamprofile	colorstreamsimulationprofile	
ColorStream comp output profile proofing	Composite output profile for ColorStream proofing	Standard			Colorstreamprofile	colorstreamsimulationprofile	Yes
Composite Output Profile	Composite Output Profile for Mauve Simulation	Standard			OutputProfileFromMauve	SimulationProfileForMauve	Yes
Normal Output Profile	Normal Output Profile, no simulation	Standard					
Uncoated	Canon iPR C10000VP series Uncoated, v1.2	Standard	Yes	Yes			
G7 Coated	Canon iPR C10000VP series G7 Coated, v1.2	G7®	Yes				
G7 Uncoated	Canon iPR C10000VP series G7 Uncoated, v1.2	G7®	Yes				
Device links	Description	Calibration	Factory defined	In use	Production profile		
ColorStream Composite DeviceLink Profile	ColorStream Device Link composite output profile	Standard			Colorstreamprofile		
Composite DeviceLink Profile	Composite DeviceLink Profile for Mauve Simulation	Standard			OutputProfileFromMauve		

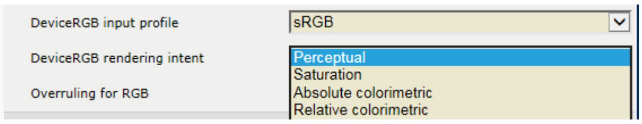
The list of output profiles can contain composite output profiles and Device Link profiles for device simulation.

The maximum C, M, Y, and K color values (expressed in ΔE) are used as calibration target during the media family calibration.

Rendering intents on the printer

When source colors fall outside the gamut of the printer, the rendering intent determines how these colors must be reproduced.

Source colors that fall outside the gamut of the printer are being mapped into in-gamut colors, often colors at the edge of the printer gamut.



Rendering intents

Composite output profile for device simulation

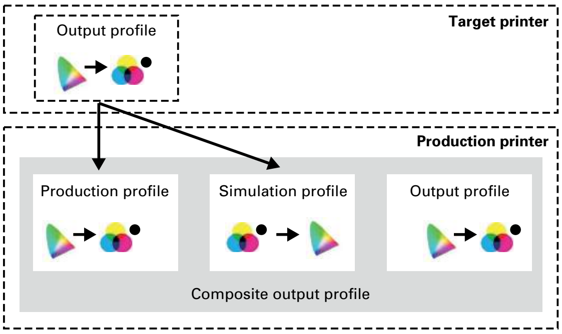
In the Graphics Arts (GA) market it is common to align the color reproduction between different printers or offset presses.

The target printer can have the following profiles:
- The **output profile**, used by the target printer to **reproduce colors approximately** on various similar media.
- An **accurate output profile** that was created for the target printer to **reproduce colors accurately** on specific media.

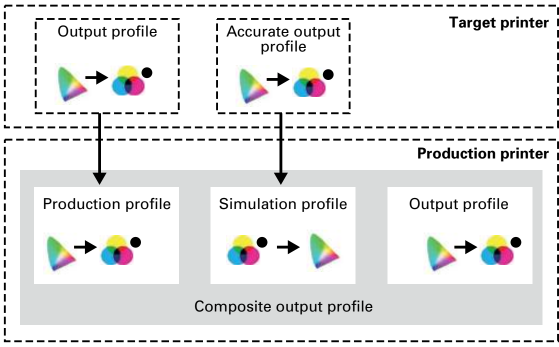
You can create a composite output profile to simulate the target printer. A composite output profile can be composed from:
- **Production profile**: refers to the output profile of the target printer.
- **Simulation profile**: refers to the output profile or accurate output profile of the target printer.

- **Output profile**: the output profile of the printer on which you are simulating.

To refer from the composite output profile to the output profiles of the target printer, the output profiles of the target printer are imported into the list of input profiles.



Composite output profile without accurate output profile

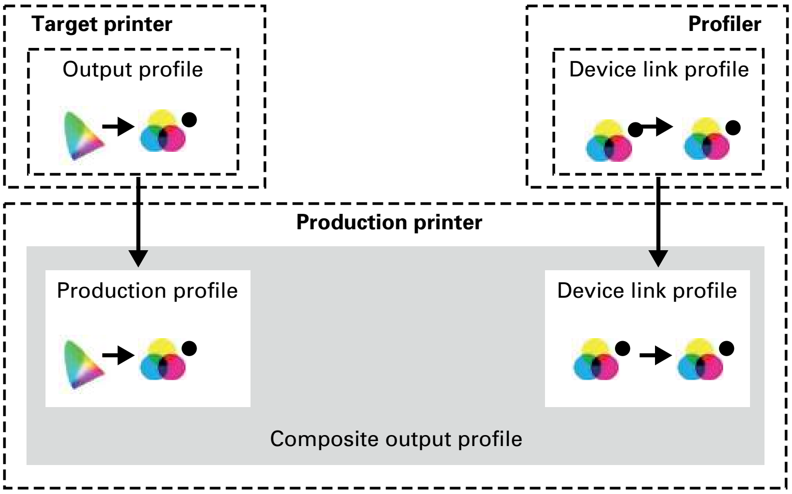


Composite output profile with accurate output profile

Composite output profile for device simulation with device link profile

A color expert can tune the color transformation with the use of a device link profile. The device link profile is typically created by an external profiler with the use of the output profile and simulation profile of the target printer.

PRISMAsync Print Server combines the output profile of the target printer with the device link profile in a **composite out profile** on the printer on which the target printer is simulated.



Composite output profile with output profile of target printer and device link profile

More information

- *Define an input profile* on page 448
- *Define an output profile* on page 450
- *Define a composite output profile for device simulation* on page 452

Define an input profile

Input and output profiles are used to link the CMYK or RGB input color spaces to the color space of the printer. The input profile is used to convert the source color space to the universal Profile Connection Space (PCS) color space.

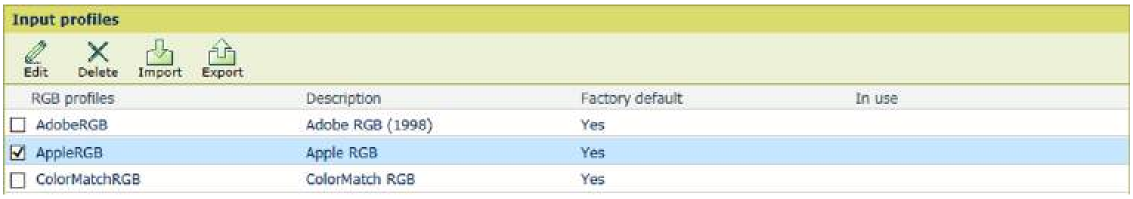
You cannot create input profiles in the Settings Editor.

For most print environments the factory defined input profiles are sufficient but, if required, you can import new input profiles.

Go to the input profiles

1. Open the Settings Editor and go to: [Color]→[Input profiles].

[Input profiles]



[Input profiles]

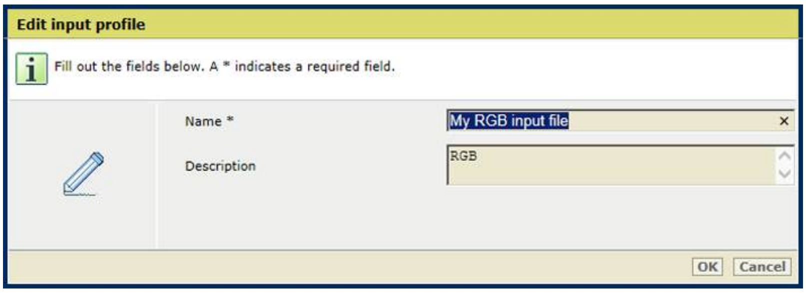
View the status of the input profiles

- [In use]: the input profile is used in a color preset or transaction setup.
- [Factory-defined]: the input profile is factory defined.

Edit an input profile

You cannot edit factory defined input profiles.

1. Select the input profile.
2. Click [Edit].
3. Change the name and description.



[Edit input profile]

Delete an input profile

You cannot delete input profiles that are in use or factory defined.

1. Select the custom input profile.
2. Click [Delete].

Import an input profile

1. Click [Import].
2. Browse to the required ICC file.
3. Define a unique name and the description.

[Import input profile]

Export an input profile

1. Select an output profile.
2. Click [Export].
3. Browse to the location to store the ICC file.

More information

Learn about input and output profiles on page 444

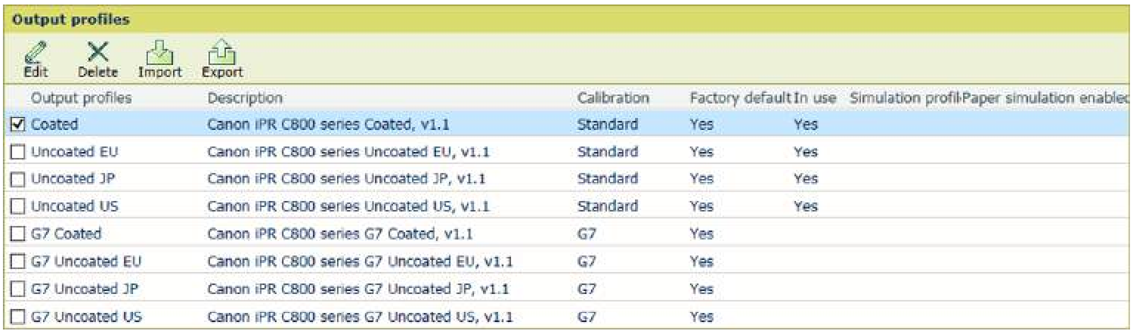
Define an output profile

Output profiles are part of the media family.

Go to the output profiles

1. Open the Settings Editor and go to: [Color]→[Output profiles].

[Output profiles] tab



Output profiles	Description	Calibration	Factory default	In use	Simulation profil	Paper simulation enablec
Coated	Canon iPR C800 series Coated, v1.1	Standard	Yes	Yes		
Uncoated EU	Canon iPR C800 series Uncoated EU, v1.1	Standard	Yes	Yes		
Uncoated JP	Canon iPR C800 series Uncoated JP, v1.1	Standard	Yes	Yes		
Uncoated US	Canon iPR C800 series Uncoated US, v1.1	Standard	Yes	Yes		
G7 Coated	Canon iPR C800 series G7 Coated, v1.1	G7	Yes			
G7 Uncoated EU	Canon iPR C800 series G7 Uncoated EU, v1.1	G7	Yes			
G7 Uncoated JP	Canon iPR C800 series G7 Uncoated JP, v1.1	G7	Yes			
G7 Uncoated US	Canon iPR C800 series G7 Uncoated US, v1.1	G7	Yes			

[Output profiles]

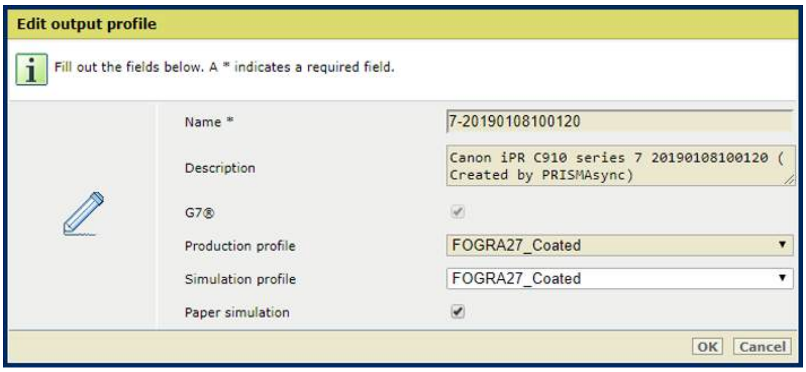
View the status of the output profiles

- [In use]: the output profile is used in a media family.
- [Factory-defined]: the output profile is factory defined.
- [Production profile]: the name of the production profile for a composite output profile or device link profile.
- [Simulation profile]: the name of the simulation profile for a composite output profile.
- [Paper simulation]: indicates if paper simulation is used for a composite output profile.

Edit an output profile

You can only edit output profiles that are not factory defined.

1. Select the output profile.
2. Click [Edit].



[Edit output profile]

3. Change the name and description.
4. Use the settings [Production profile], [Simulation profile], and [Use paper simulation] when you want to create a composite output profile.
5. Click [OK].

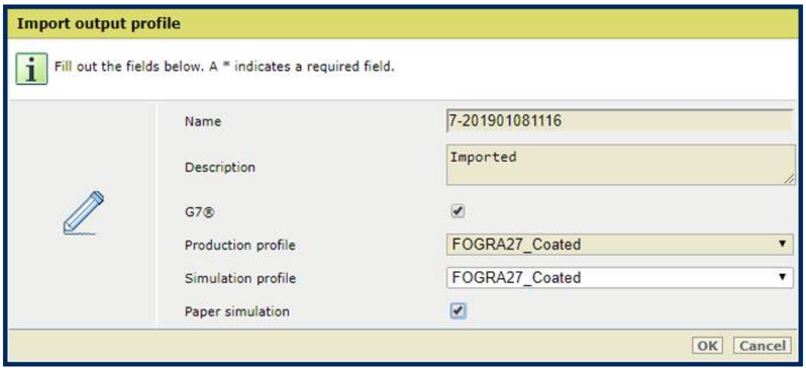
Delete an output profile

You cannot delete output profiles that are factory defined.

1. Select the output profile.
2. Click [Delete].

Import an output profile

1. Click [Import].
2. Browse to the required ICC file.
3. Define a unique name and the description.



[Import output profile]

4. Use the G7 option to import a G7 output profile.
5. Use the settings [Production profile], [Simulation profile], and [Use paper simulation] when you want to create a composite output profile.
6. Click [OK].

Export an output profile

1. Select an output profile.
2. Click [Export].
3. Browse to the location to store the output profile ICC file.

More information

Learn about input and output profiles on page 444

Define a composite output profile for device simulation

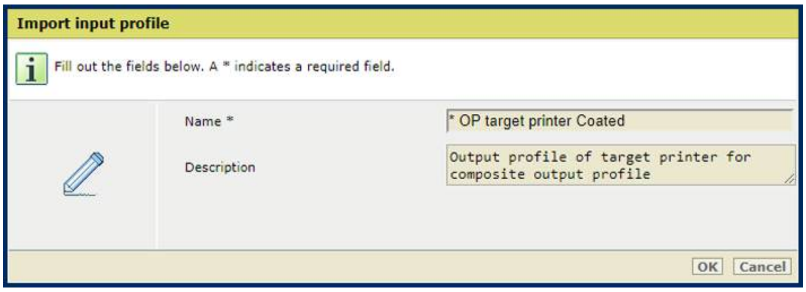
You can make a composite output profile.

Import an output profile of the target printer

1. Make sure the required production and simulation profile are available for import in the Settings Editor.
2. Open the Settings Editor and go to: [Color]→[Input profiles].

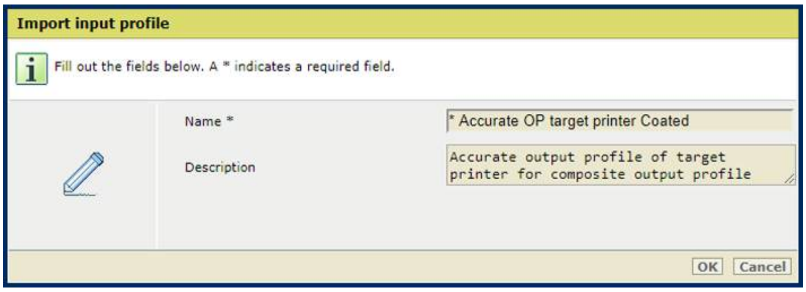
[Input profiles] tab

3. Click [Import].
4. Browse to the output profile of the target printer.
5. Enter a unique name and description.



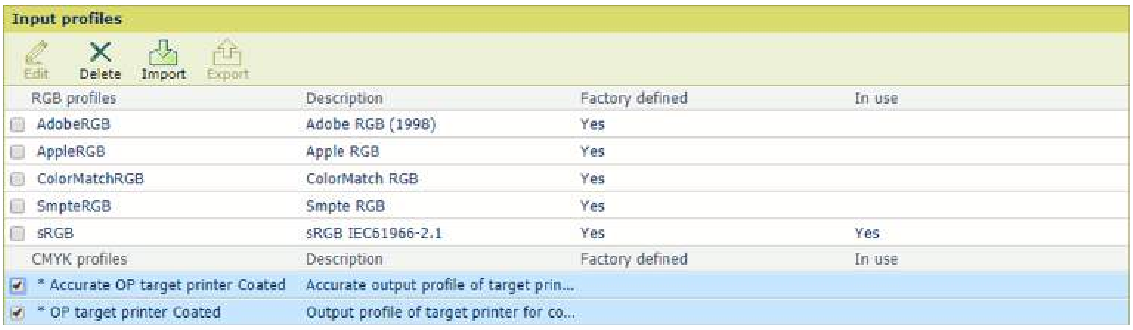
Output profile of target printer as input profile

6. Click [OK].
7. Repeat step 2 - 5 to import the accurate output profile, if required.



Profiler output profile of target printer as input profile

8. The profiles are added to the list of input profiles.

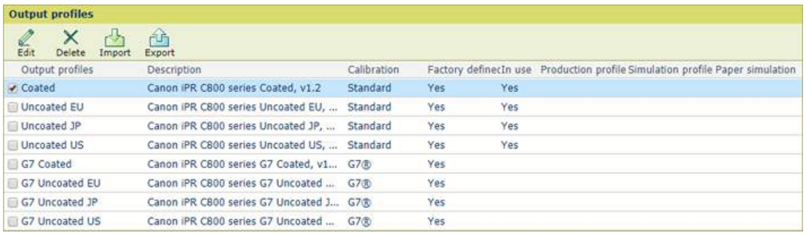


Define a composite output profile with output profiles of the target printer

1. Open the Settings Editor and go to: [Color]→[Output profiles].

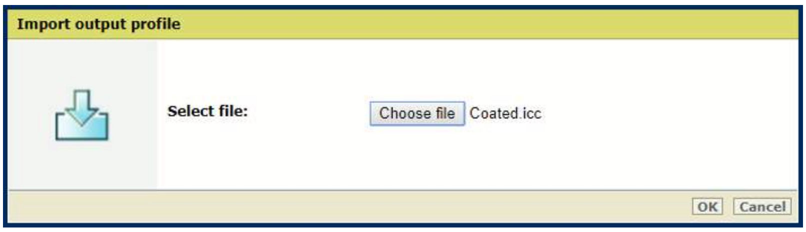
Output profiles tab

2. Select the output profile of the production printer.
3. Click [Export].

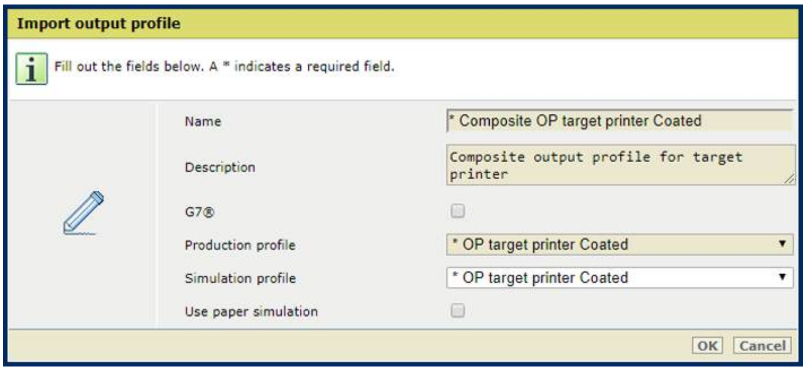


[Export] option

4. Click [Import].
5. Browse to the exported output profile.

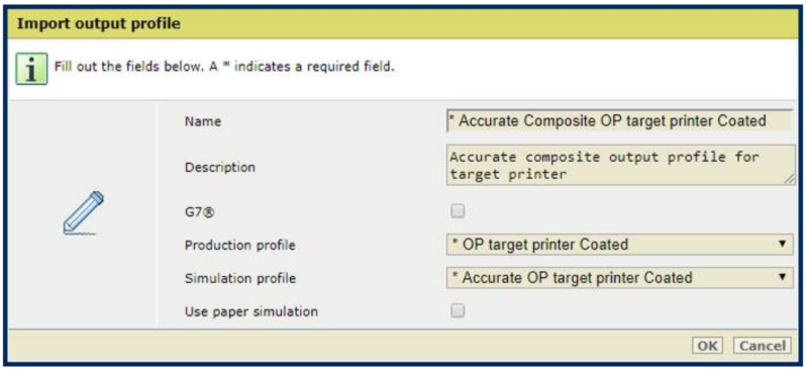


6. Browse to the output profile of the target printer.
7. Click [OK].
8. Define a unique name and the description.
9. Select the G7® check box to use G7® simulation, if required.
10. At [Production profile], select the output profile of the target printer from the list of input profiles.



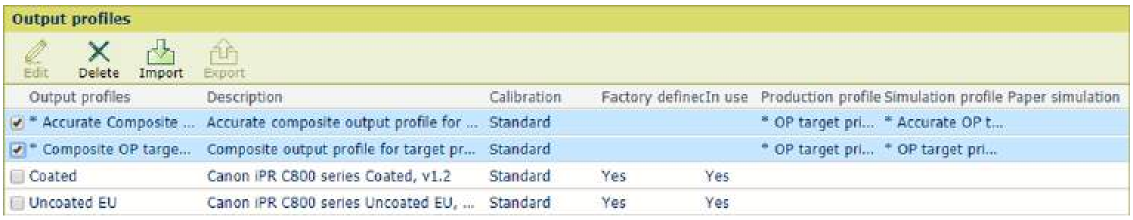
The production profile is the same as the simulation profile

11. For exact color simulation, select at [Simulation profile] the profiler output profile of the target printer from the list of input profiles.



The production profile and simulation profile differ

12. Select the [Use paper simulation] check box to enable paper simulation.
When paper simulation is enabled, the color of the media on the simulated printer is simulated by the rendering intent. absolute colorimetric The rendering intent relative colorimetric is applied with black preservation but without black point compensation.
13. Click [OK].



Composite output profiles in list of output profiles

Define a composite output profile with device link profile

1. Open the Settings Editor and go to: [Color]→[Output profiles].

Output profiles tab

2. Click [Import].
3. Browse to the device link profile.
4. Click [OK].
5. Define a unique name and the description.
6. Select the G7® check box to use G7® simulation, if required.
7. At [Production profile], select the output profile of the target printer from the list of input profiles.

8. Click [OK].
9. Composite output profiles are added to the list of output profiles.

More information

Learn about input and output profiles on page 444

Create a trapping preset

When you notice unwanted white edges between color planes, use a trapping preset to correct the connection between the color planes. A trapping preset is a collection of trapping settings for a job or job subset. Use the trapping preset editor to create your own trapping presets. You can select a trapping preset for a print job.

Printed output without trapping preset (left-hand side) and with trapping preset

The trapping preset window provides an interface to create trapping presets. The following groups of settings are available:
- Trap width: value to specify the overlap for each trap
- Trap appearance: options to control the shape of the trap joins
- Images: options how to trap images
- Trap threshold: values to specify when trapping occurs

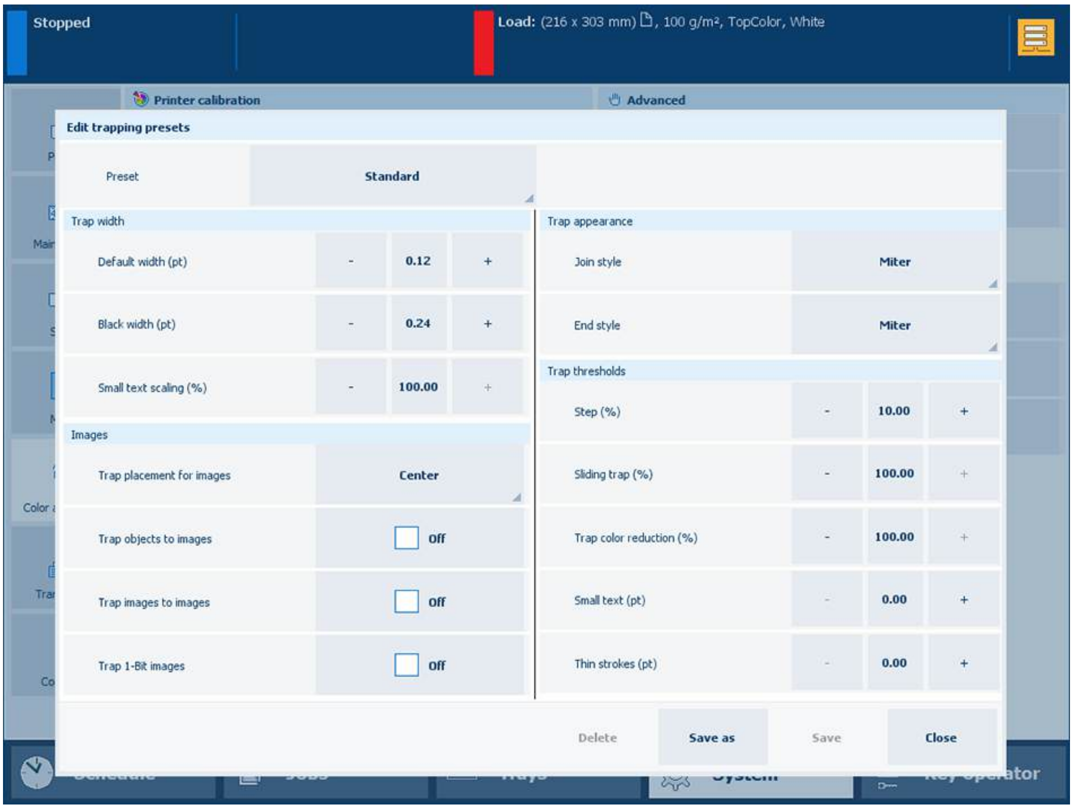
Be aware that an edge can appear at locations where color planes overlap each other.

Procedure

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Check the print quality to decide if you want to proceed.
3. Perform the automatic color mismatch correction.
4. Check the print quality to decide if you want to proceed.
5. Touch [System]→[Color adjustment]→[Adjust color].
6. Touch [Edit trapping presets].



Trapping preset editor

7. Select one of the available trapping presets.
8. Define the trapping settings in the [Trap width], [Trap appearance], [Images] and [Trap thresholds] panes.
9. Touch [Save as] to save the settings and enter a name for the preset.
10. Touch [OK].

More information

- *Learn about the print job settings* on page 299
- *Calibrate the printer* on page 368
- *Automatic color mismatch correction* on page 570

Manage spot colors

Learn about spot colors

Spot color definitions

Spot colors realize a consistent color and are often used in logos and house style colors. The printer provides a spot color editor and a set of pre-defined spot color libraries, such as PANTONE libraries and HKS libraries. The library of custom spot colors contains spot colors that users have created in the Settings Editor or on the control panel.

Custom spot color library

If a source document contains a spot color definition, the printer needs to know how to print that particular color. A spot color is the combination of a spot color name, the color value, and a tint value. The device-independent spot color definition uses a CIELAB value (Lab value). When you create a spot color for a specific media family you use a CMYK value.

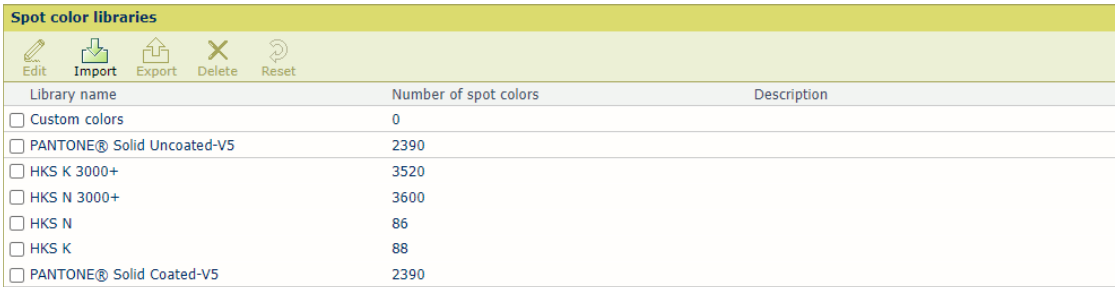
You can add more CMYK color values to a spot color definition, for other media families that print the spot color.

When you want to add a spot color that has the exact color value of a sample, for example in an offset print, you measure the spot color with the i1Pro3 spectrophotometer. The measured Lab values define the new spot color.

When you add or edit a spot color definition, you can print a patch chart to check how the set CMYK values and small variations on these values appear on media. The printed colors and the CMYK values of the patches give the best visual match to fine-tune the spot color for the given media family.

Spot color library

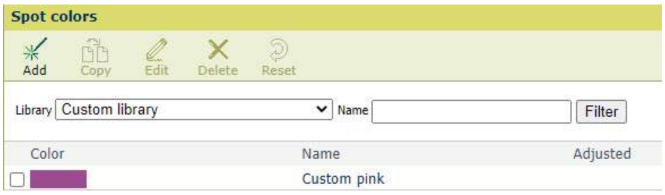
The printer provides pre-defined spot color libraries, such as PANTONE libraries and HKS libraries.



Spot color libraries in the Settings Editor

The Settings Editor can store three types of spot color libraries:
- **Pre-defined spot color libraries,** with pre-defined spot color definitions.
- **Custom spot color libraries,** with spot color definitions that have been created on the control panel or in the Settings Editor.
- **Imported spot color libraries (Named color profiles)**, with spot color definitions stored in an ICC profile.

In the Settings Editor you can import named spot color profiles and export spot color libraries that were imported as named spot color libraries.



Custom spot color library

A **named color profile** is an ICC profile that contains a list of spot colors names with color values (Lab or XYZ). When you import a named color profile, the named color profile is added as an imported spot color library. A named color profile has a standardized file format defined by the ICC (International Color Consortium).

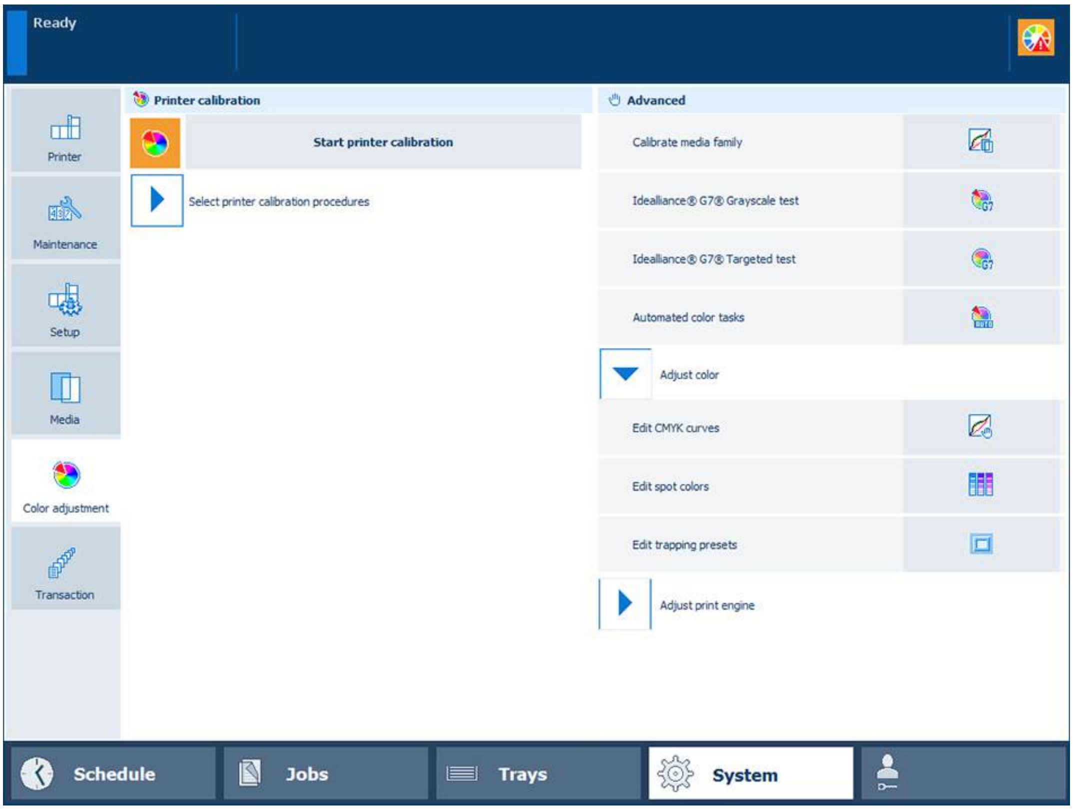
More information

- *Create a spot color from the control panel* on page 460
- *Create and edit a spot color in Settings Editor* on page 468
- *Read the spot color patch chart* on page 472
- *Define a spot color library* on page 474

Create a spot color from the control panel

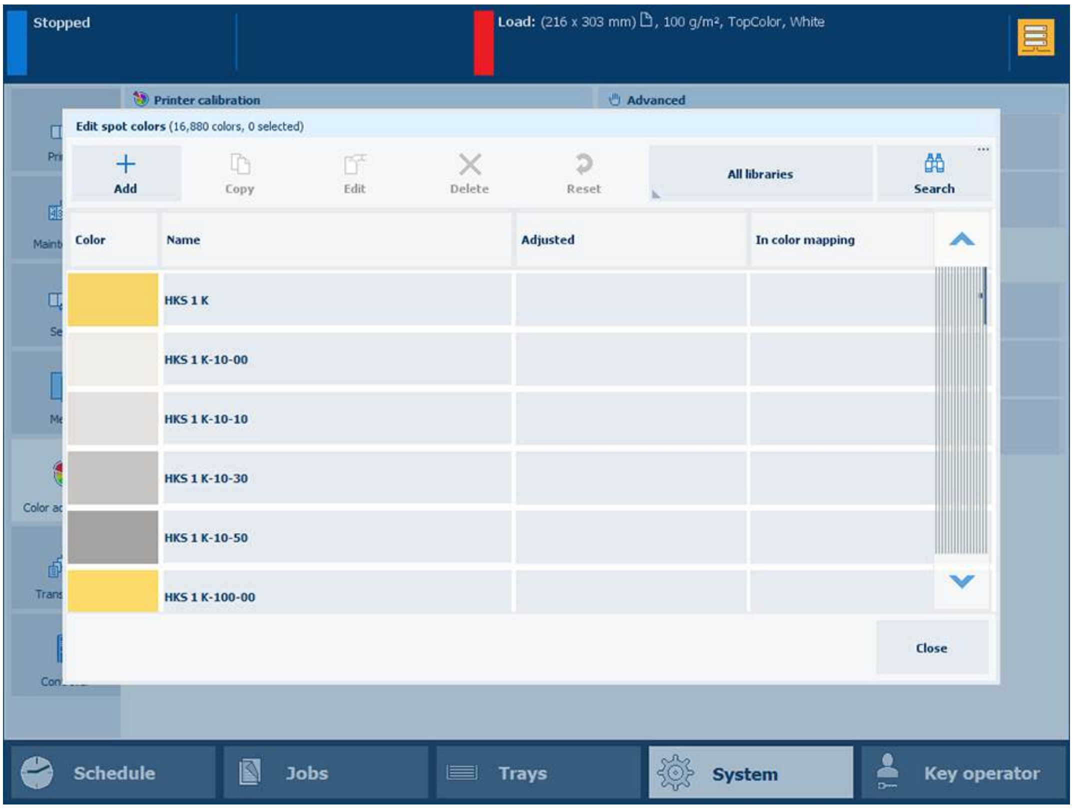
Go to the spot color editor

1. Touch [System]→[Adjust color]→[Edit spot colors].



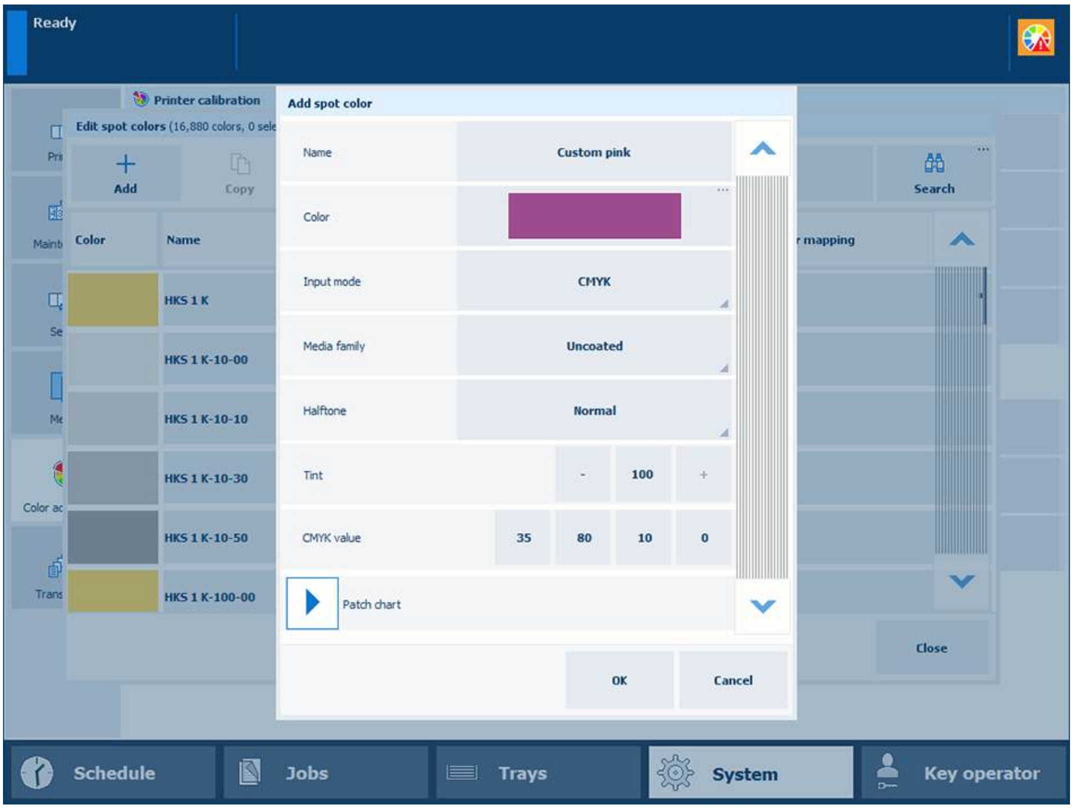
Location of spot color editor

2. To find a spot color, touch the [Search] button and enter a search string.



Create a new [CMYK] spot color

1. Touch [Add].
2. Enter a name for the spot color



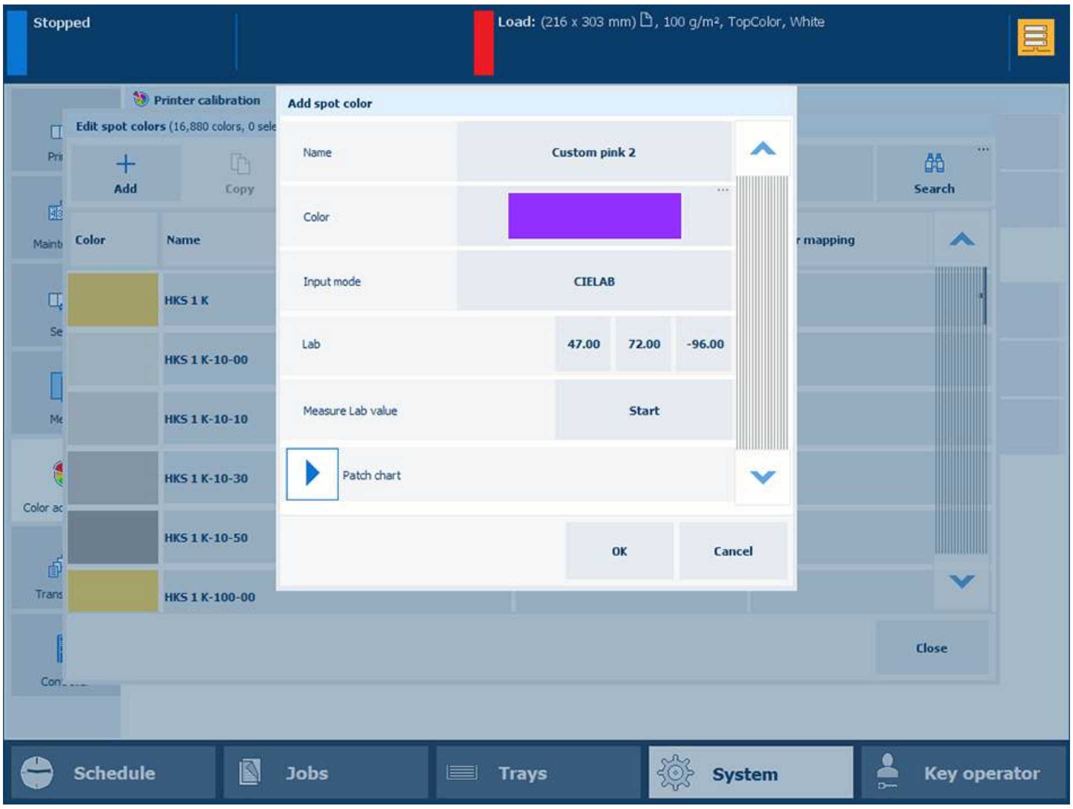
Add a spot color

3. Use the [Input mode] option to select [CMYK].
4. Use the [Media family] drop-down list to select the media family.
5. Use the [Tint] option to enter a tint value (%).
6. Enter the C, M, Y, and K values in the [CMYK value] fields.
7. Touch [OK].
   The spot color definition is stored in the custom spot color library.

Create a new [CIELAB] spot color

With the spot color measurement procedure, you measure three locations of a sample of the spot color. The printer calculates and returns the average Lab value and reports a ΔE to indicate the differences. The reported ΔE value is the largest difference between each individual measured value and the average of all measured values. A ΔE larger than 3 means a large difference. In that case you are advised to measure the spot color again.

1. Touch [Add].
2. Use the [Input mode] option to select [CIELAB].

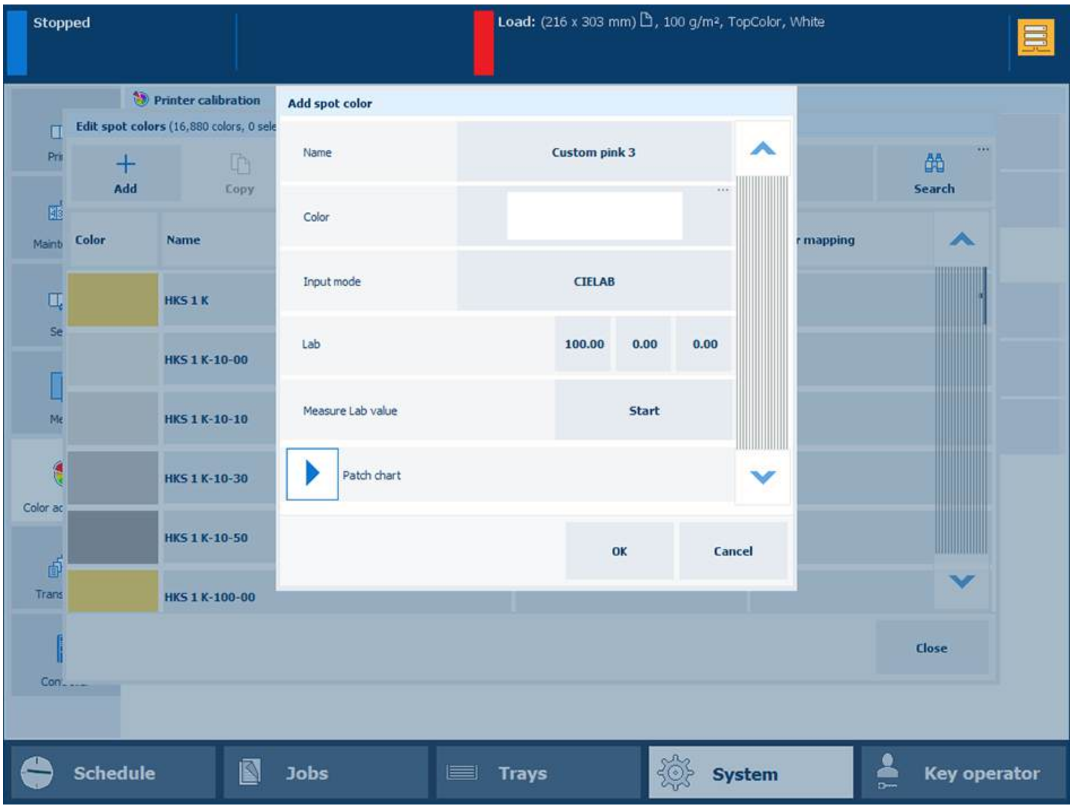


3. Enter the [L*], [a*], and [b*] values in the [Lab] option.
4. Touch [OK].
   The spot color is stored in the custom spot color library.

Measure [CIELAB] spot color values

With the spot color measurement procedure, you measure three locations of a sample of the spot color. The printer calculates and returns the average Lab value and reports a ΔE to indicate the differences. The reported ΔE value is the largest difference between each individual measured value and the average of all measured values. A ΔE larger than 3 means a large difference. In that case you are advised to measure the spot color again.

1. Touch [Add].
2. Use the [Input mode] option to select [CIELAB].



Measure Lab value

3. Touch [Start].
4. Follow the wizard instructions how to measure the color.

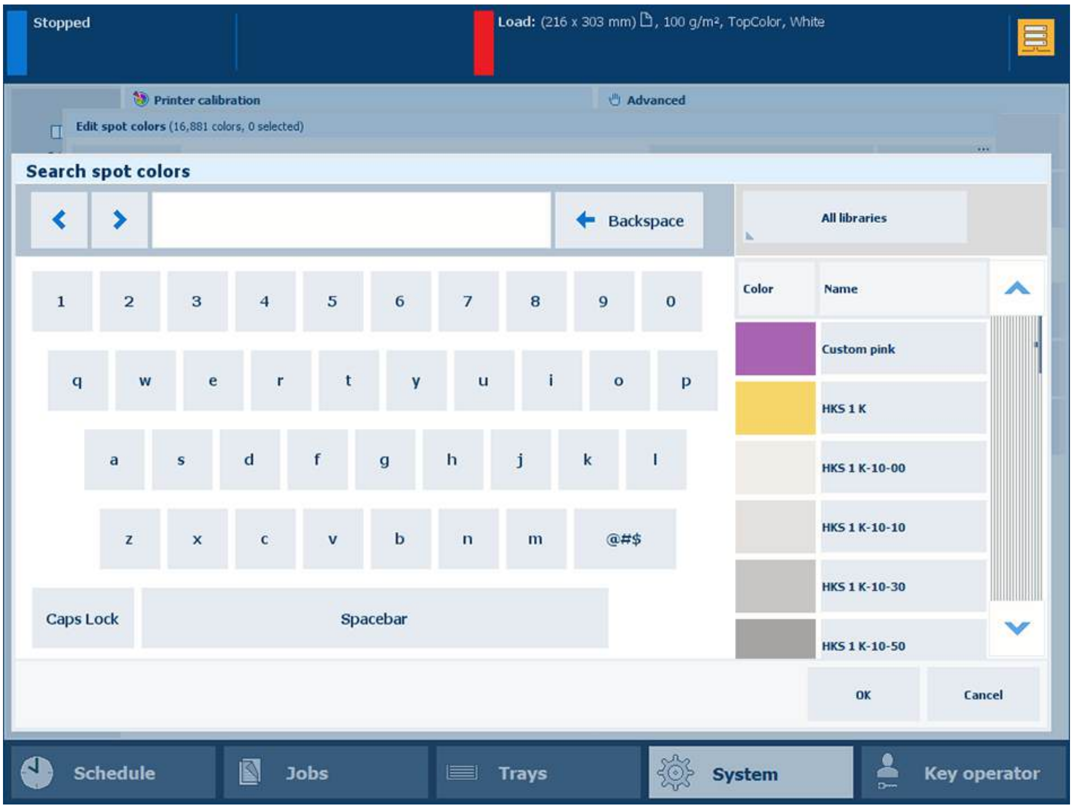
IMPORTANT
If ΔE is larger than 3, you are advised to measure the spot color again.

5. Touch [OK].
The spot color is stored in the custom spot color library.

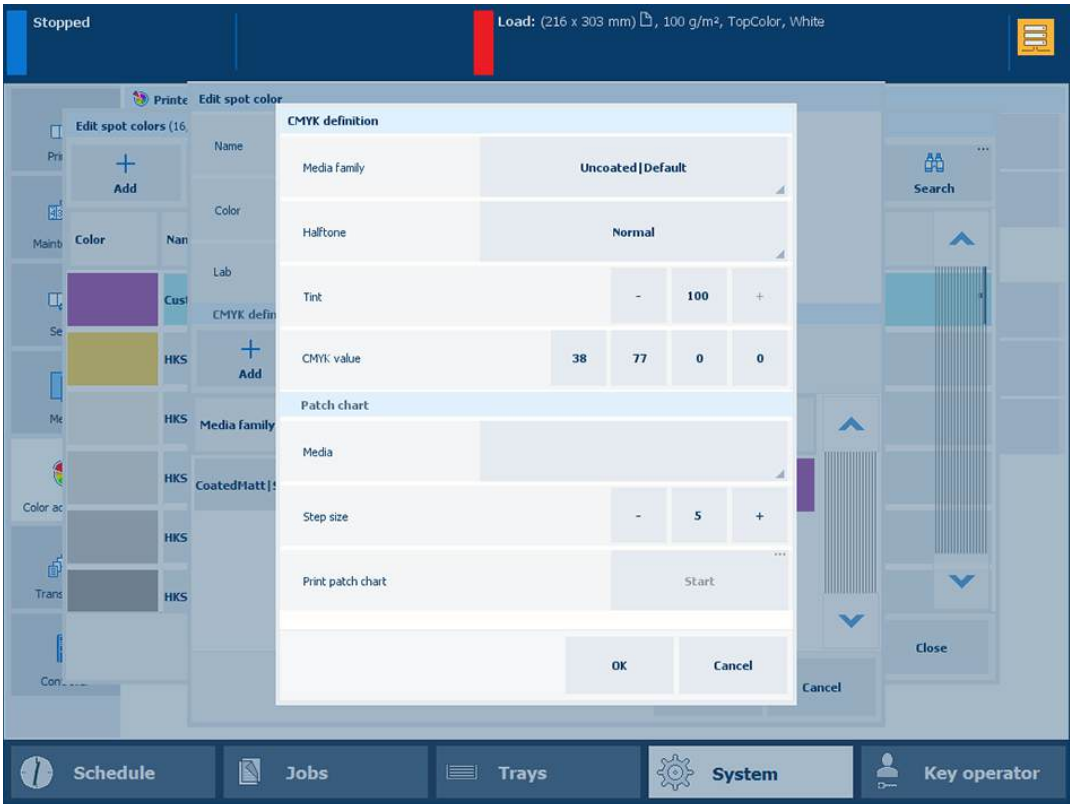
Add [CMYK] values to a spot color

You can add [CMYK] values to a spot color to apply to a specific media family.

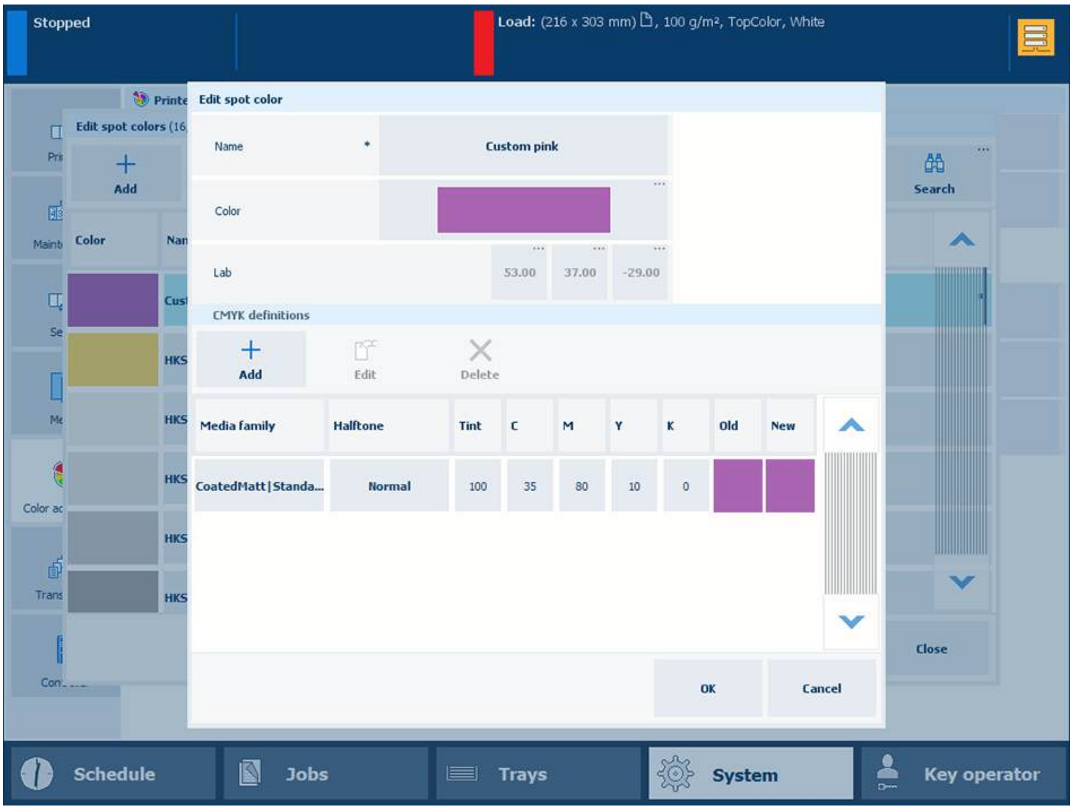
1. Select the spot color library and search the spot color.



2. Select the spot color and touch [Edit].
3. Touch [Add].



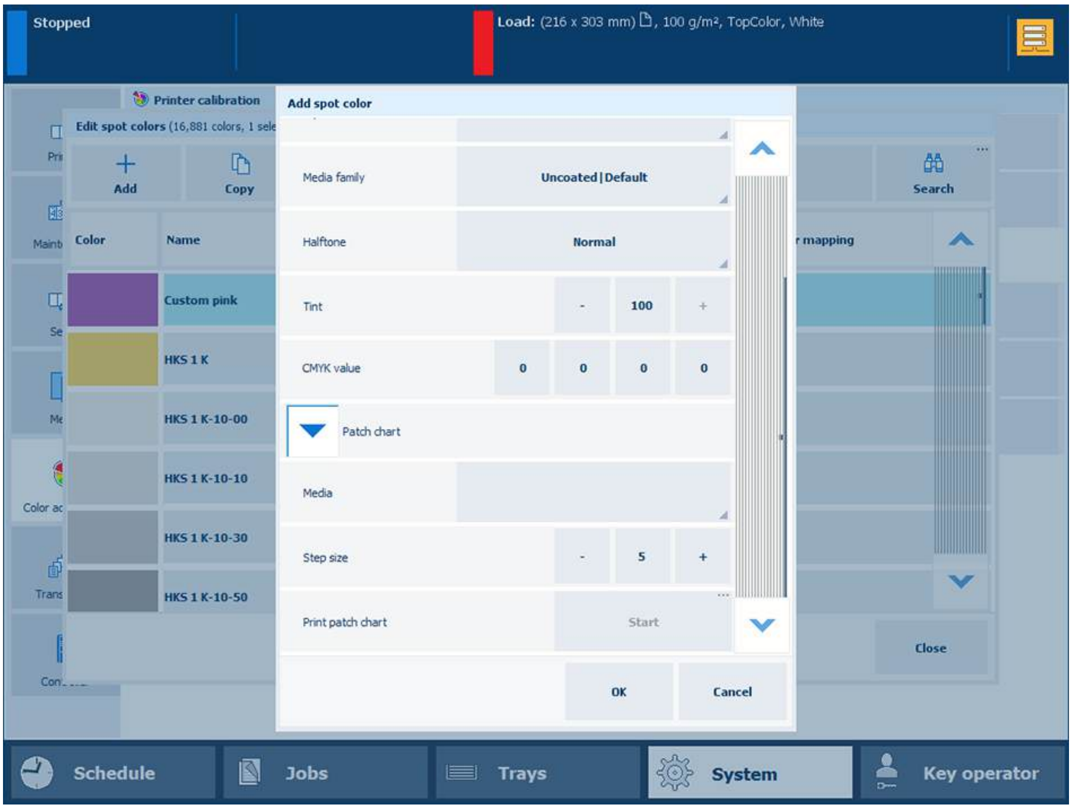
4. Use the [Media family] drop-down list to select the media family combination.
5. Use the [Tint] option to enter a tint value (%).
6. Enter the C, M, Y, and K values in the [CMYK value] fields.



7. Select a new media family.
   The new combination is listed.
   The [Old] color patch is the color rendering of the Lab definition. The [New] color patch shows the rendering of the CMYK values for the media family.
8. Touch [OK].

Print a patch chart

To evaluate the CMYK values print a chart on media for the media family.



Print patch chart

1. Expand the [Patch chart] options.
2. Use the [Media] drop-down list to select the media for the media family.
3. Use the [Step size] option to define the degree of patch variations.
4. In the [Print patch chart] field, touch [Start].

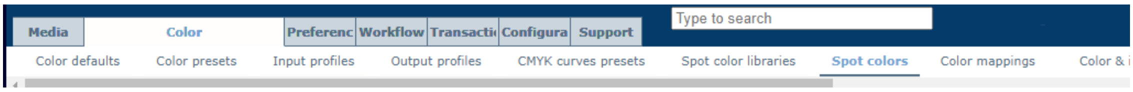
More information

- *Learn about spot colors* on page 458
- *Create and edit a spot color in Settings Editor* on page 468
- *Read the spot color patch chart* on page 472
- *Define a spot color library* on page 474

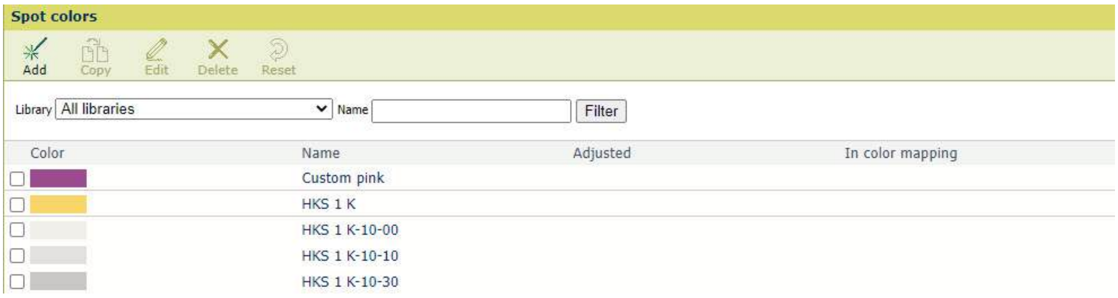
Create and edit a spot color in Settings Editor

You can create and edit spot colors in Settings Editor.

1. Open the Settings Editor and go to: [Color]→[Spot colors].



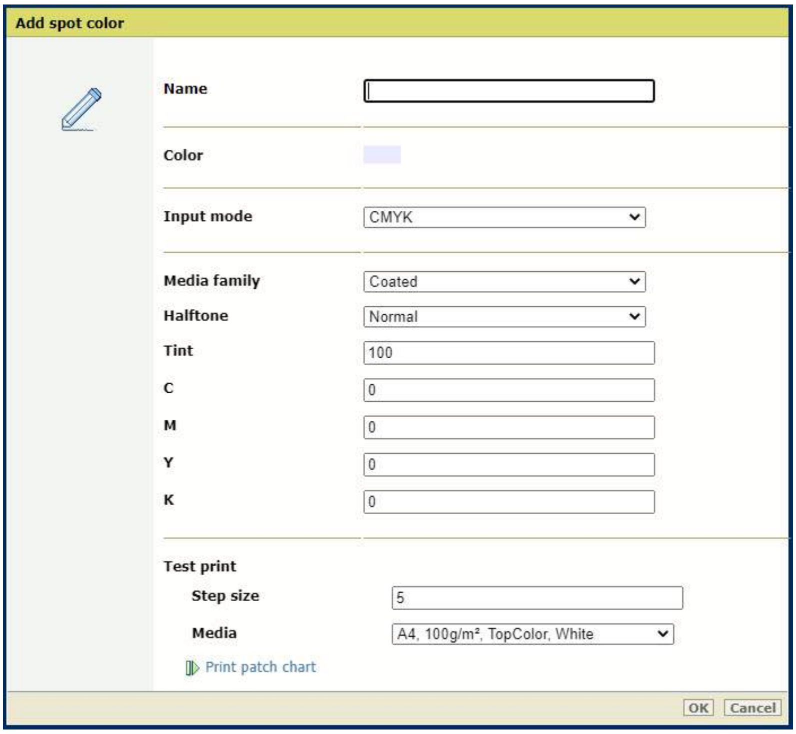
[Spot colors] tab



Spot Colors

Create a new [CMYK] spot color

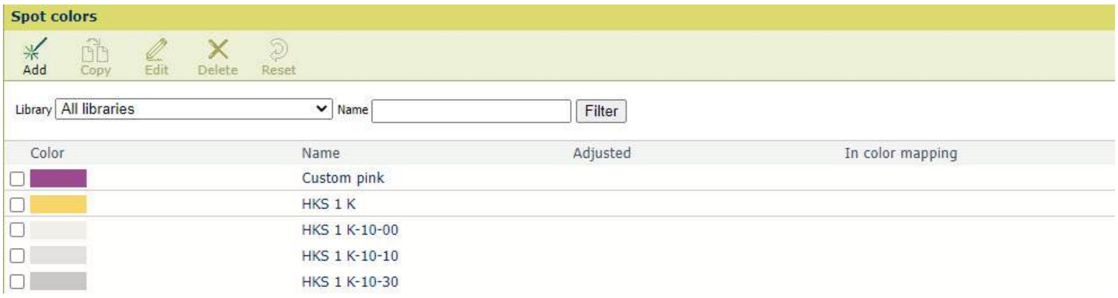
1. Click [Add].
2. Enter a name for the spot color.
3. Use the [Input mode] option to select [CMYK].



Add [CMYK] spot color

4. Use the [Media family] drop-down list to select the media family .
5. Use the [Halftone] drop-down list to select the halftone.

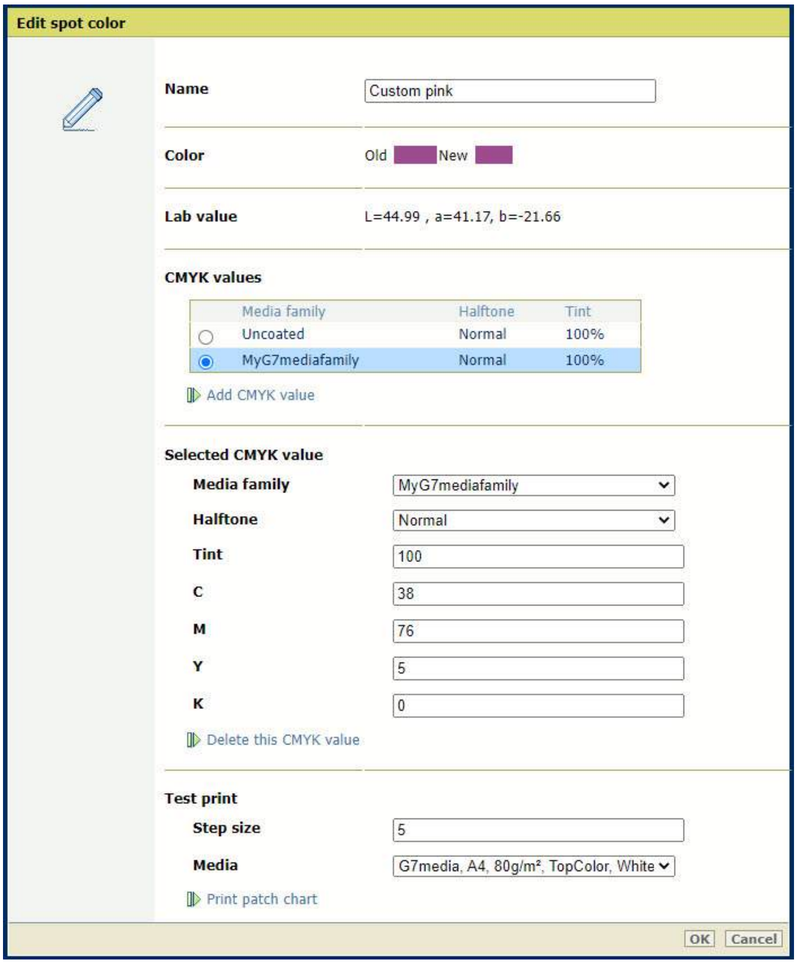
6. Enter a value (%) in the [Tint] field.
7. Enter values in the [C], [M], [Y], and [K] fields.
8. Click [OK].
   The spot color definition is stored in the custom spot color library.



Add [CMYK] values to a spot color

You can add [CMYK] values to a spot color to apply to a specific media family.

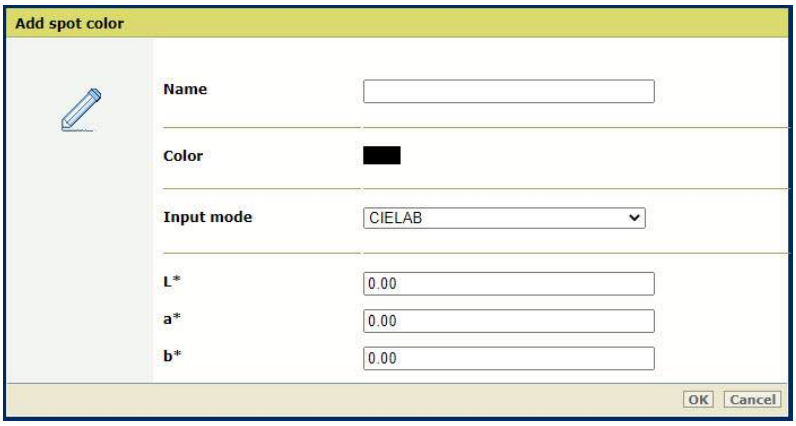
1. Click the [CMYK] spot color.
2. Click [Edit].
3. Click [Add CMYK value].
4. Select a new media family.
   The new combination is listed.



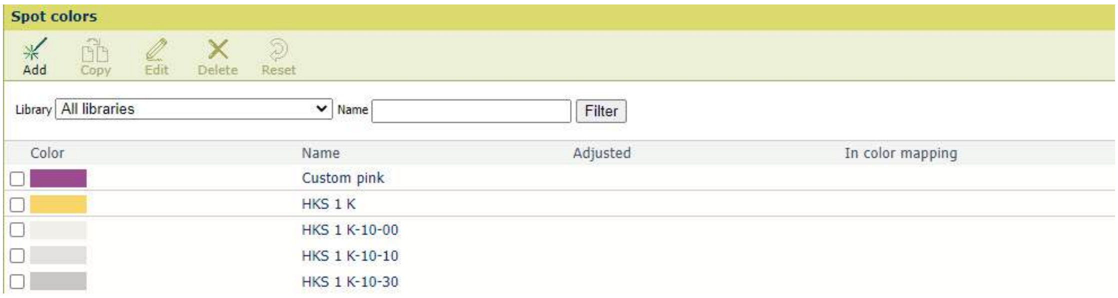
5. Enter a value (%) in the [Tint] field.
   The [Old] color patch is the color rendering of the Lab definition. The [New] color patch shows the rendering of the CMYK values for the media family.
6. Click [OK].
   The media family is added.

Create a new [CIELAB] spot color

1. Click [Add].
2. Enter a name for the spot color
3. Use the [Input mode] option to select [CIELAB].



4. Enter values in the [L*], [a*], and [b*] fields.
5. Click [OK].
   The spot color is stored in the custom spot color library.



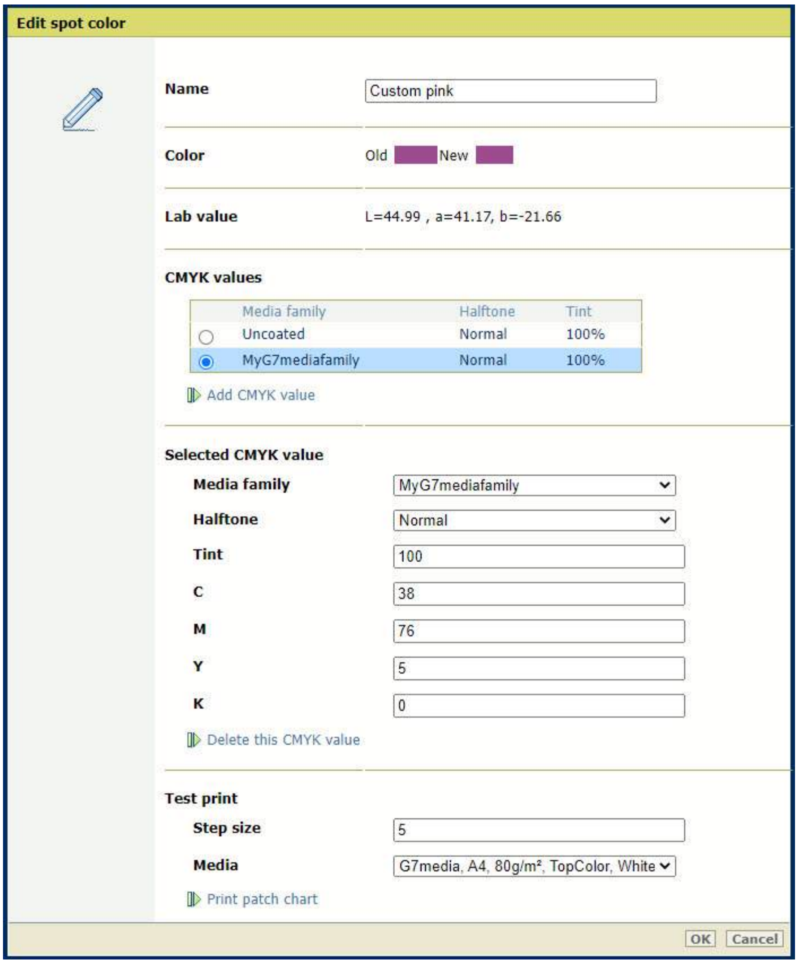
Delete a spot color

You can only delete custom spot colors.

1. Select a spot color.
2. Click [Delete]

Print a test chart

1. Click a spot color.
2. Click [Edit].
3. Use the [Media family] drop-down list to select the media family .



Print patch chart

4. Use the [Step size] option to define the degree of patch variations.
5. Use the [Media] drop-down list to select the media for the media family
6. Click [Print patch chart] to print the chart.

More information

- *Learn about spot colors* on page 458
- *Create a spot color from the control panel* on page 460
- *Read the spot color patch chart* on page 472
- *Define a spot color library* on page 474

Read the spot color patch chart

When you add or edit a spot color definition, you can print a patch chart to check how the set CMYK values and small variations on these values appear on media. The printed colors and the CMYK values of the patches give the best visual match to fine-tune the spot color for the given media family.

A patch chart shows 39 patches that each have a variation of the set CMYK value. You print a patch chart on the media that belong to the media family of the CMYK value of the spot color.

You print the patch chart on media that belong to the media family of the CMYK values. A patch chart shows 39 patches with variations of CMYK values. The chosen step size determines the degree of patch variations in percentages (1% - 20%).

<table>
<tr>
<td>

Patch variations

- The patches are arranged in three rows. All patches of a row all have the same K-value. The middle row patches represent the K-value of the spot color.
- The K-value is decreased for the top row and increased for the bottom row; the step size determines the new K-values.
- The patches within a row show variations in CMY values; the step size determines the difference between two adjacent patches.

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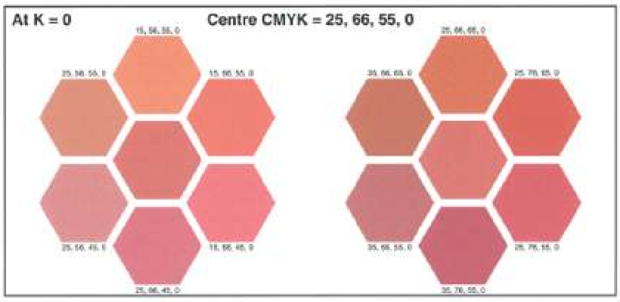
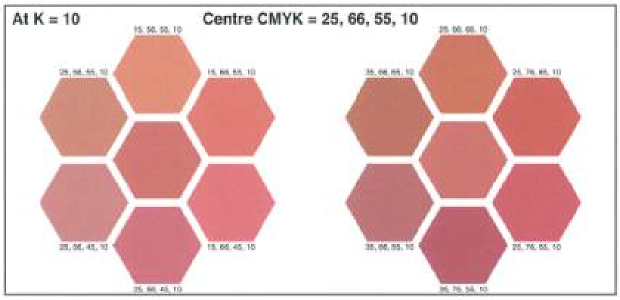
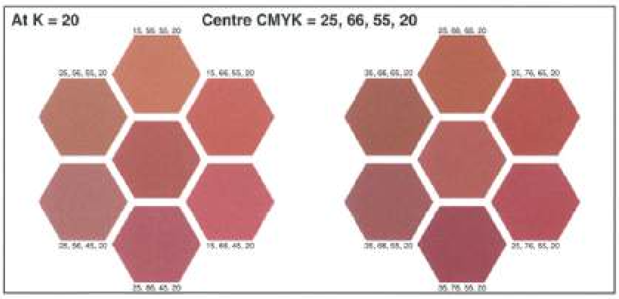
<td>







</td>
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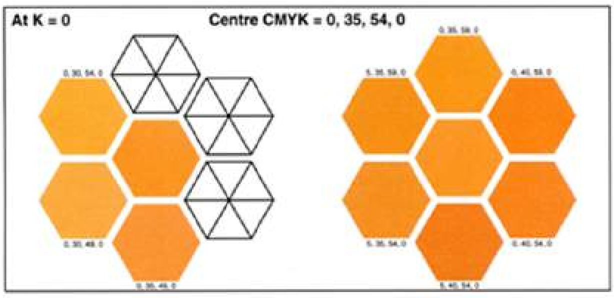
<td>

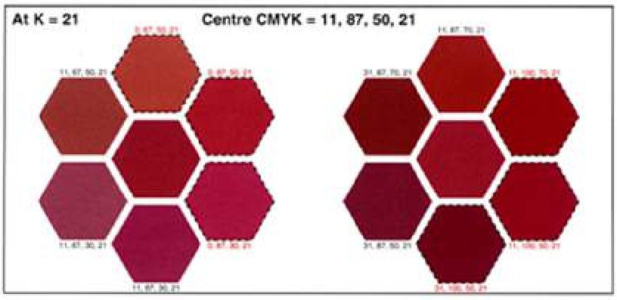
Patches with a black cross

When a color value is equal to a color value of an other patch, one of the patches appear as a white patch with a black cross.

</td>
<td>



</td>
</tr>
</table>

Patches with red text When a CMYK value comes below 0% or above 100%, the value is clipped. This new CMYK value is printed in red text and the patch has a dotted line.	

More information

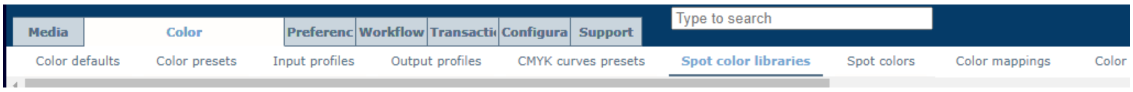
- *Learn about spot colors* on page 458
- *Create a spot color from the control panel* on page 460
- *Create and edit a spot color in Settings Editor* on page 468
- *Define a spot color library* on page 474

Define a spot color library

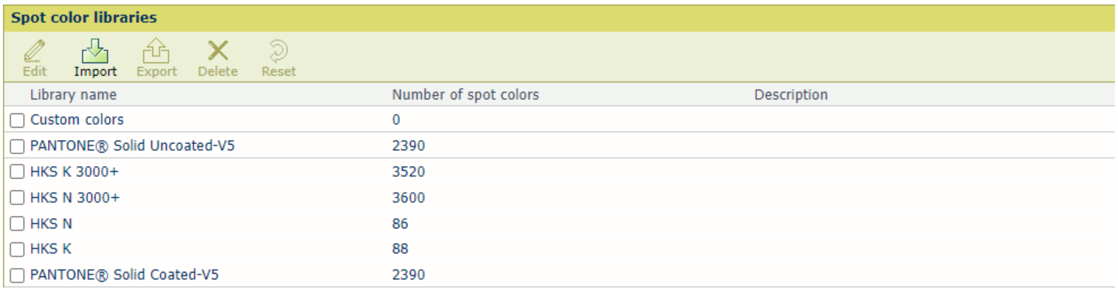
The printer provides predefined spot color libraries such as PANTONE libraries and HKS libraries.

The Settings Editor can store three types of spot color libraries:
- pre-defined spot color libraries
- custom spot color libraries
- imported spot color libraries (named color profiles).

1. Open the Settings Editor and go to: [Color]→[Spot color libraries].



[Spot color libraries] tab



Spot color libraries in the Settings Editor

Edit a spot color library

You can only edit previously imported spot color libraries (Named color profiles).
1. Select the spot color library.
2. Click [Edit].
3. Change the name and description.
4. Click [OK].

Import a spot color library

You can only import a Named color profile.
1. Click [Import].
2. Select the ICC file that contains the named color profile.
3. Enter a name and description.
4. Click [OK].

Export a spot color library

You can only export a previously imported spot color library (Named color profiles).
1. Select the spot color library.
2. Click [Export].

Delete a spot color library

You can only delete previously imported spot color libraries (Named color profiles).
1. Select the spot color library.
2. Click [Delete].

Reset a spot color library

You can only recover the original values of the spot colors of the predefined and imported spot color libraries.

IMPORTANT
When you select a custom spot color library, the [Reset] option deletes all custom spot color definitions.

1. Select the spot color library.
2. Click [Reset].
3. Click [OK].

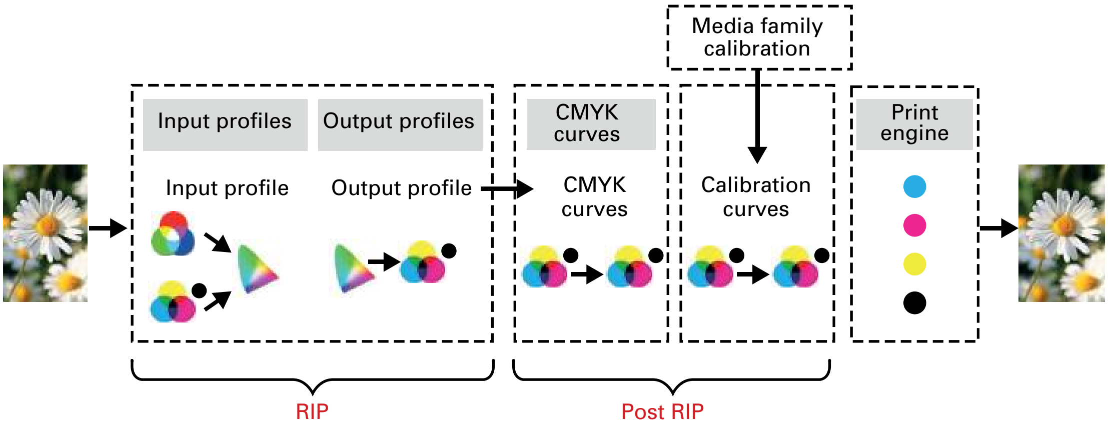
More information

- *Learn about spot colors* on page 458
- *Create a spot color from the control panel* on page 460
- *Create and edit a spot color in Settings Editor* on page 468
- *Read the spot color patch chart* on page 472

Use CMYK curves

Learn about CMYK curves

The color reproduction is defined by the document and the color management performed by the RIP. After the color management by the RIP, the calibration curves are applied.



However, when the color reproduction is not according to your expectation, you can apply CMYK curves that are stored on PRISMAsync Print Server or that you create in the CMYK curves editor.

CMYK curves presets

With the CMYK curves editor, you can adjust CMYK curves and save these as a **CMYK curves preset** for later reuse. These CMYK curves presets are available in the Settings Editor. A CMYK curves preset can be selected on **job** level or on **media family** level.

factory defined CMYK curves presets

The factory defined CMYK curves presets can be:
- Changed
- Deleted
- Attached to a media family

Exchange CMYK curves between printers

You can import and export CMYK curves presets in the Settings Editor to exchange CMYK curves presets between printers.

Import format
- Adobe Photoshop (ACV file): binary file that contains CMYK curves defined by a finite set of points. The adjustment mechanism is based on the spline interpolation method.
- Chromix Curve4 (CGATS file): tab delimited text file that contains CMYK curves defined by a finite set of points. The adjustment mechanism is based on the linear interpolation method.
- PRISMAsync Print Server (XML file)

Export format
- Adobe Photoshop (ACV file): binary file that contains CMYK curves defined by a finite set of points. The adjustment mechanism is based on the spline interpolation method.

- PRISMAsync Print Server (XML file)

More information

- *Adjust CMYK curves of a media family* on page 478
- *Adjust CMYK curves of a job* on page 481
- *Import or export CMYK curves presets* on page 485

Adjust CMYK curves of a media family

PRISMAsync Print Server offers a CMYK editor with a pixel-precise preview to adjust the color density for specific media. You can emphasize colors or other elements for images printed on media that belong to a media family. The new CMYK curves are added to the media family calibration values.

There are two modes to edit the CMYK values of a media family:
1. **Basic mode**, for simple CMYK editing with sliders for quick and easy color tweaking.
2. **Advanced mode**, for precise CMYK editing.

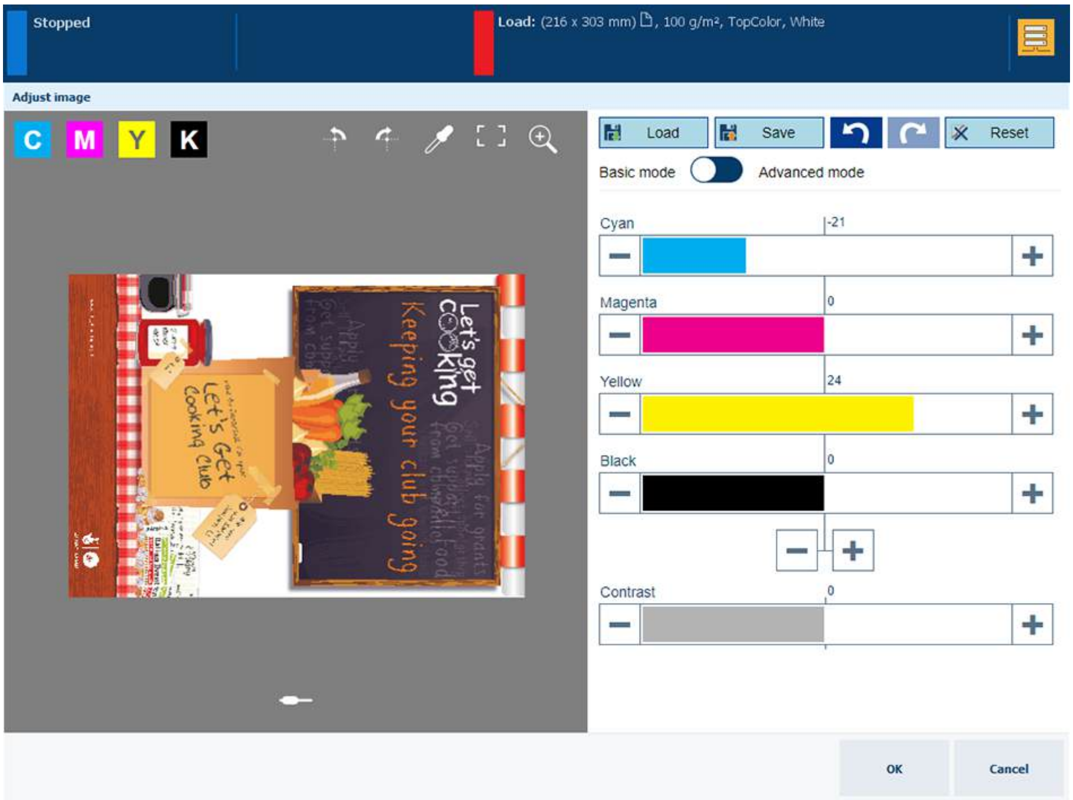
IMPORTANT
Be aware that the CMYK adjustments affect all media of the media family.

Open the CMYK editor

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Check the print quality to decide if you want to proceed.
3. Touch [System]→[Color adjustment]→[Adjust color].
4. Touch [Edit CMYK curves].
5. Select the media family.
6. Select the halftone.
7. Touch [Edit].
   The [Basic mode] window opens.



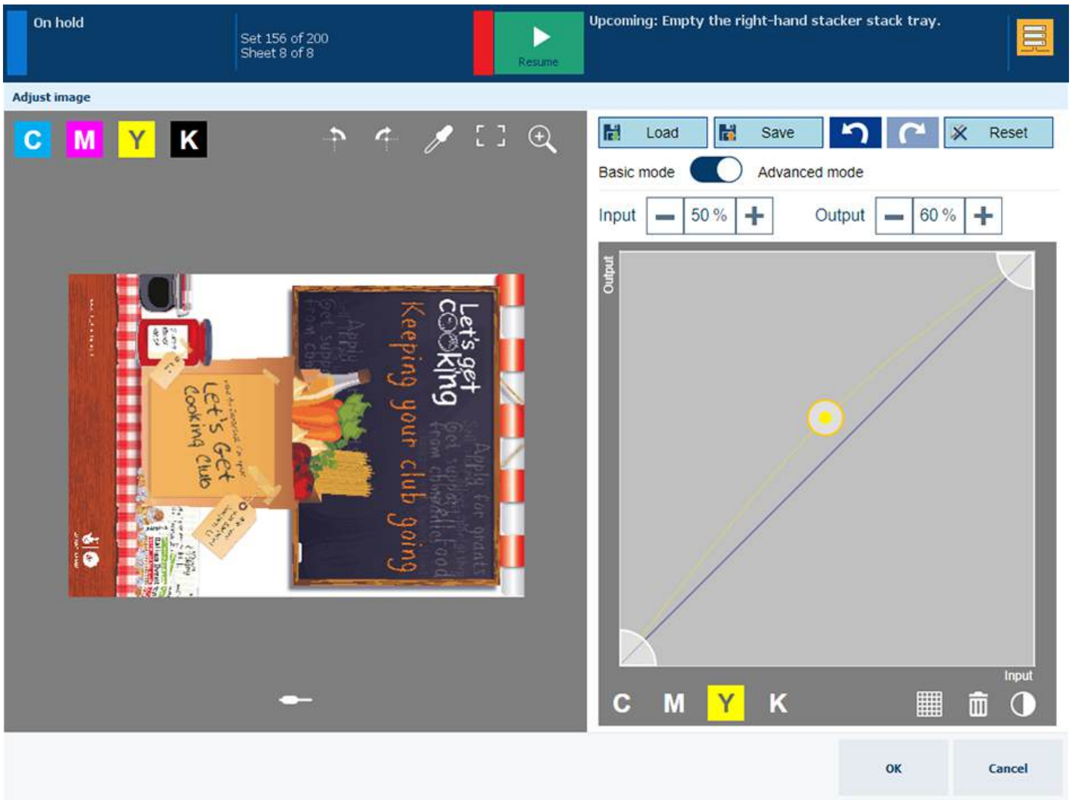
CMYK editor in basic mode

Adjust the CMYK curves in basic mode

1. Adjust the density values for the four colors.
2. Use the zoom function to check the result of the adjustment in a specific area.
3. Use your finger to pan the image.
4. Touch [OK].
5. Repeat this procedure for other halftones, if required.

Adjust the CMYK curves in advanced mode

Use the table below to adjust the CMYK curves in advanced mode.



CMYK editor in advanced mode

Task	Instruction
Go to the advanced mode	1. Touch [Advanced mode].
Preserve colors	1. Touch the dropper icon. 2. Position the target icon with your finger on the color you want to preserve. 3. Touch [Add points].
Apply a CMYK-curve preset	1. Touch the load icon. 2. Select a CMYK-curve preset. 3. Touch [Done]. 4. Check the result in the preview.
Define the color clipping range	1. Touch and drag the lower left-hand wedge . 2. Touch and drag the upper right-hand wedge .

Task	Instruction
Adjust the curves manually	1. Touch the raster icon. 2. Touch the curve to add a control point. To delete a control point, first select a control point. Then, touch the trash icon. 3. Touch the + and - signs to move the control point or drag the control point with your finger. 4. Check the result in the preview. 5. When ready, select a new color.
Adjust the contrast of all colors	1. Touch the contrast icon. 2. Use the slider to adjust the contrast. 3. Check the result in the preview.

You can perform the above tasks also for other halftones.

More information

- *Calibrate the printer* on page 368
- *Learn about CMYK curves* on page 476
- *Adjust CMYK curves of a job* on page 481
- *Import or export CMYK curves presets* on page 485

Adjust CMYK curves of a job

PRISMAsync Print Server offers a CMYK editor with a pixel-precise preview to adjust the color density for specific media. You can emphasize colors or other elements for images printed on media that belong to a job.

There are two modes to edit the CMYK values of a job:

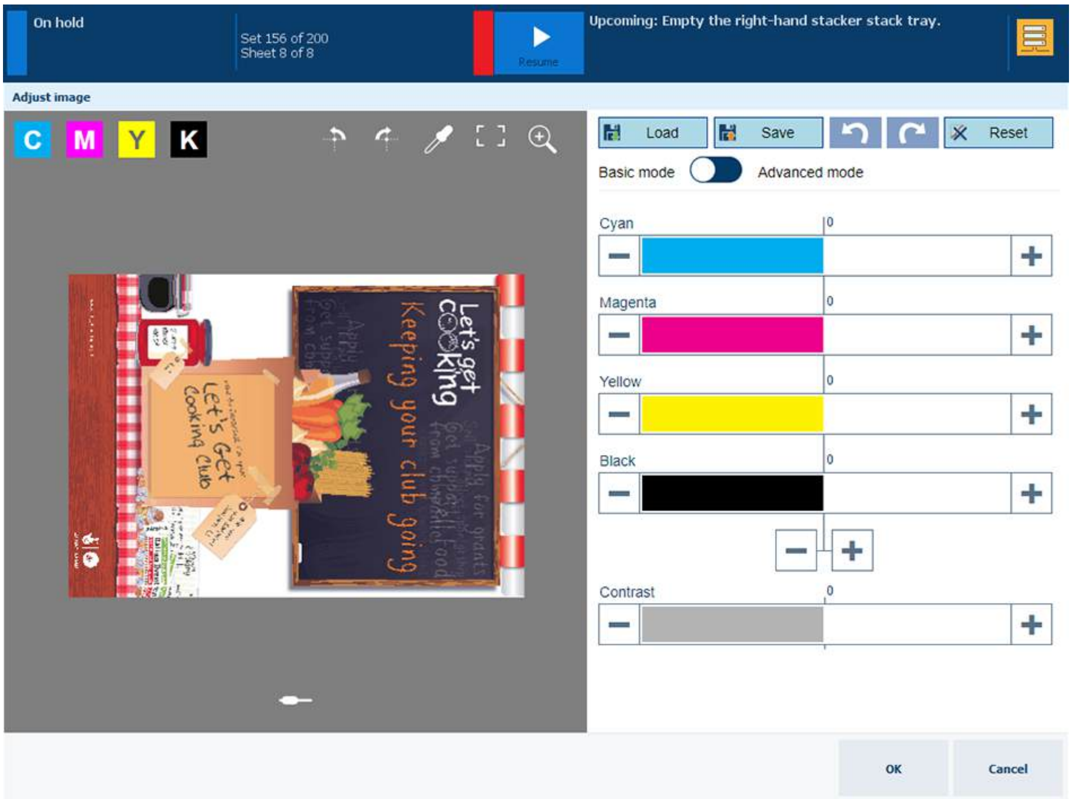
1. **Basic mode**, for simple CMYK editing with sliders for quick and easy color tweaking.
2. **Advanced mode**, for precise CMYK editing.

IMPORTANT

- Be aware that achieving a consistent reproduction of colors is difficult when you adjust CMYK curves on job level.
- First, check whether the printer calibration delivers the required color quality before you perform this procedure.

Open the CMYK curve editor with pixel precise preview

1. Touch [Jobs].
2. Go to the location of the job.
3. Select the job you want to change.
4. Touch [Edit], or double-tap the job to open the [Edit] window.
5. Touch [Adjust image].
   The CMYK curve editor with pixel precise preview opens the [Basic mode] window.



CMYK curve editor with pixel precise preview in basic mode

Prepare the pixel precise preview to monitor adjustments

1. Touch the slider icon to browse the document.

- Touch to rotate the pages.
- Touch and to enter and exit the full screen view.
- Touch the thumbnails of the page you want to examine.
- Touch and hold one thumbnail to undo the selection of multiple pages.

2. Touch the zoom function to examine the result of the adjustment in a specific area.
   - Use the zoom bar to zoom in or out.
   - Use your finger to pan the image.
   - Touch to hide the zoom bar again.
3. Touch the color picker tool to examine the result of the adjustment for a specific color.
   - Move the color picker icon to a pixel.
   - Note that the input density (%) and output density (%) of the CMYK colors are the same.

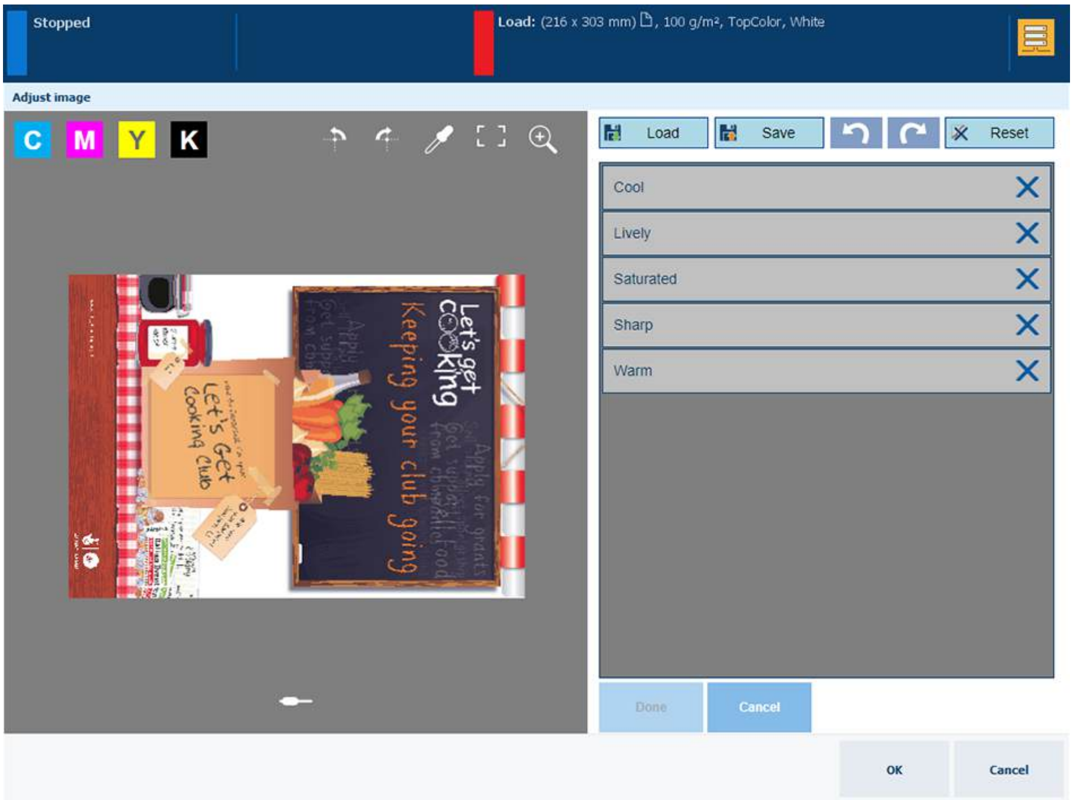
Adjust the CMYK curves in basic mode

The **Basic mode** is for simple CMYK editing with sliders for quick and easy color tweaking.

1. Adjust the colors with the color and contrast sliders.
2. Examine the results in the magnified area and color picker tool.
3. Touch the save icon to store the adjustment in a preset for later use, if required.
4. Touch [OK] to apply the adjustment to this job.

Apply an adjustment with a preset

1. Touch the load icon to open the list of presets.



Adjust image with a preset

2. Select one of the standard presets.
3. Touch [Done].

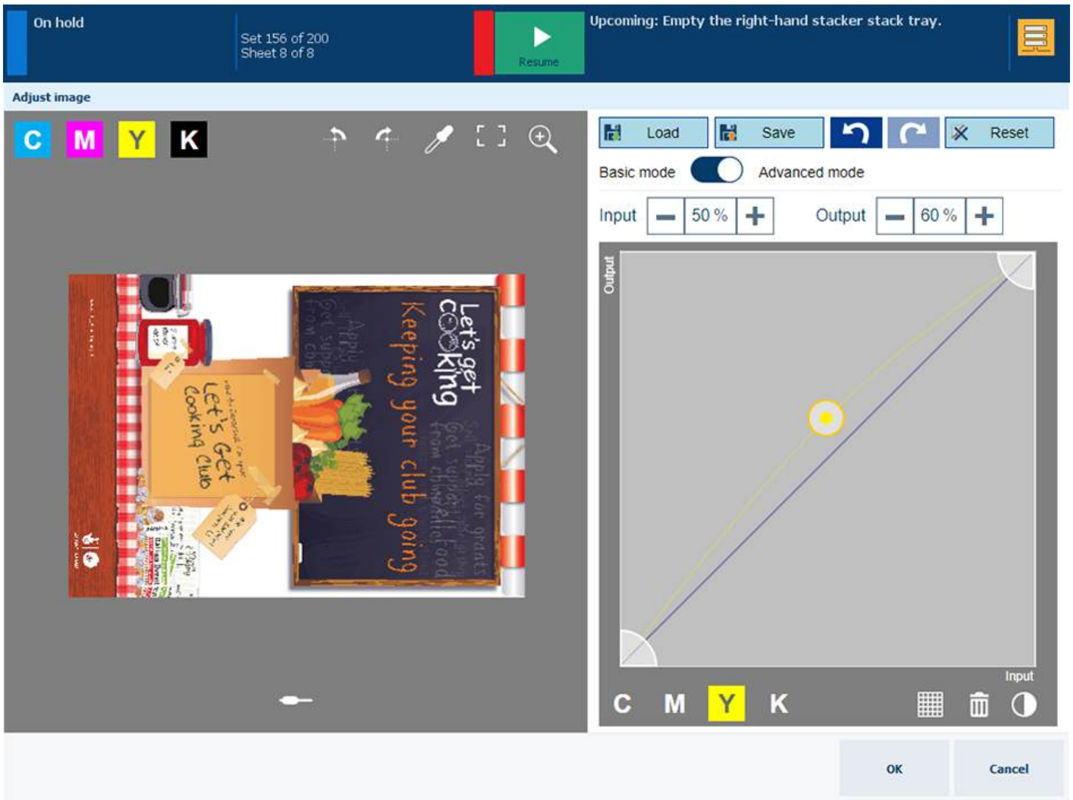
NOTE
The CMYK curve editor switches to [Advanced mode]

4. Check the result in the pixel precise preview.
5. Touch [OK] to apply the adjustment to this job.

Adjust the CMYK curves in advanced mode

The [Advanced mode] is CMYK editing for precise and reproducible adjustments

1. Touch [Advanced mode].



CMYK curve editor with pixel precise preview in advanced mode

2. Select one of the CMYK curves.
3. Add control points to adjust the tone curve.
   - Touch the curve to add a control point.
     The [Input] field represents the input density (%) of the selected control point.
     The [Output] field represents the output density (%) of the selected control point.
   - Touch the [Add points] option of the color picker tool to add control points of a specific color.
   - Touch the + and - signs to move the control point or drag the control point with your finger.
   - Touch the raster icon to locate control points more easily.
   - To delete a control point, first select a control point. Then, touch the trash icon.
4. Repeat step 3 and 4 for other curves.
5. Touch the contrast icon to adjust the contrast.
6. Clip the highlight and shadow tones.
   - Touch and drag the lower left-hand wedge.
   - Touch and drag the upper right-hand wedge.
7. Check the result in the pixel precise preview.
8. Touch the save icon to store the adjustment in a preset for later use, if required.

9. Touch [OK] to apply the adjustment to this job.

More information

- *Learn about calibration* on page 346
- *Learn about CMYK curves* on page 476
- *Adjust CMYK curves of a media family* on page 478
- *Import or export CMYK curves presets* on page 485

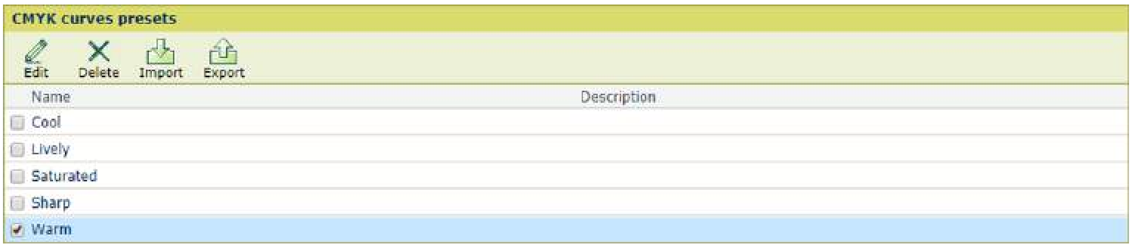
Import or export CMYK curves presets

You can edit, delete, import and export CMYK curves presets.

Go to the CMYK curves presets

1. Open the Settings Editor and go to: [Color]→[CMYK curves presets].

[CMYK curves presets] tab



[CMYK curves presets]

Edit a CMYK curves preset

1. Select a CMYK curve preset from the list.
2. Click [Edit].
3. Define a unique name and the description.

[Edit a CMYK curves preset]

4. Use the [Overwrite duplicates] setting to overwrite an existing CMYK curves preset.

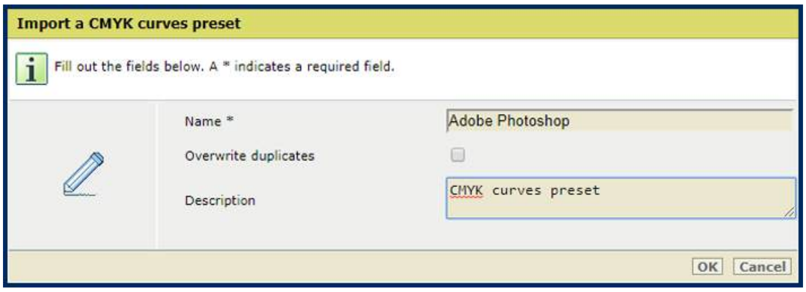
Delete a CMYK preset

1. Select a CMYK curve preset from the list.
2. Click[Delete] .

Import a CMYK curves preset

After import the CMYK curve preset is available for a media family and print job.

1. Click [Import].
2. Browse to the required file.
3. Click [OK].
4. Define a unique name and the description.

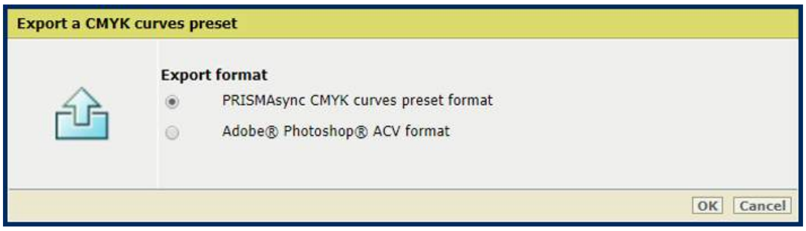


[Import a CMYK curves preset]

5. Use the [Overwrite duplicates] setting to overwrite an existing CMYK curves preset.

Export a CMYK curves preset

1. Select a CMYK curve preset from the list.
2. Click [Export].
3. Select the export format:
   - [PRISMAsync CMYK curves preset format] (XML)
   - [Adobe® Photoshop® ACV format]



[Export a CMYK curves preset]

More information

- *Learn about CMYK curves* on page 476
- *Adjust CMYK curves of a media family* on page 478
- *Adjust CMYK curves of a job* on page 481

Define color mappings

A color mapping is an element that **maps** (transforms) a source color value into a new target color value. Color mapping is applied during the job RIP process. Color mapping also applies to color bars and information bars.

You add one or more color mappings to a **color mapping group**. A color mapping group enables the combination of different color mappings. You can assign the color mapping group to a color preset.

Go to the color mappings

1. Open the Settings Editor and go to: [Color]→[Color mappings].

[Color mappings] tab

[Color mappings]

Add a color mapping group

1. Click [Add group].
2. Define a name and description.

[Add color mapping group]

3. Click [OK].

Edit a color mapping group

1. Select a color mapping group.
2. Click [Edit].
3. Define a name and description.

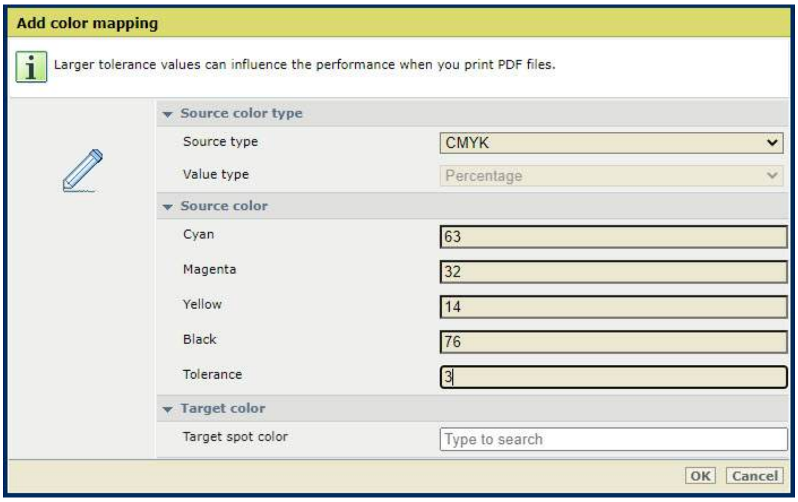
Copy a color mapping group

1. Select a color mapping group.
2. Click [Copy].

3. Define a name and description.

Add a color mapping

1. Select a color mapping group.
2. Click [Add color mapping].



[Add color mapping]

3. Define the color mapping settings, explained in the table below.

Edit a color mapping

1. Select a color mapping.
2. Click [Edit].
3. Define the color mapping settings, explained in the table below.

Copy a color mapping

1. Select a color mapping.
2. Click [Copy].
3. Define the color mapping settings, explained in the table below.

Delete a color mapping or color mapping group

1. Select one or more color mappings or color mapping groups.
2. Click [Delete].

Move a color mapping to another group

1. Select the color mapping.
2. Click [Move].
3. Select the new color mapping group.

Define the color mapping settings

Settings	Description
[Source color type]	Use the [Source type] drop-down list to select the type of source color. • RGB color value • CMYK color value • Spot color of a spot color library • Spot color of a custom spot color library CMYK colors and spot colors are specified as percentages (0-100). RGB colors can be specified as 8-bit numbers (0-255) or as percentages (0-100). Use the [Value type] to define the format of the RGB value.
[Source color]	Use the [Source color] setting to define the source color. • RGB color value. Enter values for the [Red], [Green], [Blue] fields. • CMYK color value. Enter values for the [Cyan], [Magenta], [Yellow], [Black] fields. • Spot color of a spot color library. Select the spot color • Spot color of a custom spot color library. Select the spot color. • Special spot color. Enter: *Cyan, Magenta, Yellow, Black, All, or None*. (See table below) NOTE Search in the list of spot colors, by entering a part of the spot color name.
[Tolerance]	Use the [Tolerance] setting to set the tolerance. The tolerance of a source color is expressed as a percentage point. A percentage point is the difference between two percentages. For example, the percentage point between 32% and 30% is 2. The tolerance of a spot color cannot be defined.
[Target color]	Use the [Target color] setting to define the target color. • Spot color of a spot color library. Select the spot color. • Spot color of a custom spot color library. Select the spot color. • Special spot colors. Enter: *Cyan, Magenta, Yellow, Black, All, None*. (See table below) NOTE Search in the list of installed spot colors, by entering a part of the spot color name.

Use special spot colors

Spot color	Description
Cyan, Magenta, Yellow, Black'	A mapping to one of these (exactly spelled) spot color names prints the primary color instead of the source color. No color management is applied to this target color.
All	A mapping to this (exactly spelled) spot color name prints all primary colors instead of the source color. This results in *rich black*.
None	A mapping of this (exactly spelled) spot color name makes objects invisible on the print. This can be useful to make markers invisible.

More information

- *Define a color preset* on page 440
- *Learn about spot colors* on page 458

Define a color bar

Color bars are color control strips to verify the color quality of the prints. The color bar uses the print settings of the media definition.

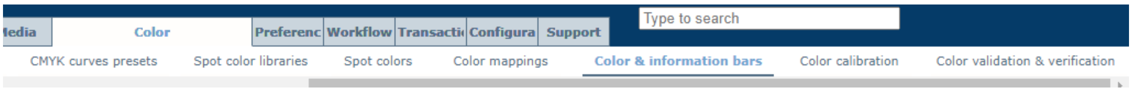
The bar is printed in the trim area of a page. Duplex sheets get the bar at both sides and simplex sheets at the print side.

There are five factory defined color bars:

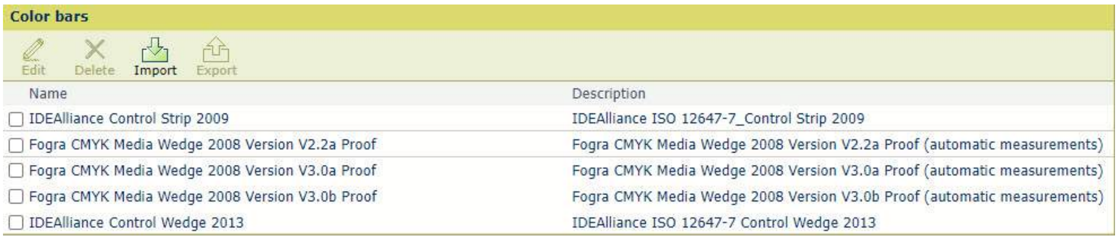
There are five default color bars:
- Idealliance Control Strip 2009
- Fogra CMYK Media Wedge 2008 Version V2.2a Proof
- Fogra CMYK Media Wedge 2008 Version V3.0a Proof
- Fogra CMYK Media Wedge 2008 Version V3.0b Proof
- Idealliance Control Wedge 2013

Go to the color bars

1. Open the Settings Editor and go to: [Color]→[Color & information bars].



[Color & information bars] tab



[Color bars]

Edit a color bar

1. Select the color bar.
2. Click [Edit].
3. Change the name and description.

[Edit color bar]

Delete a color bar

1. Select the color bar.
2. Click [Delete]

Import a color bar

1. Click [Import].
2. Select the .eps file that represents the color bar.

Export a color bar

You cannot export factory defined color bars.

1. Select the color bar.
2. Click [Export].

Define an information bar

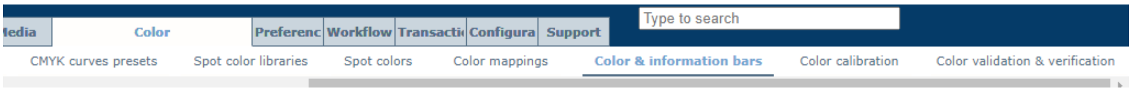
An information bar contains printer and job information. It can be selected as a job property. The bar is printed in the trim area of a page. Two-sided printed sheets get the bar at both sides and simplex sheets at the print side.

Information bars can help to evaluate the print quality according to the displayed information.

There is one factory-defined information bar: FOGRA Validation Print. You can add more custom information bars.

Go to the information bars

1. Open the Settings Editor and go to: [Color]→[Color & information bars].

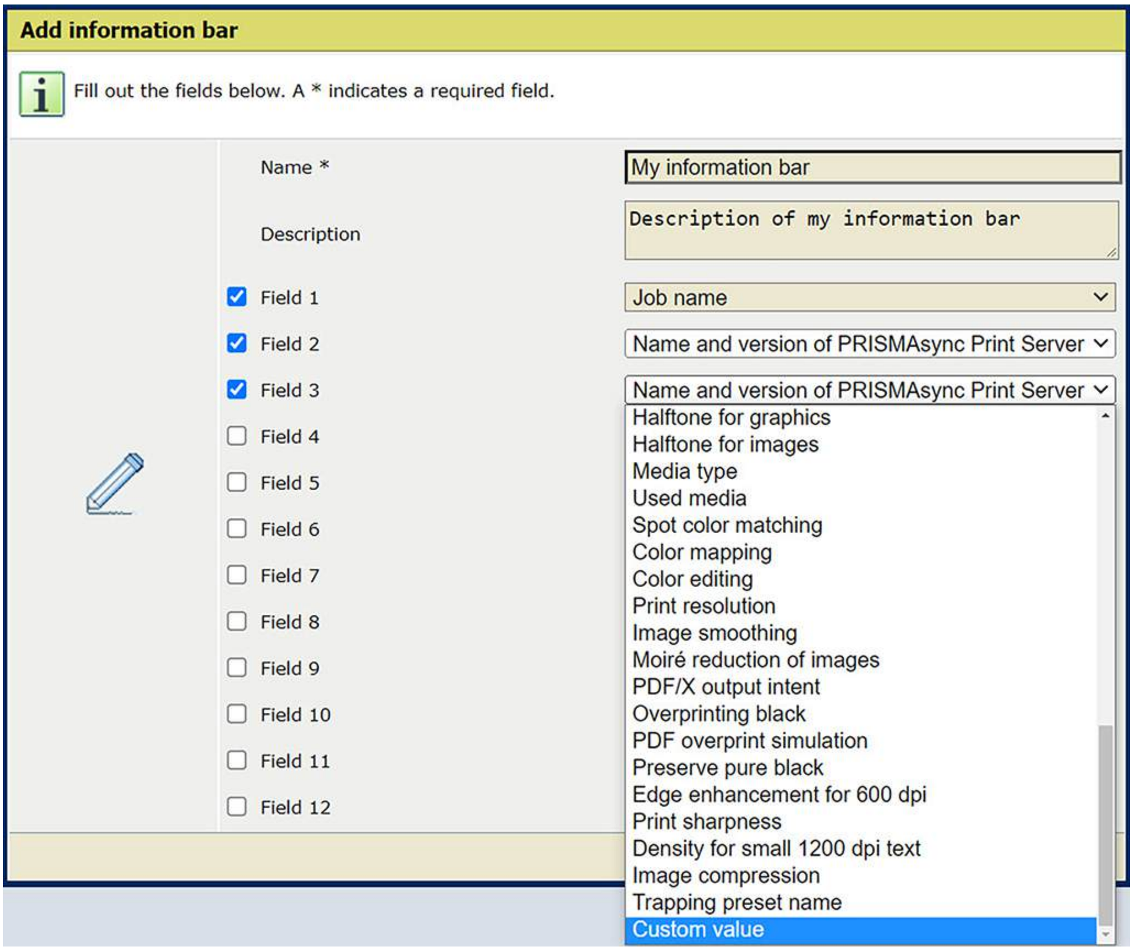


[Color & information bars] tab

[Information bars]

Add an information bar

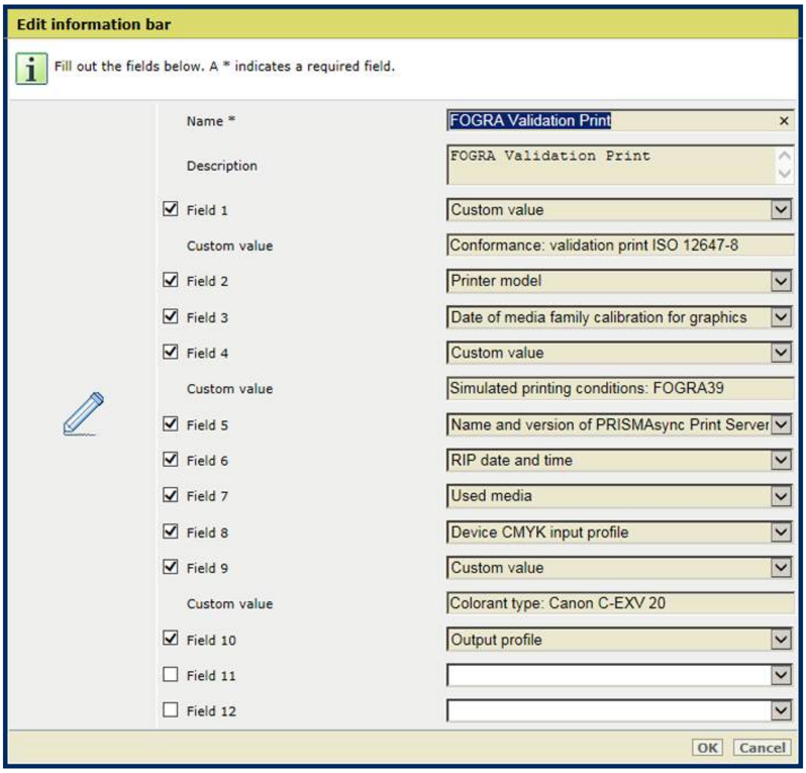
1. Click [Add].
2. Enter a name and description for the information bar.
3. Select for each field the information you want to display. Select [Custom value] for a fixed text that you can enter. You can use a maximum of 12 fields.



[Add information bar]

Edit an information bar

1. Select the information bar.
2. Click [Edit].
3. Select for each field the information you want to display. Select [Custom value] for a fixed text that you can enter. You can use a maximum of 12 fields.



[Edit information bar]

Delete an information bar

1. Select the information bar.
2. Click [Delete].

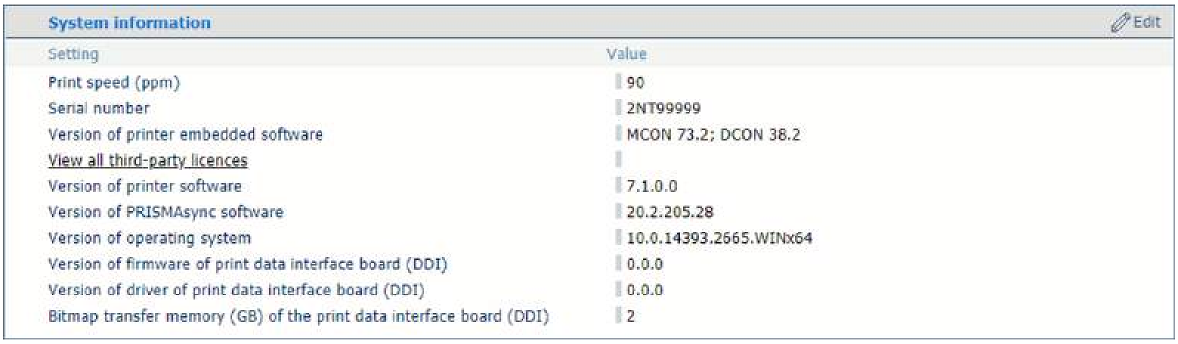
Export, import, and restore color configuration

What is color configuration

You can export the color configuration of PRISMA Print Server to store the configuration that reflects a certain printer state or date.

The color configuration ZIP file can be imported on the same printer, for example, when the existing color configuration causes problems.

The color configuration ZIP file can be imported on another printer that has the same PRISMAsync software version. In this way you can apply the color configuration of one printer on multiple printers.



Version of PRISMAsync software

It is also possible to restore the factory defined color configuration. The color configuration of the first installation is restored.

Color configuration
Input profiles
Output profiles, including composite output profiles
Color defaults
Color calibration settings
Color Presets
Media Families
CMYK curve presets
Calibration curves
Color mappings
Color bars
Information bars
Trapping presets
Validation tests and tolerance levels
Spot color libraries, including the custom spot color library

IMPORTANT
Be aware that the export and restore function can remove printer resources, such as profiles, spot color definitions, and media definitions.

NOTE
The color configuration is also part of the backup.

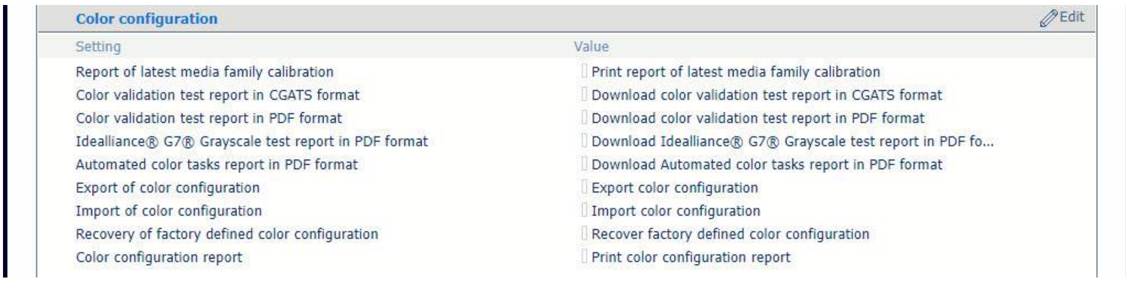
NOTE
Refer to the operation guide of the printer how to use the media family calibration and validation test reports.

Export the color configuration

1. Open the **Settings Editor** and go to: [Support]→[Troubleshooting].

[Troubleshooting] tab

2. Go to the [Color configuration] section.



[Color configuration] section

3. Click [Export color configuration].
4. Click [Export color configuration] in the [Export of color configuration] dialog box. A ZIP file is being created.

[Export color configuration]

5. Click the ZIP file and store it on an external location.
6. Click [OK] to close the dialog box.

NOTE
Refer to the operation guide of the printer how to use the other [Color configuration] reports.

Import the color configuration

1. Click [Import color configuration].
2. Click [Choose file] and browse to the ZIP file with the color configuration.

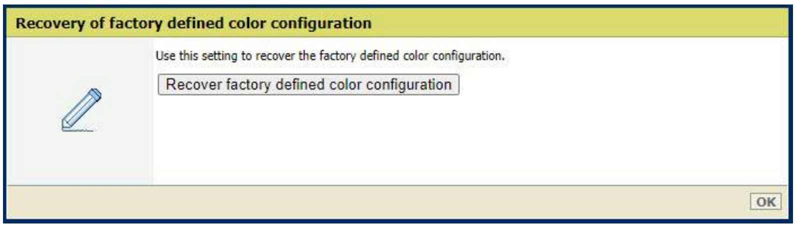
[Import of color configuration]

3. Click [Import color configuration].
4. Click [OK] to close the dialog box.

NOTE
You need to restart the system to implement the configuration.

Restore the factory defined color configuration

1. Click [Recover factory defined color configuration].



[Recover factory defined color configuration]

2. Click [Recover factory defined color configuration].
3. Click [OK].

NOTE
You need to restart the system to implement the configuration.

Print a color configuration report

The color configuration report contains the color configuration items.

1. Click [Print color configuration report].
2. Click [Print color configuration report]
3. Click [OK].

More information

- *Create, download, and print configuration report* on page 552
- *Create and download a log file* on page 546

Chapter 13

Maintain the printer

Learn about system configuration and maintenance

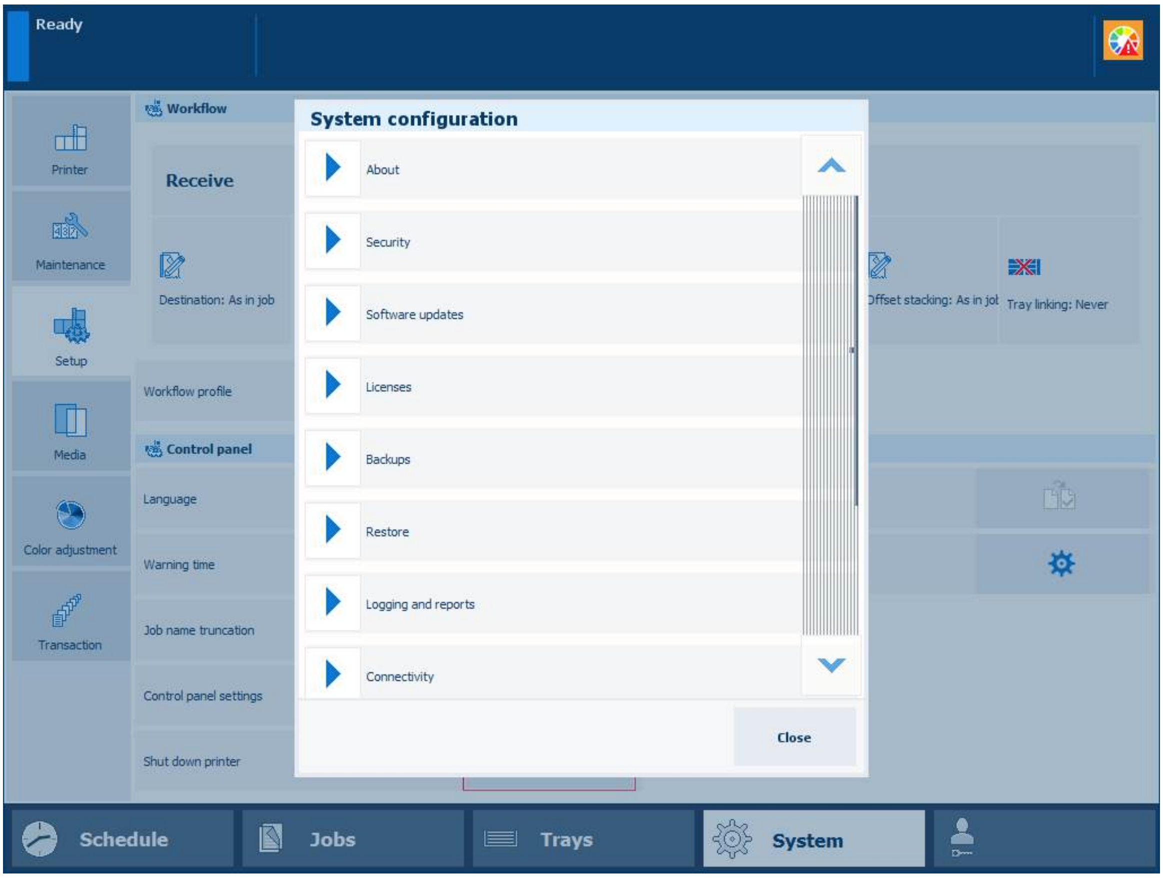
Your organization and print environment ask for a system configuration that meets production and workflow requirements. In addition, authorization, security, and sustainability guidelines are important when you set up a print system.

The installation of the print system includes most configuration tasks, such as definitions of preferences, job workflows, and print languages. The requirements with respect to media and color quality are translated into the correct system settings.

It depends on the type of jobs if it is necessary to change settings at a later moment.

Configuration tasks for the system administrator and key operator

The configuration settings are available via a web-based configuration tool, the Settings Editor. Both the key operator and the system administrator can change settings in the Settings Editor. A part of the Settings Editor settings and information is also available on the System view and Start view of the control panel. You need rights to access configuration settings to change these settings.



The system configuration settings on the control panel

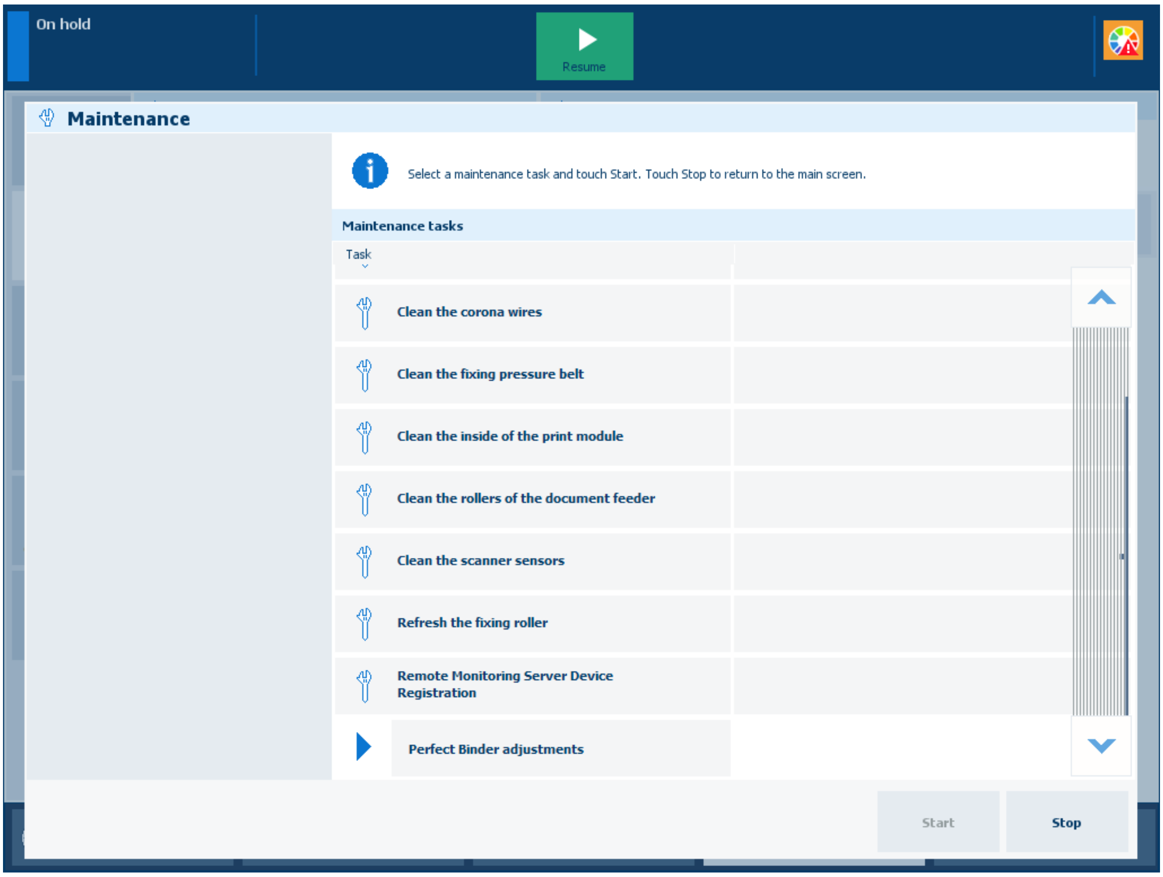
Maintenance tasks for operators

Operators are responsible for refilling the consumables, removal of staple and punch waste, and manual cleaning tasks.

Maintenance tasks for maintenance operators

Maintenance operators can perform procedures to clean machine parts or improve the image quality. Furthermore, they are responsible for mechanical maintenance tasks and maintenance procedures via the control panel.

For some procedures you must have the rights to perform maintenance tasks to prevent unauthorized use.



Maintenance tasks on the control panel

Billing counters

The Maintenance view shows the following counters that give insights into your system productivity.

- Billing counters
  Billing counters display the total number of images the print system has produced after installation.
- Day counters
  Day counters display the total number of images the print system has produced since the last reset to null. Reset the day counters at the beginning of the day or before you start a series of jobs, for example for a specific customer.

There are counters for normal media (length equal or less than 364 mm (14.3 inch) and large media (length more than 364 mm (14.3 inch)). You can read counters, reset day counters, or print the billing counters report in the [System]→[Maintenance] view.

Your Service organization can enable the visibility of toner cartridge counters.

More information

- *Maintain consumables* on page 503
- *Remove staple and punch waste* on page 520
- *Clean printer parts* on page 524

Maintain consumables

Consumables

The following consumables are available from Canon. For more information, contact your local authorized Canon dealer.

We recommend that you order paper stock and toner from your local authorized Canon dealer before your stock runs out.

Recommended paper

In addition to plain paper (A3/11 inch x 17 inch and A4/LTR size), recycled paper, color paper, transparencies (recommended for this machine), labels, and other types of paper stock are available.

CAUTION
Do not store paper in places exposed to open flames, as this may cause the paper to ignite, resulting in burns or a fire.

IMPORTANT
Some commercially available paper types are not suited for this machine. Contact your local authorized Canon dealer when you need to purchase paper.
To prevent moisture build-up, tightly wrap any remaining paper in its original package for storage.

NOTE
For high-quality printouts, use paper recommended by Canon.

Toner

If a message prompting you to replace the toner cartridge appears on the control panel, replace the used toner cartridge with a new one.

Toner comes in four colors: black, cyan, magenta, and yellow.

When replacing the toner cartridge, make sure that you replace toner of the correct color.

The toner cartridges are intended for use with this machine only.

For optimum print quality, it is recommended to use Canon genuine toner.

Product name	Supported Canon genuine toner
imagePRESS V1000	Canon imagePRESS Toner T11 Black Canon imagePRESS Toner T11 Cyan Canon imagePRESS Toner T11 Magenta Canon imagePRESS Toner T11 Yellow

WARNING
Do not burn or throw used toner cartridges into open flames, as this may cause the toner remaining inside the cartridges to ignite, resulting in burns or a fire.
Do not store toner cartridges in places exposed to open flames, as this may cause the toner to ignite, resulting in burns or a fire.
If you accidentally spill or scatter toner, carefully wipe up the loose toner with a damp, soft cloth and avoid inhaling any toner dust. Do not use a vacuum cleaner that is not equipped with safety measures to prevent dust explosions to clean up loose toner. Doing so may cause damage to the vacuum cleaner or result in a dust explosion due to static discharge.

CAUTION
Keep toner out of the reach of small children. If toner is ingested, consult a physician immediately.

IMPORTANT
Store toner cartridges in a cool location, away from direct sunlight. The recommended storage conditions are temperatures below 30°C and humidity below 80%.
Be careful of counterfeit toners
Please be aware that there are counterfeit Canon toners in the marketplace. Use of counterfeit toner may result in poor print quality or machine performance. Canon is not responsible for any malfunction, accident or damage caused by the use of counterfeit toner.
For more information, see canon.com/counterfeit.

Waste toner container

The waste toner containers are intended for use with this machine only.

Do not replace the waste toner container before the message prompting you to replace it appears on the control panel.

Product name	Supported Canon genuine waste toner container	Shape
imagePRESS V1000	WT-202	

Staple cartridge in the staple unit

Finisher	Staple cartridge	
	Name	Shape
Booklet Finisher-AG Staple Finisher-AG	Staple-N1	

Staple cartridge in the saddle-stitch unit

Finisher	Staple cartridge	
	Name	Shape
Booklet Finisher-AG	Staple-P1	

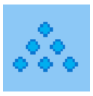
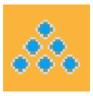
Genuine consumables

Canon continuously develops technology innovations in Canon toners, drums and cartridges, specifically designed for use in Canon multifunctional machines.

Experience the benefits of optimal print performance, print volume and high quality outputs, achieved through Canon's new advanced technologies. Therefore, the use of Canon genuine consumables is recommended for your Canon multifunctional machines.

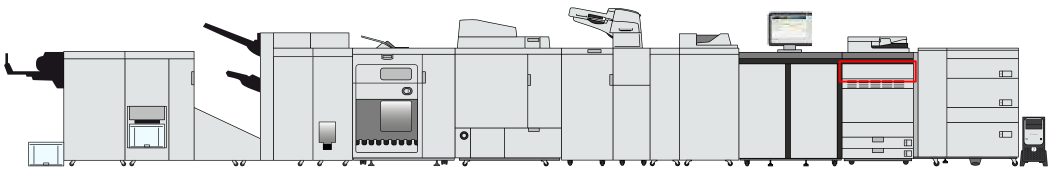
Replace a toner cartridge

The dashboard at the top of the control panel and the operation attention light warns you when a toner cartridge needs replacing. The color of the icon indicates the status of a toner cartridge. (*Learn about the printer status* on page 61)

	Toner cartridge contains sufficient toner.
	Toner cartridge has less than 25% toner.
	Toner cartridge is empty.

You can check the current status of the toner cartridge in the control panel. Location: [System]→[Printer]→[Supplies].

Toner cartridges come in four colors: black, cyan, magenta, and yellow. When a toner cartridge becomes empty during a print job, the print process resumes after you replace the toner cartridge. You find the toner cartridges in the print module.



Location of the toner cartridges

WARNING

- **Do not burn or throw used toner cartridges into open flames. This can cause toner ignition, which may result in burns or a fire.**
- **Do not store toner cartridges in places that are exposed to open flames. This can cause toner ignition, which may result in burns or a fire.**
- **When toner is spilled, carefully wipe up the loose toner with a damp, soft cloth. Avoid inhaling toner dust. To avoid explosions of toner dust due to static discharge, do not use a vacuum cleaner.**

CAUTION

- **Keep toner out of the reach of small children.**
- **If toner is ingested, consult a physician immediately.**
- **If toner gets onto your hands or clothing, immediately wash it off with cold water. Warm water will set the toner. If this happens, it will become impossible to remove the toner stains.**

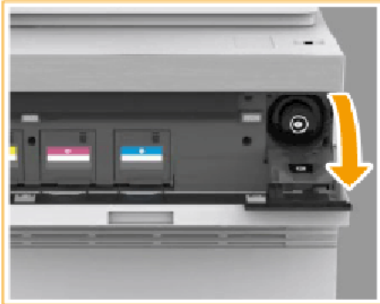
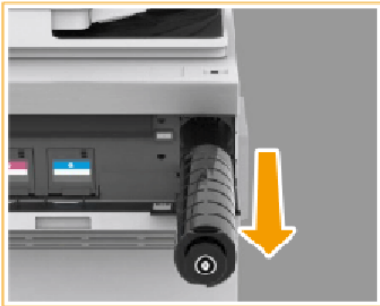
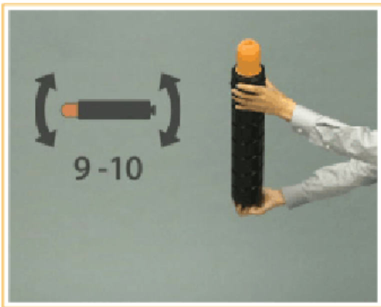
IMPORTANT

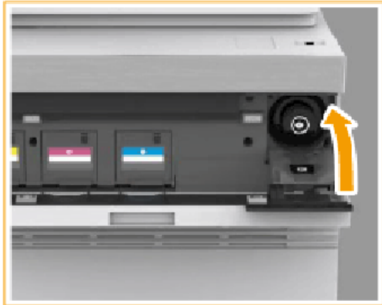
- For information on the Canon genuine toner, see *Consumables* on page 503.
- **Be careful with counterfeit toners.**
  Be aware that there are counterfeit Canon toners in the marketplace. Use of counterfeit toner can result in poor print quality or machine performance. Canon is not responsible for any malfunction, accident or damage caused by the use of counterfeit toner. For more information, see http://www.canon.com/counterfeit.
- Do not rotate the toner cartridge. This can cause the toner cartridge to leak.
- Do not replace toner cartridges until a message appears informing you that you must replace a toner cartridge.
- You can replace a toner cartridge while the print system is busy.
- The color of the toner cartridge that needs to be replaced is displayed on the control panel. If multiple toner cartridges need replacement, replace the toner cartridges in the following order: black, yellow, magenta, cyan.
- Wait to replace color toner cartridges when you want to continue copying or printing in black and white. Replace the color cartridge after the jobs are ready.
- When a color cartridge becomes empty, the print system interrupts the job and you cannot continue to copy or print in color and black and white. However, you can cancel the interrupted job and continue to copy or print in black and white.
- Never touch the tip of the toner cartridge or hit and shake the cartridge. This can cause a toner cartridge leak.

- Store toner cartridges in a cool location, away from direct sunlight. The recommended storage conditions are temperatures below 30°C and humidity below 80%.
- The order of the toner cartridge from left to right is: yellow, magenta, cyan and black.

Procedure

	Action	
1	Hold both sides of the cover of the toner compartment and open it.	
2	Go to the control panel and touch [System]→[Printer].	

	Action	
3	Select and touch one of the open buttons ⏏ in the [Supplies] pane to open the cover of the toner cartridge you want to replace.	
4	Pull out the empty toner cartridge. **WARNING** **Do not burn or throw used toner cartridges into open flames. This can cause toner ignition in the cartridge which may result in burns or a fire.**	
5	Unpack the new toner cartridge. IMPORTANT Check that the color of the toner cartridge matches the label color on the internal cover.	
6	Gently tilt the new toner cartridge up and down 10 times.	
7	Unscrew the red protective cap.	

	Action	
6	Insert the new toner cartridge as far as possible.	
7	Close the internal cover. **CAUTION** **When you close the internal cover, be careful not to get your fingers caught. This can cause personal injury.**	
8	Close the cover of the toner compartment. **CAUTION** **When you close the cover of the toner compartment, be careful not to get your fingers caught. This can cause personal injury.**	

After you finish

After you replace a color cartridge, a color difference in the output may occur. Then, perform the Auto gradation adjustment. (*Calibrate the printer* on page 368)

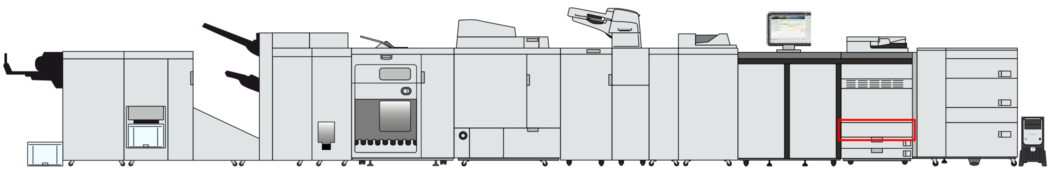
Replace the waste toner container

The dashboard at the top of the control panel and the operation attention light warn you when the waste toner container is full. The color of the icon indicates the status of the waste toner container. (*Learn about the printer status* on page 61)

	Waste toner container has sufficient space.
	Waste toner container is almost full.
	Waste toner container is full

You can check the current status of the waste toner container in the control panel. Location: [System]→[Printer]→[Supplies].

You can replace the waste toner container at a later time, but be aware that the active job or one of the next jobs will cause an error because of the full waste toner container. You can find the waste toner container in the print module.



Location of the waste toner container

WARNING

- **Do not burn or throw used waste toner containers into open flames. This can cause toner ignition, which may result in burns or a fire.**
- **Do not store toner cartridges in places that are exposed to open flames. This can cause toner ignition, which may result in burns or a fire.**
- **If toner gets onto your hands or clothing, immediately wash it off with cold water. Warm water will set the toner. If this happens, it becomes impossible to remove the toner.**
- **When toner is spilled, carefully wipe up the loose toner with a damp, soft cloth. Avoid inhaling toner dust. To avoid explosions of toner dust due to static discharge, do not use a vacuum cleaner.**

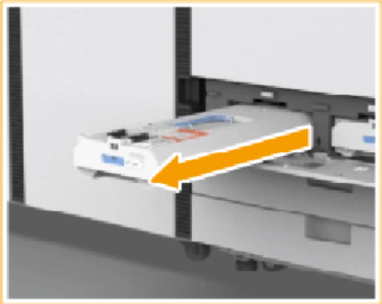
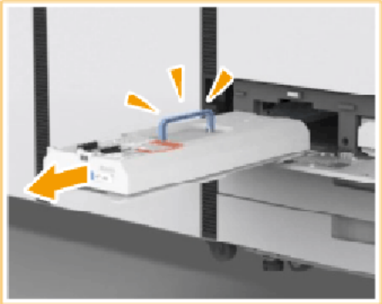
CAUTION

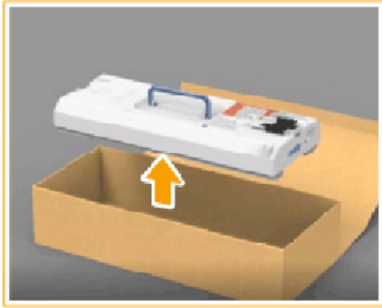
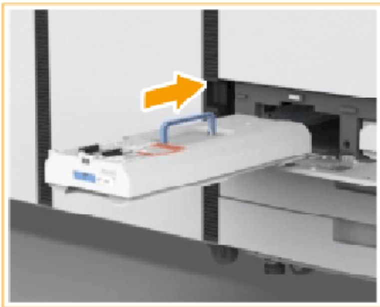
- **If toner is ingested, consult a physician immediately.**
- **If toner gets onto your hands or clothing, immediately wash it off with cold water. Warm water will set the toner. If this happens, it will become impossible to remove the toner stains.**

IMPORTANT
- For information on the Canon waste toner containers, see *Consumables* on page 503.
- Your local authorized Canon dealer will dispose of used waste toner containers.
- You cannot reuse toner. Do not mix new and used toner together.
- Only use waste toner containers intended for your print system.
- Do not replace the waste toner container before a message on the control panel appears to inform you that you must replace the container.
- When you insert the waste toner container, push the container as far as possible.

NOTE
When you replace the waste toner container during a print job, the print process will resume after the waste toner container is replaced.

Procedure

	Action	
1	Open the waste container cover. NOTE On the left-hand side of the cover is an opening where you can flip it open.	
2	Remove the full waste container. Once the waste container is halfway out, hold the handle on the top and remove the waste container completely. **CAUTION** **Do not tilt the waste toner container when you remove it. This causes toner to be spilled onto your hands or clothing.**	 

	Action	
3	Take the new waste container out of the box.	
4	Place the old waste toner container in the empty box from the new waste container. NOTE Use the plastic bag in which the new waste container was packed to pack the old waste container. Tightly close it, so toner does not spill.	
5	Insert the new waste container in its compartment. Make sure the Canon logo faces up and is at the side that slides in as last. IMPORTANT Push the waste container completely in. If the waste container is not installed correctly, the container shutter may stay open and can cause toner to scatter.	
6	Close the waste container compartment.	

Replace the staple cartridge in the staple unit of the stacker / stapler

The dashboard of the control panel indicates if the staple cartridge needs replacing. The color of the icon indicates the status of the staple cartridge. (*Learn about the printer status* on page 61)

	Staple cartridges contains sufficient staples.
	At least one staple cartridge is almost empty.
	At least one staple cartridge is empty.

You can check the current status of the staple cartridge in the control panel. Location: [System]→[Printer]→[Supplies].

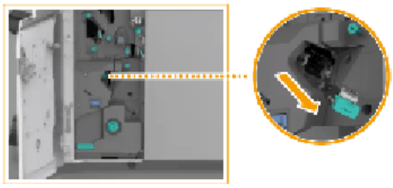
You can find the staple unit in the stacker/stapler.

IMPORTANT
- For information on the Canon genuine staple cartridges, see *Consumables* on page 503.
- Be careful when you perform maintenance tasks on optionals attached to the machine. When you perform a maintenance task, such as replace a staple cartridge, remove waste, or solve jams, other machine parts can continue with job process activities.
- Remove the seal that holds the staples together after you place the staple cartridge into the staple case.

NOTE
- We recommend ordering staple cartridges from your local authorized Canon dealer before your stock runs out.
- Only use staple cartridges intended for your print system.

Staple-N1 for Staple Finisher-AF and Booklet Finisher-AF

Procedure

	Action	
1	Open the front cover of the stacker/stapler.	
2	Hold the green tab of the staple case and pull it out. NOTE When the staple unit is at the back, it is difficult to pull out the staple case. If this happens, turn the dial on the bottom left-hand side to the left to move the staple unit.	
3	Take out the empty staple cartridge.	
4	Take a new staple cartridge from the box	

	Action	
4	Insert the new staple cartridge into the staple case. NOTE After you have put the new staple cartridge into the staple case, remove the seal that holds the staples together.	
5	Gently push the staple case into the staple unit as far as possible.	
6	Close the front cover of the stacker/stapler. **CAUTION** **When you close the front cover of the stacker/stapler, be careful not to get your fingers caught. This can cause personal injury.**	

Replace the staple cartridge in the saddle-stitch unit of the stacker/stapler

The dashboard of the control panel indicates if the staple cartridge needs replacing. The color of the icon indicates the status of the staple cartridge. (*Learn about the printer status* on page 61)

	Staple cartridges contains sufficient staples.
	At least one staple cartridge is almost empty.
	At least one staple cartridge is empty.

You can check the current status of the staple cartridge in the control panel. Location: [System]→[Printer]→[Supplies].

You can find the saddle-stitch unit in the stacker/stapler.

IMPORTANT
- For information on the Canon genuine staple cartridges, see *Consumables* on page 503.
- Be careful when you perform maintenance tasks on optionals attached to the machine. When you perform a maintenance task, such as replace a staple cartridge, remove waste, or solve jams, other machine parts can continue with job process activities.
- Remove the printed output from the booklet tray before you replace the staple cartridge in the saddle-stitch unit.

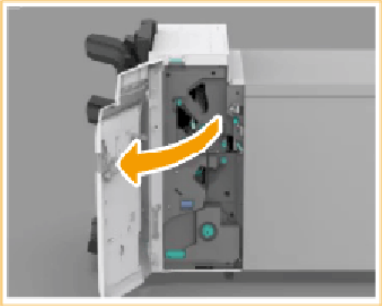
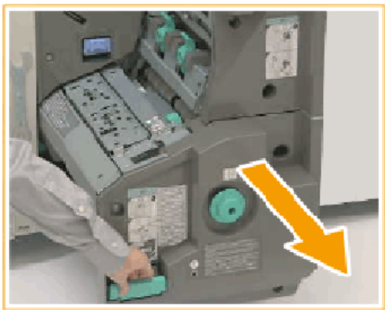
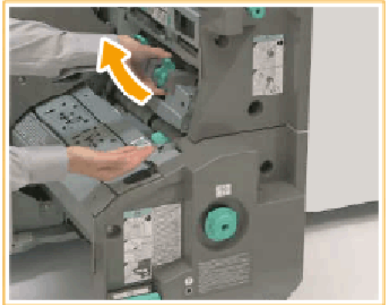
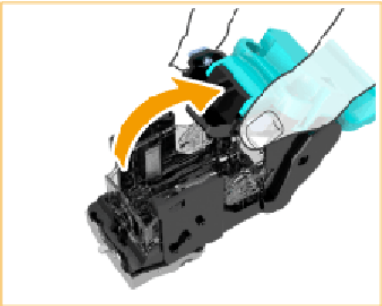
NOTE
- We recommend ordering staple cartridges from your local authorized Canon dealer before your stock runs out.
- When a staple cartridge is empty, replace both cartridges.
- Only use staple cartridges intended for your print system.

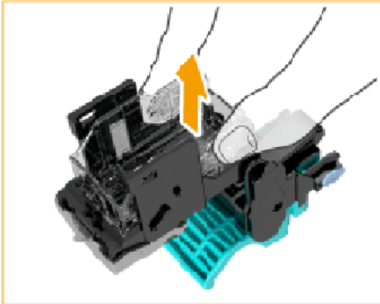
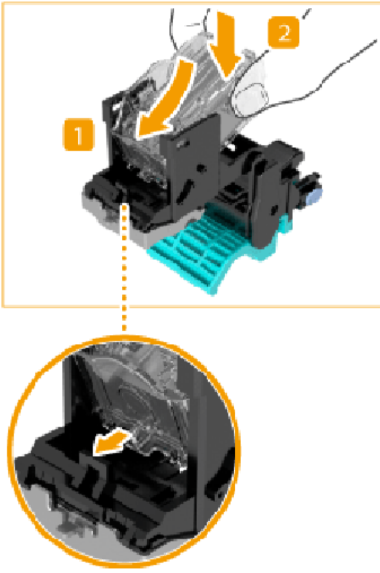
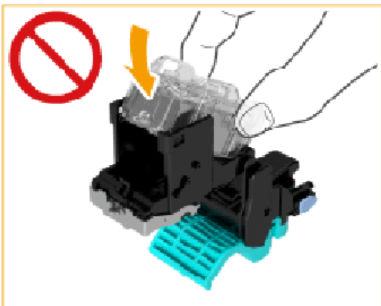
Staple-P1 for Booklet Finisher-AF

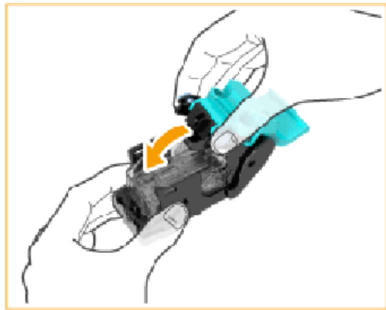
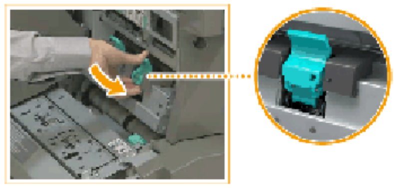
Before you begin

Remove printed output from the booklet tray.

Procedure

	Action	
1	Open the front cover of the stacker/stapler.	
2	Pull out the saddle-stitch unit with the green handle.	
3	Pull out the staple cartridges.	
4	Squeeze the light blue areas to lift the staple case cover.	

	Action	
5	Squeeze to release and lift the empty staple cartridge.	
6	Take out a new staple case from the box.	
7	Insert the new staple cartridge until it clicks into place. IMPORTANT Make sure you insert the staple cartridge in the correct direction.	 

	Action	
8	Place the staple case cover back in its original position.	
9	Return the staple case back into the saddle-stitch unit and check that the two arrows are aligned.	
10	Repeat step 2 - 7 for the other staple cartridge.	
11	Gently push the saddle-stitch unit back into the stacker/stapler as far as possible and close the front cover. **CAUTION** • **When you push the saddle-stitch unit to its original position, be careful not to get your fingers caught. This can cause personal injury.** • **When you close the front cover of the stacker/stapler, be careful not to get your fingers caught. This can cause personal injury.**	

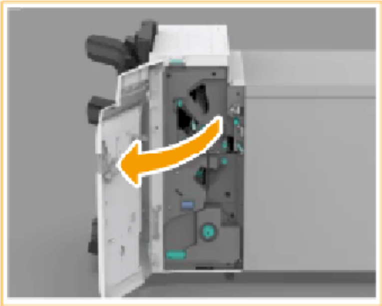
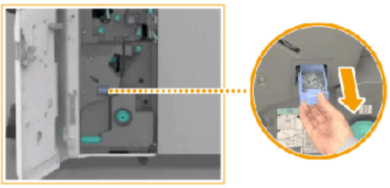
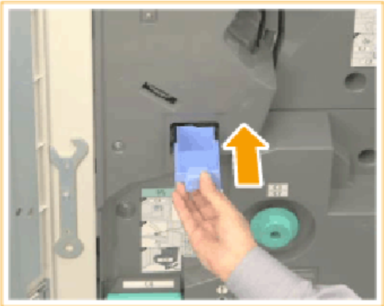
Remove staple and punch waste

Remove the staple waste

The dashboard of the control panel indicates if the staple waste tray is full. (*Learn about the printer status* on page 61)

You can find the staple waste tray in the stacker/stapler.

Procedure

	Action	
1	Open the front cover of the stacker/stapler.	
2	Pull out the staple waste tray at the bottom of the front side of the staple unit.	
3	Discard the staple waste. ⚠ **CAUTION** **When you discard the staple waste, be careful not to touch it. This can cause personal injury.**	
4	Push the staple waste tray back to its original position, or as far as possible.	

	Action	
5	Close the front cover of the stacker/stapler. ⚠ **CAUTION** **When you close the front cover of the stacker/stapler, be careful not to get your fingers caught. This can cause personal injury.**	

Remove the punch waste in the stacker /stapler

The dashboard of the control panel indicates if the punch waste tray is full. (*Learn about the printer status* on page 61)

You can check the current status of the punch waste tray at the control panel. Location: [System] →[Printer]→[Finishers].

The print system cannot punch when the punch waste tray is full. This instruction only describes the removal of punch waste of the stacker/stapler. Refer to the user manual of the professional puncher how to remove punch waste from the professional puncher.

IMPORTANT
Be careful when you perform maintenance tasks on optionals attached to the machine. When you perform a maintenance task, such as replace a staple cartridge, remove waste, or solve jams, other machine parts can continue with job process activities.

Procedure

	Action	
1	Open the front cover of the stacker/stapler.	
3	Pull out the punch waste tray. NOTE Check that the guide (F-A2) is closed. When the guide is open, you cannot pull out the punch waste tray. If applicable, close the guide.	
4	Discard the punch waste. IMPORTANT Make sure that the punch waste tray is completely emptied.	

	Action	
5	Push the punch waste tray back to its original position, or as far as possible.	
6	Close the front cover of the stacker/stapler. **CAUTION** **When you close the front cover of the stacker/stapler, be careful not to get your fingers caught. This can cause personal injury.**	

Clean printer parts

Cleaning tasks and procedures

You are advised to clean the control panel **weekly**. (*Clean the control panel* on page 525)

To keep your print system in optimal condition and prevent poor image quality, clean the following system parts monthly:

Part	Task
Glass plate	*Clean the glass plate area* on page 526
Scanning area of the automatic document feeder	*Clean the automatic document feeder scanning area* on page 528
Scanning sensors	*Clean the scanning sensors* on page 536
Automatic document feeder rollers	*Clean the rollers of the automatic document feeder* on page 535
The inside of the print module	*Clean the inside of the print module* on page 534
Corona assembly wires	*Clean the inside of the print module* on page 534
Main unit roller	*Clean the clean roller* on page 532
Fixing roller	*Refresh the fixing roller* on page 537

WARNING

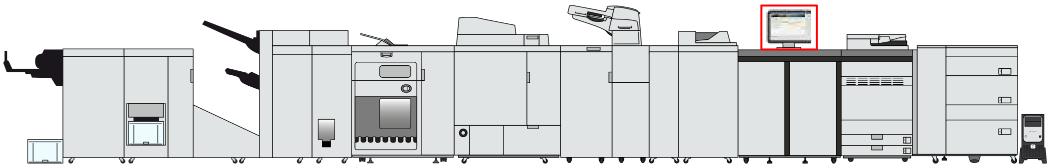
- **When you clean the machine, first turn OFF the main power switch, and disconnect the power cord. Failure to observe these steps can result in a fire or electrical shock.**
- **Disconnect the power cord from the power outlet regularly, and clean the area around the base of the power plug's metal pins and the power outlet with a dry cloth to ensure that all dust and grime is removed. If the power cord is connected for a long period of time in a damp, dusty, or smoky location, dust can build up around the power plug and become damp. This can cause a short circuit and may result in a fire.**

IMPORTANT

- Do not dampen the cloth with too much water. This can damage prints or machine parts.
- Do not use alcohol, benzene, paint thinner, or other solvents to clean machine parts. This can damage plastic machine parts.

Clean the control panel

When the screen of the control panel becomes dirty, you can simply clean the panel with water or a mild cleaning agent.



Location of the control panel

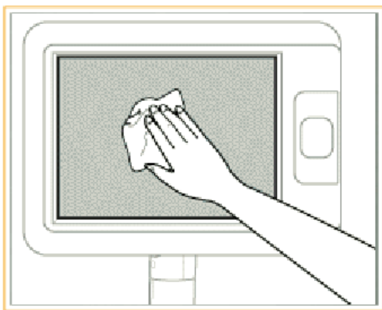
WARNING

- **When you clean the machine, first turn OFF the main power switch, and disconnect the power cord. Failure to observe these steps can result in a fire or electrical shock.**
- **Disconnect the power cord from the power outlet regularly, and clean the area around the base of the power plug's metal pins and the power outlet with a dry cloth to ensure that all dust and grime is removed. If the power cord is connected for a long period of time in a damp, dusty, or smoky location, dust can build up around the power plug and become damp. This can cause a short circuit and may result in a fire.**

IMPORTANT

- Do not dampen the cloth with too much water. This can damage prints or machine parts.
- Do not use alcohol, benzene, paint thinner, or other solvents to clean machine parts. This can damage plastic machine parts.

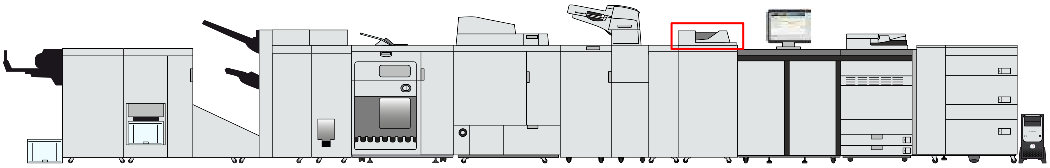
Procedure

1	Moisten the cloth with water or a mild cleaning detergent and wring the cloth out thoroughly.	
2	Clean the display of the control panel	
3	Dry the display of the control panel with a dry soft cloth.	

Clean the glass plate area

Clean the automatic document feeder and the glass plate in order that the scanner can scan the originals correctly and can detect the size of the originals.

You can use water or water mixed with a mild cleaning agent to clean. It is easy to use the cleaning cloth provided as a system accessory of the automatic document feeder. When the cleaning cloth becomes dirty, wash and rinse it, and then leave to dry.

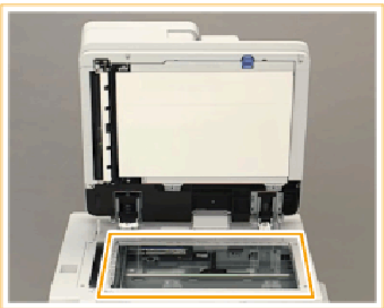


Location of the glass plate

WARNING
- **When you clean the machine, first turn OFF the main power switch, and disconnect the power cord. Failure to observe these steps can result in a fire or electrical shock.**
- **Disconnect the power cord from the power outlet regularly, and clean the area around the base of the power plug's metal pins and the power outlet with a dry cloth to ensure that all dust and grime is removed. If the power cord is connected for a long period of time in a damp, dusty, or smoky location, dust can build up around the power plug and become damp. This can cause a short circuit and may result in a fire.**

IMPORTANT
- Do not dampen the cloth with too much water. This can damage prints or machine parts.
- Do not use alcohol, benzene, paint thinner, or other solvents to clean machine parts. This can damage plastic machine parts.

Procedure

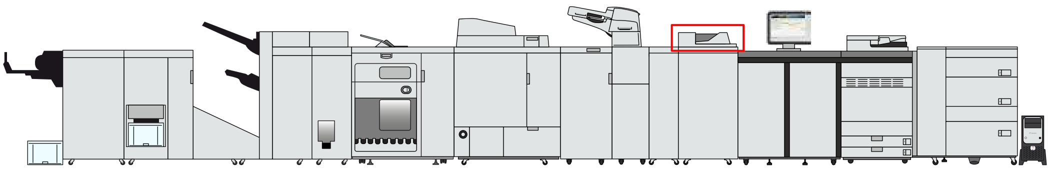
1	Moisten the cloth with water or a mild cleaning detergent and wring it out thoroughly.	
2	Clean the glass plate.	

3	Clean the underside of the automatic document feeder with the cloth.	
4	Wipe dry with a soft, dry cloth.	

Clean the automatic document feeder scanning area

Clean the automatic document feeder and the glass plate in order that the scanner can scan the originals correctly and can detect the size of the originals.

You can use water or water with a mild cleaning agent to clean this. It is easy to use the cleaning cloth provided as a system accessory of the automatic document feeder. When the cleaning cloth becomes dirty, wash and rinse it, and then leave to dry.



Location of the automatic document feeder

WARNING

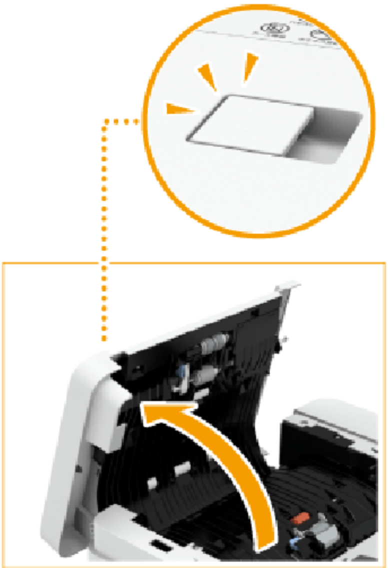
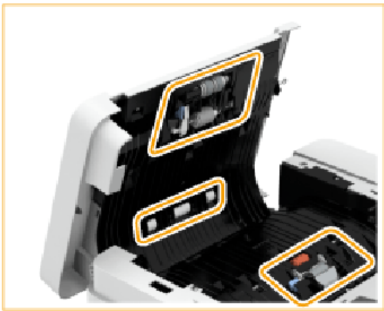
- **When you clean the machine, first turn OFF the main power switch, and disconnect the power cord. Failure to observe these steps can result in a fire or electrical shock.**
- **Disconnect the power cord from the power outlet regularly, and clean the area around the base of the power plug's metal pins and the power outlet with a dry cloth to ensure that all dust and grime is removed. If the power cord is connected for a long period of time in a damp, dusty, or smoky location, dust can build up around the power plug and become damp. This can cause a short circuit and may result in a fire.**

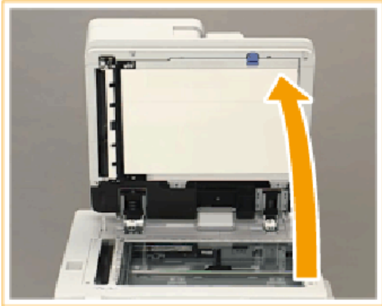
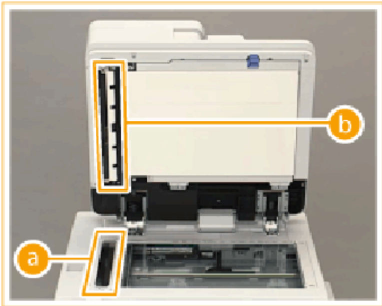
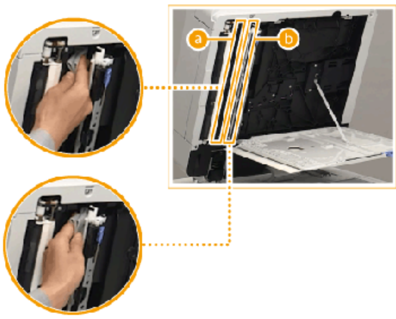
IMPORTANT

- Do not dampen the cloth with too much water. This can damage prints or machine parts.
- Do not use alcohol, benzene, paint thinner, or other solvents for cleaning. This can damage plastic machine parts.

Procedure

	Action	
1	Moisten a cloth with water or a mild cleaning detergent and wring it out thoroughly.	

	Action	
2	Pull the lever and open the cover of the automatic document feeder.	
3	Clean the rollers at the indicated areas. Then, wipe dry with a clean, dry cloth.	
4	IMPORTANT Make sure the sensor for detecting multiple sheets keeps clean.	
5	Gently close the cover.	

	Action	
6	Open the automatic document feeder.	
7	Clean the sensor strip areas (a) and (b). Then, wipe dry with a clean, dry cloth.	
8	Pull the top lever of the automatic document feeder to open the inner side of the document scan cover. IMPORTANT Open the document scan cover carefully to prevent scratches on the glass plate. Clean the thin glass strip of the scan area. Then, wipe dry with a clean, dry cloth.	
	Pull the lever at the underside of the document scan cover to open the document feed scanning area.	
	Clean the sensor strip areas (a) and (b). Then, wipe dry with a clean, dry cloth.	

	Action	
	Close the document feed scanning area, the document scan cover and the automatic document feeder. **CAUTION** • **When you close the covers and the automatic document feeder, be careful not to get your fingers caught. This can cause personal injury.** • **Be aware that the light emitted from the glass plate can be very bright when you close the automatic document feeder.**	

Clean the clean roller

If streaks appear on the printed output, perform a correction procedure to clean the roller of the print module.

The procedure takes approximately one minute.

IMPORTANT
Do not clean during printing.

Before you begin

Make sure the list of scheduled jobs is empty.

Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Clean the clean roller (print module)].
4. Follow the wizard steps.
5. Close the menu.

Clean the corona assembly wires

If streaks appear on the printed output or parts of the printed image are not visible, perform a correction procedure to clean the corona assembly wires inside the print module.

It takes approximately 35 seconds to clean the wires.

IMPORTANT
- Do not turn off the printer during cleaning.
- Do not open the front covers during cleaning.
- Do not clean during printing.

Before you begin

Make sure the list of scheduled jobs is empty.

Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Clean the corona wires].
4. Follow the wizard steps.
5. Close the menu.

After you finish

NOTE
If the procedure displays an engine error, contact your service organization.

Clean the inside of the print module

If streaks appear on the printed output or parts of the printed image are missing, clean the inside of the print module.

The control panel displays how long it takes to clean the inside of the print module.

Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Clean the inside of the print module].
4. Follow the wizard steps.
5. Close the menu.

Clean the rollers of the automatic document feeder

If your originals have black streaks or appear dirty after they have been through the automatic document feeder, clean the automatic document feeder rollers.

It takes approximately 20 seconds to clean the rollers of the automatic document feeder.

Before you begin

- Make sure the list of scheduled jobs is empty.
- Make sure you have clean A4 / LTR plain paper (60 - 80 g/m^2 / 16 lb bond - 20 lb bond).

Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Clean the rollers of the document feeder].
4. Follow the wizard steps.
5. Close the menu.

Clean the scanning sensors

If streaks appear on the printed output or you cannot scan originals properly, clean the scanner sensors.

Before you begin

For this procedure you must have the rights to perform maintenance tasks.

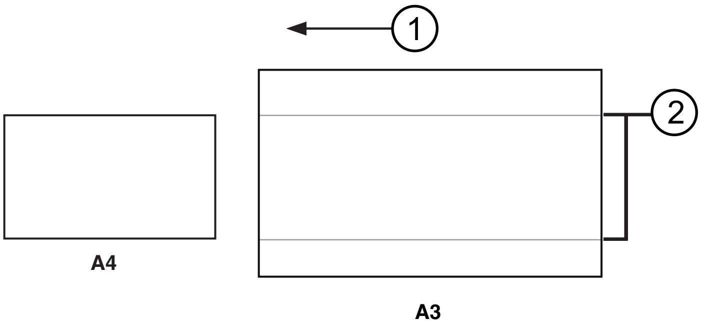
Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Clean the scanner sensors].
4. Follow the wizard steps.
5. Close the menu.

Refresh the fixing roller

After the print system has printed more than 100 sheets of a specific paper size, for example A4/LTR, and then prints on a larger paper size, for example A3/11 inch x 17 inch, luster streaks (ghost printing) can appear on both sides of the paper. Perform a correction procedure to refresh the fixing roller to correct this problem.

You can also perform this procedure when you use heavy or coated paper and patches of uneven gloss appear in the high density printed areas.



The feed direction of paper path (1) and luster streaks (2)

IMPORTANT
Perform this procedure only in urgent situations, because this procedure can largely decrease the lifetime of the fixing roller.

Before you begin

- Make sure the list of scheduled jobs is empty.
- For this procedure you must have the rights to perform maintenance tasks.

Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Refresh the fixing roller].
4. Follow the wizard steps.
5. Close the menu.

After you finish

IMPORTANT
If the procedure fails or if you do **not** get the required result, contact your service organization.

Maintain printer parts

Perform operator maintenance tasks in the print module

Parts must be replaced when indicated on the control panel. To increase the uptime of the print system, the operator can replace the indicated parts himself.

IMPORTANT
Only trained operators are allowed to perform maintenance tasks in the print module. Your service organization must prepare the system to allow maintenance in the print module. Contact your service organization for more information.

The dashboard indicates when maintenance is required with an icon.

	Maintenance is required soon.
	Maintenance is required now.

Before you begin

For this procedure you must have the rights to perform maintenance tasks.

Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Print module maintenance].
4. Follow the instructions on the control panel.

   IMPORTANT
   Be aware that this procedure resets the counters that indicate when maintenance is required.

5. Close the menu.

After you finish

After you replaced a part you must perform a printer calibration.

More information

Calibrate the printer on page 368

Lubricate a die set

You need to lubricate a die set of the professional puncher when indicated on the control panel. Perform the procedure below to lubricate a die set.

	Lubrication is required soon.

Before you begin

- User manual for the professional puncher.
- For this procedure you must have the rights to perform maintenance tasks.

Procedure

1. Touch [System]→[Maintenance].
2. Touch [Go to maintenance tasks].
3. Touch [Lubricate the die set in the professional puncher].
4. Refer to the user manual of the professional puncher how to lubricate die sets.
5. Close the wizard.

   IMPORTANT
   Confirm only if you actually lubricated the die set. A counter keeps track of when lubrication is required. This procedure resets this counter.

Operator Maintenance Application (OMApp)

Introduction

The Operator Maintenance Application (OMApp) allows trained operators to perform complicated maintenance actions without the need of a service technician. The operator can start various maintenance actions e.g. replacing and cleaning of parts.

Follow the steps below to use the Operator Maintenance Application:

Procedure

1. Go to the control panel and touch: [System]→[Maintenance]→[Go to maintenance tasks]
2. Touch [Printer maintenance]
3. Follow the instructions on the control panel

Chapter 14
Problem Solving

Find solutions for problems

Problems occur due to a print system or a job handling problem.

Below, you find an overview of solutions on problems that can occur.

Problem	Solution
Paper jam	Follow the instructions on the control panel. After you have solved a paper jam, the print process resumes from the set where the jam occurred. Check for double prints.
Paper jam with translucent media	Check the surface type of the media. (*Add media to the media catalog* on page 429)
Machine error	Follow the instructions on the control panel. The control panel informs you if a restart of the print system is necessary.
Service required	Follow the instructions on the control panel. The control panel informs if you must contact the service organization.
Unit reconditioning	Follow the instructions on the control panel. When you have solved the problem, the print process resumes from the set where the problem occurred. The control panel may ask to remove sheets.
Print system cannot determine if the output is complete.	Follow the instructions on the control panel and check the output. The print system asks you to confirm that the printed output is complete.
Printed output or originals have streaks or spots.	Perform a procedure to clean machine parts. (*Cleaning tasks and procedures* on page 524) For some procedures you must have the rights to perform maintenance tasks.
A paper tray does not open.	Check if you closed all paper trays correctly.
Printed media has curls.	Perform the curled output media correction procedure. (*Correct curled output media* on page 573)
Images on the printed output are not aligned correctly.	Perform a media registration. (*Adjust media registration* on page 575)
Images on the printed output are skewed or slightly rotated.	Perform the skew correction procedure. (*Adjust media registration* on page 575)
Similar colors for text and color panes in the source document are different on the printed output.	Perform a media calibration for all halftones. (*Calibrate a media family* on page 373)

Problem	Solution
Similar colors in the source document are different on different media.	When the different media are in two media families, calibrate both media families. (*Calibrate a media family* on page 373) When the different media are in one media family, create a new media family with new output profiles and calibrate the new media family. (*Calibrate a media family* on page 373)
CMYK colors are not correctly aligned.	Perform the automatic color mismatch procedure. (*Automatic color mismatch correction* on page 570)
Printed output has white gaps between color panes.	You can create new trapping presets. (*Create a trapping preset* on page 456)
An uneven density occurs.	1. Perform the Auto gradation adjustment procedure. 2. Perform the Shading correction procedure. (*Learn about calibration* on page 346)
Gradation problem occurs: Automatic gradation adjustment shows strange results. Automatic gradation adjustment does not work.	1. Reset the Auto gradation adjustment calibration curves. 2. Perform the Auto gradation adjustment procedure. (*Learn about calibration* on page 346)
Temperature or humidity changes significantly in the print environment.	Perform the Auto gradation adjustment procedure. (*Learn about calibration* on page 346)
Uneven color planes occur.	The Shading correction procedure assures consistent, even color planes on the output. (*Learn about calibration* on page 346)
Moiré patterns occur on the printed output.	Change the image smoothing setting. (*Learn about the print job settings* on page 299)

Learn about print defects, print errors, and printer errors

It can happen that the output on PRISMAsync Print Server-driven printers is not according to the configured settings or does not have the expected print quality. The printer can also encounter an error that leads to a printer stop. This printer error can be related to a specific job or event.

When such problems occur more often, you can contact the Service organization to solve the problem. The Service organization uses log files or job files to analyze the problem.

This User guide uses the following terminology to indicate a print or printer problem.
- A **performance problem**.
- A **print error**.
- A **printer error**.

Performance problem

In the print industry, many different performance problems can occur. The printer and finishing hardware, color and media management, inks or toner, media, and environmental circumstances all play a part in the print process. They must be in balance to realize output that meets your expectations.

In most circumstances, performance problems can be solved by color adjustments procedures, calibration procedures, changing the job media, or the use of settings to improve the print quality or the productivity.

Performance problems can be split up into three areas.

Performance areas	Description
Print, copy, or scan quality	Quality of the output compared to the quality that was originally intended and according to the printer specifications.
Job productivity	The amount of output you can achieve according to your expectations and the printer specifications. For example, paper jams can reduce the job productivity.
Finishing	Way the copies or prints are finished or behave after the delivery in the output trays. For example, specific finishing devices can affect the toner or ink layer.

When you are not able to solve a performance problem, you can manually create a log file or a file of the job so that the Service organization can analyze the problem. The log file option is available in the Settings Editor, in PRISMAsync Remote Manager and on the control panel. You can create a file of the last received job in the Settings Editor.

Print error

A **print error** is an error when the prints, copies or scans are not according to the **job or printer settings**. For example, when the job is not printed completely or a job with staples is not stapled. The printer does not automatically create a log file when a print error occurs. You must create and download the log file manually immediately after the job was finished.

IMPORTANT
To allow Service to analyze the print error, you must create and download the log file immediately after the job was finished.

Printer error

A printer error can be caused by a specific job or event. There are two types of printer errors.

- A printer error that is followed by a **blue error screen**. An error code is visible on the blue screen. A **restart** a needed to use the printer again. The printer automatically creates a log file when this printer error occurs.
- A printer error that is indicated at the status area of the **dashboard of the control panel**. The error code is visible on the dashboard. When enabled, the printer automatically creates a log file when this printer error occurs.

IMPORTANT
You can download the automatically created log file for the Service organization and download the job that caused a printer error.

More information

- *Create, download, and print configuration report* on page 552
- *Create and download a log file* on page 546

Create and download a log file

Understand log file names

The log file names are composed of several parts depending on the contents.

*<hostname>*_**TraceData**_*<yyyymmdd>*_*<hhmmss>*	Reports/trace file manually created by a user at specified time on the specified system.
*<hostname>*_**Reports**_ *<yyyymmdd>*_*<hhmmss>*	Reports file manually created by a user at specified time on the specified system.
*<hostname>*_**TraceData**_ *<printer error code>*_*<yyyymmdd>*_*<hhmmss>*	Reports/trace file created by the specified system for a printer error at specified time.

Manually create a log file with reports/trace files in Settings Editor

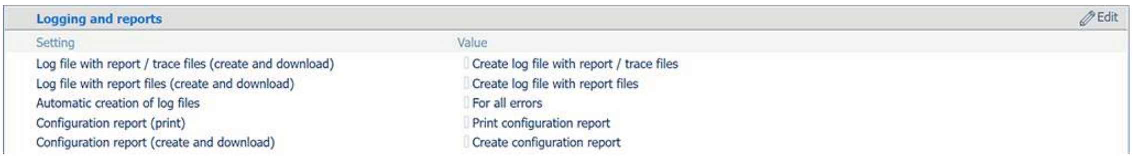
To create the log file with reports/trace files, use the **[Create log file with report / trace files]** setting. Create this log file immediately after a print error occurs.

NOTE
Create the alternative log file with **[Create log file with report files]** setting only on request by the Service organization.

1. Open the **Settings Editor** and go to: [Support]→[Troubleshooting].

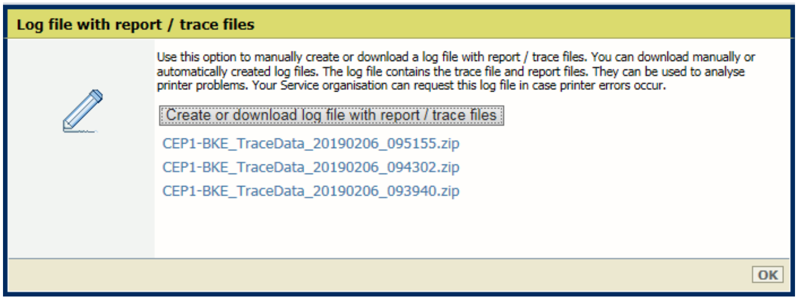
[Troubleshooting] tab

2. Go to the [Logging and reports] section.



[Logging and reports] section

3. Click [Create log file with report / trace files].
4. In the [Log file with report / trace files (create and download)] dialog box, click [Create log file with report / trace files] to **create** a new log file.



Create a log file with report/trace files

5. Click [OK].

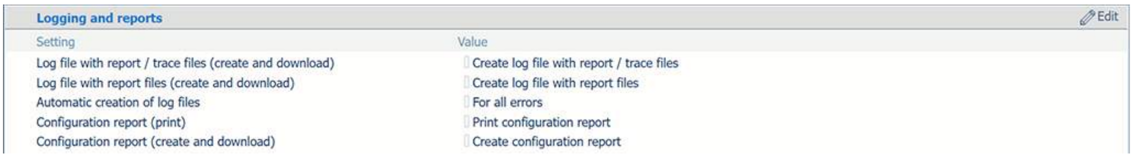
Download a log file from Settings Editor

Log files can be created manually or automatically by the printer when an error occurred.

1. Open the **Settings Editor** and go to: [Support]→[Troubleshooting].

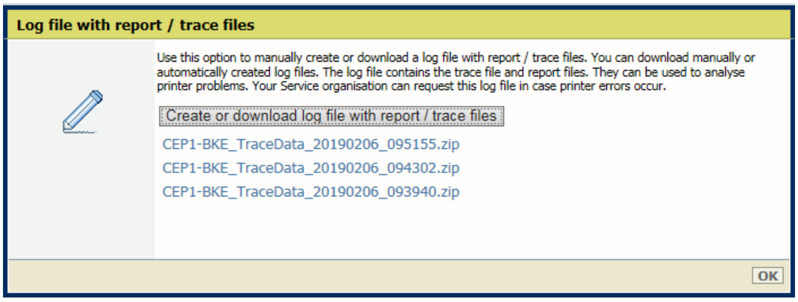
[Troubleshooting] tab

2. Go to the [Logging and reports] section.



[Logging and reports] section

3. Click [Create log file with report / trace files].
4. In the [Log file with report / trace files (create and download)] dialog box, click the required log file to **download** the file.



Create or download log file with report / trace files

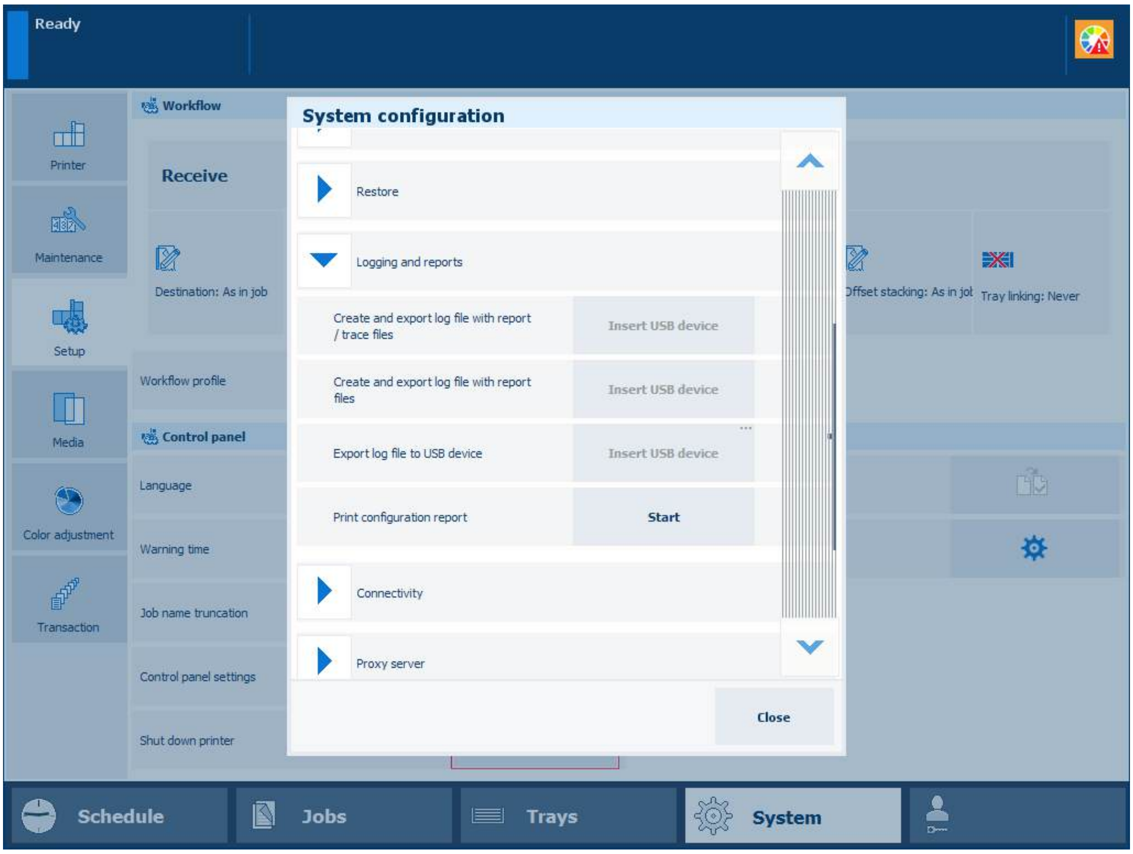
5. Store the log file on an external location.
6. Click [OK].

Create a log file on control panel

To create the log file with reports/trace files, use the **[Create log file with report / trace files]** setting. Create this log file immediately after a print error occurs.

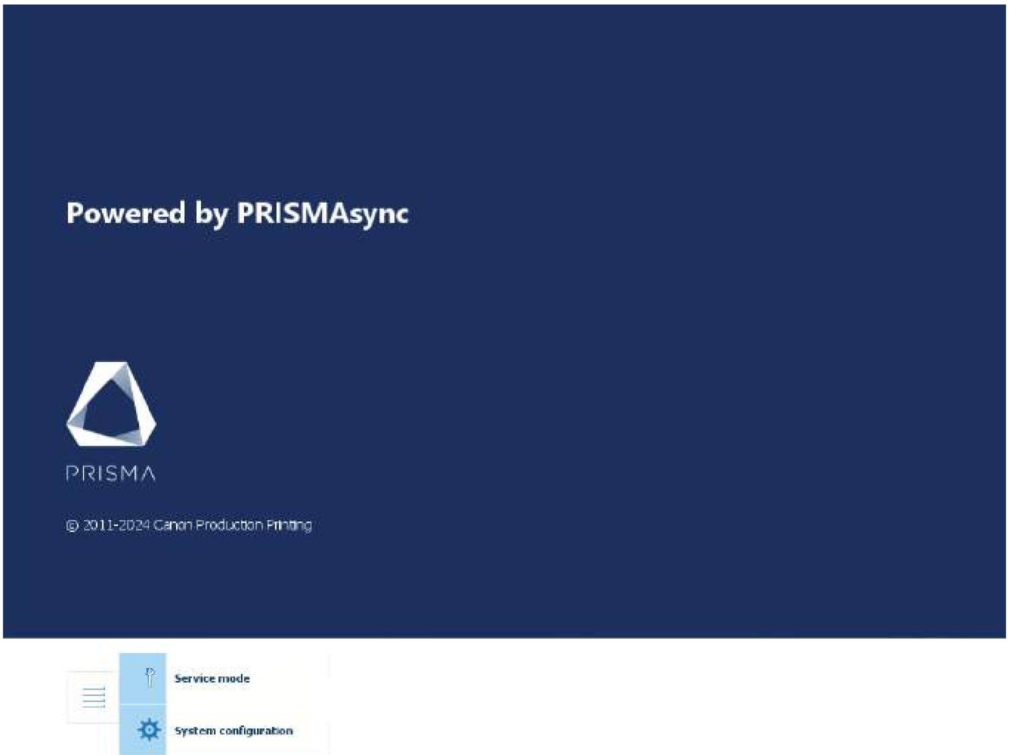
NOTE
Create the alternative log file with **[Create log file with report files]** setting only on request by the Service organization.

1. On the control panel, touch: [System]→[Setup]→[System configuration]→[Logging and reports].



System configuration

NOTE
You can also touch the button on the left-hand side of the start screen of the control panel and touch [System configuration].



Start screen of control panel

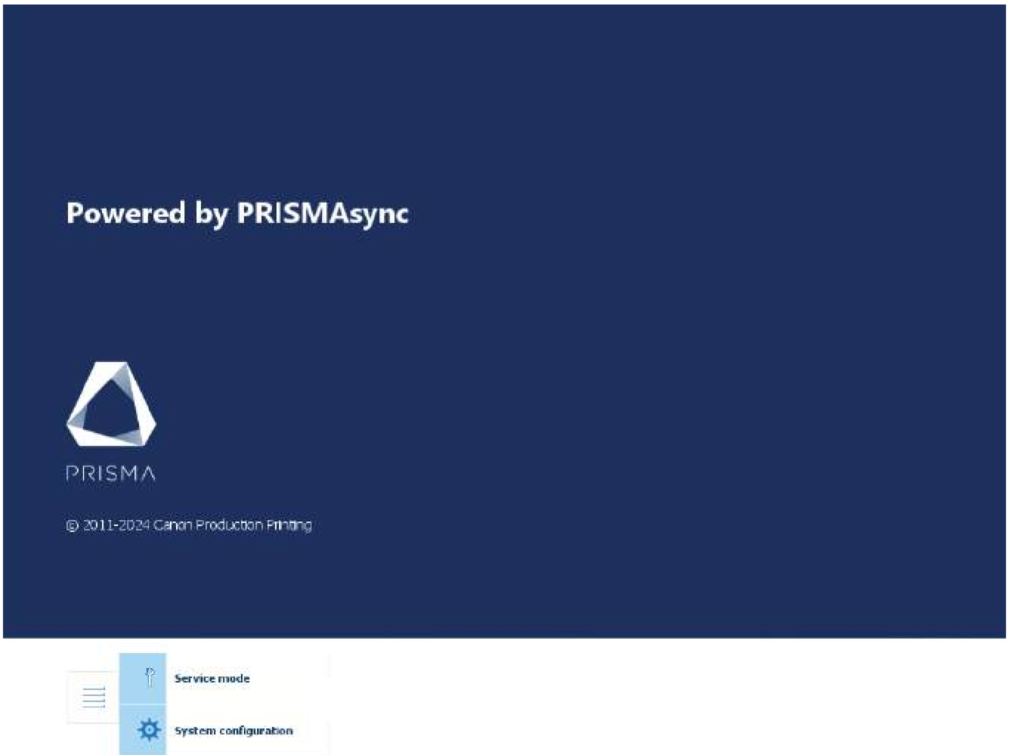
2. Insert a USB device into a USB port of the control panel.
3. Touch [Create] in the [Create and export log file with report / trace files].
4. The log file is being created.
5. The last created log file is selected in the [Export log file] dialog box.
6. Touch [OK] to export this log file to the USB device.
   Touch [Cancel] to download this log file at a later moment.

Export a log file to USB device from control panel

You can export a manually or automatically created log file to a USB device.

1. On the control panel, touch: [System]→[Setup]→[System configuration]→[Logging and reports].

NOTE
You can also touch the button on the left-hand side of the start screen of the control panel and touch [System configuration].

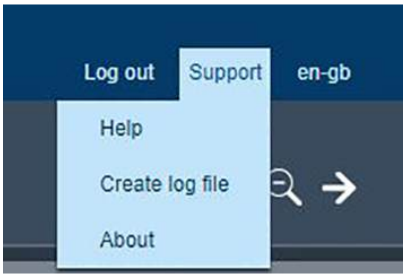


Start screen of control panel

2. Insert a USB device into a USB port of the control panel.
3. Touch [Export].
4. Select the required log file.
5. Touch [OK] to export the log file to the USB device.

Create a log file from PRISMAsync Remote Manager

1. In the top right-hand menu, click [Support]→[Create log file].

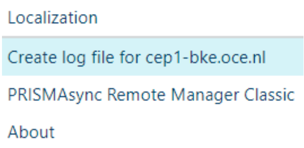


Support menu

2. You are redirected to the [Create log file with report / trace files] setting of the Settings Editor that belongs to the selected printer.
3. When requested, log in to the Settings Editor.
Use the first two instructions of this topic how to create and download the log file.

Create a log file from PRISMAremote Manager

1. To create a log file, click ⋮ in the header and select [Create log file for printer_name].

   Create log file

2. You are redirected to the [Create log file with report / trace files] setting of the Settings Editor that belongs to the selected printer.
3. Log in to the Settings Editor.
   Use the first two instructions of this topic how to create and download the log file.

Create, download, and print configuration report

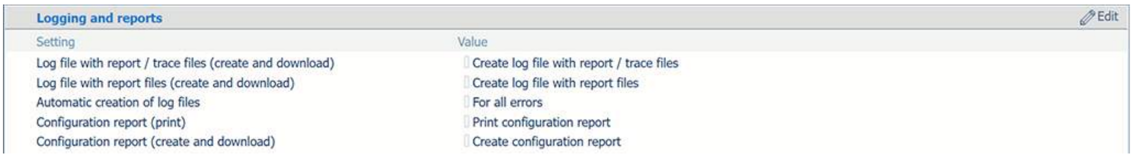
Create configuration report from Settings Editor

The configuration report provides an overview of the configuration of the printer. You can print the actual configuration report.

1. Open the **Settings Editor** and go to: [Support]→[Troubleshooting].

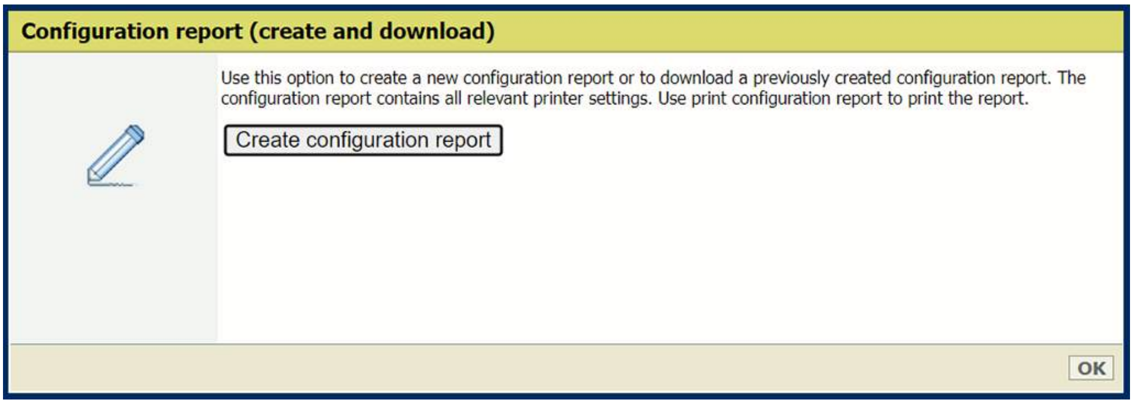
[Troubleshooting] tab

2. Go to the [Logging and reports] section.



[Logging and reports] section

3. Click the [Configuration report (create and download)] setting.
4. In the [Configuration report (create and download)] dialog box, click [Create configuration report].

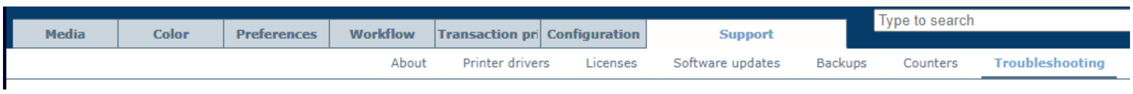


[Configuration report (create and download)] setting

5. Click the PDF file and store it on an external location.
6. Click [OK].

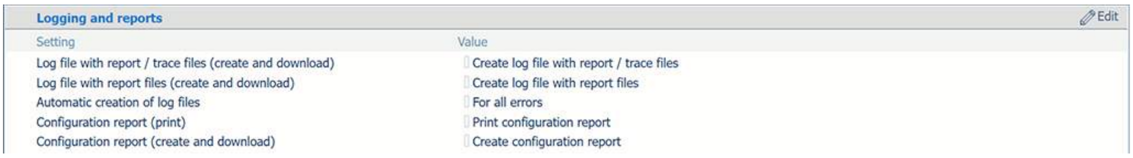
Print configuration report from Settings Editor

1. Open the Settings Editor and go to: [Support]→[Troubleshooting].



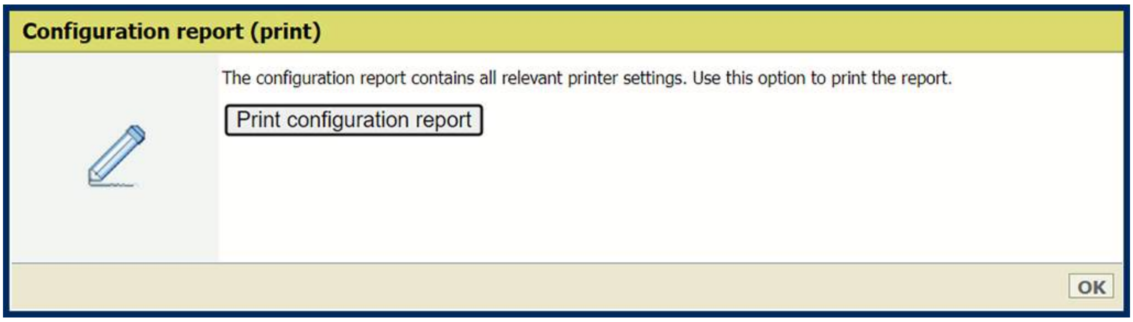
[Troubleshooting] tab

2. Go to the [Logging and reports] section.



[Logging and reports] section

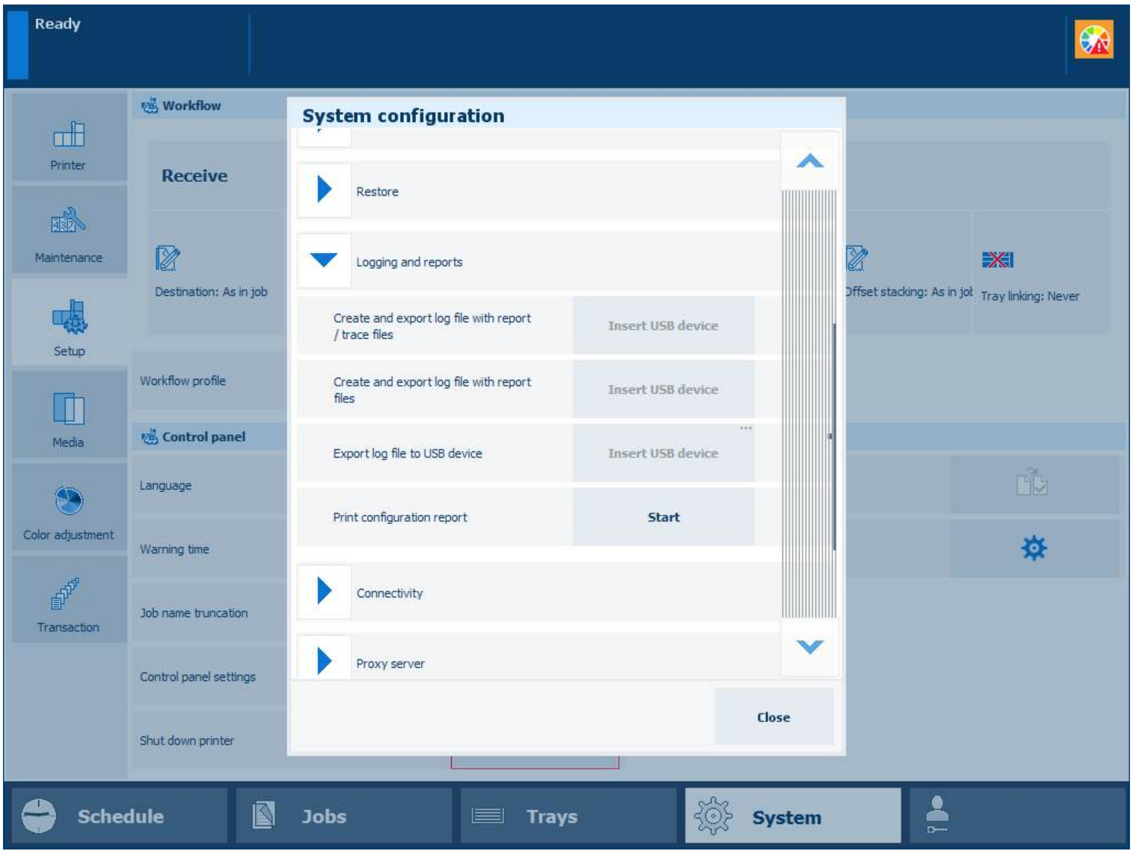
3. In the [Logging and reports] option group, click [Print configuration report].



4. Click the [Configuration report (print)] setting.
5. In the [Configuration report (print)] dialog box, click [Print configuration report].
6. Click [OK] to send the job to print the report to the print queue.

Print configuration report from control panel

1. On the control panel, touch: [System]→[Setup]→[System configuration]→[Logging and reports].



System configuration

NOTE
You can also touch the button on the left-hand side of the start screen of the control panel and touch [System configuration].

Start screen of control panel

2. In the [Print configuration report] field, touch [Start].
3. The job to print the report is sent to the print queue.

More information

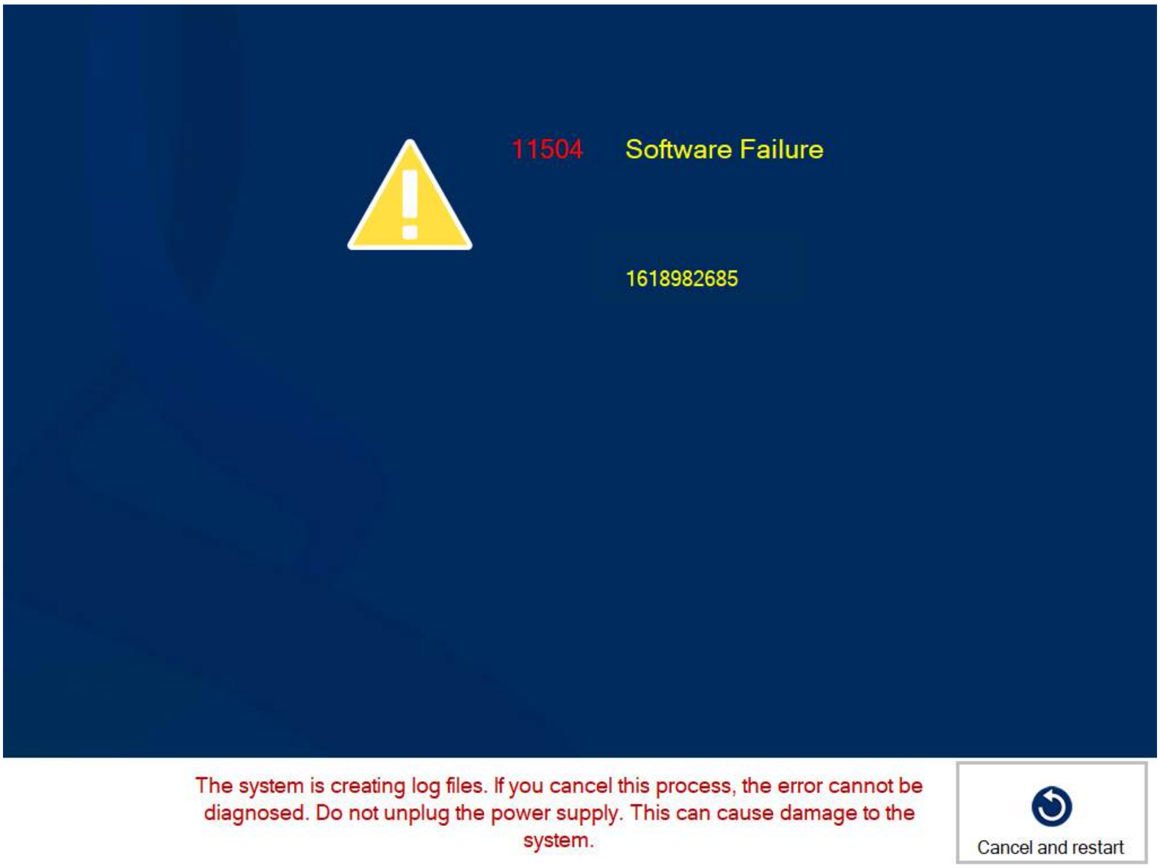
- *Create and download a log file* on page 546
- *Learn about print defects, print errors, and printer errors* on page 544

Cancel the creation of log files

When a software failure occurs, the system automatically creates log files that contain all the crucial information for your service organisation to analyse and solve the problem. Creating the log files can take up to half an hour. It is possible that the log files were already created at the previous occurrence of the error. Therefore, you might not want to wait half an hour for the log files to be generated. You can cancel the creation of log files to reduce downtime of the printer.

Cancel the creation of log files and restart the system

Touch [Cancel and Restart] to cancel the creation of log files.



Example of a software failure

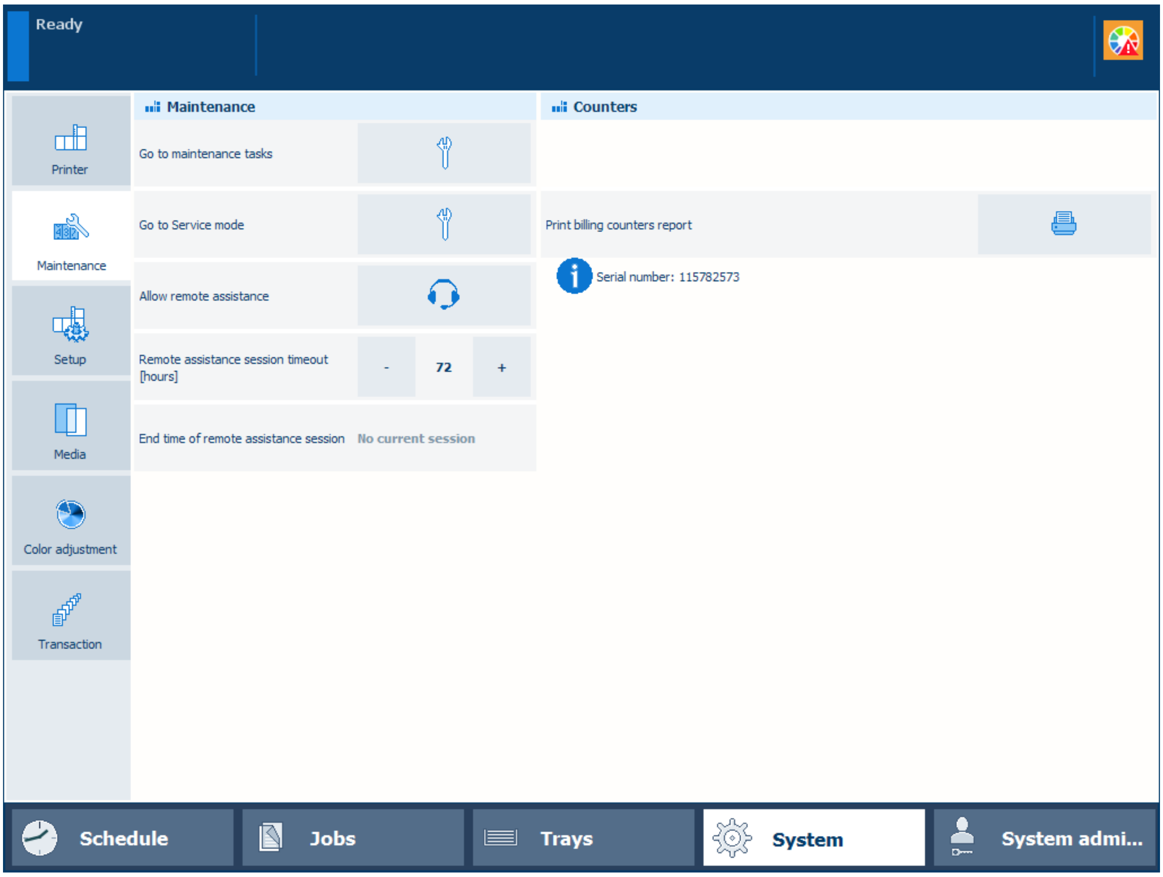
NOTE
It is advised to create log files regularly. This way it will be easier to analyse and solve the problem.

Use remote assistance

When you have a question or a problem, you can use remote assistance. The Service organization or the support helpdesk can provide support via a remote connection.

The remote assistance session is active when you confirm the assistance request and as long as the session time has not expired.

To use remote assistance, the system administrator must first enable the remote connection on the control panel. To do that, go to: [System]→[Setup]→[System configuration]→[Remote connection].



Remote assistance

Define the remote assistance session timeout

You can only define the session timeout when there is no active remote assistance session.

1. Touch [System]→[Maintenance].
2. Define the timeout in the [Remote assistance session timeout [hours]] field. The default value is 72 hours. The range is 1 - 168 hours.

Start a remote assistance session

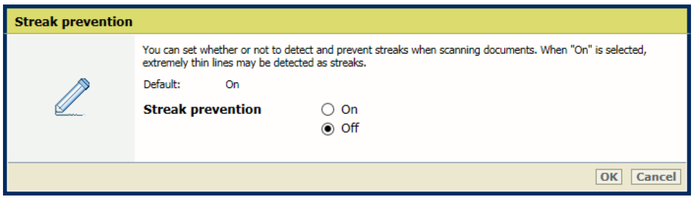
When a remote assistance session is initiated, you can establish the remote assistance connection.

1. Touch [Allow remote assistance].
2. The setting [End time of remote assistance session] shows when the session will end.

Optimize the scan quality of the print system

The print system detects photos, rasters and fine lines on the originals and makes sure that the copy and scan job have the correct layout and quality.

Below you find an overview of solutions on problems with scan and copy output.

<table>
<tr><th>Problem</th><th>Solution</th></tr>
<tr><td>A scanned page is upside down.</td><td>Place the original, with the correct orientation, into the automatic document feeder or on the glass plate, and then scan the originals again.</td></tr>
<tr><td>Every second scanned page is upside down.</td><td>Select the correct original and document layout settings, and scan the originals again.</td></tr>
<tr><td>The image position is not correct on all pages.</td><td>Place the originals with the correct orientation into the automatic document feeder, and then scan the originals again.
Select the correct original and document layout settings, and scan the originals again.
Use another application to check the results.</td></tr>
<tr><td>The information area on the page is not scaled correctly.</td><td>Select the correct original and document layout settings, and scan the originals again.
Use the correct zoom factor.</td></tr>
<tr><td>The information on the page is too light or too dark.</td><td>Select the correct [Adjust image] settings.</td></tr>
<tr><td>Extremely thin lines are detected as streaks.</td><td>[You can set whether or not to detect and prevent streaks when scanning documents. When "On" is selected, extremely thin lines may be detected as streaks.]
Open the Settings Editor and go to: [Preferences]→[System adjustments].
Disable the [Streak prevention] option.

Click [OK].</td></tr>
</table>

More information

- *Use the automatic document feeder to copy or scan* on page 178
- *Use the glass plate to copy or scan* on page 181
- *Learn about copy job templates and settings* on page 182
- *Learn about scan jobs templates and settings* on page 191
- *Zoom function* on page 586

Fix "density not within required range" error

In exceptional cases the shading correction procedure fails with a "density not within required range" message. If this happens, perform the correction procedure to fix the problem.

Procedure

1. Touch [System]→[Color adjustment]→[Select printer calibration procedures].
2. Select [Auto gradation adjustment].
3. Select [Shading correction].
4. Touch [Start printer calibration].
5. Follow the instructions on the control panel.
6. Touch [System]→[Color adjustment]→[Select printer calibration procedures].
7. Deselect [Shading correction].
8. Select [Auto gradation adjustment].
9. Touch [Start printer calibration].
10. Follow the instructions on the control panel.
11. Close the menu.

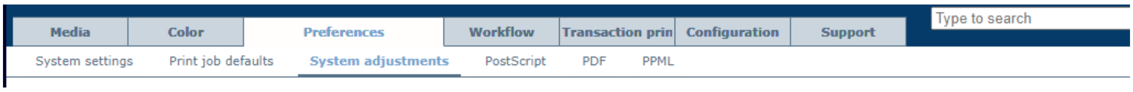
After you finish

When the [Shading correction] procedure still fails, contact your service organization.

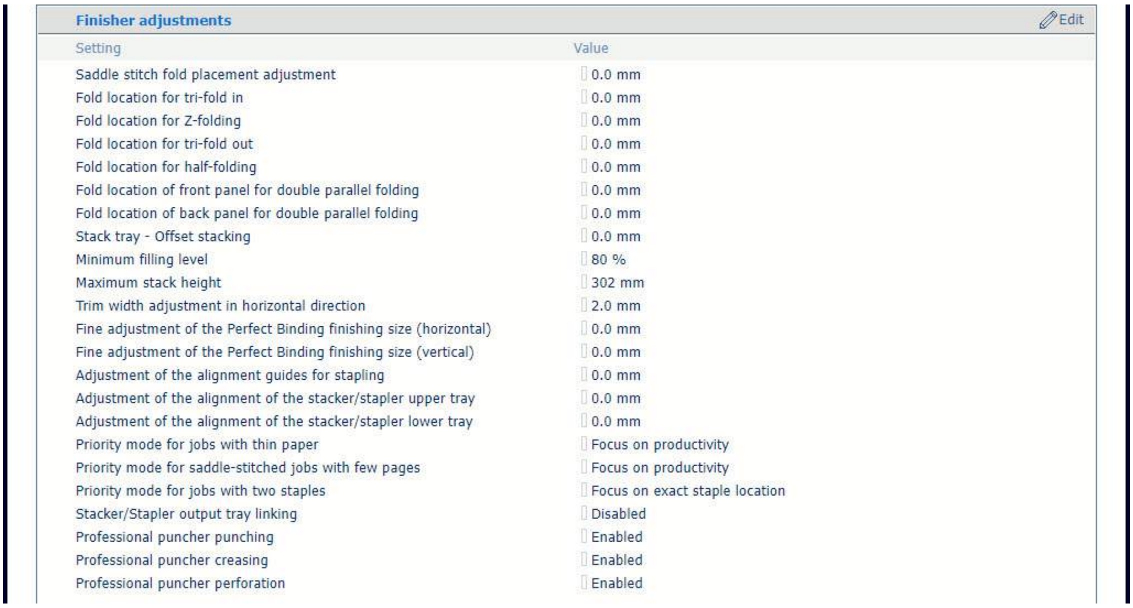
Define finisher adjustments

Note that the finisher adjustments affect all print jobs.

1. Open the Settings Editor and go to: [Preferences]→[System adjustments].



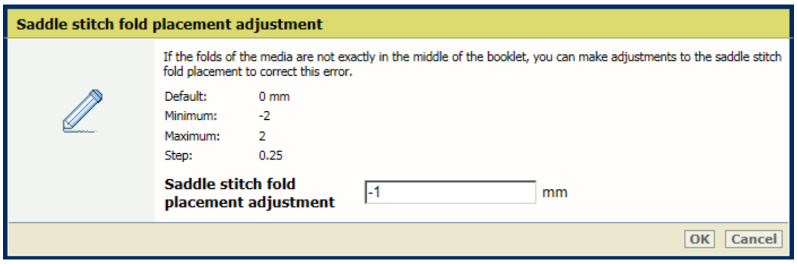
2. Go to the [Finisher adjustments] setting.



Finisher adjustment settings

Adjust the location of the fold in booklets

Use the [Saddle stitch fold placement adjustment] setting to make adjustments to the saddle stitch fold placement.



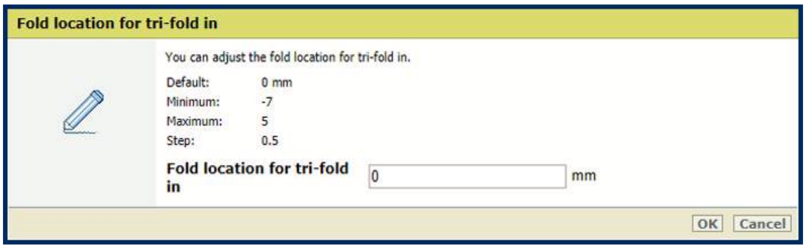
[Saddle stitch fold placement adjustment] setting

Adjust fold locations

Use the following settings to adjust the fold location:

- [Fold location for tri-fold in];
- [Fold location for Z-folding];
- [Fold location for tri-fold out];

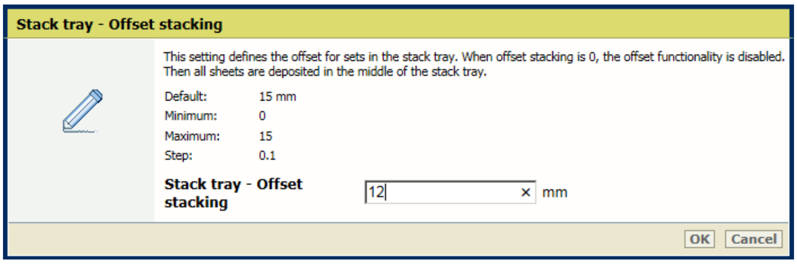
- [Fold location for half-folding];
- [Fold location of front panel for double parallel folding];
- [Fold location of back panel for double parallel folding].



[Fold location for tri-fold in] setting

Enable offset stacking in the stack tray

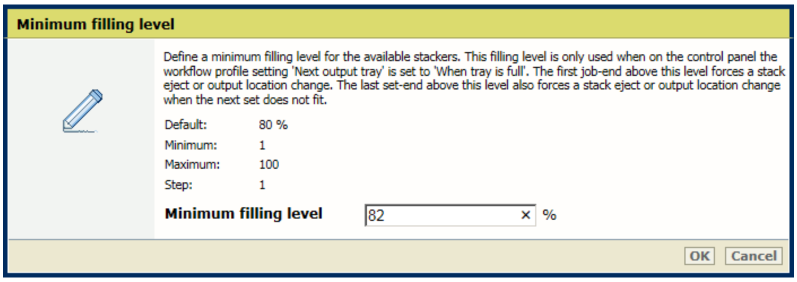
Use the [Stack tray - Offset stacking] setting to define the offset for the sets in the stack tray. When the value is set to 0, the offset functionality is disabled.



[Stack tray - Offset stacking] setting

Define the minimum filling level of the stackers

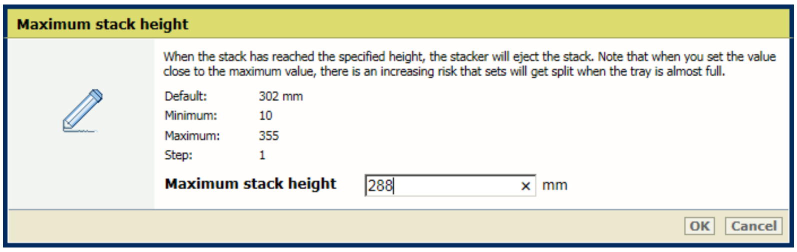
Use the [Minimum filling level] setting to define the minimum filling level for the available stackers.



[Minimum filling level] setting

Define the maximum stack height of the stackers

Use the [Maximum stack height] setting to define the maximum stack height of the stackers.

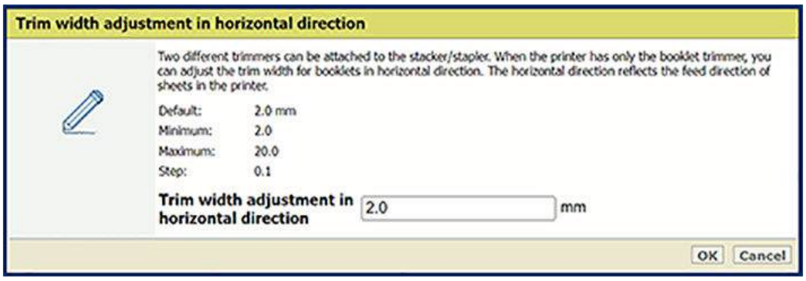


[Maximum stack height] setting

Adjust the trim width for booklets in horizontal direction

Two different trimmers can be attached to the stacker/stapler. When the printer has only the booklet trimmer, you can adjust the trim width for booklets in horizontal direction with the [Trim width adjustment in horizontal direction] setting.

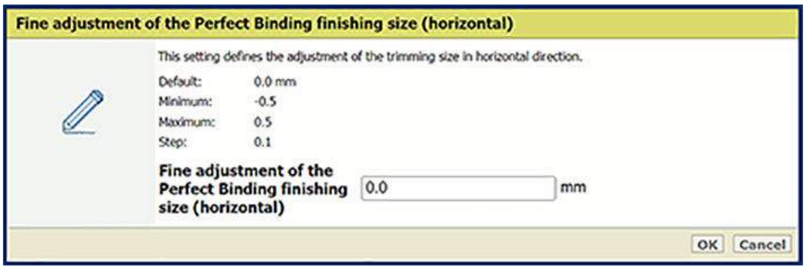
The horizontal direction reflects the feed direction of sheets in the printer.



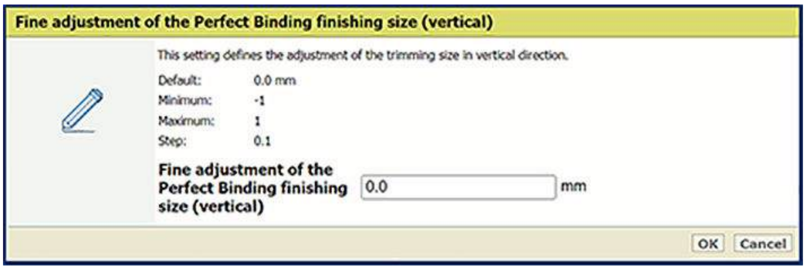
[Trim width adjustment in horizontal direction] setting

Adjust the perfect binding finishing size

Use the [Fine adjustment of the Perfect Binding finishing size (horizontal)] and [Fine adjustment of the Perfect Binding finishing size (horizontal)] settings to define the adjustment of the perfect binding finishing size in horizontal and vertical directions.



[Fine adjustment of the Perfect Binding finishing size (horizontal)] setting

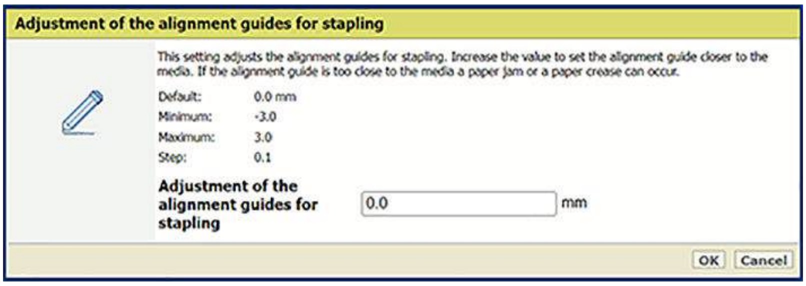


[Fine adjustment of the Perfect Binding finishing size (horizontal)] setting

Adjust the alignment guides for stapling

Use the [Adjustment of the alignment guides for stapling] setting to adjust the alignment guides for stapling. Increase the value to set the alignment guide closer to the media.

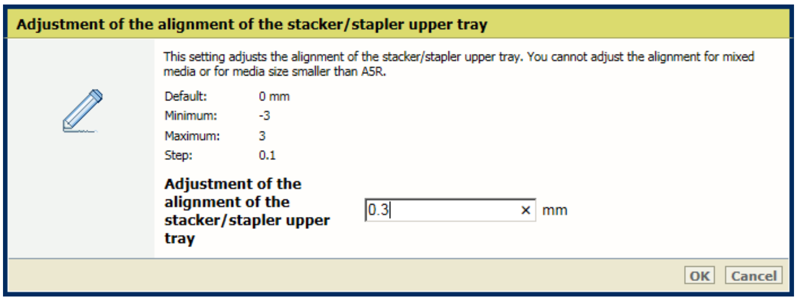
NOTE
If the alignment guide is too close to the media, a paper jam or a paper crease can occur.



[Adjustment of the alignment guides for stapling] setting

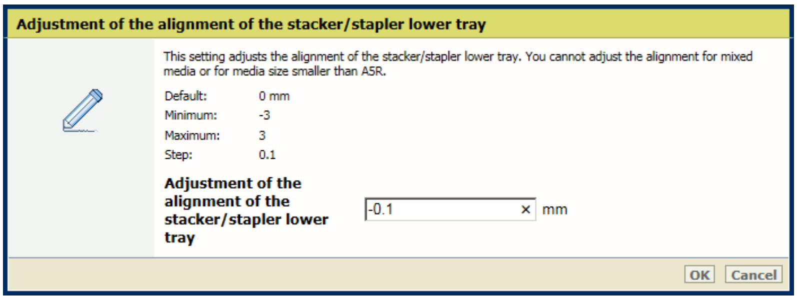
Adjust the trim width of the stacker/stapler trays

Use the [Adjustment of the alignment of the stacker/stapler upper tray] setting to adjust the alignment for the upper tray.



[Adjustment of the alignment of the stacker/stapler upper tray] setting

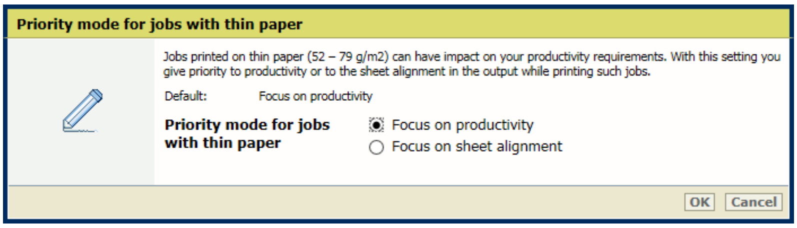
Use the [Adjustment of the alignment of the stacker/stapler lower tray] setting to adjust the alignment for the lower tray.



[Adjustment of the alignment of the stacker/stapler lower tray] setting

Productivity or sheet alignment for jobs with thin paper

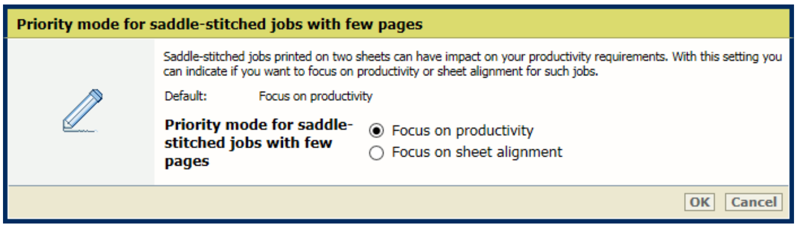
Use the [Priority mode for jobs with thin paper] setting to indicate whether priority must be given to productivity or sheet alignment in the output while printing jobs on thin paper.



[Priority mode for jobs with thin paper] setting

Productivity or sheet alignment for saddle-stitched jobs with few pages

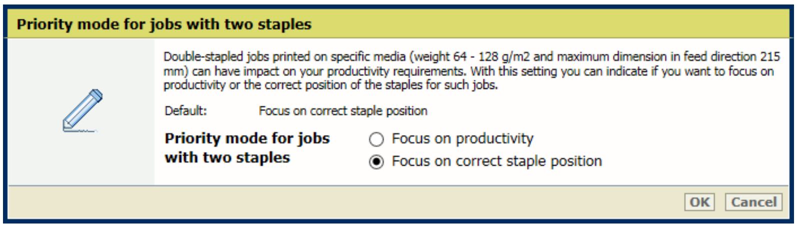
Use the [Priority mode for saddle-stitched jobs with few pages] setting to indicate whether priority must be given to productivity or sheet alignment in the output while printing the saddle-stitched jobs with few pages.



[Priority mode for saddle-stitched jobs with few pages] setting

Productivity or correct staple position for jobs with two staples

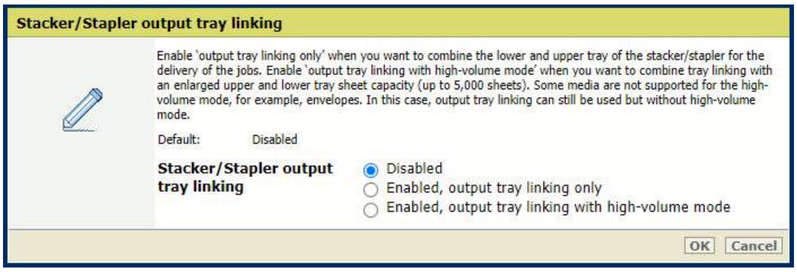
Use the [Priority mode for jobs with two staples] setting to indicate whether priority must be given to productivity or sheet alignment in the output while printing the jobs with two staples.



[Priority mode for jobs with two staples] setting

Enable stacker/stapler output tray linking

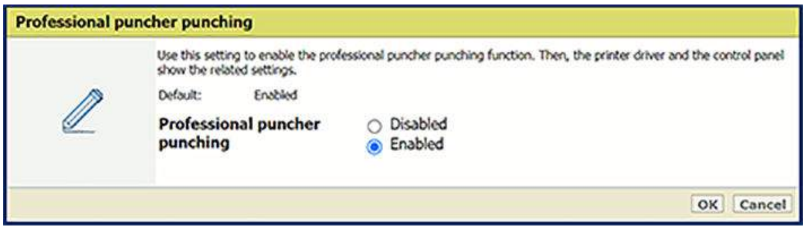
Use the [Stacker/Stapler output tray linking] setting to indicate whether the lower and upper tray of the stacker/stapler are combined for the delivery of the jobs. It is also possible to combine tray linking with an enlarged upper and lower tray sheet capacity.



[Stacker/Stapler output tray linking] setting

Enable functionalities of the professional puncher

Use the [Professional puncher punching], [Professional puncher creasing] and [Professional puncher perforating] settings to enable the punching, creasing or perforating function of the professional puncher. When these settings is enabled, the printer driver and control panel show the related settings.

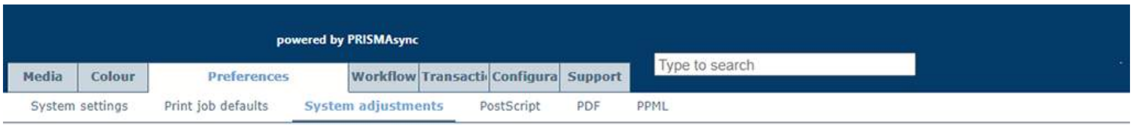


[Professional puncher punching] setting

Define print and scan quality adjustments

Note that the print and scan quality adjustments affect all print jobs.

1. Open the Settings Editor and go to: [Preferences]→[System adjustments].



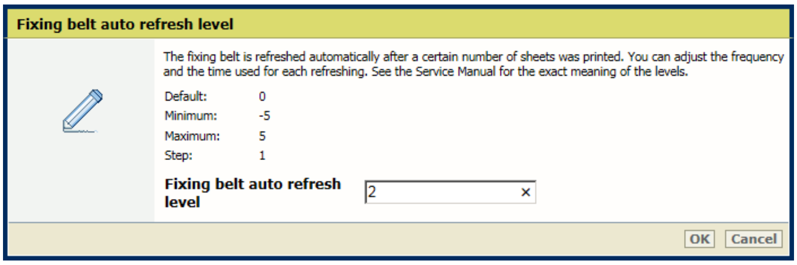
[System adjustments] tab

2. Go to the [Print quality adjustments] section.

Settings for print quality adjustments

Adjust when and how long the fixing belt needs to be refreshed

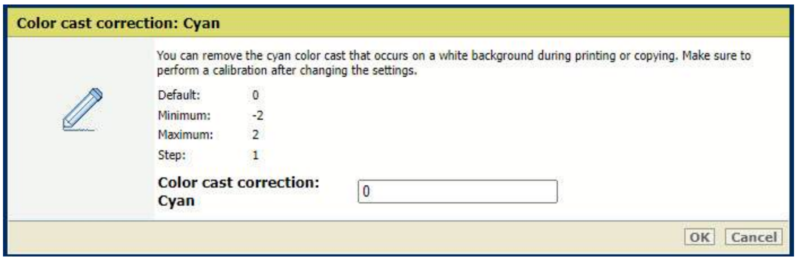
Use the [Fixing belt auto refresh level] setting to adjust the frequency and time used for refreshing of the fixing belt.



[Fixing belt auto refresh level] setting

Remove a color cast that occurs on a white background

Use the [Color cast correction: Cyan], [Color cast correction: Magenta], [Color cast correction: Yellow] and [Color cast correction: Black] settings to adjust the default correction level per color.

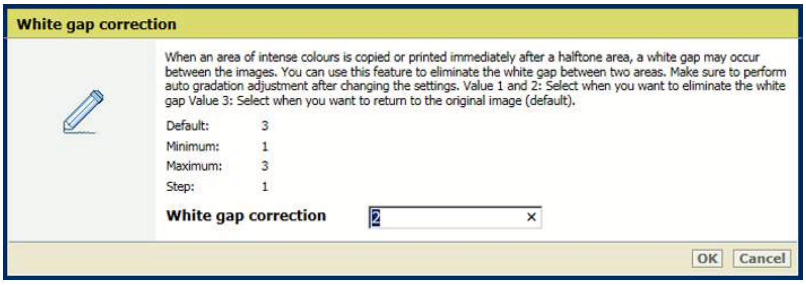


[Color cast correction: Cyan] setting

Perform a printer and media family calibration after changing these settings.

Remove white gaps between areas

Use the [White gap correction] setting to adjust the default level of the white gap correction.

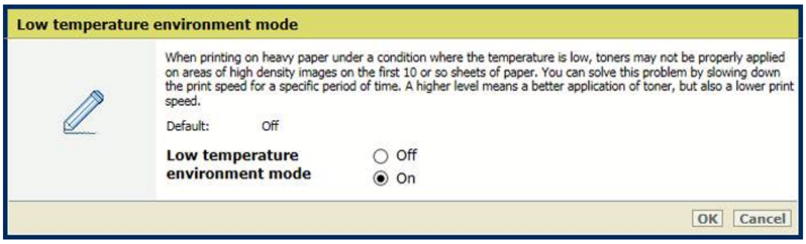


[White gap correction] setting

Perform an automatic gradation adjustment after changing this setting.

Low temperature environment mode

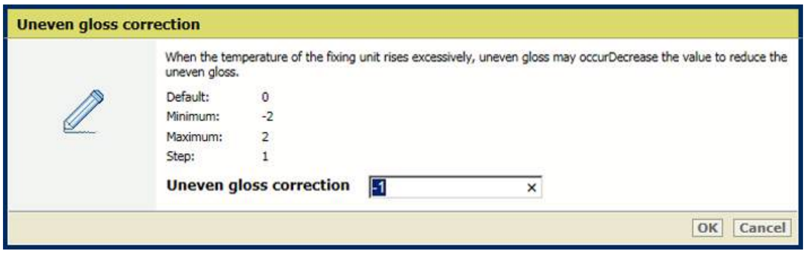
Enable the [Low temperature environment mode] setting to increase print quality for jobs on heavy paper but with loss of print speed.



[Low temperature environment mode] setting

Remove uneven gloss

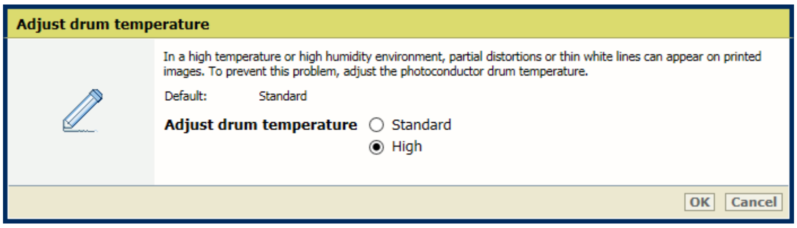
Use the [Uneven gloss correction] setting to decrease the level of uneven gloss correction to reduce the uneven gloss.



[Uneven gloss correction] setting

Adjust drum temperature

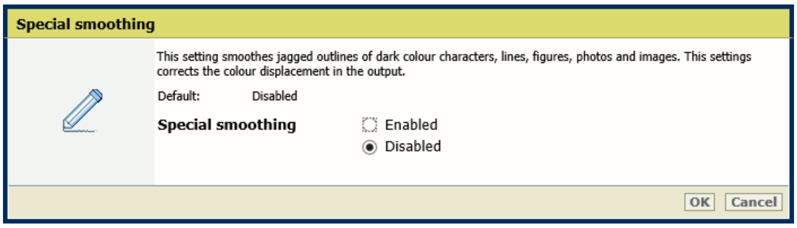
Use the [Adjust drum temperature] setting to adjust the drum temperature. Increase the drum temperature to remove partial distortions or thin white lines due in an environment with high temperature or high humidity.



[Adjust drum temperature] setting

Enable special smoothing

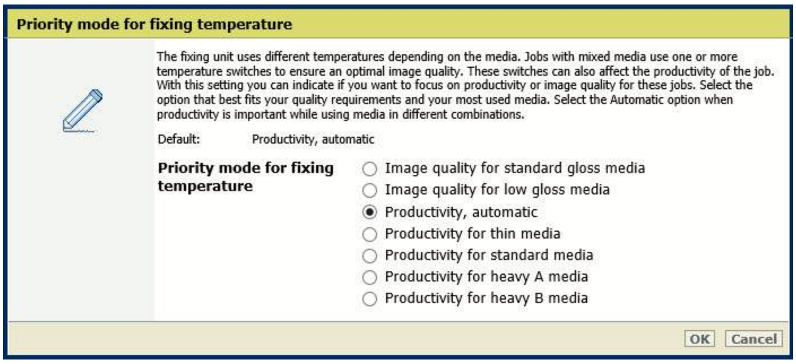
Use the [Special smoothing] setting to smoothen jagged outlines.



[Special smoothing] setting

Define the priority mode for fixing temperature

Use the [Priority mode for fixing temperature] setting to indicate if you want to focus on productivity or image quality for mixed media jobs. Select the value that best fits your quality requirements and most used media.

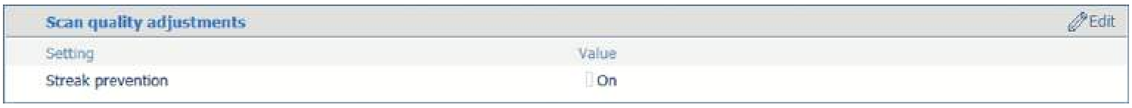


Media	Type/weight
Thin media	Uncoated 60 g/m^2 - 180 g/m^2 (16-lb bond–66-lb cover) Coated 80 g/m^2 - 180 g/m^2 (16-lb bond–66-lb cover)
Standard media	Uncoated 60 g/m^2 - 300 g/m^2 (16-lb bond–110-lb cover) Coated 106 g/m^2 - 256 g/m^2 (28-lb bond–95-lb cover)
Heavy A media	Uncoated 129 g/m^2 - 400 g/m^2 (79-lb bond–148-lb cover) Coated 129 g/m^2 - 256 g/m^2 (79-lb bond–96-lb cover)
Heavy B media	Uncoated 129 g/m^2 - 400 g/m^2 (79-lb bond–148-lb cover) Coated 129 g/m^2–400 g/m^2 (79-lb bond –148-lb cover)

Additional information about type and weight of media

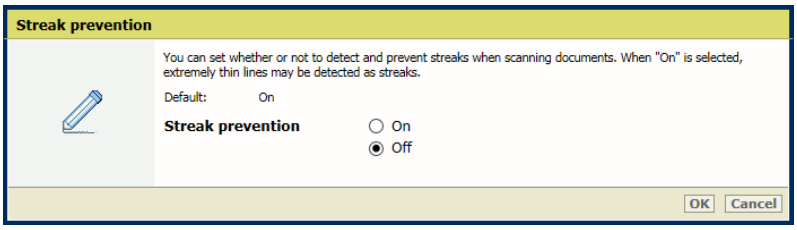
Scan documents with extremely thin lines

1. Go to the [Scan quality adjustments] section.



Settings for scan quality adjustments

2. Disable the [Streak prevention] setting to be able to scan documents with very thin lines.



[Streak prevention] setting

Automatic color mismatch correction

When the four different colors are not aligned correctly, use the automatic color mismatch correction to improve the alignment.

Incorrect alignment of colors

NOTE
You cannot perform the automatic color mismatch correction when the print system is printing.

Procedure

1. Calibrate the printer.

   IMPORTANT
   You must always perform a printer calibration before you continue with this procedure.

2. Check the print quality to decide if you want to proceed.
3. Touch [System]→[Maintenance]→[Go to maintenance tasks].
4. Press [Auto color mismatch correction].
5. Follow the instructions on the control panel.
6. Close the menu.

After you finish

If the procedure displays an engine error, contact your service organization.

More information

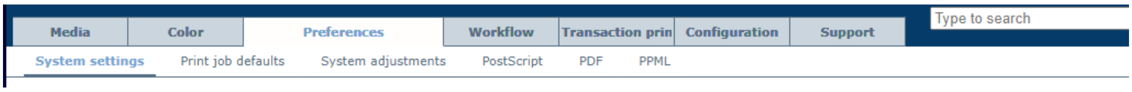
Calibrate the printer on page 368

Optimize image rotation

When you print face-up and header-down, if the front side of the sheet is landscape (LEF) or portrait (SEF), sheets are rotated by 180 degrees. This results in pages being turned upside down. If a document contains both portrait and landscape pages, some sheets in the output can have a different orientation than the rest of the document. To solve this problem, use the [Optimize image rotation] setting in the Settings Editor.

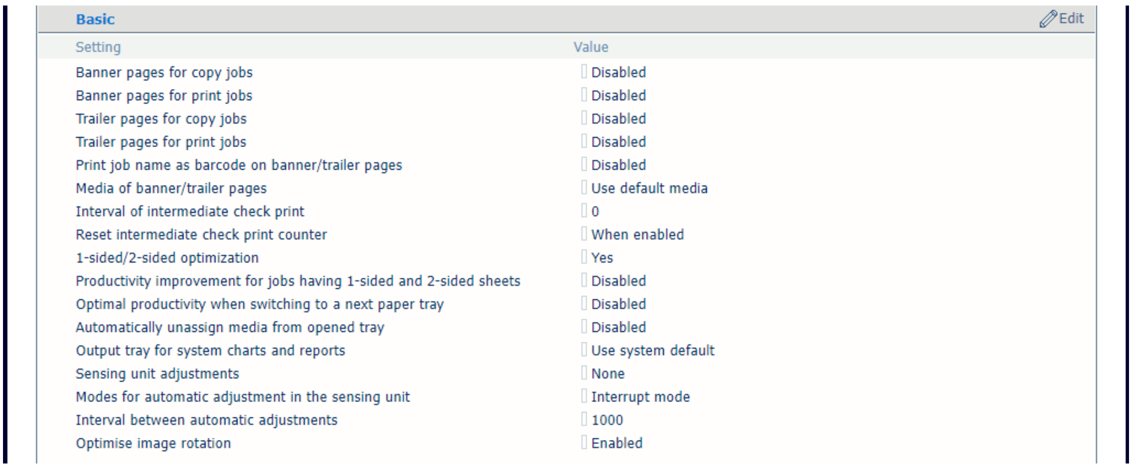
Procedure

1. Open the Settings Editor and go to: [Preferences]→[System settings] .

   [System settings] tab

2. Go to the [Basic] section.

   [Basic] section

3. Disable the [Optimize image rotation] setting. Once the setting is disabled, both face-up and face-down output will be rotated in the same way.
4. Click [OK].

Perform media adjustments

Learn about media adjustments

The print quality of the printed output depends on the media used. When you add media to the media catalog, you can define attributes that influence the print behavior of the print system for these media types.

When you notice that output media in the output tray is curled, perform a correction procedure to prevent curled sheets.

If you notice that the media quality of the printed output is not according to your expectations, perform a correction procedure to adjust the media registration.

The correction procedures change the media attributes in the media catalog, so that the adjustments are applied to all jobs that use the same media.

More information

- *Learn about the media-based workflow* on page 416
- *Correct curled output media* on page 573
- *Adjust media registration* on page 575

Correct curled output media

Media can arrive curled in the output tray. This is caused by temperature differences during the print process. You can perform a correction procedure to prevent curled sheets in the output trays. The results of the correction are immediately effective for jobs that use these media.

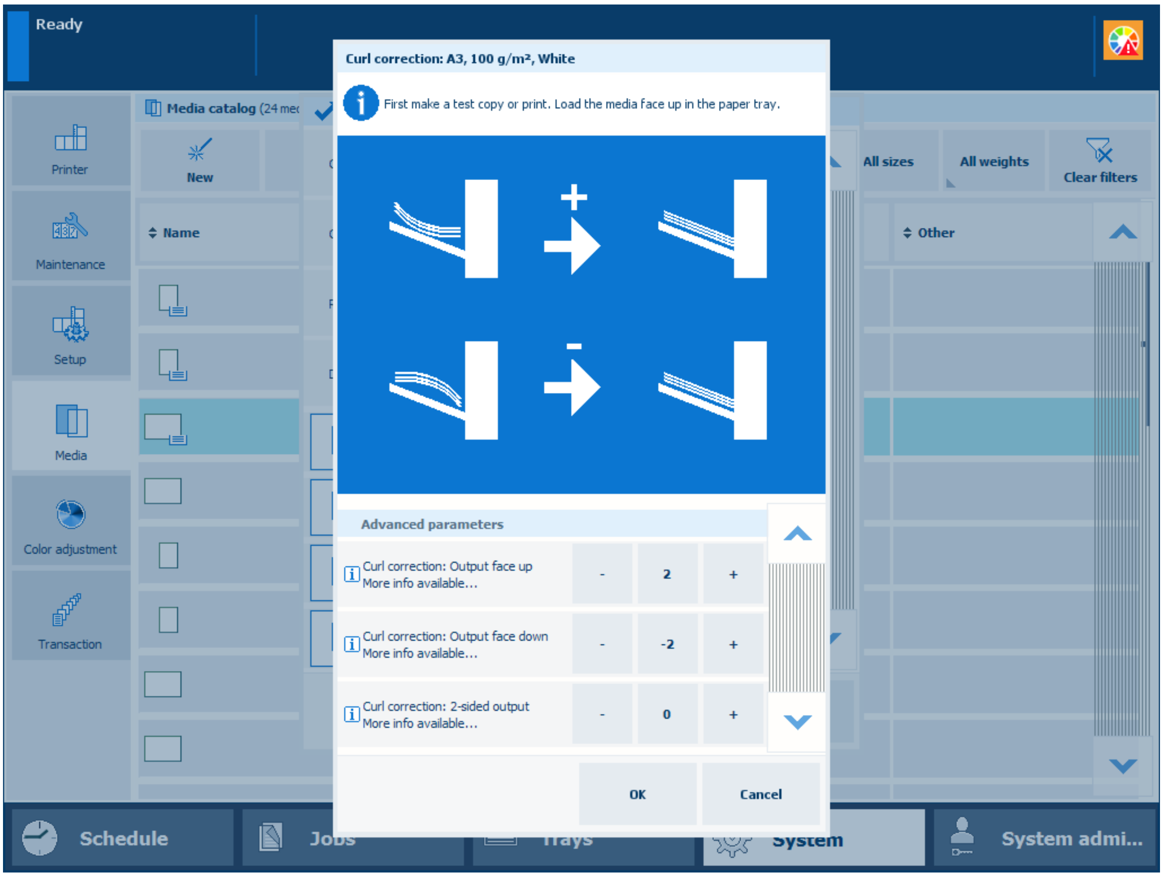
For media that need a curl correction on the length and width, create two different entrances in the media catalog. The correction procedure changes the media attributes of the media in the media catalog, so that the adjustments are applied to all jobs that use these media.

Procedure

1. Touch [System]→[Media].

   NOTE
   You can also find the curl correction procedure at the Trays view and at the media setting of the job properties.

2. Select the media from the media catalog.
3. Touch [Optimize]→[Curl correction].
4. Define the correction factor with the + and - buttons.
   Start with small increments to avoid paper jams.



Curl correction

5. Touch [OK].

Result

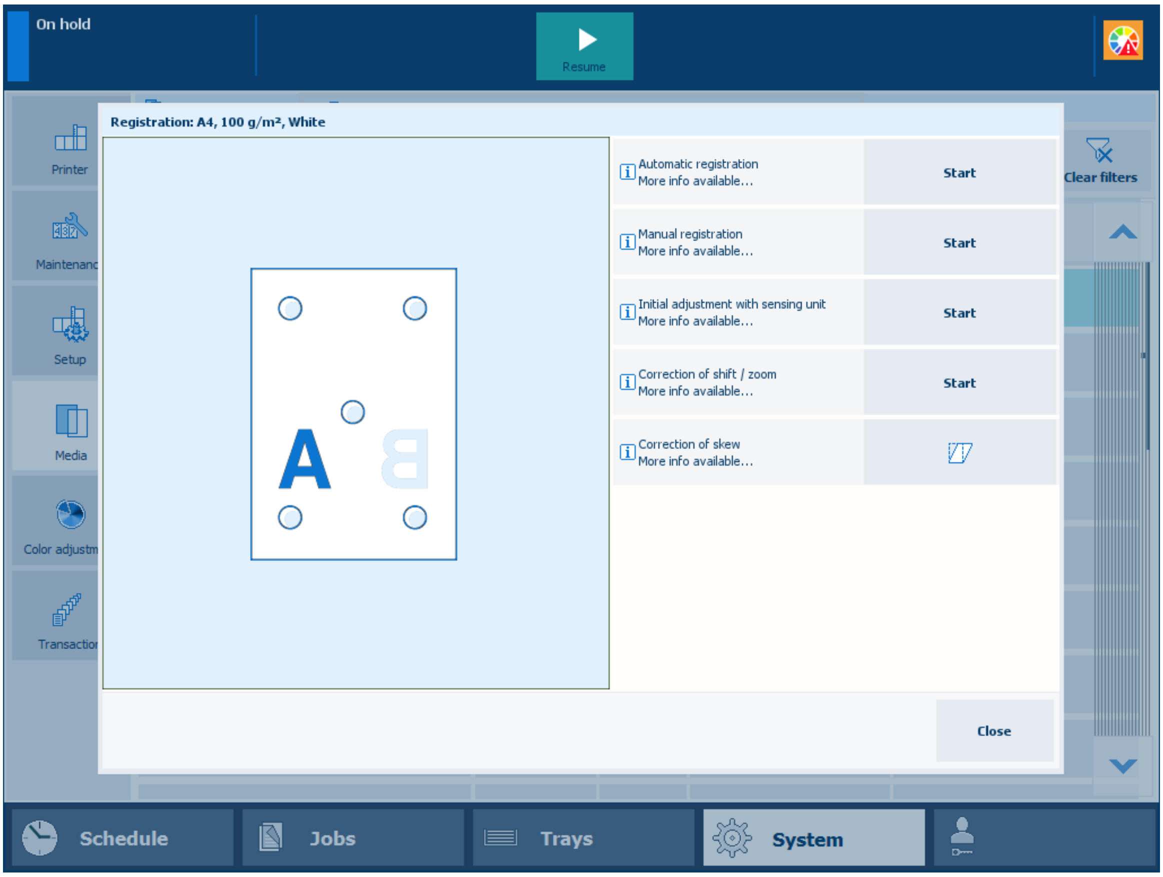
You find the correction values in the system adjustment settings on the Preferences tab of the Settings Editor.

Adjust media registration

The print system positions the front and back images precisely on both sides of the sheet during the print process. Although the media registration of the print system will mostly meet your expectations, you are able to adjust the media registration values when you notice deviations.

There are registration procedures for the following situations:

Situation	Procedure
You want optimal media registration.	Automatic registration This procedure prints registration charts that the system can measure via the glass plate or automatic document feeder. Registration values are automatically calculated and applied.
You want to adjust the media registration manually.	Manual registration This expert function has settings to manually change values of the image sheet registration.
You want optimal media registration (sensing unit).	Initial adjustment with the sensing unit (only if installed) This procedure prints registration charts that the printer can measure via the sensing unit. Registration values are automatically calculated and applied.
You notice incorrect image positions, reduced or enlarged images.	Correction of shift/zoom This procedure prints registration charts which you must measure to correct shift and zoom deviations. Registration values are automatically calculated and applied.
You notice skew deviations.	Correction of skew This procedure prints registration charts which you must measure to correct skew deviations. Registration values are automatically calculated and applied.



The media registration procedures

Required tools

- Scanner of the printer
- Ruler
- Measuring loupe
- Sensing unit (optional)
- Glass plate (optional)
- Automatic document feeder (optional)

Procedure

1. Touch [System]→[Media].

   NOTE
   You can also find the media registration procedures at the Trays view and at the media setting of the job properties.

2. Select the media from the media catalog.
3. Touch [Optimize]→[Registration].
4. Select the required correction procedure.
5. Follow the instructions of the wizard.
6. Close the menu.

Result

The media registration procedure changes the media attributes of the media in the media catalog. The adjustments are applied to all jobs that use these media.

Correct color shift or mottles for media

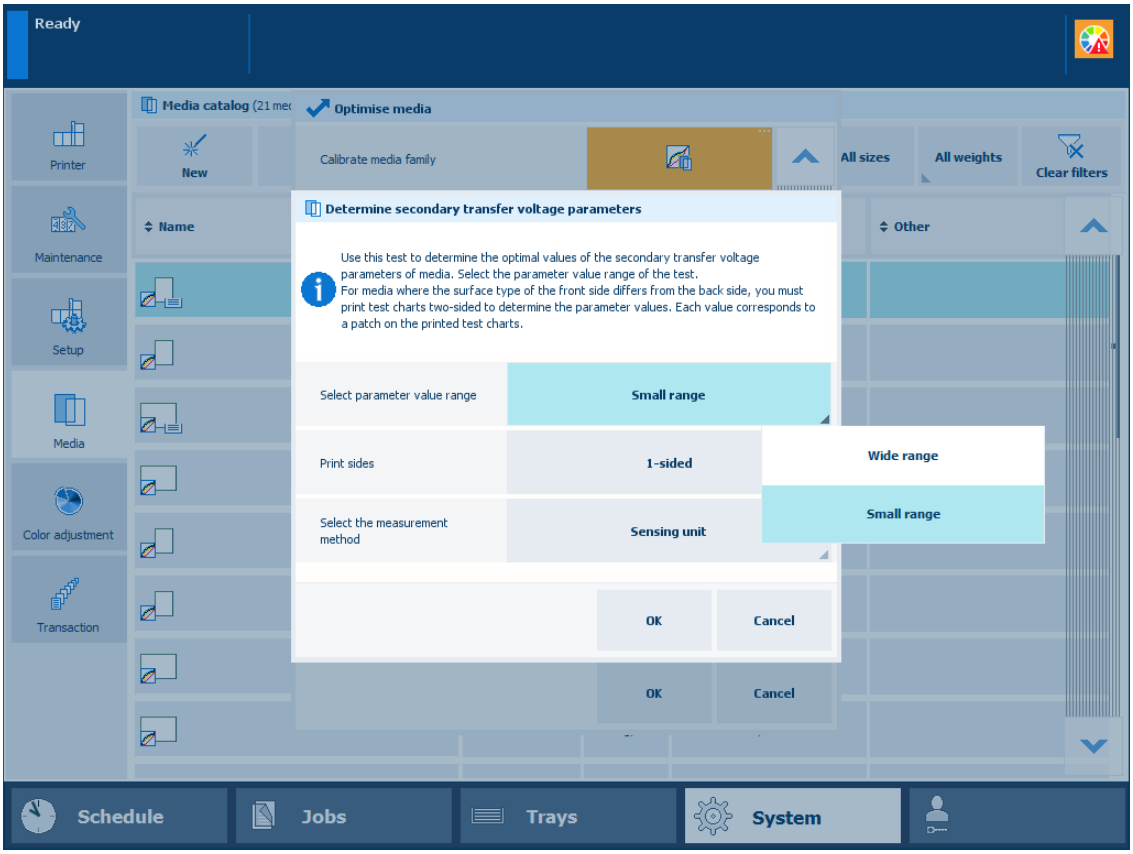
In prints of some media, a color shift or mottles can appear. You can correct this with the following procedure.

Before you begin

1. Calibrate the printer.
2. Calibrate the media family.

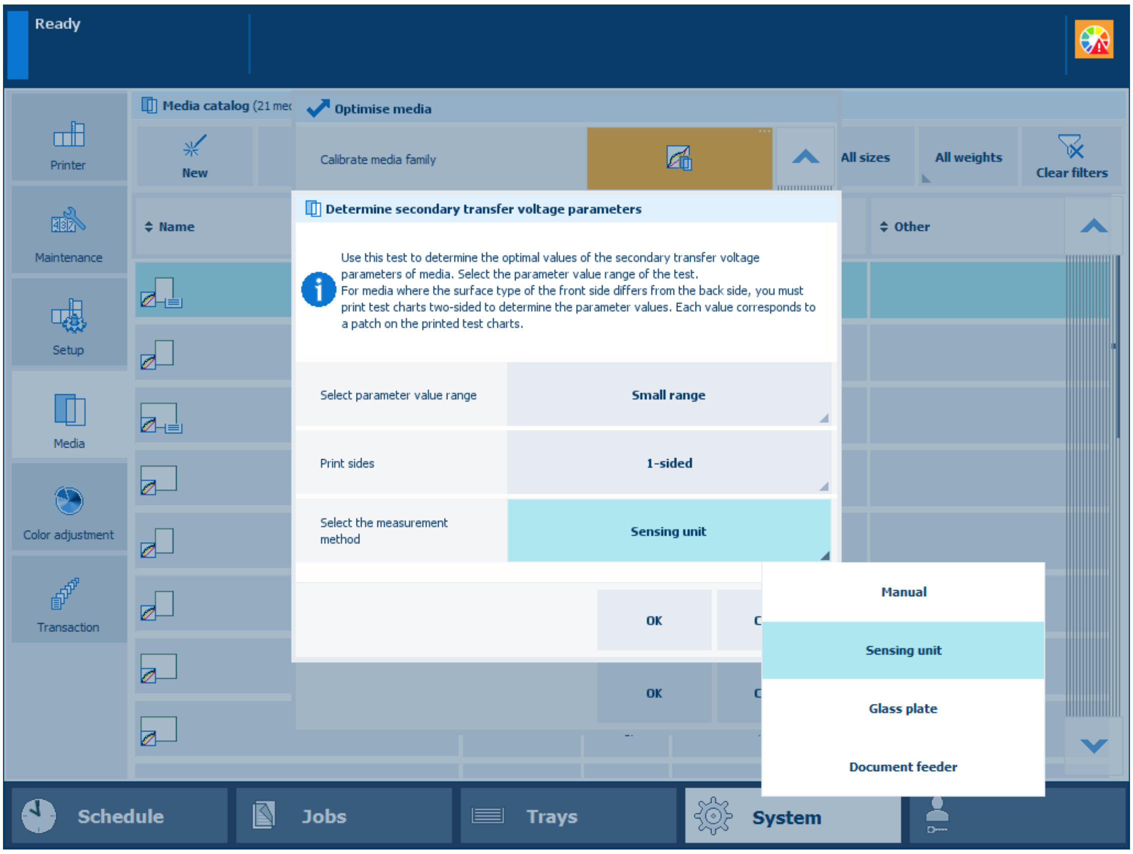
Procedure

1. Touch [System]→[Media].
2. Select the media from the media catalog.
3. Touch [Optimize].
4. Touch the [Determine secondary transfer voltage] procedure.
5. Select [Small range] in the [Select parameter value range] field to print a small range of patches with secondary transfer voltage parameters.



Parameter value range

6. Select [2-sided] in the [Print sides] field when the surface type of the front side differs from the back side.
7. Select the measurement method.



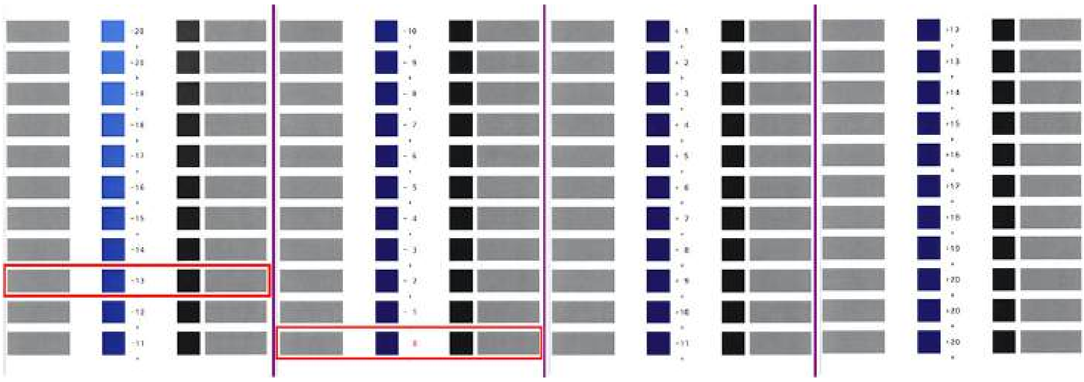
Measurement method

8. Touch [OK].

NOTE
- The sensing unit performs the procedure automatically. It measures the test charts and automatically adjusts the “Secondary transfer voltage” parameters.
- If you use the glass plate or document feeder, you need to place the printed sheets on the glass plate or in the document feeder. The rest of the procedure is performed automatically.

If you chose to determine the secondary transfer voltage parameters manually, continue with the next steps.

9. Follow the instructions on the control panel.
   1. Take the test charts.
      The current values of the secondary transfer voltage parameters are printed in a different color.

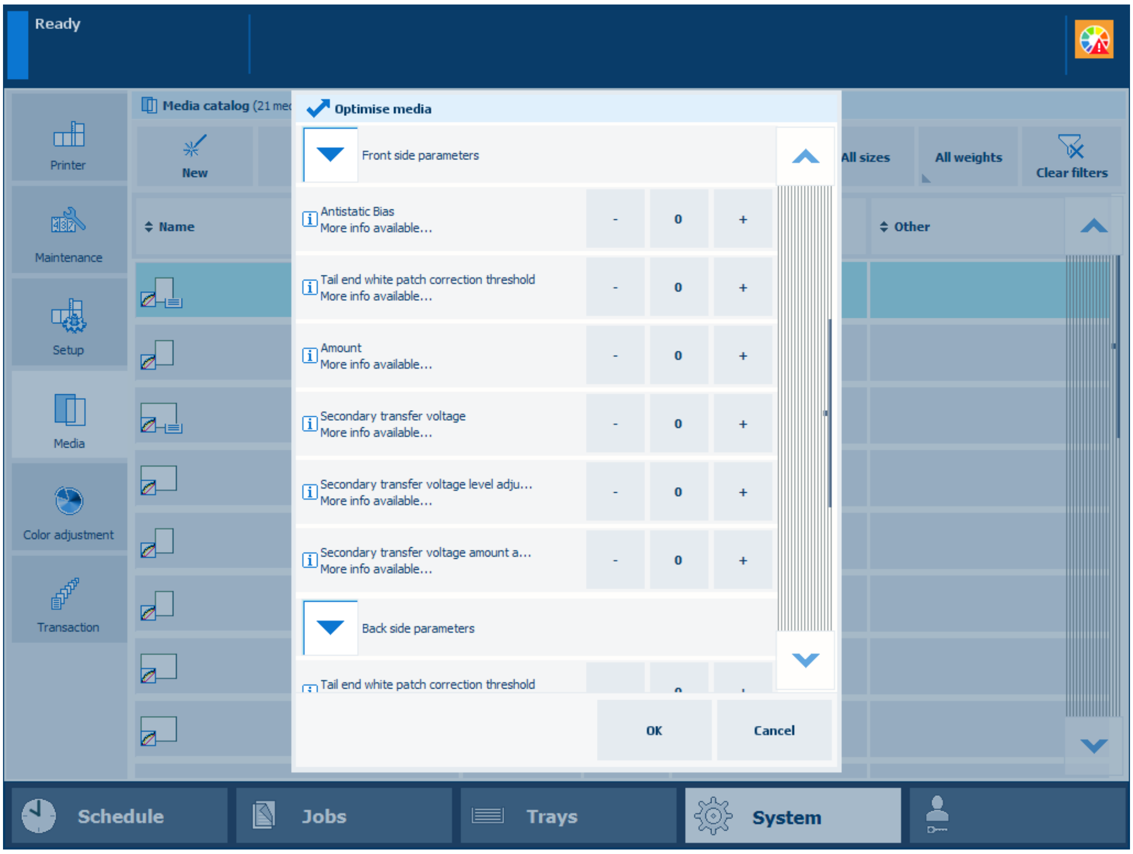


Example of test charts (wide range of patches, A3, 1-sided)

In the above figure is the current value of the secondary transfer voltage parameter 0. The patch with the optimal print quality is -13.

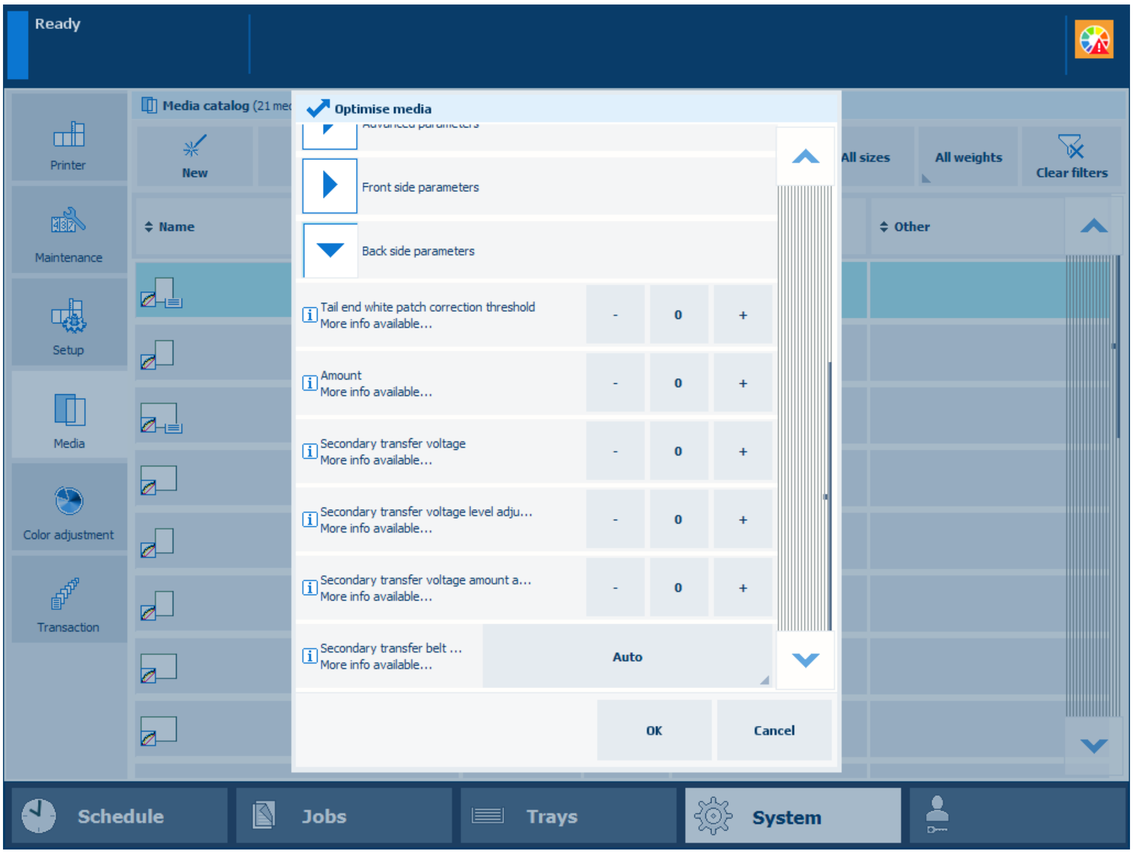
2. Identify the patches that have the optimal print quality.
   When a series of patches show the optimal print quality, take the middle value.
   When the patches do not show the optimal print quality, touch [Finish] and redo the procedure with the [Wide range] option.
   When you printed 2-sided test charts, examine the front side and back side.
   The front side charts shows a single asterisk symbol (*) between the values. The back side charts show two asterisk symbols (**) between the values. The current parameter values are printed in a different color.
3. Remember the parameter values of the patch with the optimal print quality. In this example: -13.
4. Touch [Finish] to close the wizard.
   Select the[Wide range] parameter in the [Select parameter value range] field to print a wide range of patches with secondary transfer voltage parameters.

10. Go to: [Front side parameters]→[Determine secondary transfer voltage].
11. Enter the determined parameter value. In this example: -13.



Parameter value for secondary transfer voltage of the front side

12. Go to: [Back side parameters]→[Determine secondary transfer voltage].
13. Enter the determined parameter value. In this example: -13.



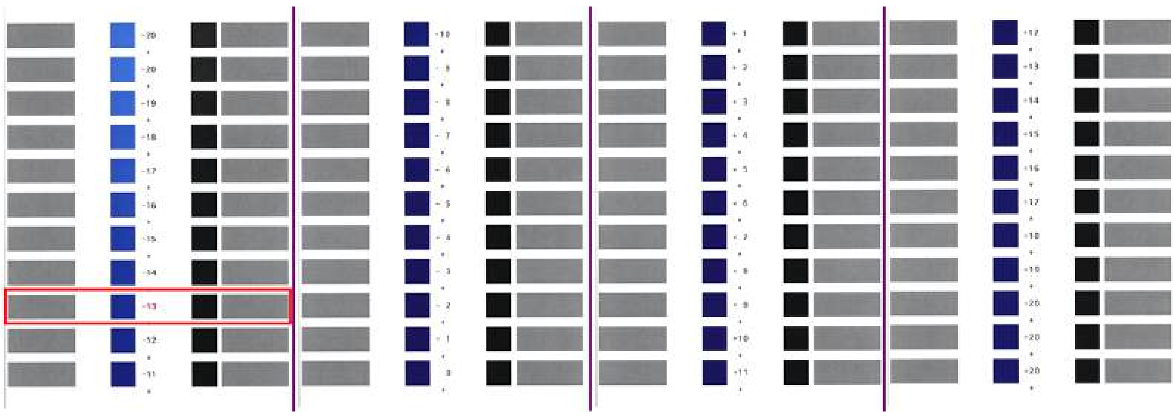
Parameter value for secondary transfer voltage of the back side

14. Touch [OK] to store the specified values of the secondary transfer voltage parameters.

IMPORTANT
When you perform the [Determine secondary transfer voltage] procedure without clicking [OK], the specified parameter values are **not** used.

Result

When you perform the [Determine secondary transfer voltage] procedure again, the new set values of the secondary transfer voltage parameters are printed in a different color.


Example of test charts (wide range of patches, A3, 1-sided) where the current parameter value is 0

More information

- *Calibrate the printer* on page 368
- *Calibrate a media family* on page 373
- *Perform shading correction* on page 370

Chapter 15
References

Zoom function

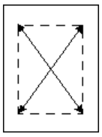
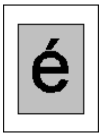
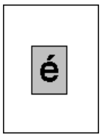
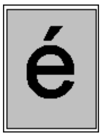
When you start a copy job or scan job, you can use the zoom function to reduce or enlarge an image.

The print system automatically scales the image according to the size of the page. To change the zoom factor manually, you can use the + and - buttons of the zoom function. When you have originals with a custom paper size, you can use the glass plate.

The zoom function of the automatic document feeder works differently from the zoom function of the glass plate.

The zoom function in the automatic document feeder

The following table shows the zoom of the automatic document feeder

Illustration	Description
	The center of the original is the origin of the image.
	Result of a scan with 100% zoom factor.
	Result of a scan with 50% zoom factor.
	Result of a scan with 200% zoom factor.

The zoom function of the glass plate

The following table shows the zoom of the glass plate.

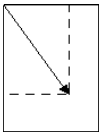
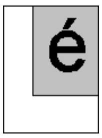
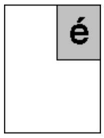
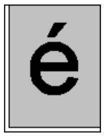
Illustration	Description
	The upper left-hand corner of the glass plate is the origin of the image.
	Result of a scan with 100% zoom factor.

Illustration	Description
	Result of a scan with 50% zoom factor.
	Result of a scan with a 200% zoom factor.

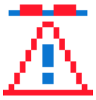
Status indicators

This topic provides an overview of the status indicators for:
- Jobs
- Paper trays
- Media
- Software updates
- Licenses
- Remote assistance

Job status indicators

The job lists have icons to indicate the job status.

Icon	Description
	The printer receives the job via the network (spooling).
	The printer must still convert the job into a raster image (raster image processing).
	The printer converts the job into a raster image (raster image processing).
	The printer prints a job.
	The printer is ready to print the job.
	The printer converts and prints the job simultaneously.
	The printer has only printed a part of the number of sets.
	The job contains a note for the print operator.
	The job is a variable data printing job.
	The job is a streaming job.

Icon	Description
	The job is a proof print.
	The job is a bundled job.
	The settings of the job can cause a conflict. Check the job settings to resume the print process.
	The job causes a raster image processing error.

Status indicators of paper trays

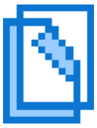
The control panel shows the following icons to indicate the status of the paper trays.

Icon	Description
	The paper tray holds media.
	The paper tray is empty.
	The paper tray is empty and a print job uses this paper tray.
	The paper tray is open.
	The paper tray contains too much media. Remove sheets from the stack to resume the print production.
	The external paper input module is not ready due to an empty paper tray, paper jam, or another reason.
	The paper tray contains media that the print system feeds in long-edge direction.
	The paper tray contains media, for example A4R / LTRR, that the print system feeds in short-edge direction.
	The paper tray is in use for the print production.
	The paper guides are set incorrectly.

Icon	Description
	The paper tray contains tab sheets.
	The paper tray of the inserter contains tab sheets as insert sheets.
	The paper tray of the inserter contains insert sheets.

Media indicators

Media is displayed on several locations of the control panel. Extra information about the media is displayed by icons.

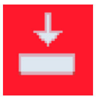
Icon	Description
	The media is required for a job in the list of scheduled jobs or list of waiting jobs.
	The media family of the media needs a media family calibration.
	The media is loaded into a paper tray.
	The media is used in a scheduled job and loaded into a paper tray.
	The warning icon indicates that the media definition needs attention. Check if all important attributes are defined.

Status indicators of software updates

When the configuration of downloading and installation of software updates is enabled in the Settings Editor, the dashboard of the control panel displays the status of the software updates.

Refer to the PRISMAsync Print Server administration guide how to configure software updates.

	One or more software updates are available for download.
	One or more software updates are ready for download at the specified time.

	One or more software updates are being downloaded.
	The download of one or more software updates has been paused.
	The download of one or more software updates failed.
	One or more software updates are ready for installation.
	One or more software updates are ready for installation at the specified time.
	One or more software updates are being installed.
	The installation of one or more software updates failed.

Status indicators of licences

The control panel shows the following icons to indicate the status of the licenses.

	A license is available.
	The introduced license is invalid.

Status indicators of remote assistance

The control panel shows the following icons to indicate the status of your remote assistance session.

	Remote assistance is connected.
	Remote assistance is enabled but has an error.

Color validation metrics

A color validation test has a series of metrics that are calculated and evaluated when the test is performed.

The tables lists all metrics that can be used in a color validation test. In practice, a color validation test uses a small subset of the available metrics.

The measurements are the measured Lab values of the patches of the printed target chart of the test.

△E00 (Delta E 2000)

Metrics	Description
Substrate ΔE00	Color difference in ΔE00 between the measured and the reference substrate white. For a side-by-side evaluation, the color control strip must be printed with the rendering intent absolute colorimetric. The reference substrate white is compared to the measured simulated substrate white of the printed media.
Maximum ΔE00	Highest color difference in ΔE00 between the measured Lab values of the color patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Average ΔE00	Average color difference in ΔE00 between the measured Lab values of the color patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
95th percentile ΔE00	The 95th percentile ΔE00 color difference between the measured Lab values of the color patches on the color control strip and the Lab values of the matching patches of the reference printing condition. This means that 95% of all measured patches have ΔE00 color differences smaller than or equal to the metric value.
Maximum ΔE00 CMYK solids	Highest color difference in ΔE00 between the measured Lab values of the CMYK solid patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔE00 CMY solids	Highest color difference in ΔE00 between the measured Lab values of the CMY solid patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔE00 RGB overprints	Highest color difference in ΔE00 between the measured Lab values of the RGB overprint patches on the color control strip and the Lab values of the matching patches of the reference printing condition.

NOTE
Delta E 2000, △E00 is the industry standard.

△E76 (Delta E 1976)

Metrics	Description
Substrate ΔE76	Color difference in ΔE76 between the measured and the reference substrate white. For a side-by-side evaluation, the color control strip must be printed with the rendering intent absolute colorimetric. The reference substrate white is compared to the measured simulated substrate white of the printed media.
Maximum ΔE76	Highest color difference in ΔE76 between the measured Lab values of the color patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Average ΔE76	Average color difference in ΔE76 between the measured Lab values of the color patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
95th percentile ΔE76	The 95th percentile ΔE76 color difference between the measured Lab values of the color patches on the color control strip and the Lab values of the matching patches of the reference printing condition. This means that 95% of all measured patches have ΔE76 color differences smaller than or equal to the metric value.
Maximum ΔE76 CMYK solids	Highest color difference in ΔE76 between the measured Lab values of the CMYK solid patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔE76 CMY solids	Highest color difference in ΔE76 between the measured Lab values of the CMY solid patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔE76 RGB overprints	Highest color difference in ΔE76 between the measured Lab values of the RGB overprint patches on the color control strip and the Lab values of the matching patches of the reference printing condition.

NOTE
Delta E 1976, △E76 is based on a less complicated and accurate distance calculation.

△H (Delta H)

Metrics	Description
Maximum ΔH CMYK solids	Highest color difference in ΔH between the measured Lab values of the CMYK solid patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔH CMYK solids and RGB overprints	Highest color difference in ΔH between the measured Lab values of the CMYK solid patches and the RGB overprint patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔH CMY solids and RGB overprints	Highest color difference in ΔH between the measured Lab values of the CMY solid patches and the RGB overprint patches on the color control strip and the Lab values of the matching patches of the reference printing condition.

Metrics	Description
Maximum ΔH CMY solids	Highest color difference in ΔH between the measured Lab values of the CMY solid patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Average ΔH composite grays	Average color difference in ΔH between the measured Lab values of the composite gray patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔH composite grays	Highest color difference in ΔH between the measured Lab values of the composite gray patches on the color control strip and the Lab values of the matching patches of the reference printing condition.

△L (Delta L)

Metrics	Description
Average |ΔL| composite grays	Average lightness difference in |ΔL| between measured Lab values of the composite gray patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum |ΔL| composite grays	Highest lightness difference in |ΔL| between measured Lab values of the composite gray patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Average weighted |ΔL| composite grays	Average lightness difference in weighted |ΔL| between measured Lab values of the composite gray patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum weighted |ΔL| composite grays	Highest lightness difference in weighted |ΔL| between measured Lab values of the composite gray patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Average |ΔL| K-only patches	Average lightness difference in |ΔL| between measured Lab values of the K-only patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum |ΔL| K-only patches	Highest lightness difference in |ΔL| between measured Lab values of the K-only patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Average weighted |ΔL| K-only patches	Average lightness difference in weighted |ΔL| between measured Lab values of the K-only patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum weighted |ΔL| K-only patches	Highest lightness difference in weighted |ΔL| between measured Lab values of the K-only patches on the color control strip and the Lab values of the matching patches of the reference printing condition.
Estimated black point |ΔL|	Lightness difference in |ΔL| between the measured darkest patch on the color control strip and the darkest patch of the reference printing condition. The darkest patch is the patch that has the lowest L* value.

NOTE
- The |ΔL| value refers to the **absolute** value of △L.
- The **weighted** |ΔL| puts more emphasis on the color differences for light grays, where the K value is lower than 50%.
- The Estimated black point |ΔL| metric is called an *estimation* because the darkest patch of the target chart is not necessarily the darkest possible color. There could be CMYK values that reproduce a darker tint. The output profile determines the real black point value.

△Ch (Delta Ch)

Metrics	Description
Average ΔCh composite grays	Average color difference in ΔCh between measured Lab values of the composite grays on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum ΔCh composite grays	Highest color difference in ΔCh between measured Lab values of the composite grays on the color control strip and the Lab values of the matching patches of the reference printing condition.
Average weighted ΔCh composite grays	Average color difference in weighted ΔCh between measured Lab values of the composite grays on the color control strip and the Lab values of the matching patches of the reference printing condition.
Maximum weighted ΔCh composite grays	Highest color difference in weighted ΔCh between measured Lab values of the composite grays on the color control strip and the Lab values of the matching patches of the reference printing condition.

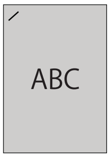
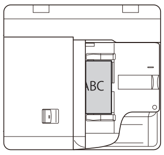
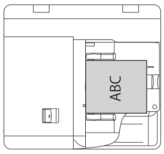
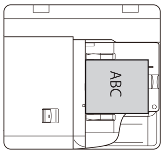
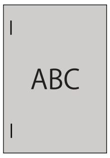
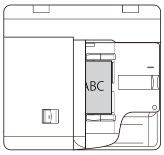
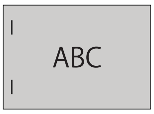
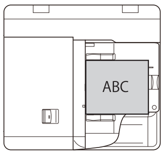
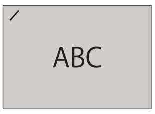
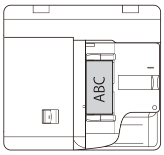
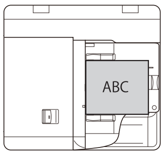
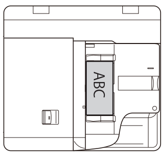
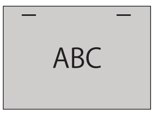
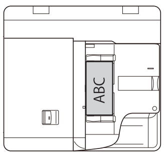
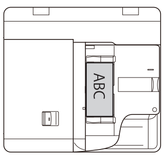
NOTE
The **weighted** ΔCh puts more emphasis on the color differences between composite grays, where C is lower than 50%.

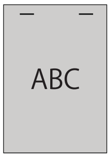
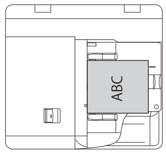
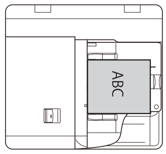
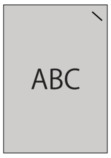
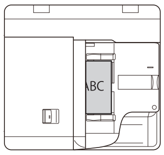
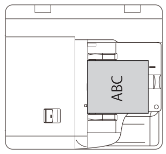
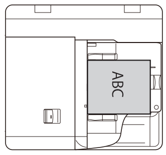
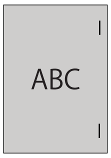
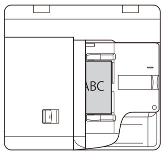
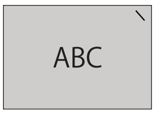
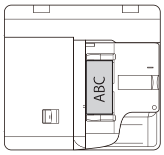
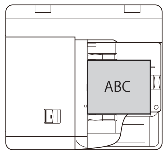
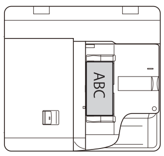
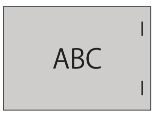
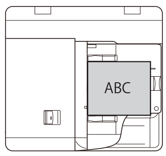
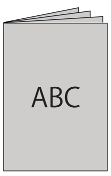
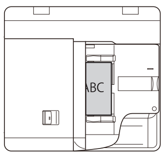
Feed instructions

Feed instruction for stapling, punching and folding

Feed instruction for stapling when you use the automatic document feeder

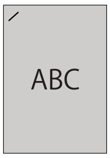
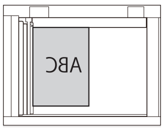
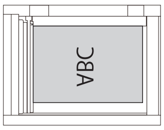
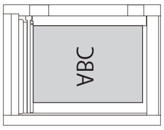
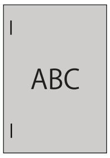
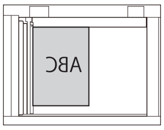
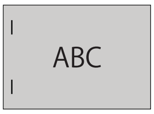
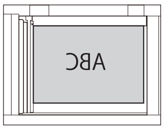
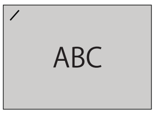
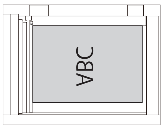
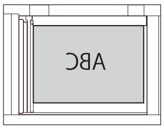
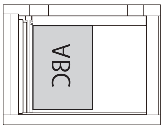
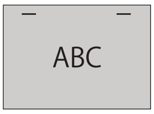
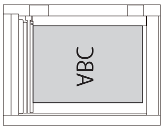
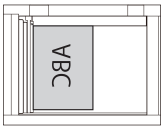
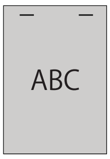
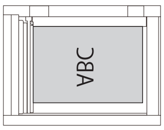
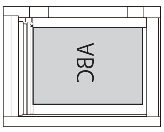
The following table shows the orientation of originals in relation to the required staple position.

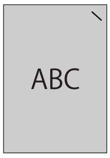
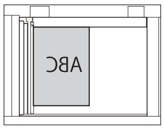
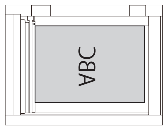
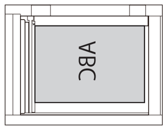
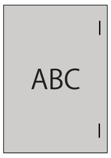
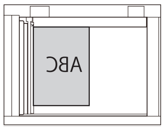
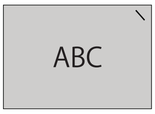
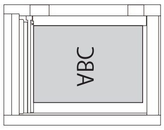
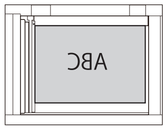
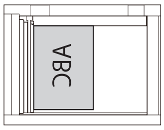
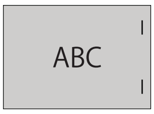
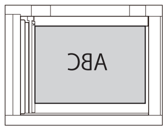
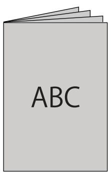
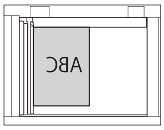
Staple position	A4 / LTR		A3 / 11"x 17"	
	Orientation	Staple setting	Orientation	Staple setting
		[1 staple] [Top left]		[1 staple] [Bottom left]
				[1 staple] [Top right]
		[2 staples] [Left edge]		
				[2 staples] [Left edge]
		[1 staple] [Bottom left]		[1 staple] [Top left]
		[1 staple] [Top right]		
		[2 staples] [Left edge]		
		[2 staples] [Right edge]		

Staple position	A4 / LTR		A3 / 11"x 17"	
	Orientation	Staple setting	Orientation	Staple setting
				[2 staples] [Left edge]
				[2 staples] [Right edge]
		[1 staple] [Top right]		[1 staple] [Top left]
				[1 staple] [Bottom right]
		[2 staples] [Right edge]		
		[1 staple] [Top left]		[1 staple] [Top right]
		[1 staple] [Bottom right]		
				[2 staples] [Right edge]
		[Saddle stitch- ing]		

Feed instruction for stapling when you use the glass plate

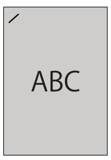
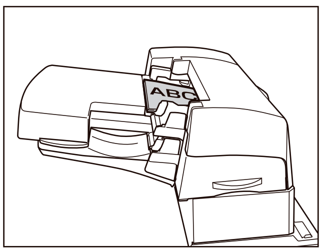
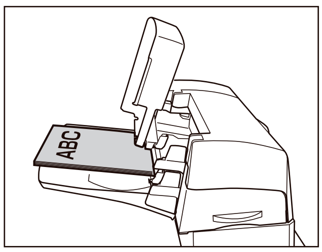
The following table shows the orientation of originals in relation to the required staple position.

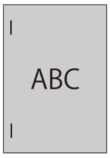
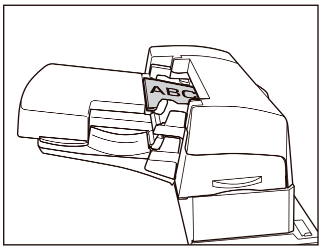
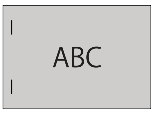
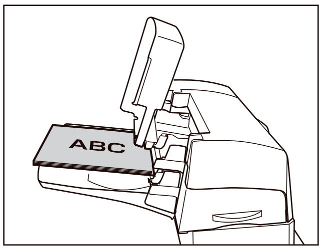
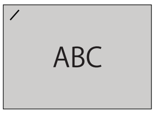
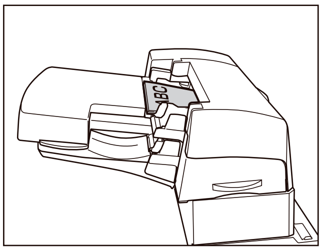
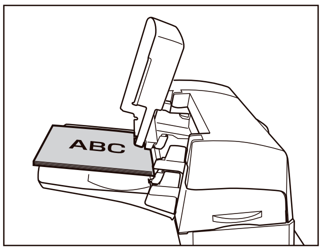
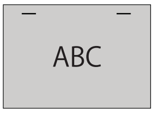
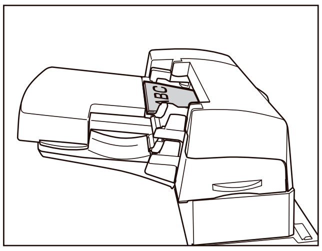
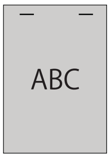
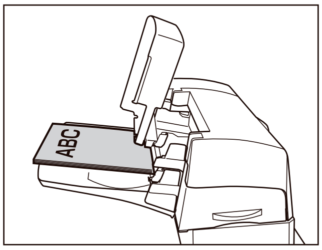
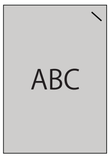
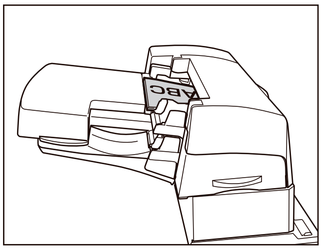
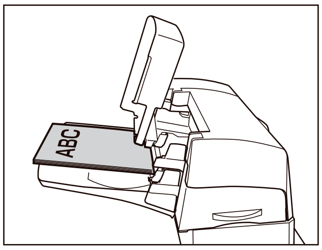
Staple position	A4 / LTR		A3 / 11"x 17"	
	Orientation	Staple setting	Orientation	Staple setting
		[1 staple] [Top left]		[1 staple] [Bottom left]
				[1 staple] [Top right]
		[2 staples] [Left edge]		
				[2 staples] [Left edge]
		[1 staple] [Bottom left]		[1 staple] [Top left]
		[1 staple] [Top right]		
		[2 staples] [Left edge]		
		[2 staples] [Right edge]		
				[2 staples] [Left edge]
				[2 staples] [Right edge]

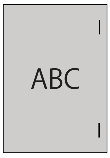
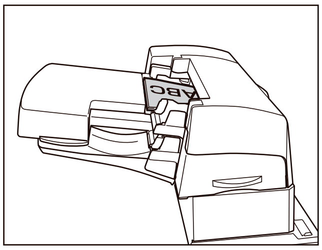
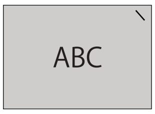
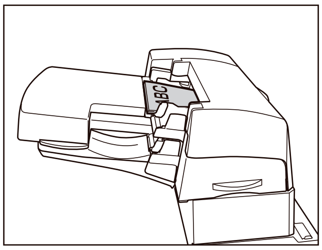
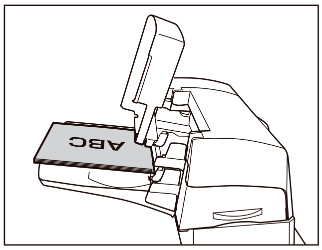
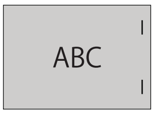
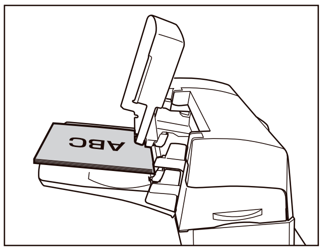
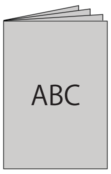
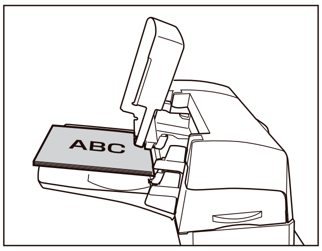
Staple position	A4 / LTR		A3 / 11"x 17"	
	Orientation	Staple setting	Orientation	Staple setting
		[1 staple] [Top right]		[1 staple] [Top left]
				[1 staple] [Bottom right]
		[2 staples] [Right edge]		
		[1 staple] [Top left]		[1 staple] [Top right]
		[1 staple] [Bottom right]		
				[2 staples] [Right edge]
		[Saddle stitch-ing]		

Feed instruction for stapling when you use the inserter

The following table shows how to feed insert sheets in relation to the staple position.

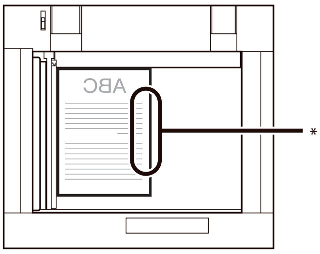
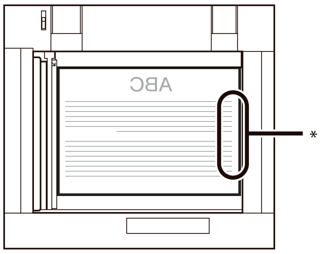
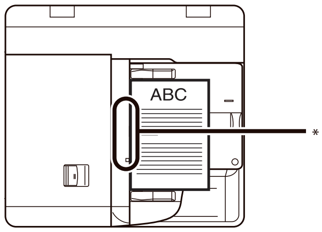
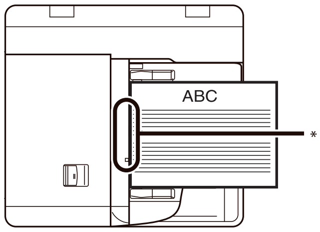
Staple position	A4 / LTR	A3 / 11"x 17"
	Orientation	Orientation
		

Staple position	A4 / LTR	A3 / 11"x 17"
	Orientation	Orientation
		
		
		
		
		
		

Staple position	A4 / LTR	A3 / 11"x 17"
	Orientation	Orientation
		
		
		
		

Feed instruction for punching

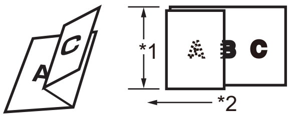
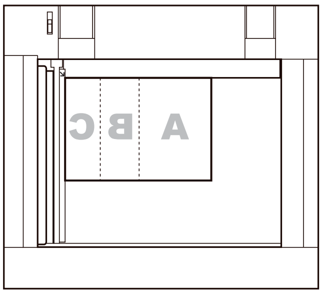
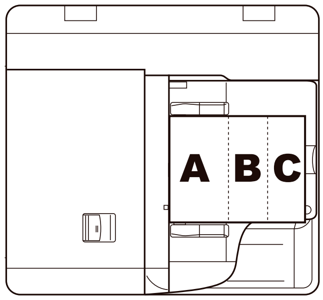
The following table shows the orientation of originals in relation to the position of punch holes.

Glass plate	Automatic document feeder
 	 

*Position of the holes in the output.

Feed instruction for [Z-fold]

The [Z-fold] setting folds A3 / 11" x 17" or A4R / LTRR paper into A4 / LTR or A5 / SMT size respectively.

Folded output	Glass plate	Automatic document feeder
 Leading-edge length (1) and feed direction (2)		

Orientation for the [Z-fold] setting

NOTE

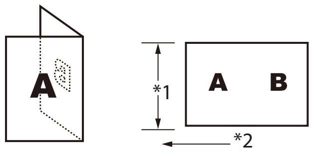
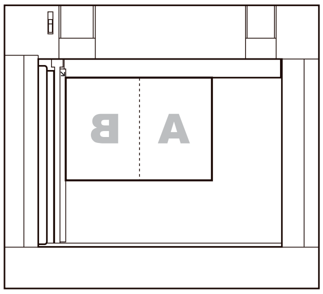
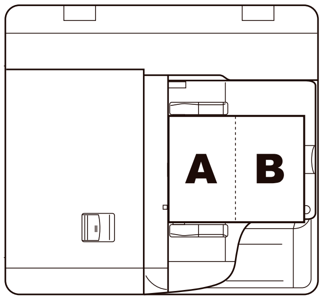
To verify whether stapling is possible, use the following formula X + 2Y +10Z, where

- X is the number of A4 sheets which are not Z-folded;
- Y is the number of A3 sheets which are not Z-folded;
- Z is the number of Z-folded sheets.

If the sum of the formula is more than 100, stapling is not possible.

Example 1: 50 A4 sheets + 4 A3 sheets + 1 A3 Z-folded sheet = 68 -> stapling is possible	Example 2: 50 A4 sheets + 7 A3 Z-folded sheets = 120 -> stapling is not possible

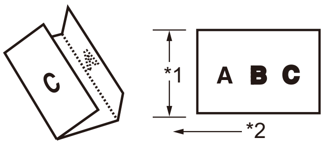
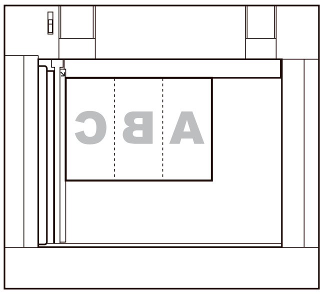
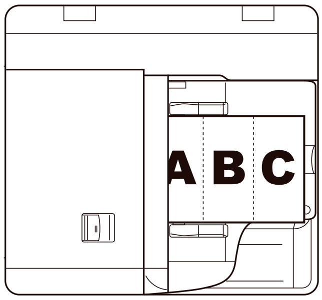
Feed instruction for [Half-fold]

Folded output	Glass plate	Automatic document feeder
 Leading-edge length (1) and feed direction (2)		

Orientation for the [Half-fold] setting

Feed instruction for [Tri-fold in]

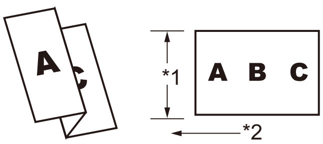
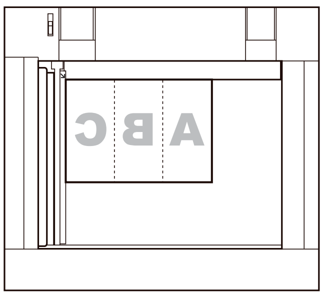
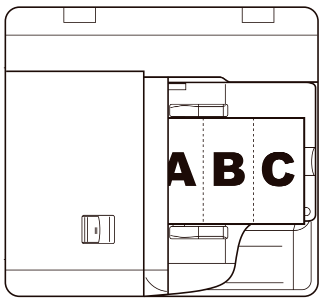
The [Tri-fold in] setting folds paper to make it suitable for envelopes. The paper gets the shape of a C.

Folded output	Glass plate	Automatic document feeder
 Leading-edge length (1) and feed direction (2)		

Orientation for the [Tri-fold in] setting

Feed instruction for [Tri-fold out]

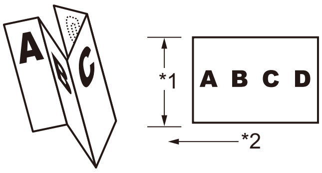
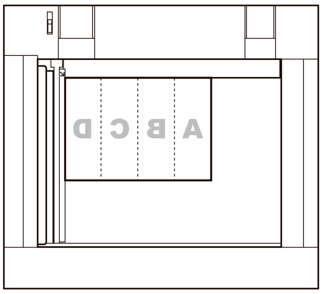
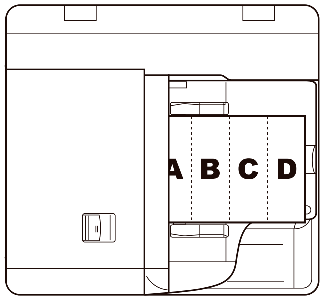
The [Tri-fold out] setting folds paper to make it suitable for envelopes. The paper is folded in the shape of a Z.

Folded output	Glass plate	Automatic document feeder
 Leading-edge length (1) and feed direction (2)		

Orientation for the [Tri-fold out] setting

Feed instruction for [Parallel fold]

The [Parallel fold] setting folds the paper in four equal parts.

Folded output	Glass plate	Automatic document feeder
 Leading-edge length (1) and feed direction (2)		

Orientation for the [Parallel fold] setting

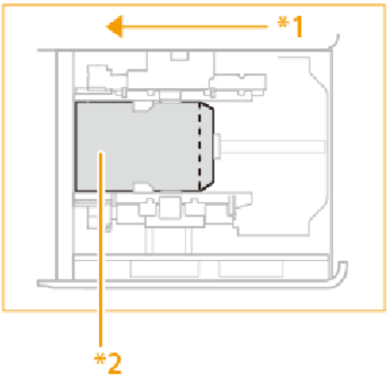
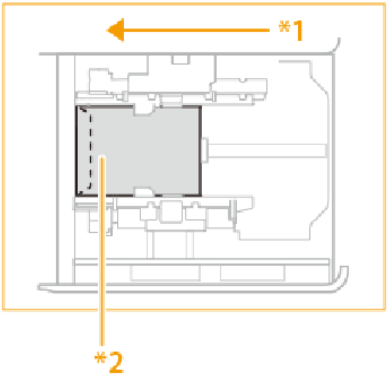
Feed instructions for envelopes and tab paper to the paper input optionals

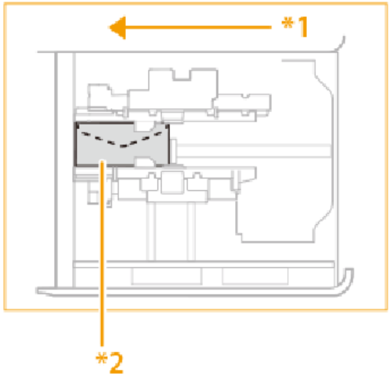
The following sections explain how to the feed envelopes in the paper trays.

Paper module

NOTE
Use in combination with Envelope Feeder Attachment-G.
Use the support tray of the Envelope Feeder Attachment-H to load the following envelope types: Kakugata 2, 9 x 12 inch and 10 x 13 inch.

The following table show how to the feed envelopes in the paper module.

Envelope type	Feed instruction
Nagagata 3, Kakugata 2, 9 x 12 inch, or 10 x 13 inch	  *1 feeding direction *2 print side up

Envelope type	Feed instruction
No. 10 (COM10), ISO-C5, DL, Monarch, Yougatanaga 3, 6 x 9 inch	 *1 feeding direction *2 print side up

The loading limit in the paper module varies depending on the envelope type.

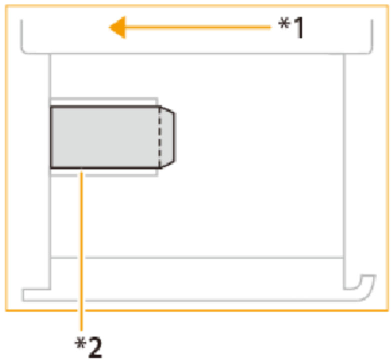
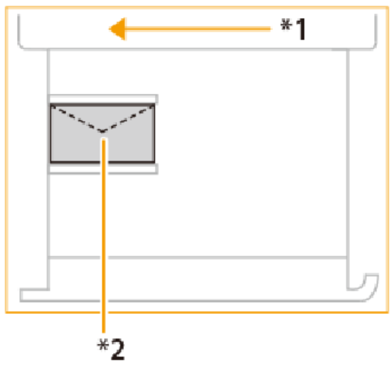
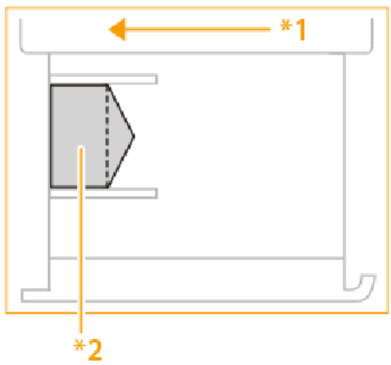
Envelope type	Loading limit (height) of upper and middle decks	Loading limit mark	Loading limit (height) of lower decks	Loading limit mark
Nagagata 3	50 mm		100 mm (3.9 inch)	
Yougatanaga 3				
Kakugata 2				
228 x 304 mm (9 x 12 inch)				
152 x 228 mm (6 x 9 inch)				
No. 10 (COM10)				
DL				
ISO-C5				
Monarch			70 mm (2.8 inch)	
254 x 330.2 mm (10 x 13 inch)	45 mm (1.8 inch)	Not applicable	45 mm (1.8 inch)	Not applicable
Kakugata 2, side seam envelope	30 mm (1.2 inch)	Not applicable	30 mm (1.2 inch)	Not applicable

Bulk paper module

NOTE
Use in combination with the Envelope Feeder Attachment-G.

Make sure that the paper stack does not exceed the loading limit line .

The following table show how to the feed envelopes in the bulk paper module.

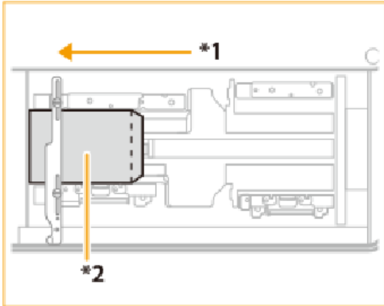
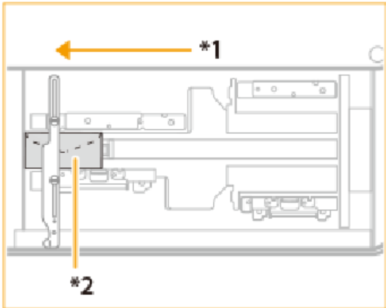
Envelope type	Feed instruction
Nagagata 3, Kakugata 2	 *1 feeding direction *2 print side up
No. 10 (COM10), ISO-C5, DL, Monarch, Yougatanaga 3	  *1 feeding direction *2 print side up

Bulk paper module XL

NOTE
Use in combination with the Envelope Feeder Attachment-G.

Make sure that the paper stack does not exceed the loading limit line ▼▼▼.

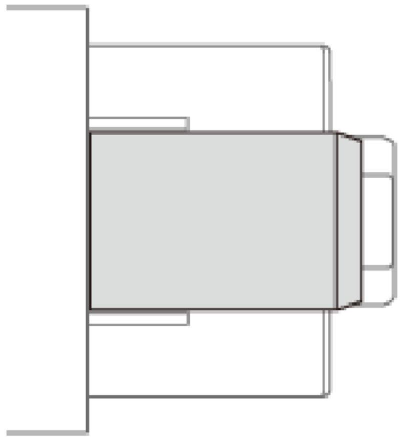
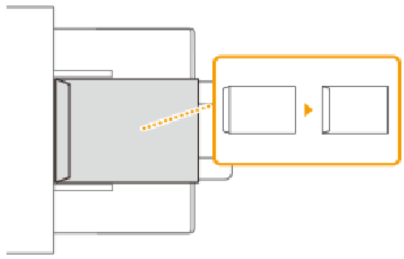
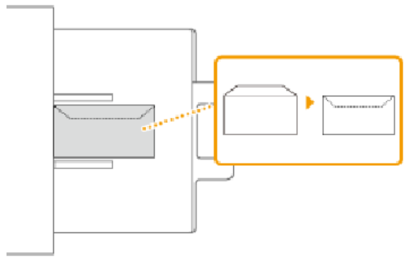
The following table show how to the feed envelopes in the bulk paper module XL.

Envelope type	Feed instruction
Nagagata 3, Kakugata 2	 *1 feeding direction *2 print side up
No. 10 (COM10), ISO-C5, DL, Monarch, Youga-tanaga 3	 *1 feeding direction *2 print side up

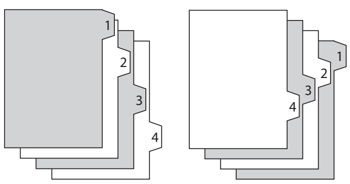
Special feeder

Make sure that the paper stack does not exceed the loading limit line .

The following table show how to the feed envelopes in the special feeder.

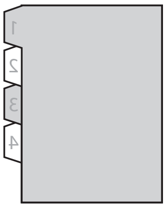
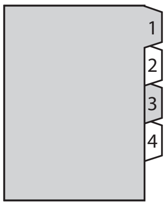
Envelope type	Feed instruction
Nagagata 3, Kakugata 2	 To change the orientation of the envelopes, close the flaps and load the envelopes so that their flaps are on the left-hand side. 
No. 10 (COM10), ISO-C5, DL, Monarch, Youga-tanaga 3	

Feed instruction for tab paper



Forward-order tab paper and reverse-order tab paper

The following table shows the orientation of tab paper in the paper module.

Orientation	Orientation and paper input optional
	Load the reverse-order tab paper face down, with the tabs at the left-hand side of the tray in: • internal paper trays of the print module; • inserter.
	Load forward-order tab paper face up, with the tabs at the right-hand side of the in tray: • paper module; • bulk paper module; • bulk paper module XL; • special feeder.

Specifications

Printer specifications

Print module

Specification	Value
Type	Digital color cut-sheet press
Resolution	2400 dpi x 2400 dpi
Device memory (SDRAM)	6 GB
Hard disk	240 GB
Warm-up time	6 minutes from power on
Print speed	100 ppm (A4/ LTR)
Copy speed	100 ppm
Scan speed	270 ipm (A4/LTR)
Power requirement	220-240 V/16 A
First print out time	18 seconds
First copy out time	20 seconds or less
Power consumption	Maximum 6000 W
Dimensions (W x D x H) (to the glass plate)	1583 mm x 872 mm x 1040 mm (62.3 inch x 34.3 inch x 40.9 inch) (without control panel and operator attention light)
Weight	Approximately 904 lbs. (410 kg)

PRISMAsync Print Server

Specification	Value
Description	Integrated high-performance PRISMAsync color print server for imagePRESS V1000
Operating system	Windows® 10
Processor	Intel® Core™ i7-8700
Memory	16 GB
Hard disk	1 x 2.5 inch SATA II, 500 GB @ 7200 rpm 1 x 3.5 inch SATA II, 1 TB @ 7200 rpm
Interface	Ethernet (10/100/1000 Base-T), TCP/IP (LPR /LPD, 9100 Socket, SMB), Static IP / Auto-IP (DHCP)
Page Description Language	Adobe® PostScript® 3 (3020), PDF 2.0 extension level 3, PDF/X-1a:2001, PDF/X-1a:2003, PDF/X-3:2002, PDF/X-3:2003, PDF/X-4, PDF/VT level 1, Optimized PostScript®, Optimized PDF, Adobe® PDF Print Engine (APPE) 6.1.8, IPDS PDF, IPDS PCL , PPML/GA level 1 versions 1.5, 2.1 and 2.2

Specification	Value
Protocols	SNMP v1/v2c/v3, Host resources MIB, System group MIB II, Printer MIB, Job Monitor MIB
Security	E-shredding, HTTPs, LDAP, SNMP v3, Integrity checker, WebDav for SMB hotfolders, Encrypted user passwords, User authentication, IPsec
Spot color libraries	HKS K, HKS K 3000+, HKS N, HKS N 3000+, Pantone® + solid-V3 (uncoated/coated)
Options	DocBox licence Page Programming licence E-shredding licence Software protection licence (integrity checker) PCL6 for transaction printing licence IPDS for transaction printing licence PDF support for IPDS licence DP Link support licence Advanced Colour Management licence: • DeviceLink support • Color simulation support • Color profile import/export • Color control and information bars • Color mapping (leave out separations) • Embedded profiling Advanced impose licence Asian font sets licenses: Korean, Japanese, Simplified Chinese, Traditional Chinese
Languages	British English, French, Italian, German, Spanish, Dutch, Finnish, Norwegian, Swedish, Danish, Polish, Czech, Hungarian, Portuguese, American English, Russian, Japanese, Simplified Chinese, Korean
Power requirement	100 V AC - 240 V AC, 10 A - 5 A , 50/60 Hz
Dimensions (W x D x H)	200 mm x 430 mm x 420 mm (7.9 inch x 16.9 inch x 16.5 inch)
Weight	Approximately 39 lbs. (18 kg)

Media specifications

Supported media types per paper input option

Media type	Internal paper tray	Paper module	Bulk paper module (XL)	Special feeder	Inserter
Thin (52 g/m² - 63 g/m²) (14 lb bond - 17 lb bond)	✓	✓	✓	✓	✓
Plain (64 g/m² - 105 g/m²) (17 lb bond - 28 lb bond)	✓	✓	✓	✓	✓
Recycled (64 g/m² - 220 g/m²) (17 lb bond - 81 lb cover)	✓	✓	✓ *1	✓	✓
Color (64 g/m² - 90 g/m²) (17 lb bond - 24 lb cover)	✓	✓	✓	✓	✓
Heavy (106 g/m² - 400 g/m²) (28 lb bond - 148 lb cover)	✓ *2	✓ *3	✓	✓	✓ *2
1-sided coated (70 g/m² - 400 g/m²) (19 lb bond - 148 lb cover)	-	✓ *3	✓ *4	✓	✓ *2
2-sided coated (70 g/m² - 400 g/m²) (19 lb bond - 148 lb cover)	-	✓ *3	✓ *4	✓	✓ *2
Matte coated (70 g/m² - 400 g/m²) (19 lb bond - 148 lb cover)	-	✓ *3	✓ *4 *5	✓	✓ *2
Textured (80 g/m² - 300 g/m²) (22 lb bond - 110 lb cover)	-	✓	✓	✓	✓
Vellum (64 g/m² - 105 g/m²) (17 lb bond - 28 lb bond)	✓	✓	✓	✓	✓
Transparency (151 g/m² - 180 g/m²) (40 lb bond - 66 lb cover)	✓	✓	✓	✓	-
Clear film (151 g/m² - 180 g/m²) (40 lb bond - 66 lb cover)	✓	✓	✓	✓	-
Translucent film (151 g/m² - 180 g/m²) (40 lb bond - 66 lb cover)	✓	✓	✓	✓	-

Media type	Internal paper tray	Paper module	Bulk paper module (XL)	Special feeder	Inserter
Prepunched (64 g/m² - 90 g/m²) (17 lb bond - 24 lb bond)	✓	✓	✓	-	✓
Tab paper (151 g/m² - 220 g/m²) (40 lb bond - 81 lb cover) *6	✓	✓	✓	✓	✓
Labels (151 g/m² - 180 g/m²) (40 lb bond - 66 lb cover)	-	✓	✓	✓	-
Bond (64 g/m² - 105 g/m²) (17 lb bond - 28 lb bond)	✓	✓	✓	✓	✓
Letterhead (151 g/m² - 180 g/m²) (40 lb bond - 66 lb cover)	✓	✓	✓	✓	✓
Synthetic (Polypropylene) (151 g/m² - 180 g/m²) (40 lb bond - 66 lb cover)	-	✓	✓	✓	✓
Synthetic (Polyester) (151 g/m² - 180 g/m²) (40 lb bond - 66 lb cover)	-	✓	✓	✓	✓
Envelope (70 g/m² - 128 g/m²) (19 lb bond - 47 lb cover)	✓	✓	✓	✓	-
Postcard (181 g/m² - 200 g/m²) (67 lb cover - 110 lb index)	✓	✓	✓	✓	-
Magnetic (900 g/m² - 1000 g/m²) (332.7 lb cover - 369.7 lb cover)		✓ *3		✓	

*1 The recycled media of 181–220 g/m² (67 lb cover–81 lb cover) is not supported.

*2 The media heavier than 300 g/m² (111 lb cover) is not supported.

*3 The media heavier than 350 g/m² (130 lb cover) can only be loaded in the lower tray.

*4 The media lighter than 80 g/m² (22 lb bond) is not supported.

*5 The media heavier than 350 g/m² (130 lb cover) is not supported.

*6 Use A4 or LTR tab paper.

Supported media sizes per paper input option

Media size	Internal paper tray	Special feeder	Bulk paper module	Bulk paper module XL	Paper module	Inserter
330 mm x 483 mm 13 inch x 19 inch	✓	✓	✓	✓	✓	✓
305 mm x 457 mm 12 inch x 18 inch	✓	✓	✓	✓	✓	✓
SRA3 320 mm x 450 mm 12.6 inch x 17.7 inch	✓	✓	✓	✓	✓	✓
A3 / 11 inch x 17 inch 297 mm x 420 mm 11.7 inch x 16.5 inch	✓	✓	✓	✓	✓	✓
A4 / LTR 210 mm x 297 mm 8.3 inch x 11.7 inch	✓	✓	✓	✓	✓	✓
A4R / LTRR 297 mm x 210 mm 11.7 x 8.3 inch	✓	✓	✓	✓	✓	✓
A5 148 mm x 210 mm 5.8 inch x 8.3 inch	✓	✓	-	-	-	-
A5R / STMTR 210 mm x 148 mm 8.3 inch x 5.8 inch	✓	✓	✓	✓	✓	-
B4 250 mm x 353 mm 9.8 inch x 13.9 inch	✓	✓	✓	✓	✓	✓
B5 176 mm x 250 mm 6.9 inch x 9.8 inch	✓	✓	✓	✓	✓	✓
B5R 250 mm x 176 mm 9.8 inch x 6.9 inch	✓	✓	✓	✓	✓	✓
16K 195 mm x 267 mm 7.7 inch x 10.5 inch	✓	✓	✓	✓	✓	✓
16KR 267 mm x 195 mm 10.5 inch x 7.7 inch	✓	✓	✓	✓	✓	✓

Media size	Internal paper tray	Special feeder	Bulk paper module	Bulk paper module XL	Paper module	Inserter
8K 270 mm x 394 mm 10.5 inch x 15.5 inch	✓	✓	✓	✓	✓	✓
LGL 215.9 mm x 279.4 mm 8.5 inch x 14 inch	✓	✓	✓	✓	✓	✓
EXEC 184 mm x 267 mm 7 inch x 10 inch	✓	✓	✓	✓	✓	✓
Custom size 98 mm x 148 mm to 330.2 mm x 487.7 mm 3.8 inch x 5.8 inch to 13 inch x 19.2 inch	✓	✓	✓ *1	✓ *1	✓ *2	✓ *3
Custom size 210 mm x 487.8 mm to 330.2 mm x 1300 mm 8.3 inch x 19.2 inch to 13 inch x 51.1 inch	-	✓	-	✓ *4	-	-

*1 Custom media sizes can only be fed if they are between 139.7 mm x 148 mm (5.5 inch x 5.8 inch) to 330.2 mm x 487.7 mm (13 inch x 19.2 inch).

*2 Custom media sizes can only be fed if they are between 139.7 mm x 182 mm (5.5 inch x 7.2 inch) to 330.2 mm x 487.7 mm (13 inch x 19.2 inch).

*3 Custom media sizes can only be fed if they are between 182 mm x 182 mm (7.2 inch x 7.2 inch) to 330.2 mm x 487.7 mm (13 inch x 19.2 inch).

*4 Custom media sizes can only be fed if they are between 210 mm x 487.8 inch (8.3 inch x 19.2 inch) to 330.2 mm x 762 mm (13 inch x 30 inch).

Supported envelope types per paper input option

Media size	Internal paper tray	Special feeder	Bulk paper module (XL)	Paper module	Inserter
ISO-C5 162 mm x 229 mm 6.4 inch x 9 inch	✓	✓	✓ *1	✓ *3	-
No. 10 (COM10) 104.7 mm x 241.3 mm 4.1 inch x 9.5 inch	✓ *2	✓	✓ *1	✓ *3	-

Media size	Internal paper tray	Special feeder	Bulk paper module (XL)	Paper module	Inserter
Monarch 98.4 mm x 190.5 mm 3.9 inch x 7.5 inch	✓ *2	✓	✓ *1	✓ *3	-
DL 110 mm x 220 mm 4.3 inch x 8.7 inch	✓ *2	✓	✓ *1	✓ *3	-
Nagagata 3 120 mm x 235 mm 4.7 inch x 9.3 inch	✓ *2	✓	✓ *1	✓ *3	-
Yougatanaga 3 235 mm x 120 mm 9.3 inch x 4.7 inch	✓ *2	✓	✓ *1	✓ *3	-
Kakugata 2 240 mm x 332 mm 9.5 inch x 13 inch	✓	✓	✓ *1	✓ *3	-

*1 The optional Envelope Feeder Attachment-G is required.

*2 The optional Envelope Feeder Attachment-F is required.

*3 The optional Envelope Feeder Attachment-H is required.

Paper input specifications

Long Sheet Tray-B

Specification	Value
Description	Optional special feeder to feed long sheets, also for 2-sided printing
Print resolution	2400 dpi x 2400 dpi
Maximum capacity	1 sheet
Media size	Custom sizes: 210 mm x 487.8 mm to 330.2 mm x 1300 mm 8.3 inch x 19.2 inch to 13 inch x 51.1 inch
Media weight	52 g/m² - 400 g/m² (14 lb bond - 148 lb cover)

Tab Feeding Attachment-F

Specification	Value
Description	Tab paper guide to guide tab paper in the internal paper tray
Maximum capacity	60 mm (2.4 inch) or 300/400 sheets
Media size	A4/LTR
Weight	Approximately 0.1 kg (0.2 lb)

Envelope Feeder Attachment-F

Specification	Value
Description	Envelope guide to guide envelopes in the internal paper tray
Supported envelopes	**Envelope types:** No. 10 (COM10), Monarch, DL, Nagagata 3, Yougatanaga 3 **Custom size envelopes:** 152.4 mm x 228.6 mm (6 inch x 9 inch)
Maximum capacity	50 sheets

Envelope Feeder Attachment-G

Specification	Value
Description	Envelope guide to guide envelopes in the bulk paper module and bulk paper module XL (POD Deck Lite-C1 and POD Deck Lite XL-A)
Supported envelopes	**Envelope types:** ISO-C5, No. 10 (COM10), Monarch, DL, Nagagata 3, Yougatanaga 3, Kakugata 2 **Custom size envelopes:** 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch), 254 mm x 330.2 mm (10 inch x 13 inch)

Specification	Value
Maximum capacity	Nagagata 3: 160 mm (6.3 inch) Nagagata 3 side seam envelope: 100 mm (3.9 inch) ISO-C5, No. 10 (COM10), DL, Yougatanaga 3, Kakugata 2: 100 mm (3.94 inch) Kakugata 2 side seam envelope: 45 mm (1.77 inch) Monarch: 70 mm (2.8 inch)

Envelope Feeder Attachment-H

Specification	Value
Description	Envelope guide to guide envelopes in the paper module (Multi-drawer Paper Deck-E)
Supported envelopes	**Envelope types:** ISO-C5, No. 10 (COM10), Monarch, DL, Nagagata 3, Yougatanaga 3, Kakugata 2 **Custom size envelopes:** 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch), 254 mm x 330.2 mm (10 inch x 13 inch)
Maximum capacity	**Upper/middle tray** • ISO-C5, No. 10 (COM10), Monarch, DL, Nagagata 3, Yougatanaga 3, Kakugata 2 (also 3 side seam), 152.4 mm x 228.6 mm, 228.6 mm x 304.8 mm (6 inch x 9 inch, 9 inch x 12 inch): 50 mm (1.9 inch) • Kakugata 2 side seam envelope: 30 mm (1.2 inch) • 254 mm x 330.2 mm (10 inch x 13 inch): 45 mm (1.8 inch) **Lower tray** • ISO-C5, No. 10 (COM10), DL, Nagagata 3, Yougatanaga 3, Kakugata 2, 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch): 100 mm (3.9 inch) • Kakugata 2 side seam envelope: 30 mm (1.2 inch) • Kakugata 2 3 side seam envelope: 50 mm (1.9 inch) • Monarch: 70 mm (2.8 inch) • 254 mm x 330.2 mm (10 inch x 13 inch): 45 mm (1.8 inch)

Stack Bypass-D

Specification	Value
Description	Special feeder to feed special media manually
Maximum capacity	250 sheets, 80 g/m² (22 lb bond)
Media size	All paper sizes supported by the printer
Media weight	52 g/m² - 1000 g/m² (14 lb bond - 369.7 lb cover)

Internal paper trays

Specification	Value
Capacity	2 x 550 sheets, 80 g/m² (22 lb bond)

Specification	Value
Media sizes	Maximum: 330.2 mm x 487.7 mm (13 inch x 19.2 inch)* Minimum: 100 mm x 148 mm (3.9 inch x 5.8 inch) Long sheet: 330.2 mm x 1300 mm (13 inch x 51.1 inch) by service mode
Media weight	52 g/m² - 400 g/m² (14 lb bond - 148 lb cover): plain, heavy 70 g/m² - 400 g/m² (18.6 lb bond - 148 lb cover): coated
Paper feed technology	• Suction feed • Air separation

Multi-drawer Paper Deck-E

Specification	Value
Description	Optional 3-tray paper input module
Input trays	**Upper/middle tray** • 1,500 sheets, 80 g/m² (22 lb bond) • 1,650 sheets, 64 g/m² (17 lb bond) **Lower tray** • 2,000 sheets, 80 g/m² (22 lb bond) • 2,200 sheets, 64 g/m² (17 lb bond)
Media sizes	**Standard sizes:** SRA3, A3, A4, A4R, A5R, B4, B5, B5R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, STMTR, EXEC, 8K, 16K, 16KR **Custom sizes:** 139.7 mm x 148 mm to 330.2 mm x 487.7 mm (5.5 inch x 5.8 inch to 13 inch x 19.2 inch) **Envelope types:** ISO-C5, No. 10 (COM), Monarch, DL, Nagagata 3, Yougatanaga 3, Kakugata 2, 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch), 254 mm x 330.2 mm (10 inch x 13 inch)
Media weight	52 g/m² - 1000 g/m² (14 lb bond - 369.7 lb cover)
Power requirement	220-240 V AC, 50/60 Hz, 3.5 A
Power consumption	Maximum 1000 W
Dimensions unit (W x D x H)	990 mm x 800 mm x 1080 mm (39 inch x 31.5 inch x 42.5 inch)
Weight unit	Approximately 178 kg (392 lb)

POD Deck Lite-C

Specification	Value
Description	Optional single-tray bulk paper module

Specification	Value
Input trays	1 x 3,500 sheets, 80 g/m² (22 lb bond) 1 x 3,660 sheets, 75 g/m² (20 lb bond) 1 x 4,000 sheets, 64 g/m² (17 lb bond)
Media sizes	**Standard sizes:** SRA3, A3, A4, A4R, A5R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, STMTR, EXEC, 8K, 16K, 16KR **Custom sizes:** 139.7 mm x 148 mm to 330.2 mm x 487.7 mm (5.5 inch x 5.8 inch to 13 inch x 19.2 inch) **Envelope types:** ISO-C5, No. 10 (COM), Monarch, DL, Nagagata 3, Yougatanaga 3, Kakugata 2, 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch), 254 mm x 330.2 mm (10 inch x 13 inch)
Media weight	52 g/m² - 350 g/m² (16 lb bond - 130 lb cover)
Paper feed technology	• Friction feed • Air separation
Power requirement	220-240 V AC, 50/60 Hz, 1.2 A
Power consumption	Maximum 480 W
Dimensions (WxDxH)	717 mm x 686 mm x 574 mm (28.2 inch x 27 inch x 22.6 inch)
Weight	Approximately 76 kg (168 lb)

POD Deck Lite XL-A

Specification	Value
Description	Optional single-tray bulk paper module for long sheets, standard size media and custom size media
Input trays	**487.7 mm (19.2 inch) or shorter** • 1 x 3,500 sheets, 80 g/m² (22 lb bond) • 1 x 3,660 sheets, 75 g/m² (20 lb bond) • 1 x 4,000 sheets, 64 g/m² (17 lb bond) **Longer than 487.7 mm (19.2 inch)** • 1 x 1,000 sheets, 80 g/m² (22 lb bond)

Specification	Value
Media sizes	**Standard sizes:** SRA3, A3, A4, A4R, A5R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, STMTR, EXEC, 8K, 16K, 16KR **Custom sizes:** 139.7 mm x 148 mm to 330.2 mm x 487.7 mm (5.5 inch x 5.8 inch to 13 inch x 19.2 inch) **Long sheets:** 210 mm x 487.8 mm to 330.2 mm x 762 mm (8.3 inch x 19.2 inch to 13 inch x 30 inch) **Envelope types:** ISO-C5, No. 10 (COM), Monarch, DL, Nagagata 3, Yougatanaga 3, Kakugata 2, 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch), 254 mm x 330.2 mm (10 inch x 13 inch)
Media weight	Long sheets: 64 g/m² - 300 g/m² (17 lb bond - 110 lb cover) Standard and custom sizes: 52 g/m² - 350 g/m² (16 lb bond - 130 lb cover)
Power requirement	220-240 V AC, 50/60 Hz, 1.2 A
Power consumption	Maximum 480 W
Dimensions (WxDxH)	1,105 mm x 686 mm x 570 mm (43.5 inch x 27 inch x 22.4 inch)
Weight	Approximately 113 kg (249 lb)

Document Insertion Unit-R

Specification	Value
Description	Optional sheet and cover insertion unit
Capacity	2 x 200 sheets, 80 g/m² (20 lb bond)
Media sizes	**Standard sizes:** SRA3, A3, A4, A4R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, 8K, 16K, 16KR **Custom sizes:** 182 mm x 182 mm to 330.2 x 487.7 mm 7.2 inch x 7.2 inch to 13 inch x 19.2 inch
Media weight	52 g/m² - 300 g/m² (14 lb bond - 110 lb cover)

Specification	Value
Media	**For inserts:** • thin paper, 52 g/m² - 63 g/m² (14 lb bond - 16 lb bond) • plain paper, 64 g/m² - 105 g/m² (17 lb bond - 28 lb bond) • recycled paper • heavy paper, 106 g/m² - 300 g/m² (29 lb bond - 110 lb cover) • coated paper, 106 g/m² - 300 g/m² (29 lb bond - 110 lb cover) • colored paper • tabs • textured paper • bond paper • prepunched paper • letterhead • tracing paper **Pass-through mode:** all media supported by the main engine
Power requirement	100-240 V AC, 50/60 Hz, 1.0 A
Power consumption	Maximum 120 W
Dimensions (W x D x H)	746 mm x 793 mm x 1407 mm (29.3 inch x 31.2 inch x 55.4 inch)
Weight	Approximately 61 kg (134 lb)

Finishers specifications

This topics describes the specifications of the following finishers:
- Paper Folding Unit-K
- Staple Finisher-AF
- Booklet Finisher-AF
- Finisher Bridge-A
- DFD Adapter-B

For specifications of the Perfect Binder-F, Booklet Trimmer-G, Two-Knife Booklet Trimmer-B, Multi Function Professional Puncher-C and High Capacity Stacker-J, refer to the manuals delivered together with these finishers.

Paper Folding Unit-K

Specification	Value
Description	Optional fold unit with various folding options
Tray capacity	Folding unit output tray: • Parallel fold: 25 sheets • Tri-fold in, tri-fold out: 40 sheets Finisher tray: • Z-fold and half fold: depends on the finisher specifications
Media weight	52 g/m² - 105 g/m² (14 lb bond - 28 lb bond) Parallel fold: 52 g/m² - 90 g/m² (14 lb bond - 24 lb bond)
Media size	Z-fold: A4R, B4, A3, LTRR, LGL, 279.4 mm x 431.8 mm (11 inch x 17 inch) Tri-fold in, tri-fold out, half fold: A4R, LTRR Parallel fold: A4R, LTRR, LGL
Media type	Thin, plain, coated (70 g/m² to 105 g/m²; 18.6 lb bond to 28 lb bond), recycled, color, bond
Power requirement	From the finisher
Power consumption	Maximum 150 W
Dimensions (W x D x H)	336 mm x 793 mm x 1,190mm (13.2 inch x 31.2 inch x 46.9 inch)
Weight	Approximately 71 kg (157 lb)

Staple Finisher-AF

Specification	Value
Description	Stacker/stapler with (offset) stacking, stapling, folding and punching

Specification	Value
Media size	**Upper tray without offset stacked** • Standard sizes: SRA3, A3, A4, A4R, A5, A5R, B4, B5, B5R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, STMT, 8K, 16K, 16KR • Custom sizes: 100 mm x 148 mm to 330.2 mm x 487.7 mm (3.9 inch x 5.8 inch to 13 inch x 19.2 inch) • Long sheet: 210 mm x 487.8 mm to 330.2 mm x 1,300 mm (8.3 inch x 19.2 inch to 13 inch x 30 inch) • Envelopes: Yougatanaga 3, Nagagata 3, Kakugata 2, No.10 (COM10), Monarch, DL, ISO-C5 • Custom envelopes: 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch), 254 mm x 330.2 mm (10 inch x 13 inch) • Z-fold: A3, A4R, B4, 297 mm x 420 mm (11 inch x 17 inch), LGL, LTRR • Half-fold: A4R, LTRR **Upper tray with offset stacked** • Standard sizes: A3, A4, A4R, A5R, B4, B5, B5R, 305 x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, STMT, 8K, 16K, 16KR • Custom sizes: 139.7 mm x 182 mm to 314.8 mm x 457.2 mm (5.5 inch x 7.2 inch to 12.4 inch to 18 inch) **Lower tray without offset stacked** • Standard sizes: SRA3, A3, A4, A4R, B4, B5, B5R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, 8K, 16K, 16KR • Custom sizes: 182 mm x 182 mm to 330.2 mm x 487.7 mm (7.2 inch x 7.2 inch to 13 inch x 19.2 inch) • Z-fold: A3, B4, 297 mm x 420 mm (11 inch x 17 inch) **Lower tray with offset stacked** • A3, A4, A4R, B4, B5, B5R, 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, 8K, 16K, 16KR • Custom sizes: 182 mm x 182 mm to 314.8 mm x 457.2 mm (7.2 inch x 7.2 inch to 12.4 inch to 18 inch) **Staple** • Standard sizes: A3, A4, A4R, B4, B5, 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, 8K, 16K • Custom sizes: 210 mm x 182 mm to 297 mm x 431.8 mm (8.3 inch x 7.2 inch to 11 inch x 17 inch)

Specification	Value
Media type	• Upper tray without offset stacked: all media that can be fed in the internal paper trays • Upper tray with offset stacked: all media that can be fed in the internal paper trays except for labels and synthetic • Lower tray without offset stacked: all media that can be fed in the internal paper trays except for synthetic • Lower tray with offset stacked: all media that can be fed in the internal paper trays except for vellum, transparency, clear film, translucent film, labels and synthetic

Upper tray capacity	**Without offset stacked**: 1,000 sheets, 80 g/m^2 (20 lb bond); 200 sheets, 301 g/m^2 to 400 g/m^2 (110 lb cover to 148 lb cover)
	• **Envelopes**
	  • Kakugata 2: 130 mm (5.1 inch) or 300 sheets
	  • Other envelope types: 130 mm (5.1 inch) or 150 sheets
	• **Z-fold**: 30 sheets
	• **Half-fold**: 50 sheets
	• **Long sheet** when Long Sheet Tray-B is attached, 487.8 mm to 762 mm (8.3 inch to 30 inch):
	  • 64 g/m^2 to 79 g/m^2 (17 lb bond - 21 lb bond): 100 sheets
	  • 80 g/m^2 to 150 g/m^2 (22 lb bond - 40 lb bond): 50 sheets
	  • 151 g/m^2 to 300 g/m^2 (40 lb bond - 110 lb cover): 25 sheets
	  • 301 g/m^2 to 350 g/m^2 (110 lb cover - 130 lb cover): 20 sheets
	• **Long sheet** when Finisher Long Sheet Catch Tray-B is attached, 487.8 mm to 762 mm (8.3 inch to 30 inch):
	  • 64 g/m^2 to 150 g/m^2 (17 lb bond - : 300 sheets
	  • 151 g/m^2 to 300 g/m^2 (40 lb bond - 110 lb cover): 150 sheets
	  • 301 g/m^2 to 350 g/m^2 (110 lb cover - 130 lb cover): 120 sheets
	• **Long sheet** when Finisher Long Sheet Catch Tray-B is attached, 762 mm to 1,300 mm (30 inch to 51.1 inch):
	  • 10 sheets
	
	With offset stacked: 1,000 sheets, 80 g/m^2 (20 lb bond); 200 sheets, 301 g/m^2 to 400 g/m^2 (110 lb cover to 148 lb cover)
	• **Z-fold**: 30 sheets
	• **Half-fold**: 50 sheets

Lower tray capacity	A4, B5, LTR, EXEC: 2,000 sheets, 80 g/m^2 (20 lb bond)
	Other sizes: 1,000 sheets, 80 g/m^2 (20 lb bond)
	Media weight between 301 g/m^2 and 400 g/m^2 (110 lb cover and 148 lb cover): 200 sheets
	• **Z-fold**: 30 sheets
	• **High volume mode**:
	  • A4, B5, LTR, EXEC: 4,000 sheets
	  • A4R, LTRR, B5R: 2000 sheets
	  • A3, B4, LGL: 1,500 sheets; coated media: 1000 sheets
	  • SRA3, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch): 1000 sheets

Specification	Value

Maximum stapling capacity	**A4, B5, LTR, EXEC, 16K**
	• Uncoated media
	  • 60 g/m² to 80 g/m² (16 lb bond to 20 lb bond): 100 sheets
	  • 80.1 g/m² to 81.4 g/m² (20 lb bond to 21.6 lb bond): 80 sheets
	  • 81.5 g/m² to 105 g/m² (21.6 lb bond to 27.9 lb bond): 60 sheets
	  • 105.1 g/m² to 200 g/m² (27.9 lb bond to 74 lb cover): 20 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.
	• Coated media
	  • 70 g/m² to 105.9 g/m² (18.6 lb bond - 28.2 lb bond): 40 sheets
	  • 106 g/m² to 200 g/m² (28.2 lb bond - 74 lb cover): 15 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.
	
	A3, A4R, B4, B5R, 297 mm x 420 mm (11 inch x 17 inch), LGL, LTRR, 8K, 16KR
	• Uncoated media
	  • 60 g/m² to 81.4 g/m² (16 lb bond to 21.6 lb bond): 50 sheets
	  • 81.5 g/m² to 105 g/m² (21.6 lb bond to 27.9 lb bond): 30 sheets
	  • 105.1 g/m² to 200 g/m² (27.9 lb bond to 74 lb cover): 10 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.
	• Coated media
	  • 70 g/m² to 105.9 g/m² (18.6 lb bond - 28.2 lb bond): 30 sheets
	  • 106 g/m² to 200 g/m² (28.2 lb bond - 74 lb cover): 10 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.

Punching	**Media size**
	• Puncher Unit-BT
	  • 2-hole punch: A3, A4, A4R, A5, B4, B5, B5R, Custom (182 mm x 182 mm to 297 mm x 432 mm; 7.2 inch x 7.2 inch to 11 inch x 17 inch)
	  • 4-hole punch: A3, A4, A5, B4, B5, Custom (257 mm x 182 mm to 297 mm x 432 mm; 10.1 inch x 7.2 inch to 11 inch x 17 inch)
	• Puncher Unit-BU
	  • 4-hole punch: A3, A4, A5, B4, B5, 297 mm x 420 mm (11 inch x 17 inch), LTR, EXEC, 8K, 16K, Custom (257 mm x 182 mm to 297 mm x 432 mm; 10.1 inch x 7.2 inch to 11 inch x 17 inch)
	
	Media type
	• Thin, plain, recycled, color, heavy (106 g/m² to 300 g/m²; 28 lb bond to 110 lb cover), coated (70 g/m² to 209 g/m²; 18.6 lb bond to 77 lb cover), bond, letterhead, index

Power requirement	120-240 V AC, 50/60 Hz, 8 A
Power consumption	Maximum 500 W
Dimensions (W x D x H)	800 mm x 792 mm x 1,239 mm (31.5 inch x 31.2 inch x 48.8 inch)
Weight	Approximately 133 kg (293 lb)

Booklet Finisher-AF

Specification	Value
Description	Stacker/stapler with (offset) stacking, stapling, folding and punch-ing
Media size	**Upper tray without offset stacked** • Standard sizes: SRA3, A3, A4, A4R, A5, A5R, B4, B5, B5R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, STMT, 8K, 16K, 16KR • Custom sizes: 100 mm x 148 mm to 330.2 mm x 487.7 mm (3.9 inch x 5.8 inch to 13 inch x 19.2 inch) • Long sheet: 210 mm x 487.8 mm to 330.2 mm x 1,300 mm (8.3 inch x 19.2 inch to 13 inch x 30 inch) • Envelopes: Yougatanaga 3, Nagagata 3, Kakugata 2, No.10 (COM10), Monarch, DL, ISO-C5 • Custom envelopes: 152.4 mm x 228.6 mm (6 inch x 9 inch), 228.6 mm x 304.8 mm (9 inch x 12 inch), 254 mm x 330.2 mm (10 inch x 13 inch) • Z-fold: A3, A4R, B4, 297 mm x 420 mm (11 inch x 17 inch), LGL, LTRR • Half-fold: A4R, LTRR **Upper tray with offset stacked** • Standard sizes: A3, A4, A4R, A5R, B4, B5, B5R, 305 x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, STMT, 8K, 16K, 16KR • Custom sizes: 139.7 mm x 182 mm to 314.8 mm x 457.2 mm (5.5 inch x 7.2 inch to 12.4 inch to 18 inch) **Lower tray without offset stacked** • Standard sizes: SRA3, A3, A4, A4R, B4, B5, B5R, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, 8K, 16K, 16KR • Custom sizes: 182 mm x 182 mm to 330.2 mm x 487.7 mm (7.2 inch x 7.2 inch to 13 inch x 19.2 inch) • Z-fold: A3, B4, 297 mm x 420 mm (11 inch x 17 inch) **Lower tray with offset stacked** • A3, A4, A4R, B4, B5, B5R, 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, 8K, 16K, 16KR • Custom sizes: 182 mm x 182 mm to 314.8 mm x 457.2 mm (7.2 inch x 7.2 inch to 12.4 inch to 18 inch) **Staple** • Standard sizes: A3, A4, A4R, B4, B5, 297 mm x 420 mm (11 inch x 17 inch), LGL, LTR, LTRR, EXEC, 8K, 16K • Custom sizes: 210 mm x 182 mm to 297 mm x 431.8 mm (8.3 inch x 7.2 inch to 11 inch x 17 inch)

Specification	Value
Media type	• Upper tray without offset stacked: all media that can be fed in the internal paper trays • Upper tray with offset stacked: all media that can be fed in the internal paper trays except for labels and synthetic • Lower tray without offset stacked: all media that can be fed in the internal paper trays except for synthetic • Lower tray with offset stacked: all media that can be fed in the internal paper trays except for vellum, transparency, clear film, translucent film, labels and synthetic

Upper tray capacity	**Without offset stacked**: 1,000 sheets, 80 g/m² (20 lb bond); 200 sheets, 301 g/m² to 400 g/m² (110 lb cover to 148 lb cover)
	• **Envelopes**
	  • Kakugata 2: 130 mm (5.1 inch) or 300 sheets
	  • Other envelope types: 130 mm (5.1 inch) or 150 sheets
	• **Z-fold**: 30 sheets
	• **Half-fold**: 50 sheets
	• **Long sheet** when Long Sheet Tray-B is attached, 487.8 mm to 762 mm (8.3 inch to 30 inch):
	  • 64 g/m² to 79 g/m² (17 lb bond - 21 lb bond): 100 sheets
	  • 80 g/m² to 150 g/m² (22 lb bond - 40 lb bond): 50 sheets
	  • 151 g/m² to 300 g/m² (40 lb bond - 110 lb cover): 25 sheets
	  • 301 g/m² to 350 g/m² (110 lb cover - 130 lb cover): 20 sheets
	• **Long sheet** when Finisher Long Sheet Catch Tray-B is attached, 487.8 mm to 762 mm (8.3 inch to 30 inch):
	  • 64 g/m² to 150 g/m² (17 lb bond - : 300 sheets
	  • 151 g/m² to 300 g/m² (40 lb bond - 110 lb cover): 150 sheets
	  • 301 g/m² to 350 g/m² (110 lb cover - 130 lb cover): 120 sheets
	• **Long sheet** when Finisher Long Sheet Catch Tray-B is attached, 762 mm to 1,300 mm (30 inch to 51.1 inch):
	  • 10 sheets
	
	With offset stacked: 1,000 sheets, 80 g/m² (20 lb bond); 200 sheets, 301 g/m² to 400 g/m² (110 lb cover to 148 lb cover)
	• **Z-fold**: 30 sheets
	• **Half-fold**: 50 sheets

Lower tray capacity	A4, B5, LTR, EXEC: 2,000 sheets, 80 g/m² (20 lb bond)
	Other sizes: 1,000 sheets, 80 g/m² (20 lb bond)
	Media weight between 301 g/m² and 400 g/m² (110 lb cover and 148 lb cover): 200 sheets
	• **Z-fold**: 30 sheets
	• **High volume mode**:
	  • A4, B5, LTR, EXEC: 4,000 sheets
	  • A4R, LTRR, B5R: 2000 sheets
	  • A3, B4, LGL: 1,500 sheets; coated media: 1000 sheets
	  • SRA3, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch): 1000 sheets

Specification	Value

Maximum stapling capacity	**A4, B5, LTR, EXEC, 16K**
	• Uncoated media
	  • 60 g/m² to 80 g/m² (16 lb bond to 20 lb bond): 100 sheets
	  • 80.1 g/m² to 81.4 g/m² (20 lb bond to 21.6 lb bond): 80 sheets
	  • 81.5 g/m² to 105 g/m² (21.6 lb bond to 27.9 lb bond): 60 sheets
	  • 105.1 g/m² to 200 g/m² (27.9 lb bond to 74 lb cover): 20 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.
	• Coated media
	  • 70 g/m² to 105.9 g/m² (18.6 lb bond - 28.2 lb bond): 40 sheets
	  • 106 g/m² to 200 g/m² (28.2 lb bond - 74 lb cover): 15 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.
	
	A3, A4R, B4, B5R, 297 mm x 420 mm (11 inch x 17 inch), LGL, LTRR, 8K, 16KR
	• Uncoated media
	  • 60 g/m² to 81.4 g/m² (16 lb bond to 21.6 lb bond): 50 sheets
	  • 81.5 g/m² to 105 g/m² (21.6 lb bond to 27.9 lb bond): 30 sheets
	  • 105.1 g/m² to 200 g/m² (27.9 lb bond to 74 lb cover): 10 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.
	• Coated media
	  • 70 g/m² to 105.9 g/m² (18.6 lb bond - 28.2 lb bond): 30 sheets
	  • 106 g/m² to 200 g/m² (28.2 lb bond - 74 lb cover): 10 sheets
	  • Media heavier than 200 g/m² (74 lb cover) can only be used for front and back covers.

Saddle stitch	**Media size**: SRA3, A3, A4R, B4, 330 mm x 483 mm (13 inch x 19 inch), 305 mm x 457 mm (12 inch x 18 inch), 297 mm x 420 mm (11 inch x 17 inch), LGL, LTRR, Custom (210 mm x 279.4 mm to 330.2 mm x 487.7 mm; 8.3 inch x 11 inch to 13 inch x 19.2 inch) **Media type**: thin, plain, recycled, color, heavy (106 g/m² to 300 g/m²; 28 lb bond to 110 lb cover), coated (70 g/m² to 300 g/m²; 18.6 lb bond to 110 lb cover), bond **Media weight for body**: 60 g/m² to 220 g/m² (16 lb bond to 81 lb cover) **Media weight for cover**: 64 g/m² to 300 g/m² (17 lb bond to 110 lb cover) NOTE The media used for covers should be the as heavy or heavier than the media for the body.

Specification	Value

Maximum saddle stitch capacity	• Coated (including cover)
	  • 70 g/m^2 to 128 g/m^2 (18.6 lb bond to 34 lb bond): 2 to 10 sheets
	  • 129 g/m^2 to 209 g/m^2 (34 lb bond to 77 lb cover): 2 to 5 sheets
	  • 210 g/m^2 to 220 g/m^2 (77 lb cover to 81 lb cover): 2 to 4 sheets
	• Uncoated (including cover)
	  • 60 g/m^2 to 80 g/m^2 (16 lb bond to 20 lb bond): 2 to 25 sheets
	  • 81 g/m^2 to 105 g/m^2 (21.6 lb bond to 27.9 lb bond): 2 to 15 sheets
	  • 106 g/m^2 to 209 g/m^2 (28.2 lb bond to 77 lb cover): 2 to 5 sheets
	  • 210 g/m^2 to 220 g/m^2 (77 lb cover to 81 lb cover): 2 to 4 sheets
	• Saddle fold
	  • 60 g/m^2 to 209 g/m^2 (16 lb bond to 77 lb cover): 5 sheets
	  • 210 g/m^2 and heavier (77 lb cover): 1 sheet

Punching	**Media size**
	• Puncher Unit-BT
	  • 2-hole punch: A3, A4, A4R, A5, B4, B5, B5R, Custom (182 mm x 182 mm to 297 mm x 432 mm; 7.2 inch x 7.2 inch to 11 inch x 17 inch)
	  • 4-hole punch: A3, A4, A5, B4, B5, Custom (257 mm x 182 mm to 297 mm x 432 mm; 10.1 inch x 7.2 inch to 11 inch x 17 inch)
	• Puncher Unit-BU
	  • 4-hole punch: A3, A4, A5, B4, B5, 297 mm x 420 mm (11 inch x 17 inch), LTR, EXEC, 8K, 16K, Custom (257 mm x 182 mm to 297 mm x 432 mm; 10.1 inch x 7.2 inch to 11 inch x 17 inch)
	Media type
	• Thin, plain, recycled, color, heavy (106 g/m^2 to 300 g/m^2; 28 lb bond to 110 lb cover), coated (70 g/m^2 to 209 g/m^2; 18.6 lb bond to 77 lb cover), bond, letterhead, index

Power requirement	120-240 V AC, 50/60 Hz, 8 A
Power consumption	Maximum 500 W
Dimensions (W x D x H)	800 mm x 792 mm x 1,239 mm (31.5 inch x 31.2 inch x 48.8 inch)
Weight	Approximately 183 kg (403 lb)

Finisher Bridge-A

Specification	Value
Description	With the bridge, you have the benefits of having a stacker/stapler and DFD finishing

Specification	Value
Media size	Up to 76.2 cm (30 inch) Folded sheets and envelopes are not supported. The [Finisher adapter] turns the sheets, therefore, it is not possible to use tabs. When you use oriented media or prepunched media, the print performance of duplex jobs becomes 40% lower for A4-like media and 50% lower for A3-like media.
Media weight	52 g/m² - 300 g/m² (14 lb bond - 110 lb cover)
Power supply	Power from the finisher adapter
Safety	Compliant with OTS8002, TüV-GS, c-UL-us, CSA
Dimensions (W x D x H)	600 mm x 745 mm x 150 mm (23.6 inch x 29.3 inch x 5.9 inch)
Weight	Approximately 30 kg (66 lb)

DFD Adapter-B

Specification	Value
Description	Finisher adapter to connect a wide range of DFD finishers at lower costs and footprint. No need to use a stacker to connect DFD finishers.
Media size	Up to 76.2 cm (30 inch) Folded sheets and envelopes are not supported. The [Finisher adapter] turns the sheets, therefore, it is not possible to use tabs. When you use oriented media or prepunched media, the print performance of duplex jobs becomes 40% lower for A4-like media and 50% lower for A3-like media.
Media weight	52 g/m² - 300 g/m² (14 lb bond - 110 lb cover)
3rd party finishing interface height	Default output height to support downstream Canon finishers: 860 mm (33.9 inch). With the optional Height-kit the DFD output-height is 1,002 mm (39.4 inch).
Power supply	Own power supply 90 - 264 V, 47 - 63 Hz
Safety	Compliant with OTS8002, TüV-GS, c-UL-us, CSA
Dimensions (W x D x H)	300 mm x 745 mm x 1040 mm (11.8 inch x 29.3 inch x 40.9 inch)
Weight	Approximately 60 kg (132 lb)

Settings Editor specifications

URL to access the Settings Editor	*http(s)://<IP address>* or *http(s)://<hostname>*
Recommended resolution of your computer screen	Minimum 640 x 480 pixels
Supported web browsers	• Microsoft® Internet Explorer® 11 • Microsoft® Edge 101 and higher • Mozilla Firefox® 100 and higher • Google Chrome™ 101 and higher • Apple Safari® 15.5 and higher • Microsoft® Edge Chromium 87 and higher
Language	As defined in your browser as long as the printer supports this language

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Canon Latin America Inc.
cla.canon.com

Canon Australia Pty. Ltd.
canon.com.au

Canon China Co. Ltd.
canon.com.cn

Canon Singapore Pte. Ltd.
sg.canon

Canon Hongkong Co. Ltd.
hk.canon

Canon Production Printing Systems Inc.
cweb.canon.jp/eng/corporate

Canon Korea Inc.
kr.canon